arXiv:1704.03477v2 [cs.NE] 16 Apr 2017

A Neural Representation of Sketch Drawings

David Ha
Google Brain
hadavid@google.com

Douglas Eck
Google Brain
deck@google.com

Abstract

We present `sketch-rnn`, a recurrent neural network (RNN) able to construct stroke-based drawings of common objects. The model is trained on thousands of crude human-drawn images representing hundreds of classes. We outline a framework for conditional and unconditional sketch generation, and describe new robust training methods for generating coherent sketch drawings in a vector format.

1 Introduction

Recently, there have been major advancements in generative modelling of images using neural networks as a generative tool. Generative Adversarial Networks (GANs) [7], Variational Inference (VI) [19], and Autoregressive (AR) [24] models have become popular tools in this fast growing area. Most of the work, however, has been targeted towards modelling rasterized images represented as a two dimensional grid of pixel values. While these models are currently able to generate realistic, low resolution pixel images, a key challenge for many of these models is to generate images with coherent structure. For example, these models may produce amusing images of cats with three or more eyes, or dogs with multiple heads [7]. In this paper we investigate a much lower-dimensional vector-based representation inspired by how people draw.

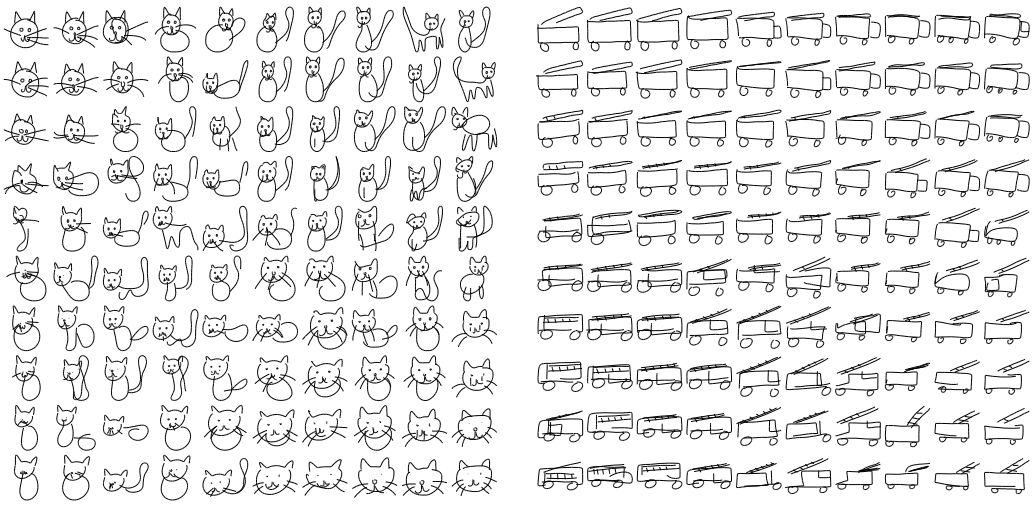


Figure 1: Latent space of cats and firetrucks generated by `sketch-rnn`.

As humans, we do not understand the world as a grid of pixels, but rather develop abstract concepts to represent what we see. From a young age, we develop the ability to communicate what we see by drawing on paper with a pencil or crayon. In this way we learn to represent an image based on a short sequence of strokes. For example, an archetypical child-drawn house – i.e. a triangle on top of

a square with door and window added – can be achieved with a few line strokes. Drawings like this may not resemble reality as captured by a photograph, but they do tell us something about how people represent and reconstruct images of the world around them. Objects such as man, woman, eyes, face, cat, dog, etc. can be clearly communicated with simple drawings. Arguably even emotions can be conveyed via line drawings made from only a few strokes. Finally, abstract visual communication is a key part of how humans communicate with each other, with many writing systems composed of symbols that originated from simple drawings known as pictograms.

Our goal is to train machines to learn to draw and generalize abstract concepts in a manner similar to humans. As a first step towards this goal, in this work we train neural networks to learn from a dataset of crude hand drawn sketches. The sketches are represented in a vector format, not as pixels, and are encoded as a list of discrete pen strokes. By training a generative recurrent neural network to model discrete pen strokes, we can observe the limits of its ability to learn a generative representation of an object that is both compact (only a few pen strokes) and in some ways abstract in that these stroke sequences capture many different strategies for drawing the same object. For example, see Figure 1 (Left), where the model learns several very different ways to draw a cat.

This paper makes the following contributions: We outline a framework for both unconditional and conditional generation of vector images composed of a sequence of lines. Our recurrent neural network-based generative model is capable of producing sketches of common objects in a vector format. We develop a training procedure unique to vector images to make the training more robust. In the conditional generation model, we explore the latent space developed by the model to represent a vector image. We demonstrate that enforcing a prior on the latent space helps the model generate more coherent images. We also discuss potential creative applications of our methodology. We are working towards making available a large dataset of simple hand drawings to encourage further development of generative models, and we will release an implementation of our model as an open source project called `sketch-rnn`.

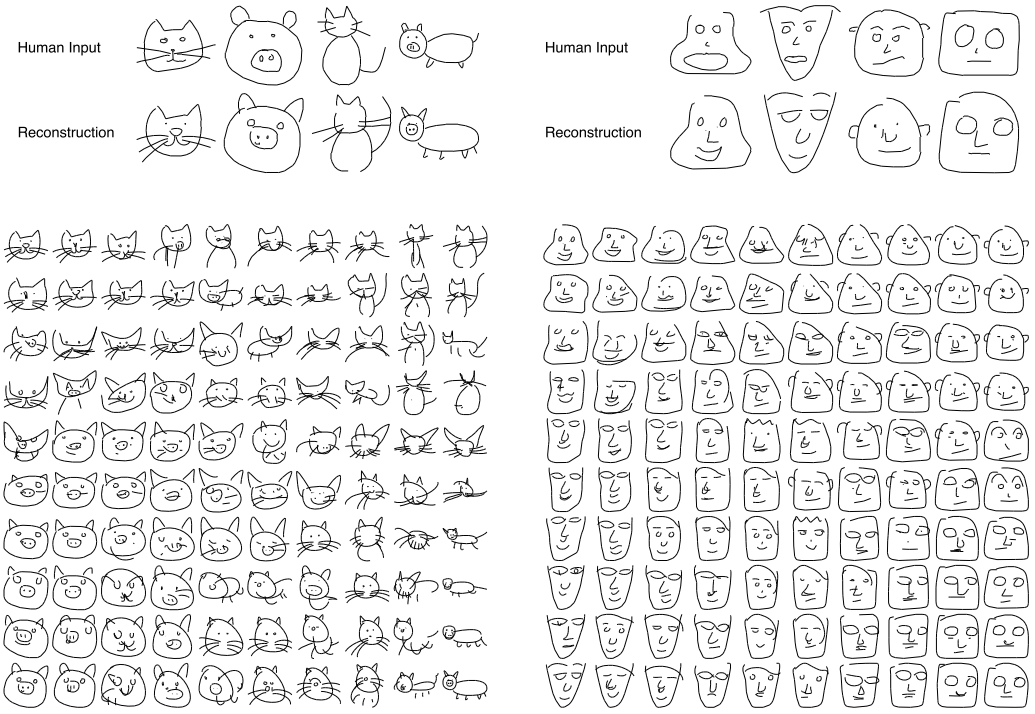


Figure 2: Example input sketches and `sketch-rnn` generated reproductions (Top), Latent space interpolation between the four reproduced sketches (Bottom).

2 Related Work

There is a long history of work related to algorithms that mimic painters. One such work is Portrait Drawing by Paul the Robot [28, 30], where an underlying algorithm controlling a mechanical robot arm sketches lines on a canvas with a programmable artistic style to mimic a given digitized portrait of a person. Reinforcement Learning based-approaches [30] have been developed to discover a set of paint brush strokes that can best represent a given input photograph. These prior works generally attempt to mimic digitized photographs, rather than develop generative models of vector images.

Neural Network-based approaches have been developed for generative models of images, although the majority of neural network-related research on image generation deal with pixel images [7, 13, 15, 18, 24, 29]. There has been relatively little work done on vector image generation using neural networks. Graves' original work on handwriting generation with recurrent neural networks [8] laid the groundwork for utilizing mixture density networks [2] to generate continuous data points. Recent works of this approach attempted to generate vectorized Kanji characters unconditionally [9] and conditionally [31] by modelling Chinese characters as a sequence of pen stroke actions.

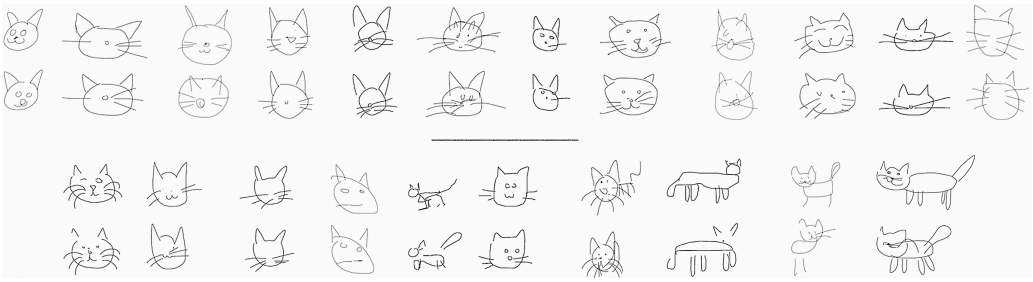


Figure 3: Human sketches of cats, and their generated reconstructions below each human sketch.

In addition to unconditionally generating sketches, we also explore encoding existing sketches into a latent space of embedding vectors. We can then conditionally generate sketches based on such latent vectors, like the examples in Figure 3. Previous work [3] outlined a methodology to combine Sequence-to-Sequence models with a Variational Autoencoder to model natural English sentences in latent vector space. A related work [20], utilizes probabilistic program induction, rather than neural networks, to perform one-shot modelling of the Omniglot dataset containing vector images of simple symbols.

One of the factors limiting research development in the space of generative vector drawings is the lack of publicly available datasets. Previously, the Sketch dataset [6], consisting of 20K vector sketches, was used to explore feature extraction techniques. A subsequent work, the Sketchy dataset [25], provided 70K vector sketches along with corresponding pixel images for various classes. This allowed for a larger-scale exploration of human sketches. ShadowDraw [22] is an interactive system that predicts what a finished drawing looks like based on a set of incomplete brush strokes from the user while the sketch is being drawn. ShadowDraw used a dataset of 30K raster images combined with extracted vectorized features. In this work, we use a much larger dataset of vector sketches that will be made publicly available.

3 Methodology

3.1 Dataset

We constructed a dataset from sketch data obtained from The Quickdraw A.I. Experiment [14], an online demo where the users are asked to draw objects belonging to a particular object class in less than 20 seconds. We have selected 75 classes from the raw dataset to construct the `quickdraw-75` dataset. Each class consists of a training set of 70K samples, in addition to 2.5K samples each for validation and test sets.

In our experiments, we attempt to model the entire dataset, in addition to modelling individual classes or multiple classes selected out of the 75 classes. We are currently looking at ways to release the full version of this dataset to the research community.

3.2 Data Format

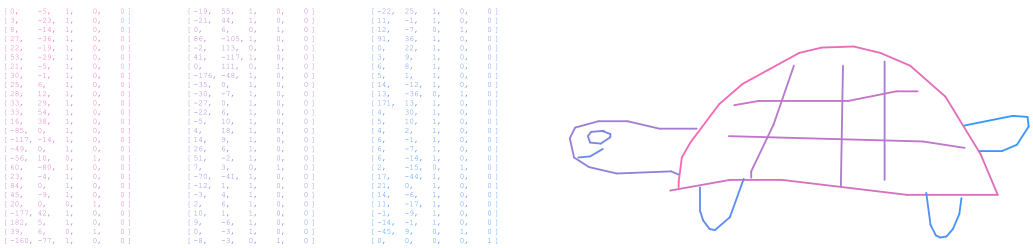


Figure 4: A sample sketch, as a sequence of $(\Delta x, \Delta y, p_1, p_2, p_3)$ points and in rendered form. In the rendered sketch, the line color corresponds to the sequential stroke ordering.

We use a data format that represents a sketch as a set of pen stroke actions. This representation is an extension of the handwriting format used in [8]. Our format extends the binary pen stroke event into a multi-state event. In this data format, the initial absolute coordinate of the drawing is located at the origin.

Figure 4 shows an example. A sketch is a list of points, and each point is a vector consisting of 5 elements: $(\Delta x, \Delta y, p_1, p_2, p_3)$. The first two elements are the offset distance in the x and y directions of the pen from the previous point. The last 3 elements represents a binary one-hot vector of 3 possible states. The first pen state, p_1, indicates that the pen is currently touching the paper, and that a line will be drawn connecting the next point with the current point. The second pen state, p_2, indicates that the pen will be lifted from the paper after the current point, and that no line will be drawn next. The final pen state, p_3, indicates that the drawing has ended, and subsequent points, including the current point, will not be rendered.

3.3 `sketch-rnn`

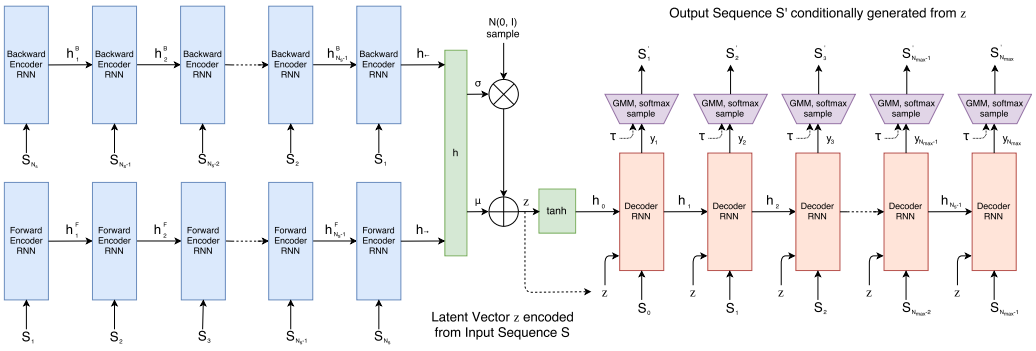


Figure 5: Schematic diagram of `sketch-rnn`.

Our model is a Sequence-to-Sequence Variational Autoencoder (VAE), similar to the architecture described in [3, 19]. Our encoder is a bidirectional RNN [26] that takes in a sketch as an input, and outputs a latent vector of size N_z. Specifically, we feed the sketch sequence, S, and also the same sketch sequence in reverse order, $S_{\texttt{reverse}}$, into two encoding RNNs that make up the bidirectional RNN, to obtain two final hidden states:

$$\begin{aligned} h_{\rightarrow} &= \texttt{encode}_{\rightarrow}(S) \\ h_{\leftarrow} &= \texttt{encode}_{\leftarrow}(S_{\texttt{reverse}}) \\ h &= [\, h_{\rightarrow} \;;\; h_{\leftarrow} \,] \end{aligned} \tag{1}$$

We take this final concatenated hidden state, h, and project it into two vectors μ and $\hat{\sigma}$, each of size N_z, using a fully connected layer. We convert $\hat{\sigma}$ into a non-negative standard deviation parameter σ using an exponential operation. We use μ and σ, along with $\mathcal{N}(0, I)$, a vector of IID Gaussian

variables of size N_z, to construct a random vector, $z \in \mathbb{R}^{N_z}$, as in the approach for a Variational Autoencoder [19]:

$$\begin{aligned} \mu &= W_\mu h + b_\mu \\ \hat{\sigma} &= W_\sigma h + b_\sigma \\ \sigma &= \exp\left(\frac{\hat{\sigma}}{2}\right) \\ z &= \mu + \sigma \odot \mathcal{N}(0, I) \end{aligned} \tag{2}$$

Under this encoding scheme, the latent vector z is not a deterministic output for a given input sketch, but a random vector conditioned on the input sketch.

Our decoder is an autoregressive RNN that samples output sketches conditional on a given latent vector z. The initial hidden states h_0, and optional cell states c_0 (if applicable) of the decoder RNN is the output of a single layer network:

$$[\ h_0\ ;\ c_0\ ] = \tanh(W_z z + b_z) \tag{3}$$

At each step i of the decoder RNN, we feed the previous point, S_{i-1} and the latent vector z in as a concatenated input x_i. Note that S_0 is defined as $(0, 0, 1, 0, 0)$. The output at each time step are the parameters for a probability distribution of the next data point S_i. In Equation 4, we model $(\Delta x, \Delta y)$ as a Gaussian Mixture Model (GMM) with M normal distributions as in [2, 8], and (q_1, q_2, q_3) as a categorical distribution to model the ground truth data (p_1, p_2, p_3), where $(q_1 + q_2 + q_3 = 1)$ as done in [9] and [31]. Unlike Graves [8], our generated sequence is conditioned from a latent code z sampled from our encoder, which is trained end-to-end alongside the decoder.

$$p(\Delta x, \Delta y) = \sum_{j=1}^{M} \Pi_j\, \mathcal{N}(\Delta x, \Delta y \mid \mu_{x,j}, \mu_{y,j}, \sigma_{x,j}, \sigma_{y,j}, \rho_{xy,j}),\ \text{where}\ \sum_{j=1}^{M} \Pi_j = 1 \tag{4}$$

$\mathcal{N}(x, y|\mu_x, \mu_y, \sigma_x, \sigma_y, \rho_{xy})$ is the probability distribution function for a Bivariate Normal distribution. Each of the M Bivariate Normal distributions consist of five parameters: $(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho_{xy})$, where μ_x and μ_y are the means, σ_x and σ_y are the standard deviations, and ρ_{xy} is the correlation parameter of each Bivariate Normal distribution. An additional vector Π of length M, also a categorical distribution, are the mixture weights of the Gaussian Mixture Model. Hence the size of the output vector y is $5M + M + 3$, which includes the 3 logits needed to generate (q_1, q_2, q_3).

The next hidden state of the RNN, generated with its forward operation, projects into the output vector y_i using a fully-connected layer:

$$\begin{aligned} x_i &= [\ S_{i-1}\ ;\ z\ ] \\ [\ h_i\ ;\ c_i\ ] &= \texttt{forward}(x_i,\ [\ h_{i-1}\ ;\ c_{i-1}\ ]) \\ y_i &= W_y h_i + b_y,\ \text{where}\ \ y_i \in \mathbb{R}^{6M+3} \end{aligned} \tag{5}$$

The vector y_i is broken down into the parameters of the probability distribution of the next data point:

$$[\ (\hat{\Pi}_1\ \mu_x\ \mu_y\ \hat{\sigma}_x\ \hat{\sigma}_y\ \hat{\rho}_{xy})_1\ ...\ (\hat{\Pi}_1\ \mu_x\ \mu_y\ \hat{\sigma}_x\ \hat{\sigma}_y\ \hat{\rho}_{xy})_M\ (\hat{q}_1\ \hat{q}_2\ \hat{q}_3)\ ] = y_i \tag{6}$$

As in [8], we apply $\exp$ and $\tanh$ operations to ensure the standard deviation values are non-negative, and that the correlation value is between -1 and 1:

$$\begin{aligned} \sigma_x &= \exp(\hat{\sigma}_x) \\ \sigma_y &= \exp(\hat{\sigma}_y) \\ \rho_{xy} &= \tanh(\hat{\rho}_{xy}) \end{aligned} \tag{7}$$

The probabilities for the categorical distributions are calculated using the outputs as logit values:

$$q_k = \frac{\exp(\hat{q}_k)}{\sum_{j=1}^{3} \exp(\hat{q}_j)}, k \in \{1,\ 2,\ 3\}$$
$$\Pi_k = \frac{\exp(\hat{\Pi}_k)}{\sum_{j=1}^{M} \exp(\hat{\Pi}_j)}, k \in \{1,\ \ldots,\ M\} \tag{8}$$

A key challenge is to train our model to know when to stop drawing. Because the probabilities of the three pen stroke events are highly unbalanced, the model becomes more difficult to train. The probability of a p_1 event is much higher than p_2, and the p_3 event will only happen once per drawing. The approach developed in [9] and later followed by [31] was to use different weightings for each pen event when calculating the losses, such as a hand-tuned weighting of $(1, 10, 100)$. We find this approach to be inelegant and inadequate for our dataset of diverse image classes.

We develop a simpler, more robust approach that works well for a broad class of sketch drawing data. In our approach, all sequences are generated to a length of $N_{\max}$ where $N_{\max}$ is the length of the longest sketch in our training dataset. In principle $N_{\max}$ can be considered a hyper parameter. As the length of S is usually shorter than $N_{\max}$, we set S_i to be $(0, 0, 0, 0, 1)$ for $i > N_s$. We discuss the training in detail in the next section.

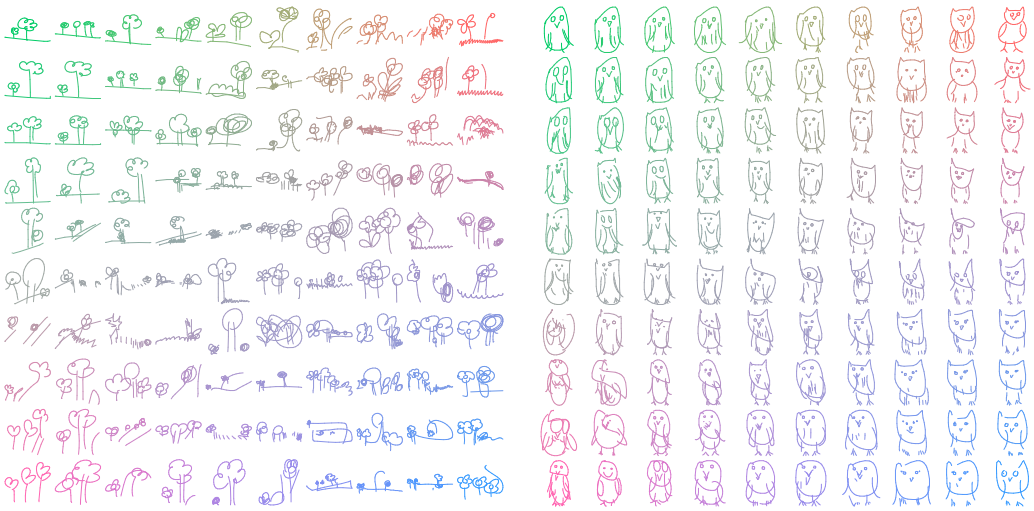


Figure 6: Latent space interpolation between four generated gardens and owls.
We also interpolate between four distinct colors for visual effect.

After training, we can sample sketches from our model. During the sampling process, we generate the parameters for both GMM and categorical distributions at each time step, and sample an outcome S'_i for that time step. Unlike the training process, we feed the sampled outcome S'_i, rather than the ground truth S_i, as the input for the next time step. We continue to sample until $p_3 = 1$, or when we have reached $i = N_{\max}$. Like the encoder, the sampled output is not deterministic, but a random sequence, conditioned on the input latent vector z. We can control the level of randomness we would like our samples to have during the sampling process by introducing a temperature parameter τ:

$$
\begin{aligned}
q_k &= \frac{\exp(\frac{\hat{q}_k}{\tau})}{\sum_{j=1}^{3}\exp(\frac{\hat{q}_j}{\tau})}, j \in \{1,\ 2,\ 3\} \\
\Pi_k &= \frac{\exp(\frac{\hat{\Pi}_k}{\tau})}{\sum_{j=1}^{M}\exp(\frac{\hat{\Pi}_j}{\tau})}, k \in \{1,\ \ldots,\ M\} \\
\sigma_x^2 &\rightarrow \sigma_x^2\tau \\
\sigma_y^2 &\rightarrow \sigma_y^2\tau
\end{aligned}
\tag{9}
$$

We can scale the softmax parameters of the categorial distribution and also the σ parameters of the Bivariate Normal distribution by a temperature parameter τ, to control the level of randomness in our samples. τ is typically set between 0 and 1. In the limiting case as $\tau \rightarrow 0$, our model becomes deterministic and samples will consist of the most likely point in the probability density function. Figure 7 illustrates of effect of sampling sketches with various temperature parameters.

3.4 Unconditional Generation

Figure 7: Unconditional sketch generation of firetrucks and yoga positions with varying τ.

As a special case, we can also train our model to generate sketches unconditionally, where we only train the decoder RNN module, without any input or latent vectors. By removing the encoder, the decoder RNN as a standalone model is an autoregressive model without latent variables. In this use case, the initial hidden states and cell states of the decoder RNN are initialized to zero. The inputs x_i of the decoder RNN at each time step is only S_{i-1} or S'_{i-1}, as we do not need to concatenate a latent vector z. In Figure 7, we sample various sketch images generated unconditionally by varying the temperature parameter from $\tau = 0.2$ at the top in blue, to $\tau = 0.9$ at the bottom in red.

3.5 Training

Our training procedure follows the approach of the Variational Autoencoder [19], where the loss function is the sum of two terms: the Reconstruction Loss and the Kullback-Leibler divergence loss. We train our model to optimize this two-part loss function. The Reconstruction Loss term, described in Equation 10, maximizes the log-likehood of the generated probability distribution to explain the training data S. We can calculate this reconstruction loss, L_R, using the generated parameters of the

pdf and the training data S. L_R is composed of the sum of the log loss of the offset terms $(\Delta x, \Delta y)$, denoted as L_s, and the log loss of the pen state terms (p_1, p_2, p_3):

$$
\begin{aligned}
L_s &= -\frac{1}{N_{\max}} \sum_{i=1}^{N_s} \log \Big(\sum_{j=1}^{M} \Pi_{j,i}\, \mathcal{N}(\Delta x_i, \Delta y_i \mid \mu_{x,j,i}, \mu_{y,j,i}, \sigma_{x,j,i}, \sigma_{y,j,i}, \rho_{xy,j,i}) \Big) \\
L_p &= -\frac{1}{N_{\max}} \sum_{i=1}^{N_{\max}} \sum_{k=1}^{3} p_{k,i} \log(q_{k,i}) \\
L_R &= L_s + L_p
\end{aligned}
\tag{10}
$$

Note that we discard the pdf parameters modelling the $(\Delta x, \Delta y)$ points beyond N_s when calculating L_s, while L_p is calculated using all of the pdf parameters modelling the (p_1, p_2, p_3) points until $N_{\max}$. Both terms are normalized by the total sequence length $N_{\max}$. We found this methodology of loss calculation to be more robust and allows the model to easily learn when it should stop drawing, unlike the earlier mentioned method of assigning importance weightings to p_1, p_2, and p_3.

The Kullback-Leibler (KL) divergence loss term measures the difference between the distribution of our latent vector z, to that of an IID Gaussian vector with zero mean and unit variance. Optimizing for this loss term allows us to minimize this difference. We use the result in [19], and calculate the KL loss term, L_{KL}, normalized by number of dimensions N_z of the latent vector:

$$
L_{KL} = -\frac{1}{2N_z}\Big(1 + \hat{\sigma} - \mu^2 - \exp(\hat{\sigma})\Big) \tag{11}
$$

The loss function in Equation 12 is a weighted sum of the reconstruction loss term L_R and the KL loss term L_{KL}:

$$
Loss = L_R + w_{KL} L_{KL} \tag{12}
$$

There is a tradeoff between optimizing for one term over the other. As $w_{KL} \to 0$, our model approaches a pure autoencoder, sacrificing the ability to enforce a prior over our latent space while obtaining better reconstruction loss metrics. Note that for unconditional generation, where our model is the standalone decoder, there will not be a L_{KL} term as we only optimize for L_R.

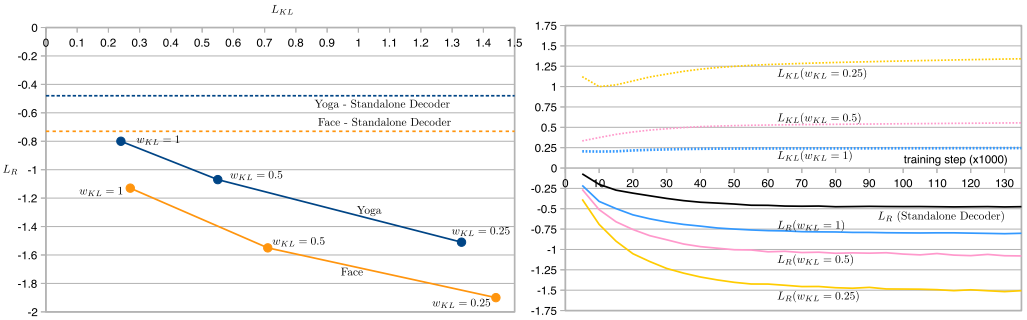


Figure 8: Tradeoff between L_R and L_{KL}, for two models trained on single class datasets (Left). Validation Loss Graph for models trained on the Yoga dataset using various w_{KL}. (Right)

Figure 8 illustrates the tradeoff between different settings of w_{KL} and the resulting L_R and L_{KL} metrics on the test set, along with the L_R metric on a standalone decoder RNN for comparison. As the unconditional model does not receive any prior information about the entire sketch it needs to generate, the L_R metric for the standalone decoder model serves as an upper bound for various conditional models using a latent vector. In our experiments, we explore more closely the qualitatively implications of this tradeoff for sketch generation.

While the loss function in Equation 12 can be used during training, we find that annealing the KL term in the loss function (Equation 13) produced better results. This modification is only used for

model training, and the original loss function in Equation 12 is still used to evaluate validation and test sets, and for early stopping.

$$\begin{aligned} \eta_{\texttt{step}} &= 1 - (1 - \eta_{\texttt{min}})R^{\texttt{step}} \\ Loss_{\texttt{train}} &= L_R + w_{KL}\ \eta_{\texttt{step}}\ \max(L_{KL}, KL_{\texttt{min}}) \end{aligned} \tag{13}$$

We find that annealing the KL Loss term generally results in better losses. Annealing the L_{KL} term in the loss function directs the optimizer to first focus more on the reconstruction term in Equation 10, which is the more difficult loss term of the model to optimize for, before having to deal with optimizing for the KL loss term in Equation 11, a far simpler expression in comparison. This approach has been used in [3, 15, 18]. Our annealing term $\eta_{\texttt{step}}$ starts at $\eta_{\texttt{min}}$ (typically 0 or 0.01) at training step 0, and converges to 1 for large training steps. R is a term close to, but less than 1.

If the distribution of z is close enough to $\mathcal{N}(0, I)$, we can sample sketches from the decoder using randomly sampled z from $\mathcal{N}(0, I)$ as the input. In practice, we find that going from a larger L_{KL} value ($L_{KL} > 1.0$) to a smaller L_{KL} value of ~ 0.3 generally results in a substantial increase in the quality of sampled images using randomly sampled $z \sim \mathcal{N}(0, I)$. However, going from $L_{KL} = 0.3$ to L_{KL} values closer to zero does not lead to any further noticeable improvements. Hence we find it useful to put a floor on L_{KL} in the loss function by enforcing $\max(L_{KL}, KL_{\texttt{min}})$ in Equation 13.

The $KL_{\texttt{min}}$ term inside the $\max$ operator is typically set to a small value such as 0.10 to 0.50. This term will encourage the optimizer to put less focus on optimizing for the KL Loss term L_{KL} once it is low enough, so we can obtain better metrics for the reconstruction loss term L_R. This approach is similar to the approach described in [18] as *free bits*, where they apply the $\max$ operator separately inside each dimension of the latent vector z.

4 Experiments

We conduct several experiments with `sketch-rnn` for both conditional and unconditional vector image generation. In our experiments, we train our model either on an individual image class, multiple image classes, or all 75 image classes. The class information is not part of the input into the model. We train the models on various settings for w_{KL} and record the breakdown of losses.

The `sketch-rnn` model treats the RNN cell as an abstract component. In our experiments, we use the Long Short-Term Memory (LSTM) [11], as the encoder RNN. For the decoder RNN, we use the HyperLSTM, as this type of RNN cell excels at sequence generation tasks [10]. The ability for HyperLSTM to spontaneously augment its own weights enables it to adapt to many different regimes in a large diverse dataset. Our encoder and decoder RNNs consist of 512 and 2048 nodes respectively.

In our model, we use $M = 20$ mixture components for the decoder RNN. The latent vector z has $N_z = 128$ dimensions. We apply Layer Normalization [1] to our model, and during training apply recurrent dropout [27] with a keep probability of 90%. We train the model with batch sizes of 100 samples, using Adam [17] with a learning rate of 0.0001 and gradient clipping of 1.0. All models are trained with $KL_{\texttt{min}} = 0.20$, $R = 0.99999$. During training, we perform simple data augmentation by multiplying the offset columns $(\Delta x, \Delta y)$ by two IID random factors chosen uniformly between 0.90 and 1.10. Unless mentioned otherwise, all experiments are conducted with $w_{KL} = 1.00$.

4.1 Experimental Results

In our experiments, we train `sketch-rnn` on datasets consisting of individual classes. To experiment with a diverse set of image classes with varying complexities, we select the cat, pig, face, firetruck, garden, owl, mermaid, mosquito, and yoga classes out of the 75 classes. We also construct datasets consisting of multiple classes, such as (cat, pig), and (crab, face, pig, rabbit), in addition to the dataset with all 75 classes. The results for test set evaluation on various datasets are displayed in Table 1.

The relative loss numbers are consistent with our expectations. We see that the reconstruction loss term L_R decreases as we relax the w_{KL} parameter controlling the weight for the KL Loss term, and meanwhile the KL Loss term L_R increases as a result. The reconstruction loss term for the conditional model is strictly less than the unconditional, standalone decoder model.

Dataset	$w_{KL} = 1.00$		$w_{KL} = 0.50$		$w_{KL} = 0.25$		Decoder Only
	L_R	L_{KL}	L_R	L_{KL}	L_R	L_{KL}	L_R
cat	-0.98	0.29	-1.33	0.70	-1.46	1.01	-0.57
pig	-1.14	0.22	-1.37	0.49	-1.52	0.80	-0.82
cat, pig	-1.02	0.22	-1.24	0.49	-1.50	0.98	-0.75
crab, face, pig, rabbit	-0.91	0.22	-1.04	0.40	-1.47	1.17	-0.67
face	-1.13	0.27	-1.55	0.71	-1.90	1.44	-0.73
firetruck	-1.24	0.22	-1.26	0.24	-1.78	1.10	-0.90
garden	-0.79	0.20	-0.81	0.25	-0.99	0.54	-0.62
owl	-0.93	0.20	-1.03	0.34	-1.29	0.77	-0.66
mermaid	-0.64	0.23	-0.91	0.54	-1.30	1.22	-0.37
mosquito	-0.67	0.30	-1.02	0.66	-1.41	1.54	-0.34
yoga	-0.80	0.24	-1.07	0.55	-1.51	1.33	-0.48
everything	-0.61	0.21	-0.79	0.46	-1.02	0.98	-0.41

Table 1: Loss figures (L_R and L_{KL}) for various w_{KL} settings.

More difficult classes, such as (mermaid, mosquito, everything), generally have higher reconstruction loss numbers compared to the easier classes, such as (pig, face, firetruck). However, these numbers can only be used approximately to compare different classes, as the maximum number of strokes parameter $N_{\max}$ is different for the various datasets, which will affect the definition of L_R in our framework. For instance, the pig dataset has $N_{\max} = 182$, while the everything dataset has $N_{\max} = 200$.

In Figure 8 (Right), we plot validation-set loss graphs for on the yoga class for models with various w_{KL} settings. As L_R decreases, the L_{KL} term tends to increase due to the tradeoff between reconstruction loss L_R and KL loss L_{KL}.

4.2 Conditional generation

4.2.1 Conditional Reconstruction

In addition to the loss figures in Table 1, we qualitatively assess the reconstructed sketch S' given an input sketch S. In Figure 9, we sample several reconstructions at various levels of temperature τ using a model trained on the single cat class, starting at 0.01 on the left and linearly increasing to 1.0 on the right. The reconstructed cat sketches have similar properties to the input image, and occasionally add or remove details such as a whisker, a mouth, a nose, or the orientation of the tail.

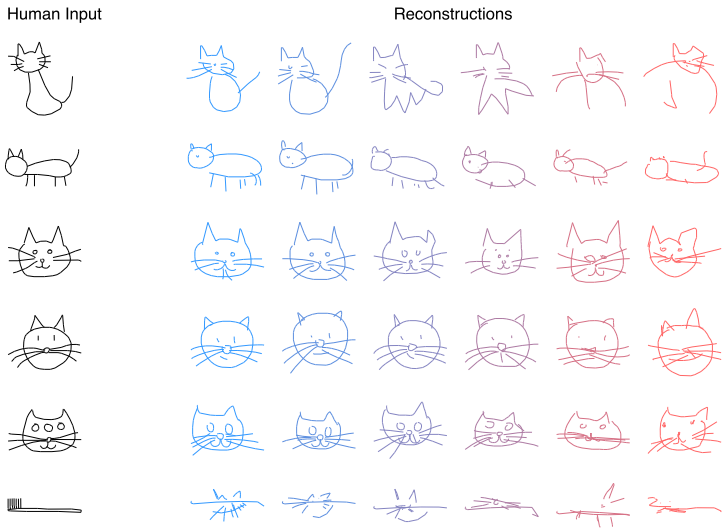


Figure 9: Conditional generation of cats.

When presented with a non-standard image of a cat, such as a cat's face with three eyes, the reconstructed cat only has two eyes. If we input a sketch from another image class, such a toothbrush, the model seemingly generate sketches with similar orientation and properties as the toothbrush input image, but with some cat-like features such as cat ears, whiskers or feet.

We perform a similar experiment with a model trained on the pig class in Figure 10. We see that the model is capable of generating pig faces or entire pigs. However, when presented with a sketch of a pig with many more legs than a normal pig, the reconstructed images generally look like a similar pig with two to four legs. The reconstructions of the eight-legged pig seems to contain extra artifacts on the pig's body or a pair of eyes that was not in the original drawing, as the decoder RNN may be trying to mimic the input image's stroke count. When the input is an image of a truck, the generated drawing looks like a pig with similar features as the truck.

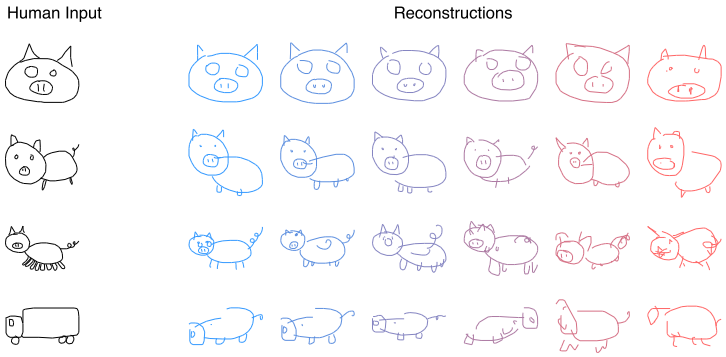


Figure 10: Conditional generation of pigs.

4.2.2 Latent Space Interpolation

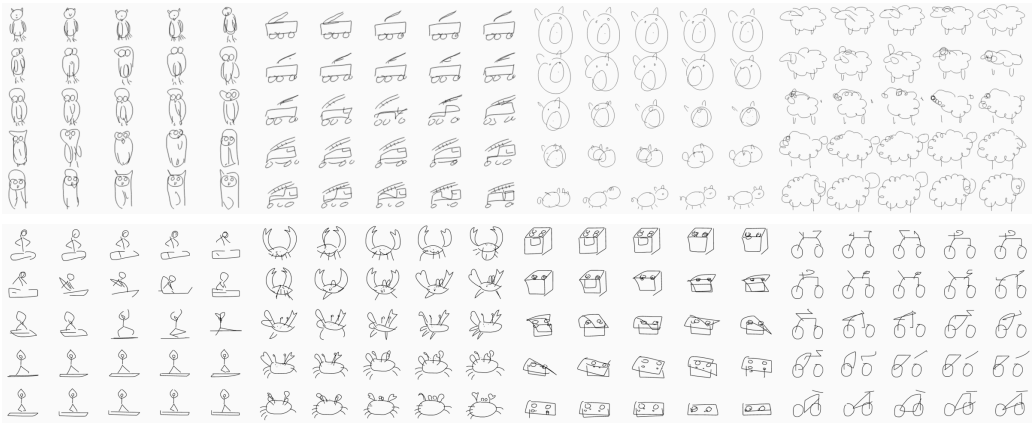


Figure 11: Example of conditional generated sketches with single class models. Latent space interpolation from left to right, and then top to bottom.

As we enforce a Gaussian prior on the latent space, we expect fewer *gaps* in the space between two encoded latent vectors. By interpolating between latent vectors, we can visualize how one image morphs into another image by visualizing the reconstructions of the interpolations. We see some examples of interpolating within models trained on single-class datasets in Figure 11.

To demonstrate the latent space interpolation effect, we trained various models on a dataset consisting of both the cat and pig classes, and we used two images from the test set as inputs - a cat face and a full pig. In Figure 12, we demonstrate the reconstructed images of the model with various settings of w_{KL}, and also reconstruct the spherically interpolated [29] latent vectors between the two encoded latent vectors.

Figure 12: Latent space interpolation between cat and pig using models with various w_{KL} settings.

Models trained with higher w_{KL} settings, resulting in lower L_{KL} losses, appear to produce more meaningful interpolated latent vectors. The reconstructions of the interpolated latent space from models trained with lower w_{KL} settings are less meaningful in comparison, and the reconstructed images are arguably no better, despite having a much lower L_R.

We would like to question the relative importance of the reconstruction loss term L_R, relative to the KL loss term L_{KL}, when our goal is to produce higher quality image reconstructions.

4.2.3 Which Loss Controls Image Coherency?

While our reconstruction loss term L_R optimizes for the log-likelihood of the set of strokes that make up a sketch, this metric alone does not give us any guarantee that a model with a lower L_R number will produce higher quality reconstructions compared to a model with a higher L_R number.

For example, imagine a simple sketch of an face, ☺, where most of the data points of S are be used to represent the head, and only a minority of points represent facial features such as the eyes and mouth. It is possible to reconstruct the face with incoherent facial features, and yet still score a lower L_R number compared to another reconstruction with a coherent and similar face, if the edges around the incoherent face are generated more precisely.

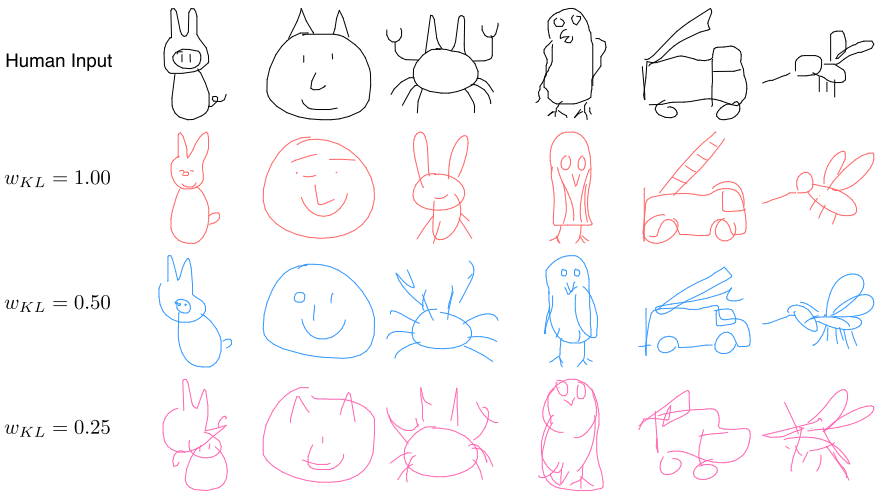


Figure 13: Reconstructions of sketch images using models with various w_{KL} settings.

In Figure 13, we compare the reconstructed images generated using models trained with various w_{KL} settings. In the first three examples from the left, we train our model on a dataset consisting of four image classes (crab, face, pig, rabbit). We deliberately sketch input drawings that contain features of two classes, such as a rabbit with a pig mouth and pig tail, a person with animal ears, and a rabbit with crab claws. We see that the model trained using higher w_{KL} weights, tend to generate sketches with features of a single class that look more coherent, despite having lower L_{KL} numbers. For instance, the model with $w_{KL} = 1.00$ omit pig features, animal ears, and crab claws from its reconstructions. In contrast, the model with $w_{KL} = 0.25$, with higher L_{KL}, but lower L_R numbers tries to keep both inconsistent features, while generating sketches that look less coherent.

In the last three examples in Figure 13, we repeat the experiment on models trained on single-class images, and see similar results even when we deliberately choose input samples from the test set with noisier lines.

As we saw from Figure 12, models with better KL loss terms also generate more meaningful reconstructions from the interpolated space between two latent vectors. This suggests the latent vector for models with lower L_{KL} control more meaningful parts of the drawings, such as controlling whether the sketch is an animal head only or a full animal with a body, or whether to draw a cat head or a pig head. Altering such latent vectors can allow us to directly manipulate these animal features. Conversely, altering the latent codes of models with higher L_{KL} results in scattered movement of individual line segments, rather than alterations of meaningful conceptual features of the animal.

This result is consistent with incoherent reconstructions seen in Figure 13. With a lower L_{KL}, the model is likely to generate coherent images given any random z. Even with a non-standard, or noisy, input image, the model will still encode a z that produces coherent images. For models with lower L_{KL} numbers, the encoded latent vectors contain conceptual features belonging to the input image, while for models with higher L_{KL} numbers, the latent vectors merely encode information about specific line segments. This observation suggests that when using `sketch-rnn` on a new dataset, we should first try different w_{KL} settings to evaluate the tradeoff between L_R and L_{KL}, and then choose a setting for w_{KL} (and $KL_{\texttt{min}}$) that best suit our requirements.

4.2.4 Sketch Drawing Analogies

The interpolation example in Figure 12 suggests the latent vector z encode conceptual features of a sketch. Can we use these features to augment other sketches without such features – for example, adding a body to a cat's head? Indeed, we find that sketch drawing analogies are possible for models trained with low L_{KL} numbers. Given the smoothness of the latent space, where any interpolated vector between two latent vectors results in a coherent sketch, we can perform vector arithmetic using the latent vectors encoded from different sketches and explore how the model organizes the latent space to represent different concepts in the manifold of generated sketches.

As an experiment, we train a model on a dataset of cat and pig classes, and encode sketches of cat heads, pig heads, full cats, full pigs into latent vectors z. We then perform vector arithmetic on $z_{\texttt{cat head}}$, $z_{\texttt{pig head}}$, $z_{\texttt{full cat}}$, and $z_{\texttt{full pig}}$.

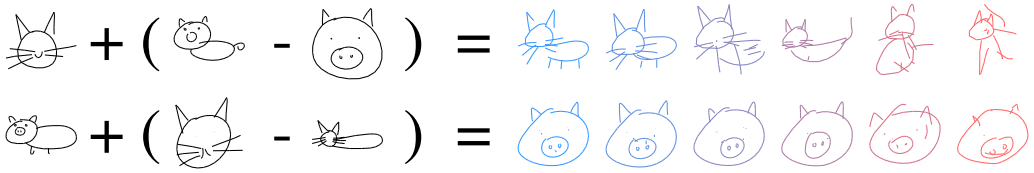


Figure 14: Sketch Drawing Analogies.

In Figure 14, we construct two new latent vectors. The first latent vector, $\tilde{z}_1$, is constructed by adding the difference of two vectors $z_{\texttt{full pig}} - z_{\texttt{pig head}}$ to $z_{\texttt{cat head}}$. Using $\tilde{z}_1$ as in input to the RNN decoder, we sample images of varying temperature τ and obtain sketches of full cats. In the second example, the second latent vector, $\tilde{z}_2$, is constructed by adding the latent vector of a full pig to the difference between the latent vectors a cat head and a full cat, resulting in sketches of pig heads.

4.2.5 Multi-Sketch Drawing Interpolation

In addition to interpolating between two sketches, we can also visualize the interpolation between four sketches in latent space to gain further insight from the model.

Figure 15: Interpolation of generated pig, rabbit, crab and face (Left),
Latent space of generated yoga positions (Right).

The left side of Figure 15 visualizes the interpolation between a full pig, a rabbit's head, a crab, and a face, using a model trained on these four classes. In certain parts of the space between a crab and a face is a rabbit's head, and we see that the ears of the rabbit becomes the crab's claws. Applying the model on the yoga class, it is interesting to see how one yoga position slowly transitions to another via a set of interpolated yoga positions generated by the model. For visual effect, we also interpolate between four distinct colors, and color each sketch using a unique interpolated color.

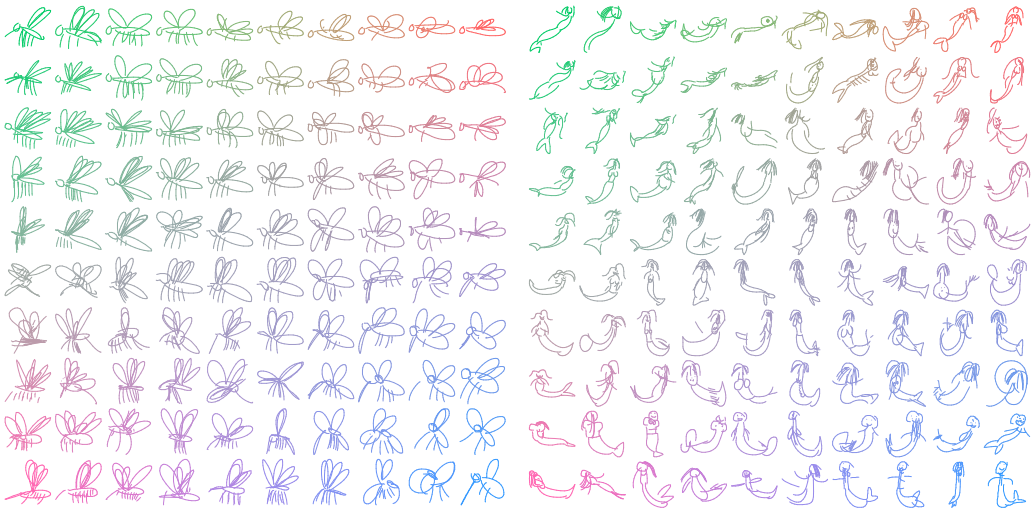


Figure 16: Latent space of generated mosquitoes and mermaids.

We also construct latent space interpolation examples for the mosquito class and the mermaid class, in Figure 16. We see that the model can interpolate between concepts such as style of wings, leg counts, and orientation.

Figure 17 is a delightful example where we use the model trained on the cat class to encode four different latent vectors from four different sketch drawings of chairs as inputs to obtain four sketches

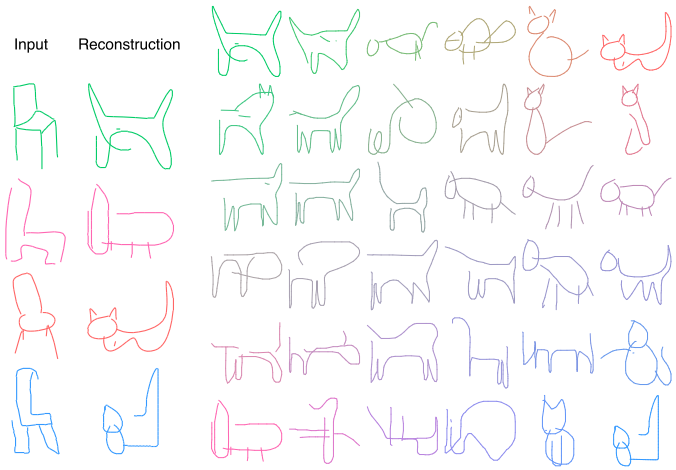


Figure 17: Latent space of generated cats conditioned on sketch drawings of chairs.

of cat-styled chairs. We can then interpolate and visualize the latent subspace of these four chair-like cats using the same model.

4.3 Unconditional generation

As a special case of `sketch-rnn`, we can use the decoder RNN as a standalone model to generate sketches without any inputs or latent vectors. This unconditional model is generally easier to train, as we only need to optimize for L_R. Furthermore, we do not need to train the complementary encoder system to a generate a meaningful vector space for the decoder. For unconditional generation with `sketch-rnn`, we find that lower L_R scores correspond directly with higher quality sketches.

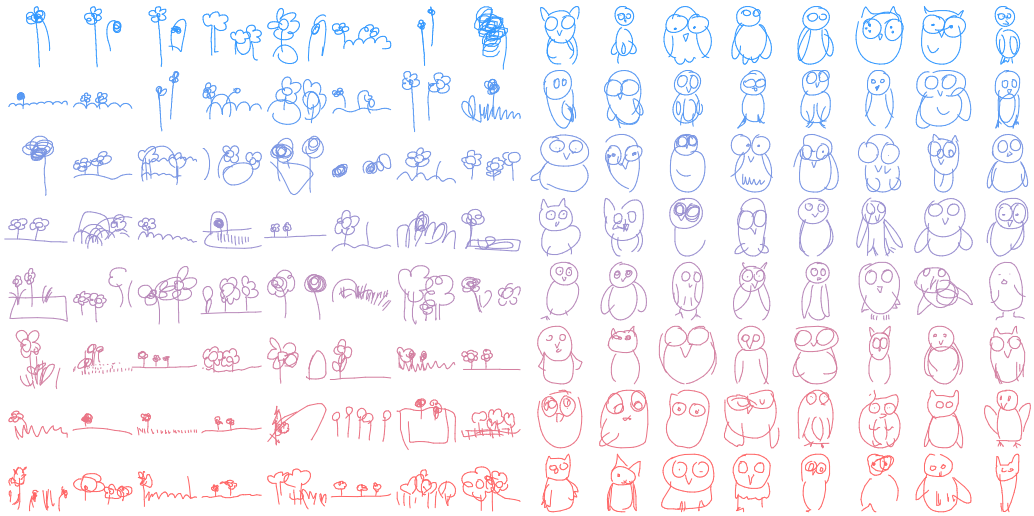


Figure 18: Unconditional generated sketches of gardens and owls.

Figure 18 displays a grid of samples from models trained on individual image classes with various temperature settings, varying from $\tau = 0.2$ in the blue region to $\tau = 0.9$ in the red region. Sketches generated using lower temperatures tend to be simpler, and constructed with smoother and more deterministic lines. Occasionally when the temperature is too low, the model stays within a single mixture of the GMM and simply draws repeated spirals until $N_{\max}$ points of the drawing have been sampled and it is forced to stop. With higher temperatures, the model can escape more easily from one mixture to another mixture of the GMM, leading to more diverse generated sketches.

4.3.1 Predicting Different Endings of Incomplete Sketches

We can use `sketch-rnn` to finish an incomplete sketch. By using the decoder RNN as a standalone model, we can generate a sketch that is conditional on the previous points. To do this, we use the decoder RNN to first encode an incomplete sketch into a hidden state h. Afterwards, we generate the remaining points of the sketch using h as the initial hidden state.

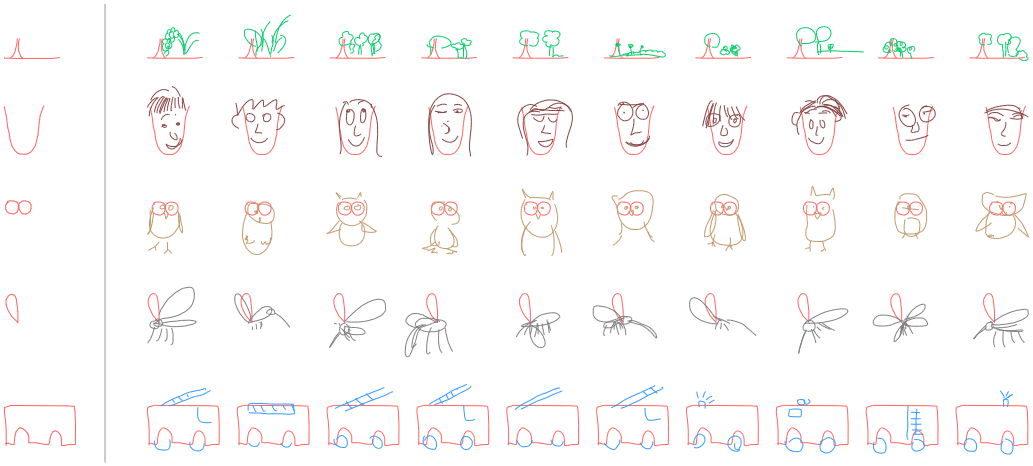


Figure 19: `sketch-rnn` predicting possible endings of various incomplete sketches (the red lines).

In Figure 19, we perform this experiment by training the standalone decoder on individual classes, and sample several different endings for the same incomplete sketch, using a constant temperature setting of $\tau = 0.8$. In the garden example, the model sketches other plants above the ground area that are consistently spaced, and occasionally draws the crown area on top of our tree trunk. When we draw the outline of a face, the model populates the sketch with remaining facial features such as eyes, nose and hair which are coherently spaced relative to the initial outline. The model also generates various mosquitoes at different orientations given the same starting wing, and various types of owls and firetrucks given the same set of eyes or truck body.

5 Discussion

5.1 Limitations

Although `sketch-rnn` can model a large variety of sketch drawings, there are several limitations in the current approach we wish to highlight. For most single-class datasets, `sketch-rnn` is capable of modelling sketches up to around 300 data points. The model becomes increasingly difficult to train beyond this length. For our dataset, we applied the Ramer–Douglas–Peucker algorithm [5] to simplify the strokes of the sketch data to less than 200 data points while still keeping most of the important visual information of each sketch.

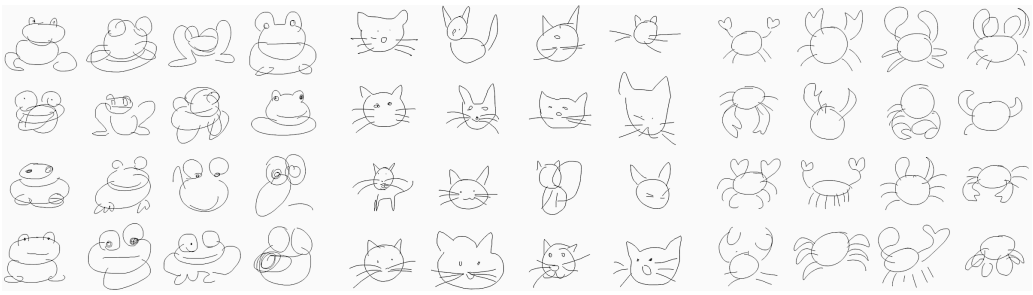


Figure 20: Unconditional generated sketches of frogs, cats, and crabs at $\tau = 0.8$.

For more complicated classes of images, such as mermaids or lobsters, the reconstruction loss metrics are not as good compared to simpler classes such as ants, faces or firetrucks. The models trained on these more challenging image classes tend to draw smoother, more circular line segments that do not resemble individual sketches, but rather resemble an averaging of many sketches in the training set. We can see some of this artifact in the frog class, in Figure 20. This smoothness may be analogous to the blurriness effect produced by a Variational Autoencoder [19] that is trained on pixel images. Depending on the use case of the model, smooth circular lines can be viewed as aesthetically pleasing and a desirable property.

While both conditional and unconditional models are capable of training on datasets consisting of several classes, such as (cat, pig), and (crab, face, pig, rabbit), `sketch-rnn` is not able to effectively model all 75 classes in the dataset. In Figure 21, we sample sketches using an unconditional model trained on all 75 classes, and a model trained on 4 classes. The samples generated from the 75-class model are incoherent, with individual sketches displaying features from multiple classes. The four-class unconditional model usually generates samples of a single class, but occasionally also combines features from multiple classes. For example, we can find some interesting generated sketches of crabs with the body of a face.

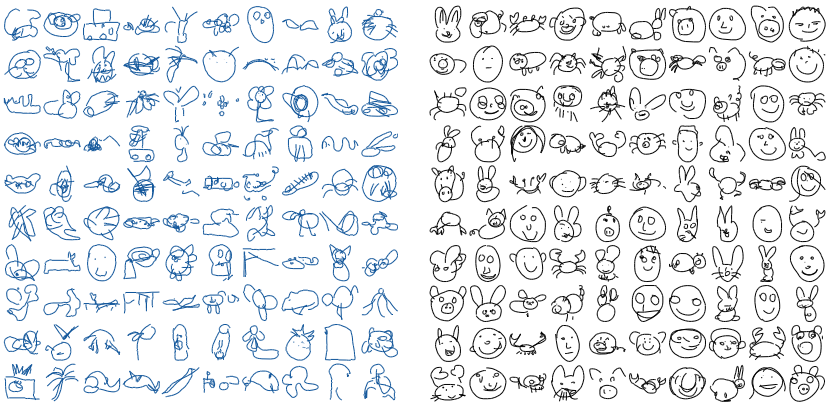


Figure 21: Unconditional generations from model trained on all classes (Left), From model trained on crab, face, pig and rabbit classes (Right).

5.2 Applications and Future Work

We believe `sketch-rnn` will find many creative applications. A `sketch-rnn` model trained on even a small number of classes can assist the creative process of an artist by suggesting many possible ways of finishing a sketch, helping the artist expand her imagination. A creative designer may find value in exploring the latent space between different image classes to find interesting intersections and relationships between two very different objects. Pattern designers can conditionally generate a large number of similar, but unique designs, for textile or wallpaper prints.

A more sophisticated model trained on higher quality sketches may find its way into educational applications that can help teach students how to draw. Even with the simple sketches in `quickdraw-75`, the authors of this work have become much more proficient at drawing animals, insects, and various sea creatures after conducting these experiments. A related application is to encode a poorly sketched drawing to generate more aesthetically looking reproductions by using a model trained with a high w_{KL} setting and sampling with a low temperature τ, as described in Section 4.2.3. In the future, we can also investigate augmenting the latent vector in the direction that maximizes the aesthetics of the drawing by incorporating user-rating data into the training process.

It will be interesting to explore various methods to improve the quality of the generated sketches. The choice of a Gaussian as the prior distribution of our Variational Autoencoder may be too simple to model a large dataset, and we may get better gains by incorporating more advanced variational inference models such a Ladder-VAEs [15] or Inverse Autoregressive Flow [18]. The addition of a GAN-style discriminator, or a critic network as a post processing step may lead to more realistic line styles and better reconstruction samples. This type of approach has been tested on VAE-GAN [21] hybrids on pixel images, and has recently been tested on text generation [12].

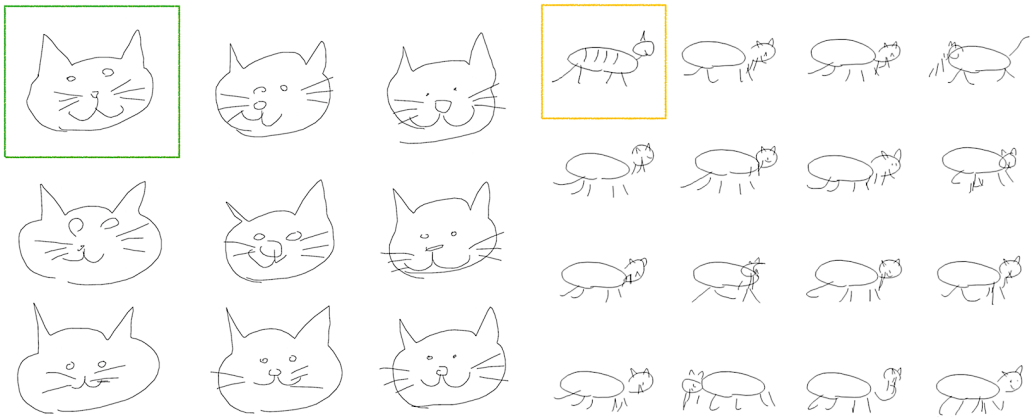


Figure 22: Generating similar, but unique sketches based on a single human sketch in the box.

Combining hybrid variations of sequence-generation models with unsupervised, cross-domain pixel image generation models, such as Image-to-Image models [4, 16, 23], is another exciting direction that we can explore. We can already combine this model with supervised, cross-domain models such as Pix2Pix [13], to occasionally generate photo realistic cat images from generated sketches of cats. The opposite direction of converting a photograph of a cat into an unrealistic, but similar looking sketch of a cat composed of a minimal number of lines seems to be a more interesting problem.

6 Conclusion

In this work, we develop a methodology to model sketch drawings using recurrent neural networks. `sketch-rnn` is able to generate possible ways to finish an existing, but unfinished sketch drawing. Our model can also encode existing sketches into a latent vector, and generate similar looking sketches conditioned on the latent space. We demonstrate what it means to interpolate between two different sketches by interpolating between its latent space. After training on a particular class of image, our model is able to correct features of an input it deems to be incorrect, and output a similar image with corrected properties. We also show that we can manipulate attributes of a sketch by augmenting the latent space, and demonstrate the importance of enforcing a prior distribution on the latent vector for coherent vector image generation.

7 Acknowledgements

We thank Ian Johnson, Jonas Jongejan, Martin Wattenberg, Mike Schuster, Thomas Deselaers, Ben Poole, Kyle Kastner, Junyoung Chung and Kyle McDonald for their help with this project. This work was done as part of the Google Brain Residency program (`g.co/brainresidency`).

References

[1] J. L. Ba, J. R. Kiros, and G. E. Hinton. Layer normalization. *NIPS*, 2016.

[2] C. M. Bishop. Mixture density networks. *Technical Report*, 1994.

[3] S. R. Bowman, L. Vilnis, O. Vinyals, A. M. Dai, R. Józefowicz, and S. Bengio. Generating Sentences from a Continuous Space. *CoRR*, abs/1511.06349, 2015.

[4] H. Dong, P. Neekhara, C. Wu, and Y. Guo. Unsupervised Image-to-Image Translation with Generative Adversarial Networks. *ArXiv e-prints*, Jan. 2017.

[5] D. H. Douglas and T. K. Peucker. Algorithms for the reduction of the number of points required to represent a digitized line or its caricature. *Cartographica: The International Journal for Geographic Information and Geovisualization*, 10(2):112–122, Oct. 1973.

[6] M. Eitz, J. Hays, and M. Alexa. How Do Humans Sketch Objects? *ACM Trans. Graph. (Proc. SIGGRAPH)*, 31(4):44:1–44:10, 2012.

[7] I. Goodfellow. NIPS 2016 Tutorial: Generative Adversarial Networks. *ArXiv e-prints*, Dec. 2017.

[8] A. Graves. Generating sequences with recurrent neural networks. *arXiv:1308.0850*, 2013.

[9] D. Ha. Recurrent Net Dreams Up Fake Chinese Characters in Vector Format with TensorFlow. `http://blog.otoro.net/`, 2015.

[10] D. Ha, A. M. Dai, and Q. V. Le. HyperNetworks. In *ICLR*, 2017.

[11] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 1997.

[12] Z. Hu, Z. Yang, X. Liang, R. Salakhutdinov, and E. P. Xing. Controllable Text Generation. *ArXiv e-prints*, Mar. 2017.

[13] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros. Image-to-Image Translation with Conditional Adversarial Networks. *ArXiv e-prints*, Nov. 2016.

[14] J. Jongejan, H. Rowley, T. Kawashima, J. Kim, and N. Fox-Gieg. The Quick, Draw! - A.I. Experiment. *https://quickdraw.withgoogle.com/*, 2016.

[15] C. Kaae Sønderby, T. Raiko, L. Maaløe, S. Kaae Sønderby, and O. Winther. Ladder Variational Autoencoders. *ArXiv e-prints*, Feb. 2016.

[16] T. Kim, M. Cha, H. Kim, J. Lee, and J. Kim. Learning to Discover Cross-Domain Relations with Generative Adversarial Networks. *ArXiv e-prints*, Mar. 2017.

[17] D. Kingma and J. Ba. Adam: A method for stochastic optimization. In *ICLR*, 2015.

[18] D. P. Kingma, T. Salimans, and M. Welling. Improving variational inference with inverse autoregressive flow. *CoRR*, abs/1606.04934, 2016.

[19] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. *ArXiv e-prints*, Dec. 2013.

[20] B. M. Lake, R. Salakhutdinov, and J. B. Tenenbaum. Human-level concept learning through probabilistic program induction. *Science*, 350(6266):1332–1338, Dec. 2015.

[21] A. B. L. Larsen and S. K. Sønderby. Generating Faces with Torch. `http://torch.ch/blog/2015/11/13/gan.html`, 2015.

[22] Y. J. Lee, C. L. Zitnick, and M. F. Cohen. Shadowdraw: Real-time user guidance for freehand drawing. In *ACM SIGGRAPH 2011 Papers*, SIGGRAPH '11, pages 27:1–27:10, New York, NY, USA, 2011. ACM.

[23] M.-Y. Liu, T. Breuel, and J. Kautz. Unsupervised Image-to-Image Translation Networks. *ArXiv e-prints*, Mar. 2017.

[24] S. Reed, A. van den Oord, N. Kalchbrenner, S. Gómez Colmenarejo, Z. Wang, D. Belov, and N. de Freitas. Parallel Multiscale Autoregressive Density Estimation. *ArXiv e-prints*, Mar. 2017.

[25] P. Sangkloy, N. Burnell, C. Ham, and J. Hays. The Sketchy Database: Learning to Retrieve Badly Drawn Bunnies. *ACM Trans. Graph.*, 35(4):119:1–119:12, July 2016.

[26] M. Schuster, K. K. Paliwal, and A. General. Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 1997.

[27] S. Semeniuta, A. Severyn, and E. Barth. Recurrent dropout without memory loss. *arXiv:1603.05118*, 2016.

[28] P. Tresset and F. Fol Leymarie. Portrait drawing by paul the robot. *Comput. Graph.*, 37(5):348–363, Aug. 2013.

[29] T. White. Sampling Generative Networks. *ArXiv e-prints*, Sept. 2016.

[30] N. Xie, H. Hachiya, and M. Sugiyama. Artist agent: A reinforcement learning approach to automatic stroke generation in oriental ink painting. In *ICML*. icml.cc / Omnipress, 2012.

[31] X. Zhang, F. Yin, Y. Zhang, C. Liu, and Y. Bengio. Drawing and Recognizing Chinese Characters with Recurrent Neural Network. *CoRR*, abs/1606.06539, 2016.

A Appendix

A.1 Dataset Details

Figure 23: Example sketch drawings from `quickdraw-75` dataset.

The `quickdraw-75` dataset consists of 75 classes. Each class consists of 70K training samples and 2.5K validation and test samples. Stroke simplification using the Ramer–Douglas–Peucker algorithm [5] with a parameter of $\epsilon = 2.0$ has been applied to simplify the lines. The maximum sequence length is 200 data points. The data was originally recorded in pixel-dimensions, so we normalized the offsets $(\Delta x, \Delta y)$ using a single scaling factor. This scaling factor was calculated to adjust the offsets in the training set to have a standard deviation of 1. For simplicity, we do not normalize the offsets $(\Delta x, \Delta y)$ to have zero mean, since the means are already relatively small.

alarm clock	ambulance	angel	ant	barn
basket	bee	bicycle	book	bridge
bulldozer	bus	butterfly	cactus	castle
cat	chair	couch	crab	cruise ship
dolphin	duck	elephant	eye	face
fan	fire hydrant	firetruck	flamingo	flower
garden	hand	hedgehog	helicopter	kangaroo
key	lighthouse	lion	map	mermaid
octopus	owl	paintbrush	palm tree	parrot
passport	peas	penguin	pig	pineapple
postcard	power outlet	rabbit	radio	rain
rhinoceros	roller coaster	sandwich	scorpion	sea turtle
sheep	skull	snail	snowflake	speedboat
spider	strawberry	swan	swing set	tennis racquet
the mona lisa	toothbrush	truck	whale	windmill

Table 2: `quickdraw-75` dataset class list.

UNDERSTANDING TEACHER PERSPECTIVES AND EXPERIENCES AFTER DEPLOYMENT OF AI LITERACY CURRICULUM IN MIDDLE-SCHOOL CLASSROOMS

Prerna Ravi, Annalisa Broski, Glenda Stump, Hal Abelson, Eric Klopfer, Cynthia Breazeal

Massachusetts Institute of Technology (UNITED STATES)

Abstract

Artificial Intelligence (AI) and its associated applications are ubiquitous in today's world, making it imperative that students and their teachers understand how it works and the ramifications arising from its usage. In this study, we investigate the experiences of seven teachers following their implementation of modules from the MIT RAICA (Responsible AI for Computational Action) curriculum. Through semi-structured interviews, we investigated their instructional strategies as they engaged with the AI curriculum in their classroom, how their teaching and learning beliefs about AI evolved with the curriculum as well as how those beliefs impacted their implementation of the curriculum. Our analysis suggests that the AI modules not only expanded our teachers' knowledge in the field, but also prompted them to recognize its daily applications and their ethical and societal implications, so that they could better engage with the content they deliver to students. Teachers were able to leverage their own interdisciplinary backgrounds to creatively introduce foundational AI topics to students to maximize engagement and playful learning. Our teachers advocated their need for better external support when navigating technological resources, additional time for preparation given the novelty of the curriculum, more flexibility within curriculum timelines, and additional accommodations for students of determination. Our findings provide valuable insights for enhancing future iterations of AI literacy curricula and teacher professional development (PD) resources.

Keywords: Artificial Intelligence Education, Teacher Professional Development, AI Integration, Disciplinary Classrooms, Diverse Teacher Backgrounds.

1 INTRODUCTION

There is currently a massive proliferation of digital platforms that utilize Artificial Intelligence (AI) stemming from the rapid growth of data and computational power. While this rapidly evolving digital landscape opens up new avenues for collaboration, it also raises several societal ramifications that need to be addressed. Scholars from recent studies have raised concerns about the extent to which K12 students are aware of the applications as well as consequences of AI in their daily lives [1, 2, 3, 4, 5]. This echoes the need for integration of AI in K12 education curriculum to satisfy the need for computational and scientific literacy among the youth and by extension, their educators. Our work investigates teachers' experiences with the introduction to and deployment of three different modules within a novel, project-based AI curriculum.

The Responsible AI for Social Empowerment and Education Initiative (RAISE) at the Massachusetts Institute of Technology (MIT) seeks to equitably educate a broad spectrum of K12 students, an inclusive workforce, and lifelong learners, cultivating them to act responsibly and actively participate in an increasingly AI-driven society. One avenue for this support is the Responsible AI for Computational Action (RAICA) curriculum, which lays a foundation for AI knowledge at the middle-school level through a series of AI learning modules. RAICA comprises a series of project-based modules that progressively increase not only AI knowledge, but also competence in using AI tools. Curriculum modules focus on ethical thinking as well as computational action [6], the idea that learners should be given the opportunity to create projects that have benefit to themselves or their communities. The modules also use a design thinking framework to lead students through the problem-solving process.

Although it is important to provide teachers and students with an AI curriculum that is enriching and engaging, the task of deploying these in classrooms can be daunting for teachers, especially when they lack familiarity in AI topics. For many teachers, curricular prompting and extensive professional development (PD) sessions are important so that they are not only able to successfully understand and deliver AI curricula in varied classroom contexts, but they are able to do so without having a strong prior

STEM background. Instructors recruited to teach AI often have a strong formal background in teaching, but very little computer science (CS) knowledge, and vice versa [7]. Teacher beliefs like high self-efficacy and teaching efficacy are also correlated with integration of content and curriculum in the case of information technology [8]. Past work from Ayanwale and their colleagues has shown that teachers' confidence in teaching AI is correlated with their intention and readiness to teach it [9]. Previous research has also proposed that the idea of AI for social good may serve as a motivating factor driving teachers to embrace AI literacy curricula [10]. Since teachers who look to teach an AI curriculum come from many different backgrounds, introducing professional development (PD) sessions is crucial for the success of the curriculum. Even CS teachers who have strong technical backgrounds in other CS domains may not have the familiarity with AI to jump into an unknown curriculum without additional instruction.

It is important to consider teachers' evolving perspectives when developing effective teaching concepts and material [11]. Prior K12 AI education research in regular classroom settings has primarily involved researchers rather than K12 teachers as the instructors [1, 12, 13, 14]. These studies hence lack the valuable feedback that can be collected from experts who work directly in classrooms [15]. There is also a dearth of information regarding PD elements needed for teaching AI literacy [16]. Our work seeks to fill these gaps. In our curriculum, teachers were offered ongoing PD during their module implementation, they interacted with RAICA researchers during weekly debrief sessions, and they completed semi-structured interviews post-module implementation. These interactions were designed to increase teacher feedback and input on the curriculum and were an important step in the codesign process.

Our research aims to explore and evaluate the impact of AI literacy curriculum (specifically RAICA) on middle-school educators. We shed light on those facets of the curriculum that enabled our teachers' rapid comprehension of new AI concepts and empowered them to draw from their prior interdisciplinary teaching experiences. This allowed them to effectively and confidently modify and adapt the provided curriculum materials to address the unique requirements of their classrooms. We also outline the tools and resources that empowered our teachers to deliver this content with a strong emphasis on fostering socially responsible AI use. We seek to use their feedback and perspectives to inform design changes in future iterations of the curriculum and teacher PD and identify elements from the current deployment that can be retained to strengthen the future landscape of K12 AI literacy.

2 METHODOLOGY

Researchers and designers of the RAICA curriculum periodically examined how teachers implemented the curriculum in classrooms. For the purposes of this study, we focussed on eight recorded interviews from seven teachers with interdisciplinary backgrounds in art, business, and computer science. Each teacher implemented one of three modules from the RAICA curriculum: in one module, students build an interactive art installation for their school, in another, students create a chatbot that could inform, entertain, or persuade members of their community, and in the third module, students design an app to help their community discover new things about their school. The curriculum materials consist of a Teacher's Guide, PowerPoint slides for presentation of content, a Student Workbook, and assessments. A kanban board[1] is also available for use as a project management tool.

Following implementation of the modules, we conducted semi-structured remote interviews with each of the seven teachers. We sought to answer three major research questions:

- RQ1: How do teachers adapt AI literacy modules based on their individual classroom contexts and personal backgrounds?
- RQ2: What are teachers' attitudes and beliefs about AI before, during, and after module implementation?
- RQ3: How do teachers' attitudes and beliefs about teaching AI literacy change after using the RAICA curriculum and associated professional development support?

Our study was approved by the Institutional Review Board at Massachusetts Institute of Technology (MIT). Per IRB protocol, we obtained written consent from all interview participants, detailing the purpose of our study and allowing us to record the interview.

[1] https://en.wikipedia.org/wiki/Kanban_board

We created an interview protocol with potential questions that could be asked during the interview. The protocol was designed to gather feedback on various aspects of module implementation. Our goal was to investigate teachers' experiences of engaging with the AI curriculum in their classroom, how their teaching and learning beliefs about AI evolved with the curriculum as well as how those beliefs impacted their implementation of the curriculum. During the interviews, we assessed how well teachers and students comprehended the content and their overall impressions of the module. We also identified any challenges they had to overcome during the course of the module, as well as their understanding of AI concepts. We gathered insights into the assessment methods teachers employed and the pedagogical approaches they used within each module to enhance student learning outcomes. The interview questions also inquired about how the module influenced teachers' approaches to teaching and their perceptions of teaching and learning. Through these questions, we aimed to understand how teachers' knowledge of AI had evolved as a result of their experience with the curriculum. We explored any concerns teachers might have had regarding teaching AI in future classes and how this aligned with their personal views on teaching and learning. We also discussed teachers' approach to planning for teaching the AI curriculum.

We transcribed the interviews to initiate the coding process. We followed a thematic data analysis process, with regular check-ins after every interview to discuss findings and emerging themes [17]. Thereafter, we conducted several rounds of deductive coding on the transcribed data. The codes closely followed the text. We identified three major themes to classify codes into: 1) Attitudes and beliefs towards AI literacy 2) Attitudes and beliefs towards teaching and learning, and 3) Teacher recommendations. We also identified subcategories within these three themes to outline more specific trends in the data.

3 RESULTS

We now outline the findings that emerged in our data. We present our results around the three primary themes that emerged in the deductive coding process above: attitudes and beliefs towards AI literacy, attitudes and beliefs towards teaching and learning more generally, and recommendations proposed by teachers.

3.1 Attitudes and Beliefs towards AI Literacy

This theme explores how teachers and students perceived AI literacy and its various facets. It also investigates specific practices they followed in classrooms to master the complex AI concepts embedded into the curriculum.

3.1.1 AI Learning

3.1.1.1 For Teachers:

One of the primary goals of our curriculum was to provide teachers with the foundation needed to understand fundamental AI concepts and ideas, so that they could ultimately relay this understanding to diverse student groups in inclusive learning environments. Additionally, teachers need to feel empowered to co-design, implement, and integrate AI curriculum into classrooms with the emphasis on socially responsible use of AI. During our interviews, we investigated teachers' perspectives on learning novel AI concepts. Teachers observed that their engagement with our curriculum facilitated a valuable exploration of AI principles. The pedagogical instruction and lessons within the curriculum played a pivotal role in elucidating the foundational aspects of AI and its broader implications. These experiences led to a transformative shift in their initial perception of AI, which had primarily been influenced by depictions in science fiction films. Their enthusiasm for acquiring this new knowledge stemmed from its practicality in understanding the underlying workings of ubiquitous AI-powered tools employed in their daily lives. For example, teachers engaged in discussions regarding biases inherent in AI systems. They were able to articulate the mechanisms through which bias propagates through datasets in light of emerging AI technologies. They also recognized how their personal experiences and beliefs shaped their students' engagement with AI and their own comprehension of the field. They were hence able to view these from a positive lens:

> *"AI can help solve people's problems, but it can also bring harm to a community depending on the group who is making it… but through this curriculum, I feel like we can see these through a positive lens."*

Some teachers were also ambivalent about using AI tools: they recognized its dual potential to yield both benefits and harm. These conversations demonstrated their growing awareness of some ethical dimensions of AI. Teachers also valued the peer support and collaboration networks established amongst their fellow educators using the curriculum. It allowed them to exchange and draw inspiration from each other's instructional methods.

However, for all our teachers, AI education exemplified the challenges usually faced during integration of complex unfamiliar technology in classrooms. Teachers ran into technical issues when acquainting themselves with AI platforms like Scratch[2] and Teachable Machines[3]. They advocated the need for one-on-one code reviews in PD sessions, highlighting the significance of personalized guidance when navigating the complexities of AI tools.

3.1.1.2 For Students

Our teachers also spoke about how their students felt about AI and technology after being exposed to the curriculum. They were particularly impressed by their students' deliberation on, and grasp of some complex, abstract ideas related to AI. Their comments were centered around one of the curriculum modules that tasked students with designing and developing a chatbot. As part of this, students delineated the nature and specifics of ideas generated by chatbots, e.g., Kuki[4] and Hector[5]. Both teachers and students gained a comprehensive understanding of the positive roles that chatbots play in informing, persuading, and entertaining users. Students evaluated the chatbot's conversational abilities, emphasizing its capacity to employ human-like language. During these sessions, the primary focus was to gain insight into the students' conceptions of AI and whether they attributed 'consciousness' to these digital entities based on factors like chatbots' friendliness and informativeness. The effectiveness of the chatbot in sustaining meaningful conversations constituted a vital component of the assessment. Some students articulated a viewpoint suggesting that consciousness entailed self-awareness, encompassing attributes such as independent thought and even physiological processes like breathing. This perspective equated consciousness with a tangible, human-like essence. Other students posited that AI, while distinct from human consciousness, possessed the capacity to acquire knowledge and learn over time:

> *"Some students said if you're conscious, it means that you're thinking for yourself and your breathing...this means AI is not conscious in the same way a human is but it can learn things… that is a part of the curriculum."*

Thus, they perceived AI systems as non-human entities capable of learning, in a manner distinct from human consciousness. This perspective led to contemplations on the programming aspect of AI and its capacity for knowledge acquisition.

3.1.2 Teaching about AI

3.1.2.1 Strategies

Our teachers also mentioned being concerned about the resistance they would receive from students when introduced to new, complex AI concepts. They employed several innovative methods to introduce fundamental AI concepts to increase classroom engagement and playful learning. One instructional strategy employed by them involved encouraging students to adopt an AI-oriented mindset: thinking like the AI does. This approach aimed to immerse students in the thought processes inherent to AI, fostering a nuanced understanding of the technology. Teachers also tried to draw connections between the lessons they were learning and teaching to other non-stem fields, which helped transcend conventional subject boundaries. For example, one of the teachers commented on the interdisciplinary approach of the module they implemented, noting that it blurred the lines between art/design and computer science classes:

> *"The class was an art class disguised as a CS class and a CS class disguised as an art class."*

This creative approach enriched student learning experiences by integrating diverse fields of study. Other teachers navigated between different instructional elements. Informal formative assessment played a pivotal role in ascertaining students' comprehension of technology concepts. Acknowledging student apprehensions and skepticism towards AI, teachers struck a balance by providing tangible examples. This approach effectively mitigated students' concerns and facilitated a smoother learning

[2] https://scratch.mit.edu/

[3] https://teachablemachine.withgoogle.com/

[4] https://chat.kuki.ai/

[5] https://ochatbot.com/hector/

process in classrooms. Teachers also prioritised scaffolding instruction and support for students, since they deemed it necessary to guide students through demystifying complex AI concepts.

3.1.2.2 Ethical Considerations in AI

The multifaceted nature of AI's impact was a central theme in the curriculum. Teachers incorporated examples into their pedagogy that exposed students to the dichotomy of using AI tools that were capable of both transformative benefits and causing potential harm. This holistic perspective empowered students to critically evaluate the implications of AI applications across various domains, which aligned well with teachers' own perspectives on responsible use of AI (that they had gained from learning about AI through the PD sessions). When guiding students in the creation of innovative AI-driven solutions for everyday challenges, teachers emphasized the significance of considering a broad spectrum of users. They focussed on three major domains of ethical considerations–awareness and consideration of stakeholders, intended and unintended impacts, and industry-aligned design values. These gave students the opportunity to retrieve and use prior learning. These activities also gave authentic avenues to consider the societal ramifications of new information that students were learning.

Teachers encouraged students to investigate the pivotal role of biases and personal experiences in shaping AI interactions. Students explored how inherent biases and stereotypes, coupled with their own unique life experiences, could influence AI's responses and behaviors. This heightened awareness underscored the ethical imperative of mitigating bias and promoting inclusivity in AI design, as supported by the curriculum themes of ethical thinking (ET), design for inclusivity, and equity in AI education.

3.1.2.3 Positive Outcomes and Benefits of AI Education

Teachers articulated several benefits from engaging with the AI curriculum, that reflected not only in their eagerness to deliver lessons but also in the responses they elicited from students. Teachers placed a significant emphasis on harnessing technology for positive social impact. This focus resonated with teachers and students alike, as they recognized the potential of AI to bring about meaningful changes in society. As part of the curriculum in one module implementation, teachers brought in AI experts to talk to students about the field and their own career trajectories, and to answer any questions they had. This real-time interaction between students and AI professionals proved to be a powerful aspect of the program. The direct engagement piqued students' interests by providing them with insights into the diverse array of AI careers and the multifaceted skills (like creativity and problem-solving) required in the field, extending beyond computer science programming and mathematics:

> *"The curriculum was really helpful in piquing their interest about what their possibilities for careers are… one of the students mentioned that you don't necessarily have to be a computer programmer to get into the field, which I loved.. I thought that was important and we talked about the different types of skills you need… you can have creativity, you can be a good problem solver. I thought it was really powerful that they could see themselves in that career path in the future."*

These conversations also shattered gender stereotypes surrounding AI careers, fostering a sense of inclusivity and encouraging students to envision themselves pursuing these opportunities regardless of their social and economic backgrounds and identities. They gained confidence in their abilities to contribute to the field of AI, not just as users, but active developers of AI systems. Discussions with these experts not only broadened students' horizons but also provided teachers with valuable insights into the real-world applications of AI and their underlying mechanisms. This deepened insight enabled them to impart knowledge with greater confidence, which was in turn instrumental in kindling students' enthusiasm and curiosity for AI. Additionally, the curriculum shed light on the vital role of AI in the workforce, making students aware of its relevance and applicability in the professional world. This awareness served to underscore the importance of AI literacy and its integration into classrooms.

3.1.2.4 Integration of AI Literacy into School Curriculum

Our teachers advocated the need for integrating AI education into existing school curricula. They spoke about devising numerous strategies to stimulate the interest of various educational stakeholders to enable this change. One of the teachers used AI careers as a significant point of interest for students, recognizing the inherent allure of the dynamic field. They also placed a strong emphasis on democratizing access to AI education, promoting creativity in AI endeavors, and highlighting practical applications of AI. Their commitment to making AI accessible to all students, regardless of their prior knowledge, aligned with the goal of achieving AI for all, fostering inclusivity and diversity in AI education.

Teachers also firmly believed that AI education played a pivotal role in shaping students' lives in multifaceted ways. This encompassed not only technical skills but also ethical thinking, responsibility, and inclusivity. They championed the idea that students should create projects that hold genuine meaning, prompting them to think critically about the technology they employ and its potential consequences. Additionally, teachers underscored the importance of design thinking, empathy, and community-focused AI projects. They viewed these aspects as essential components of AI literacy, encouraging students to consider the broader impact of their AI creations on society. In their pedagogical approach, they utilized AI tools such as Kuki as a springboard for AI-related conversations. This strategy enabled students to delve into AI topics from various angles, fostering a deeper understanding and appreciation of the field.

3.1.2.5 Challenges

Integrating AI curriculum into classrooms presented a range of challenges for teachers, as evidenced by their experiences and feedback, which encompassed technical issues, varied levels of student knowledge, the need for improved teaching materials, and concerns about potential harm associated with AI. Several teachers found themselves dropping AI components of projects due to limited time and technical issues, emphasizing the importance of extensive teacher training and support to prepare educators for the challenges and opportunities presented by future iterations of the curriculum. In section 3.1.1.1, we spoke about the technical difficulties that teachers ran into when using new AI tools. Frustrations stemming from these difficulties underscored the need for robust technical support and resources to facilitate a seamless AI learning and teaching experience amongst teachers. These problems compounded when students came in with diverse levels of prior technical knowledge. These disparities necessitated the need for tailored approaches and one-on-one code reviews to accommodate individual learning needs. Teachers also discussed refining the AI tools presented to them and their students to ensure that they were user-friendly and effective. The challenges in grasping concepts like Teachable Machines and the recognition of potential harm associated with AI highlighted the significance of comprehensive training and pedagogical support. Teachers acknowledged how their individual teaching methods impacted student learning outcomes in very different ways. It also brought out the need for adaptable resources that could cater to the evolving needs of both educators and students.

3.2 Attitudes and Beliefs towards Teaching and Learning

This theme covers how teaching and learning attitudes (more generally) evolved with the curriculum and how those beliefs influenced its implementation in classrooms.

3.2.1 Teaching Approaches and Strategies

Teachers employed a variety of instructional strategies when implementing AI curriculum in the classroom, drawing from their prior experiences and insights, thus reflecting their overall attitudes and beliefs about teaching and learning. They encouraged students to engage in brainstorming sessions to foster creativity and independence. They also drew on real-world examples to supplement theoretical lessons in classrooms. Teachers showed a strong preference for visual aids for enhancing student understanding. They also encouraged flexibility and diverse self-expression that enabled students to explore AI concepts in novel ways. There were instances when teachers used structured approaches to delivering content, often involving the use of PowerPoint presentations, since it helped guide students systematically through the material. Teachers simplified complex topics by breaking down lessons into smaller chunks and coupled them with student discussions to promote effective communication and presentation skills. Additionally, prototyping and building systems physically played a crucial role in enhancing students' comprehension and engagement by allowing them to apply their learnings to real-world settings.

Teachers took special measures to reinforce the notions of community and identity amongst student groups. They allocated time to discuss student perceptions on these ideas:

> *"We covered the whole idea of community and identity several times and reminded students of it…these conversations helped them come up with some fantastic ideas."*

Tools like kanban boards played a crucial role in instilling a sense of community in the classroom. Group prototyping sessions served as another avenue for fostering community-oriented thinking. These sessions allowed students to exercise empathy as they brainstormed novel concepts and identified stakeholders for their final projects, simultaneously acknowledging the influence of their own personal experiences in shaping these ideas.

3.2.2 Assessment and Evaluation

Teachers employed diverse assessment tools to gauge students' progress and comprehension. To streamline assessments, some teachers created straightforward criteria checklists and used desired learning outcomes to evaluate students' performance and understanding of AI concepts. Other teachers focused primarily on project completion and adherence to its predefined criteria. Most of the teachers used the kanban board to evidence students' planning and design thinking process, but only few of them used the more playful assessments that were designed for the curriculum. Teachers also created opportunities for student peer teaching and evaluations to foster a collaborative learning environment and provide students with diverse perspectives and constructive criticism on their work. They encouraged students to link specific tasks associated with their project to the overarching driving question or goal that they had initially started with. This was however challenging to accomplish. Given the open-ended structure inherent in the project-based learning model of the curriculum, students found the driving question to be too vague, leading to its limited utility in guiding project work. Consequently, they directed their efforts towards fulfilling assigned tasks, without engaging in critical reflection on the task's overarching relevance:

> *"Once students got into the project, they just wanted to work and there's very little thought in the process or the reasoning behind it… and so you have to really keep pulling them back into that line of thinking… this was an ongoing battle throughout."*

In response, teachers consistently posed probing questions to redirect students' focus towards the central driving question throughout the learning process. They also motivated students to engage in reflections to nurture their understanding of AI concepts and ethical considerations.

3.2.3 Curriculum Flexibility and Student-Centered Learning

The implementation of our curriculum underscored the importance of adopting student-driven learning approaches that cater to diverse learning preferences and interpretations. Teachers noted the provisions made within the curriculum to incorporate various modalities of student expression, fostering creativity, and amplifying student voices. While some students preferred assimilating and learning new material through text, some others thrived in participatory classroom discussions and valued interactive visualizations:

> *"Students had really nice thoughts and they expressed themselves differently… sometimes it was written, sometimes they shared them through discussions out loud… other times through visual representations… they felt comfortable as a result… it goes back to the flexibility of the program where it does not have to be just via writing."*

These variations also influenced their capacity to retain freshly acquired information, with certain students opting to meticulously transcribe their learnings for future reference and recall. The adaptability of the curriculum also empowered teachers to draw upon their distinct pedagogical backgrounds, including those in non-STEM disciplines such as art and business, to establish real-world connections when elucidating new concepts to students. They seamlessly integrated elements of computational thinking and design thinking, encouraging students to independently explore and generate innovative ideas. They promoted ideas of independent problem-solving through tinkering, experimentation, and troubleshooting errors. They showed students how they could harness the feedback they received from teacher guidance and support to iteratively refine their project designs, thus enhancing their overall efficacy and social impact.

3.3 Teacher Recommendations

The eight semi-structured interviews brought to light many recommendations that teachers thought would be useful when integrating the curriculum into their own classrooms. These recommended changes can be broken down into two categories: curriculum and material improvements and teaching and support enhancements.

3.3.1 Curriculum and Material Improvements

The teachers made many suggestions regarding improving the curriculum and associated material. The RAICA modules come with a Student Workbook for students to answer questions and document their design thinking process, keeping track of ideas they are formulating when learning about AI. One of our teachers thought that the workbook was too long and required their students to do too much writing,

conflicting with their typical hands-on teaching style. Another teacher made a more general request for more opportunities for hands-on experiences for their students.

One teacher modified the curriculum to include more peer feedback during the prototyping stage and felt this to be incredibly useful in their classroom. They suggested that more peer feedback be incorporated into the curriculum:

> *"Within those sessions, we could ideate more [and] solve each other's problems as well."*

Conversely, another teacher wanted the curriculum to dive right into practice, making prototyping with feedback a less crucial step. Two teachers found the use of kanban boards for planning to be *"too difficult"* for their students, abandoning their use in their classrooms, while another made the kanban board a collaborative effort between students to promote engagement with the tool.

Teachers requested more supplemental materials for both themselves and students. One teacher wanted to share more resources with their class, requesting additional examples for students.

Some teachers found that the arrangement of the materials presented challenges. One teacher articulated that a central location with links to all necessary materials would have helped them stay more organised, and that students needed a clear place to curate their ideas and projects.

Although one teacher found the curriculum to be too open-ended, they wanted to stress that teachers provide different directions that their students could take if they felt challenged by the open-endedness. This same teacher also pointed out that they would like to see the curriculum made more accessible to students with differential learning needs, which was seconded by another.

3.3.2 Teaching Support

Teachers suggested a plethora of methodologies for enhancing teaching and their feelings of support. Technology was a challenge for some teachers. One teacher had a dedicated tech support person in their classroom and found this to be a valuable tool for their students. Another teacher stated that it was very important for teachers to have more experience with *"real-world AI applications"* to succeed with the deployment of the curriculum.

Teachers wanted to help their students see themselves as part of the future of AI augmented technology, one stating that it is important for teachers to be able to guide their students in making meaningful connections between their projects and their personal lives.

Teachers at times mentioned feeling underprepared to deliver curriculum material and ideas in classrooms. One teacher found the teacher's guides to be *"too vague"* and they wanted more specific directions of what to do in their classroom; another expressed a lack of confidence in addressing students' fears and concerns of AI in a positive manner. A different teacher suggested revisiting past ideas on their own practices of design thinking and empathy to apply to this unfamiliar AI context.

One of our teachers recommended including useful resources to walkthroughs and tutorials with the curriculum for other educators to use when increasing their knowledge base. A second teacher added to this idea by requesting videos to enhance their understanding of AI tools:

> *"I found [video tutorials] super helpful to kind of understand the walkthrough and to help the kids with the technology, because I had done it first."*

One of our teachers even went as far as to suggest that more instruction prior to teaching the modules would be useful, stating that they would have loved an extra month to prepare and understand the content before diving into teaching this material.

4 CONCLUSIONS

Taken together, our teachers provided valuable information and feedback necessary to inform future curriculum design changes to this context. We summarize the key takeaways from our study below:

- Designing AI literacy curriculum that is flexible and conducive to varied classroom contexts, including diverse student and teacher backgrounds is critical to ensuring its success and long-term sustainability in the classroom. Since some teachers requested more directions and specifications within our materials to help them navigate novel concepts better, future iterations can provide more examples of hands-on activities (that teachers could organise in classrooms) and scaffolding questions (that they can use to stimulate discussions amongst students). The

open-ended nature of our curriculum empowered teachers to leverage their unique backgrounds and prior teaching experiences when developing materials to explain new AI concepts. This is an aspect that we would like to preserve in future iterations.

- Teacher support and professional development (PD) sessions should precede the implementation of new AI curricula to ensure that teachers with no prior experience are equipped with the resources needed to comfortably deliver foundational AI knowledge. Some teachers were apprehensive about facilitating challenging conversations about AI and its ethical implications with their students. Preparing teachers for engaging in discussions about fears students have of traversing an unknown technical landscape would require additional PD training and examples on moderating difficult conversations in class. This approach has been utilised in other work resulting in teachers' increased comfort in having conversations about AI ethics as well as using AI tools [18]. The inclusion of such materials in the RAICA curriculum would also likely increase teachers' competence and self-confidence in multiple areas, adding to their productivity and efficacy for students.
- Inclusivity is another key factor to consider when deploying AI curricula. Hands-on and unplugged activities (that aren't reliant on device availability amongst students) can increase the accessibility of the curriculum for classrooms with limited access to technology [19], allowing students to learn something about AI without programming themselves. Additional teaching resources and accommodations for students of determination can also increase the impact of lessons, by catering to a wider audience.

Our work throws light on important considerations when developing an AI literacy curriculum and associated PD. It also underscores the need and benefit of continuous feedback during development and initial deployment. Our interviews help us understand the impact of the curriculum in different classroom environments and encourage us to place a stronger emphasis on accessibility and addressing the varied needs of students in future iterations. Teachers' perspectives are invaluable in achieving these goals.

ACKNOWLEDGEMENTS

This work was supported by funding from DP World. The funding body played no role in the study design or in the analysis and interpretation of data. We wish to acknowledge and thank our partner schools and their teachers who generously gave of their time to implement our curriculum and provide feedback.

REFERENCES

[1] DiPaola D, Payne BH, Breazeal C. Decoding design agendas: an ethical design activity for middle school students. In Proceedings of the interaction design and children conference 2020 Jun 21 (pp. 1-10).

[2] Zhang H, Lee I, Ali S, DiPaola D, Cheng Y, Breazeal C. Integrating ethics and career futures with technical learning to promote AI literacy for middle school students: An exploratory study. International Journal of Artificial Intelligence in Education. 2023 Jun;33(2):290-324.

[3] Ali S, DiPaola D, Lee I, Hong J, Breazeal C. Exploring generative models with middle school students. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems 2021 May 6 (pp. 1-13).

[4] Ali S, DiPaola D, Breazeal C. What are GANs?: introducing generative adversarial networks to middle school students. In Proceedings of the AAAI Conference on Artificial Intelligence 2021 May 18 (Vol. 35, No. 17, pp. 15472-15479).

[5] Druga S, Williams R, Breazeal C, Resnick M. "Hey Google, is it ok if I eat you?" Initial explorations in child-agent interaction. In Proceedings of the 2017 conference on interaction design and children 2017 Jun 27 (pp. 595-600).

[6] Tissenbaum M, Sheldon J, Abelson H. From computational thinking to computational action. Communications of the ACM. 2019 Feb 21;62(3):34-6.

[7] Yadav A, Gretter S, Hambrusch S. Challenges of a computer science classroom: Initial perspectives from teachers. In Proceedings of the workshop in primary and secondary computing education 2015 Nov 9 (pp. 136-137).

[8] Shang F. Research on the Influencing Factors of Teachers' Beliefs in Intelligent Education. The Educational Review.. 2021 Aug 25;5(8):287-94.

[9] Ayanwale MA, Sanusi IT, Adelana OP, Aruleba KD, Oyelere SS. Teachers' readiness and intention to teach artificial intelligence in schools. Computers and Education: Artificial Intelligence. 2022 Jan 1;3:100099.

[10] Chai CS, Lin PY, Jong MS, Dai Y, Chiu TK, Huang B. Factors influencing students' behavioral intention to continue artificial intelligence learning. In2020 international symposium on educational technology (ISET) 2020 Aug 24 (pp. 147-150). IEEE.

[11] Lindner A, Romeike R, Jasute E, Pozdniakov S. Teachers' perspectives on artificial intelligence. In12th International conference on informatics in schools,"Situation, evaluation and perspectives", ISSEP 2019.Th

[12] Burgsteiner H, Kandlhofer M, Steinbauer G. Irobot: Teaching the basics of artificial intelligence in high schools. In Proceedings of the AAAI conference on artificial intelligence 2016 Mar 5 (Vol. 30, No. 1).

[13] Druga S. Growing up with AI: Cognimates: from coding to teaching machines (Doctoral dissertation, Massachusetts Institute of Technology).

[14] Lin P, Van Brummelen J, Lukin G, Williams R, Breazeal C. Zhorai: Designing a conversational agent for children to explore machine learning concepts. InProceedings of the AAAI Conference on Artificial Intelligence 2020 Apr 3 (Vol. 34, No. 09, pp. 13381-13388).

[15] Lin P, Van Brummelen J. Engaging teachers to co-design integrated AI curriculum for K-12 classrooms. In Proceedings of the 2021 CHI conference on human factors in computing systems 2021 May 6 (pp. 1-12).

[16] Lee I, Perret B. Preparing High School Teachers to Integrate AI Methods into STEM Classrooms. In Proceedings of the AAAI Conference on Artificial Intelligence 2022 Jun 28 (Vol. 36, No. 11, pp. 12783-12791).

[17] Saldaña J. Coding and analysis strategies.

[18] Lee I, Zhang H, Moore K, Zhou X, Perret B, Cheng Y, Zheng R, Pu G. AI Book Club: An Innovative Professional Development Model for AI Education. In Proceedings of the 53rd ACM Technical Symposium on Computer Science Education-Volume 1 2022 Feb 22 (pp. 202-208).

[19] Williams R, Ali S, Devasia N, DiPaola D, Hong J, Kaputsos SP, Jordan B, Breazeal C. AI+ ethics curricula for middle school youth: Lessons learned from three project-based curricula. International Journal of Artificial Intelligence in Education. 2023 Jun;33(2):325-83.

Teaching Students to Question the Machine

Short-Term Effects of an AI Literacy Workshop on Middle-School Students' Regulation of LLM Interaction

Olivier Clerc[1,*], Rania Abdelghani[2], Chloé Desvaux[1], Eliott Poisson[1], Pierre-Yves Oudeyer[1,†] and Hélène Sauzéon[1,†]

[1]INRIA research center University of Bordeaux, Talence, France
[2]Hector Research Institute, University of Tübingen, Germany

Abstract

Generative artificial intelligence (GenAI) systems are increasingly used by students, making AI literacy an urgent educational goal. In the context of large language models (LLMs), AI literacy involves not only understanding how these systems work and fail, but also applying that understanding to monitor prompts, evaluate responses, and decide when to revise, reject, or continue an interaction. Yet classroom evidence on how to support these applied aspects of AI literacy in middle school remains limited.

We report a quasi-experimental classroom study examining whether a two-hour AI literacy workshop shaped students' regulation of LLM use during science problem-solving tasks. A total of 116 students in grades 8–9 completed six LLM-supported science problems. The intervention group attended a workshop two days earlier that introduced how LLMs work and fail, together with practical guidance on prompting and response evaluation; the control group completed the same tasks without attending the workshop.

Students who attended the workshop showed more adaptive interaction behaviors: they were less likely to accept underspecified prompts, more likely to ask follow-up questions after insufficient responses, and more sensitive to prompt specification when judging response correctness. They also achieved modestly higher final task-performance scores. In contrast, GenAI and metacognitive self-report scores did not robustly predict behavioral regulation or final task performance. These findings suggest that brief AI literacy instruction can shape middle-school students' short-term regulation of LLM-supported science problem solving, while also highlighting the need for longer-term studies of retention, transfer, and learning beyond immediate task performance.

Keywords

Artificial intelligence in education, AI literacy, Generative AI, Self-regulation, Intervention

1. Introduction

Generative artificial intelligence (GenAI) systems are increasingly entering students' everyday learning practices. Recent surveys show rapid uptake of GenAI tools across age groups, including teenagers and students in secondary science, technology, engineering, and mathematics (STEM) contexts [1, 2]. Students now use large language model (LLM)-based chatbots for activities such as generating explanations, asking questions, and solving subject-specific problems [3], creating new opportunities for adaptive support and personalized learning [4, 5]. At the same time, these opportunities depend on how students engage with model outputs. As LLMs become more fluent, useful, and persuasive, students may be increasingly willing to delegate key reasoning steps to the model rather than generate, evaluate, or revise explanations themselves. For learning, this cognitive offloading may be especially consequential, because it can reduce effortful processing and weaken opportunities to develop critical thinking and self-regulated inquiry [6, 7, 8]. The educational value of GenAI therefore depends not only on what these systems generate, but also on how learners regulate their interaction with them [9, 10].

This challenge has made AI literacy an increasingly important educational objective. AI literacy is commonly conceptualized as a set of competencies that includes understanding how AI systems work, recognizing their limitations, using them effectively, and evaluating their social and ethical

ALIT4ALL 2026: 2nd International Workshop on AI Literacy Education For All
*Corresponding author.
†Equal supervision.
olivier.clerc@inria.fr (O. Clerc)
© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

implications [11, 12]. In LLM-supported problem solving, this requires more than declarative knowledge: learners must translate goals into prompts, evaluate outputs, and decide whether to revise, reject, or continue the exchange. In this paper, we use *regulation of LLM interaction* to refer to this task-level monitoring-and-control process. This process draws on metacognitive monitoring and control [13, 14], but is specific to the demands of generative AI interaction, where learners must regulate both their own problem-solving process and the system's contributions. Our definition is deliberately narrower than the full AI literacy construct. It does not cover all forms of GenAI regulation, such as detecting sycophantic behavior, deciding when not to use AI at all, or verifying factual accuracy with external sources. These competencies are important, but they were outside the scope of the present intervention and behavioral measures.

Middle-school students may particularly need support in developing this applied dimension of AI literacy. Their critical evaluation and self-regulatory skills are still developing, and persuasive LLM outputs may further encourage over-trust or shallow engagement [8, 15]. Consistent with this concern, Abdelghani et al. [16] found that middle-school students often struggled to select effective prompts or challenge unsatisfactory answers when using ChatGPT for science problem solving. Despite having access to an LLM, students produced deep and accurate final explanations in only about half of the tasks on average, suggesting that tool access alone was insufficient. Students with more positive attitudes toward AI were less able to distinguish effective from underspecified prompts. In contrast, self-reported metacognitive regulation and more critical perceptions of AI were associated with behaviors linked to higher-quality final answers, such as better prompt discrimination and more follow-up questioning.

These findings point to a need for AI literacy interventions that are practical, classroom-feasible, and behaviorally assessable. Because students already use LLMs but often struggle to regulate their interactions, instruction should go beyond declarative knowledge. Yet controlled classroom evidence on brief, scalable AI literacy interventions remains limited in K–12 settings [17], and existing school-based studies are often exploratory, small, or lack comparison groups [18, 19].

To address this gap, we conducted a quasi-experimental classroom study in a French middle school ($N = 116$, grades 8–9, ages 13–15) to examine whether a two-hour AI literacy workshop shaped students' regulation of LLM use during science problem solving. The workshop combined conceptual content on how LLMs work and fail with practical activities designed to preserve students' agency during interaction. Importantly, the workshop did not rehearse the experimental science tasks. Instead, students practiced checking whether an initial prompt was aligned with the intended task, whether the LLM response satisfied that goal, and whether they needed to revise the prompt or continue the interaction. Two days later, students completed six science problem-solving tasks with access to a GenAI system; a control group completed the same session without having attended the workshop. In these tasks, suggested prompts were either well-specified or underspecified for the science problem, without students being informed of this manipulation. The outcome session therefore tested whether students transferred a general regulatory stance—keeping agency over prompting, response evaluation, and follow-up—to the experimental task. This design allowed us to observe whether students accepted or reformulated suggested prompts, whether they asked follow-up questions, and how accurately they judged the correctness of LLM responses.

We treated these behavioral indicators as the primary outcomes of interest. Final answer quality (students' exercise scores) was analyzed as a downstream task-performance measure. We also examined whether GenAI self-reports and a general metacognitive awareness questionnaire predicted task performance. This allowed us to compare behavioral indicators of LLM-interaction regulation with offline self-report measures.

The study was guided by three research questions:

1. Does a brief AI literacy workshop change middle-school students' observable regulation of LLM interaction during science problem-solving tasks?
2. Are students' regulation behaviors associated with their final task performance?
3. Do GenAI self-reports and general metacognitive awareness predict students' behavioral regulation and final task performance?

2. Method

2.1. Design and participants

The study was conducted in one French middle school during regular class hours. Because the intervention was implemented within existing school schedules, individual random assignment was not feasible. Students were assigned to the intervention or control condition at the class level according to timetable availability. Both groups were drawn from the same school and were tested during the same period.

A total of 162 students in grades 8 and 9 (typical age range 13–15 years) initially participated. Due to end-of-year scheduling constraints, 27 students who attended the workshop were absent during the test phase and were excluded from the main analyses. An additional 19 students were excluded because they did not complete all six exercises. The final sample comprised 116 students: 40 in the control group (24 grade 8, 16 grade 9) and 76 in the intervention group (50 grade 8, 26 grade 9). The unequal group sizes reflect the use of intact classes and differences in attendance across the workshop and test sessions.

Within the intervention group, two students provided inconsistent identifiers between the pre- and post-intervention GenAI questionnaires. These students were retained for the main behavioral and task-performance analyses, but excluded from pre–post questionnaire analyses. Accordingly, questionnaire analyses were conducted on $n = 40$ control students and $n = 74$ intervention students.

2.2. Procedure overview

Control students completed a 90-min exercise session individually, solving science problems with access to an LLM as a support tool. Intervention students completed the same exercise session, preceded two days earlier by a two-hour AI literacy workshop. A timeline and classroom procedure are provided in the Supplementary Material (SM Figs. S1–S2).

Before the exercise session, all students completed self-report questionnaires measuring GenAI perceptions and metacognitive awareness. During the exercise session, students completed six science problem-solving tasks. For each task, they decided whether to use a suggested prompt, interacted with the LLM, evaluated the response, optionally asked a follow-up question, and wrote a final answer in their own words.

2.2.1. AI literacy workshop

The intervention was a two-hour, classroom-feasible AI literacy workshop designed to strengthen both conceptual understanding of LLM-based GenAI systems and applied regulation of LLM interaction. The conceptual component introduced age-appropriate explanations of how LLMs generate text, why they can produce incomplete or incorrect answers, and why fluent responses should not be accepted uncritically. The practical component did not rehearse the later experimental science tasks. Instead, it focused on preserving students' agency during interaction with an LLM. Students first completed a collective response-evaluation activity in which they discussed common problems in LLM outputs. Students then worked in small groups on prompting activities from domains distinct from the experimental science tasks. For each activity, they received a task goal and an initial prompt, predicted whether the prompt was likely to elicit a satisfactory response, queried the LLM, evaluated the answer, and revised the prompt or continued the interaction when needed. These activities trained students to maintain agency during their LLM interaction. Examples of workshop materials are provided in the Supplementary Material (SM S3), and the full workshop slide deck and facilitation script are available in the accompanying OSF repository.

2.2.2. Science problem-solving tasks and LLM environment

The exercise session was based on the science problem-solving tasks described in Abdelghani et al. [16]. Each student completed six science exercises within 90 minutes (Fig. 1). The six exercises were randomly

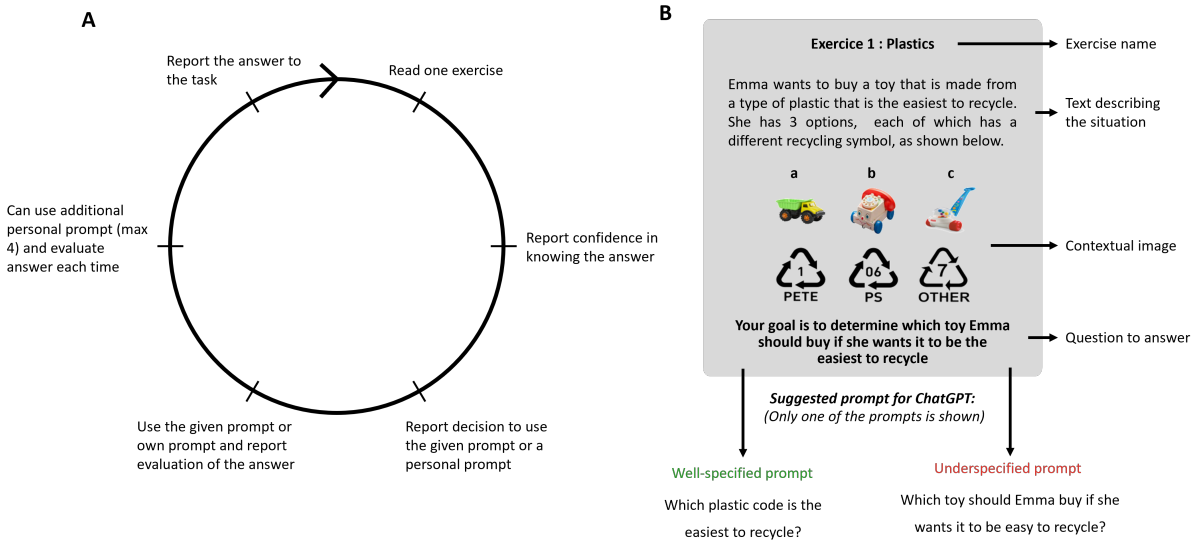


Figure 1: Student–LLM workflow and exercise structure. (A) For each exercise, students rated confidence (3-point), decided whether to use the suggested prompt, iteratively queried and evaluated the LLM (3-point; optional follow-up), and wrote a final answer; repeated across six exercises. **(B)** Example sheet and prompt manipulation. The suggested prompt was either well-specified or underspecified (one shown per exercise).

selected from a pool of twelve tasks and presented in random order to limit collaborative solving between students.

Each exercise included a single suggested prompt. Suggested prompts were either well-specified, meaning that they contained the key contextual elements needed for the exercise, or underspecified, meaning that they omitted important information and were likely to elicit generic or incomplete responses. Prompt specification was pseudorandomized such that each student encountered three well-specified and three underspecified prompts. Students were not informed about this manipulation. Exercises were provided on paper to prevent direct copy-paste of the full problem statement into the chatbot interface.

Students worked individually at a computer with two browser tabs open: an LLM-based chatbot interface and a web questionnaire guiding the procedure. For each exercise, students could interact iteratively with the LLM and evaluate its responses before producing a final written answer in their own words.

Because access constraints sometimes required students to use different public interfaces, the LLM environment was not perfectly homogeneous across task attempts. They primarily used GPT-4o mini *via* DuckDuckAI; when this interface failed, some used other public chatbot interfaces running GPT-4o, GPT-4.1-nano, or GPT-3.5-turbo (robustness checks and interface use: SM S4.2–S4.4). To address this, we conducted robustness checks on the prompt manipulation itself, verifying that prompts designed as underspecified still elicited low-quality responses and well-specified prompts high-quality responses independently of the possible models used by the students. We report interface use in SM S4.4, Table S5.

2.3. Measures

2.3.1. Behavioral indicators of LLM-interaction regulation

We operationalized regulation of LLM interaction through three behavioral indicators collected during each exercise. First, *query-level control* was measured through prompt acceptance: students decided whether to use the suggested prompt or replace it with their own formulation. Rejecting an underspecified prompt was interpreted as adaptive regulation, whereas accepting it indicated lower query-level control. Second, *response evaluation* was measured through students' correctness judgments after receiving the LLM answer. Third, *interaction control* was measured through follow-up questioning: students

could ask additional questions after the initial LLM response, a behavior expected to be especially useful when the initial response was insufficient. These measures were treated as behavioral indicators of the applied regulatory dimension of AI literacy, not as measures of the full AI literacy construct.

2.3.2. Final answer quality

Students' final written answers were scored on a 0–2 rubric: 0 = incorrect, 1 = partially correct, and 2 = correct. Exercise-level scores were used in analyses linking regulation behaviors to answer quality. Student-level final task performance was computed by summing the six exercise scores and converting the total to a 0–20 scale.

2.3.3. Self-reported measures

All students completed two questionnaires prior to the exercise phase. The first was a GenAI questionnaire adapted from Bernabei et al. [20] following Abdelghani et al. [16]. It covered six dimensions: Attitude, Trust, Social influence, Fairness & ethics, Usefulness, and Effort & ease of use (24 items; 4-point Likert scale; maximum score = 96). It captured students' familiarity with ChatGPT, perceived reliability of its answers, and perceived usefulness for school tasks; for example, items asked whether students knew the limits of ChatGPT and whether they considered its answers reliable. Because this adapted version has not been formally validated for this age group, GenAI self-reports were treated as secondary and interpreted cautiously.

The second questionnaire was the Jr. MAI [21], a middle-school measure of metacognitive knowledge and regulation of cognition (18 items; 5-point Likert scale; maximum score = 90). It was included as a broad measure of general metacognitive awareness, not as an LLM-specific regulation instrument. Full item wording and scoring details are provided in SM S2.

Questionnaires served two purposes. First, GenAI and metacognitive self-report scores were used as individual-difference predictors to test whether students' self-perceptions were associated with behavioral regulation and final task performance. Second, in the intervention group only, GenAI self-reports were collected both before and after the workshop as a secondary outcome to assess whether the workshop changed these self-reported dimensions.

2.4. LLM-as-judge validation and final-answer scoring

We used GPT-4o as an LLM-as-judge to validate the prompt manipulation and to score students' final written answers. In both cases, we first evaluated the annotation setup on a random $\approx 10\%$ subset independently rated by human annotators, and applied GPT-4o to the full dataset only after meeting predefined reliability and non-inferiority criteria. Agreement was assessed with Krippendorff's α, appropriate for multi-rater ordinal ratings, and with the advantage-probability test of Calderon et al. [22], which compares the model against held-out human annotators.

For prompt-response annotation, agreement was higher when GPT-4o was included ($\alpha = .825$) than for human-only annotations ($\alpha = .763$), and GPT-4o outperformed each held-out human annotator ($\omega = 1.00$). For student-answer scoring, GPT-4o was non-inferior to human annotators ($\alpha = .752$ with GPT-4o included vs. $\alpha = .740$ human-only; $\omega = 1.00$). Full validation details are reported in SM S4.

2.5. Data analysis

Analyses were conducted in Python with two-sided tests and $\alpha = .05$. For repeated task-level observations, generalized linear models used cluster-robust standard errors at the student level. We report odds ratios (ORs) for logistic models, unstandardized coefficients for linear models, and effect sizes where appropriate. False-discovery-rate (FDR) correction was applied to families of questionnaire follow-up tests. Full details are reported in the Supplementary Material, and the anonymized analysis notebook used to generate the results and figures is provided in the accompanying OSF repository.

We first tested whether the intervention affected the three behavioral indicators of LLM-interaction regulation. Query-level control was analyzed with a logistic GLM predicting prompt acceptance from condition, prompt specification, their interaction, and pre-query confidence. Response evaluation was analyzed among accepted-prompt trials with a logistic GLM predicting correctness judgments from condition, prompt specification, and their interaction. Interaction control was analyzed among accepted underspecified-prompt trials by comparing student-level follow-up-question rates between groups.

We then tested whether regulation behaviors were associated with final answer quality. For each prompt type, we computed within-student reject-minus-accept contrasts in answer quality and compared these contrasts between groups. We also tested whether follow-up-question rates after accepted underspecified prompts predicted mean answer quality differently by condition.

Because final task scores were bounded and non-normal in the intervention group, the main group comparison used a Mann–Whitney test. We also computed signal-detection indices (d' and response bias β) to quantify students' selectivity in accepting well-specified rather than underspecified prompts, and tested whether these indices predicted final task performance. Finally, we examined whether GenAI and metacognitive self-reports predicted behavioral regulation or final task performance.

2.6. Ethics and data protection

The study was approved by an institutional ethics committee; identifying details are withheld for double-blind review. Parent/guardian consent and student assent were obtained before participation. No personally identifying data were collected. Students did not use personal accounts to access the LLM interface and were instructed not to enter personal information into the chatbot.

To preserve students' privacy, we did not collect or access the full transcripts of their interactions with the LLM. The dataset only contained students' responses to the study questionnaire, including their prompt-use decisions, correctness judgments, follow-up-question indicators, confidence ratings, and final written answers. To preserve anonymity and comply with feasibility constraints in the school context, we did not collect gender, race/ethnicity, or other demographic identifiers beyond grade level. As a result, the dataset could not be used to test whether intervention effects differed across demographic subgroups.

3. Results

3.1. The intervention changed students' regulation of LLM interaction

We first examined whether the workshop affected the three behavioral indicators of LLM-interaction regulation.

Query-level control: A logistic GLM predicting prompt acceptance showed that intervention students were less likely than control students to accept suggested prompts overall (OR $= 0.47$, $p = .022$; average marginal effect $= -.155$, $p = .018$; Fig. 3C). Well-specified prompts were more likely to be accepted than underspecified prompts (OR $= 1.85$, $p = .040$). The condition $\times$ prompt-specification interaction was not significant ($p = .228$), but follow-up contrasts showed that the group difference was concentrated on underspecified prompts: intervention students accepted these prompts less often than controls (51.5% vs. 66.7%; OR $= 0.47$, $p = .022$), whereas groups did not differ reliably for well-specified prompts (74.4% vs. 79.2%; $p = .385$).

Response evaluation: Students in both groups were more accurate at judging the correctness of AI responses when prompts were well-specified rather than underspecified (OR $= 2.06$, $p = .001$), but this effect was larger in the intervention group (interaction OR $= 2.52$, $p = .012$; Fig. 2B), indicating that intervention students' evaluation judgments were more accurate.

Interaction control: Among accepted underspecified-prompt trials, intervention students asked follow-up questions more often than controls (59.2% vs. 27.9%; Welch $t = 3.80$, $p < .001$, $d = 0.80$; Fig. 2C).

Additional analyses are provided in SM S5.3–S5.5.

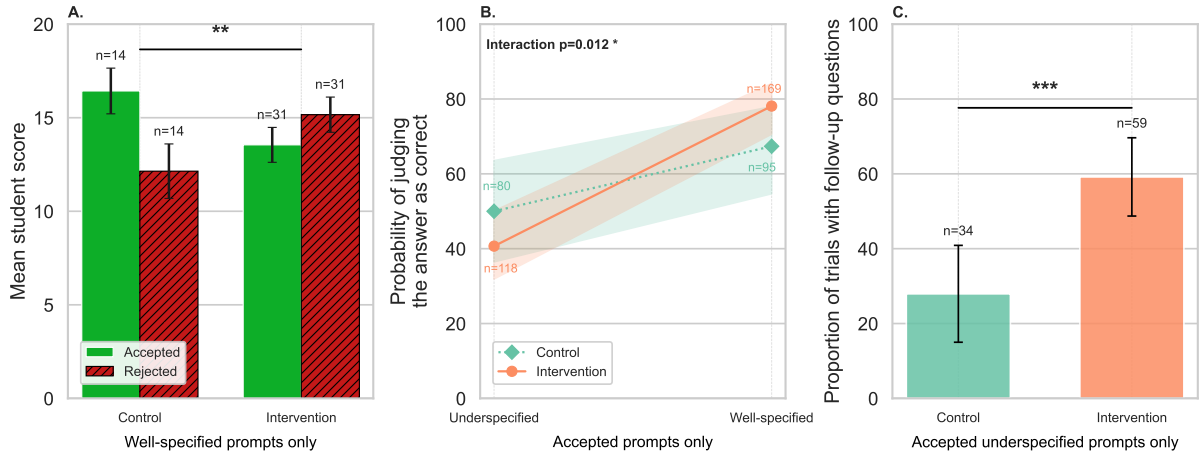


Figure 2: Effects of the intervention on prompt replacement, follow-up behavior, and correctness judgments. (A) Within-student performance by accept vs. reject on well-specified prompts. Rejecting well-specified prompts was more costly in the control group. **(B)** Probability of judging an AI answer as correct (GLM; 95% CI) by prompt specification. Trained students made more accurate correctness judgments. **(C)** Follow-up questioning after accepted underspecified prompts by group. Trained students asked more follow-up questions after underspecified prompts.

3.2. Regulation behaviors were associated with final answer quality

We next examined whether students' regulatory behaviors were associated with final answer quality. For well-specified prompts, rejecting rather than accepting the suggested prompt was costly in the control group but not in the intervention group. The reject-minus-accept contrast differed significantly between groups ($M = -4.29$ in control vs. $M = 1.61$ in intervention; Welch $t = 2.83$, $p = .009$, $d = 0.91$; Fig. 2A). This suggests that control students often replaced well-specified prompts with less effective alternatives, whereas intervention students could reformulate prompts without a comparable performance penalty.

For underspecified prompts, rejected trials scored higher than accepted trials descriptively in both groups, but the between-group difference in this contrast was not significant ($p = .194$). Because follow-up questions are especially relevant when students initially accept an underspecified prompt, we examined this subset directly. Among students who accepted underspecified prompts, follow-up-question rate was more strongly associated with final answer quality in the intervention group than in the control group ($b = 6.43$, 95% CI [0.36, 12.50], $p = .038$). Within the intervention group, this behavior was associated with higher-quality answers (approximately $+5$ points out of 20), whereas no such benefit was observed in the control group.

3.3. Final task performance showed a modest group difference

Final task-performance scores differed between groups, with higher scores in the intervention group than in the control group ($M = 11.38$, $SD = 3.87$, $n = 76$ vs. $M = 10.29$, $SD = 3.04$, $n = 40$; Mann–Whitney $U = 1871$, $p = .040$, $r_z = .19$, rank-biserial $r = .23$; Fig. 3A). This effect was modest, and absolute performance remained limited in both groups despite access to a GenAI system. However, maximum scores were high (18.3/20), indicating that the tasks were feasible under the scoring and time constraints.

To examine whether final answer quality was related to students' selectivity in accepting well-specified versus underspecified prompts, we computed signal-detection indices. Sensitivity (d') was numerically higher in the intervention group than in the control group, but this difference was not significant ($M = 0.53$ vs. 0.28; Welch $t = 1.45$, $p = .150$, $d = 0.27$). Response bias (β) did not differ between groups ($p = .746$). Across students, higher sensitivity predicted higher final task-performance scores ($b = 1.76$, 95% CI [1.06, 2.46], $p < .001$), as did a more conservative response bias ($b = 1.65$, 95%

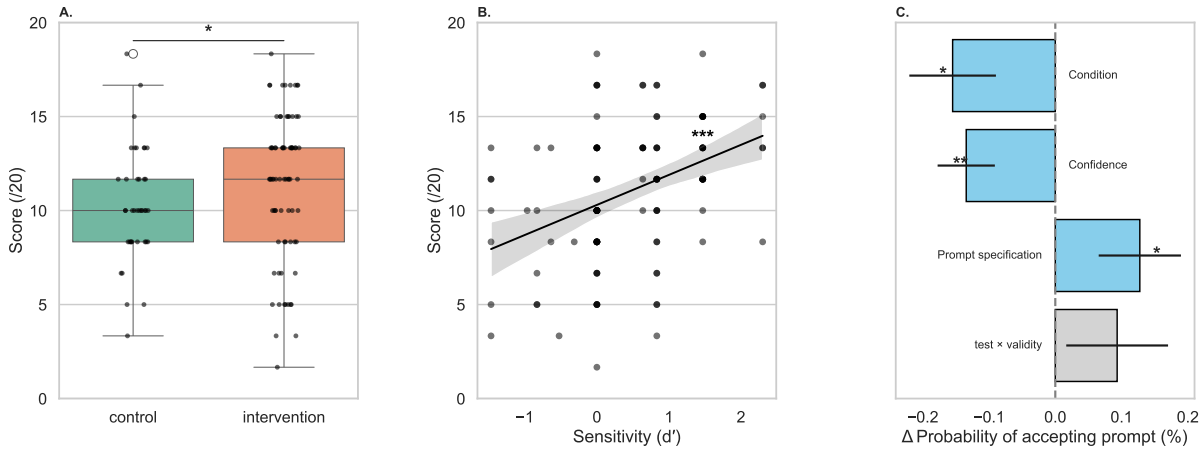


Figure 3: Workshop effects on performance and prompt acceptance. (A) Final-answer performance (0–20) by group. Intervention students achieved higher final-answer scores. **(B)** Relationship between performance and prompt-discrimination sensitivity (d'). Performance increased with prompt discrimination. **(C)** Predictors of accepting the suggested prompt (GLM).

CI [0.25, 3.05], $p = .021$). Once d' and β were included in the model, the direct condition effect was no longer significant ($p = .279$). The condition $\times d'$ interaction was positive but did not reach significance ($b = 1.36$, $p = .072$). Exploratory within-group models showed that both d' and β predicted final task performance in the intervention group, whereas neither predictor was reliable in the control group. These results suggest that the workshop may have helped students act on prompt-discrimination cues, rather than simply improving their ability to detect underspecified prompts. In other words, the intervention may have strengthened the link between noticing a problem and selecting an appropriate regulatory action, such as reformulating the prompt or asking a follow-up question.

3.4. GenAI self-report scores changed only weakly

Baseline GenAI self-report dimensions did not differ reliably between groups. A one-way MANOVA on pre-test dimensions was not significant (*Pillai's trace* = .083, $F(6, 107) = 1.61$, $p = .151$), and FDR-corrected univariate comparisons showed no significant baseline group differences (all $p_{\text{FDR}} \geq .189$).

Within the intervention group, a multivariate test of pre–post change scores showed a small overall shift in GenAI self-reports (*Pillai's trace* = .195, $F(6, 68) = 2.75$, $p = .019$). However, no individual dimension survived FDR correction. Attitude and Fairness & ethics showed small uncorrected increases (Attitude: $M_\Delta = 0.11$, $p = .018$, $p_{\text{FDR}} = .054$; Fairness & ethics: $M_\Delta = 0.13$, $p = .011$, $p_{\text{FDR}} = .054$), but these effects should be interpreted cautiously. Thus, the workshop's effects were clearer for behavioral indicators of LLM-interaction regulation than for self-reported GenAI perceptions.

3.5. Self-report measures did not robustly predict behavioral regulation or final task performance

We first tested whether GenAI and metacognitive self-report scores predicted the student-level behavioral indicators of LLM-interaction regulation described above. None of the associations survived FDR correction (all $p_{\text{FDR}} \geq .580$). Within the intervention group, changes in the six GenAI self-report dimensions explained little variance in prompt-discrimination sensitivity ($R^2 = .11$), and dimension-level predictors were not robust after correction.

Self-report scores also did not explain final task performance. Across students, GenAI self-report scores were not correlated with final task-performance scores ($r = .01$, $p = .881$, $R^2 < .001$, $n = 116$). Metacognitive self-report scores were likewise unrelated to final task-performance scores ($r = .04$, $p = .649$, $R^2 = .002$, $n = 116$). Together with the behavioral analyses above, these results suggest that

observable regulation during LLM use was more informative than offline self-reports for explaining students' final answer quality.

4. Discussion

This study examined whether a brief AI literacy workshop could shape middle-school students' regulation of LLM use during science problem solving. The main finding is that the workshop changed students' observable interaction behaviors. Compared with control students, intervention students were less likely to accept underspecified prompts, their response-evaluation judgments were more sensitive to prompt specification, and they asked more follow-up questions when initial prompts were underspecified. These behavioral changes were accompanied by a modest downstream difference in final task performance, but absolute scores remained limited in both groups. Thus, the study should not be interpreted as showing durable learning effects. Rather, it provides classroom-based evidence that short AI literacy instruction can affect how students regulate LLM-supported problem solving under realistic school constraints.

4.1. AI literacy as regulatory action

The workshop was designed to support an applied dimension of AI literacy: maintaining agency while interacting with an LLM. The results suggest that this regulatory stance transferred to the science problem-solving tasks. Intervention students were more selective with underspecified prompts, their correctness judgments were more sensitive to prompt specification, and they asked more follow-up questions when the initial prompt was likely to produce an insufficient answer.

These effects appear to reflect changes in regulatory action more than large improvements in prompt-discrimination ability. Although intervention students accepted underspecified prompts less often, their overall sensitivity to prompt specification was only numerically higher than controls'. In contrast, prompt-discrimination sensitivity and response bias predicted final task performance more reliably in the intervention group. This suggests that the workshop may have strengthened the link between noticing a problem and acting on it. The fact that control students suffered a performance cost when replacing well-specified prompts further suggests that unguided prompt revision can be harmful when students do not know what information needs to be preserved.

4.2. Task performance is improved but is not evidence of durable learning

Intervention students achieved modestly higher final task-performance scores than control students, but this outcome should be interpreted carefully. Students completed the tasks with access to an LLM, and the test occurred only two days after the workshop. The study therefore shows improved regulation during LLM-supported problem solving, not durable learning gains.

The modest absolute performance in both groups also reinforces this point. Even with access to GenAI, students' average final scores remained around 11–12 out of 20. This suggests that access to an LLM alone is not sufficient for producing high-quality answers. Students still need to identify what information is relevant, evaluate whether the model's response is adequate, and synthesize a final answer in their own words. These demands are challenging for middle-school students [23] and may be especially important when GenAI outputs appear fluent and authoritative. Because the activity was low-stakes (not graded), reduced motivation may also have contributed [24] to this; importantly, these constraints applied to both groups and therefore do not undermine the between-group effect.

4.3. Self-reports were weak indicators of applied AI literacy

GenAI and metacognitive self-report scores did not predict students' behavioral regulation or final task performance. It suggests that offline self-reports may capture students' perceived competence and not actual regulation during interaction. This interpretation is consistent with recent evidence that

self-report measures of metacognition often correlate weakly with online monitoring judgments during task performance, whereas behavioral indicators of monitoring and control can be more informative for explaining learning-related outcomes [25]. It also aligns with broader evidence that self-assessments can overestimate competence [26], including in the domain of digital skills [27].

The workshop also produced only limited changes in GenAI self-reports. Although the multivariate pre–post test suggested a small overall shift, no individual dimension survived correction for multiple comparisons. In contrast, clearer effects were observed in students' behavior during the LLM-supported task. This pattern suggests that changes in applied AI literacy may not be well captured by short-term changes in self-reported attitudes, confidence, or perceived knowledge.

At the same time, the null self-report findings should be interpreted cautiously. The GenAI questionnaire was adapted from prior work and has not been formally validated for this age group. The Jr. MAI measures general metacognitive awareness, not LLM-specific regulation. Future work should develop and validate AI-literacy instruments that distinguish conceptual knowledge, attitudes, ethical awareness, perceived competence, and applied regulation during interaction.

4.4. Implications for classroom AI literacy education

The study has three implications for AI literacy education in middle school. First, it suggests that brief, classroom-feasible interventions can change students' LLM interaction behaviors, at least in the short term. This extends Abdelghani et al. [16], who showed that middle-school students often struggle to select effective prompts and challenge unsatisfactory answers when using ChatGPT for science problem solving. The control group in the present study showed similar difficulties, consistent with evidence that non-experts often struggle to design robust prompts without structured support [28]. A two-hour workshop might not be expected to produce stable expertise, but it can introduce regulatory habits that students may begin to apply during LLM-supported tasks.

Second, AI literacy interventions should target not only conceptual knowledge about how AI systems work, but also practical regulation skills and learner agency. This matters because students may possess general questioning and explanation-evaluation skills in non-GenAI contexts [29, 30], but may not spontaneously apply them when confronted with fluent and confident-looking model outputs [8]. In this sense, the goal of AI literacy education is not simply to teach students about AI, but to help them remain active agents during AI-supported work.

Third, AI literacy interventions should be evaluated with behavioral measures in addition to self-reports. The present findings suggest that self-reports may miss important differences in actual use. This is especially important if AI literacy is understood as a competence that includes procedural knowledge, not only declarative knowledge.

4.5. Limitations and future directions

Several limitations should be noted. First, the study was quasi-experimental rather than individually randomized. Groups came from the same school, were tested during the same period, and did not differ on pre-test GenAI self-reports, but students were assigned by class schedules and group sizes were unequal. Unmeasured baseline differences, including science ability, therefore cannot be ruled out.

Second, the study measured short-term effects only, as the test occurred two days after the workshop. Future work should examine retention, repeated practice, transfer to new tasks, and performance without LLM support. Third, the study was conducted in one school and focused on science problem solving, so generalization to other subjects, schools, and student subgroups remains to be tested.

Fourth, the prompt-acceptance task does not fully mirror ecological student uses of LLMs. In many classroom contexts, students would write their own prompts from scratch rather than decide whether to accept a suggested prompt. Prompt acceptance should therefore be interpreted as an experimental probe of whether students noticed missing task information. By contrast, evaluating the adequacy of the model's response and deciding whether to ask a follow-up question are likely to remain central forms of regulation whenever students use LLMs for learning.

Fifth, the LLM environment was not fully homogeneous because students used different public interfaces when access constraints required it. However, this heterogeneity reflects ecological conditions, and our prompt-validation procedure confirmed that the experimental manipulation was robust across models. Finally, our behavioral measures captured broad regulatory decisions but not the linguistic content of prompts or follow-up questions. Future work could combine these indicators with privacy-preserving transcript analysis.

Overall, the findings suggest that brief, classroom-feasible AI literacy instruction can support students' applied regulation of LLM interaction, but longer-term and broader studies are needed to determine whether such regulation leads to durable learning benefits.

Supplementary Materials, Data, Code, and Workshop Materials

Supplementary materials, workshop materials, analysis code, and anonymized data are available in an OSF repository: https://osf.io/fyu7c/overview?view_only=e254db4360f34da8a14f44e9c468dd59. The repository is organized into three folders: `supplementary_materials`, `workshop_materials`, and `data_and_code`. The `workshop_materials` folder includes the workshop slide deck and facilitation script, currently available in French. The `data_and_code` folder includes the anonymized dataset and the code used to reproduce the analyses and figures reported in the paper.

Declaration on Generative AI

During the preparation of this work, the author(s) used generative AI tools for language editing and formatting assistance. After using these tools, the author(s) reviewed and edited the content as needed and take full responsibility for the publication's content.

References

[1] O. Sidoti, E. Park, J. Gottfried, About a quarter of U.S. teens have used ChatGPT for schoolwork—double the share in 2023, Pew Research Center, 2025. URL: https://www.pewresearch.org/short-reads/2025/01/15/about-a-quarter-of-us-teens-have-used-chatgpt-for-schoolwork-double-the-share-in-2023/.

[2] F. Valeri, P. Nilsson, A. M. Cederqvist, Exploring students' experience of chatgpt in stem education, Computers and Education: Artificial Intelligence 8 (2025). doi:10.1016/j.caeai.2024.100360.

[3] P. Zhang, G. Tur, A systematic review of chatgpt use in k-12 education, 2024. doi:10.1111/ejed.12599.

[4] M. M. Rahman, Y. Watanobe, Chatgpt for education and research: Opportunities, threats, and strategies, Applied Sciences (Switzerland) 13 (2023). doi:10.3390/app13095783.

[5] H. Li, T. Xu, C. Zhang, E. Chen, J. Liang, X. Fan, H. Li, J. Tang, Q. Wen, Bringing generative ai to adaptive learning in education, preprint arXiv:2402.14601 (2024).

[6] E. F. Risko, S. J. Gilbert, Cognitive offloading, Trends in cognitive sciences 20 (2016) 676–688.

[7] M. Gerlich, Ai tools in society: Impacts on cognitive offloading and the future of critical thinking, Societies 15 (2025). doi:10.3390/soc15010006.

[8] C. Zhai, S. Wibowo, L. D. Li, The effects of over-reliance on ai dialogue systems on students' cognitive abilities: a systematic review, Smart Learning Environments 11 (2024). doi:10.1186/s40561-024-00316-7.

[9] Y. Wardat, M. A. Tashtoush, R. AlAli, A. M. Jarrah, Chatgpt: A revolutionary tool for teaching and learning mathematics, Eurasia Journal of Mathematics, Science and Technology Education 19 (2023). doi:10.29333/ejmste/13272.

[10] T. Adiguzel, M. H. Kaya, F. K. Cansu, Revolutionizing education with ai: Exploring the transformative potential of chatgpt., Contemporary educational technology 15 (2023).

[11] T. K. Chiu, Z. Ahmad, M. Ismailov, I. T. Sanusi, What are artificial intelligence literacy and competency? a comprehensive framework to support them, Computers and Education Open 6 (2024) 100171.
[12] I. H. Y. Yim, A critical review of teaching and learning artificial intelligence (ai) literacy: Developing an intelligence-based ai literacy framework for primary school education, Computers and Education: Artificial Intelligence 7 (2024) 100319.
[13] L. Tankelevitch, V. Kewenig, A. Simkute, A. E. Scott, A. Sarkar, A. Sellen, S. Rintel, The metacognitive demands and opportunities of generative ai, in: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 2024, pp. 1–24.
[14] S. N. Yeh, C. J. R. Siah, Students' attitudes toward llms and its association with metacognitive abilities: A cross-sectional study, Nurse Education in Practice 88 (2025). doi:10.1016/j.nepr.2025.104567.
[15] N. Bessas, E. Tzanaki, D. Vavougios, V. P. Plagianakos, The role of chatgpt in junior high school physics education: Insights from teachers and students and guidelines for optimal use, Social Sciences and Humanities Open 11 (2025). doi:10.1016/j.ssaho.2025.101610.
[16] R. Abdelghani, K. Murayama, C. Kidd, H. Sauzéon, P.-Y. Oudeyer, The illusion of understanding: How middle-schoolers fail to regulate inquiry with chatgpt in a science task, preprint arXiv:2505.01106v2 (2026).
[17] F. Guo, T. Li, C. J. Cunningham, One year in the classroom with chatgpt: empirical insights and transformative impacts, Frontiers in Education 10 (2025). doi:10.3389/feduc.2025.1574477.
[18] P. Bitzenbauer, Chatgpt in physics education: A pilot study on easy-to-implement activities, Contemporary Educational Technology 15 (2023). doi:10.30935/cedtech/13176.
[19] B. Puppart, J. Aru, Short-term ai literacy intervention does not reduce over-reliance on incorrect chatgpt recommendations, preprint arXiv:2503.10556 (2025).
[20] M. Bernabei, S. Colabianchi, A. Falegnami, F. Costantino, Students' use of large language models in engineering education: A case study on technology acceptance, perceptions, efficacy, and detection chances, Computers and Education: Artificial Intelligence 5 (2023) 100172.
[21] B. Kim, B. Zyromski, M. Mariani, S. M. Lee, J. C. Carey, Establishing the factor structure of the 18-item version of the junior metacognitive awareness inventory, Measurement and Evaluation in Counseling and Development 50 (2017) 48–57.
[22] N. Calderon, R. Reichart, R. Dror, The alternative annotator test for LLM-as-a-judge: How to statistically justify replacing human annotators with LLMs, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vienna, Austria, 2025, pp. 16051–16081. doi:10.18653/v1/2025.acl-long.782.
[23] P. N. Winograd, Strategic difficulties in summarizing texts, Reading research quarterly (1984) 404–425.
[24] C. E. DeMars, Test stakes and item format interactions, Applied Measurement in Education 13 (2000) 55–77.
[25] P. Seban, R. Sikl, K. Urban, Relationship between monitoring judgments and self-report measures of metacognition in educational research, International Journal of Educational Research 131 (2025) 102578.
[26] D. Dunning, C. Heath, J. M. Suls, Flawed self-assessment: Implications for health, education, and the workplace, Psychological science in the public interest 5 (2004) 69–106.
[27] M. Palczyńska, M. Rynko, Ict skills measurement in social surveys: Can we trust self-reports?, Quality and Quantity 55 (2021) 917–943. doi:10.1007/s11135-020-01031-4.
[28] J. D. Zamfirescu-Pereira, R. Y. Wong, B. Hartmann, Q. Yang, Why johnny can't prompt: how non-ai experts try (and fail) to design llm prompts, in: Proceedings of the 2023 CHI conference on human factors in computing systems, 2023, pp. 1–21.
[29] C. M. Mills, C. H. Legare, M. G. Grant, A. R. Landrum, Determining who to question, what to ask, and how much information to ask for: The development of inquiry in young children, Journal of Experimental Child Psychology 110 (2011) 539–560.

[30] C. M. Mills, K. R. Sands, S. P. Rowles, I. L. Campbell, "i want to know more!": Children are sensitive to explanation quality when exploring new information, Cognitive Science 43 (2019) e12706.

A SURVEY OF HUMAN-IN-THE-LOOP FOR MACHINE LEARNING

Xingjiao Wu[1,2], *Luwei Xiao*[2], *Yixuan Sun*[3], *Junhang Zhang*[2], *Tianlong Ma*[1,2], *Liang He*[1,2,*]

1 Shanghai Key Laboratory of Multidimensional Information Processing, East China Normal University, Shanghai, China

2 School of Computer Science and Technology, East China Normal University, Shanghai, China

3 Fudan University, Shanghai, China

{wuxingjiao2885, louisshaw008, madmaxkgb, junhangzhang68}@gmail.com, {tlma, lhe}@cs.ecnu.edu.cn

arXiv:2108.00941v3 [cs.LG] 26 Apr 2022

ABSTRACT

Machine learning has become the state-of-the-art technique for many tasks including computer vision, natural language processing, speech processing tasks, *etc.* However, the unique challenges posed by machine learning suggest that incorporating user knowledge into the system can be beneficial. The purpose of integrating human domain knowledge is also to promote the automation of machine learning. Human-in-the-loop is an area that we see as increasingly important in future research due to the knowledge learned by machine learning cannot win human domain knowledge. Human-in-the-loop aims to train an accurate prediction model with minimum cost by integrating human knowledge and experience. Humans can provide training data for machine learning applications and directly accomplish tasks that are hard for computers in the pipeline with the help of machine-based approaches. In this paper, we survey existing works on human-in-the-loop from a data perspective and classify them into three categories with a progressive relationship: (1) the work of improving model performance from data processing, (2) the work of improving model performance through interventional model training, and (3) the design of the system independent human-in-the-loop. Using the above categorization, we summarize the major approaches in the field; along with their technical strengths/ weaknesses, we have a simple classification and discussion in natural language processing, computer vision, and others. Besides, we provide some open challenges and opportunities. This survey intends to provide a high-level summarization for human-in-the-loop and to motivate interested readers to consider approaches for designing effective human-in-the-loop solutions.

Index Terms— Human-in-the-loop, machine learning, deep learning.

1. INTRODUCTION

Deep learning is a frontier for artificial intelligence, aiming to be closer to its primary goalartificial intelligence. Deep learning has seen great success in a wide variety of applications, such as natural language processing, speech recognition, medical applications, computer vision, and intelligent transportation system [1]. The great success of deep learning is due to the larger models [2]. The scale of these models has included hundreds of millions of parameters. These hundreds of millions of parameters allow the model to have more degrees of freedom enough to awe-inspiring description capability.

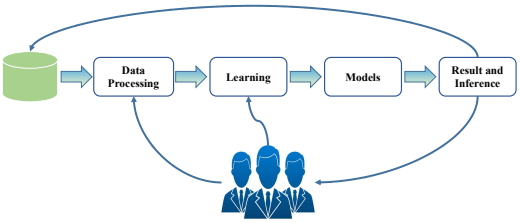


Fig. 1: The development cycle of model.

However, the large number of parameters requires a massive amount of training data with labels [3]. Improving model performance by data annotation has two crucial challenges. On the one hand, the data growth rate is far behind the growth rate of model parameters, which has primarily hindered the further development of the model. On the other hand, the emergence of new tasks has far exceeded the speed of data updates, and annotating all samples is labor-intensive and time-consuming. To tackle this challenge, many researchers build new datasets by generating samples, thereby speeding up model iteration and reducing the cost of data annotation [4, 5, 6]. Besides, a group of researchers devise pre-trained models and transfer learning to solve this challenge [7, 8, 9], such as Transformers [10], BERT [11] and GPT [12]. These works have achieved incredible results.

Unfortunately, the generated data is only used as base data to initialize the model. To obtain a high-precision usable model, labeling and updating specific data is often necessary. So various work based on weak supervision has been proposed [13, 14]. A great many researchers have proposed using few-shot to push the model to learn from fewer sam-

*Corresponding author.

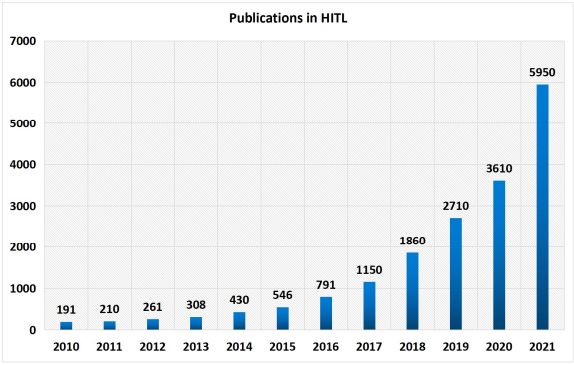


Fig. 2: The increasing research interest in the human-in-the-loop, obtained through Google scholar search with keywords: "human-in-the-loop" and "machine learning".

ples [15].

1.1. Significance of Human-in-the-loop

Integrated a priori knowledge in the learning framework is an effective means to deal with sparse data, as the learner does not need to induce the knowledge from the data [16]. More recently, an ever-growing number of researchers have made efforts to incorporate pre-training knowledge into their learning framework [17, 18, 19]. As special agents, humans have rich prior knowledge. If the developer can encourage the machine to engage with learning human wisdom and knowledge, it would help deal with sparse data, especially in medical fields like clinical diagnosis and lack of training data [20, 21, 22, 23]. Furthermore, recent advances in cognitive science and human-machine interaction have suggested that human-related elements, such as emotional state and practical capability, impact human teaching performance and machine learning results on different tasks.

A multitude of researchers have proposed utilizing a conception called "human-in-the-loop" to tackle the above challenges, mainly addressing these issues by incorporating human knowledge into the modeling process [24]. Human-in-the-loop conception is an extensive area of research that covers the intersection of computer science, cognitive science, and psychology.

As illustrated in Fig. 2, human-in-the-loop (namely "human-in-the-loop" and "machine learning") is an active research topic in machine learning, and there has been a rich publication in the past ten years.

As shown in Fig. 1, a conventional machine-learning algorithm typically consists of three parts [25]. The first is data preprocessing, the second is data modeling, and the last is the developer modifying the existing process to improve performance. Generally speaking, the performance and results of machine learning models are unpredictable, which leads to a large degree of uncertainty in which part of the machine-human interaction is capable of bringing the best learning effect. Different researchers focus on manual intervention in distinguishable parts. In this paper, we investigate existing studies on human-in-the-loop technology via various implementations of human-in-the-loop from different practical perspectives(e.g. Data Processing, Model Training and Inference, and System construction and Application). It is crucial to explore how interaction type interplays with other components of a human-in-the-loop pipeline to affect the intelligent systems' learning outcomes. In addition, more research focuses on the design of independent systems to help complete the improvement of the model. In this paper, we first discuss the work of improving model performance from data processing. Next, we discuss the work of improving model performance through interventional model training. Finally, we discuss the configuration of the system independent "human-in-the-loop".

We explore the following questions around human-in-the-loop for machine learning:

- What is the human-in-the-loop for machine learning challenges, and what are the possible solutions to advance this research?
- From a data perspective, what is the research status of human-in-the-loop for machine learning, research challenges, and what are the future directions?
- From the perspective of model training, what is the research status of human-in-the-loop for machine learning, research challenges, and what are the future directions?
- From an application perspective, what is the research status of human-in-the-loop for machine learning, research challenges, and what are the future directions?

1.2. Importance of this survey

This paper is a systematic summary and analysis of the research area of human-in-the-loop with a focus on the following essential aspects:

- We had a comprehensive summary of the work of human-in-the-loop so far, and divided these papers into CV, NLP and other application areas. We connected these papers in series according to data preprocessing, data annotation, and model training and inference from the perspective of data flow; and finally, we focused on the application based on human-in-the-loop;
- We have classified and compared various methods of human-in-the-loop. Through classification and comparison, we have summarized the challenges currently encountered by human-in-the-loop and put forward some discussions on solving these challenges;
- We also conducted qualitative evaluations and comparisons between different methods to evaluate them con-

sistently. This would assist readers in deciding which method is appropriate for the problem at hand;
- We have also identified a series of important milestones achieved by methods in this area;
- In addition to the methodological summary and analysis of human-in-the-loop's work, we also discussed the system construction and application of human-in-the-loop. We analyze the human-in-the-loop on system construction divided by system components and applications following the baseline of engineering requirements.

Before our work, there were also quiet a few excellent works reviewing human-in-the-loop [26, 27, 28, 29, 30, 31], but we are the first review of human-in-the-loop for machine learning. Much of the previous work is from the perspective of hardware and robotics, while we are talking about machine learning. In our research process, we also follow on previous experiences and ideas. Certainly, a single article may not cover all methods in this growing field. Nevertheless, we have sought to make this survey as comprehensive as possible. To help with this goal, we researched and analyzed a good many related references and documents. In addition, we also tried a variety of classification methods to divide the article structure, and finally, we chose to connect the whole work from the perspective of data.

1.3. Organization of contents

Since there has been no similar review before, we explore all papers (since 1990) containing the keywords "human-in-the-loop" and "machine learning" by Google Scholar. Firstly, we screen out relevant articles by reading the titles and abstracts. Secondly, we briefly classify the screened articles. Moreover, we read the screen articles in detail and constantly revised the classification. Finally, we summarize and form the final content according to the classification.

In Section 2, we investigate the data processing method based on human-in-the-loop and then discuss data preprocessing, data annotation, and iterative labeling. In Section 3, we summarize and analyze research on model training and reasoning based on human-in-the-loop, and discuss human-in-the-loop from natural language processing and computer visual perspectives, respectively. In Section 4, we review the human-in-the-loop on system construction divided by system components and applications and discuss human-in-the-loop from the software and hardware integrated perspective, respectively. In Section 5, we propose a list of challenges based on the results of the survey. Finally, we conclude our work in Section 6.

2. DATA PROCESSING

At present, deep learning has played an irreplaceable role in many fields [1, 32]. The great success of deep learning is due to larger-scale models, which include hundreds of millions of parameters [2]. Such a large amount of parameters empower the model with enough degrees of freedom to obtain awe-inspiring description capability. A massive amount of training data with labels are required to deal with a considerable number of parameters [3]. However, making annotations requiring much labor is likely to lag behind the growth in model capacity, and available datasets are quickly becoming outdated in size and density [33]. So the methods of utilizing unlabeled data to improve the model capability have increasingly gained much attention [3, 34, 35, 36]. The most significant difficulty is based on the fact that unlabeled data usually include incorrect samples, such as disturbing images, defective statements, and violations of constraints. Suppose these inaccurate samples are exactly sampled as the key one, the errors brought about will be fatal [37]. To tackle this challenge, numerous researchers focus on exploring the way to generate a more rich sample space [4, 38, 5], trying to develop a universal model such as Transformers [39], BERT [11] or GPT [12] so that the model can learn features more effectively. Based on these successful methods, researchers then consider the further step: adopting little data to obtain more satisfactory results. So these models are employed for more tasks by fine-tuning and performing incredible results [40, 41, 42]. Although these methods still need to annotate a lot of data, which brings unnecessary trouble, we still noticed that interference in the model performance only some critical samples in the new datasets. Here goes to a critical issue that needs to be solved urgently, *How do we find out the key samples, and can we annotate key samples more easily?*

The intuitive idea to solve this concern with a specific method is in three steps: Select a few samples which models can not recognize. (1) Use the particular approach to annotate selected samples. (2) Push the model to learn features from the latest annotated samples. (3) This idea allows the model to make the most of the data information at the least cost.

Multiple researchers try to apply human-in-the-loop-based methods to optimize models from the perspective of data. According to surveys, scientists spend about 80% of their time on data processing compared to model building [30]. We investigated the data processing methods based on human-in-the-loop and established a pipeline as shown in Fig. 3. We reviewed the representative works in data processing and showed the classification result as shown in Table 1. This section explores the strengths and deficiencies of data processing with human-in-the-loop by demonstrating data preprocessing, data annotation, and iterative labeling.

2.1. Data Preprocessing

It is well-known that deep learning is a process of modeling data. The success of deep learning largely depends on the quality of data, and data analysis plays an irreplaceable role in building a more effective model. However, it is not easy

Table 1: A overview of representative works in data processing. DP: data preprocessing; DA: data Annotation; IL: iterative labeling; CV: Computer Vision; NLP: Natural Language Processing; SP: Speech Processing.

Work	Data Processing			Year	Area				Task	Quantitative results
	DP	DA	IL		CV	NLP	SP	Other		
Yu *et al.* [33]			✓	2015	✓				Scene and Object Categories	$0.82- > 0.88(mAP)$
He *et al.* [43]			✓	2016		✓			CCG Parser	$84.2\%- > 85.9\%(F_1)$
Self *et al.* [44]	✓			2016				✓	Data Analysis	$--$
Zhuang *et al.* [45]	✓			2017				✓	Knowledge Bases Integration	$84.3\%- > 86.6\%(F_1)$
Li *et al.* [46]	✓			2017				✓	Data Integration	$--$
KIM *et al.* [47]		✓		2018			✓		Finding Sound Events	$--$
Doan *et al.* [48]	✓			2018				✓	Data Analysis	$--$
Dong *et al.* [49]	✓			2018		✓		✓	Data Fusion;DOM extraction	$--$
Gentile *et al.* [50]	✓	✓		2019		✓			Dictionary Expansion	$41.17\% \uparrow$
Zhang *et al.* [51]		✓	✓	2019		✓			Entity Extraction	$0.8376- > 0.8644(F_1)$
Laure *et al.* [52]	✓			2019	✓	✓	✓	✓	Data Preparation	$--$
Gurajada *et al.* [53]	✓			2019		✓			Entity Resolution	$--$
Lou *et al.* [54]	✓			2019		✓			Knowledge Graph Programming	$98.78\%(top-10\ hit)$
Liu *et al.* [55]		✓	✓	2019	✓				Person Re-Identification	$45.55\%- > 71.52\%(mAP)$
Wallace *et al.* [56]		✓		2019		✓			Question Answering	$--$
Fan *et al.* [57]			✓	2019	✓			✓	Network Anomaly Detection	$84.71\%(FPR)$
Krokos *et al.* [58]			✓	2019	✓			✓	Knowledge Discovery	$67.7\%- > 88.2\%(Acc)$
Klie *et al.* [59]		✓		2020		✓			Entity Linking	$35\% \uparrow (annotation\ speed)$
Chai *et al.* [30]	✓			2020				✓	Outlier Detection	$88\%- > 94\%(recall)$
Butler *et al.* [60]		✓		2020	✓				Facial Expressions	$--$
Ristoski *et al.* [61]	✓			2020		✓			Relation Extraction	$0.02- > 0.0321(F_1)$
Qian *et al.* [62]	✓			2020		✓			Entity Name Understanding	$--$
Le *et al.* [63]		✓	✓	2020	✓				Self-Annotation For Video Object	$34.1\%- > 56.6\%(mIoU)$
Bartolo *et al.* [64]		✓		2020		✓			Reading Comprehension	$39.1\%(F_1)$
Cutler *et al.* [65]	✓			2021		✓			Entity Recognition	$--$
Meng *et al.* [66]		✓		2021	✓				3D Point Cloud Object Detection	$--$
Zhang *et al.* [67]		✓		2021	✓				Screentone and Manga Processing	$--$
Adhikari *et al.* [68]		✓		2021	✓				Object Detection	$--$

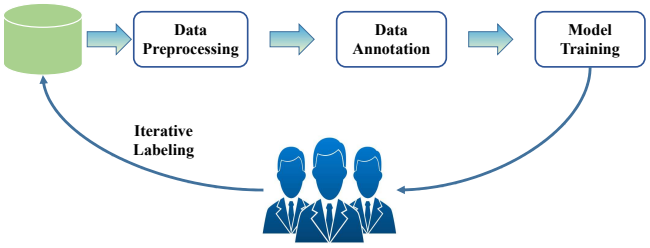


Fig. 3: A human-in-the-loop data processing pipeline.

to find a static method for data analysis, which means data scientists must analyze existing data by employing experts' experience. The greatest challenge in data analysis lies in the complexity of high-dimensional data, which makes it hard for models to discover data structure. Furthermore, adjusting these parameters dramatically depends on the knowledge of data experts or domain experts. Motivated by this phenomenon, Self *et al.* [44] proposed a human-model interactive parameter adjustment mode to facilitate user participation by bridging the gaps between a user's intention and the parameters of a weighted multidimensional scaling model. Therefore, Doan *et al.* thought establishing a benchmark is an effective means to solve this challenge [48]. Besides, data analysis inevitably involves two considerations: how to carry out automated parameter analysis methods, and the other is how to explore the ability to establish a specific benchmark. Considering these two issues simultaneously, Laure [52] expanded based on Learn2Clean. They developed automated machine learning approaches (AutoML) that can optimize the hyperparameters of a considered ML model with a list of by-default preprocessing methods. This method is devoted to proposing a principled and adaptive data preparation approach to help and learn from the user to select the optimal sequence of data preparation tasks. With the rapid development of research, researchers are no longer satisfied with solving specific problems in human-in-the-loop data analysis (HILDA). They are more concerned with several "big picture" questions regarding HILDA. Current data analysis technology can correctly obtain the necessary information and knowledge by constructing a knowledge base or graph. However, for the HILDA community or tools, the degree of attention is not enough. Researchers should pay more attention to such issues and make them more popular in the user community to develop data repositories and tools.

Utilizing human-in-the-loop methods to deal with natural language data has inherent advantages over other kinds of data (*i.e.*, speech recognition, medical applications, computer vision, and intelligent transportation system). Most of the human-in-the-loop methods are used in the informa-

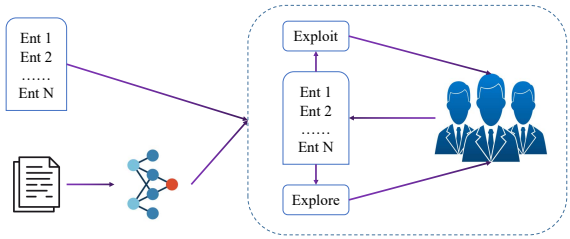


Fig. 4: The architecture of explore and exploit.

tion extraction stage. As Fig. 4, Gentile *et al.* [50] propose an interactive dictionary expansion tool using two neural language models. Many researchers add humans to NLP tasks(such as entity analysis, knowledge graphs, etc.) by using crowdsourcing [50, 52, 54]. Ristoski *et al.* [61] introduced a method of extracting instances from various web resources, which dramatically improves the performance of the system by introducing human-recycling components. Besides, this method can integrate the human experience and knowledge to empower machines' accurate intelligence. Consistent with the method of dictionary expansion proposed before, the core idea of this method is also realized by expanding the existing dictionary. In addition to the direct annotation, since there are semantic disambiguation phenomena in a few NLP tasks, exploiting the model alone cannot accomplish these tasks. However, the performance of these tasks can be significantly improved by adopting the human knowledge intervention accumulated in unconscious learning. Qian *et al.* [62] put forward a deep learning-based entity name understanding system called PARTNER, which provided a more reasonable way of interaction. PARTNER is based on active learning and a weak supervision method. It is necessary to apply data screening technology for sample selection to find those error-prone samples in the human-in-the-loop process. Cutler *et al.* [65] presented a method that marked potentially incorrect labels with high sensitivity in the named entity recognition corpus.

Our summary of the previous papers finds that most of the existing human-in-the-loop preprocessing studies concern extracting and analyzing complex information in the real world. Nevertheless, there is still very little work that utilizes human-in-the-loop technology to perform data preprocessing on CV tasks. We conjuncture that the nature of this phenomenon is that there is a lack of a perfect mode to integrate the human experience into image processing, detailed discussion can be found in Section 5.

2.2. Data Annotation

For new tasks, annotating data is a complex but crucial task to realize artificial intelligence. A considerable number of researchers have proposed employing a human-in-the-loop-based method for fast and precise (compared to fragile labeling) operations, especially in NLP and CV domains.

In NLP tasks, data annotation is bifurcated into two categories. One is the annotation of specific task datasets, such as entity extraction [50, 51], entity linking [59], and the other is more abstract tasks, such as Q&A tasks [56] and reading comprehension tasks [64].

The entity processing task is critical in NLP, and its success or failure directly affects the performance of NLP [69]. Currently, there are two main methods for entity extraction, one is to formulate regular expressions for automatic extraction, and the other is the entity mentioned in the manual tagged document. However, neither of these two strategies can extract entities efficiently and accurately. Zhang *et al.* [51] devised a human-in-the-loop-based entity extraction method to obtain the best return on investment in a limited time. With the deepening of research, we need to handle a growing base of tasks. The emergence of new schemes is beyond our expectations. Regular expressions can help handle common data, but there is no expected magic for new data never seen before. To mitigate this issue, a few studies proposed approaches to solve the cross-domain problem in entity links. They find the entities mentioned in the text and filter and discriminate them according to entities sorting information. This method is especially suitable for semantic disambiguation tasks [59].

How to cope with more complicated tasks is also a focus of research now. Researchers attempt to integrate human experience and knowledge to endow machines with more intelligence. More specifically, to what extent do neural network models understand the natural language, and can they be further enhanced? To explain this problem and explore more interpretability of the neural network model. Wallace *et al.* [56] developed an open application system that contains an interactive interface to talk with the machine, thereby generating more Q&A language materials to collect more research data and help the researcher explain the model predictions. Bartolo *et al.* [64] tried three different sets of annotation methods in the reading comprehension task to build a gradually more robust model in the annotation cycle. Significantly, they created a challenging dataset by collecting 36,000 samples. However, with the enhancement of the cyclic model, the performance gradually deteriorates. In contrast, the more robust model can still learn from the data intensively collected by the weaker model in the loop [70].

As for CV, human-in-the-loop specifically explored how to exploit weak labeling to provide feedback at present. Besides, it also analyzed how to provide users with a unified intervention experience. It is involved in numerous tasks, such as person re-identification, face recognition, 3D point cloud object detection, and object detection. While considerable current pedestrian re-identification (Re-ID) methods can achieve superior results under the training of a large amount of labeled data, these models cannot produce an exceptional performance as in the experiment when deployed in a natu-

ral environment. Moreover, so much data is new in a natural environment because these data have not been in the training set. The trickier part is that new data will constantly accumulate over time, which can cause the model to fail to work. To tackle this problem, Liu *et al.* [55] proposed a human-in-cycle model based on reinforcement learning, which released the limitation of pre-labeling and upgraded the model through continuously collected data. The goal is to minimize human annotation work meanwhile maximizing the performance of Re-ID. In addition to directly using reinforcement learning for dynamic learning, researchers also pay attention to expanding and refining data on a new task. Facial expression recognition is an exciting task in CV, which is of great help to sentiment analysis and behavior analysis tasks. Traditional facial expression recognition can only deal with the seven simplest facial expressions (*i.e.*happiness, sadness, fear, anger, disgust, surprise, and contempt). In real life, it is additionally important to deal with more micro-expressions. More specifically, it is an interesting task that builds more refined micro expression processing datasets based on existing expression identification. Butler *et al.* [60] exploited a micro-expression recognition method based on the human-in-the-loop system. This method provides a flexible interface for manual proofreading of automatically processed tags, thereby ensuring the accuracy and usability of the extended dataset. In addition to directly constructing new datasets, it is also of great significance to explore existing datasets, especially for tasks that are difficult to label, such as target detection tasks, the labeling workload is enormous. To reduce the labor and time cost of annotation of the bounding box of video objects, Le *et al.* [63] applied an efficient and straightforward interactive self-annotation framework based on cyclic self-supervised learning. The entire framework consists of automatic model learning and interactive processes. The automatic learning process makes the model learn faster and speeds up the interaction process. In the interactive recursive annotation, the detector receives feedback from the human annotator to process the human loop annotation scene. Moreover, to save labeling time, they proposed a new level correction module, which strengthens the utility of neighbor frames by CNN by reducing the distance of annotated frames at each time step. Based on the framework of Le *et al.*, Adhikari *et al.* [68] modified the framework to be completed in one stage, and the most significant work of humans in it has become to correct errors instead of performing full annotations, which further improved the user experience.

Using the above two approaches is ineffective for more intricate image tasks, such as 3D point cloud labeling tasks. Because of the limited effect of employing only one stage for labeling, Meng *et al.* [66] designed a multi-stage human-in-the-loop labeling method based on predecessors. However, the previous work only started from the perspective of data annotation. It ignored integrating human experience and knowledge into the model to the greatest extent to incorporate human knowledge and intelligence effectively. Zhang *et al.* [67] considered specific talents and skills of humans in painting, and these skills cannot be fully quantified as rules and knowledge. If the model can learn painting skills, it would undoubtedly help human-in-the-loop application takes a significant step forward. They present a data-driven framework for generating comics from digital illustrations. To further create high-quality comics, these three components are humanely annotated by the artist.

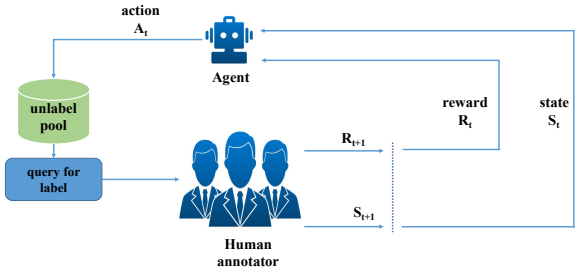


Fig. 5: The human-in-the-loop framework based on reinforcement learning.

2.3. Iterative Labeling

At present, there is still a high degree of coupling between deep learning tasks and data processing, and the performance of deep learning largely depends on the quality of the data. A large amount of high-quality labeled data is needed for a new task to obtain better performance. However, labeling large-scale data requires a lot of labor and takes a long time, while many iterations of tasks cannot afford such a cost and wait such a long time. Unlike weak annotate and automatically annotate, human-in-the-loop-based methods emphasize finding the essential samples that play a decisive factor in new sample data.

Unlike the data annotation mentioned above in 2.2, data iterative labeling pays more additional attention to user experience, not just directly allowing users to perform data annotation. From annotation to iterative labeling, the goal has been changed in the following two aspects: to focus on adding knowledge and experience to the learning system. The other is to focus on the interaction with users.

Yu *et al.* [33] utilized a partially automated labeling scheme for annotation, which free up human labor by using deep learning of human-in-the-loop. This constitutes the basic prototype of simple iterative annotation. Recently, with the proliferation of reinforcement learning, Liu *et al.* [55] developed a representative human-in-the-loop system based on reinforcement learning, which applied reinforcement learning to carry out iterative labeling. A typical framework of human-in-the-loop for reinforcement learning is shown in Fig. 5. This novel attempt extends the practical usability of human-in-the-loop to the field of reinforcement learning for

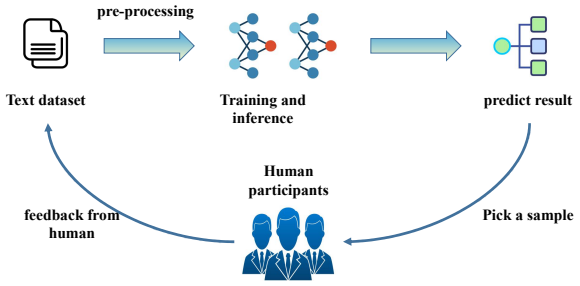


Fig. 6: The model training and inferencing workflow of human-in-the-loop in Natural Language Processing. The human participants provide various feedback in the stage of model training and inferencing according to specific tasks to boost the performance of NLP models.

the first time, which brings a valuable contribution to the human-in-the-loop community. In addition to implementing the simple manual intervention, they take person Re-Identification as a research task and explore how to minimize human annotation work while optimizing the performance of Re-ID. Fan *et al.* [57] set out to solve the data challenge in the scheme of network anomaly detection to allow users to intervene in data labeling rather than implement simple user labeling. They introduce a new intelligent labeling method, and the method combines active learning and visual interaction to detect network abnormalities through the iterative labeling process of users. The difference is that they began to pay attention to the connection between the algorithm and the visual interface, and the algorithm and the optical interface are tightly integrated.

3. MODEL TRAINING AND INFERENCE

In many fields of Artificial Intelligence, such as Natural Language Process (NLP) and Computer Vision (CV), there are a variety of approaches that leverage human intelligence to train and infer experimental results. For both NLP and CV, related research spans deep learning [71] techniques and human-machine hybrid methods. These heuristic methods have taken the diverse quality of human creativity into account to achieve high-quality results.

3.1. Natural Language Process

Fig. 6 briefly illustrates the cooperation between the individuals and the model training and inferencing process in the Natural Language Processing Loop. The continuous executive loop develops a more reliable human-AI partnership to a certain extent, contributing to higher accuracy and stronger robustness of the NLP system.

3.1.1. Text Classification

Text Classification (TC) is a fundamental NLP task that categorizes a sentence/text into its corresponding category. Karmakharm *et al.*[72] propose a rumor classification system. The core idea of this system is to obtain additional manual feedback from the journalists to retrain a more accurate machine learning model. This framework first exploits a Rumour Classification System to classify collected social media posts and sends these information back to the journalists. As most state-of-the-art text classification approaches are dominated by the deep neural network [73, 74], which is generally considered as "black boxes" by end-users, another motivation for the researcher to construct the human-in-the-loop framework for TC is to overcome the opaqueness of those models, making them more explainable. To achieve this purpose, Arous *et al.* [75] put forward a hybrid human-AI framework that gives a moral idea to reinforce human reliability in merging human rationales into a deep learning algorithm. Their work presents MARTA, a Bayesian framework that jointly learns and updates the model parameters and human reliability via an iterative way, enabling the learning processes of parameters and human reliability to benefit from each other until the label and rationales reach agreements.

3.1.2. Syntactic and Semantic Parsing

Besides text classification, human-in-the-loop approaches for syntactic and semantic parsing are also promising. Syntactic parsing is a process to obtain the valid syntactic structure of input sentences. The goal of semantic parsing is to map natural language to formal domain-specific semantic representations. A human-in-the-loop parsing method [43] is proposed to improve the parsing accuracy of CCG parsing by employing non-expert to answer simple what-questions generated from the parser's output. These answers are treated as soft constraints when re-training the model. This work is the first attempt at introducing human-in-the-loop for syntactic parsing. However, most parsing technologies still face several challenges: (1) the purpose or expression of users can be ambiguous or vague under some circumstances, posing obstacles for them to get the ground truth in one shot, (2) in the real-world scenario, the performance of state-of-the-art parsers are generally not high enough, and (3) since the mainstream neural network-based models are known as "black-box" that indicates the lack of explainability, it is difficult for end-users to verify the parsing results independently. Currently, Yao *et al.* [76] propose allowing the semantic parsers system to ask end-users clarification questions and produce an If-Then program simultaneously. Although recent works successfully verified the effectiveness of interactive semantic parsing in practice, they are generally restricted to a specific type of formal language. Furthermore, Yao *et al.* [77] develop a model-based interactive semantic parsing (MISP) as the general principle for interactive semantic parsing.

Table 2: A brief overview of representative works in human-in-the-loop NLP. Each row represents one work. Works are sorted by task types (TC: Text Classification. SSP: Syntactic and Semantic Parsing. TS: Text Summarization. QA: Question Answering. SA: Sentiment Analysis). Each column corresponds to a dimension from the two subsections (task, motivation).

Work	Task					Motivation			Quantitative results
	TC	SSP	TS	QA	SA	Performance	Interpretability	Usability	
Arous et al. (2021) [75]	✓					✓	✓		$0.840- > 0.960(Acc)$
Karmakharm et al. (2019) [72]	✓					✓		✓	$--$
Yao et al. (2019) [76]		✓				✓	✓	✓	$0.640- > 0.968(Acc)$
Yao ZiYu et al. (2019) [77]		✓				✓	✓		$0.615- > 0.729(Acc)$
Ziegler et al. (2019) [78]			✓			✓			$--$
Stiennon et al. (2020) [79]			✓			✓			$--$
Hancock et al. (2019) [80]				✓		✓			$0.447- > 0.463(Acc)$
Wallace et al. (2019) [56]				✓		✓		✓	$0.4- > 0.6(Acc)$
Liu et al. (2021) [81]					✓		✓		$0.71- > 0.82(Precision)$

3.1.3. Text Summarization

Besides applying a human-in-the-loop framework to topic modeling, researchers also use it to generate new texts. Text Summarization (TS) generates a shorter version of a given sentence/text while preserving its meaning [82]. In recent years, there have been some significant breakthroughs in this field. For instance, Ziegler *et al.* [78] fine-tune pre-trained language models with reinforcement learning by exploiting a reward model trained from human preferences. Then the model is used to generate summaries over Reddit TL, DR, and CNN/DM datasets. However, one limitation of their framework is that there are low agreement rates between labelers and researchers. Stiennon *et al.* [79] propose to gather a dataset composed of human preferences between pairs of summaries as the first step. Then the prediction of the human-preferred summary is generated by a reward model (RM) trained via supervised learning. Lastly, the score produced by the RM is maximized as much as possible by a policy trained via reinforcement learning (RL). Their method ensures a relatively higher labeler-researcher agreement through the above steps and successfully separates the policy and value networks.

3.1.4. Question Answering

Recently, various human-in-the-loop related frameworks have been designed to apply dialogue and Question Answering (QA). The purpose of this task is to allow chatbots/agents to have a conversation with users. These human-in-the-loop dialogue intelligent systems can be bifurcated into two main categories: online feedback loop and offline feedback loop [83]. For the online feedback loop, human feedback is utilized to update the model continuously. Compared with traditional approaches that mismatch the training set and online use case for dialogue systems, researchers have demonstrated that the application of online reinforcement learning can improve the model with human feedback. For instance, a lifetime learning framework is proposed by Hancock *et al.* [80]. The self-feeding mechanism in this framework enables the chatbot to generate new examples when the conversation with users goes well, and these new examples are exploited to re-train itself continuously. For the offline feedback loop, a large set of human feedback needs to be collected as a training set, then this training set is used to update the model. For instance, Wallace *et al.* [56] employ "trivia enthusiasts" to creatively generate adversarial examples that can confuse their QA system. These examples are finally implemented for negative training. Since some of the end-user feedback can be misleading, offline methods is more appropriate for improving the robustness of the model.

3.1.5. Sentiment Analysis

Sentiment Analysis (SA) is one of the attractive research branches of Opinion Mining (OM). The research scope of SA is the computational study of individuals' opinions and attitudes toward entities mentioned in a text. The entities generally refer to individuals or events. Recently numerous neural network-based approaches have been widely utilized and demonstrated their effectiveness in solving sentiment analysis tasks [73, 84, 85]. Most deep learning-based methods for SA use accuracy and F1-score as evaluation metrics. Since these metrics can only evaluate the predictive performance, they lack the mechanisms to explain when and why the sentiment models give false predictions in run-time [86]. Liu *et al.* [81] introduce an explainable human-in-the-loop SA framework for sentiment analysis task. The execution of their framework is segmented into three steps: First, the human-in-the-loop SA model analyzes local feature contributions. This goal is achieved by executing a data perturbation process. Next, local features are aggregated to calculate the explainable global-level features and humans participate in this loop to assess the relevance of the top-ranked global features to the ground truth and report the errors they find in this process. Finally, the system calculates an erroneous score based on global-level and local-level sentimental features for each instance. Scores higher than a specific threshold are indicated as wrong predictions.

3.1.6. Summarization for Human-in-the-Loop in NLP

A brief overview of representative works in human-in-the-loop NLP is shown in Table 2. For most of the surveyed papers above, their original purpose is to apply human-in-the-loop techniques to various NLP tasks for better performance. The effectiveness of the approaches proposed by these surveyed papers is evaluated via multiple metrics. The experimental results in the documents we investigate show that a relatively small set of human feedback can dramatically boost model performance. For instance, the human-in-the-loop technique improves the classification accuracy for text classification [72]. Similar situations occur in dialogue and question answering where the QA systems have higher ranking metric hits [80]. Besides, human-in-the-loop techniques have also enhanced the model's robustness and generalization [79]. In addition to improving model performance, some studies have demonstrated that human-in-the-loop methods enable models to be more interpretable and usable in solving NLP problems. For instance, Arous *et al.* [75] incorporate human rationales into an attention-based Bayesian framework reasonably while weighing worker reliability, thus providing a more human-understandable interpretation of classification results and enhancing the model performance at the same time. Liu *et al.* [81] chose uni-grams as the explainable feature for LIME [87]; thus the proposed system allows the end-users to understand better the overall contribution of each word to the final sentiment classification made by the model. Wallace *et al.* [56] invite "trivia enthusiasts" to creatively generate specific adversarial questions that confuse the intelligent question answering system. These questions can be treated as probes further to explore the inherent characteristics of the underlying model behaviors.

3.2. Computer Vision

In recent years, neural network-based Deep Learning methods (DL) have emerged as the state-of-the-art technique for performing many computer vision tasks [88, 89]. To further improve the performance of these methods, feedback from humans has been integrated into the deep learning architecture to make the whole system more intelligent in solving complex cases that can not be handled politely by the model. A typical human-in-the-loop framework for Computer Vision is outlined in Fig. 7.

3.2.1. Object Detection

Object detection, as one of the most fundamental and challenging problems in computer vision [90], has received significant attention in recent years [91]. The goal of object detection is to detect instances of visual objects of a specific class (such as individuals, vehicles, or other creatures) in digital images. Yao *et al.* [92] point out that iterations between queries may be expensive and time-consuming, making it unrealistic for executing interaction with end-users. They present an interactive object detection architecture to employ individuals to correct a few annotations proposed by a detector for the un-annotated image with the maximum predicted annotation cost. However, it is still difficult to detect some occluded objects, tiny objects, and blurred objects by these approaches. Madono *et al.* [93] put forward an efficient human-in-the-loop object detection framework composed of bi-directional deep SORT [94] and annotation-free segment identification (AFSID). Humans' role in this architecture is to verify the object candidates that bi-directional deep SORT can not detect automatically. Then train the model over the supplementary objects annotated by individuals.

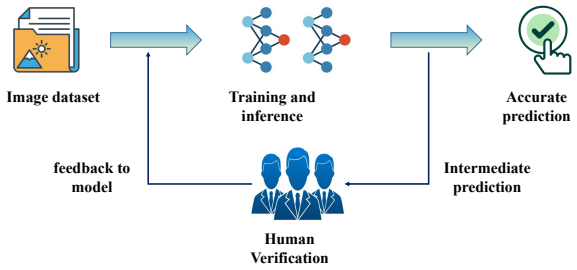


Fig. 7: Overview of general human-in-the-loop frameworks for model training and inferencing in Computer Vision.

3.2.2. Image Restoration

Image restoration (IR) aims to recover the preliminary version of damaged images [95]. The image inpainting frameworks proposed by previous studies can be bifurcated as exemplar-based [96] approaches and deep learning-based [97] methods. Although profound learning-based works are the mainstream and show decent results, neural network-based approaches constantly suffer from over-fitting when only a relatively small training set is available on a large dataset. Besides, in real-world application, the restored images are often filled with unknown artifacts like uneven texture or monotone color due to the missing crucial semantic information in severely corrupted areas. Weber *et al.* [98] propose an interactive machine learning system for image restoration based on Deep Image Prior (DIP) [99]. Their proposed human-in-the-loop framework embeds human knowledge into the training process by the following steps. Initially, the images from the dataset are sent to the automated DIP for preliminary restoration. Secondly, the operators actively refine the images via a pre-designed user interface. Thirdly, the refined images are sent back to the input of DIP again for further polishing. Finally, the whole loop continues until the restoration reaches the user's expectation. In the field of Electron Microscopy, one drawback of automation is that it generally ignores the expertise of the microscopy user that comes with

manual analysis. To alleviate such a challenging problem, Roels *et al.* [100] propose a hybrid human-in-the-loop system that incorporates expert microscopy knowledge with the power of large-scale parallel computing to enhance the Electron Microscopy image quality by exploiting image restoration algorithms.

3.2.3. Image Segmentation

Image segmentation (i.e. semantic segmentation) is a crucial step in most image studies. Image segmentation (IS) aims at assigning a class label to each pixel in the image [101]. This field has recently become explosive popularity because it plays a crucial role in a wide range of computer vision applications [102]. However, few works explore how to effectively expose failures of top-performing semantic segmentation models and rectify the models by utilizing such counter-examples reasonably. Wang *et al.* [103] present a two-step hybrid system with human efforts for troubleshooting pixel-level image labeling models. The hybrid system first automatically picks up un-labeled images from a large pool. These selected unlabelled images are used to compose an unlabeled set, which is the most informative in exposing the weaknesses of the target model. To reduce the number of false positives, individuals filter the unlabeled set to obtain a smaller straight set. In the second step, they fine-tune and re-train the target model to study from the counter-examples contained in the refining set without ignoring previously seen examples. Data annotation is always complicated and expensive in the medical image process domain [104]. Ravanbakhsh *et al.* [105] introduce a training protocol based on combining the conditional Generative Adversarial Network (cGAN) and human workers interactively. For complex cases, human experts are responsible for annotating them. These newly annotated images are used to continue the training and inference procedure.

3.2.4. Image Enhancement

As one of the challenging issues in computer vision, the purpose of Image Enhancement (IE) is to process an image and generate a new and improved one that is more suitable for a specific application [110]. The research field of image enhancement has attracted ample attention from researchers in recent years, especially after the emergence of deep neural network algorithms [111]. However, most current frameworks have ignored the user preferences and experiences, enhancing the image only via a black-box style, which can leave end-users with sub-optimal results that are not suitable for their specific taste. Murata *et al.* [106] take user preference into account. The user first provides an example to his system; the image enhancement functions in the framework are applied to the example image via randomly selected parameters. Several objective photos are produced, and the end-user needs to score each of the images. Then the RankNet [112] is exploited to learn the user's preference from these scores. During the learning process, based on the scores given by users, parameters are optimized to make the generated enhanced images suitable for the taste of users. Fischer *et al.* [107] devise Neural Image Correction and Enhancement Routine (NICER). A component called Image Manipulator in the NICER first exploits a series of learned image operations (e.g. contrast, brightness) with variable magnitude onto the original image provided by users. Another module named Quality Assessor is followed to evaluate the final enhancement quality by generating related scores. This system iteratively optimizes the parameters of the image enhancement functions to maximize the scores given by the Quality Assessor. Users can modify the parameters of the Image Manipulator before, during, and after the optimization process, guiding the optimization procedure towards more satisfying local optima.

3.2.5. Video Object Segmentation

The goal of Video Object Segmentation (VOS) is to segment a particular object instance in the entire video sequence of the object mask on a manual or automatic first frame [113]. This research area has become popular in the computer vision community [114]. Since videos have intrinsic characteristics such as motion blur, bad composition, occlusion, etc., it is harder for fully automatic approaches to segment more complex sequences accurately. Employing user to input for each frame is unrealistic due to its expensive costs and time consumption. Thus, the human-in-the-loop framework is adopted for solving such problems. Benard *et al.* [108] introduce a novel interactive video object segmentation method based on [115]. The core idea of their human-in-the-loop framework is to utilize the current segmentation mask as an additional input. A practical framework for interactive segmentation scenario is designed by Oh *et al.* [109], named Interaction-and-Propagation Networks (IPN). The IPN is composed of two modules and the critical architecture of these two modules is deep convolutional neural networks. The primary operations of these two modules are interaction and propagation, respectively. Individuals are allowed to interact with the proposed model several times; meanwhile, the feedback is provided in scribbles on multiple frames during this interactive procedure.

3.2.6. Summarization for Human-in-the-Loop in CV

A brief overview of representative studies in human-in-the-loop CV is displayed in Table 3. It can be observed from Table 3 that the motivation of all the surveyed human-in-the-loop works for computer vision is to boost the model performance. From the experiment results of all these surveyed papers, although the evaluation criteria are different, the system that incorporates the human-in-the-loop method performs

Table 3: A brief overview of representative works in human-in-the-loop CV. Each row represents one work. Works are sorted by their task types (OD: Object Detection. IR: Image Restoration. IS: Image Segmentation. IE: Image Enhancement. VOS: Video Object Segmentation). Each column corresponds to a dimension from the two subsections (task, motivation).

Work	Task					Motivation			Quantitative results
	OD	IR	IS	IE	VOS	Performance	Interpretability	Usability	
Yao et al. (2012) [92]	✓					✓			−−
Madono et al. (2020) [93]	✓					✓			−−
Roels et al. (2019) [100]		✓				✓	✓		−−
Weber et al. (2020) [98]		✓				✓			$0.2816- > 0.2227(DSSIM)$
Wang et al. (2020) [103]			✓			✓			$0.1365- > 0.4233(mIoU)$
Ravanbakhsh et al. (2020) [105]			✓			✓			$0.645- > 0.846(Acc)$
Murata et al. (2019) [106]				✓		✓			−−
Fischer et al. (2020) [107]				✓		✓			−−
Benard et al. (2017) [108]					✓	✓		✓	$0.504- > 0.822(IoU)$
Oh et al. (2019) [109]					✓	✓		✓	$0.555- > 0.691(AUC)$

better than without combining it. Taking Madono *et al.* [93] as an example, they conduct experiments for pedestrian detection and the results have proven at least two advantages for this task: For one thing, the proposed approach boosts the recall rate by 11% at most over deep SORT. For the other, the amount of unlabeled samples that need manual annotation is decreased by 67% at most compared with bi-directional deep SORT without AFSID, which dramatically improves the overall model performance. Associating the contents in Table 2, this phenomenon in CV is similar to NLP, which demonstrates that the core motivation of almost all human-in-the-loop studies in both CV and NLP serves the purpose of boosting model performance. We also notice that in Table 3, only one work [100] tries to bring interpretability for the model. Roels *et al.* [100] have validated the potential enhancements that DenoisEM can provide in 3D EM image interpretation by denoising SBF-SEM image data of an Arabidopsis thaliana root tip. Besides, human-in-the-loop conception can also improve the usability of CV models. For instance, Madono *et al.* [93] have proven that their framework is advantageous/practical in scenarios where obtaining annotations is a costly affair. Oh *et al.* [109] validate the usefulness and robustness of their Interaction-and-Propagation Networks with real interactive cutout use-cases. Hudec *et al.* [116] propose a new classification of aggregate functions in terms of mixed behavior by the variability of ordinal sums of associative and disjunctive functions, which play an important role for model fusing the knowledge of domain experts.

4. SYSTEM CONSTRUCTION AND APPLICATION

Previously, we summarized and reviewed the human-in-the-loop workaround of 'data processing' and 'model training and inference'. Moreover, we noticed that some researchers focus on how to build human-centric applications in a broad spectrum of application scenarios. We selected representative studies from four application scenarios of Security Systems, Code Production Tools, Simulation Systems, and Search Engines to summarize, as shown in Table 4.

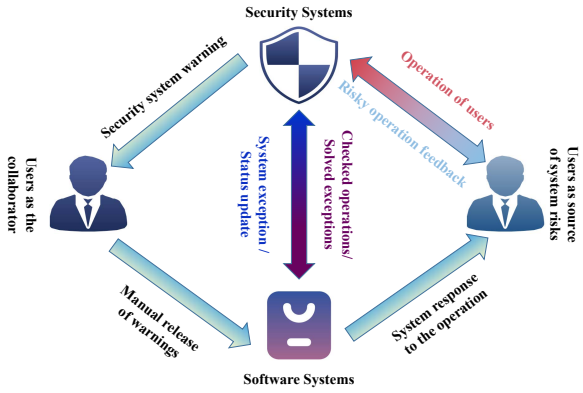


Fig. 8: The workflow of the human-in-the-loop security system.

4.1. Security System

The security system is an application system with solid coupling with people (Fig. 8). With the development of deep learning algorithm, researchers begin to explore more effective human-machine cooperation modes to replace the manual intervention process gradually. Humans usually fail to execute certain events due to some objective reasons (*i.e.*inattentiveness, increasing age, nonproficiency). At present, many safety systems are highly coupled to people, which would lead to the continuous amplification of errors. To address this challenge, humans should maximize the chances of operators successfully performing their safety-critical functions. To this end, Cranor [118] proposed a human-in-the-loop-based reasoning framework, which provided a systematic approach to identifying potential causes of human failures, thereby helping system designers use the framework to identify problem areas and proactively address defects before building systems. Similarly, in order to stop operator error in time, Singh et al. [128] started by address-

Table 4: An overview of representative applications for software-based human-in-the-loop systems. (SECS: Security Systems. CP: Code Production. SIMS: Simulation System. SE: Search Engine.)

Systems	Application					Year	Role of human				Description	Quantitative results
	SECS	CP	SIMS	SE	Others		Supervisor	Supervisee	Collaborator	User		
Brostoff & Sasse [117]	✓					2001		✓		✓	Security system for traditional tasks	––
Cranor [118]	✓					2008		✓		✓	Security system for traditional tasks	––
MacHiry *et al.* [119]		✓				2013			✓		Software testing	47%
Kovashka *et al.* [120]				✓		2015				✓	Image-based searching engine	––
Louis Rosenberg [121]					✓	2016			✓		Crowdsourcing	––
Yan *et al.* [122]		✓				2017	✓		✓		Software testing	53.45%
Wogalter [123]	✓					2018			✓	✓	Security system for traditional tasks	––
MA [124]		✓				2018	✓				AI model optimizing in training	60%(*reduction*)
Salam *et al.* [125]		✓				2019	✓				AI model optimizing in testing	––
Plummer *et al.* [126]				✓		2019				✓	Image-based searching engine	82.28%(*Acc*)
Fredrik Wrede [127]					✓	2019	✓		✓		Stochastic gene regulatory	––
Singh and Mahmoud [128]	✓					2020		✓	✓	✓	Security system for traditional tasks	75.43%(*Acc*)
Demartini*et al.* [129]	✓					2020	✓		✓		Security system for modern tasks	––
ODEKERKEN & BEX [130]	✓					2020			✓		Security system for modern tasks	––
Bohme *et al.* [131]		✓				2020	✓				Program repairing	75%(*Acc*)
Renner [132]		✓				2020	✓		✓		AI model optimizing in the whole steps	––
Davidson *et al.* [133]			✓			2021			✓	✓	Simulation system for decision making	––
Demirel [134]			✓			2020				✓	Simulation system for process forecast	––
Metzner *et al.* [135]			✓			2020				✓	Simulation system for procedure control	––
Polisetty & Avinesh [136]				✓		2020				✓	Searching engine	––
Zhu *et al.* [137]					✓	2020	✓		✓		Renal Pathology	57%(*reduction*)
Li *et al.* [138]					✓	2020			✓		Model checking	––

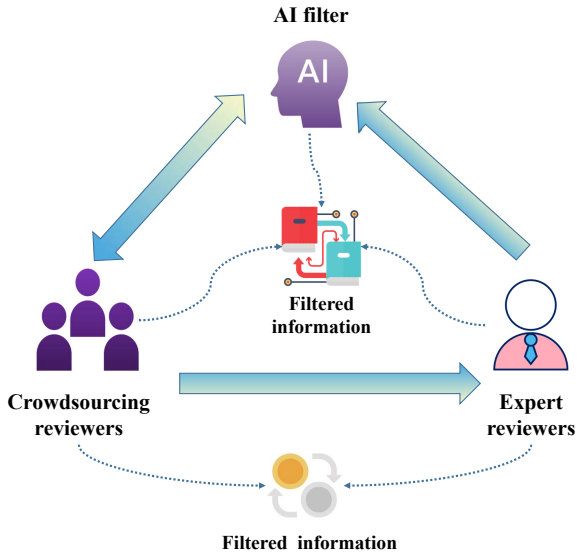


Fig. 9: The workflow of the human-in-the-loop security system in modern tasks.

ing the operational safety of nuclear power plants (NPP) and the commercial aviation industry. They first used the interaction between operator and control panel (HMI) state, where a visual feedback loop is used to predict the near-term expected state. The HMI state sequence prediction is transformed into a language translation problem and finely modeled in a human-in-the-loop fashion. The proposed method is validated by a dataset of natural factory scenes, demonstrating the effectiveness of the method.

The workflow of the human-in-the-loop security system can be summarized as Fig. 9 in which the artificial intelligence algorithm and humans (experts or crowd-sourcing workers) collaborate all together. Demartini *et al.* [129] devised a solution to combat online misinformation by combining machine learning algorithms, crowd-sourcing workers and experts, and their introduction to combining automatic and manual fact-checking methods to the challenges and opportunities in combating the spread of online misinformation. They pointed out that by addressing the main question of "who should do what", the cost and performance of the three roles can be balanced, along with the improvement of credibility and safety of the human-in-the-loop system. For the same purpose, ODEKERKEN *et al.* [130] introduced a proxy framework for transparent human-in-the-loop classification that combines dynamic arguments with legal case-based reasoning to create a system that can elaborate its decisions and adapt to new situations. In this system, human analysts can add new factors to update system results and make recommendations to the classification algorithm as supervisors.

Across that span of time, with the further development of Internet Technology, security systems become an urgent need. However, the use of human-in-the-loop is not limited to virus detection and fraudulent information filtering and has also shown remarkable performance in issues such as privacy protection, authentication attack prevention, and spam filtering.

4.2. Code Production Tools

The programming of programs and the training of models are essential subtasks of artificial intelligence, and this two tasks are highly coupled with humans. However, with the establishment of the open-source community, various codes and model resources have been disclosed, making simple program writing and model training a repetitive work and thus leading to an automated basis for the model. The fundamental difficulty is making integration tools learn to splice existing components, which requires integration tools to learn hu-

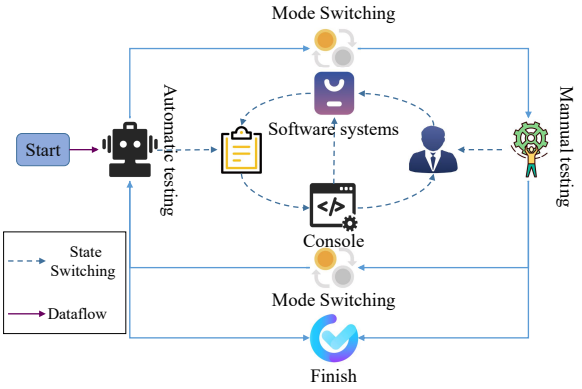


Fig. 10: With the help of human-in-the-loop system, human can collaborate a project with computer.

man integration experience. Under the guidance of this idea, there have been some tools for automation. With the help of these practical tools, developers can collaborate on existing projects, rather than writing projects from scratch (Figure 10).

Software testing was first addressed. MacHiry *et al.* [119] designed a test generation system called Dynodroid for fuzzing unmodified Android applications. Dynodroid decomposed an application into a set of event-driven programs. Dynodroid employed the Android framework to generate a series of events and automatically executed these generated events to interact with the surrounding environment. It collected interactive feedback to program functions. The most worth mentioning is that Dynodroid would use manual intervention mode when necessary, and also tried to exploit input feedback to adjust the frame to generate new output dynamically. Through the above approach, Dynodroid implemented fuzz testing of unmodified Android applications in a human-machine hybrid manner. Yan *et al.* [122] put forward a human& tool-centric vulnerability analysis system that leveraged humans (with varying levels of expertise) to perform well-defined subtasks. The vulnerability analysis system had a larger scale of available programs and higher labor utilization than previous work. Besides, in the practical application situation, the coders pay attention to the test of the code and want to see the application that can make the program correction. Bohme *et al.* [131] proposed the first human-in-the-loop semi-automatic program repair framework called LEARN2FIX. LEARN2FIX repaired bugs by negotiating with users to observe errors, trained an automatic error oracle through the bug samples marked in the negotiation and finally realized the automatic repair of the program. LEARN2FIX learned a sufficiently accurate automatic oracle with relatively low labeling effort (e.g. 20 queries). Productization tools of human-in-the-loop systems in software engineering bring great convenience to programmers. In the future, the application of human-in-the-loop in this field will expand from debugging and software testing to most coding programs. In particular, with the introduction of pre-trained

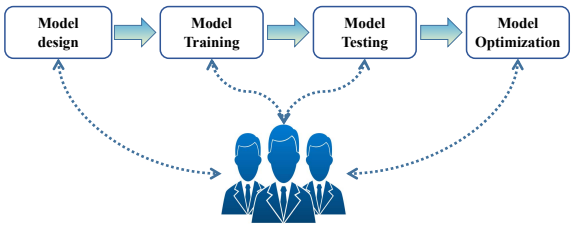


Fig. 11: human-in-the-loop system in machine learning algorithm optimizing.

models such as BERT [11] and GPT [12], human-machine collaborative programming is becoming a new focus.

With the development of machine learning, the training of deep models has become another prevalent task. With the continuous exploration of model training, more researchers focus on constructing semi-automatic or even fully automatic training tools. The researchers make efforts to incorporate human knowledge during various stages of model training (Fig. 11). To solve the problem that the underlying structure of the model is unknown in the model design process, Salam *et al.* [125] applied a semi-automated human-in-the-loop attribute design framework to assist human analysts in converting original attributes into classifications of valid derived properties of the question. The framework first provided a human analyst with k buckets containing promising original attribute selections based on a random walk-based heuristic. It then iteratively allowed human analysts to deal with attributes involving attributes based on a scalable and efficient greedy heuristic algorithm. The top-l derived attribute finally got a designed model frame, and the designer completed the detailed design interactively. To address the model training process, the ultimate goal of an ML system becomes the problem of reducing the time it takes to get a deployable model from scratch. MA *et al.* [124] exploited a human-machine collaboration system called Helix which optimized execution across iterations by appropriately reusing or recomputing intermediate results. It is worth pointing out that Helix contained a workflow management module as well as a visualization module to interact with people, thus enhancing the usability of the tool. To solve the problem that transparency and operability are ignored in the process of model optimization, Renner *et al.* [132] focused on both transparency (how the system is explained to humans) and operability (how users provide feedback or coach the system) in interactive machine learning. It combined novel topic visualization techniques with a human-centered interactive topic modeling system. Furthermore, it revealed how users understand and interact with machine learning models and guided the further development of human-in-the-loop systems.

4.3. Simulation System

The simulation system generally utilizes virtual systems

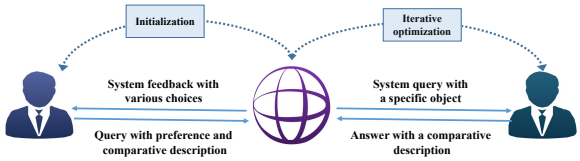


Fig. 12: The human-in-the-loop usage for image search system.

to simulate an object or workflow, and is widely applied in the decision-making of system construction, process forecast, and safety control. Due to the feature of these applications, interaction with a human is indispensable. A good many researchers have been working towards the goal of improving the usability of the human-in-the-loop in predicting and optimizing the integration of soldier systems. Davidson *et al.* [133] presented a semi-structured interview method with key informants, and in the review method, eight critical requirements for improving human-in-the-loop simulation systems were identified. Addressing essential requirements can improve the ability of current human-in-the-loop simulation tools to accommodate the military's need for human consideration early in the design process. In addition, the system can also be used for other planning problems such as cargo, transportation problems, and traffic management problems. Other researchers are concerned with the simulation of processes, the most representative of integrating human-machine dialogue into a system. Demirel *et al.* [134] proposed a novel human-computer interaction method to evaluate the safety and performance of human-product interactions. This method modeled humans into a computationally designed environment through digital human modeling technology. With this system, the form and function of the characters in the workflow could be pre-designed, and ergonomic evaluation indicators could be used to implement the work cycle. In terms of program control, Metzner *et al.* [135] proposed a simulated human-robot collaboration method that combined virtual reality, motion tracking, and standard simulation software for industrial robots. It used a virtual reality system to simulate the workplace of a robot and its collaborators to design a motion tracking system. The system captured human movements by fusing two above-mentioned subjects. Among other things, this approach would evaluate the performance of the human-robot collaboration system safety controls and the fulfillment of the defined requirements.

4.4. Search Engine

The automation and intelligence of search engines are also important research areas. We discover that current search engine is still positioned as an human assistant, but if the search engine can further model the human work, it can be more practical. At present, the researches on search engines are mainly divided into recommendation system and image search. Polisetty *et al.* [136] introduced a new joint recommender system that is updated by continuously learning from user feedback. Recommender systems better prepare information from the three main research directions of information aggregation, enrichment, and recommendation by placing humans in the loop. The system features review summarization and rating prediction, with a web-based interface for improving methods through human-computer interaction. With the development of computer vision, searching through images has become an important task and the image search task has been well solved due to the fast development of deep learning. However, the visual-text semantic matching task remains a significant challenge that affects interaction performance and search results. Since humans have an inherent advantage in handling image search tasks, trying to introduce users into the image search loop (Figure 12) has excellent potential. Kovashka *et al.* [120] devised a new feedback model for image search that allowed users to interact, communicated preferences through visual comparisons, and ultimately determined user searches. First, the user initiates a query; then the system initiates feedback on the user's query, and the user provides feedback; then the system collects user feedback and responds until the system no longer receives user feedback; finally, the system learns from the user feedback and updates the system. Compared to traditional passive and active methods, the system can provide more accurate search results with less user interaction through an iterative learning process. Building on Kovashka *et al.*, Plummer *et al.* [126] put forward an attribute-based interactive image search method, which can iteratively refine image search results using human-in-the-loop feedback. Unlike before, the system trains a deep reinforcement model to learn which images are informative, rather than relying on manual measurements commonly exploited in previous work. Besides, this work extends conditional similarity networks to incorporate global similarity into training visual embeddings, resulting in more natural transitions when users explore learned similarity embeddings. The system is self-updating to provide more accurate image search results. With the help of human-in-the-loop, we believe that search engines can continuously adapt to different changes through interactive learning to meet searchers' personalized needs better.

In addition to the above scenarios, human-in-the-loop systems are also used in other fields [127, 137], such as bioinformatics, supervisory healthcare, and crowd-sourcing work. For example, Fredrik Wrede *et al.* [127] utilized human-in-the-loop semi-supervised learning for stochastic gene regulation. Rosenberg *et al.* [121] applied an artificial swarm intelligence scheduling system for crowd-sourcing tasks, Li *et al.* [138] employed a model checking method for human-in-the-loop system. With the further development of the systems based on human-computer interaction and the expansion of application scenarios, human-computer interaction

will be applied on more occasions.

5. DISCUSSION AND FUTURE DIRECTIONS

In this section, we first discuss some existing challenges and key issues of human-in-the-loop for machine learning. Next, we show future directions on CV, NLP, and applications based on human-in-the-loop.

5.1. The Challenges and Discussion

How to add human experience and knowledge to computer vision tasks? By reviewing the previous works, we find that most human-in-the-loop researches only focus on natural language processing. Analyzing the reasons, it is not easy to directly allow people to interact with images effectively(except for direct labeling), and add human experience and knowledge to the model throughout the cycle. With the development of multi-modal technology, utilizing multi-modality for image representation can be an effective way [139]. It is particularly worth pointing out that Holzinger *et al.* [140] adopted Graph Neural Networks as a method of choice, enabling information fusion for multi-modal capability, which is a milestone. Besides, applying inverse reinforcement learning also seems to be a feasible and practical solution [141].

How does the model learn human knowledge and experience from a higher dimension? The goal of human-in-the-loop is to connect humans to the model loop in a specific way, so that the machine can learn human knowledge and experience during the loop. Most current methods achieve this goal through human data annotation which is only the most basic realization process. As the saying goes, it is better to teach a man fishing than to give him fish. Researchers are supposed to take how to help agents acquire this knowledge effectively into account [142]. Language is an experience accumulated in the human learning process. Researchers currently focus on employing human intervention in dialogue to enable machines to learn human knowledge and intelligence from dialogue procedures iteratively [143]. In addition, numerous reasoning tasks contain higher-dimensional knowledge. By integrating humans into the reasoning loop, the machines can also learn more about human experience [118]. Image quality evaluation and design tasks are a higher level of human activity. Although human aesthetics and design inspiration can constitute the theory; however, more inspiration and aesthetics still come from human experience [144]. If we find a productive way to allow the model to learn more expert experience, the model's improvement can be dramatic.

How to select key samples? The critical technology for the human-in-the-loop is obtaining essential samples and labeling them with human intervention. At present, researchers mainly exploit confidence-based methods to obtain critical samples. This method plays an irreplaceable role in classification tasks [145, 146]. However, for other tasks (e.g. semantic segmentation, regression, and target detection tasks), confidence is the evaluation that is not so noticeable and thus cannot reflect the improvement of the system. Active learning aims to train an accurate prediction model with the least cost by marking the examples that provide the most information. There are multiple mature and worthy reference methods in selecting criteria, and perhaps researchers can obtain inspiration from these methods [147].

How to construct an evaluation benchmark? To develop the entire community, providing an effective test benchmark is important. At present, there is no uniform standard for human-in-the-loop research benchmarks. To better explore this research topic, it is essential to study how to develop evaluation methods and benchmarks for human-in-the-loop systems. Moreover, the formation of a unified benchmark is also conducive to the further refinement of research [30]. The current human-in-the-loop-based research is a more influential direction for exploring ways that are more conducive to human-in-the-loop. In addition to creating the standards for these interaction methods, restricting and theorization are also particularly important.

How to implement a general multitasking framework by human-in-the-loop? The real-world task is complex and in its current form, so it is not easy to completely solve it with one characterization [148]. With the emergence of a unified large-scale pre-training model [149, 39], we have seen the hope of achieving a universal model through human-in-the-loop fine-tuning. In particular, the current machine learning models are not as intelligent as humans, so it may be the next direction to consider using a suitable way to introduce human knowledge into large models.

5.2. Future Directions

To facilitate more researchers developing more advanced Human-in-the-loop systems, we summarize the following concrete future directions for Human-in-the-loop NLP, Human-in-the-loop CV, and Human-in-the-loop Real-world Applications.

5.2.1. Future Directions for Human-in-the-loop NLP Systems

- For systems like a chatbot, automatic summarization tool, or commercial machine translation, when interacting with them, individuals can only give a reward signal to the one output that is sent to them, which leads to the sparsity in feedback concerning the size of the output space [150].
- In syntactic parsing tasks, exploring more intelligent questions about other types of parsing uncertainties based on human-in-the-loop and scaling the method to large unlabelled corpora or other languages are important [43].

- In user-centered design and evaluation of a human-in-the-loop topic modeling task, the key is considering trust or confidence [151].
- In terms of AI safety, among existing human-in-the-loop techniques, some of them also allow malicious individuals to efficiently train models that serve their purpose, which may cause damage to all aspects of society. For example, they could exploit human feedback to fine-tune a language model to be more persuasive and manipulate humans' beliefs, instill radical ideas, commit fraud, etc. [79].

5.2.2. *Future Directions for Human-in-the-loop CV Systems*

- In the image restoration task, it is necessary to pay attention to predictive parameter optimization based on supervised regression models and scientifically analyzing correlations between the parameters of different algorithms based on the human-in-the-loop methods [100].
- In image enhancement tasks, using active learning to help users obtain a better estimation of cluster membership with the fewest image enhancements of images is important [152].

5.2.3. *Future Directions both NLP and CV Systems*

- Human supervision may be preferable due to various levels of expertise and with the increase of work overload, erroneous decisions are potential to occur [153].
- Collect and share more human feedback datasets for different tasks of NLP and CV [30].
- We should consider user credibility to affect the influence of their annotations by analyzing the quality of provided feedback[72].
- More rigorous in-depth user studies need to be designed and conducted to evaluate the effectiveness and robustness of human-in-the-loop frameworks in addition to model performance [151].
- For generative tasks, an explicit function can be defined by user feedback to evaluate and collect the generated signals [150].
- It is vital to find an efficient way to dynamically pick up the most representative and valuable feedback to collect [154].
- It is crucial to have a more friendly manner of showing what the model has learned from feedback and what kind of feedback. Specifically, it is possible to explore the process of changing the model in an attempt to visualize the results of the manual feedback [155].

5.2.4. *Future Directions in Real-world Applications*

- It is fundamental to choose an appropriate artificial intervention time, especially the tasks with solid demand for reliability and safety [156].
- For a system with human-computer interaction, users' expectations of experience usually take precedence over performance [157, 158].
- Modeling the sensor signal and solving the unified coding of abstract and concrete information is an essential problem in the process of human-computer interaction [159].
- Human intervention remains on superficial judgments (such as acceptance/rejection or direction), and exploring more complex feedback is also a critical issue in human-in-the-loop applications [160].
- Human-in-the-loop-based systems should have high robustness and generalization ability to domain changes, disturbances, and "out-of-range" samples [161].

6. CONCLUSION

In this paper, we review existing studies in human-in-the-loop techniques for machine learning. We first discuss the work of improving model performance from data processing. We divide this section into three parts: Data Preprocessing, Data Annotation, and Iterative Labeling according to the human-in-the-loop data processing pipeline. The core question is how to achieve more significant performance with fewer samples from a data perspective. Through the investigation of human-in-the-loop-based data processing methods, we find that the current human-in-the-loop-based data processing methods focus more on how to use semi-supervised methods for data collection and annotation than on how to select and identify key samples in the dataset. Then, we discuss the work of improving model performance through interventional model training. In this part, we divide into sub-module focusing on NLP and CV according to the target task. The core issue is how to promote the breakthrough of the model to crucial problems by adding human high-dimensional knowledge to the model. By investigating the work of human-in-the-loop in model training, we found that most human-in-the-loop exists in model training by participating in simple data increments. Exploring model training methods that effectively integrate human knowledge is the core of solving such problems. Methods such as inverse reinforcement learning and multimodal learning seem to be effective solutions. Finally, we discuss the design of the system independent "human-in-the-loop". In this section, we provide an overview of several human-in-the-loop real-world applications tasks, and we divide the content into Security Systems, Code Production Tools, Simulation Systems, and Search Engines by task type. Through the investigation of human-in-the-loop applications, we find that the coordination

of considerable variables in practical applications is more complicated than a single variable in academic research. In actual production, although the system integrating human-in-the-loop has advantages, how to quickly integrate high-level human knowledge in actual production and maintain the robustness of the system is a problem worthy of further consideration. Besides, we provide open challenges and opportunities and introduce some exciting questions. We first discuss some challenges and problems of the current human-in-the-loop in machine learning. On this basis, we further clarify the future directions in terms of CV, NLP, and applications. The establishment of artificial intelligence models that can effectively integrate human high-dimensional knowledge has development potential, especially with the research of large-scale pre-training models, the human-machine hybrid method based on few-shot has broad prospects.

Acknowledgment. This work was supported in part by the 2020 East China Normal University Outstanding Doctoral Students Academic Innovation Ability Improvement Project (YBNLTS2020-042), the Science and Technology Commission of Shanghai Municipality (19511120200).

7. REFERENCES

[1] S. Dong, P. Wang, and K. Abbas, "A survey on deep learning and its applications," *Computer Science Review*, vol. 40, p. 100379, 2021.

[2] A. Brutzkus and A. Globerson, "Why do larger models generalize better? a theoretical perspective via the xor problem," in *Proceedings of International Conference on Machine Learning (ICML)*, 2019, pp. 822–830.

[3] B. Zhou, A. Lapedriza, J. Xiao, A. Torralba, and A. Oliva, "Learning deep features for scene recognition using places database," in *Advances in Neural Information Processing Systems (NIPS)*, 2014, pp. 487–495.

[4] J. Li, J. Yang, A. Hertzmann, J. Zhang, and T. Xu, "LayoutGAN: Synthesizing graphic layouts with vector-wireframe adversarial networks," *IEEE Transaction on Pattern Analysis and Machine Intelligence*, vol. 43, no. 7, pp. 2388–2399, 2021.

[5] S. Zhao, Z. Liu, J. Lin, J.-Y. Zhu, and S. Han, "Differentiable augmentation for data-efficient gan training," *Advances in Neural Information Processing Systems (NIPS)*, vol. 33, pp. 7559–7570, 2020.

[6] H. T. Shen, X. Zhu, Z. Zhang, S.-H. Wang, Y. Chen, X. Xu, and J. Shao, "Heterogeneous data fusion for predicting mild cognitive impairment conversion," *Information Fusion*, vol. 66, pp. 54–63, 2021.

[7] X. Qiu, T. Sun, Y. Xu, Y. Shao, N. Dai, and X. Huang, "Pre-trained models for natural language processing: A survey," *Science China Technological Sciences*, pp. 1–26, 2020.

[8] M. Zaib, Q. Z. Sheng, and W. Emma Zhang, "A short survey of pre-trained language models for conversational ai-a new age in nlp," in *ACSW*, 2020, pp. 1–4.

[9] S. Bahrami, F. Dornaika, and A. Bosaghzadeh, "Joint auto-weighted graph fusion and scalable semi-supervised learning," *Information Fusion*, vol. 66, pp. 213–228, 2021.

[10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *NIPS*, 2017.

[11] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *NAACL*, 2019, pp. 4171–4186.

[12] A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever, "Improving language understanding by generative pre-training," 2018.

[13] M. Habermann, W. Xu, M. Zollhofer, G. Pons-Moll, and C. Theobalt, "Deepcap: Monocular human performance capture using weak supervision," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 5052–5063.

[14] Y. Wang, W. Yang, F. Ma, J. Xu, B. Zhong, Q. Deng, and J. Gao, "Weak supervision for fake news detection via reinforcement learning," in *AAAI*, vol. 34, no. 01, 2020, pp. 516–523.

[15] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, 2021.

[16] M. Diligenti, S. Roychowdhury, and M. Gori, "Integrating prior knowledge into deep learning," in *ICMLA*, 2017, pp. 920–923.

[17] S. Chen, Y. Leng, and S. Labi, "A deep learning algorithm for simulating autonomous driving considering prior knowledge and temporal information," *Computer-Aided Civil and Infrastructure Engineering*, vol. 35, no. 4, pp. 305–321, 2020.

[18] Y. Lin, S. L. Pintea, and J. C. van Gemert, "Deep hough-transform line priors," in *European Conference on Computer Vision (ECCV)*, 2020, pp. 323–340.

[19] G. Hartmann, Z. Shiller, and A. Azaria, "Deep reinforcement learning for time optimal velocity control using prior knowledge," in *ICTAI*, 2019, pp. 186–193.

[20] X. Zhang, S. Wang, J. Liu, and C. Tao, "Towards improving diagnosis of skin diseases by combining deep neural network and human knowledge," *BMC medical informatics and decision making*, vol. 18, no. 2, pp. 69–76, 2018.

[21] R. Zhang, F. Torabi, L. Guan, D. H. Ballard, and P. Stone, "Leveraging human guidance for deep reinforcement learning tasks," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2019.

[22] A. Holzinger, M. Plass, M. Kickmeier-Rust, K. Holzinger, G. C. Crişan, C.-M. Pintea, and V. Palade, "Interactive machine learning: experimental evidence for the human in the algorithmic loop," *Applied Intelligence*, vol. 49, no. 7, pp. 2401–2414, 2019.

[23] Y.-t. Zhuang, F. Wu, C. Chen, and Y.-h. Pan, "Challenges and opportunities: from big data to knowledge in ai 2.0," *Frontiers of Information Technology & Electronic Engineering*, vol. 18, no. 1, pp. 3–14, 2017.

[24] V. Kumar, A. Smith-Renner, L. Findlater, K. Seppi, and J. Boyd-Graber, "Why didn't you listen to me? comparing user control of human-in-the-loop topic models," in *ACL*, 2019.

[25] D. Xin, L. Ma, J. Liu, S. Macke, S. Song, and A. Parameswaran, "Accelerating human-in-the-loop machine learning: Challenges and opportunities," in *Proceedings of the second workshop on data management for end-to-end machine learning*, 2018, pp. 1–4.

[26] S. Budd, E. C. Robinson, and B. Kainz, "A survey on active learning and human-in-the-loop deep learning for medical image analysis," *Medical Image Analysis*, vol. 71, p. 102062, 2021.

[27] W. Jung and F. Jazizadeh, "Human-in-the-loop hvac operations: A quantitative review on occupancy, comfort, and energy-efficiency dimensions," *Applied Energy*, vol. 239, pp. 1471–1508, 2019.

[28] S. Agnisarman, S. Lopes, K. C. Madathil, K. Piratla, and A. Gramopadhye, "A survey of automation-enabled human-in-the-loop systems for infrastructure visual inspection," *Automation in Construction*, vol. 97, pp. 52–76, 2019.

[29] L. Benedikt, C. Joshi, L. Nolan, R. Henstra-Hill, L. Shaw, and S. Hook, "Human-in-the-loop ai in government: a case study," in *IUI*, 2020, pp. 488–497.

[30] C. Chai and G. Li, "Human-in-the-loop techniques in machine learning," *Data Engineering*, p. 37, 2020.

[31] B. M. Tehrani, J. Wang, and C. Wang, "Review of human-in-the-loop cyber-physical systems (hilcps): The current status from human perspective," *Computing in Civil Engineering 2019: Data, Sensing, and Analytics*, pp. 470–478, 2019.

[32] Z. Y. Khan, Z. Niu, S. Sandiwarno, and R. Prince, "Deep learning techniques for rating prediction: a survey of the state-of-the-art," *Artificial Intelligence Review*, vol. 54, no. 1, pp. 95–135, 2021.

[33] F. Yu, A. Seff, Y. Zhang, S. Song, T. Funkhouser, and J. Xiao, "Lsun: Construction of a large-scale image dataset using deep learning with humans in the loop," 2015.

[34] O. Siméoni, M. Budnik, Y. Avrithis, and G. Gravier, "Rethinking deep active learning: Using unlabeled data at model training," in *Proceedings of the International Conference on Pattern Recognition (ICPR)*, 2021, pp. 1220–1227.

[35] Y. Wang, L. Zhang, Y. Yao, and Y. Fu, "How to trust unlabeled data instance credibility inference for few-shot learning," *IEEE Transaction on Pattern Analysis and Machine Intelligence*, pp. 1–1, 2021.

[36] Y. Shi and A. K. Jain, "Boosting unconstrained face recognition with auxiliary unlabeled data," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 2795–2804.

[37] Z. Ren, R. Yeh, and A. Schwing, "Not all unlabeled data are equal: Learning to weight data in semi-supervised learning," *Advances in Neural Information Processing Systems (NIPS)*, vol. 33, 2020.

[38] S. Niu, B. Li, X. Wang, and H. Lin, "Defect image sample generation with gan for improving defect recognition," *IEEE Transactions on Automation Science and Engineering*, vol. 17, no. 3, pp. 1611–1622, 2020.

[39] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Computing Surveys (CSUR)*, 2021.

[40] T. D. Pham, "Classification of covid-19 chest x-rays with deep learning: new models or fine tuning?" *Health Information Science and Systems*, vol. 9, no. 1, pp. 1–11, 2021.

[41] S. Chen, Y. Hou, Y. Cui, W. Che, T. Liu, and X. Yu, "Recall and learn: Fine-tuning deep pretrained language models with less forgetting," in *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 7870–7881.

[42] G. Wang, W. Li, M. A. Zuluaga, R. Pratt, P. A. Patel, M. Aertsen, T. Doel, A. L. David, J. Deprest, S. Ourselin *et al.*, "Interactive medical image segmentation using deep learning with image-specific fine tuning," *IEEE transactions on medical imaging*, vol. 37, no. 7, pp. 1562–1573, 2018.

[43] L. He, J. Michael, M. Lewis, and L. Zettlemoyer, "Human-in-the-loop parsing," in *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016, pp. 2337–2342.

[44] J. Z. Self, R. K. Vinayagam, J. Fry, and C. North, "Bridging the gap between user intention and model parameters for human-in-the-loop data analytics," in *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*, 2016, pp. 1–6.

[45] Y. Zhuang, G. Li, Z. Zhong, and J. Feng, "Hike: A hybrid human-machine method for entity alignment in large-scale knowledge bases," in *CIKM*, 2017, pp. 1917–1926.

[46] G. Li, "Human-in-the-loop data integration," *Proceedings of the VLDB Endowment*, vol. 10, no. 12, pp. 2006–2017, 2017.

[47] B. Kim and B. Pardo, "A human-in-the-loop system for sound event detection and annotation," *ACM Transactions on Interactive Intelligent Systems (TiiS)*, vol. 8, no. 2, pp. 1–23, 2018.

[48] A. Doan, "Human-in-the-loop data analysis: a personal perspective," in *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*, 2018, pp. 1–6.

[49] X. L. Dong and T. Rekatsinas, "Data integration and machine learning: A natural synergy," in *COMAD*, 2018, pp. 1645–1650.

[50] A. L. Gentile, D. Gruhl, P. Ristoski, and S. Welch, "Explore and exploit. dictionary expansion with human-in-the-loop," in *European Semantic Web Conference*, 2019, pp. 131–145.

[51] S. Zhang, L. He, E. Dragut, and S. Vucetic, "How to invest my time: Lessons from human-in-the-loop entity extraction," in *KDD*, 2019, pp. 2305–2313.

[52] L. Berti-Equille, "Reinforcement learning for data preparation with active reward learning," in *International Conference on Internet Science*, 2019, pp. 121–132.

[53] S. Gurajada, L. Popa, K. Qian, and P. Sen, "Learning-based methods with human-in-the-loop for entity resolution," in *CIKM*, 2019, pp. 2969–2970.

[54] Y. Lou, M. Uddin, N. Brown, and M. Cafarella, "Knowledge graph programming with a human-in-the-loop: Preliminary results," in *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*, 2019, pp. 1–7.

[55] Z. Liu, J. Wang, S. Gong, H. Lu, and D. Tao, "Deep reinforcement active learning for human-in-the-loop person re-identification," in *Proceedings of the IEEE international conference on computer vision (ICCV)*, 2019, pp. 6122–6131.

[56] E. Wallace, P. Rodriguez, S. Feng, I. Yamada, and J. Boyd-Graber, "Trick me if you can: Human-in-the-loop generation of adversarial examples for question answering," *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 387–401, 2019.

[57] X. Fan, C. Li, X. Yuan, X. Dong, and J. Liang, "An interactive visual analytics approach for network anomaly detection through smart labeling," *Journal of Visualization*, vol. 22, no. 5, pp. 955–971, 2019.

[58] E. Krokos, H.-C. Cheng, J. Chang, B. Nebesh, C. L. Paul, K. Whitley, and A. Varshney, "Enhancing deep learning with visual interactions," *ACM Transactions on Interactive Intelligent Systems (TiiS)*, vol. 9, no. 1, pp. 1–27, 2019.

[59] J.-C. Klie, R. E. de Castilho, and I. Gurevych, "From zero to hero: Human-in-the-loop entity linking in low resource domains," in *ACL*, 2020, pp. 6982–6993.

[60] C. Butler, H. Oster, and J. Togelius, "Human-in-the-loop ai for analysis of free response facial expression label sets," in *IVA*, 2020, pp. 1–8.

[61] P. Ristoski, A. L. Gentile, A. Alba, D. Gruhl, and S. Welch, "Large-scale relation extraction from web documents and knowledge graphs with human-in-the-loop," *Journal of Web Semantics*, vol. 60, p. 100546, 2020.

[62] K. Qian, P. C. Raman, Y. Li, and L. Popa, "Partner: Human-in-the-loop entity name understanding with deep learning," in *The AAAI Conference on Artificial Intelligence*, vol. 34, no. 09, 2020, pp. 13 634–13 635.

[63] T.-N. Le, A. Sugimoto, S. Ono, and H. Kawasaki, "Toward interactive self-annotation for video object bounding box: Recurrent self-learning and hierarchical annotation based framework," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2020, pp. 3231–3240.

[64] M. Bartolo, A. Roberts, J. Welbl, S. Riedel, and P. Stenetorp, "Beat the AI: Investigating adversarial human annotation for reading comprehension," *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 662–678, 2020.

[65] K. Muthuraman, F. Reiss, H. Xu, B. Cutler, and Z. Eichenberger, "Data cleaning tools for token classification tasks," in *Proceedings of the Second Workshop on Data Science with Human in the Loop: Language Advances*, 2021, pp. 59–61.

[66] Q. Meng, W. Wang, T. Zhou, J. Shen, Y. Jia, and L. Van Gool, "Towards a weakly supervised framework for 3d point cloud object detection and annotation," *IEEE Transaction on Pattern Analysis and Machine Intelligence*, pp. 1–1, 2021.

[67] L. Zhang, X. Wang, Q. Fan, Y. Ji, and C. Liu, "Generating manga from illustrations via mimicking manga creation workflow," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 5642–5651.

[68] B. Adhikari and H. Huttunen, "Iterative bounding box annotation for object detection," in *Proceedings of the International Conference on Pattern Recognition (ICPR)*, 2021, pp. 4040–4046.

[69] J. L. Martinez-Rodriguez, A. Hogan, and I. Lopez-Arevalo, "Information extraction meets the semantic web: a survey," *Semantic Web*, vol. 11, no. 2, pp. 255–335, 2020.

[70] H. Ye, W. Shao, H. Wang, J. Ma, L. Wang, Y. Zheng, and X. Xue, "Face recognition via active annotation and learning," in *ACM International Conference on Multimedia*, 2016, pp. 1058–1062.

[71] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[72] T. Karmakharm, N. Aletras, and K. Bontcheva, "Journalist-in-the-loop: Continuous learning as a service for rumour analysis," in *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 115–120.

[73] Y. Song, J. Wang, T. Jiang, Z. Liu, and Y. Rao, "Targeted sentiment classification with attentional encoder network," in *ICANN*. Springer, 2019, pp. 93–103.

[74] X. Bai, P. Liu, and Y. Zhang, "Investigating typed syntactic dependencies for targeted sentiment classification using graph attention neural network," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 503–514, 2020.

[75] I. Arous, L. Dolamic, J. Yang, A. Bhardwaj, G. Cuccu, and P. Cudré-Mauroux, "Marta: Leveraging human rationales for explainable text classification," in *The AAAI Conference on Artificial Intelligence*, vol. 35, no. 7, 2021, pp. 5868–5876.

[76] Z. Yao, X. Li, J. Gao, B. Sadler, and H. Sun, "Interactive semantic parsing for if-then recipes via hierarchical reinforcement learning," in *The AAAI Conference on Artificial Intelligence*, vol. 33, no. 01, 2019, pp. 2547–2554.

[77] Z. Yao, Y. Su, H. Sun, and W.-t. Yih, "Model-based interactive semantic parsing: A unified formulation and a text-to-sql case study," in *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

[78] D. M. Ziegler, N. Stiennon, J. Wu, T. B. Brown, A. Radford, D. Amodei, P. Christiano, and G. Irving, "Fine-tuning language models from human preferences," 2019.

[79] N. Stiennon, L. Ouyang, J. Wu, D. Ziegler, R. Lowe, C. Voss, A. Radford, D. Amodei, and P. F. Christiano, "Learning to summarize with human feedback," *Advances in Neural Information Processing Systems (NIPS)*, vol. 33, pp. 3008–3021, 2020.

[80] B. Hancock, A. Bordes, P.-E. Mazare, and J. Weston, "Learning from dialogue after deployment: Feed yourself, chatbot!" in *ACL*, 2019, pp. 3667–3684.

[81] Z. Liu, Y. Guo, A. A. AI, and J. Mahmud, "When and why does a model fail? a human-in-the-loop error detection framework for sentiment analysis," *NAACL-HLT 2021*, p. 170, 2021.

[82] S. Chopra, M. Auli, and A. M. Rush, "Abstractive sentence summarization with attentive recurrent neural networks," in *NAACL*, 2016, pp. 93–98.

[83] Z. J. Wang, D. Choi, S. Xu, and D. Yang, "Putting humans in the natural language processing loop: A survey," in *Proceedings of the First Workshop on Bridging Human–Computer Interaction and Natural Language Processing*, 2021, pp. 47–52.

[84] L. Xiao, X. Hu, Y. Chen, Y. Xue, D. Gu, B. Chen, and T. Zhang, "Targeted sentiment classification based on attentional encoding and graph convolutional networks," *Applied Sciences*, vol. 10, no. 3, p. 957, 2020.

[85] L. Xiao, X. Hu, Y. Chen, Y. Xue, B. Chen, D. Gu, and B. Tang, "Multi-head self-attention based gated graph convolutional networks for aspect-based sentiment classification," *Multimedia Tools and Applications*, pp. 1–20, 2020.

[86] B. Nushi, E. Kamar, and E. Horvitz, "Towards accountable ai: Hybrid human-machine analyses for characterizing system failure," in *The AAAI Conference on Artificial Intelligence*, vol. 6, no. 1, 2018.

[87] M. T. Ribeiro, S. Singh, and C. Guestrin, "" why should i trust you?" explaining the predictions of any classifier," in *Annual ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 1135–1144.

[88] X. Wu, Y. Zheng, T. Ma, H. Ye, and L. He, "Document image layout analysis via explicit edge embedding network," *Information Sciences*, vol. 577, pp. 436–448, 2021.

[89] X. Wu, B. Xu, Y. Zheng, H. Ye, J. Yang, and L. He, "Fast video crowd counting with a temporal aware network," *Neurocomputing*, vol. 403, pp. 13–20, 2020.

[90] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision (ICCV)*, 2015, pp. 1440–1448.

[91] Z. Zou, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," 2019.

[92] A. Yao, J. Gall, C. Leistner, and L. Van Gool, "Interactive object detection," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012, pp. 3242–3249.

[93] K. Madono, T. Nakano, T. Kobayashi, and T. Ogawa, "Efficient human-in-the-loop object detection using bi-directional deep sort and annotation-free segment identification," in *APSIPA ASC*, 2020, pp. 1226–1233.

[94] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *IEEE International Conference on Image Processing (ICIP)*, 2017, pp. 3645–3649.

[95] M. R. Banham and A. K. Katsaggelos, "Digital image restoration," *IEEE signal processing magazine*, vol. 14, no. 2, pp. 24–41, 1997.

[96] A. Criminisi, P. Perez, and K. Toyama, "Object removal by exemplar-based inpainting," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, vol. 2, 2003, pp. II–II.

[97] G. Liu, F. A. Reda, K. J. Shih, T.-C. Wang, A. Tao, and B. Catanzaro, "Image inpainting for irregular holes using partial convolutions," in *European Conference on Computer Vision (ECCV)*, 2018, pp. 85–100.

[98] T. Weber, H. Hußmann, Z. Han, S. Matthes, and Y. Liu, "Draw with me: Human-in-the-loop for image restoration," in *IUI*, 2020, pp. 243–253.

[99] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Deep image prior," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 9446–9454.

[100] J. Roels, F. Vernaillen, A. Kremer, A. Gonçalves, J. Aelterman, H. Q. Luong, B. Goossens, W. Philips, S. Lippens, and Y. Saeys, "A human-in-the-loop approach for semi-automated image restoration in electron microscopy," *bioRxiv*, p. 644146, 2019.

[101] S. Minaee, Y. Y. Boykov, F. Porikli, A. J. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE Transaction on Pattern Analysis and Machine Intelligence*, pp. 1–1, 2021.

[102] V. Badrinarayanan, A. Kendall, and R. Cipolla, "Segnet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Transaction on Pattern Analysis and Machine Intelligence*, vol. 39, no. 12, pp. 2481–2495, 2017.

[103] H. Wang, T. Chen, Z. Wang, and K. Ma, "Efficiently troubleshooting image segmentation models with human-in-the-loop," pp. 1–1, 2020.

[104] A. Taleb, C. Lippert, T. Klein, and M. Nabi, "Multimodal self-supervised learning for medical image analysis," in *IPMI*, 2021, pp. 661–673.

[105] M. Ravanbakhsh, V. Tschernezki, F. Last, T. Klein, K. Batmanghelich, V. Tresp, and M. Nabi, "Human-machine collaboration for medical image segmentation," in *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020, pp. 1040–1044.

[106] Y. Murata and Y. Dobashi, "Automatic image enhancement taking into account user preference," in *CW*, 2019, pp. 374–377.

[107] M. Fischer, K. Kobs, and A. Hotho, "Nicer: Aesthetic image enhancement with humans in the loop," in *The Thirteenth International Conference on Advances in Computer-Human Interactions*, 2020, pp. 357–362.

[108] A. Benard and M. Gygli, "Interactive video object segmentation in the wild," 2017.

[109] S. W. Oh, J.-Y. Lee, N. Xu, and S. J. Kim, "Fast user-guided video object segmentation by interaction-and-propagation networks," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5247–5256.

[110] K. N. Shukla, A. Potnis, and P. Dwivedy, "A review on image enhancement techniques," *IJEACS*, vol. 2, no. 7, pp. 232–235, 2017.

[111] X. Fu, J. Yan, and C. Fan, "Image aesthetics assessment using composite features from off-the-shelf deep models," in *IEEE International Conference on Image Processing (ICIP)*, 2018, pp. 3528–3532.

[112] C. Burges, T. Shaked, E. Renshaw, A. Lazier, M. Deeds, N. Hamilton, and G. Hullender, "Learning to rank using gradient descent," in *Proceedings of International Conference on Machine Learning (ICML)*, 2005, pp. 89–96.

[113] R. Yao, G. Lin, S. Xia, J. Zhao, and Y. Zhou, "Video object segmentation and tracking: A survey," *ACM Transactions on Intelligent Systems and Technology (TIST)*, vol. 11, no. 4, pp. 1–47, 2020.

[114] S. Caelles, K.-K. Maninis, J. Pont-Tuset, L. Leal-Taixé, D. Cremers, and L. Van Gool, "One-shot video object segmentation," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 221–230.

[115] N. Xu, B. Price, S. Cohen, J. Yang, and T. S. Huang, "Deep interactive object selection," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 373–381.

[116] M. Hudec, E. Mináriková, R. Mesiar, A. Saranti, and A. Holzinger, "Classification by ordinal sums of conjunctive and disjunctive functions for explainable ai and interpretable machine learning solutions," *Knowledge-Based Systems*, vol. 220, p. 106916, 2021.

[117] S. Brostoff and M. A. Sasse, "Safe and sound: a safety-critical approach to security," in *Proceedings of the 2001 workshop on New security paradigms*, 2001, pp. 41–50.

[118] L. F. Cranor, "A framework for reasoning about the human in the loop," in *Proceedings of the 1st Conference on Usability, Psychology, and Security*, 2008, pp. 1–15.

[119] A. Machiry, R. Tahiliani, and M. Naik, "Dynodroid: An input generation system for android apps," in *Proceedings of the 2013 9th Joint Meeting on Foundations of Software Engineering*, 2013, pp. 224–234.

[120] A. Kovashka, D. Parikh, and K. Grauman, "Whittlesearch: Interactive image search with relative attribute feedback," *International Journal of Computer Vision*, vol. 115, no. 2, pp. 185–210, 2015.

[121] L. Rosenberg, "Artificial swarm intelligence, a human-in-the-loop approach to ai," in *The AAAI Conference on Artificial Intelligence*, vol. 30, no. 1, 2016.

[122] Y. Shoshitaishvili, M. Weissbacher, L. Dresel, C. Salls, R. Wang, C. Kruegel, and G. Vigna, "Rise of the hacrs: Augmenting autonomous cyber reasoning systems with human assistance," in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017, pp. 347–362.

[123] M. S. Wogalter, "Communication-human information processing (c-hip) model," in *Forensic Human Factors and Ergonomics*, 2018, pp. 33–49.

[124] L. Ma, "Towards understanding and simplifying human-in-the-loop machine learning," pp. 1–1, 2018.

[125] M. A. Salam, M. E. Koone, S. Thirumuruganathan, G. Das, and S. Basu Roy, "A human-in-the-loop attribute design framework for classification," 2019, pp. 1612–1622.

[126] B. A. Plummer, M. H. Kiapour, S. Zheng, and R. Piramuthu, "Give me a hint! navigating image databases using human-in-the-loop feedback," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2019, pp. 2048–2057.

[127] F. Wrede and A. Hellander, "Smart computational exploration of stochastic gene regulatory network models using human-in-the-loop semi-supervised learning," *Bioinformatics*, vol. 35, no. 24, pp. 5199–5206, 2019.

[128] H. V. Singh and Q. H. Mahmoud, "Human-in-the-loop error precursor detection using language translation modeling of hmi states," in *SMC*, 2020, pp. 2237–2242.

[129] G. Demartini, S. Mizzaro, and D. Spina, "Human-in-the-loop artificial intelligence for fighting online misinformation: Challenges and opportunities," *The Bulletin of the Technical Committee on Data Engineering*, vol. 43, no. 3, pp. 1–10, 2020.

[130] D. Odekerken and F. Bex, "Towards transparent human-in-the-loop classification of fraudulent web shops," in *Legal Knowledge and Information Systems*, 2020, pp. 239–242.

[131] M. Böhme, C. Geethal, and V.-T. Pham, "Human-in-the-loop automatic program repair," in *ICST*, 2020, pp. 274–285.

[132] A. Renner, "Designing for the human in the loop: Transparency and control in interactive machine learning," Ph.D. dissertation, University of Maryland, College Park, 2020.

[133] J. B. Davidson, R. B. Graham, S. Beck, R. T. Marler, and S. L. Fischer, "Improving human-in-the-loop simulation to optimize soldier-systems integration," *Applied Ergonomics*, vol. 90, p. 103267, 2021.

[134] H. O. Demirel, "Digital human-in-the-loop framework," in *International Conference on Human-Computer Interaction*, 2020, pp. 18–32.

[135] M. Metzner, D. Utsch, M. Walter, C. Hofstetter, C. Ramer, A. Blank, and J. Franke, "A system for human-in-the-loop simulation of industrial collaborative robot applications," in *CASE*, 2020, pp. 1520–1525.

[136] A. Polisetty Venkata Sai, "Information preparation with the human in the loop," Ph.D. dissertation, TU Darmstadt, 2020.

[137] Z. Zhu, Y. Lu, R. Deng, H. Yang, A. B. Fogo, and Y. Huo, "Easierpath: An open-source tool for human-in-the-loop deep learning of renal pathology," in *Interpretable and Annotation-Efficient Learning for Medical Image Computing*, 2020, pp. 214–222.

[138] N. Li, S. Adepu, E. Kang, and D. Garlan, "Explanations for human-on-the-loop: A probabilistic model checking approach," in *Proceedings of the IEEE/ACM 15th International Symposium on Software Engineering for Adaptive and Self-Managing Systems*, 2020, pp. 181–187.

[139] P. Wiriyathammabhum, D. Summers-Stay, C. Fermüller, and Y. Aloimonos, "Computer vision and natural language processing: recent approaches in multimedia and robotics," *ACM Computing Surveys (CSUR)*, vol. 49, no. 4, pp. 1–44, 2016.

[140] A. Holzinger, B. Malle, A. Saranti, and B. Pfeifer, "Towards multi-modal causability with graph neural networks enabling information fusion for explainable ai," *Information Fusion*, vol. 71, pp. 28–37, 2021.

[141] S. Arora and P. Doshi, "A survey of inverse reinforcement learning: Challenges, methods and progress," *Artificial Intelligence*, vol. 297, p. 103500, 2021.

[142] A. Doan, A. Ardalan, J. Ballard, S. Das, Y. Govind, P. Konda, H. Li, S. Mudgal, E. Paulson, G. P. Suganthan *et al.*, "Human-in-the-loop challenges for entity matching: A midterm report," in *Proceedings of the 2nd workshop on human-in-the-loop data analytics*, 2017, pp. 1–6.

[143] J. Li, A. H. Miller, S. Chopra, M. Ranzato, and J. Weston, "Dialogue learning with human-in-the-loop," *International Conference on Learning Representations (ICLR)*, pp. 1–23, 2016.

[144] H. Amirpourazarian, A. Pinheiro, E. Fonseca, M. Ghanbari, and M. Pereira, "Quality evaluation of holographic images coded with standard codecs," *IEEE Transactions on Multimedia*, pp. 1–1, 2021.

[145] S. Wan, Y. Hou, F. Bao, Z. Ren, Y. Dong, Q. Dai, and Y. Deng, "Human-in-the-loop low-shot learning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 7, pp. 3287–3292, 2021.

[146] L. Yang, Q. Sun, N. Zhang, and Z. Liu, "Optimal energy operation strategy for we-energy of energy internet based on hybrid reinforcement learning with human-in-the-loop," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, pp. 1–11, 2020.

[147] Y. Fu, X. Zhu, and B. Li, "A survey on instance selection for active learning," *Knowledge and information systems*, vol. 35, no. 2, pp. 249–283, 2013.

[148] J. Zhang, P. Fiers, K. A. Witte, R. W. Jackson, K. L. Poggensee, C. G. Atkeson, and S. H. Collins, "Human-in-the-loop optimization of exoskeleton assistance during walking," *Science*, vol. 356, no. 6344, pp. 1280–1284, 2017.

[149] Y. Tay, M. Dehghani, D. Bahri, and D. Metzler, "Efficient transformers: A survey," 2020.

[150] J. Kreutzer, S. Riezler, and C. Lawrence, "Offline reinforcement learning from human feedback in real-world sequence-to-sequence tasks," in *SPNLP*, 2021, pp. 37–43.

[151] A. Smith, V. Kumar, J. Boyd-Graber, K. Seppi, and L. Findlater, "Closing the loop: User-centered design and evaluation of a human-in-the-loop topic modeling system," in *IUI*, 2018, pp. 293–304.

[152] A. Kapoor, J. C. Caicedo, D. Lischinski, and S. B. Kang, "Collaborative personalization of image enhancement," *International Journal of Computer Vision*, vol. 108, no. 1-2, pp. 148–164, 2014.

[153] J.-S. Jwo, C.-S. Lin, and C.-H. Lee, "Smart technology–driven aspects for human-in-the-loop smart manufacturing," *The International Journal of Advanced Manufacturing Technology*, vol. 114, no. 5, pp. 1741–1752, 2021.

[154] B. Settles, "Closing the loop: Fast, interactive semi-supervised annotation with queries on features and instances," in *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2011, pp. 1467–1478.

[155] T. Y. Lee, A. Smith, K. Seppi, N. Elmqvist, J. Boyd-Graber, and L. Findlater, "The human touch: How non-expert users perceive, interpret, and fix topic models," *International Journal of Human-Computer Studies*, vol. 105, pp. 28–42, 2017.

[156] N. M. Marquand, "Automated modeling of human-in-the-loop systems," Ph.D. dissertation, Purdue University Graduate School, 2021.

[157] J. J. Dudley and P. O. Kristensson, "A review of user interface design for interactive machine learning," *ACM Transactions on Interactive Intelligent Systems (TiiS)*, vol. 8, no. 2, pp. 1–37, 2018.

[158] K. Shilton, "Values and ethics in human-computer interaction," *Foundations and Trends® in Human–Computer Interaction*, vol. 12, no. 2, 2018.

[159] A. Jolfaei, M. Usman, M. Roveri, M. Sheng, M. Palaniswami, and K. Kant, "Guest editorial: Computational intelligence for human-in-the-loop cyber physical systems," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 6, no. 1, pp. 2–5, 2022.

[160] W. Xu, M. J. Dainoff, L. Ge, and Z. Gao, "Transitioning to human interaction with ai systems: New challenges and opportunities for hci professionals to enable human-centered ai," *International Journal of Human–Computer Interaction*, pp. 1–25, 2022.

[161] K. Zhou, Z. Liu, Y. Qiao, T. Xiang, and C. Change Loy, "Domain generalization: A survey," 2021.

Human-in-the-loop Machine Learning: A Macro-Micro Perspective

Jiangtao Wang[1] Bin Guo[2] Luke Chen[3]
1. Coventry University, Coventry, United Kingdom
2. Northwestern Polytechnical University, Xi'an, China
3. Ulster University, Newtownabbey, United Kingdom
Jiangtao.wang@conventry.ac.uk, guobin.keio@gmail.com, l.chen@ulster.ac.uk

Abstract: Though technical advance of artificial intelligence and machine learning has enabled many promising intelligent systems, many computing tasks are still not able to be fully accomplished by machine intelligence. Motivated by the complementary nature of human and machine intelligence, an emerging trend is to involve humans in the loop of machine learning and decision-making. In this paper, we provide a macro-micro review of human-in-the-loop machine learning. We first describe major machine learning challenges which can be addressed by human intervention in the loop. Then we examine closely the latest research and findings of introducing humans into each step of the lifecycle of machine learning. Finally, we analyze current research gaps and point out future research directions.

1. Introduction

Advances in machine learning (ML) technologies have led to an explosion of major breakthroughs in the field of artificial intelligence (AI), which in turn has given rise to ubiquitous intelligent systems in our daily life. Typical applications include autonomous vehicles, game playing (e.g., AlphaGo), precise medical diagnosis and prescription, assistive robots, face recognition, and so forth. Despite the increasing usage and power of ML in these AI applications, there is still a large spectrum of computing tasks that pure machine intelligence cannot fully handle due to challenges such as lack of large-scale dataset, low data quality, insufficient training labels, and explanability and reliability issues of ML-based decision-making.

Motivated by the complementary nature of human and machine intelligence [Kamar 2016] [Wang 2019], an approach that exploits human's cognitive power to help address these ML challenges has emerged recently, which is referred to as "Human-in-the-loop Machine Learning (HML)". HML is intended to integrate relevant human competences into the whole cycle of ML, ranging from data collection, algorithm tuning, parameters selection, to the usage of the outcomes of ML to actuate the physical world. The fundamental goal of this emerging topic is to re-examining and reframing the ML workflow from a human-centered perspective. The study of HML has the following significance. First, it can generate more usable and human-centric ML tools for AI engineers or even application end users. Second, it can lead to an in-depth understanding about how the power of human intelligence and machine intelligence can be combined to solve real-world computing problems. Inspired by the aforementioned visions, a number of studies for HML have been recently presented in top conference or prestigious journals, which has become a trendy research topic across multiple overlapping research communities (e.g., AI, Human Computer Interaction, crowd and social computing, data mining, and so on).

There are several literature reviews or tutorials summarizing the state of the art with each focusing on different topical area, see Table 1. Nevertheless, they only touch individual specific research problem within the scale of HML. To the best of our knowledge, there are no survey papers to present a global picture about how human cognitive power can be appropriately integrate into ML processes. A systematic study and classification of the research problems in the HML research domain is still missing.

To this end, we attempt to fill the gap by systematically studying and classifying the research work of HML from a *macro-micro* perspective. Instead of discussing a particular aspect such as interface design [Dudley 2018] or the explainable models [Riccardo 2019], this paper is to provide a high-level overview of HML for those researchers who want to step into this topic area for the first time. More specifically, this paper makes the following contributions in this paper.

- From a *macro* perspective of the complementary nature of machine and human, we identify the main bottlenecks of ML, and then provide deep insights about why and how human can collaborate with machine so as to address these challenges.
- We present a *macro-micro* review by dividing the representative studies into several stages of the ML lifecycle. Here, in order to help the readers to establish a clearer and more understandable picture, the reviews are organized in our proposed structure/framework called Human4ML.
- We analyze existing research gaps in this field with a number of future research directions and proposals, which may contribute to new ideas and visions.

It should be noted that this paper does not aim to give a comprehensive review of HML due to the page limit. We provide a general framework into which other existing works can be easily added by the readers themselves if they intend to extend it to a more comprehensive one.

Table 1: Relevant Reviews/Tutorials and Their Focused Aspects

Relevant Survey	**Topical Areas**
[Dudley 2018]	Interface design techniques in the interactive machine learning
[Riccardo 2019]	Models and approaches to make AI more explainable
[Zhang 2018]	Visualization approaches for interpreting deep-learning-based models
[Wang 2019]	Optimization for crowdsourcing and ML (e.g., labeling and inference)
[Zhang 2019]	Leveraging Human Guidance for Deep Reinforcement Learning Tasks

2. The Needs for Human Involvement

Although ML is powerful in developing a variety of smart systems, it still encounters a number of inevitable challenges under real-world practical situations. While these challenges can be addressed in many different ways, in this paper we focus on those challenges that can be potentially alleviated by introducing human intelligence into the process of ML.

Challenge 1: Dirty and Incomplete Data.

The most fundamental ingredient of an AI system is data for training the learning model, because ML is essentially the process of automatically learning knowledge from data. The quality of data is crucially important for the performance of a learning model. Unfortunately, datasets acquired in the real world can be low quality. Dirty or incomplete data, such as missing values, typos, inconsistent values, out-of-date records, usually leads to inaccurate

analysis and learning. Several automatic data cleaning approaches have been developed to enhance data quality, but their performance is still far from meeting the requirement of applications [Xu 2016]. In many scenarios, we still need the help from human workers to detect, repair or eliminate dirty or incomplete data.

Challenge 2: Heterogeneity and Isolation

Many intelligent applications need data feed from multiple domains, which are usually originated from heterogenous and isolated ICT systems. For example, if we attempt to develop an AI-based system to predict population health status in different regions/countries in EU, the first challenge is data integrations and linkage from different information systems to build a comprehensive training dataset for model training. However, this task is non-trivial because the data structure and database design might be different for systems with similar semantics, which increase the difficulty and workload of data linkage. For example, if we want to get an overall picture of heart attacks in Europe, we need to integrate and link relevant data from heterogenous information system of many hospitals from multiple countries. Although a lot of data integration tools such as schema matching have been proposed and proved to be effective to some extent, they are often uncertain about schema matchings they suggest, and this uncertainty is inherent since it arises from the inability of the schema to fully capture the semantics of the represented data [Zhang 2018]. Human common sense can often help in this process.

Challenge 3: Insufficient Labels in Training Data

In recent years, the real-world impact of ML has grown in leaps and bounds. However, ML models heavily rely on massive sets of hand-labeled training data, and the cost of building such labeled datasets is one of the key bottlenecks to large-scale ML adoption. Although some weak supervised learning approaches [Zhou 2017] can ease this problem in some cases, it is still preferable in general if we can obtain sufficient labels for training. Since labeling a large amount of data is time-consuming, there is an emerging trend to leverage the power of the crowdsourcing workers to accomplish this task.

Challenge 4: Feature Construction with Limited Data

Constructing feature set is an essential but uneasy step to ML. There are two common approaches for feature discovery or extraction: the automatic methods, e.g., deep learning and domain-knowledge-based methods (e.g., feature engineering). However, if we are only given a limited amount of training data, both approaches become ineffective as irrelevant features extracted from limited datasets can lead to the problem called overfitting. For example, it is easy to distinguish a dog from a cat in an image with deep learning as the relevant labeled training data on the Web is huge. However, the classification between “Northern Flicker” and “Red Bellied Woodpecker” [Deng 2016]) is much more challenging when there are insufficient training data and fewer discriminative features.

Challenge 5: Trust and Transparency

ML-based AI systems sometimes are completely treated as black boxes without human supervision, where the learning models sometimes learn undesirable tricks that do an optimal job of satisfying pre-programmed goals (e.g. minimizing loss function) on the training data without reflecting the complicated implicit desires of the human system designers. In some ML-enabled applications, the black box issue doesn’t matter if the task is not very critical. However, this issue becomes very sensitive when the application domain is critical (e.g., medical treatment, and security/safety control), where knowing the reasons behind the machine’s decision is of high importance. Thus, it is of vital significance to open that black box to achieve a more transparent, explainable, and trustworthy AI and ML with human in the loop [Riccardo 2019].

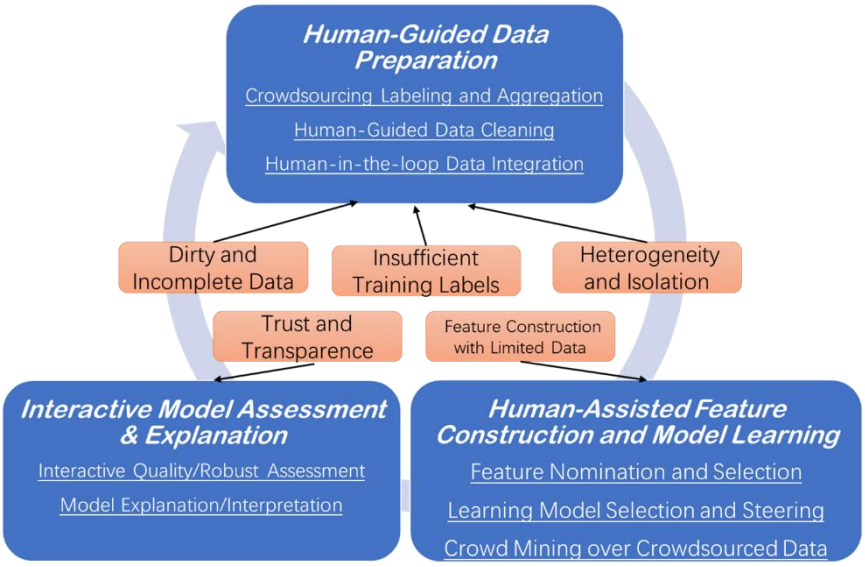


Fig 1. Human4ML framework: organizing representative studies

3. Structure Overview: Lifecycle Perspective

The typical ML process consists of three iterative stages: data pre-preparing (e.g., data collection and pre-processing), feature construction and model learning, model assessment. Human intelligence can be integrated into each of the three stages. To this end, we present an overview of the state-of-the-art HML research using a general framework named Human4ML following the ML lifecycle, as shown in Fig.1. In this framework, corresponding human-computation-based techniques are adopted to improve certain stage or task of ML. From Figure 1, we can see an overview of the utilization of human capabilities in each stage of ML. Here, we characterize the features of each stage by using the logic of "2W1H" (What, Who, and How) in Table 2. "What" refers to what problems or challenges each component aims to support, "How" refers to the general idea to achieve the goal, and "who" refers to the stakeholders who are involved in this stage.

Phase 1: Human-Guided Data Preparation

Human intelligence can be leveraged in data collection and pre-processing procedures (we call them together as data preparation in this paper) to address the challenge of low quality and insufficiency of training dataset mentioned above. To our best knowledge, human computation techniques (e.g., crowdsourcing) has primarily adopted in three tasks in data preparation phase, including data labeling and aggregation [Liu 2018], data cleaning [Xu 2016], and data integration [Li 2017]. Some representative research works related to this component will be introduced in the Section 4 with more details.

Phase 2: Human-Assisted Feature Construction and Model Learning

For either automatic or domain expert-based approaches, feature construction and model learning is still a challenge for fine-grained features with limited training data. Recent studies show the potential in inviting human (either experts or crowdsourcing workers) to collaborate with machine algorithms to better elicit good features and perform model training. The key insights lie in two aspects. One is that human workers are better at nominating subtle behavioral signals than machines [Cheng 2015], so that they can complement machine intelligence in feature extraction. The other is that the selection of learning models are significantly correlated with training data and

learning target, and human can help compare, select, and steer different ML models (e.g., choosing the right classifiers in the classification task) [Das 2019]. We will introduce related work for this component in Section 5.

Table 2. Using "2W1H" to characterize Human4ML Framework

Stages	**WHAT**	**HOW**	**WHO**
Human-guided Data Preparation	To obtain more labels, clean data, and integrate data (address challenge 1,2 and 3)	Use crowdsourcing or invite domain experts to help	Workers in crowdsourcing platform, domain experts, source data managers
Human-Assisted Feature Construction and Model Learning	Collaborate with machine to construct feature set and perform model training (address challenge 4)	Nominate/select features (based on crowdsourcing), select/steer models by designing suitable visualization interface, and perform crowd-based data mining over crowdsourced data	AI engineers, workers in crowdsourcing platforms, domain experts
Interactive Model Assessment and Explanation	Quality assurance, present the reasons behind the decision of ML models (address challenge 5)	Design human-computer interface, visualization, adopt explainable models	AI engineers, domain experts

Phase 3: Interactive Model Assessment and Explanation

Only when we have adequate trust, can we use a ML model in real world, especially for critical systems (e.g., medical, military, financial, and so on). Although good performance in simple indicators on testing sets (e.g., precision, recall, etc.) can provide some belief to some extent, human themselves, also serving as the end user of the ML applications, cannot be excluded out of the evaluation process. There are two key research issues human can actively participate in this stage, namely (1) enabling quality and robustness assessment with human in the loop, and (2) providing model interpretability by designing effective human-computer interfaces to bridge the gap between machine and human. Representative studies for this component will be introduced in Section 6.

4. Human-Guided Data Preparation

In this paper, the data preparation consists of key operations such as data collection, data cleaning, and data integration. Accordingly, this section introduces how human intelligence is utilized in these operations respectively.

4.1 Crowdsourcing Labeling and Aggregation

In the data preparing phase, the most studied task is to use crowdsourcing [Howe 2006] in the data labeling and aggregation. The naïve way for label aggregation is majority vote, where the aggregated value is regarded as the opinion of the majority of the crowd workers. However, in realistic scenarios, workers are heterogenous in skills, expertise, and reputation, and so on. Therefore, weight-based aggregation methods are proposed, where the label of a worker with high weight would count more. Aggregation methods that consider the diversity and dependency of workers are proposed in recent years to improve the label aggregation accuracy [Nushi 2015] [Venanzi 2016]. In addition to the aggregation approach, worker selection and task assignment is also important. For example, authors

in [Liu 2018] studied in an active learning setting where training samples are adaptively selected to be labeled, and a learner can query a set of crowdsourcing workers with unknown expertise level for label information.

4.2 Human-Guided Data Cleaning

Detecting and repairing dirty data is one of the perennial challenges in data analytics, and failure to do so can result in inaccurate analytics and unreliable decisions. In the data preparation phase, another task that human workers can help is data cleaning. For example, the authors present CrowdCleaner in [Tong 2014], a smart system to clean multi-version data on the Web. In this system, the authors leverage active and passive crowdsourcing methods together for rectifying errors for multi-version data. In another example, [Chu 2015], the authors use both human computation and open knowledge base to generate top-k possible repairs for incorrect data. More extensive review for this technical aspect can be referred to [Xu 2016].

4.3 Human-in-the-loop Data Integration

Human intelligence and crowdsourcing techniques are extensively used in many data integration tasks in recent years.

(1) *Schema matching in Database*. Schema matching is a central challenge for data integration systems. Automated tools are often uncertain about schema matchings they suggest, and this uncertainty is inherent since it arises from the inability of the schema to fully capture the semantics of the represented data. Inspired by the popularity and the success of easily accessible crowdsourcing platforms, a number of studies such as [Li 2017] explore the technique of human computation and crowdsourcing to reduce the uncertainty of schema matching with the consideration of cost and quality. (2) *Entity linkage in Knowledge Base/Graph*. In entity linking, words of interest (names of persons, locations and companies) are mapped from an input text to corresponding unique entities in a target knowledge base (KB), which serves as a basic component of many ML and AI applications. Several automatic linking methods have been proposed. However, due to the inconsistency and uncertainty of large-scale KBs, automatic techniques for entity linkage achieve low quality. Thanks to the open crowdsourcing platforms, studies such as [Zhuang 2017] propose solutions to harness the human workers to improve the linkage quality.

5. Human-Assisted Feature Construction and Model Learning

Human intelligence can be used to nominate and select features in ML, and further help in model selection and steering.

5.1 Feature Nomination and Selection

Crowdsourcing and human computation can be utilized to discover (nominate or select) features for supervised/unsupervised learning. For example, the study in [Cheng 2015] presented a hybrid crowd-machine learning framework called Flock. The basic idea of the framework is that: it starts with a written description of a learning goal, uses the crowd to suggest predictive features, and then weighs these features using ML to produce models that are accurate and use human-understandable features. Specifically, Flock uses comparative examples to help crowd workers better nominate subtle features, and then adopts a clustering approach to re-organize these features. Flock is shown to be more effective than pure ML approaches. In the work of [Deng 2016], the authors introduced a novel approach for fine-grained image classification tasks (e.g., distinguish between “Northern Flicker” and “Red Bel-lied Woodpecker”). The authors proposed an online crowdsourcing game that elicits discriminative features by crowd worker (player). During the game, the player can choose to reveal details of circular regions (called "bubbles"), with certain incentive mechanism. Additionally, crowd workers can help discover features for unsupervised learning. For example, the authors in [Zou 2015] introduce an unsupervised approach to effectively discover features in a dataset via photo crowdsourcing. This approach first asks crowd workers to articulate a feature that is common to two out of three displayed photos and then requires the crowd workers to provide binary labels

to the remaining photos based on the discovered features. It adaptively repeats the above process based on the labels of the previously discovered features.

5.2 Learning Model Selection and Steering

In additional to constructing a set of discriminative features, selecting appropriate models (e.g., the adopted classifiers in classification tasks) are also important for the performance of ML. Interactive model steering can help people incrementally build machine learning models that are tailored to their domains and tasks. Studies such as [Das 2019] aim to help ML engineers to select the most suitable models for corresponding data and learning tasks, where human-computer-interaction (HCI) tools are built to enable the interactive comparison and selection of different models. Specifically, it proposed a technique to allow users to inspect and steer multiple machine learning models. The technique steers and samples models from a broader set of learning algorithms and model types. More extensive review can be found in the survey [Riccardo 2019].

5.3 Crowd Mining over Crowdsourced Data

With the development of mobile social networks and mobile crowd sensing (MCS), more and more crowdsourced data are generated on the Web or collected from real-world sensing. The fragmented, heterogeneous, and noisy nature of online/offline crowdsourced data, however, makes it difficult to be understood. Traditional content-based analysis methods suffer from issues such as computational intensiveness and poor performance. To address this, a new research direction of HML, entitled CrowdMining [Guo 2019], has emerged. Based on the observation that the knowledge hidden in the process of data generation, including individual/crowd behavior patterns (e.g., mobility patterns, community contexts such as social ties and structure) and crowd-object interaction patterns (flickering or tweeting patterns), are neglected in crowdsourced data mining, it advocates to leverage the so-called '*crowd intelligence*' for crowdsourced data mining and understanding.

Various types of crowd intelligence are embedded in the data sensing or content generation process. These include interaction contexts based on human perception, decision making and opinions, and community contexts related to individual traits, community structure, and social/individual behavior patterns. They are often used indirectly for crowdsourced data understanding, i.e., used as features or parameter inputs of MI. There are various data mining tasks that can be performed by means of crowdsourcing, such as data filtering, classification, and clustering. For example, FlierMeet [Guo 2014] is an MCS app for public information (distributed fliers in the city) sensing and tagging. It identifies a novel set of crowd-object interaction hints (e.g., crowd-object interaction entropy, temporal patterns of interaction, interaction frequency) to predict the semantic tags of crowdsensed flier pictures.

6. Interactive Model Assessment & Explanation

In addition to data preparing and model learning phases, human can also be heavily involved in the model evaluation and explanation process.

6.1 Interactive Quality/Robustness Assessment

The training and refinement of ML models is an interactive process. The goal of AI engineers is to train the model to some level of acceptable accuracy. Therefore, the quality of ML models should be appropriately evaluated to determine if the model learning can be stopped. Amershi et al. [Saleema 2014] demonstrate visualization of the current model accuracy as an interface element present during the standard model steering task. In their work, the authors suggest that this visualization allows users to perform different strategies and evaluate their impact on accuracy. The quality assessment task may be interactive such as in the method proposed by [Saleema 2011], which provides a working environment where users can interact with the confusion matrix and express preferred classification requirements. In additional to the traditional quality evaluation on models, researchers recently started to focus on the robustness and safety of ML and AI systems (especially for deep-learning-based models), where a

ML system is interactively tested by the cooperation of human and machine under deliberately designed adversary cases [Ruan 2019].

6.2 Model Explanation/Interpretation

Understanding the reasons behind the suggestions (e.g., predictions or recommendations) made by AI systems is important in assessing trust, which helps end users to make final decisions (e.g., take an action or not). Understanding how a model works can also help refine an untrustworthy model into a trustworthy one.

A number of research works has started to study how users can evaluate and refine trained models when learning-based AI systems make incorrect decisions (e.g., classification, prediction, and recommendation). There are several methods or frameworks proposed in recent years to interpret ML models. Ribro et al. [Tulio 2016] developed a framework called LIME, which is able to explain predictions by learning an interpretable model locally around the prediction. For example, a model predicts that a patient has ae flu, and LIME highlights the symptoms in the patient's history that led to the prediction. Sneeze and headache are portrayed as contributing to the "flu" prediction, while "no fatigue" is evidence against it. With these, a doctor can make an informed decision about whether or not to trust the model's prediction. The authors in [Teso 2018] proposed the novel framework of explanatory interactive learning. In each step, the learner explains its interactive query to the user, and queries any active classifier for visualizing explanations of the corresponding predictions. In their study, the authors demonstrate that this framework can boost the predictive and explanatory powers of and the trust into the learned model.

Deep Learning models are powerful but being criticized the least interpretable. To this end, progress in this field have been accomplished by producing interpretation of the features learned at each layer of a Neural Network [Yosinski 2015] [Olah 2018]. A more extensive literature review about interpretability in deep learning can be found in [Zhang 2018].

7. Opportunities and Proposals

While various techniques have been proposed to improve ML with the human in the loop, several challenges still remain. In this section, we will highlight some existing gaps and propose new visions that can stimulate the research community to pursue new directions (i.e., new research problems).

P1: Human-Machine Collaboration Fundamentals

Although there are increasing number of technical researches on human-machine collaboration in the context of ML, the fundamental collaboration model and mechanism is still unclear. For example, what is the collaboration mode for human and machine? For the parallel mode, how will the output of machine and human (e.g., machine-extracted and human-nominated features) be integrated? For the sequential mode, how will two sides handoff to each other on different learning tasks. Besides, for different tasks, who are the human collaborators? Ordinary citizens or someone with specialized expertise? How many human workers should be involved? By successfully answering these questions, we can derive a clear picture on the model and mechanism of how human and machine is cooperated in the context of ML.

P2: Multi-Objective Optimization

When introducing human into the multiple tasks of the ML process, we have to consider multiple factors to optimize the utilization process itself. For example, from the perspective of the human workers, we have to consider factors such as intrusiveness, motivations, and their expertise/reputation. From the perspective the ML-related task requestors, task quality, budget, completion time, etc. In general, we formulate such research problem as an optimization with ML-task-specific goals and constraints. For example, a novel research problem can be how to select a minimum number of human workers to nominate features while ensuring a certain level of model prediction

accuracy before the required deadline. Specifically, we need to further characterize and model the above factors in the context of ML applications and understand the difference compared with other types of crowdsourcing/crowdsensing tasks (e.g., urban environment sensing, software development, etc.)

P3: Bias and Ethics Issue

Task requesters usually overlook human workers behind tasks, this can lead to issues of ethics (e.g., unfair payment) and amplification of human biases, which are transferred into training data and affect machine learning in the real world. Research works such as [Barbosa 2019] have started to look into these issues and proposed feasible solutions, but most of them only focus on the crowdsourcing labeling tasks, while ignoring other tasks in the lifecycle of HML. Besides, we still need to understand human biases in HML, and interesting research topics could consist of effect of cultural, gender and ethnic biases, effect of human in the loop training and past experiences, effect of human expertise vs interest, bias in experts vs bias in crowdsourcing, bias in task selection, and task assignment/recommendation for reducing bias.

P4: Integration of Human-Machine-Hybrid Features

Human intelligence can help generate human-understandable features, and further by combining them with machine-extracted features, we have the potential to build a more powerful learning model [Cheng 2015]. However, there are two key critical remaining challenges that existing works do not address in this step. *First, how to obtain the value of human-nominated features?* As many of these features are subtle, machines cannot automatically assign values to them. Instead, humans have to handle many of these labeling tasks. This feature value acquisition task is far more costly than the classification labeling task, because for a certain training record (say, for a classification task), there is only one classification label but with numerous features. Therefore, it is crucial to study how to obtain the value of crowd-nominated features with the budget concern in mind. One promising research direction is to study the selection of most informative features with the objective of achieving a good tradeoff between the total feature labeling cost and model classification/prediction performance. *Second, how to integrate human and machine features?* When combining the human-generated features with machine-extracted ones, a naïve approach is to directly connect these two types of features as a longer feature vector. However, an alternative approach could be training machine and human classifiers separately, then combining them by adopting ensemble learning techniques to make a more accurate prediction.

P5: From Individual to Crowd

Most existing studies in HML focus on the interaction of one single human worker with a ML system. With the popularity of social networks (e.g., twitter, Facebook, etc.) and crowdsourcing platforms in different domains (e.g., product design, software engineering, etc.), there is now an opportunity of recruiting large number of crowd workers collaboratively in a ML system. Therefore, techniques and tools should be developed to support the collaborative work in the context of building ML models or applications. For instance, a collaborative working environment should be developed to help workers to discuss, comment, share, and nominate more efficiently. We have already witnessed substantial progress in crowdsourcing-based labeling tasks. Nevertheless, further research on how the crowd can work collaboratively in other higher-level tasks such as feature selection, evaluation, and interpretations of models, is still needed.

8. Conclusion

This paper presents a review of ML with human in the loop. Specifically, by organizing existing studies along the multi-stage lifecycle of ML, we present various kinds of strategies exploiting human intelligence in ML. in addition, we point introduce some research gaps and directions that may further integrate the power of human and machine intelligence to build more cost-effective, reliable, and robust intelligent systems.

References

[Barbosa 2019] Barbosa, et al. "Rehumanized Crowdsourcing: A Labeling Framework Addressing Bias and Ethics in Machine Learning." CHI 2019, p. 543. ACM, 2019.

[Cheng 2015] Cheng, J., et al. (2015, February). Flock: Hybrid crowd-machine learning classifiers. In Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (pp. 600-611).

[Chu 2015] Chu, X., et al. (2015, May). Katara: A data cleaning system powered by knowledge bases and crowdsourcing. In Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data (pp. 1247-1261).

[Das 2019] Das, Subhajit, et al. "BEAMES: Interactive Multi-Model Steering, Selection, and Inspection for Regression Tasks." IEEE computer graphics and applications (2019).

[Deng 2016] Deng, J., et al. (2016). Leveraging the wisdom of the crowd for fine-grained recognition. IEEE transactions on pattern analysis and machine intelligence, 38(4), 666-676.

[Dudley 2018] Dudley, et al. "A review of user interface design for interactive machine learning." ACM Transactions on Interactive Intelligent Systems, 8, no. 2 (2018): 8.

[Guo 2014] Guo, Bin, et al. "FlierMeet: a mobile crowdsensing system for cross-space public information reposting, tagging, and sharing." IEEE Transactions on Mobile Computing 14.10 (2014): 2020-2033.

[Guo 2019] Guo, Bin, et al. "From crowdsourcing to crowdmining: using implicit human intelligence for better understanding of crowdsourced data." World Wide Web (2019): 1-25.

[Kamar 2016] Kamar E. Directions in Hybrid Intelligence: Complementing AI Systems with Human Intelligence. In IJCAI 2016, Jul 9, pp. 4070-4073.

[Olah 2018] C. Olah, et al. The Building Blocks of Interpretability. Distill, 2018. doi: 10.23915/distill.00010

[Li 2017] Li G, et al. Crowdsourced data management: Overview and challenges. In Proceedings of the 2017 ACM International Conference on Management of Data, 2017 May 9 (pp. 1711-1716). ACM.

[Liu 2018] Yang Liu et al. Doubly Active Learning: when Active Learning meets Active Crowdsourcing. AAAI 2018.

[Howe 2006] Howe J. The rise of crowdsourcing. Wired magazine. 2006 Jun 6;14(6):1-4.

[Nushi 2015] Nushi, B., et al. (2015, September). Crowd access path optimization: Diversity matters. In Third AAAI Conference on Human Computation and Crowdsourcing.

[Riccardo 2019] Guidotti, Riccardo, et al. "A survey of methods for explaining black box models." ACM computing surveys (CSUR) 51, no. 5 (2019): 93.

[Saleema 2014] Amershi, Saleema, et al.. "Power to the people: The role of humans in interactive machine learning." AI Magazine 35, no. 4 (2014): 105-120.

[Saleema 2011] Amershi, Saleema, et al.. "Effective end-user interaction with machine learning." In Twenty-Fifth AAAI Conference on Artificial Intelligence. 2011.

[Ruan 2019] Ruan, Wenjie, et al. "Global Robustness Evaluation of Deep Neural Networks with Provable Guarantees for the Hamming Distance." IJCAI, 2019.

[Teso 2018] S. Teso and K. Kersting. ”Why Should I Trust Interactive Learners?” Explaining Interactive Queries of Classifiers to Users. arXiv preprint, 5 2018.

[Tong 2014] Tong, Y., et al. (2014, March). Crowdcleaner: Data cleaning for multi-version data on the web via crowdsourcing. In Data Engineering (ICDE), 2014 IEEE 30th International Conference on (pp. 1182-1185). IEEE.

[Tulio 2016] Ribeiro, et al.. "Why should I trust you?: Explaining the predictions of any classifier." In Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, pp. 1135-1144. ACM, 2016.

[Venanzi 2016] Venanzi, et al. (2016). Time-sensitive bayesian information aggregation for crowdsourcing systems. Journal of Artificial Intelligence Research, 56, 517-545.

[Wang 2019] Wang J, et al.. Crowd-Assisted Machine Learning: Current Issues and Future Directions. Computer. 2019 Mar 13;52(1):46-53.

[Xu 2016] Chu, Xu, et al.. "Data cleaning: Overview and emerging challenges." In Proceedings of the 2016 international conference on Management of Data, pp. 2201-2206. ACM, 2016.

[Yosinski 2015] J. Yosinski, et al. Understanding Neural Networks Through Deep Visualization. In ICML Workshop on Deep Learning, 2015.

[Zhang 2018] Zhang C, et al. Reducing uncertainty of schema matching via crowdsourcing with accuracy rates. IEEE Transactions on Knowledge and Data Engineering. 2018.

[Zhang 2018] Zhang QS, Zhu SC. Visual interpretability for deep learning: a survey. Frontiers of Information Technology & Electronic Engineering. 2018 Jan 1;19(1):27-39.

[Zhang 2019] Ruohan Zhang, et al.. Leveraging Human Guidance for Deep Reinforcement Learning Tasks. In Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI), August 2019.

[Zhou 2017] Zhou ZH. A brief introduction to weakly supervised learning. National Science Review. 2017 Aug 25;5(1):44-53.

[Zhuang 2017] Zhuang Y, et al. Hike: A hybrid human-machine method for entity alignment in large-scale knowledge bases. In Proceedings of the 2017 ACM on Conference on Information and Knowledge Management 2017 Nov 6 (pp. 1917-1926). ACM.

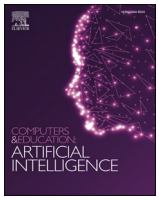
[Zou 2015] Zou, J. Y., et al. (2015). Crowdsourcing feature discovery via adaptively chosen comparisons. Proceedings of the 31st International Conference on Machine Learning, Lille, France

Computers and Education: Artificial Intelligence 3 (2022) 100074

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence



Explainable Artificial Intelligence in education

Hassan Khosravi [a,*], Simon Buckingham Shum [b], Guanliang Chen [c], Cristina Conati [d], Yi-Shan Tsai [c], Judy Kay [e], Simon Knight [b], Roberto Martinez-Maldonado [c], Shazia Sadiq [a], Dragan Gašević [c]

[a] The University of Queensland, Brisbane, Australia
[b] University of Technology Sydney, Australia
[c] Monash University, Australia
[d] The University of British Columbia, Australia
[e] The University of Sydney, Australia

ARTICLE INFO

Keywords:
Explainable AI
AI in Education
Open learner models

ABSTRACT

There are emerging concerns about the Fairness, Accountability, Transparency, and Ethics (FATE) of educational interventions supported by the use of Artificial Intelligence (AI) algorithms. One of the emerging methods for increasing trust in AI systems is to use eXplainable AI (XAI), which promotes the use of methods that produce transparent explanations and reasons for decisions AI systems make. Considering the existing literature on XAI, this paper argues that XAI in education has commonalities with the broader use of AI but also has distinctive needs. Accordingly, we first present a framework, referred to as XAI-ED, that considers six key aspects in relation to explainability for studying, designing and developing educational AI tools. These key aspects focus on the stakeholders, benefits, approaches for presenting explanations, widely used classes of AI models, human-centred designs of the AI interfaces and potential pitfalls of providing explanations within education. We then present four comprehensive case studies that illustrate the application of XAI-ED in four different educational AI tools. The paper concludes by discussing opportunities, challenges and future research needs for the effective incorporation of XAI in education.

1. Introduction

Artificial intelligence (AI) has a large and increasing role in education. One important case is of personalised teaching systems which are already well established, with growing evidence of their effectiveness for improving learning (VanLehn, 2011; Kulik & Fletcher, 2016; Steenbergen-Hu & Cooper, 2014, 2013; Ma, Adesope, Nesbit, & Liu, 2014; du Boulay, 2016). AI in education (AIED) systems may also make diverse and sophisticated use of AI to create the interface that is so important for the learning experience. For example, the interface may use natural language processing and generation, speech interfaces, avatars, video analysis of the learner to judge their attention and emotion (see Fig. 12).

These systems collect data about learners as they use the system. This may be collected from the interaction with the teaching interface as part of learning activities, while other more recent systems collect data beyond keyboard/mouse/screen actions, for example, from cameras, microphones and wearable devices. Learners may be more or less aware of the nature of the data being collected. European legislation in the General Data Protection Regulation (GDPR)[1] has led the world in codifying societal values around the governance of data and its use, reflecting a growing concern that people should be in control of technology and its use of their data (Knijnenburg et al., 2022; Wang, Yang, Abdul, & Lim, 2019; Wang et al., 2019). This means that learners should be able to determine how AI works, how that may affect them, and whether it is trustworthy (Drachsler & Greller, 2016; Holmes et al., 2021). It follows that AIED systems are one important context for the need for eXplainable AI (XAI).

While the role and need for XAI in education shares much in common with broader uses of AI (including accountability for accuracy, fairness and privacy management), education has distinctive needs for XAI and the nature of its data poses distinctive challenges. In particular, learning

* Corresponding author.
E-mail address: h.khosravi@uq.edu.au (H. Khosravi).
[1] https://gdpr-info.eu, visited March 2022.

https://doi.org/10.1016/j.caeai.2022.100074
Received 1 August 2021; Received in revised form 20 March 2022; Accepted 20 April 2022
Available online 13 May 2022
2666-920X/© 2022 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

data has many *sources* of noise, and *reasoning* about such data is noisy at many levels (Kahneman, Sibony, & Sunstein, 2021). A significant body of work emphasises the importance of giving learners agency and responsibility for their own learning (Berger, Rugen, Woodfin, & Education, 2014; Winne, 1995), and in this context, XAI has the potential to assist (Kay, 2001), with some learning tools harnessing data to create interfaces that support the learner's metacognitive processes of self-monitoring, reflection and planning (Bull, 2020; Bull & Kay, 2013). There are also well-known phenomena that are of particular importance for education, such as the ways that a flawed AI system may introduce misconceptions or encourage students to game a system (Baker et al., 2009).

This paper tackles the particular challenges and mechanisms for XAI in education. It makes three main contributions. Firstly, in Section 3, we present the *XAI in Education (XAI-ED)* framework, that draws on the fields of AI, Human-Computer Interaction and the Cognitive and Learning Sciences. XAI-ED characterises the nature of XAI in Education in terms of questions about six key aspects: *the people involved (stakeholders) and the benefits to each group; how to deliver the explanation; the widely used classes of models used in education; the human-centred design of the AI and interfaces to* support *explanation; and the potential pitfalls of providing explanations.* The second contribution is a set of four case studies, presented in Section 4. These illustrate the importance of explainability in state-of-the art examples with reference to XAI-ED. The third contribution in Section 5 articulates a list of opportunities, challenges and future research needs for advancing XAI and its effective incorporation in education.

2. Background

This section provides an overview of the XAI literature and its association with education. Section 2.1 begins by considering how the AIED field is responding to the broad challenges of FATE. Section 2.2 reviews the current landscape of XAI. Section 2.3 provides an overview of the importance of explanations in human-human interactions in the context of education. Section 2.4 provides an overview of the work on open learner models (OLMs), which is arguably the most well-established application of explanations in human-machine interactions and XAI in education.

2.1. The FATE of AI in education

A range of analyses are now appearing in the literature, which provide conceptual frameworks for understanding the challenge of FATE for AIED. In a commentary on the field, Holmes et al. (2021) surveyed AIED researchers on their perceptions of the challenge of FATE, from which they distill a set of thematic challenges for the community. In more technical detail, Kizilcec and Lee (2022) present an incisive introduction to, and analysis of, the concept of algorithmic "fairness" in education, using the three key steps of "measurement (data input), model learning (algorithm), and action (presentation or use of output)" to illustrate where this can break down. Focusing specifically on machine learning-based AIED, Baker and Hawn (2021) develop a more detailed taxonomy that summarises the various sources of algorithmic bias, and how they can be mitigated in terms of "a framework for moving from unknown bias to known bias and from fairness to equity". We also refer the reader to the recently edited collection by Porayska-Pomsta, Woolf, Holmes, and Holstein (2021), and the forthcoming volume by Holmes & Porayska-Pomsta, (In Press).

To illustrate the kinds of biases that have been studied, we introduce a few examples. It is known that when modelling students' learning performance and identifying at-risk students, AI techniques may provide more accurate predictions for a group of students than the others simply based on the different demographic attributes of the students, e.g., female vs. male (Gardner, Brooks, & Baker, 2019). The authors propose that the differential prediction accuracy between different groups of students can be quantified by using a metric called Absolute Between-ROC Area (ABROCA) to measure the unfairness of a predictive model. With ABROCA, Gardner et al. (2019) evaluated the unfairness displayed by five AI models used to predict to what extent students in massive open online courses were likely to accomplish their studies. Similarly, Hutt, Gardner, Duckworth, and D'Mello (2019) assessed the unfairness of models predicting on-time graduation based on the data provided by students in the setting of college applications. A very different form of analysis by Sha et al. (2021) reported that AI models may be unfair to students when characterizing their posts in discussion forums. For instance, the posts by English-*as*-first-language students were more likely to be accurately classified than those made by non-native English speakers.

To summarise, FATE covers an extremely broad range of concerns, to which the AIED field is developing a range of responses. A powerful form of mitigation that is often referred in FATE literature is *explainability*, and it is to this intriguing concept that we now turn our attention.

2.2. Explainable AI

One of the emerging methods for increasing trust in AI systems is to use XAI, which promotes the use of methods that "enable human users to understand, appropriately trust, and effectively manage the emerging generation of artificially intelligent partners" (Gunning, 2017). The initial focus of XAI was predominantly algorithm-centred. At a high level, machine learning models can be categorised based on their level of interpretability, which can be defined as the degree to which a human can understand the cause of a decision or be able to exactly reproduce what the model does (Miller, 2019). As a design criterion, having models with high interpretability is desirable since in principle, this can help mitigate against partiality in decision-making (detecting and correcting various forms of bias), increase robustness against adversarial perturbations (noise added to the input to fool the system while being quasi-imperceptible for humans (Moosavi-Dezfooli, Fawzi, Fawzi, & Frossard, 2017)) that could change the prediction, and improve the employment of meaningful variables and truthful causality in the model reasoning (Arrieta et al., 2020). Some models such as general additive models, rule-based models and decision trees, due to their relatively simple structure, are considered interpretable by design, making it easy to explain how these models work. For example in the case of a simple decision tree, the rules can be explained to a human such that they understand and have the ability to reproduce the decision made by the model. In contrast, some models such as tree ensembles, support vector machines and deep neural networks have complex structures that are not readily interpretable. To make these models explainable to humans, there has been extensive work in the XAI research community on developing post-hoc explainability techniques, which aim to explain how a model produces its predictions without elucidating the structure of the model (Lipton, 2018) (See Section 3.2 for some examples).

Recent advances in XAI have seen a shift and a developmental step towards socially-situated XAI by introducing and exploring a socio-technically informed perspective that incorporates the socio-organizational context into explaining AI-mediated decision-making (Ehsan, Liao, Muller, Riedl, & Weisz, 2021). This shift has leveraged insights from the social sciences where an explanation is seen not just as a product, but as a process that requires social interactions and a knowledge transfer process from an explainer to an explained (Miller, 2019). Srinivasan and Chander (2020) offer a helpful review of types of explanations from a cognitive science perspective, including *trust, troubleshooting or design, education, action, justification, aesthetics*, and *communication* as key aims in explanation. They note that in deciding what kind of explanation to give, the specific stakeholder and task should be considered, highlighting the need for human-centred approaches to generating explanations. The shift has therefore benefited from user-centred approaches and methodologies from the HCI community, which have demonstrated that a range of XAI techniques were

not as effective as assumed in *assisting in sensemaking* (Alqaraawi, Schuessler, Weiβ, Costanza, & Berthouze, 2020), *enhancing user trust* (Yang, Huang, Scholtz, & Arendt, 2020), or *enabling actionable decisions* (Liao, Gruen, & Miller, 2020). These views suggest that XAI in education needs to be sensitive to the learning context and consider the different actors in a learning community as potential audiences.

2.3. Explanations in education

Explanations in educational contexts come in diverse forms depending on the stakeholders and their particular aims or tasks. The need for explanations arises since educators must be accountable (e.g. to students, parents, or the government); when providing individual feedback to students; in giving teachers diagnostic feedback to understand where a class of students needs increased focus; and in parental consultations to help them support their child's learning.

In particular, feedback for students and teachers are a form of explanation in education that is key to educational outcomes. For students, the most common forms of explanations are provided by teachers regarding student performance on particular tasks, suggestions to improve, prompts for self-monitoring and directing, and affect-level comments (Hattie & Timperley, 2007). The main purposes of this type of feedback are to scaffold learning including the development of domain knowledge, self-regulated skills, and a sense of being. Moreover, feedback for students is seen as a relational process through which teachers may encourage positive motivation and help learners build confidence and self-esteem (Price, Handley, Millar, & O'donovan, 2010; Nicol & Macfarlane-Dick, 2006).

For teachers, feedback tends to serve the purpose of assessing the effectiveness of teaching approaches, learning design, and areas of support (particular) students may need. Common materials that may aid teaching reflections include grade distribution of students, class engagement (e.g., attendance), student surveys, parent-teacher communications, and peer evaluation. Feedback for teachers is crucial to the scholarship of teaching (Boyer, 1990), which positions teachers as learners who, through reflections on the feedback, construct instructional knowledge (e.g., instructional design), pedagogical knowledge (e.g., knowing how students learn), and curricular knowledge (e.g., rationales of curriculum design) (Kreber, 2005) that together allow teachers to perform effective teaching.

Another form of explanation that is important to the operation of an educational entity, and its ongoing improvement, involves the use of data to present an overall profile and performance of an institution, e.g., student enrolment data, academic performance (e.g., teaching & research quality), staff profile, performance, and retention, research outputs, institution income, teacher-student ratio, and others. Business intelligence has been employed broadly to manage and gain insights from this form of explanations, which particularly target administrators and managers as key audiences (Drake & Walz, 2018). Across these contexts, the aim is to support people in decision making, and to develop their judgement capacities.

2.4. Open and scrutable learner models

A defining feature of AIED has been the personalisation of teaching (Self et al., 1999). A driving goal for foundational AIED was to create intelligent tutoring systems that could achieve the huge learning benefits of an expert one-to-one human tutor (Bloom, 1984). There is still huge appeal in creating personalised teaching software that can do this at scale.

Such personalisation is driven by a *learner model*. This is often defined as the machine's set of beliefs about the learner's knowledge, goals, preferences and other attributes. Some learner models are detailed cognitive models. Notable cases are the cognitive tutors (Anderson, Corbett, Koedinger, & Pelletier, 1995) which were some of the first widely deployed intelligent tutoring systems (Koedinger, Anderson, Hadley, Mark et al., 1997) and constraint based tutors (Mitrovic, 2003) which were also rigorously evaluated and widely deployed (Mitrovic, 2012). Both these have quite complex learner models and reasoning about the learner. Many other personalised teaching systems have used much simpler, more ad-hoc learner models where the system designer determines the set of Knowledge Components (KCs) (Koedinger, Corbett, & Perfetti, 2012) to model. Commonly, KCs form an ontology, defined by the relationships between them. For example, a genetic graph relates KCs in terms of the generalisation, correction, and refinement that learners make as they progress (Goldstein, 1979). Other ontologies have relationships for prerequisites and hierarchical structure. The learner model also typically represents the value of each KC as a level, such as Bloom (Bloom et al., 1956), SOLO (Biggs & Collis, 2014) or a small integer value. As the learner interacts with a personalised teaching system, the learning data drives changes in the learner model and this, in turn, drives personalised learning.

In many AIED systems, aspects of the learner model are available to the learner via interfaces. Such Open Learner Models (OLMs) typically present a view of key KCs so that the student can monitor their learning progress. OLMs have been designed for multiple purposes (Bull & Kay, 2016). From an XAI perspective, some of these purposes are particularly important for education. The most important is to achieve improved learning outcomes, as demonstrated in work such as (Brusilovsky, Somyürek, Guerra, Hosseini, & Zadorozhny, 2015; Long & Aleven, 2017; Mitrovic & Martin, 2007) and many other examples reviewed in (Bull, 2020). A second valuable role for this form of XAI, is as a foundation for giving learners greater control and responsibility over their learning (Abdi, Khosravi, Sadiq and Gasevic, 2019, 2020).

One class of open learner models and personalised systems has been described as *scrutable* (Kay, 2006). This are designed so that a learner can delve into the way they work by answering questions such as: What "raw data" is kept about me? Where does that data come from? How do I control this inflow? How do I volunteer data about a KC? Similarly, the learner can scrutinise the learner model to answer questions like: What is modelled about me? How does the modelling process work? A more complete set is available in (Kay & Kummerfeld, 2019). There are two main reasons to describe this form of XAI as scrutability. Firstly, scrutability acknowledges the real effort needed to *scrutinise* a system to find the answers to such questions. Secondly, in sophisticated systems, even after scrutinising, the learner may gain understanding of just some aspects but this may fall short of the breadth of understanding implied by terms like transparency and explainability.

The unfairness issues witnessed in AI models, beyond question, greatly slow down the pace of wide adoption of AI in education, and thus calls for more work on XAI in education. In addition to unfairness, it has been reported that instructors often lack enough understanding of AI and so are concerned about it use (Chounta, Bardone, Raudsep, & Pedaste, 2021). This, again, calls for the development of relevant XAI in education presentation methods to communicate the prediction results with instructors and enhance their perception of AI and support them to tackle the challenges they face while using AI for teaching.

3. The XAI-ED framework

This section contextualises the use of XAI in education.

As previously discussed, the role and need for XAI in education has much in common with its broader use. Therefore, as shown in Fig. 1, XAI in education draws insights and best practices from the fields of AI, Human-Computer Interaction and the interdisciplinary and emerging field of Human-Centred AI where much of the initial research on XAI has been conducted. However, there are also distinctive needs for XAI in education that are grounded in theories from cognitive and learning sciences. Some of these needs and their applications have been under exploration in the interdisciplinary and emerging fields of AI in education and learning analytics.

Drawing insights from these related fields, this section characterises

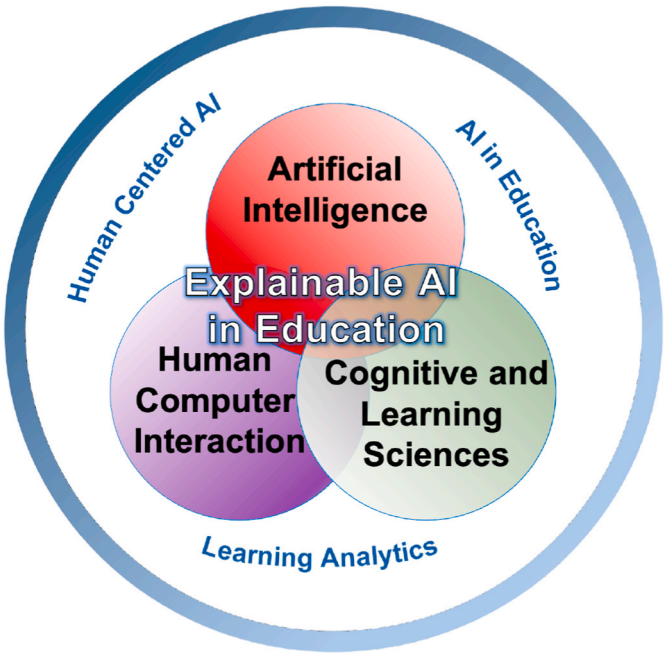


Fig. 1. Related fields to explainable AI in Education.

the nature of XAI in education in the XAI-ED framework that responds to fundamental questions about six key aspects. Section 3.1 presents an overview of the main stakeholders in education and how they may benefit from XAI in their roles and responsibilities. Section 3.2 presents some common approaches for presenting explanations in education. Section 3.3 presents some representative and widely used classes of models for implementing XAI in education. Section 3.4 discusses points for consideration in a human-centred design of the AI and interfaces to support explanation. Finally, Section 3.5 presents common pitfalls and potential shortcomings of incorporating XAI in education and recommendations to avoid them. Fig. 2 visualises the framework and its dimensions.

The XAI-ED aims to provide the means for educational tool developers and researchers to consider these six dimensions and "fill" in the parts of the framework for their particular context, which we hope can contribute to the development of more effective educational XAI systems.

3.1. Stakeholders and potential benefits

Much of the research in XAI has been devoted to providing explanations to system engineers or data scientists; however human-in-the-loop factors are crucial for the enactment of explanations in any given context (Srinivasan & Chander, 2020). The particular people seeking explanations, their intentions, and the explanations are important to the development of understanding of and trust in AI and the decisions AI feeds into (Páez, 2019).

Educational stakeholders include technologists and researchers, including learners, parents, teachers, educational administrators and policy makers, with each diverse and often different needs. All want educational benefits of AI that must also be accountable and trusted. Below, we explore some of the potential benefits of XAI for stakeholders in education.

Agency. Explanations of AI can facilitate conversation among these different stakeholders, further enabling processes of co-design and value co-creation (Dollinger, Liu, Arthars, & Lodge, 2019; Dollinger & Lodge, 2018). In particular, explanations of AI (including the choice of language and details to include) should allow students, teachers, and parents to see personal relevance (Srinivasan & Chander, 2020), thus empowering them to make conscious decisions as to whether or not to adopt AI and how to use AI in ways that may optimise values. Having functional understanding (Páez, 2019) – being able to interpret a decision made by the AI and the relation between an input and an output – will increase the confidence of these stakeholders in AI, and allow them to exert agency through AI (Selwyn, 2021), such as reasoning the credibility of a recommendation by AI and deciding whether or not to act on it.

Student-teacher interactions. On the other hand, XAI is important to facilitate the socio-cultural process of learning where interactions between teachers and students are fundamental to guide learners through zones of proximal development (Rieber & Carton, 1987). Although AI can significantly improve efficiency by automating some mundane work of teaching, the social role of teachers in a learning process remains unique and not replaceable by AI (Miao, Holmes, Huang, & Hui, 2021; Selwyn, 2019). For example, dialogic pedagogies (Lyle, 2008), Moore's theory of transactional distance (Moore, 2013), and the community of inquiry model (Garrison, 2016) highlight the importance of student-teacher interactions in building a learning community where learners can feel supported and belonging. Explanations of AI can prompt dialogue between learners and teachers on insights and implications of AI decisions and possibilities of such decisions to be challenged and improved, which can also feed back to the AI system for further improvement (Moore & Paris, 1992).

AI literacy. While AI-based innovations, such as learning analytics,

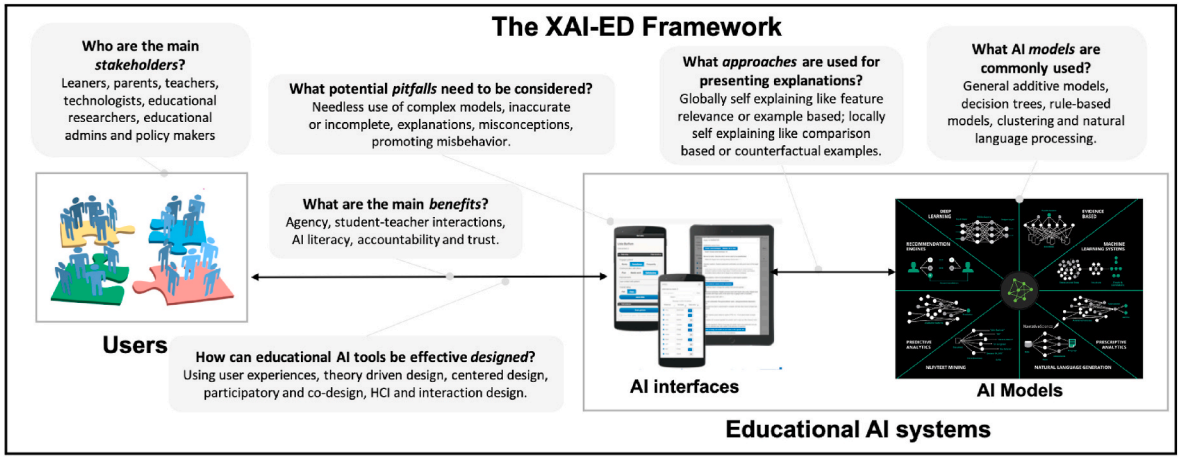


Fig. 2. The XAI-ED framework.

open up new ways to understand and support learning, the huge influence (often seen as disruptive) of AI on how people think and act calls for a new set of skills important to our navigation in a world of AI - AI literacy. Broadly speaking, AI literacy includes an understanding of what AI is, the ability to learn with AI, and the skill to collaborate with AI in a world that is increasingly integrated with AI (Miao et al., 2021). Long and Magerko (2020) defines AI literacy as a set of 17 competencies that enable "individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace" (p. 598). They propose 14 design considerations to support AI developers and educators in creating learner-centred AI. Among these, explainability is the first to consider when designing an AI system, e.g., through the use of graphic visualisations, simulations, explanations of agent decision-making process, and iterative demonstrations to aid the learner's understanding of AI. Explainability of AI is also important for the development of AI literacy among teachers who play an orchestrator role in deciding when and how to best use AI tools to support learning in the classroom (Miao et al., 2021).

Accountability and trust. Motivations behind AI adoption in education are not purely educational but also socio-political. Higher education, in particular, faces constant pressure to provide evidence of education quality not only for quality assurance, but also to compete locally and internationally to attract students, staff, and funding. As a result, AI is not only perceived as promising in helping students develop essential skills as described above, but also to enhance institutional performance in areas such as teaching quality, student progression and retention, student satisfaction, graduate employment, and international reputation. This political agenda has attracted various parties outside the education domain (e.g., think tanks and EdTech industry) to offer solutions and shape education futures, resulting in outcries about education being manipulated by non-education experts (Williamson, 2018). In addition to this, several socio-political issues related to the way data is sourced, collected, processed, and transformed into seemly arbitrary decisions have caused fear and distrust in AI. Prominent issues include surveillance, privacy, bias, and context relevance (e.g., educational constructs) among others. Learning analytics, for example, were found to suffer from trust issues with reasons such as numbers being subjective, threat to agency and autonomy, and the design and implementation of learning analytics failing to meet needs or negatively affecting student well-being (Tsai, Whitelock-Wainwright, & Gašević, 2021). In light of the potential danger of AI, UNESCO published Beijing Consensus (UNESCO, 2019) outlining 44 recommendations to harness AI in education.

In summary, XAI can be a catalyst for several educational benefits and desirable futures of education. It is important to consider whom the explanations are for, what the purposes are, and how to effectively communicate the explanations to different stakeholders who need varying levels of understanding (Páez, 2019) to benefit directly or help others benefit from AI. We posit that XAI can play a crucial role in enabling collective efforts of various stakeholders to shape desirable education futures through responsible, fair, and effective use of AI. In particular, XAI needs to communicate algorithmic transparency effectively to educational administrators (e.g., institutional leaders and legal officers) who may ensure model compliance and downstream implications, to policy makers who are responsible for the governance and continuous monitoring of ethical, legal, and desirable use of AI, and to teachers who can provide feedback to enhance AI systems based on their domain expertise. In this way, XAI can help keeping educational entities and service providers accountable and address some trust issues around the use of AI, thus realising and scaling the potential benefits.

3.2. Approaches

Here we present some of the common approaches to explainability in AI that can help various stakeholders gain insights into the details of machine learning models. These approaches often rely on simplification techniques to generate proxies with reduced complexity. Fig. 3 presents an overview of these approaches.

Approaches to explainability in AI are generally classified according to two main criteria: (1) global approaches that explain the entire model (approaches a, b, and c) vs. local approaches that explain an individual prediction (approaches d, e, and f) and (2) self-explainable models that have a single structure (a and d) vs. post-hoc approaches which explain how a model produces its predictions without elucidating the structure of the model (b, c, e, f) (Arrieta et al., 2020). and chapters 5 and 6 of (Molnar, 2019) provide a more detailed view of these common XAI approaches.

Global explanations. This approach focuses on explaining how different features/variables affect predictions. Global explanations are generally model-specific and work best for models that are interpretable by design. The example given in Fig. 3-a illustrates a decision tree that can be used to predict the risk level of every student in a class. Many teacher-facing systems (e.g. (Lakkaraju et al., 2015)) use global explanations to inform instructors of the performance of their students. For models that are less interpretable by design, one approach is to implement a global surrogate by learning an interpretable model to approximate the predictions of a black box model. However, this method needs to be used with care as the surrogate model may provide an accurate representation for one subset of the dataset, but diverge widely for another subset.

Feature relevance. One common XAI approach presents the computed relevance, in terms of a score, for each feature in the prediction process (as shown in Fig. 3-b. A comparison of the scores among different features demonstrates the importance granted by the model to each of the features when producing its output. The scores can be computed using a variety of models such as linear or logistic regression or more recently Shapley values (Shapley, 2016) that take a coalitional game theory approach to fairly distribute the "payout" among the features. The relevance of each feature can be shared by simply reporting the scores or visualising them in various graphs. This method has been used within predictive learning analytics applications (e.g. (Lakkaraju et al., 2015)) to present the importance of each feature in determining the "at-riskness" of the learners.

Example-based. Example-based explainable approaches select particular instances to explain the model. A main advantage of these explanations is that they are generally model-agnostic and can be used to make any machine learning model more interpretable. They can provide both local explanations (described below) and global explanations. Fig. 3-c illustrates this approach for identifying boundaries of a risk model based on students' midterm grades at a global level. This is done by visualising the entire set of instances, highlighting the examples and articulating them.

Local explanations. This approach focuses on explaining a particular instance, independent of what might be happening at the model level. Local explanations are only generally model-agnostic; however, they work particularly well with rule-based models and decision trees. The example in Fig. 3-d illustrates how risk factor for an individual student is computed using a interpretable model. Many student-facing systems and open learner models (e.g., Abdi et al. (2020)) use local explanations to inform students of their progress without overwhelming the students about the entire model and system. For models that are less interpretable by design, one approach is to implement a series of local surrogate models by learning an interpretable model locally around the prediction (Ribeiro, Singh, & Guestrin, 2016).

Comparison. This approach selects particular instances to locally explain the outcome of other instances. It is particularly well suited to non-parametric models, such as K nearest neighbours, where the model structure is determined by the instances. The example in Fig. 3-e compares essays of two students with high similarity for academic integrity purposes. Many plagiarism detection systems use comparison to show the evidence for their prediction (Foltýnek, Meuschke and Gipp, 2019).

Counterfactual explanation: This approach generally describes a

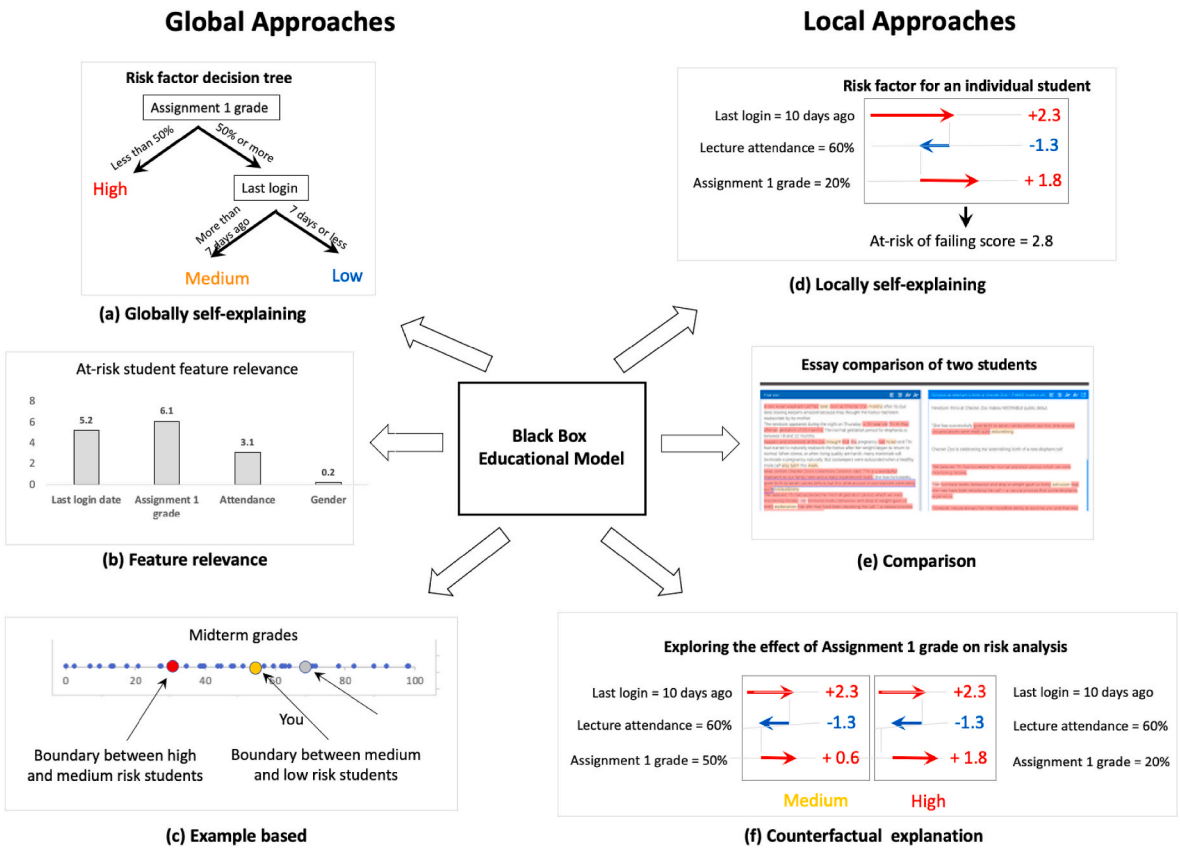


Fig. 3. Common explainability approaches.

causal situation in the form: "If X had not occurred, Y would not have occurred". Counterfactual explanations can be used to explain and demonstrate the smallest change to the feature values that changes the prediction to a predefined output. The example given in Fig. 3-f demonstrates the case of two students with similar values across features with the exception of their Assignment 1 grade which has led to one being categorised at medium risk while the other is categorised as high risk. The difference in outcome explains and provides evidence that the Assignment 1 grade plays an important part in the student risk modelling. There are two main challenges associated with counterfactual explanation (Molnar, 2019): (1) Identification of instances that provide counterfactual explanations can sometimes be a challenging task and (2) there may be multiple counterfactual explanations where each tells a different "story" of the importance of other features which may come across as contradictory or confusing.

3.3. Models

In this subsection, we present some representative AI models that are commonly used education. Some of the most promising use cases of AI in education include adaptive learning systems that tailor instruction based on student needs (Aleven, McLaughlin, Glenn, & Koedinger, 2016), adaptive testing systems that tailor assessment items based on students' mastery (Mills, Potenza, Fremer, & Ward, 2005), learner models and in particular open learner models that allow students to better regulate their learning (Bull, 2020), automated feedback tools that provide instant feedback on students' academic writing tasks (Knight, Abel, et al., 2020; Roscoe & McNamara, 2013) as well as intelligent learning analytics dashboards that help teachers with sense-making and identifying students in need of attention (Khosravi, Shabaninejad, et al., 2021). It should be noted that the majority of the examples we identified in the literature incorporated interpretable models for explainability rather than using post-hoc explanations alongside non-interpretable models.

Generalized Additive Model (GAM) assumes a linear relationship between the dependent variable and independent variables, and attempts to predict the value of the dependant variable by aggregating a number of smooth functions which take the independent variables as input. Given the underlying linear nature of GAM, we can easily verify the importance of an independent variable by evaluating how its corresponding smooth function affects the predicted value. For example, Dikaya, Avanesian, Dikiy, Kirik, and Egorova (2021) aimed to depict the relationship between the psychological traits of students and their attitudes toward remote learning during the COVID-19 pandemic. By using GAM, it was illustrated that the interpersonal communicative skills of students (e.g., being manipulative or shy) correlated highly with their attitudes towards remote learning. Many of the well-established interpretable learner models such as Adaptive Factor Models (AFM) (Cen, Koedinger, & Junker, 2006) and Performance Factor Analysis (PFA) (Pavlik Jr, Cen and Koedinger, 2009) fall into this category. It is worth noting that there is a of class learner models that use deep learning such as deep knowledge tracing (DKT) (Piech et al., 2015) which are not interpretable.

Decision Trees are one of the most widely-used machine learning techniques in the field of learning analytics and educational data mining. Different from GAM, decision trees aim to learn a set of decision rules, which are organized in a hierarchical tree-like manner, to determine the value of the dependant variable. For instance, when predicting students' competence in collaborative problem solving, Cukurova, Zhou, Spikol, and Landolfi (2020) and Pardo et al. (2016) built decision trees to provide a set of useful decision rules to capture factors that were essential to students' performance. Due to their hierarchical structure, decision trees have been widely regarded as a useful technique to enable XAIED. However, if the model has a large number of complex features,

the size of decision tree can be very large and incomprehensible to a human user.

Rule-based learning is similar to decision trees and also aims at using a set of decision rules to generate predictions (Liu, Gegov, & Cocea, 2015). However, the rules are not necessarily structured in a hierarchical tree-like manner. The rules can take the form of simple conditional statements (e.g., *"if X then Y"*) plus more complex combinations. For instance, Engin et al. (2014) applied rule-based learning to develop two educational expert systems to recommend courses and scholarships to students. The prediction accuracy of rule-based learning often depends on the coverage (or amount) and the specificity (length) of the adopted rules. Rule-based systems can explain their entire chain of reasoning, which can be used to generate explanations that an analyst can use to diagnose and debug, and which in suitably simplified form can be understood by non-technical stakeholders. However, the more rules a model uses, the less understandable it becomes to regular educational stakeholders; similarly, the specificity of the rules also tends to hinder its users from interpreting how it works.

Clustering methods such as K-means clustering (Krishna & Murty, 1999) group data points into multiple clusters by calculating their similarities with each other, which is often measured based on the distance between these data points. These clusters can be used to reveal and explain different data patterns, e.g., exploratory learning behaviours specific to different groups of students. For instance, the FUMA framework presented in Section 4.2 and Conati, Barral, Putnam, and Rieger (2021) incorporate clustering to generate interpretable and personalised hints that guide student learning.

Natural Language Processing has commonly been used for detecting potential plagiarism and identifying misconduct cases. For instance, Turnitin (Heckler, Rice, & Hobson Bryan, 2013) is a well-recognized tool used for plagiarism detection. It works by comparing one student submission against an archive of relevant documents (e.g., internet articles and academic publications) and produces a report indicating where the text within the submission matches another source. This can be viewed by instructors to help determine whether it is a plagiarism case or not. It can also be presented to a student to improve the submission. Batane (2010) demonstrated that, with the aid of Turnitin, fewer plagiarism cases occurred in an undergraduate class.

3.4. XAI designs

Providing appropriate information to help people understand AI can be considered a human-AI interaction *design* challenge (Liao et al., 2020). Simply opening the algorithmic 'black box' is not enough to understand the implications of AI in the sociotechnical system at large. Some of the most accurate AI models (e.g. deep learning algorithms) are very complex and hard even for data scientists to understand (Sejnowski, 2020). It is also very challenging for people without formal data analysis training to comprehend and trust even simple, rule-based algorithms. This is precisely the case for most end-users of data-intensive educational innovations (i.e., students and their teachers). In this subsection, we present some design approaches and areas of research and development that can be considered when creating AIED systems.

User experience (UX) is the process design teams follow to create products and systems that provide meaningful and relevant experiences to *end-users* (Hassenzahl & Tractinsky, 2006). User experience itself is subjective as it includes people's perceptions of utility, accessibility, efficiency and ease of use about a system. Yet, the design attributes that contribute to such user experience are objective. This means that researchers and designers can maximise the opportunities for people to have an effective experience with a system. UX design is key for the development of effective XAI systems. In fact, a recent study demonstrated that effectively designing causability and explanatory cues in AI systems help people understand the decision-making process of algorithms by bringing transparency and accountability into such systems (Shin, 2021). However, designers of AI innovations are facing unique challenges that are not present when designing in other contexts. For example, UX designers need to collaborate effectively with AI developers to create a joint vision of the ideal impact of the models, their evaluation and the technical feasibility in addressing end-user needs (Yang, Scuito, Zimmerman, Forlizzi, & Steinfeld, 2018). The current reality is that UX design is not always a key component in the development of AI algorithms. To address this, UX designers need to be included in the development cycle of the algorithms to understand the capabilities of such algorithms and improve the explainability of the system to end-users (Dove, Halskov, Forlizzi, & Zimmerman, 2017). This suggests specific AI literacy capabilities that need to be developed along with design core knowledge and practices.

User-centred design (UCD). It is starting to be acknowledged that a prerequisite to create effective AI and learning analytics innovations in education is to understand the authentic needs of students, educators and other educational stakeholders while designing AI innovations (Buckingham Shum, Ferguson, & Martinez-Maldonado, 2019), which is now establishing itself as an active stream of learning analytics research. Yet, it is critical to understand to what extent the voices of these key educational stakeholders can influence the design of XAI. From a classic bottom-up, UCD perspective, the target user (e.g. teachers or students) is an object of study (Sanders & Stappers, 2008). The researcher or designer observes or interviews the educational stakeholder, often bringing knowledge from theories. Then the person in charge of the design receives this knowledge, adds an understanding of the technology capabilities, and creates a product. In the context of XAIED systems, there is no guarantee that the target users will be able to understand the XAI explanations. According to UCD, a design needs to provide a comprehensible explanation about the AI algorithm or its output, based on target users' needs and capabilities (Xu, 2019). Thus, the same XAI version targeted at researchers or developers cannot be the same as that targeted at non-expert users.

Top-down, theory driven design. Top-down guidelines have been proposed to help designers choose relevant XAI techniques that may work effectively in a certain domain. For example, Arrieta et al. (2020) proposed taxonomies of explainability techniques related to specific machine learning algorithms which can be used by other researchers working with such algorithms. This way, there is no need to deeply engage with the user values in making design decisions about XAI. Educational theory can provide the necessary foundations to inform the design of AIED and learning analytics research and the interpretation of its results (Er et al., 2021, pp. 196–206). It may therefore be possible to align XAI design guidelines with well-established educational theories to effectively design XAI techniques that support explanation and reasoning. For example, Wang et al. (2019) drew on educational psychology theory to map from conceptualisations about how people reason and explain to the ways to design XAI to support such processes.

Participatory design and co-design are aimed at giving an active voice to the end-users. In contrast to seeing end-users as passive objects of study, they can have an active role in the design process and they are given the position of 'expert of their own experience' (Sanders & Stappers, 2008). In the development of AIED and learning analytics systems, there has been a growing call for considering educational stakeholders as equal partners instead of simply conducting consultation or evaluation sessions with them (Prieto-Alvarez, Martinez-Maldonado, & Anderson, 2018). Yet, this bring additional challenges in terms of how users without data analysis training can effectively communicate needs that can be addressed through XAI innovations. A way to address this is by identifying the types of explanations needed by different stakeholders and, then involving such stakeholders in the design process of the XAI itself. For example, Liao, Pribić, Han, Miller, and Sow (2021) proposed a question-driven process for XAI UX design to engage end-users in the formulation of modelling solutions. Toolkits to conduct design studies with users to identify XAI needs and capabilities are also emerging for XAI practitioners (e.g., https://www.uxai.design/).

HCI and Interaction Design (IxD). While XAI research has so far been dominated by AI and machine learning experts, a rapidly growing community of HCI and IxD researchers/professionals is forming to contribute to the design of the XAI interfaces and bring psychological theories of explanations to XAI research (Xu, 2019). HCI and IxD professionals can take advantage of their interdisciplinary approaches to help AI experts to create effective UIs, contribute to design algorithms that enable explainability and involve users in the design process, as highlighted above. Inroads are being made through human-computer interaction (HCI) studies and interaction design (IxD) experiences that are trying to bridge the AI and human perspectives (e.g. Gunning, 2017). AI innovations in education require a balance between learning theory, data science and design (Gašević, Kovanović, & Joksimović, 2017). We need to consider what is possible with current XAI methods to be used throughout the design process. At the same time, there is value in identifying the educational stakeholders of the sociotechnical system from early stages in the design process. Only then, would it be possible to give an active voice to learners, educators and other potential end-users, and identify what to explain and how to explain it.

3.5. Pitfalls and how to avoid them

While there are many benefits in increasing the explainability of educational AI systems, there are also challenges and pitfalls that need to be considered. This section describes some of the key pitfalls as well as pointing out potential solutions and principles for avoiding them.

Needless use of complex models. A common misjudgment is to use over-complex models in cases where the use of an interpretable model would have delivered a comparable or even superior performance. As an example, in the context of student modelling and knowledge tracing, Gervet, Koedinger, Schneider, Mitchell et al. (2020) report that logistic regression outperforms deep learning models on datasets of moderate size or containing a very large number of interactions per student. Therefore, designers should start with simple interpretable models and increase complexity in a step-wise manner where performance in both accuracy and interpretability are measured and compared (Molnar et al., 2020). Nevertheless, in some cases more complex models would significantly outperform interpretable models. To follow the same example, Gervet et al. (2020) report that the use of deep knowledge tracing approaches outperform more interpretable approaches on datasets of large size or where precise temporal information matters most. In these cases, one approach would be to complement the complex model with interpretable models for explainability. As an example, Ghosh, Heffernan, and Lan (2020) propose a learner model which couples complex attention-based neural network models with a series of novel, interpretable model components inspired by cognitive and psychometric models for explainability.

Inaccurate explanations. Explanations that are inaccurate or even incorrect may lead to unfavourable outcomes for the stakeholders. A common underlying cause of inaccurate explanations is the use of models that are poor, potentially due to under- or over-fitting or noisy data. For example, a lenient open learner model that overestimates the mastery level of students may encourage them to develop a false sense of confidence of their abilities, resulting in poor performance in summative assessments and exams. Given that an interpretation can only be as good as its underlying model, it is crucial to develop the model rigorously by conducting best practices for model selection, hyperperameter tuning and evaluation before the adoption of AI tools (Molnar et al., 2020). Additionally, with the increasing demand for model explainability, some AI tool designers may be tempted to provide plausible and convincing interpretations that may be inaccurate or even incorrect. One way to combat this issue is to only incorporate sound explanations based on the underlying mathematical foundations which are truthful in describing the underlying system (Kulesza, Burnett, Wong, & Stumpf, 2015).

Incomplete explanations. Given the complexity of many AI models, some system developers may be tempted to provide incomplete and simplified explanations that disguise the entire complexity of the model. While the incorporation of incomplete explanations may appeal to a broad base of users, it provides them with a false sense of comprehension which may again lead to unfavourable outcomes. One way to combat this issue is to only incorporate complete explanations that expose all aspects of the relevant mechanisms (Kulesza et al., 2015). An alternative method would be to clearly flag their incompleteness, and the rationale, to warn the user.

Misconceptions. Given the complexity of many AI systems, it is reasonable to assume that some users may misinterpret or not fully understand sound and complete explanations of the system. Kulesza et al. (2015) propose the use of iterations for providing sound and complete explanations without overwhelming the users. They suggest that "explanations could take the form of concise, easily consumable "bites" of information—if a user is interested in learning more about the system, he or she can attend to many of these explanations to iteratively build a higher-fidelity mental model" (p. 127). Iterative cycle of explanations can focus on breadth (i.e., various aspects of the model) or depth (i.e., drilling down into the details of the model).

Promoting dysfunctional behaviour. The challenge in many of the previously mentioned pitfalls was due to users developing a false sense understandably. However, another line of concern is that if users have access to information about how a decision or recommendation has been made, they may be able to game the system by altering their behaviour to gain a more favourable outcome. For example, by having access to multiple rounds of submitting assignments to plagiarism detection software, students may use a strategy of making minimal changes to "bypass" the system rather than developing the disposition not to engage in academic misconduct. In this case, the issue is with how the system is implemented rather than its AI functionality or interpretability. Taking best practices from game theory mechanism design may help in developing systems in which users are incentivised to avoid malicious gaming and other dysfunctional behaviours (Maskin, 2008).

4. Case studies

In this section, we present four comprehensive case studies that illustrate the application of XAI-ED for studying a diverse range of AIED systems. Section 4.1 discusses the use of explanations within an adaptive learning system powered by high-quality learnersourced content. Section 4.2 discusses the effect of adding explanation functionality to an adaptive system that helps students learn an algorithm for constraint satisfaction problems. Section 4.3 discusses the use of explanations in the context of feedback within a writing analytics tool. Finally, Section 4.4 discusses the case of using explanations for data storytelling through team work analytics in educational healthcare systems. Table 1 demonstrates the relation between each of the case studies and the dimensions of XAI-ED.

4.1. RiPPLE: A learnersourced adaptive educational system

4.1.1. Overview

Adaptive educational systems (AESs) make use of data about students and learning processes to adapt the level or type of instruction for each student (Aleven et al., 2016). To provide such adaptivity, AESs require access to a large pool of instructional material and learning resources. These resources are commonly created by domain experts, which makes AESs expensive to develop and challenging to scale (Aleven et al., 2016). RiPPLE (Khosravi, Kitto, & Joseph, 2019) is an AES that takes the crowdsourcing approach of partnering with students, also referred to as learnersourcing (Khosravi, Demartini, Sadiq, & Gasevic, 2021; Kim, 2015), to create the resource repository.

4.1.2. Stakeholders and benefits

RiPPLE is predominantly a course-level platform that instructors can incorporate in their teaching. Therefore, the platform should first and

Table 1
Relationship between the case studies and dimensions of XAI-ED.

	Main Stakeholders	XAI Benefits	Approaches	Models	XAI design	Pitfalls
RiPPLE	- Educators - Students - Educational researchers	- Accountability and trust - AI literacy - Agency	- Comparison - Local explanations - Example based	- NLP functions - Graph-based trust propagation - Content-based recommender systems - Elo rating system	- Co-design - User experience	- Disengagement - Dysfunctional behaviour
FUMA	- Educators - Students	- Trust - Sensemaking	- Global explanations - Local explanations	- Clustering - Classification - Association rule mining	- User-centred design	- Overly complexed models - May not benefit all students
AcaWriter	- Educators - Students	- Agency - Trust - AI literacy	- Local explanations - Comparisons	- Rule-based NLP	- Co-design in design-research cycles	- Narrowness of rules-based systems - Context sensitivity
TeamWork Analytics	- Educators - Students	- Accountability and Trust - Agency - Sensemaking	- Global explanations - Local explanations	- Rule-based learning	- Co-design	- Incomplete explanations - Dysfunctional behaviour

foremost appeal to the *instructors*. However, the majority of the features of the platform are designed to be student-facing and to help *students* with their learning. A secondary mission of developing RiPPLE is to support *educational researchers* in conducting ethical, sound, large scale educational research. As detailed below, several attempts have been made to employ XAI in various parts of RiPPLE. The intention has been to (1) help instructors make sense of how the platform is operating, *trust* its decisions, and provide oversight and raise concerns as needed; (2) enable students to develop *AI literacy* and to have *agency* to regulate their learning and trust the system; and (3) support educational researchers to evaluate the platform and propose changes to improve the system and increase *accountability*.

4.1.3. Approaches and models

Here we discuss various ways used to incorporate XAI in RiPPLE.

1. *Explainable automated feedback on poor quality peer reviews.* To effectively utilise a learnersourced repository of content, there is a need for a selection process to separate high-quality from low-quality resources. RiPPLE uses an evaluation process, which relies on crowdsourcing, where students review and evaluate existing resources. A common challenge with incorporating peer review and feedback is that some students may not have the incentive or the ability to provide high-quality feedback. To increase the quality of the provide reviews, RiPPLE uses a range of *NLP models* (Darvishi, Khosravi, & Sadiq, 2020) and *comparison* based explanation approaches to flag comments that are too similar to previously provided reviews or lack the required depth (as shown in Fig. 4-left). A set of tips (as shown in Fig. 4-right) provide training on providing better feedback. Data collected by system shows that around 35% of users whose reviews are flagged revise do go on to resubmit their comments while the other 65% ignore the hint and submit anyway (Darvishi, Khosravi, Abdi, Sadiq, & Gasevic, 2022).
2. *Explainable consensus approach.* RiPPLE assigns each resource for evaluation to multiple moderators, which then requires a consensus approach of optimally integrating the decisions made by multiple people towards an accurate final decision. Summary statistics such as mean or median aggregation are a common explainable consensus approach. However, these models are quite fragile against users with diverse abilities or interests (El Maarry, Güntzer, & Balke, 2015). To increase the accuracy, RiPPLE uses a *graph-based trust propagation approach* (Darvishi, Khosravi, & Sadiq, 2021) that infers the reliability of each moderator. The final decision is computed as a weighted average of the evaluation ratings given by the peer evaluators, which is easy to interpret. Fig. 5 shows an example of a *locally self-explaining* approach for sharing the evaluations and the inferred outcome with the author and moderators. They are first invited to vote on the helpfulness of each moderation. They are then asked to determine whether they agree with the outcome and provide further feedback. Interestingly, data captured by the platform shows that only 2.2% of the submitted responses disagree with the inferred outcome (Darvishi, Khosravi, Sadiq, & Gasevic, 2022).
3. *Explainable spot-checking.* Despite efforts to increase the quality of the peer reviews, the outcome of the consensus approach based on judgements from students, as experts-in-training, cannot be wholly trusted. Given the limited availability of instructors, RiPPLE incorporates a *content-based recommender system* that identifies and recommends resources that most need expert review (Darvishi et al., 2021). When flagging a resource for spot checking, RiPPLE uses absolute and relative points of *comparison* to help instructors make sense of the recommendation. Fig. 6 is an example of a resource

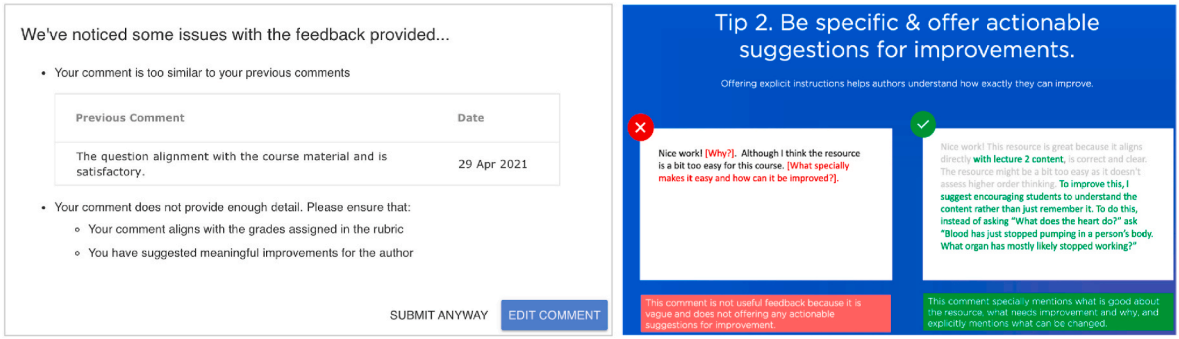


Fig. 4. Explainable suggestions for improving quality of reviews.

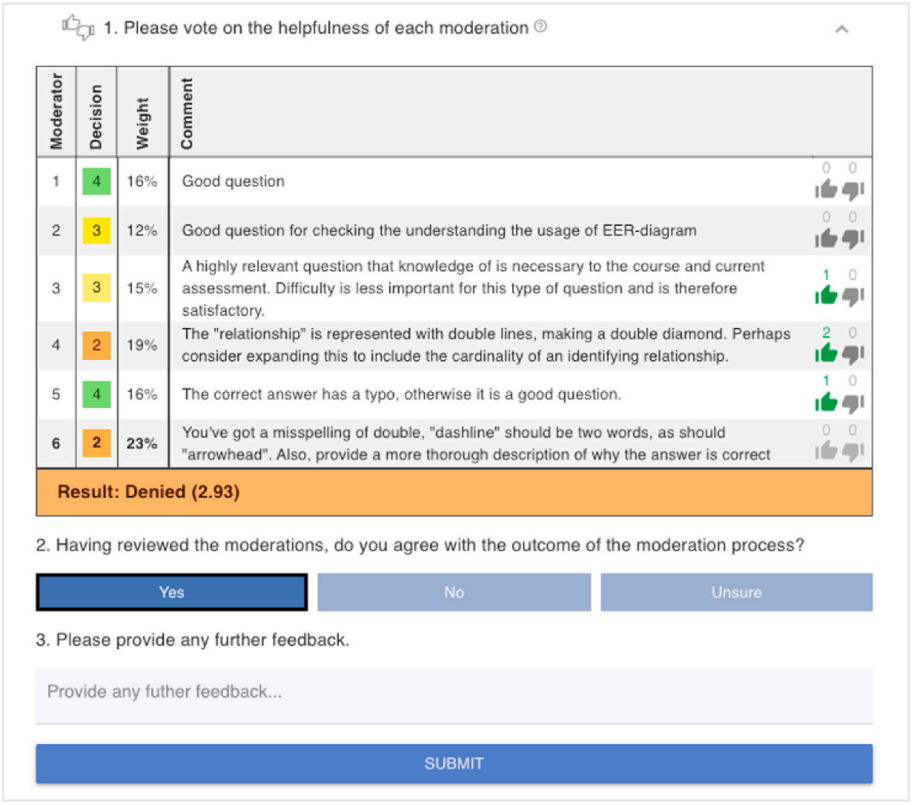


Fig. 5. Explainable consensus approach.

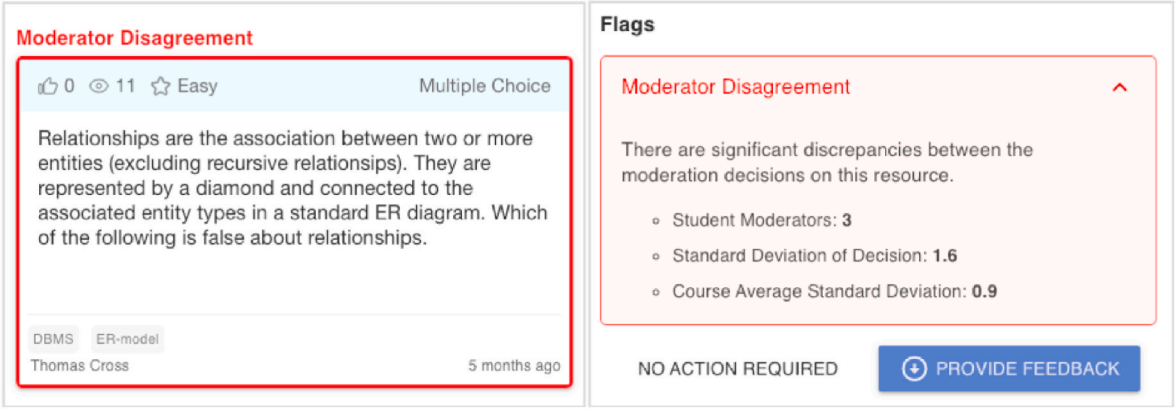


Fig. 6. Explainable spot-checking.

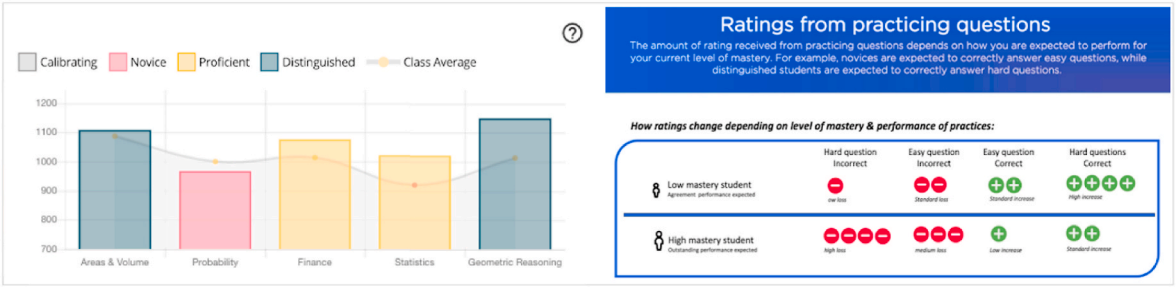


Fig. 7. Explainable learner modelling.

flagged due to moderator disagreement and the relative points of comparison provided to support this decision.

4. *Explainable and open learner model.* RiPPLE employs an extension of the *Elo rating system* to estimate a student's competence in each topic (knowledge component) of a course (Abdi, Khosravi, Sadiq, & Gasevic, 2019) (Other open learner models based on the Elo (Abdi, Khosravi, & Sadiq, 2020) Clicko (Abdi, Khosravi, & Sadiq, 2021) ratings have also been used in the system). The left side of Fig. 7 shows this learner model visualised as a bar chart. The colour of the bars categorises competencies into three levels: novice, proficient, and distinguished. "Calibrating" is used when less than a threshold number of resources are answered on a topic. The model uses an *example based* explanation approach to show the average competency of the entire cohort over each knowledge unit using a line graph. The right side of the figure shows RiPPLE's explanation of the model for students. The explanation can be accessed by clicking on the question mark on the top right hand side of the model.
5. *Explainable resource recommendations.* The adaptive engine of RiPPLE makes use of the learner model to recommend resources at the right level of difficulty for each student. In particular, it recommends easier questions on topics where students are developing mastery and harder questions on topics where the student has already demonstrated mastery. Fig. 8 demonstrates how RiPPLE has included both the recommendation and the learner model on the same page to help with transparency of how resources are recommended to students. Results from a randomised controlled experiment suggest that complementing the recommender system with the learner model can have a positive effect on student engagement and their perception of the effectiveness of the system (Abdi et al., 2020).

4.1.4. XAI designs

RiPPLE has taken a *co-design* approach of partnering with students and instructors across the conceptualisation, development, validation and deployment phases. The initial conceptualisation was done with eight academics with diverse disciplinary backgrounds including computer science, engineering, medicine, pharmacy and education and one member from the central IT systems and support of the university. During the development phase, the RiPPLE team worked closely with academics who trialled early pilots of the platform and a group of four paid student partners who provided advice and helped with usability studies. The validation was based on field studies in close partnership with instructors and students who have given consent. As an example, Abdi et al. (2020) explored the benefits of complementing recommender systems with explainable learner models. Finally, since deployment, frequent short surveys through the platform and interviews with students and instructors have been conducted to better understand the *user experience* and ensure that the stakeholders' concerns and feedback are heard and considered.

4.1.5. Potential pitfalls

By and large, the attempts taken to employ XAI in RiPPLE have been well received; however, there have also been pitfalls and side effects in terms of promoting misbehaviour that are worth discussing. (1) Most students have reported that the availability of the open learner model increased their engagement with RiPPLE as it motivated them to improve their competency. However, feedback from some students revealed that the model can also act as a source of disengagement. In particular, some students indicated that they tend not to attempt questions if they are unsure of the answer to avoid 'taking a hit' in their rating. (2) In the case of flagging similar feedback submissions, instead of trying to address the issue, a strategy taken by some students was to try to *game the system* by making minor changes and resubmitting with the hope of no longer being flagged.

4.2. FUMA a framework for user modeling and adaptation

4.2.1. Overview

FUMA is a framework for modelling and supporting students in learning for exploratory, open-ended learning environments (OLE), such as interactive simulations, educational games and MOOCS (Conati et al., 2021). FUMA uses a range of machine learning models to discover classes of exploratory learning behaviours. These behaviours can then be used to classify users based on their learning needs and to generate personalised hints in real-time that guide the student towards a more effective usage of the tools and affordances available in the OLE.

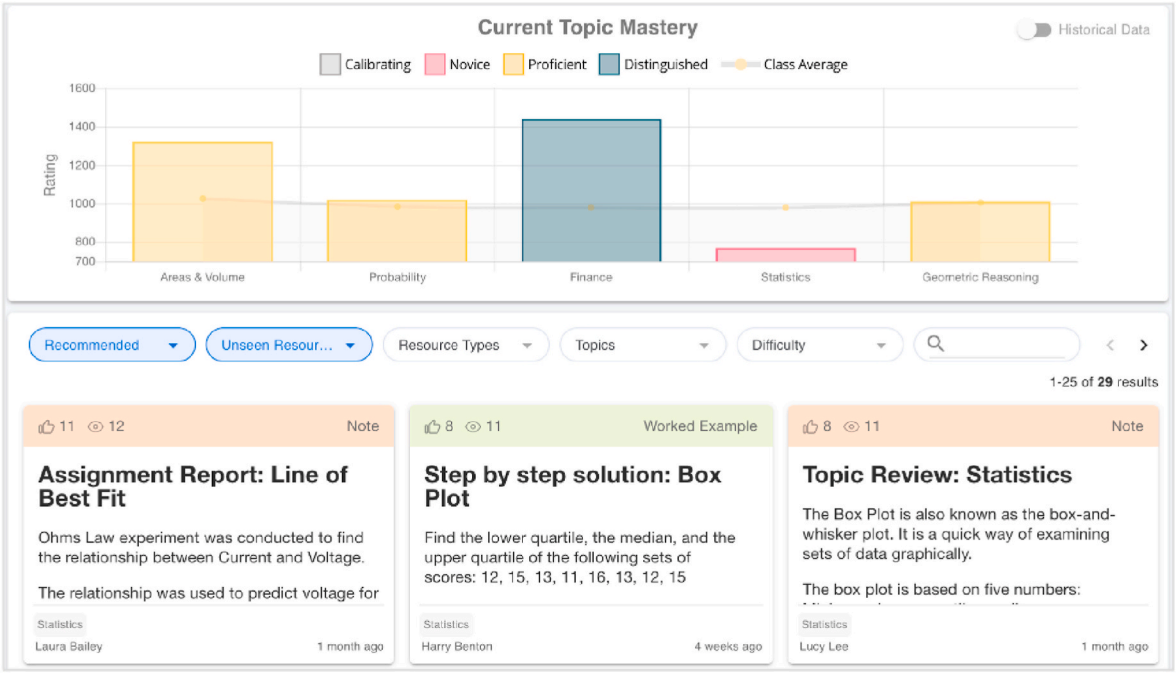


Fig. 8. Explainable recommendation.

4.2.2. Stakeholders and benefits

The primary stakeholders for this technology are *students*, especially those who need guidance during less structured educational activities. Various conducted studies (Fratamico, Conati, Kardan, & Roll, 2017; Kardan & Conati, 2015; Lallé & Conati, 2020) have shown that FUMA can identify clusters of user behaviours that map onto different learning outcomes, and predict when a new student is not learning well early on during the interaction. FUMA has been shown to foster better student learning (Kardan & Conati, 2015) and higher trust in hints (Conati et al., 2021). A second category of stakeholders are *instructors* who can use FUMA for *sensemaking* and to gain a better understanding of how the students learn (or not) from the environment (Conati et al., 2021).

4.2.3. Approaches and models

FUMA uses unsupervised *clustering* and *association rule mining* on existing data of students interacting with a target OLE to discover classes of exploratory behaviours conducive or detrimental to learning. Supervised machine learning is applied to the resulting clusters and association rules to build *classifiers* that predict in real-time whether a student is learning from the exploratory interaction and, if not, what behaviours are responsible for this outcome. These behaviours can then be used to generate personalised hints that guide the student towards a more effective usage of the tools and affordances available in the OLE. Fig. 9 includes detailed steps for each of these phases, to give a sense of the depth of the AI mechanisms embedded in FUMA.

To ascertain if the complex AI underlying the FUMA-driven hints can be externalised to students to further increase the hints uptake and effectiveness, an explanation interface was implemented to convey to the students the motivations (why) and processes used (how) for each of the hints they receive. These explanations aim to help users gain a *globally self-explaining* understanding of the AI driving the hints, as well as a *locally self-explaining* understanding of the specific hints generated. The explanation interface is structured around three tabs, each providing a self-contained, incremental part of the explanation for a given hint, as shown in Fig. 10 (A-C). The states in Fig. 9 are used to justify specific aspects of the rationale for hint computation (why explanations) and the processes to explain how some of the relevant algorithm components work. These tabs display the following three why explanations: 1) "Why am I delivered this hint?" 2) "Why am I predicted to be lower learning?", and 3) "Why are the rules used for classification?". In addition, for the second of these why explanations (Fig. 10 (B)), the user can access more details on how three specific aspects were computed (Fig. 10 (D–F)): "How was this score computed?", "How was this specific hint chosen?", and "How was my hint's rank calculated?" (see Fig. 11).

4.2.4. XAI designs

FUMA takes a user-centred design approach for implementing the explanations of the hints, and follows three guiding principles from (Kulesza et al., 2015), aiming to make the explanations be iterative, namely accessible at different levels of detail, sound, and not overwhelming. Determining how to convey coherent clear and non-overwhelming information on the elements of the FUMA model was supported by a rigorous process of iterative design and pilot evaluations, which led to the development of the interfaces presented in Fig. 10. Essentially, there is a trade-off between having complete explanations and explanations that are not overwhelming. Enabling iterative access to the explanations, as shown in the figure, is an important means to achieve this trade-off, and it is the criterion we used for the explanation functionality.

4.2.5. Potential pitfalls

Although the association rules learned by the FUMA student models have been shown to have a degree of inherent interpretability, they can

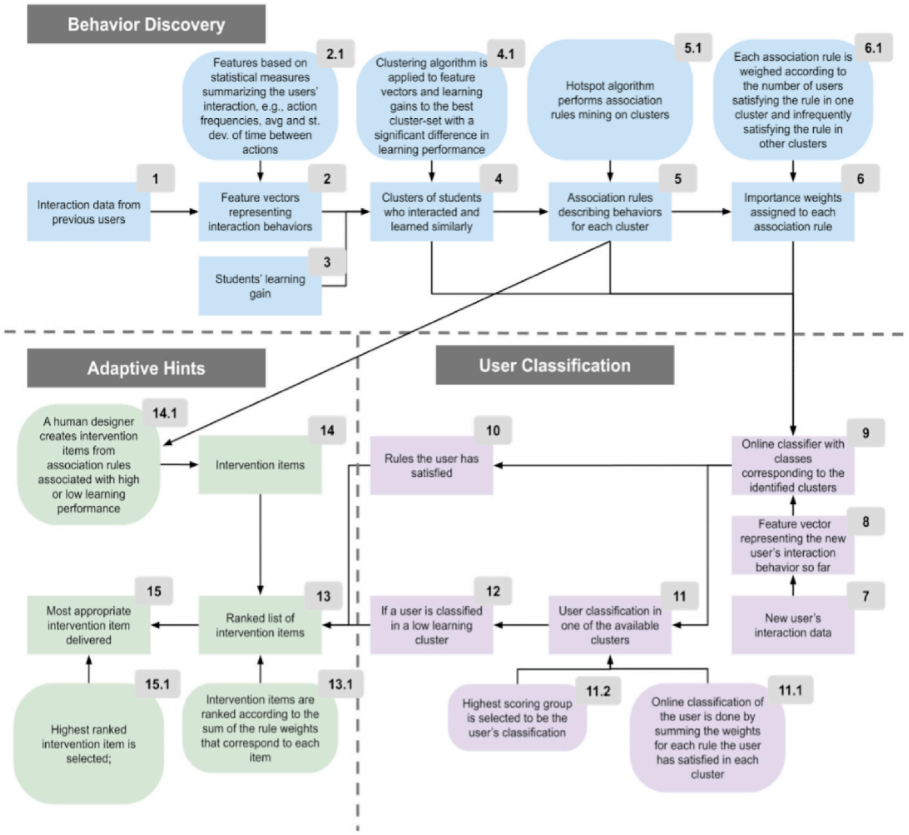


Fig. 9. User Modeling Framework broken down into three phases: Behavior Discovery, User Classification, and Adaptive Hints; rectangular nodes represent inputs and states, oval nodes represent processes.

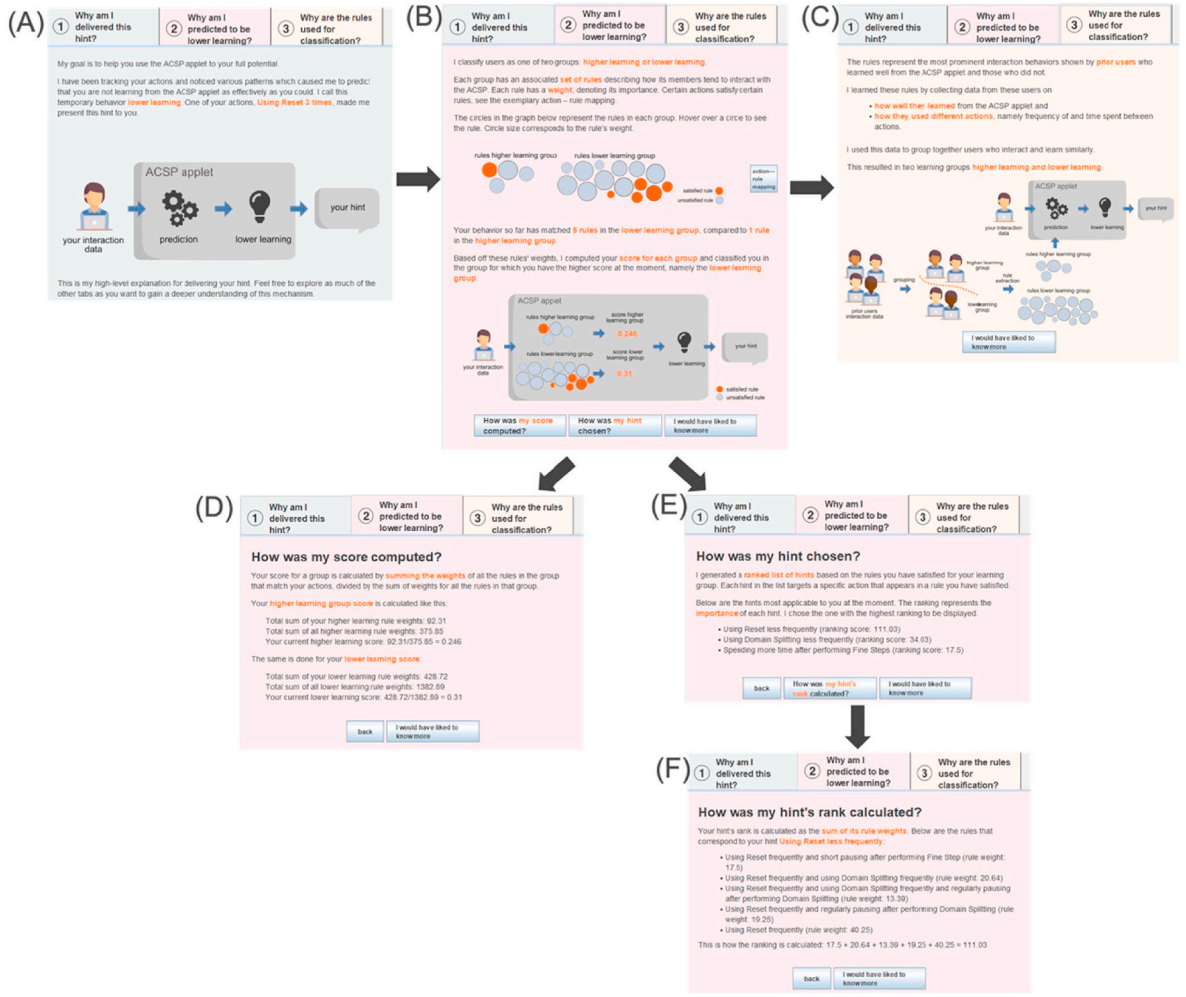


Fig. 10. Flow chart of explanation navigation.

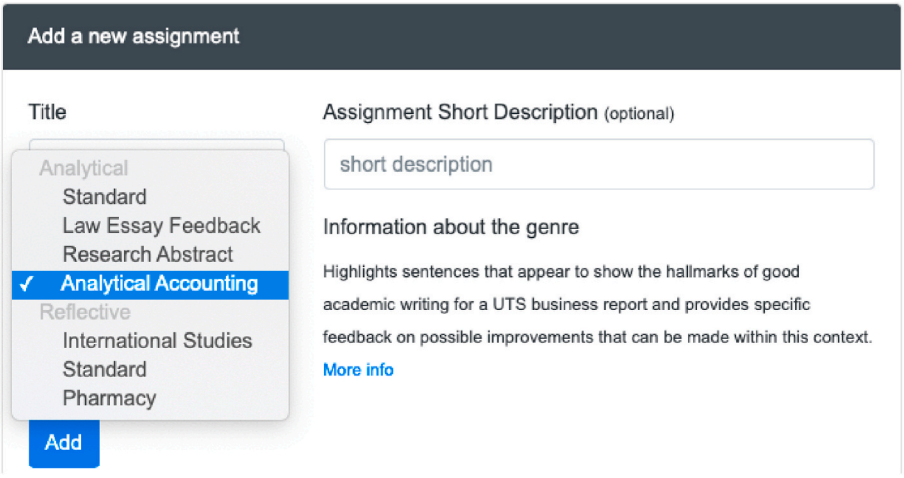


Fig. 11. Sample AcaWriter 'add a new assignment' screenshot; the tool is designed to facilitate instructors in aligning their tasks with genre-based feedback. Figure reproduced from Knight, Shibani, et al. (2020).

become exceedingly complex when dealing with sophisticated OLE that support a large set of actions. Thus, it is important to investigate, for each FUMA application, what the minimal set of features is that can capture relevant student behaviours and provide acceptable model accuracy without hindering rule interpretability. Additionally, a formal evaluation of the explanation interface shows that whether the explanations improve student learning depends to some extent on a student's level of conscientiousness, a personality trait defining one's tendency to follow rules and instructions, where students with low conscientiousness benefit from having the explanations whereas their high-level

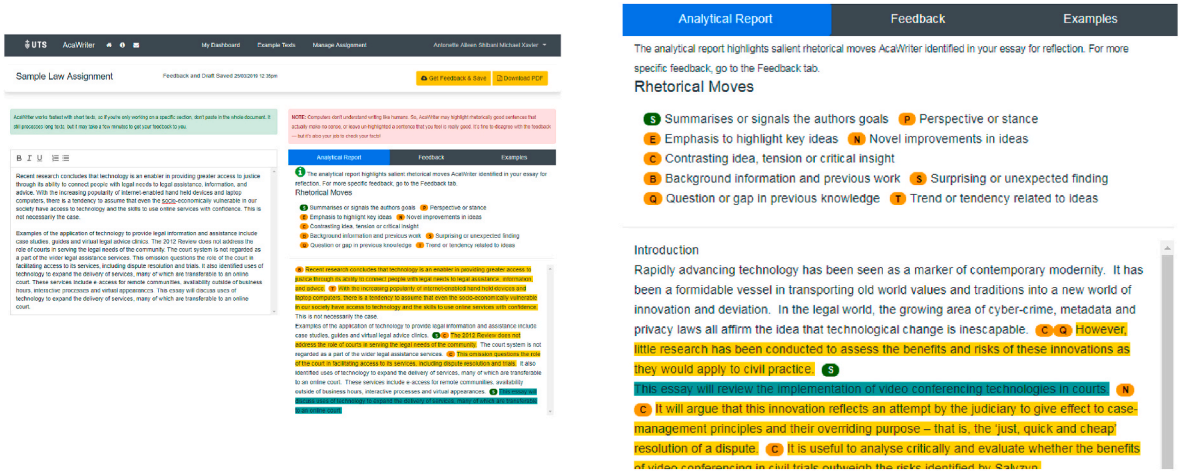


Fig. 12. Sample analytical report on a law essay in AcaWriter highlighting rhetorical moves in the writing. Left image shows the whole screen layout with the editor and feedback; right image shows the detailed feedback. Figures reproduced from Knight, Shibani, et al. (2020).

counterparts do not (Conati et al., 2021). These findings suggest personalizing the explanation functionality so that it proactively encourages users who are known to have low contentiousness or high reading proficiency to access explanations. They also uncover the need to investigate how to reduce the possibly negative effects of explanations on users with high contentiousness and low reading proficiency, for instance by discouraging explanation access or by understanding how to make them useful for these users.

4.3. AcaWriter

4.3.1. Overview

Writing Analytics tools apply analytic techniques drawing on process and sequence mining, natural language processing, and machine learning approaches to understand and provide feedback on features of the writing (process or product) to support that writing (see, e.g. Buckingham Shum, Knight, McNamara, Allen, Betik and Crossley, 2016). The focus of these tools has tended to be on formative feedback in tasks that do not have pre-created (and validated) questions, building on a longer body of work in developing Automated Essay Scoring (AES) systems for automated summative assessment and Automated Writing Evaluation (AWE) using similar approaches for formative purposes (Warschauer & Grimes, 2008) and intelligent tutoring systems that guide students through strategies to address well-defined constructed response writing tasks (Roscoe & McNamara, 2013). AcaWriter is a *theory driven* writing analytics tool that provides feedback primarily to *university students*, on the rhetorical structures in either their academic scholarly, or reflective, writing (Knight, Shibani, et al., 2020).

4.3.2. Stakeholders and benefits

The AcaWriter tool has primarily been targeted at university level writing, through both integration into specific courses (e.g., Law and Accounting for the analytical parser, and Pharmacy for the reflective parser), wider rollout to all students and supporting materials for academics to implement the tool in their own context, and a higher-degree research (HDR) version both as a standalone and integrated into an online course for learning how to write an abstract (Abel, Kitto, Knight, & Buckingham Shum, 2018). The AcaWriter tool has been designed to provide low stakes *formative feedback*, in which the user is engaged in tasks to develop their understanding of their subject, including through their writing and the incorporation of argumentative structures – or rhetorical moves – into this writing.

4.3.3. Approaches and models

Feedback is provided using a rule-based natural language processing system, which identifies syntactic relations between rhetorical concepts ('in contast', 'previous research', etc.) and key terms or topics in the writing. The feedback provided is coupled with a close alignment to the task's *learning design*, in order to *augment* wider assessment structures with this formative feedback (Knight, 2020; Knight, Shibani, & Buckingham Shum, 2018). It is through this alignment that *explanation* is drawn. Elements of the feedback provided are editable for the particular learning context (Shibani, Knight, & Buckingham Shum, 2019), with materials available to guide instructors in aligning the feedback to their assignments. The intent is to provide methods to align feedback (or explanations) to learning contexts through design that encodes expectations of what is to be learned and structures - including XAI - to support this learning (Knight, Gibson, & Shibani, 2020).

The implementation approach for AcaWriter has aligned learning design and analytics. This implementation is evaluated by operationalising what it would mean for the feedback to have impact, or to put it another way, for the explanation to achieve understanding in the student. For example, we have investigated whether revisions to the text introduce particular target features. At its simplest, this has involved analysing drafts and redrafts of the same text. In other contexts a guided process was implemented that builds up the students' knowledge of the constituent parts of the feedback and its relation to their assessment tasks, targeting those assessment tasks and the students evaluative judgement capacities (Shibani, Knight, Buckingham Shum, & Ryan, 2017).

4.3.4. XAI designs

The feedback information provided through the tool takes several forms:

1. *Annotation at the sentence level*, with iconic labels and highlighting indicating the type of sentence identified by the tool (and by extension, unlabelled sentences show that no rhetorical move is detected);
2. Through a 'feedback' tab that provides *document level feedback commentary* regarding issues such as a typical sequence of moves, or the presence of particular moves in introductory sections;
3. A third tab offers reminders of *how the moves map to the writing assignment rubric*, plus *examples* of the kinds of moves students can make in their writing; these can be customised to the writing task, although there is potential to automate selection of examples that match the genre of the text provided (Knight, Abel, et al., 2020).

The tool includes a number of student guides regarding its effective use, as well as a notice that users should "Remember, AcaWriter does not

really understand your writing the way people do […] If you think it got it wrong, that's fine – now you're thinking about more than spelling, grammar, and plagiarism.", or, as the webpage introduction says "UTS isn't here to tell you what to think, but to help you learn how to think. Similarly, AcaWriter won't tell you what to write, but will help you learn how to say it in the most rigorous, effective way." These aim to build *trust*, *agency* and *AI literacy* in understanding the uses and pitfalls of the tool.

In summary, the AcaWriter tool instantiates XAI through implementing feedback on units of analysis — sentences and documents — in the context of design for learning that couples feedback and formative learning tasks, and resources that seek to explain the features on which feedback is provided (while also acknowledging the inherent limitations of such automated analyses).

AcaWriter feedback is *theory driven*, underpinned by literature in writing theory, which has been instantiated in a number of broadly comparable tools (see Knight, Abel, et al., 2020) in this case into a rule-based NLP system. Early versions of a user interface onto the rhetorical move detection were evaluated through both student and instructor *user experience evaluation* and *co-design*, with subsequent evaluation involving more formal testing with quasi-experimental designs (Knight, Buckingham Shum, Ryan, Sándor, & Wang, 2018; Knight, Shibani, et al., 2020). The tool was scaled through piloting in one discipline context, and adapting these learning designs to other discipline contexts. In these evaluations, we demonstrate that the tool adds value to perceived 'usefulness' of *formative writing tasks*, and that students with access to the tool do indeed make more revisions and include more rhetorical moves (Knight, Buckingham Shum, Ryan, Sándor, & Wang, 2018; 2020c).

User-centred co-design with academics has been key in this implementation and disciplinary contextualisation (Knight, Gibson, & Shibani, 2020; Shibani, Knight, & Buckingham Shum, 2020). The tool is now deployed at scale, with students and instructors able to access the tool and in the academic case create their own 'assignment' code for students to receive one of the pre-created feedback options. This work points to the potential of these tools, with students who receive the feedback more likely to develop draft texts, and to incorporate rhetorical moves (Knight, Shibani, et al., 2020).

4.3.5. Potential pitfalls

While the tool is designed to augment effective design for learning, its implementation raises a number of pitfalls. Chief among these is that for *explanation* to be achieved, users (primarily students) must of course understand the characteristics of the features or moves for which feedback is received. As a result, the tool and effectiveness of explanation cannot be considered outside the context of use. In those contexts, the tool also provides feedback only on a narrow range of features (rhetorical structures) for particular types of writing. In addition, it does so imperfectly, particularly where other features (such as spelling and grammar) may interfere in the text processing. These imperfections may provide some learning opportunity (Kitto, Buckingham Shum, & Gibson, 2018; Knight, Gibson, & Shibani, 2020) and the interface itself does encourage users to reflect on the feedback to build their own judgements. Nevertheless, opportunity costs (time spent on x, rather than y) may be present in any system. These pitfalls highlight the importance of human-in-the-loop decision making, and task design for learning in implementing learning analytics tools. The importance of empirically studying students' use of automated feedback cannot be over-stated; it is only by seeing their engagement (or lack thereof) that we can tell if the feedback and explanations are actioned appropriately. The observation that students needed scaffolding in their use of AcaWriter has led, for instance, to the creation of online tutorials that prepare them (Abel, 2022), and activities that promote deeper critical reflection on the feedback (Shibani, Knight, & Buckingham Shum, 2022).

4.4. Data storytelling through teamwork analytics in healthcare

4.4.1. Overview

Immersive, high-fidelity simulations are a common pedagogical approach used in healthcare to support students' and practitioners' learning. In these, participants are posed with an authentic challenge for them to decide on the potential course of action under time pressure. Simulations are commonly conducted in learning spaces instrumented with various physical and digital devices that resemble those available in real hospital rooms. The simulations are followed by a debrief in which participants engage in deep reflection about errors they may have made, areas of improvement and how stressed they may have felt during the simulation (Palominos, Levett-Jones, Power, & Martinez-Maldonado, 2019). In the debriefs, an educator leads the reflection based on direct observations or video-recordings of the simulation. However, the educator and learners commonly do not have objective evidence to discuss as replaying the video-recordings can be time consuming and unpractical. This has motivated the use of multimodal learning analytics to augment the data capture capabilities of the simulation rooms to identify salient aspects of team activity and make these available during the debrief (Martinez-Maldonado et al., 2017). The data is collected using indoor positioning sensors, microphones, physiological wristbands and an observation console (Martinez-Maldonado, Echeverria, Fernandez Nieto, & Buckingham Shum, 2020).

4.4.2. Stakeholders and benefits

The multimodal analytics are aimed at supporting *educators* and *undergraduate nursing students* enrolled in clinical units. These students do not have the time to delve into the analysis of their own data because they also need to reflect on the clinical case they had to address. Based on co-design sessions (Prieto-Alvarez, Martinez-Maldonado, & Shum, 2018), it was identified that both kinds of users would benefit from explanatory visualisations by enabling them to reflect on evidence automatically captured during the group activities. This can increase *accountability* of students who can demonstrate how their actions reflect the application of their clinical knowledge. Exposing the algorithms used to generate the visual interfaces aims at enhancing *trust* in the system and ensure the final users have the *agency* to make their own teaching and learning decisions.

4.4.3. Approaches and models

The concept of data storytelling was adopted to use narrative to communicate insights extracted from the multimodal data. The idea of data storytelling has been suggested, both by the general XAI research community (El-Assady et al., 2019) and the emerging XAI sub-community within learning analytics (De Laet et al., 2018), as a way to automatically structure explanations from data analysis and to emphasise some data points that are relevant to particular stakeholders or users. Here we discuss two ways used to incorporate data storytelling in the multimodal learning analytics interfaces for healthcare simulation.

1. *Explaining key insights from multimodal data.* A number of data storytelling interfaces were created to support *sensemaking* during the debrief. The multimodal data captured during the teamwork sessions can be complex. For example, actions can be logged by the digital devices located in the room and also by an observer. Even though these logged actions are discrete and easy to recognise by an educator or learners (e.g., checking vital signs or providing medication to the patient), sensemaking challenges emerge if they try to focus on assessing such actions for reflection. For example, it is not enough for learners just to 'see' their actions but they need to assess these actions in light of domain knowledge, guidelines or regulations, to understand if and how they can improve their practice. In other words, learners can get lost in the interpretation of such actions (see Fig. 13).

In response, the data storytellling approach adopted in this case consisted in providing layered *local explanations* based on the learning goals of the simulation. A data storytelling layer is created for each learning goal, which consists in combination of data visualisations and textual narratives. Fig. 14 (left) a presents an example of a data storytelling layer that emphasises only certain data points that are relevant to a learning goal (i.e., see Fig. 14, A). In this case, one of the learning goals is for students to reflect whether they checked the vital signs of the patient every 10 min after complaining from chest pain and clearly deteriorating (B). The interface is then improved by adding narrative that *explains* why certain data points are relevant (A) or to indicate the absence of critical data points (C). An explanatory title describes, in lay language, the main take-away message from that data layer. If the goal is to reflect on stress or arousal as automatically captured by the physiological wristband, the students or the educator can press a button (E) and labels appear on top of the visualisation to indicate in text if mild, high or very high arousal was detected by the electrodermal activity sensor (F), which can be indicative of high stress or cognitive load. At the top of the visualisation, a short summary (D) is provided, indicating which roles in the team experienced more arousal during the simulation. The purpose is to spark discussion by providing brief but clear explanations from the data analysis in ways that users without multimodal data analysis training can make sense of. Inroads are currently being made to also create layers that communicate insights from the analysis of x-y indoor positioning traces (Fernandez-Nieto, Martinez-Maldonado, et al., 2021).

2. *Explaining the algorithm used for narrative visualisation. Rule-based algorithms* were used to generate the storytelling layers presented above. These can receive parameters as input or analyse sensor data and compare it across the whole dataset. For example, to assess whether an action was performed in the right time frame, a parameter can reflect official indicators according to healthcare guidelines. For the case of physiological data, the data streams from a student are compared across the arousal levels experienced by other students in the past and therefore determine the level of arousal for the present student. The data stories hide the underlying algorithm during the reflection.

Fig. 15 illustrates how these rules have been opened for participants to provide them with *global explanations* about how the different visual and narrative elements appear on the interface. For example, Fig. 15 (A) show how the narrative that appears at the top of the data storytelling depends on a parameter that reflect one healthcare guideline ("Students should check the vital signs of the patient every 2 min"). Similar parameters are used to automatically add the orange shaded areas to indicate that certain actions were missing during the simulation (B). A different rule assessed the order of certain actions (e.g., indicating that calling to the doctor should happen before a key event).

An evaluation of manipulability and transparency of the teamwork analytics dashboard showed that educators wanted to gain control over the way the data stories are programmed to be able to trust in them (Martinez-Maldonado et al., 2020). They want to be aware of the parameters used to generate the stories for the purpose of compliance with

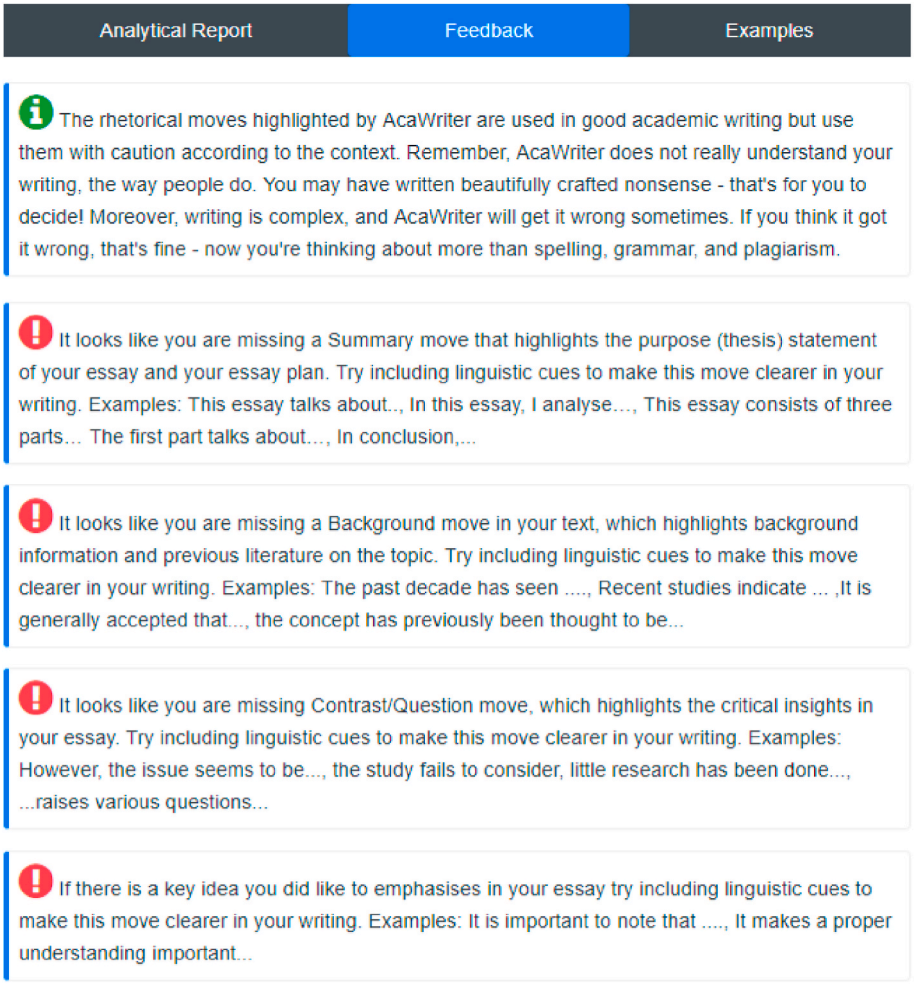


Fig. 13. Sample feedback messages for law from AcaWriter on missing rhetorical moves. Figures reproduced from Knight, Shibani, et al. (2020).

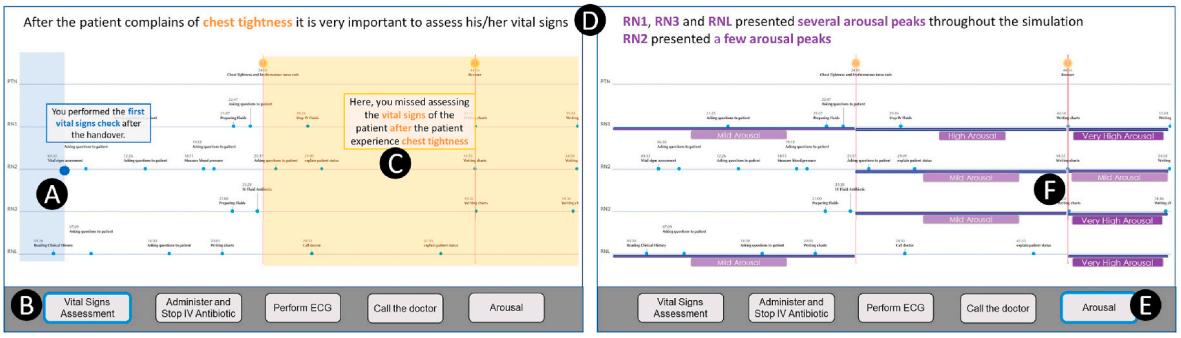


Fig. 14. Two example data storytelling layers explaining activity logs (left) and physiological data (right), adapted from (Fernandez-Nieto, Echeverria-Barzola, et al., 2021, pp. 1–15).

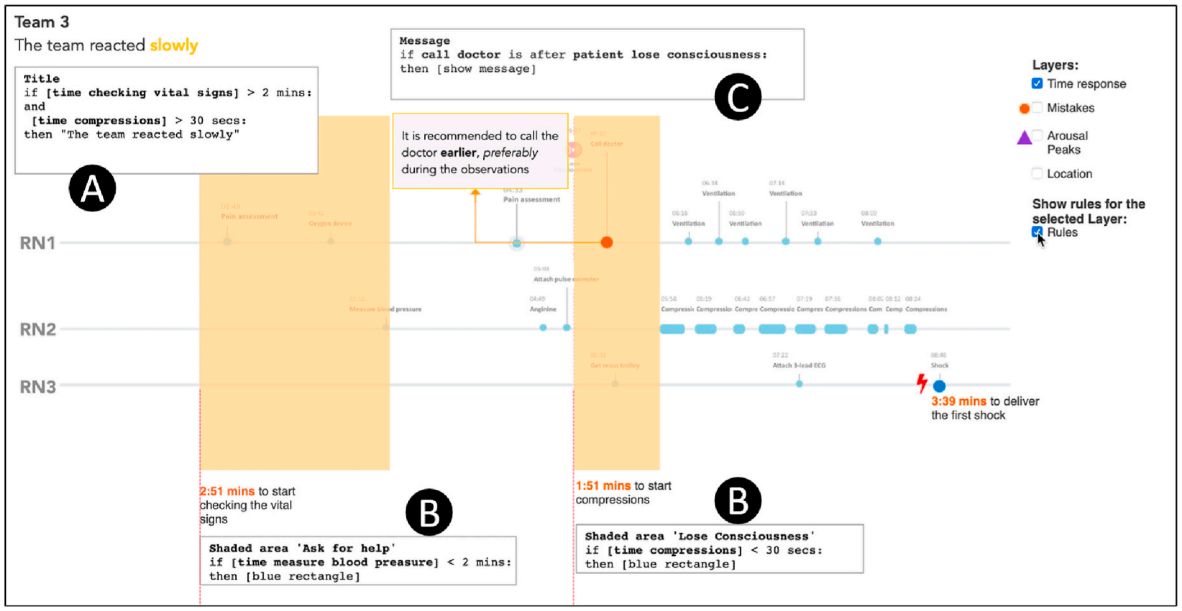


Fig. 15. Explaining the algorithm used for narrative visualisation, adapted from (Echeverria, Martinez-Maldonado, & Buckingham Shum, 2019).

accreditation goals, and be able to adapt/change these parameters according to the cohort of students (for example, by being more relaxed with first year students compared to final year students). By contrast, nursing students did not feel in the position to understand or change the parameters, but they appreciated being able to understand the intended goals of the task based on those parameters.

4.4.4. *XAI designs*

A bottom-up, human centred approach was followed in this case to co-create data stories with relevant stakeholders. A number of *co-design* techniques such as persona identification, user journeys and rapid prototyping were conducted in various cycles of design with educators and nursing students (Prieto-Alvarez, Martinez-Maldonado, & Shum, 2018). In fact, the intention behind this case is that the characteristics of the learning design drive the kinds of explanations that are offered by each data story and, therefore, the **UX design**. The narratives added to the visualisations are a reflection of the kind of feedback a teacher would provide to students and can be edited by them in design time. A new simulation would require the adaptation of the interface according to the learning goals, to emphasise the data points that are relevant to the task at hand. With effective authoring interfaces, the tool can become an explanatory learning analytics interface that augments the kind of feedback that can be provided by the educator during the debrief.

4.4.5. *Potential pitfalls*

The tool has been well received by educators and learners. They can see the potential of augmenting the evidence they can reflect upon and also how the same evidence could be used more widely to improve clinical practice in authentic scenarios. However, the data storytelling approach can open up an interesting debate around *incomplete explanations*. This approach addressed the problem of data abundance by extracting what is important. However, what if important data and insights get unintentionally hidden? Since teamwork simulation is a very complex activity, many other salient aspects may not be covered by the current data capture capabilities. The data stories would emphasise the data relevant to the initial pedagogical goals. A derived potential issue is that *misbehaviour* could be inadvertently promoted if feedback and reflection is only focused on the data that can be automatically captured. For example, sensors may be able to capture if certain medical procedure was performed but not necessarily how it was performed. This could even encourage students to game the system. Keeping the educator in the loop is therefore critical to guide and complement the use of evidence for promoting reflective practices. This naturally occurs for the case of teacher-led debriefs in health care. However, if data storytelling is applied in interfaces used by students alone, these potential pitfalls should be carefully considered and mitigated.

5. Opportunities, challenges and future research needs

This section discusses opportunities, challenges and future research needs for advancing the effective incorporation of XAI in education.

5.1. Actionable explanations

Explainable AI has often been associated with the aim to provide actionable explanations for stakeholders with the growing importance of big data in education and learning analytics. While actionable explanations are commonly related to data-informed decision making in education, a formal definition of actionable explanations as *actionable insights* has been proposed: "should be interpreted as data that allows a corrective procedure, or feedback loop, to be established for a set of actions" (Jørnø & Gynther, 2018, p. 198). For explainable AI in education this means that the emphasis should not only be on explaining the inner workings of an algorithm and how certain results are computed, but that there should be a purposeful design consideration of an AI system that can guide the user to take a certain action (Winne, 2021). As shown in the current paper and the literature, certain classes of AIED system seek to provide *actionable* explanations to promote learning, such as provision of formative feedback (Knight, 2020), triggering reflection and metacognitive monitoring, explainable recommendation of learning resources to engage with (Abdi et al., 2020; Barria-Pineda et al., 2021), (Palominos et al., 2019), revision of course content (Ali, Hatala, Gašević, & Jovanović, 2012), assess algorithmic bias and fairness (Baker & Hawn, 2021; Kizilcec and Lee, In Press), optimal use of instructors time to review student work (Darvishi et al., 2021) or provide support for those who need it the most (Khosravi, Shabaninejad, et al., 2021), or more generally taking a wide range of pedagogical actions (Pardo et al., 2016).

Future research on XAI to provide actionable insights should address several critical challenges. First, a recent systematic review of dashboards (Matcha, Gašević, Pardo et al., 2020), a form of explainable AI systems in learning analytics, showed significant limitations related to data used along with challenges related to design and evaluation of systems. No study on dashboards had data about the key component of self-regulated learning —- cognitive and metacognitive tactics and strategies. As Winne (2021) argues, the absence of data about cognitive and metacognitive tactics and strategies does not allow for providing guidance to learners about how to approach their learning. That is, even if the algorithms that are used can offer a great deal of explainability, we should also work on mechanisms for purposeful collection of data and measurement of constructs on which actionable insights are to be provided (Gašević, Dawson, & Siemens, 2015; Winne, 2020).

5.2. Personalised explanations

Existing research on XAI suggests that having AI systems explain their inner workings to their end users can help foster transparency, interpretability, and trust. However, there are also results suggesting that such explanations are not always valued by or beneficial for all users. For instance, in Section 4.2 we discussed findings showing how some user characteristics modulate the effect of explanations on both student's learning and perception of the adaptive hints available in the ACSP tutoring system, providing insights on how personalisation might bolster explanation effectiveness in this context. These results are in line with previous work that showed the impact of user differences on explanation effectiveness in domains different to education (e.g. (Millecamp, Htun, Conati and Verbert, 2020, 2019; Kleinerman, Rosenfeld, & Kraus, 2018; Kouki, Schaffer, Pujara, O'Donovan, & Getoor, 2019)), and calls for research further exploring the value of personalised XAI in education. The vision is that of a personalised XAI, endowing AI-driven pedagogical systems with the ability to understand when, and how to provide explanations to their end users (e.g., students, teachers, parents) (Conati et al., 2021). Specifically, it is important to identify: *what types of explanations different end users need* (e.g., why or how the system generated a specific outcome); *how the explanations should be delivered* in order to be informative and non-intrusive; and whether these factors might depend on *individual differences* (e.g., long-term abilities and traits, short-term cognitive and affective states, preferences) in order to enable delivery of personalised explanations that accounts for user diversity and bolster inclusiveness of outcomes.

5.3. Human-centred AI design in education

The process for the design of AI and learning analytics systems is underexplored in spite of promising results that have been demonstrated in the recent literature. A recent systematic literature review (Bodily & Verbert, 2017) showed that only about one in ten studies reported a needs analysis before designing a learning analytic dashboard or recommender system. An important research direction is to work on methods that will build design partnerships between developers, researchers and the stakeholders who are typically the people who are ultimately expected to use the tools. Particular attention should be paid to stakeholders from underrepresented groups to ensure they have a meaningful voice (Buckingham Shum et al., 2019; Dollinger et al., 2019). Design considerations should also focus on the way we envision interaction of stakeholders with explainable AI as another form of actionability. Opening learner models can also be a form of invitation for learners to 'push back' against some of information that is incomplete or doubtful (Bull, 2016), and thus contribute to validation of an AI system, while promoting learning engagement.

5.4. Evaluating explanations

We have already noted the importance of studying XAI in use, to see whether in fact the explanations are *understood*, and *appropriately actioned*. Without such evidence, no claim to have designed effective XAI can be made. HCI offers diverse range of techniques can be used to gather such evidence, each of which brings its own strengths and weaknesses (Olson & Kellogg, 2014). These include usability laboratory studies to gain insight from learner reflections in combination with high definition interaction data; activity log analysis to test at scale whether learner actions appear to have addressed feedback and explanations; retrospective cued recall that uses images/video to prompt users to share their thinking; and qualitative self-report data on the level of trustworthiness learners/educators have in the tool/infrastructure.

5.5. Towards trustworthy AI educational systems

Ultimately, we might state our goal to be the creation of *trustworthy*, *AI-augmented*, *sociotechnical* systems. Firstly, the emphasis on *trustworthiness* points to the fact that this is not the same as creating AIED that people trust —- after all, people place misguided trust in all sorts of technology. Secondly, the emphasis on *AI-augmented* draws attention to our belief that while it is certainly possible to learn some things as an isolated individual in front of a computer, the most effective, engaging forms of learning are relational, human activities, involving emotions, other learners, and a relationship in which you trust your teacher/coach or peers to both support and challenge you. AI augments this system. Connected to this, thirdly, *sociotechnical* systems reminds us that we need to frame AIED design not solely as the creation of a digital tool, but as the design of an overall educational system involving other people with different roles, working in an organisational contexts (e.g. a high school in an economically struggling suburb; an elite university; a training unit in a large multinational), in a society that may introduce further constraints (e.g. a national curriculum, certain kinds of teachers, assessment expectations, parental expectations). These contexts and constraints define the 'life-chances' of a given AIED tool. As so many educational technology cases have demonstrated over the years (Scanlon et al., 2013), failure to take these into account simply adds to the

graveyard of promising tools that never achieved sustained adoption.

This paper has begun to demonstrate that what counts as an 'adequate, trustworthy' *explanation* will vary drastically, depending on whether that person is an AIED researcher, a learning scientist, a school teacher or principal, a parent or a student. Possibly only the researcher would know what they were even looking at if permitted to look 'inside the algorithmic black box'. An important line of empirical work is therefore opening up, to clarify what stakeholders consider to be trustworthy explanations, and beyond that of course, what it takes to trust the software as a whole, which will turn on all its behaviour, not just its explanations.

However, as noted above, through no fault of their own, people may be misguided in placing trust in a system they only partially understand. We need as a field, therefore, to move beyond just the empirical question, and ask a more normative question: By what criteria *should* we (as educational and AI experts) declare an AIED system to be trustworthy? By analogy, this is the difference between answering the question, *Are people willing to fly on this plane?* and *Is this plane fit for purpose, as judged by engineering and other civil aviation standards?* The difficulty that AIED currently faces as a field, is that the societal 'trust infrastructure' that we have around mature engineering fields is simply not in place yet. While one key element to building trust is the software's capacity to explain its behaviour adequately to a given stakeholder, in fact, a chain of diverse arguments underpins the claim that the software is trustworthy; developing criteria to test the integrity of each link in that chain is the focus of one or more disciplines, which together establish a notion of overall 'system integrity' (Buckingham Shum, 2019).

6. Conclusion

Advancements in AI are impacting on most if not all sectors and education is no exception. Given that the development of AI tools and technologies is outpacing the social and even legal aspects of the implications of wide scale adoption, it is understandable that there is a degree of public mistrust. Explainable AI is a growing area of research wherein the concerns of fairness, accountability, transparency and ethics can be mitigated. In the educational setting, the importance and need for Explainable AI is further heightened due to the issues related to learner autonomy and agency, support for learner metacognitive processes and reflective processes and broader issues relating to authentic assessment, credentialing and academic integrity.

In this paper we have examined the multiple and complex aspects of Explainable AI in education. We have outlined a framework called XAI-ED with which educational AI tools can be studied, designed and developed. XAI-ED covers six fundamental aspects of XAI in education relating to stakeholders, benefits, approaches, models, designs and pitfalls. The comprehensive discussion of these six aspects allows for a detailed understanding of current state-of-art and open challenges. We have further anchored these six aspects into four diverse case studies relating to leanersourced adaptive systems, open-ended learning environments, writing analytics tools, team based learning to support knowledge transfer. There is a wealth of experiential knowledge embedded in these case studies, which we have deliberated upon to expose a number of XAI opportunities and challenges. Finally, we have used the analysis of the six fundamental aspects and experiences from the case studies to synthesise an agenda for future research for XAI in education. We highlight five areas of particular importance, namely the need for development of actionable and personalised explanations, further incorporation of human-centred design in development of educational tools, rigorous evaluation of the impact of incorporating XAI in education and ultimately advancing towards development of trustworthy AI educational systems.

We conclude that XAI is a critical element that is indispensable to fully avail the opportunities and benefits that AIED systems present for education, human capital development and learning sciences We call upon the research and practitioner community in AIED systems to review, critique, champion and advance the opportunities, challenges and future research needs we have outlined in this paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was supported through multiple granting agencies including: the Australian Research Council's Industrial Transformation Training Centre for Information Resilience (CIRES) (IC200100022), and Discovery Projects (DP210100060 and DP220101209), Jacobs Foundation Center for Learning and Living with AI (CELLA) and Connecting the EdTech Research Ecosystem (CERES), the National Science and Engineering Research Council of Canada (NSERC) and the Economic and Social Research Council of the United Kingdom (ES/S015701/1).

References

Abdi, S., Khosravi, H., & Sadiq, S. (2020). *Modelling learners in crowdsourcing educational systems, in: International Conference on ArtificialIntelligence in Education* (pp. 3–9). Springer.

Abdi, S., Khosravi, H., Sadiq, S., & Gasevic, D. (2019). A multivariate elo-based learner model for adaptive educational systems. In *Proceedings of the educational data mining conference* (pp. 462–467).

Abdi, S., Khosravi, H., & Sadiq, S. (2021). Modelling learners in adaptive educational systems: A multivariate glicko-based approach. In *LAK21:11th International Learning Analytics and Knowledge Conference* (pp. 497–503). New York, NY, USA: Association for Computing Machinery.

Abdi, S., Khosravi, H., Sadiq, S., & Gasevic, D. (2020). Complementing educational recommender systems with open learner models. In *Proceedings of the tenth international conference on learning analytics & knowledge* (pp. 360–365).

Abel, S. (2022). *Writing an abstract (online open access course*. University of Technology Sydney. URL: https://open.uts.edu.au/uts-open/study-area/communication-media/writing-an-abstract.

Abel, S., Kitto, K., Knight, S., & Buckingham Shum, S. (2018). Designing personalised, automated feedback to develop students' research writing skills. In *35th international conference on innovation, practice and research in the use of educational technologies in tertiary education (ASCILITE)* (pp. 15–24). ASCILITE, 00000.

Aleven, V., McLaughlin, E. A., Glenn, R. A., & Koedinger, K. R. (2016). Instruction based on adaptive learning technologies. *Handbook Research on Learning and Instruction*, 522–560.

Ali, L., Hatala, M., Gašević, D., & Jovanović, J. (2012). A qualitative evaluation of evolution of a learning analytics tool. *Computers & Education, 58*, 470–489.

Alqaraawi, A., Schuessler, M., Weiß, P., Costanza, E., & Berthouze, N. (2020). Evaluating saliency map explanations for convolutional neural networks: A user study. In *Proceedings of the 25th international conference on intelligent user interfaces* (pp. 275–285).

Anderson, J. R., Corbett, A. T., Koedinger, K. R., & Pelletier, R. (1995). Cognitive tutors: Lessons learned. *The Journal of the Learning Sciences, 4*, 167–207.

Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., et al. (2020). Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion, 58*, 82–115.

Baker, R. S., De Carvalho, A., Raspat, J., Aleven, V., Corbett, A. T., & Koedinger, K. R. (2009). Educational software features that encourage and discourage "gaming the system". In *Proceedings of the 14th international conference on artificial intelligence in education* (pp. 475–482).

Baker, R. S., & Hawn, A. (2021). Algorithmic bias in education. *International Journal of Artificial Intelligence in Education*. https://doi.org/10.1007/s40593-021-00285-9

Barria-Pineda, J., Akhuseyinoglu, K., Želem-Ćelap, S., Brusilovsky, P., Milicevic, A. K., & Ivanovic, M. (2021). Explainable recommendations in a personalized programming practice system. In *International conference on artificial intelligence in education* (pp. 64–76). Springer.

Batane, T. (2010). Turning to turnitin to fight plagiarism among university students. *Journal of Educational Technology & Society, 13*, 1–12.

Berger, R., Rugen, L., Woodfin, L., & Education, E. (2014). *Leaders of their own learning: Transforming schools through student-engaged assessment*. John Wiley & Sons.

Biggs, J. B., & Collis, K. F. (2014). *Evaluating the quality of learning: The SOLO taxonomy (structure of the observed learning outcome)*. Academic Press.

Bloom, B. S. (1984). The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring. *Educational Researcher, 13*, 4–16 (Publisher: Sage Publications Sage CA: Thousand Oaks, CA).

Bloom, B. S., et al. (1956). *Taxonomy of educational objectives, 1* pp. 20–24). New York: McKay: Cognitive domain.

Bodily, R., & Verbert, K. (2017). Review of research on student-facing learning analytics dashboards and educational recommender systems. *IEEE Transactions on Learning Technologies, 10*, 405–418.

du Boulay, B. (2016). Recent meta-reviews and meta–analyses of AIED systems. *International Journal of Artificial Intelligence in Education, 26*, 536–537.

Boyer, E. L. (1990). *Scholarship reconsidered: Priorities of the professoriate*. ERIC.

Brusilovsky, P., Somyürek, S., Guerra, J., Hosseini, R., & Zadorozhny, V. (2015). The value of social: Comparing open student modeling and open social student modeling. In *International conference on user modeling, adaptation, and personalization* (pp. 44–55). Springer.

Buckingham Shum, S. (2019). Black box learning analytics? Beyond algorithmic transparency (video tutorial, learning analytics summer school, society for learning analytics research). URL: https://youtu.be/VJBt1qpPfXw.

Buckingham Shum, S., Ferguson, R., & Martinez-Maldonado, R. (2019). Human-centred learning analytics. *Journal of Learning Analytics, 6*, 1–9.

Buckingham Shum, S., Knight, S., McNamara, D., Allen Laura, K., Betik, D., & Crossley, S. (2016). Critical perspectives on writing analytics. In *6th ACM learning analytics and knowledge conference* (pp. 481–483). ACM. http://dl.acm.org/citation.cfm?id=2883854.

Bull, S. (2016). Negotiated learner modelling to maintain today's learner models. *Research and Practice in Technology Enhanced Learning, 11*, 1–29.

Bull, S. (2020). There are open learner models about. *IEEE Transactions on Learning Technologies*.

Bull, S., & Kay, J. (2013). Open learner models as drivers for metacognitive processes. In *International handbook of metacognition and learning technologies* (pp. 349–365). Springer.

Bull, S., & Kay, J. (2016). Smili: A framework for interfaces to learning data in open learner models, learning analytics and related fields. *International Journal of Artificial Intelligence in Education, 26*, 293–331.

Cen, H., Koedinger, K., & Junker, B. (2006). Learning factors analysis–a general method for cognitive model evaluation and improvement. In *International conference on intelligent tutoring systems* (pp. 164–175). Springer.

Chounta, I. A., Bardone, E., Raudsep, A., & Pedaste, M. (2021). Exploring teachers' perceptions of artificial intelligence as a tool to support their practice in Estonian k-12 education. *International Journal of Artificial Intelligence in Education*, 1–31.

Conati, C., Barral, O., Putnam, V., & Rieger, L. (2021). Toward personalized XAI: A case study in intelligent tutoring systems. *Artificial Intelligence, 298*, Article 103503.

Cukurova, M., Zhou, Q., Spikol, D., & Landolfi, L. (2020). Modelling collaborative problem-solving competence with transparent learning analytics: Is video data enough?. In *Proceedings of the tenth international conference on learning analytics & knowledge* (pp. 270–275). New York, NY, USA: Association for Computing Machinery. URL: https://doi-org.ezproxy.lib.monash.edu.au/10.1145/3375462.3375484.

Darvishi, A., Khosravi, H., Abdi, S., Sadiq, S., & Gasevic, D. (2022a). Incorporating training, self-monitoring and ai-assistance to improve peerfeedback quality. In *9th International Conference on Learning at Scale, ACM* https://doi.org/10.1145/3491140.3528265

Darvishi, A., Khosravi, H., & Sadiq, S. (2020). Utilising learnersourcing to inform design loop adaptivity. In C. Alario-Hoyos, M. J. Rodríguez-Triana, M. Scheffel, I. Arnedillo-Sánchez, & S. M. Dennerlein (Eds.), *Addressing global challenges and quality education* (pp. 332–346). Cham: Springer International Publishing.

Darvishi, A., Khosravi, H., & Sadiq, S. (2021). Employing peer review to evaluate the quality of student generated content at scale: A trust propagation approach. In *Proceedings of the eighth ACM conference on learning@ scale* (pp. 139–150).

Darvishi, A., Khosravi, H., Sadiq, S., & Gasevic, D. (2022b). Incorporating ai and learning analytics to build trustworthy peer assessment systems.British. *Journal of Educational Technology*. https://doi.org/10.1111/bjet.13233

De Laet, T., Broos, T., Duorado, R., De Croon, R., Millecamp, M., & Verbert, K. (2018). Explainable learning analytics: Challenges and opportunities of this emerging research line. In *Adjunt proceedings of the international learning analytics & knowledge conference* (pp. 1–7).

Dikaya, L. A., Avanesian, G., Dikiy, I. S., Kirik, V. A., & Egorova, V. A. (2021). How personality traits are related to the attitudes toward forced remote learning during covid-19: Predictive analysis using generalized additive modeling. *Frontiers in Education, 6*, 108. URL: https://www.frontiersin.org/article/10.3389/feduc.2021.629213.

Dollinger, M., Liu, D., Arthars, N., & Lodge, J. (2019). Working together in learning analytics towards the co-creation of value. *J. Learn. Analy., 6*, 10–26.

Dollinger, M., & Lodge, J. M. (2018). Co-creation strategies for learning analytics. In *Proceedings of the 8th international conference on learning analytics and knowledge* (pp. 97–101).

Dove, G., Halskov, K., Forlizzi, J., & Zimmerman, J. (2017). Ux design innovation: Challenges for working with machine learning as a design material. In *Proceedings of the 2017 chi conference on human factors in computing systems* (pp. 278–288).

Drachsler, H., & Greller, W. (2016). Privacy and analytics: it's a delicate issue a checklist for trusted learning analytics. In *Proceedings of the sixth international conference on learning analytics & knowledge* (pp. 89–98).

Drake, B. M., & Walz, A. (2018). Evolving business intelligence and data analytics in higher education. *New Direct. Institut. Res.*, 39–52, 2018.

Echeverria, V., Martinez-Maldonado, R., & Buckingham Shum, S. (2019). Towards collaboration translucence: Giving meaning to multimodal group data. In *Proceedings of the 2019 CHI conference on human factors in computing systems* (pp. 1–16).

Ehsan, U., Liao, Q. V., Muller, M., Riedl, M. O., & Weisz, J. D. (2021). *Expanding explainability: Towards social transparency in AI systems*. New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3411764.3445188. URL:.

El Maarry, K., Güntzer, U., & Balke, W. T. (2015). A majority of wrongs doesn't make it right-on crowdsourcing quality for skewed domain tasks. In *International conference on web information systems engineering* (pp. 293–308). Springer.

El-Assady, M., Jentner, W., Kehlbeck, R., Schlegel, U., Sevastjanova, R., Sperrle, F., et al. (2019). Towards xai: Structuring the processes of explanations. In *ACM workshop on human-centered machine learning*.

Engin, G., Aksoyer, B., Avdagic, M., Bozanlı, D., Hanay, U., Maden, D., et al. (2014). Rule-based expert systems for supporting university students. URL: *Procedia Computer Science, 31*, 22–31. https://doi.org/10.1016/j.procs.2014.05.241, 2nd International Conference on Information Technology and Quantitative Management, ITQM 2014 https://www.sciencedirect.com/science/article/pii/S1877050914004189.

Er, E., Villa-Torrano, C., Dimitriadis, Y., Gasevic, D., Bote-Lorenzo, M. L., Asensio-Pérez, J. I., et al. (2021). *Theory-based learning analytics to explore student engagement patterns in a peer review activity* (pp. 196–206). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3448139.3448158. URL:.

Fernandez-Nieto, G., Echeverria-Barzola, V., Buckingham Shum, S., Mangaroska, K., Kitto, K., Palominos, E., et al. (2021). *Storytelling with learner data: Guiding student reflection on multimodal team data* (pp. 1–15). IEEE Transactions on learning technologies in press.

Fernandez-Nieto, G., Martinez-Maldonado, R., Echeverria, V., Kitto, K., An, P., & Buckingham Shum, S. (2021). What can analytics for teamwork proxemics reveal about positioning dynamics in clinical simulations?. In , *5. Proceedings of the ACM on Human-Computer Interaction* (pp. 1–24).

Foltỳnek, T., Meuschke, N., & Gipp, B. (2019). Academic plagiarism detection: A systematic literature review. *ACM Computing Surveys, 52*, 1–42.

Fratamico, L., Conati, C., Kardan, S., & Roll, I. (2017). Applying a framework for student modeling in exploratory learning environments: Comparing data representation granularity to handle environment complexity. *International Journal of Artificial Intelligence in Education, 27*, 320–352.

Gardner, J., Brooks, C., & Baker, R. (2019). Evaluating the fairness of predictive student models through slicing analysis. In *Proceedings of the 9th international conference on learning analytics &; knowledge* (pp. 225–234). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3303772.3303791. URL:.

Garrison, D. R. (2016). *E-Learning in the 21st century: A community of inquiry framework for research and practice*. New York and London: Routledge.

Gašević, D., Dawson, S., & Siemens, G. (2015). Let's not forget: Learning analytics are about learning. *TechTrends, 59*, 64–71.

Gašević, D., Kovanović, V., & Joksimović, S. (2017). Piecing the learning analytics puzzle: A consolidated model of a field of research and practice. *Learning: Research and Practice, 3*, 63–78.

Gervet, T., Koedinger, K., Schneider, J., Mitchell, T., et al. (2020). When is deep learning the best approach to knowledge tracing? *JEDM| Journal Educational Data Mining, 12*, 31–54.

Ghosh, A., Heffernan, N., & Lan, A. S. (2020). Context-aware attentive knowledge tracing. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 2330–2339). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3394486.3403282. URL:.

Goldstein, I. P. (1979). The genetic graph: A representation for the evolution of procedural knowledge. *International Journal of Man-Machine Studies, 11*, 51–77.

Gunning, D. (2017). *Explainable artificial intelligence (xai)*. Defense Advanced Research Projects Agency (DARPA). nd Web 2.

Hassenzahl, M., & Tractinsky, N. (2006). User experience-a research agenda. *Behaviour & Information Technology, 25*, 91–97.

Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research, 77*, 81–112.

Heckler, N. C., Rice, M., & Hobson Bryan, C. (2013). Turnitin systems: A deterrent to plagiarism in college classrooms. *Journal of Research on Technology in Education, 45*, 229–248.

Holmes, W., Porayska-Pomsta, K., (in press). Ethics in artificial intelligence in education. Taylor & Francis.

Holmes, W., Porayska-Pomsta, K., Holstein, K., Sutherland, E., Baker, T., Buckingham Shum, S., et al. (2021). Ethics of ai in education: Towards a community-wide framework. *International Journal of Artificial Intelligence in Education*, 1–23.

Hutt, S., Gardner, M., Duckworth, A. L., & D'Mello, S. K. (2019). *Evaluating fairness and generalizability in models predicting on-time graduation from college applications*. International Educational Data Mining Society.

Jørnø, R. L., & Gynther, K. (2018). What constitutes an "actionable insight" in learning analytics? *Journal of Learning Analytics, 5*, 198–221.

Kahneman, D., Sibony, O., & Sunstein, C. R. (2021). *Noise: A flaw in human judgment*. Brown: Little.

Kardan, S., & Conati, C. (2015). Providing adaptive support in an interactive simulation for learning: An experimental evaluation. In *Proceedings of the 33rd annual ACM conference on human factors in computing systems* (pp. 3671–3680).

Kay, J. (2001). Learner control. *User Modeling and User-Adapted Interaction, 11*, 111–127.

Kay, J. (2006). Scrutable adaptation: Because we can and must. In *International conference on adaptive hypermedia and adaptive web-based systems* (pp. 11–19). Springer.

Kay, J., & Kummerfeld, B. (2019). From data to personal user models for life-long, life-wide learners. *British Journal of Educational Technology, 50*, 2871–2884.

Khosravi, H., Demartini, G., Sadiq, S., & Gasevic, D. (2021). Charting the design and analytics agenda of learnersourcing systems. In *LAK21: 11th international learning analytics and knowledge conference* (pp. 32–42). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3448139.3448143.

Khosravi, H., Kitto, K., & Joseph, W. (2019). Ripple: A crowdsourced adaptive platform for recommendation of learning activities. *Jouranl of Learning Analytics, 6*, 91–105.

Khosravi, H., Shabaninejad, S., Bakharia, A., Sadiq, S., Indulska, M., & Gašević, D. (2021). Intelligent learning analytics dashboards: Automated drill-down recommendations to support teacher data exploration. *Jouranl of Learning Analytics, 8*, 133–154.

Kim, J. (2015). *Learnersourcing: Improving learning with collective learner activity*. Massachusetts Institute of Technology. Ph.D. thesis.

Kitto, K., Buckingham Shum, S., & Gibson, A. (2018). Embracing imperfection in learning analytics. In *Proceedings of the 8th international conference on learning analytics and knowledge* (pp. 451–460). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3170358.3170413.

Kizilcec, R., & Lee, H. (2022). *Algorithmic fairness in education*. ArXiv, 2007.05443 https://arxiv.org/abs/2007.05443.

Kleinerman, A., Rosenfeld, A., & Kraus, S. (2018). Providing explanations for recommendations in reciprocal environments. In *Proceedings of the 12th ACM conference on recommender systems* (pp. 22–30).

Knight, S. (2020). *Augmenting assessment with learning analytics. 7*. Springer. https://doi.org/10.1007/978-3-030-41956-1_10. of The Enabling Power of Assessment. URL:.

Knight, S., Abel, S., Shibani, A., Yoong Kuan, G., Conijn, R., Gibson, A., et al. (2020). Are you being rhetorical? A description of rhetorical move annotation tools and open corpus of sample machine annotated rhetorical moves. *Jouranl of Learning Analytics, 7*, 138–154. https://doi.org/10.18608/jla.2020.73.10

Knight, S., Buckingham Shum, S., Ryan, P., Sándor, A., & Wang, X. (2018). Academic writing analytics for civil law: Participatory design through academic and student engagement. *International Journal of Artificial Intelligence in Education, 28*, 1–28. https://doi.org/10.1007/s40593-016-0121-0

Knight, S., Gibson, A., & Shibani, A. (2020). Implementing learning analytics for learning impact: Taking tools to task. *Internet and Higher Education*. https://doi.org/10.1016/j.iheduc.2020.100729

Knight, S., Shibani, A., Abel, S., Gibson, A., Ryan, P., Sutton, N., et al. (2020). Acawriter: A learning analytics tool for formative feedback on academic writing. *Journal of Writing Research, 12*, 299–344.

Knight, S., Shibani, A., & Buckingham Shum, S. (2018). Augmenting formative writing assessment with learning analytics: A design abstraction approach. In J. Kay, & R. Luckin (Eds.), *13th international conference of the learning sciences: Rethinking learning in the digital age. Making the learning sciences count* (pp. 1783–1790). International Society of the Learning Sciences.

Knijnenburg, B. P., Page, X., Wisniewski, P., Lipford, H. R., Proferes, N., & Romano, J. (2022). *Modern socio-technical perspectives on privacy*.

Koedinger, K. R., Anderson, J. R., Hadley, W. H., Mark, M. A., et al. (1997). Intelligent tutoring goes to school in the big city. *International Journal of Artificial Intelligence in Education, 8*, 30–43.

Koedinger, K. R., Corbett, A. T., & Perfetti, C. (2012). The knowledge-learning-instruction framework: Bridging the science-practice chasm to enhance robust student learning. *Cognitive Science, 36*, 757–798.

Kouki, P., Schaffer, J., Pujara, J., O'Donovan, J., & Getoor, L. (2019). Personalized explanations for hybrid recommender systems. In *Proceedings of the 24th international conference on intelligent user interfaces* (pp. 379–390).

Kreber, C. (2005). Reflection on teaching and the scholarship of teaching: Focus on science instructors. *Higher Education, 50*, 323–359.

Krishna, K., & Murty, M. N. (1999). Genetic k-means algorithm. In , *29. IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)* (pp. 433–439).

Kulesza, T., Burnett, M., Wong, W. K., & Stumpf, S. (2015). Principles of explanatory debugging to personalize interactive machine learning. In *Proceedings of the 20th international conference on intelligent user interfaces* (pp. 126–137).

Kulik, J. A., & Fletcher, J. (2016). Effectiveness of intelligent tutoring systems: A meta-analytic review. *Review of Educational Research, 86*, 42–78.

Lakkaraju, H., Aguiar, E., Shan, C., Miller, D., Bhanpuri, N., Ghani, R., et al. (2015). A machine learning framework to identify students at risk of adverse academic outcomes. In *Proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1909–1918).

Lallé, S., & Conati, C. (2020). A data-driven student model to provide adaptive support during video watching across moocs. In *International conference on artificial intelligence in education* (pp. 282–295). Springer.

Liao, Q. V., Gruen, D., & Miller, S. (2020). Questioning the ai: Informing design practices for explainable ai user experiences. In *Proceedings of the 2020 CHI conference on human factors in computing systems* (pp. 1–15).

Liao, Q. V., Pribić, M., Han, J., Miller, S., & Sow, D. (2021). *Question-driven design process for explainable ai user experiences*. arXiv preprint arXiv:2104.03483.

Lipton, Z. C. (2018). The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. *ACM Queue, 16*, 31–57.

Liu, H., Gegov, A., & Cocea, M. (2015). *Rule based systems for big data: A machine learning approach, 13*. Springer.

Long, Y., & Aleven, V. (2017). Enhancing learning outcomes through self-regulated learning support with an open learner model. *User Modeling and User-Adapted Interaction, 27*, 55–88.

Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI conference on human factors in computing systems* (pp. 1–16).

Lyle, S. (2008). Dialogic teaching: Discussing theoretical contexts and reviewing evidence from classroom practice. *Language and Education, 22*, 222–240.

Ma, W., Adesope, O. O., Nesbit, J. C., & Liu, Q. (2014). Intelligent tutoring systems and learning outcomes: A meta-analysis. *Journal of Educational Psychology, 106*, 901.

Martinez-Maldonado, R., Echeverria, V., Fernandez Nieto, G., & Buckingham Shum, S. (2020). From data to insights: A layered storytelling approach for multimodal learning analytics. In *Proceedings of the 2020 chi conference on human factors in computing systems* (pp. 1–15).

Martinez-Maldonado, R., Power, T., Hayes, C., Abdiprano, A., Vo, T., Axisa, C., et al. (2017). Analytics meet patient manikins: Challenges in an authentic small-group healthcare simulation classroom. In *Proceedings of the seventh international learning analytics & knowledge conference* (pp. 90–94).

Maskin, E. S. (2008). Mechanism design: How to implement social goals. *The American Economic Review, 98*, 567–576.

Matcha, W., Gašević, D., Pardo, A., et al. (2020). A systematic review of empirical studies on learning analytics dashboards: A self-regulated learning perspective. *IEEE Transactions on Learning Technologies, 13*, 226–245.

Miao, F., Holmes, W., Huang, R., & Hui, Z. (2021). *AI and education: Guidance for policy-makers*. Technical Report. UNESCO. URL: https://unesdoc.unesco.org/ark:/48223/pf0000376709.

Millecamp, M., Htun, N. N., Conati, C., & Verbert, K. (2019). To explain or not to explain: The effects of personal characteristics when explaining music recommendations. In *Proceedings of the 24th international conference on intelligent user interfaces* (pp. 397–407).

Millecamp, M., Htun, N. N., Conati, C., & Verbert, K. (2020). What's in a user? Towards personalising transparency for music recommender interfaces. In *Proceedings of the 28th ACM conference on user modeling* (pp. 173–182). Adaptation and Personalization.

Miller, T. (2019). Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence, 267*, 1–38.

Mills, C. N., Potenza, M. T., Fremer, J. J., & Ward, W. C. (2005). *Computer-based testing: Building the foundation for future assessments*. Routledge.

Mitrovic, A. (2003). An intelligent sql tutor on the web. *International Journal of Artificial Intelligence in Education, 13*, 173–197.

Mitrovic, A. (2012). Fifteen years of constraint-based tutors: What we have achieved and where we are going. *User Modeling and User-Adapted Interaction, 22*, 39–72.

Mitrovic, A., & Martin, B. (2007). Evaluating the effect of open student models on self-assessment. *International Journal of Artificial Intelligence in Education, 17*, 121–144.

Molnar, C. (2019). Interpretable machine learning. https://christophm.github.io/interpretable-ml-book/.

Molnar, C., König, G., Herbinger, J., Freiesleben, T., Dandl, S., Scholbeck, C. A., et al. (2020). *Pitfalls to avoid when interpreting machine learning models*. arXiv preprint arXiv:2007.04131.

Moore, M. G. (2013). The theory of transactional distance. In M. G. Moore (Ed.), *Handbook of distance education* (pp. 84–103). New York and London: Routledge.

Moore, J. D., & Paris, C. L. (1992). Exploiting user feedback to compensate for the unreliability of user models. *User Modeling and User-Adapted Interaction, 2*, 287–330.

Moosavi-Dezfooli, S. M., Fawzi, A., Fawzi, O., & Frossard, P. (2017). Universal adversarial perturbations. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1765–1773).

Nicol, D. J., & Macfarlane-Dick, D. (2006). Formative assessment and self-regulated learning: A model and seven principles of good feedback practice. *Studies in Higher Education, 31*, 199–218.

Olson, J. S., & Kellogg, W. A. (2014). *Ways of knowing in HCI*. Springer Publishing Company, Incorporated.

Páez, A. (2019). The pragmatic turn in explainable artificial intelligence (XAI). *Minds and Machines, 29*, 441–459. https://doi.org/10.1007/s11023-019-09502-w. URL:.

Palominos, E., Levett-Jones, T., Power, T., & Martinez-Maldonado, R. (2019). Healthcare students' perceptions and experiences of making errors in simulation: An integrative review. *Nurse Education Today, 77*, 32–39.

Pardo, A., Mirriahi, N., Martinez-Maldonado, R., Jovanovic, J., Dawson, S., & Gašević, D. (2016). Generating actionable predictive models of academic performance. In *Proceedings of the sixth international conference on learning analytics & knowledge* (pp. 474–478).

Pavlik, P. I., Jr., Cen, H., & Koedinger, K. R. (2009). *Performance factors analysis–a new alternative to knowledge tracing*. Online Submission.

Piech, C., Bassen, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L. J., et al. (2015). Deep knowledge tracing. In *Advances in neural information processing systems* (pp. 505–513).

Porayska-Pomsta, K., Woolf, B., Holmes, W., & Holstein, K. (2021). The FATE of AI in education: Fairness, accountability, transparency, and ethics. *International Journal of Artificial Intelligence in Education*. URL: https://link.springer.com/journal/40593/topicalCollection/AC_dcac58fbbf2e68a27dd420b8fa69ba47.

Price, M., Handley, K., Millar, J., & O'donovan, B. (2010). Feedback: All that effort, but what is the effect? *Assessment & Evaluation in Higher Education, 35*, 277–289.

Prieto-Alvarez, C. G., Martinez-Maldonado, R., & Anderson, T. D. (2018). Co-designing learning analytics tools with learners. In J. M. Lodge, J. C. Horvath, & L. Corrin (Eds.), *Learning analytics in the classroom* (1st ed., pp. 93–110). Abingdon, Oxon ; New York, NY: Routledge. https://doi.org/10.4324/9781351113038-7. Routledge, 2019.

Prieto-Alvarez, C. G., Martinez-Maldonado, R., & Shum, S. B. (2018). Mapping learner-data journeys: Evolution of a visual co-design tool. In *Proceedings of the 30th Australian conference on computer-human interaction* (pp. 205–214).

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1135–1144).

Rieber, R. W., & Carton, A. S. (1987). The collected works of ls vygotsky. *Problems of General Psychology., 1*, 325–339.

Roscoe, R. D., & McNamara, D. S. (2013). Writing pal: Feasibility of an intelligent writing strategy tutor in the high school classroom. *Journal of Educational Psychology, 105*, 1010.

Sanders, E. B. N., & Stappers, P. J. (2008). Co-creation and the new landscapes of design. *CoDesign, 4*, 5–18.

Scanlon, E., Sharples, M., Fenton-O'Creevy, M., Fleck, J., Cooban, C., Ferguson, R., et al. (2013). *Beyond prototypes: Enabling innovation in technology-enhanced learning*. milton keynes: Open university.

Sejnowski, T. J. (2020). The unreasonable effectiveness of deep learning in artificial intelligence. *Proceedings of the National Academy of Sciences, 117*, 30033–30038. https://doi.org/10.1073/pnas.1907373117. arXiv:https://www.pnas.org/content/117/48/30033.full.pdf.

Self, J., et al. (1999). The defining characteristics of intelligent tutoring systems research: Itss care, precisely. *International Journal of Artificial Intelligence in Education, 10*, 350–364.

Selwyn, N. (2019). *Should robots replace teachers?: AI and the future of education*. John Wiley & Sons.

Selwyn, N. (2021). There is a danger we get too robotic": an investigation of institutional data logics within secondary schools. *Educational Review*, 1–17.

Shapley, L. S. (2016). 17. a value for n-person games. In *Contributions to the theory of games (AM-28), II* pp. 307–318). Princeton University Press.

Sha, L., Raković, M., Whitelock-Wainwright, A., Carroll, D., Yew, V. M., Gasevic, D., et al. (2021). Assessing algorithmic fairness in automatic classifiers of educational forum posts. In *International conference on artificial intelligence in education* (pp. 381–394). Springer.

Shibani, A., Knight, S., & Buckingham Shum, S. B. (2019). Contextualizable learning analytics design: A generic model and writing analytics evaluations. event-place: Tempe, AZ, USA). In *Proceedings of the 9th international conference on learning analytics & knowledge* (pp. 210–219). ACM. https://doi.org/10.1145/3303772.3303785. URL:.

Shibani, A., Knight, S., & Buckingham Shum, S. (2020). Educator perspectives on learning analytics in classroom practice. *Internet and Higher Education*. https://doi.org/10.1016/j.iheduc.2020.100730.00000

Shibani, A., Knight, S., & Buckingham Shum, S. (2022). Questioning learning analytics? Cultivating critical engagement as student automated feedback literacy. In *LAK22: 12th international learning analytics and knowledge conference* (pp. 326–335). New York, NY, USA: Association for Computing Machinery. https://doi.org/10.1145/3506860.3506912.

Shibani, A., Knight, S., Buckingham Shum, S., & Ryan, P. (2017). Design and implementation of a pedagogic intervention using writing analytics. In W. Chen, J. C. Yang, A. F. Mohd Ayub, S. L. Wong, & A. Mitrovic (Eds.), *25th international conference on computers in education, Asia-pacific society for computers in education* (pp. 306–315), 00000.

Shin, D. (2021). The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable ai. *International Journal of Human-Computer Studies, 146*, Article 102551.

Srinivasan, R., & Chander, A. (2020). Explanation perspectives from the cognitive sciences—a survey. In *Proceedings of the twenty-ninth international joint conference on artificial intelligence, international joint conferences on artificial intelligence organization* (pp. 4812–4818). URL: https://www.ijcai.org/proceedings/2020/670.

Steenbergen-Hu, S., & Cooper, H. (2013). A meta-analysis of the effectiveness of intelligent tutoring systems on k–12 students' mathematical learning. *Journal of Educational Psychology, 105*, 970.

Steenbergen-Hu, S., & Cooper, H. (2014). A meta-analysis of the effectiveness of intelligent tutoring systems on college students' academic learning. *Journal of Educational Psychology, 106*, 331.

Tsai, Y. S., Whitelock-Wainwright, A., & Gašević, D. (2021). More than figures on your laptop: (dis)trustfulimplementation of learning analytics. *J. Learn. Analy., 8*, 81–100.

UNESCO. (2019). *Beijing consensus on artificial intelligence and education*. URL: UNESCO. Technical Report https://unesdoc.unesco.org/ark:/48223/pf0000368303.

VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist, 46*, 197–221.

Wang, D., Yang, Q., Abdul, A., & Lim, B. Y. (2019). Designing theory-driven user-centric explainable AI. In *Proceedings of the 2019 CHI conference on human factors in computing systems* (pp. 1–15).

Warschauer, M., & Grimes, D. (2008). Automated writing assessment in the classroom. *Pedagogies: International Journal, 3*, 22–36. https://doi.org/10.1080/15544800701771580. URL:.

Williamson, B. (2018). The hidden architecture of higher education: Building a big data infrastructure for the 'smarter university. *International Journal of Technologies in Higher Education, 15*, 1–26.

Winne, P. H. (1995). Inherent details in self-regulated learning. *Educational Psychologist, 30*, 173–187.

Winne, P. H. (2020). Construct and consequential validity for learning analytics based on trace data. *Computers in Human Behavior, 112*, Article 106457.

Winne, P. H. (2021). Open learner models working in symbiosis with self-regulating learners: A research agenda. *International Journal of Artificial Intelligence in Education, 31*, 446–459. https://doi.org/10.1007/s40593-020-00212-4. URL:.

Xu, W. (2019). Toward human-centered AI: A perspective from human-computer interaction. *Interactions, 26*, 42–46.

Yang, F., Huang, Z., Scholtz, J., & Arendt, D. L. (2020). How do visual explanations foster end users' appropriate trust in machine learning?. In *Proceedings of the 25th international conference on intelligent user interfaces* (pp. 189–201).

Yang, Q., Scuito, A., Zimmerman, J., Forlizzi, J., & Steinfeld, A. (2018). Investigating how experienced ux designers effectively work with machine learning. In *Proceedings of the 2018 designing interactive systems conference* (pp. 585–596).

arXiv:2312.12751v1 [cs.CY] 20 Dec 2023

HUMAN-CENTRED LEARNING ANALYTICS AND AI IN EDUCATION: A SYSTEMATIC LITERATURE REVIEW

A PREPRINT

Riordan Alfredo*
riordan.alfredo@monash.edu

Vanessa Echeverria*†
vanessa.echeverria@monash.edu

Yueqiao Jin*
ariel.jin@monash.edu

Lixiang Yan*
lixiang.yan@monash.edu

Zachari Swiecki*
zach.swiecki@monash.edu

Dragan Gašević*
dragan.gasevic@monash.edu

Roberto Martinez-Maldonado*
roberto.martinezmaldonado@monash.edu

ABSTRACT

The rapid expansion of Learning Analytics (LA) and Artificial Intelligence in Education (AIED) offers new scalable, data-intensive systems but also raises concerns about data privacy and agency. Excluding stakeholders—like students and teachers—from the design process can potentially lead to mistrust and inadequately aligned tools. Despite a shift towards human-centred design in recent LA and AIED research, there remain gaps in our understanding of the importance of human control, safety, reliability, and trustworthiness in the design and implementation of these systems. We conducted a systematic literature review to explore these concerns and gaps. We analysed 108 papers to provide insights about i) the current state of human-centred LA/AIED research; ii) the extent to which educational stakeholders have contributed to the design process of human-centred LA/AIED systems; iii) the current balance between human control and computer automation of such systems; and iv) the extent to which safety, reliability and trustworthiness have been considered in the literature. Results indicate some consideration of human control in LA/AIED system design, but limited end-user involvement in actual design. Based on these findings, we recommend: 1) carefully balancing stakeholders' involvement in designing and deploying LA/AIED systems throughout all design phases 2) actively involving target end-users, especially students, to delineate the balance between human control and automation, and 3) exploring safety, reliability, and trustworthiness as principles in future human-centred LA/AIED systems.

Keywords Human-centered AI · Human-centered learning analytics · AI in education · Stakeholders involvement · Education technology · Ethical considerations

1 Introduction

Advancements in Artificial Intelligence (AI) are rapidly changing how we carry out our daily activities (Chakraborty et al., 2022). In educational contexts, AI and learning analytics (LA) innovations are prompting a significant transformation by both enabling innovative teaching and learning strategies (Markauskaite et al., 2022) and, at the same time, challenging current assessment practices (Swiecki et al., 2022). Emerging LA and AI in Education (AIED) systems offer novel data-intensive solutions that promise to enable adaptive and personalised teaching and learning experiences at scale. For instance, various intelligent tutoring systems and LA solutions provide personalised learning support and automated feedback (e.g. Maier and Klotz, 2022; Cavalcanti et al., 2021). LA dashboards are providing educators with new means to track student progress and offer targeted support, potentially leading to improved student outcomes (e.g.

*Centre for Learning Analytics Monash - Monash University, 20 Exhibition Walk, Clayton, 3800, VIC, Australia
†Escuela Superior Politécnica del Litoral, 30.5 Via Perimetral, Guayaquil, Ecuador

Williamson and Kizilcec, 2022; Fernandez Nieto et al., 2022). LA and AIED systems, paired with novel interaction innovations, such as gamification and mixed reality (e.g. Carter and Egliston, 2023), are also enabling new pedagogical strategies that can potentially make learning more engaging and interactive for students. In sum, educational providers are increasingly adopting LA/AIED systems because data-intensive technologies hold the promise of making learning more accessible, scalable, effective, and personalised for students (Macfadyen, 2022) by providing various forms of teacher-facing and student-facing user interfaces (Mavrikis and Cukurova, 2021; Buckingham Shum and Luckin, 2019).

Yet, the proliferation of LA/AIED systems raises key concerns about the privacy of students and security of educational data (Viberg et al., 2022), as well as the potential for algorithms to perpetuate biases and discrimination (Uttamchandani and Quick, 2022). The lack of involvement of students and teachers in the design and development of LA/AIED systems can potentially lead to a lack of understanding and trust in the technology (Shibani et al., 2022; Dollinger et al., 2019; Sarmiento and Wise, 2022; Alzahrani et al., 2023). This lack of involvement also raises questions about the accountability and transparency of AI in education (Tsai et al., 2020). Furthermore, increased student adoption of generative AI systems may recurrently challenge existing assessment practices and blur the lines of academic integrity (Moya et al., 2023). It is thus essential to address these concerns and incorporate the perspectives and authentic needs of students and teachers in designing and implementing LA/AIED systems. This can potentially contribute to creating ethical, effective, and inclusive ways to adopt LA/AIED systems (Williamson and Kizilcec, 2021, 2022).

In response to the above challenges, there has been a growing interest in adopting human-centred design (HCD) approaches that prioritise human needs, values, and perspectives in the design and deployment of LA/AIED systems (Lang and Davis, 2023; Viberg et al., 2023; Luckin et al., 2006; Buckingham Shum et al., 2019). Beyond education, there is also a growing interest in applying similar human-centred principles in the emerging discipline of Human-Centred AI (HCAI). An HCAI approach views AI as systems aimed at serving human interests rather than as a means of achieving technical goals or replacing humans (Shneiderman, 2022; Usmani et al., 2023). Thus, the design of HCAI systems is to be guided by a set of ethical principles and design guidelines that empower end-users. The aim is to ensure that technologies providing high automation capabilities are created in ways that guarantee their trustworthiness, transparency, and benefit to society (Usmani et al., 2023; Shneiderman, 2020a; Ozmen Garibay et al., 2023).

In recent years, the concept of HCAI has gained significant attention in LA and AIED with a growing recognition of the importance of considering educational stakeholders' needs and perspectives in the design and deployment of data-intensive innovations (Buckingham Shum et al., 2019; Yang et al., 2021; Sarmiento and Wise, 2022; Holmes et al., 2022; Kloos et al., 2022; Alfredo et al., 2024). This includes, for example, the use of participatory design and co-design methods to involve teachers (e.g. Ahn et al., 2021; Holstein et al., 2019), students (e.g. Prieto-Alvarez et al., 2018; Sarmiento et al., 2020), and other educational stakeholders (such as educational decision-makers, learning designers, IT developers and educational researchers) (e.g. Prieto et al., 2019; Schmitz et al., 2022) in the co-creation of LA/AIED systems, the development of ethical frameworks particularly tailored to LA and AIED (e.g. Dollinger et al., 2019; Holmes et al., 2022; Brossi et al., 2022), the exploration of new methods for incorporating teachers perspectives and experiences into machine learning algorithms and end-user LA interface designs (Luckin et al., 2006; Wise et al., 2021), and ways to open the AI algorithmic "black box" to provide insight to teachers and students into the analytics or communicate analysis outputs to them in ways they can easily understand (Khosravi et al., 2022).

Given the rapidly growing interest in applying human-centred design principles in LA/AIED research and practice, a timely systematic literature review (SLR) in this area offers a crucial opportunity to synthesise and evaluate the existing body of research. This can inform the direction of future research and ensure that new work is building upon the foundations established by previous studies. The closest work to ours was presented by Sarmiento and Wise (2022), who conducted an initial, non-systematic review of the literature in participatory and co-design of LA. The authors surveyed the design systems and techniques used in the participatory design of learning analytics. To the best of our knowledge, no previous research has conducted *comprehensive* and *systematic review* of the body of work focused on applying human-centred design and HCAI principles in LA/AIED literature.

This paper presents a SLR that goes beyond previous works and aims to address the knowledge gap about the state of the art of human-centredness in LA/AIED systems. We used the human-centred AI framework (Shneiderman, 2022) as a lens to motivate and structure our SLR. This framework helped us to focus on key themes of human-centred design, balance human control and computer automation, safety, reliability, and trustworthiness (SRT), and to identify relevant studies that address these themes in the context of stakeholder involvement in LA and AIED. The findings of this SLR contribute insights into gaps in the existing research, highlighting areas where further investigation is needed and methodological challenges that need to be addressed for human-centred LA and AI in education data-intensive systems to remain relevant and potentially become part of the mainstream practices in the foreseeable future.

2 Background

2.1 Foundations of Human-Centred AI and Learning Analytics in Education

While AI and analytics technologies exhibit potential for augmenting human decision-making, concerns over transparency, accountability, algorithmic bias, discrimination, and potential threats to human autonomy and agency can mitigate their benefits (van Berkel et al., 2022). Consequently, governments – such as the G20 members (Jelinek et al., 2021) – and large tech companies (e.g., see review by Jobin et al., 2019) have proposed guidelines for AI application design centred on human values (e.g., safety, reliability, and trustworthiness). In this context, HCAI is emerging as a new research discipline that can be broadly defined as an approach to creating "*software designs that give users high levels of understanding and control over their AI-enabled systems to preserve human agency*" (Shneiderman, 2021, p.56). HCAI has its roots in fields such as human-computer interaction, human-centred design, and human factors engineering (Chignell et al., 2022), which focus on understanding and incorporating human perspectives and experiences into technology design (Grudin, 2009). HCAI aims to extend human-computer interaction and HCD principles to address unique issues and unforeseen impacts of AI autonomy (Xu et al., 2023).

Echoing Engelbart's (1962) seminal research on augmenting intelligence, an HCAI approach emphasises the complementarity of humans and machines, aiming to design AI systems that amplify, augment, and empower individuals by considering SRT principles (Shneiderman, 2022). In educational contexts, Buckingham Shum et al. (2019) and Luckin et al. (2006) were among the pioneers advocating for the integration of human-computer interaction and HCD approaches (Giacomin, 2014) in the fields of LA and AIED, respectively. They emphasised the need to fully comprehend educational stakeholders' needs, preferences and challenges, often necessitating their inclusion in some stages of the design process by using HCD methods. This reinforces the relevance of HCAI and HCD methods in designing the technical, social, and data-related aspects of LA and AIED, mitigating potential social harms related to the spectrum of analytics use, from rule-based and descriptive to AI and machine learning-driven predictive and prescriptive forms (Davenport, 2018).

2.1.1 Human and Computer Automation complementarity.

To articulate the balance between humans and AI, Shneiderman (2022) developed a two-dimensional framework. This framework, designed for a broad audience, distinguishes between varying levels of ***human control*** and ***computer automation***. These factors are not considered mutually exclusive.

The notion of ***human control*** is closely related to the *sense of agency*. This subjective experience stems from perceived control over one's actions to make decisions and influence events (Moore, 2016). Viewed through the HCD lens, sense of agency is crucial in designing interfaces that support an internal locus of control (i.e., individuals' perceptions of controlling their outcomes). In an educational setting, human control can range widely, from learners or teachers exercising a high level of agency over the outcomes of LA/AIED systems to simply receiving and understanding information (Hooshyar et al., 2023). A *high* level of human control may involve learners using their expertise to strategically make informed decisions and take appropriate actions in their learning environment. In contrast, a *low* level of control might occur when individuals receive and understand information but lack autonomy in their learning or teaching environment. This notion of human control encompasses key elements of intentional, purposeful, and meaningful learning (Jääskelä et al., 2021) and teaching (Biesta et al., 2015).

The notion of ***computer automation*** in an HCAI context refers to the characteristics of systems that use computer technology or algorithms to perform tasks automatically, streamlining and expediting operations that were previously completed by human actors (e.g., teachers or students) (Parasuraman et al., 2000). The level of automation is determined by its complexity (e.g., ranging from simple rule-based algorithms to complex machine learning and AI models) and the *number* of information process stages the automation supports (i.e., acquiring information, information analysis, decision-making, or action implementation) in the decision-making process (Onnasch et al., 2014). In educational setting, a *high* level of automation commonly refers to LA/AIED system capabilities that can automatically capture learner data and make predictions, commonly achieved using multiple algorithms or AI techniques, such as machine learning, computer vision, natural language processing, or intelligent agents (e.g. Dogan et al., 2023; Jamalova et al., 2022). On the other hand, a *low* level of computer automation typically involves a system where information is manually prepared or a system follows a fixed set of rules strictly deterministically. Examples can include non-data-intensive solutions (e.g., presentation slide decks) and manually pre-configured visualisations in LA dashboards (Fernandez Nieto et al., 2022). Such systems often lack autonomy and *adaptability* when faced with situations beyond their programmed capabilities.

In turn, a single LA/AIED system can provide various features that combine different levels of human control and computer automation, catering to different tasks and intended users (Holstein et al., 2020). This results in the following

four quadrants (two-dimensional HCAI framework) contextualised for LA/AIED systems, each representing a different combination of human control and computer automation:

Q1: **LOW human control & LOW computer automation**. This quadrant represents systems with limited user control or configuration and minimal or no automation. After receiving information from the system, end-users can only comprehend it but cannot personalise or modify it. Some examples in this quadrant include *learning resources* in the learning management system; *student-facing reporting systems* that generate reports for awareness or reflection in an asynchronous manner (e.g., Bodily and Verbert, 2017); and *rule-based visual data stories* (e.g., Echeverria et al., 2018, 2024; Fernandez-Nieto et al., 2024), which provide feedback to students about their collaboration process and task completion after a learning activity.

Q2: **HIGH human control & LOW computer automation**. This quadrant represents systems where end-users can personalise and configure aspects of the information process and maintain a sense of agency over the learning environment. In contrast, the system facilitates exploration through manual operation with minimal or no automation. Examples include *personalised visualisation dashboards* (e.g., Muslim et al., 2016), which allow end-users to control or co-configure the descriptive analytics that learners can use to reflect on the achievement of learning goals; and *educator-driven data analytics systems* (e.g., Fernandez Nieto et al., 2022), which may rely on educators' experience, intuition, and informal observations for educators themselves to make decisions about instructional strategies and feedback.

Q3: **LOW human control & HIGH computer automation**. This quadrant represents systems that heavily rely on automation with minimal end-user control. Here, automated processes and algorithms handle decision-making and action-taking processes. Examples include systems that capture interactions of students or teachers and utilise them in *predictive analytics* components to provide fully *automated feedback* (e.g., Ochoa et al., 2018; Maier and Klotz, 2022) and *automated grading systems* (e.g., Shetty et al., 2022), which may fully or partly replace the teacher's role in specific assessment tasks.

Q4: **HIGH human control & HIGH computer automation**. This quadrant represents systems that enable manual operation while benefiting from automated assistance to enhance the decision-making process in the learning environment for teachers or learners. Examples in this quadrant include *intelligent teaching assistants* that support classroom orchestration (e.g., Lawrence et al., 2023)), *recommender systems* that promote the development of metacognitive skills (e.g., Khosravi et al., 2019; Darvishi et al., 2024), and sophisticated modelling/predictive features in *analytics reports or dashboards* that provide guidance and support (e.g., Khachatryan et al., 2014). Here, teachers or students can utilise the highly automated features of the tool to perceive and make sense of learning data and make informed pedagogical decisions about the next course of action (Holstein et al., 2019).

2.1.2 The HCAI principles of safety, reliability, and trustworthiness

In addition to defining these levels of control, designing any LA/AIED systems that offer a significant level of human control and computer automation must consider the fundamental principles of safety, reliability, and trustworthiness to ensure ethical practices and reduce the risk of data misuse (Cavalcante Siebert et al., 2023; Shneiderman, 2022). In educational contexts, LA/AIED systems should embrace a *safety* culture, which implies that researchers, designers, and system operators should establish strong relationships with end-users and relevant educational stakeholders, as well as implement strict data privacy measures (Holmes et al., 2022). These may include involving stakeholders in data-sharing decisions, ensuring data collection transparency, and granting data access only to specific users (Drachsler and Greller, 2016).

An LA/AIED system is deemed *reliable* is when it delivers *accurate* information as anticipated by the user while interacting with it (Shneiderman, 2022). However, no predictive model can achieve such perfection (Baker, 2016). Therefore, LA/AIED systems designers may benefit from accepting the existence of *imperfection* by studying how users validate and respond to whether such systems may adversely affect their learning (Kitto et al., 2018). This can potentially be done by educating humans about AI capabilities and *human biases*, allowing for a balanced delegation of automation tasks while incorporating elements of human oversight (Pinski et al., 2023). Finally, an LA/AIED system is considered *trustworthy* when users have confidence (and, therefore, it is regarded as 'trusted' by users). Aiming to foster greater trust and confidence among users, the system could actively seek feedback from users, learn from its mistakes, and adapt to improve its performance while aligning with user expectations (Usmani et al., 2023). Designers ought to adopt a practical approach that prioritises transparency (Nazaretsky et al., 2022a,c) and accountability (Pardo and Siemens, 2014) and aim to understand the features of a system that would increase users' trust in it.

2.1.3 Human-centred design in LA/AIED

An HCAI approach strongly advocates for the participation of pertinent stakeholders, such as end-users with *lived experience*, potential future users, policymakers, and experts in ethics and human values, in the design and deployment of AI systems (Usmani et al., 2023; Shneiderman, 2020b). HCD thus plays a crucial role in HCAI as it focuses on the needs and requirements of the people for whom the system is intended, rather than the designer's creative process or the technology capabilities (Giacomin, 2014). As a participatory practice, HCD involves an iterative process of understanding the context, identifying end-user requirements, and involving stakeholders with lived experiences in the design and evaluation of the system, ideally engaging with stakeholders as equal partners (Xu et al., 2023). From Hanington and Martin's (2012) perspective, the HCD process includes five multifaceted phases: (1) *planning, scoping, and definition* – clearly defining the parameters of the project; (2) *exploration, synthesis, and design implications* – conducting immersive research and design ethnography to gather information and derive insights that will guide design choices; (3) *concept generation and early prototype iteration* – generating ideas, concepts, or creating an early version of prototypes with stakeholders; (4) *evaluation, refinement, and production* – testing and gathering feedback from stakeholders to refine designs, ensuring they meet the desired standards in the production; and (5) *launch and monitor* – conducting quality assurance testing of the design to ensure readiness for public use, and ongoing analysis to make necessary adjustments if needed.

Numerous HCD techniques are available to assist designers and researchers in the aforementioned phases (Giacomin, 2014; Maguire, 2001). These techniques can enable interaction with stakeholders and aid in the identification of their meanings, desires, and needs, which can be achieved through verbal techniques (e.g., ethnographic interviews, questionnaires, cognitive tasks, think-aloud, persona crafting, and brainstorming) or non-verbal techniques (e.g., probing, observations, body language analysis, and physiological analysis). A growing set of more speculative techniques (e.g., real fiction, role-playing, para-functional prototypes, what-if scenarios and fabulation) are used for simulating opportunities and envisaging possible futures for designing future-looking features. Nowadays, some of these HCD techniques are starting to be used to design contemporary LA/AIED systems or to envisage potential future scenarios of AI application in education (e.g., Prestigiacomo et al., 2020; Holstein et al., 2019; Echeverria et al., 2023).

Regarding stakeholder involvement, Sarmiento and Wise (2022) conducted an initial review of HCD methods in the LA literature, particularly in participatory design and co-design. They pointed out the increased use of these two methods in higher education and highlighted detailed descriptions of research techniques that were often lacking. From this point of view, the extent of stakeholders' involvement can be categorised into *active* or *passive*. **Active involvement** refers to possessing agency in shaping the outcomes that arise from the design processes (Dollinger et al., 2019). Stakeholders engage in conversations that contribute to creating designs, assisting in testing, evaluating, and providing feedback on designs, or being actively involved in decision-making processes that shape the design of LA/AIED systems. On the other hand, **passive involvement** refers to stakeholders playing a role as more consultative or advisory in the design activities (Edelenbos and Klijn, 2006). They provide input, feedback, or opinions but may have limited decision-making authority or direct influence over the outcomes.

2.2 Related Works and Research Gaps

To better understand how human-centredness has been considered in the design of LA/AIED systems, it is crucial to examine the key *characteristics* of current research in this area. These include the education levels mainly targeted, the types of research methods used (such as qualitative, quantitative, or mixed), and the most commonly applied HCD techniques like co-design, focus groups, and prototyping. For instance, previous reviews of LA/AIED generally have focused on higher education (e.g., Williamson and Kizilcec, 2022; Leitner et al.; Dogan et al., 2023), K-12 (Zhang and Aslan, 2021; Granić, 2022; Lin et al., 2023), or informal learning (Granić, 2022). However, it remains unclear which education level has most thoroughly considered human-centredness. Moreover, there is a gap in current research regarding the diverse research methodologies used in human-centred LA/AIED research and the specific design phases (listed in Section 2.1.3) where they have been applied. At the same time, previous non-systematic reviews that initially explored human-centredness in LA (e.g., Sarmiento and Wise, 2022) and AIED (e.g., Khosravi et al., 2022) literature have not identified the specific HCD techniques employed at different design stages. Together, these issues motivate our first research question:

> **RQ1.** What is the current state of human-centred LA/AIED research, specifically through the lens of education levels targeted, research methodologies employed, design phases covered, and HCD techniques utilised?

Arguably (Lang and Davis, 2023), the involvement of educational stakeholders is paramount in the human-centred design process, as highlighted in recent human-centred LA (e.g., Sarmiento and Wise, 2022; Barreiros et al., 2023) and AIED (e.g., Li and Gu, 2023; Lin et al., 2023) works. However, this involvement can take various forms (Lang

and Davis, 2023). It remains unclear to what extent educational stakeholders' involvement, whether *active* or *passive* (see Section 2.1.3), has been considered in the various phases of the design process, from inception to implementation and evaluation of existing human-centred LA/AIED systems. Additionally, none of these works have discussed how extensively the perspectives of each educational stakeholder (i.e., teachers, students, and experts) have been incorporated at the various design phases. This highlights a pressing need to explore the extent of each educational stakeholder's contribution to the design process of current human-centred LA/AIED systems. This motivates our second research question:

> **RQ2:** To what extent have educational stakeholders contributed to the design process (phases) of current human-centred LA/AIED systems?

Although the involvement of stakeholders in the design of LA/AIED systems has received significant attention (e.g., Lawrence et al., 2023; Holstein et al., 2019; Kaliisa et al., 2023) other key aspects of human-centredness, such as balancing system features for end-users against fully automated features focused on technical goals or replacing human activity, are also important (Lang and Davis, 2023; Usmani et al., 2023). LA/AIED researchers have suggested various ways to empower users, such as offering control over automated recommendations (e.g., Ma et al., 2022; Lawrence et al., 2023) or enhancing the explainability of AI outcomes (e.g., Khosravi et al., 2022; Kloos et al., 2022). However, the balance between human control and computer automation in existing human-centred LA/AIED systems remains underexplored. Additionally, the importance of empowering stakeholders in light of rapid AI advancements was emphasised in a systematic review by Dogan et al. (2023), which focused on the implementation of AI in online learning but not in other LA/AIED system types like visualisation/dashboards, intelligent tutoring systems, and recommender systems (Kaliisa et al., 2023; Williamson and Kizilcec, 2022; Lin et al., 2023; da Silva et al., 2023). To our knowledge, no studies have examined the level of human control and computer automation in designing existing human-centred LA/AIED systems. The two-dimensional HCAI framework proposed by Shneiderman (2020b) (see Section 2.1.1) provides a frame of reference to analyse the human control/automation balance. This motivates our third research question:

> **RQ3:** What levels of human control and computer automation have been considered in various types of human-centred LA/AIED systems designed with the involvement of stakeholders?

As mentioned above, some researchers have increasingly recognised the importance and potential of HCAI in the current LA/AIED literature landscape (e.g., Renz and Vladova, 2021; Zhao et al.). One critical aspect that has garnered considerable attention within the HCAI domain is the assurance of safety, reliability, and trustworthiness in such systems. For example, Renz and Vladova (2021) highlighted the importance of incorporating human values and ethical considerations while developing AI systems for personalised learning environments. The authors visualised the applications of AIED systems (Renz and Vladova, 2021, fig. 4, p.12) in an adapted Shneiderman's (2020b) two-dimensional HCAI framework, by adding a *trustworthiness dimension* perpendicular to human-AI augmentation and types of machine learning. In a more recent LA study, Zhao et al. proposed a framework based on HCAI to identify the most effective learning strategies highlighting the significance of *reliability* of the AI algorithms. In AIED literature, Yang et al. (2021) argued that AI systems should be designed to be transparent, explainable, and accountable to reduce the risk of algorithmic bias and misuse of AI. While these examples recognise the critical role of investigating *safety, reliability, and trustworthiness* principles, there is a lack of understanding of how these principles have been considered when designing human-centred LA/AIED systems. This motivates our fourth research question:

> **RQ4:** How and to what extent have the HCAI principles of safety, reliability, and trustworthiness been discussed in existing human-centred LA/AIED systems?

3 Method

3.1 Review Procedures

To conduct the systematic literature review, we followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol (Page et al., 2021), which has four phases and aims to promote transparent reporting. We searched four reputable bibliographic databases, including Scopus, ACM Digital Library, IEEE Xplore, and Web of Science, to find high-quality peer-reviewed publications on HCAI in LA, and where the relevant LA/AIED (e.g., *Learning Analytics and Knowledge* – LAK and *Artificial Intelligence in Education* – AIED) conferences are commonly published (i.e., ACM and Scopus, respectively). The initial query that was used for the title, abstract, and keyword search of peer-reviewed publications included the following groups of keywords:

- *(human-cent*red OR user-cent*redness OR "value-sensitive design" OR "co-design" OR "participatory design" OR "co-creation" OR design process) AND*
- *("learning analytics" OR education* OR student* OR teacher* OR classroom) AND*
- *("artificial intelligence" OR "intelligent augmentation" OR tool* OR system OR automation OR AI OR analytic* OR algorithm* OR visuali*ation OR "dashboard" OR "interface").*

To focus on studies published in the established research field of LA and AIED in the last decade, a publication year constraint was used to search for relevant publications from January 1, 2012 to November 1, 2023. By focusing on the past 10 years, this SLR captures the most recent advancements in LA/AI technologies and their educational applications, providing an up-to-date understanding of the field's current state. The database search initially yielded 1,678 articles with 511 duplicates removed (14 duplicates were identified and merged manually), leaving 1167 articles for the title and abstract screening process (see Figure 1). To ensure a thorough and accurate screening, three researchers independently reviewed the titles and abstracts of eligible articles according to three predetermined inclusion and exclusion criteria, as follows:

1. First, we included articles that reported empirical studies for developing LA/AIED systems (long papers). These included studies proposing a design framework that may use tools/interfaces for illustration/explanation in practice. We *excluded* non-empirical studies, such as theoretical, opinion/positioning, dataset, literature review, and short papers.
2. Second, we included articles that reported on the studies that aimed to design or develop data-intensive education systems/tools/interfaces for end-users and excluded studies that only mentioned "education", "learning", "LA", or "AI" as an example of other more general topics.
3. Third, we only *included* articles that involved stakeholders (e.g., teachers, students, administrative staff, and designers) in the study and *excluded* studies that focus merely on advancing technical aspects of the technology (e.g., improving the accuracy of AI algorithms) without directly studying the implications for human learning.

At least two researchers reviewed each article (R1-R2), and a third researcher (R3) resolved the conflicts through in-depth simultaneous discussions until a consensus was reached. After the title and abstract screening, a total of 272 articles (as seen in Figure 1) were identified as candidates for full-text review. The inter-rater reliability among the three researchers was 0.80 (R1&R2), 0.88 (R1&R3), and 0.74 (R2&R3), as measured by Cohen's kappa, indicating a substantial to high agreement among the reviewers (Page et al., 2021).

After the full-text review, 164 articles were excluded following our exclusion criteria (detailed above), consisting of *not aiming to design or develop LA/AIED systems for end-users* (n=81), *non-empirical study* (n=53), *merely mentioned "education"/"learning"/"AI"/"LA" as examples* (n=15), *focused on advancing algorithms/technologies without learning implications* (n=9), *lack of full-text access* (n=6). This means that 108 articles were chosen for data extraction, and the inter-rater reliability (Cohen's kappa) for the full-text screening was 0.71 (R1&R2) and 0.67 (R1&R3), indicating a substantial agreement between the researchers. Conflicting decisions were resolved through in-depth discussions among researchers (R1–R3) or consulting a fourth researcher to reach a consensus (Yan et al., 2022). The following subsections describe the data extraction and analysis processes to address each research question.

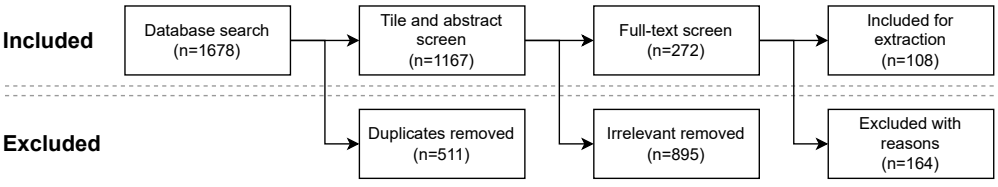


Figure 1: The PRISMA protocol as applied in the current systematic literature review

3.2 Data extraction and analysis

To address **RQ1**, we collected the study characteristics, including targeted educational levels (i.e., K-12, middle school, high school, and university/college) and employed research methodologies (i.e., qualitative, quantitative, or mixed). Next, we employed the five multifaceted design phases by Hanington and Martin (2012, see Section 2.1.3) to identify in which phase(s) stakeholders' views or inputs were considered in the studies included in the review. We also collected information about what HCD techniques were utilised in the study (e.g., co-design, prototyping, interviews, and storyboards). It is important to note that each study could employ multiple HCD research techniques throughout

different design phases. We presented our findings in the form of a Sankey Diagram to inductively analyse relationships between characteristics, design phases, and HCD techniques.

To address **RQ2**, we extracted the information on stakeholders' involvement as active or passive, aligning with their definitions presented above (in Section 2.1.3). We further identified the level of involvement based on stakeholders' roles, such as students, teachers, subject experts, and administrators. To ascertain the impact of stakeholder involvement on the design of the LA/AIED system, we mapped their active or passive involvement status in each of five multifaceted design phases proposed by Hanington and Martin (2012, see Section 2.1.3). For example, Holstein et al. (2019) actively involved teachers in developing a classroom orchestration system for their use, representing active involvement. On the other hand, Fernandez Nieto et al. (2022) consulted teachers in designing a system for student use, in which students who would be the intended users were not involved in any of the phases of the design process. All this information was subjected to quantitative analysis by comparing and employing descriptive statistics as the primary analytical approach for reporting results. The percentages presented in the next section were calculated based on the total of the included studies (n=108).

To address **RQ3**, first, the proposed human-centred LA/AIED system described in each article was categorised based on the definition of the two-dimensional HCAI framework (Shneiderman, 2022), which included the four quadrants of human control and computer automation outlined in Section 2.1.1. Then, we further mapped these systems with the aforementioned stakeholder involvement (active or passive, see 2.1.3). We also collected the type of LA/AIED system (i.e., visualisation/dashboard, intelligent tutoring system, learning design tool, recommender system, prediction system, information retrieval assistant, and evaluation of course essay system) presented in each article (Zhang and Aslan, 2021; Granić, 2022; da Silva et al., 2023). This gathered information underwent inductive analysis through a comparative approach, incorporating descriptive statistics and supported by illustrative examples.

Lastly, regarding **RQ4**, any discussion about HCAI principles within the scope of safety, reliability and trustworthiness was extracted with a question *"How did authors discuss safety, reliability, or trustworthiness?"*. We qualitatively analysed these discussions using inductive thematic analysis (Braun and Clarke, 2012). We prepared and applied an initial coding scheme that pertained to safety, reliability and trustworthiness (as defined in Section 2.1.2). In this case, data *safety* included the subcodes: data privacy, data sharing, and transparency in data collection. *Reliability* encompassed the subcodes: data accuracy, completeness, bias, validity, and consistency. *Trustworthiness* included the subcodes: trust, transparency, and accountability. The coded data was cross-checked between two researchers concurrently, and any emerging codes or conflicts were jointly agreed upon in iterative discussion sessions until consensus was reached. Lastly, the emerging themes were reported by classifying them into the context of safety, reliability and trustworthiness.

4 Results

4.1 Current research on human-centred LA/AIED systems (RQ1)

Most studies were reported at a University/college level (53%), followed by K-12 (34%) and informal learning (13%). The type of research method reported in these studies was distributed as follows: qualitative (44%), mixed-methods (43%), and quantitative (13%). Regarding the design phases of the HCD process, we found that 18% of the studies addressed Phase 1 (planning, scoping, and definition). Phase 2 (exploration, synthesis, design implications) was addressed in 53% of the studies, while Phase 3 (concept generation and early prototype iteration) accounted for 51%. Phase 4 (evaluation, refinement, and production) received the highest attention at 56%. However, only a small minority, 9% of the studies, considered Phase 5 (launch and monitor).

Next, common HCD techniques used in the design process include interviews (52%), questionnaires (47%), co-design (28%), prototype validation (26%), focus groups (10%), observations (10%), and surveys (8%). Less used techniques included workshops (5%), storyboards (4%), personas (4%), and card sorting (3%). Very few studies used other techniques that are common in wider HCD research, such as Wizard-of-Oz (Schulz et al., 2022; Vinella et al., 2022; Echeverria et al., 2023), speed dating (Holstein et al., 2019; Tenório et al., 2022), think-aloud (Conijn et al., 2020b; Ahn et al., 2021), eye tracking (Lallé et al., 2017; Mangaroska et al., 2018), and user journeys (Weigand and Kindsmüller, 2021).

Figure 2 illustrates the characteristics of current research focused on human-centred LA/AIED systems. In summary, we found that most studies have been conducted at the university level, and studies have mostly involved qualitative or mixed-method research. More attention has been given to Phases 2, 3, and 4 of the HCD process, while Phases 1 and 5 received less attention. There was a shift from a more qualitative methodology approach in Phase 2 to a more mixed methodology in Phase 4 as the design tended to mature. Quantitative research has not been employed in Phase 1. Yet, some quantitative methods are used at later stages of the design process (phases 4 and 5), particularly in the form of

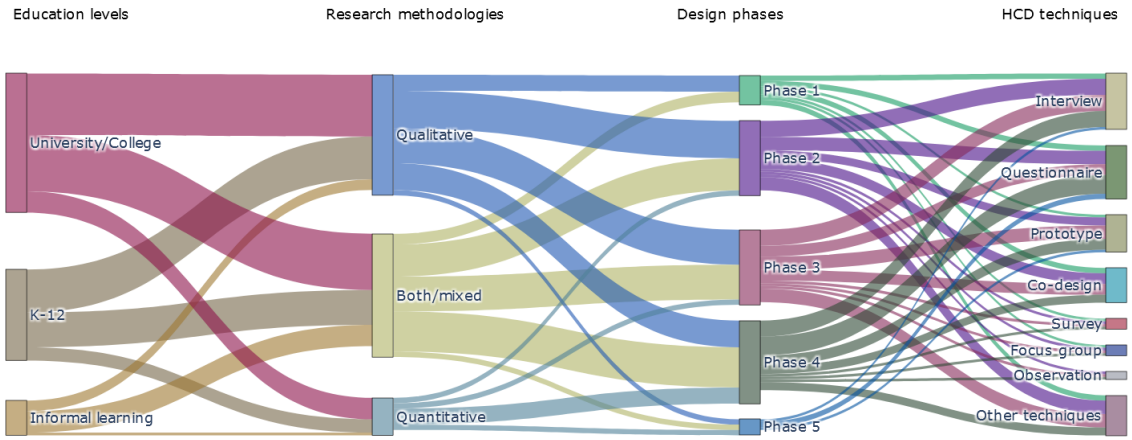


Figure 2: The distribution of key characteristics in the current research state of human-centred LA/AIED systems, depicted through a Sankey diagram. Categories are sorted by frequency. The diagram illustrates the distribution of education levels, the utilisation of research methodologies (qualitative, quantitative, or mixed), the progression through five multifaceted design phases, and the application of HCD techniques. The varying thickness of the flow path represents the significance of each transition.

evaluation questionnaires. Last but not least, interviews have been the most commonly used HCD technique (20%) in most design phases, except for Phase 4 (5%), where questionnaires were more prevalent (7%).

4.2 The extent of stakeholder involvement in design (RQ2)

Concerning stakeholder involvement in the design process, stakeholders were classified into active and passive roles (see definitions in Section 2.1.3). Figure 3 depicts each design phase with the percentage of included studies covering stakeholders' active/passive involvement.

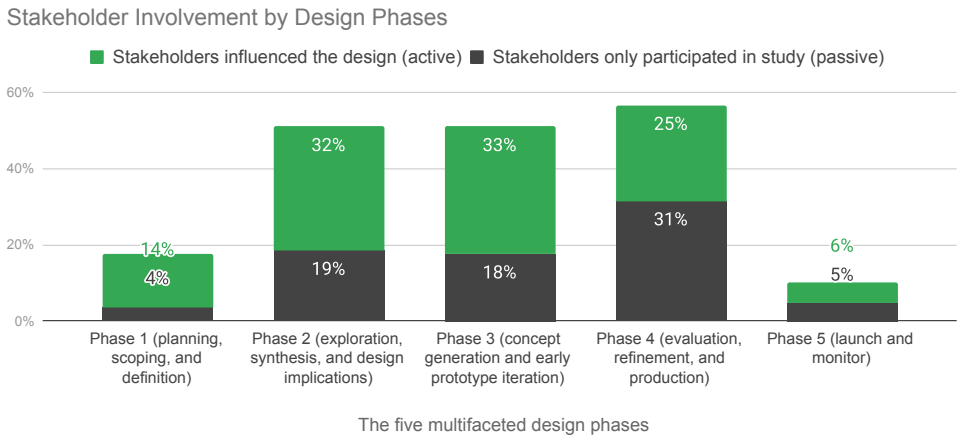


Figure 3: Distribution of stakeholder involvement varies across various design phases. The data demonstrates shifts in passive (black) and active (green) involvement throughout design phases, with more passive stakeholders in Phase 4.

Overall, lower (both active and passive together) stakeholder involvement has been observed in Phases 1 (18% – 4% passive and 14% active) and 5 (10% – 5% passive and 6% active), while there has been more stakeholder involvement in other design phases (Phase 2 – 51%, Phase 3 – 51%, and Phase 4 – 56%). Our findings also indicated a higher **active** stakeholder involvement in Phase 2 (32%) and Phase 3 (33%) compared to the **passive** involvement. Notably, the design **Phase 4** – evaluation, refinement, and production – was the only phase that had higher passive (31%) compared to active (25%) stakeholder involvement. This finding suggests that while most of the works are concentrated on evaluating the design of their LA/AIED systems, there is a tendency for the outcomes produced by stakeholders not to be further considered for improving these designs. For example, Khosravi et al.'s (2019) study illustrates this *passive* stakeholder

involvement, where the authors only reported on students' evaluation of a system through surveys and lab experiments. In contrast, Wiley et al.'s (2023) work illustrates *active* involvement in which teachers participated in several iterative evaluation sessions to re-design an LA dashboard based on their continuous input and feedback.

Furthermore, our analysis revealed that students and teachers were involved in 71% and 59%, respectively of the total number of included studies (see Figure 4). Subject experts made up 17%, administrators 8%, and educational designers 4%. The remaining 9% included professionals such as software developers (e.g., Wiley et al., 2023), evaluators (e.g., Ocumpaugh et al., 2017), and counsellors (e.g., Cukurova et al., 2017).

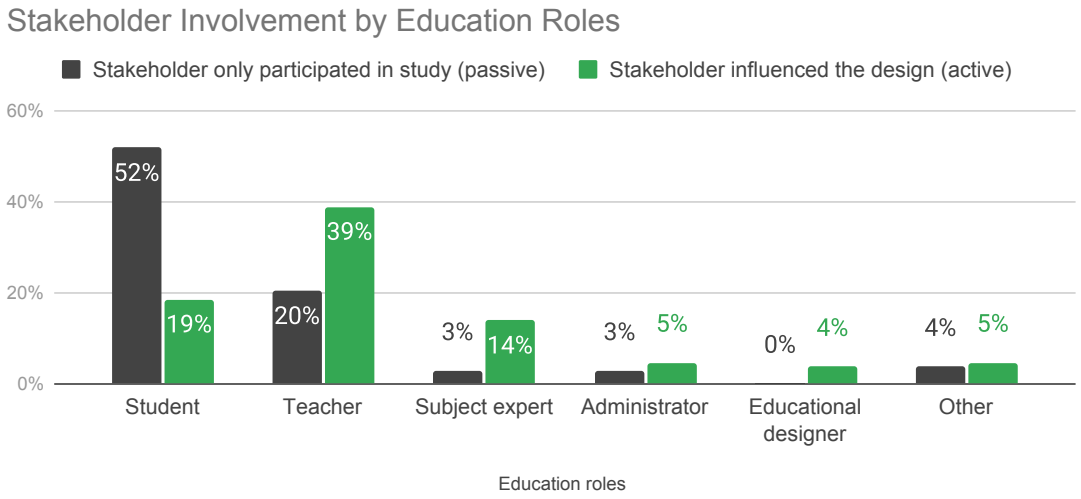


Figure 4: Distribution of stakeholder involvement in the design of human-centred LA/AIED systems. The green bar represents the percentage of stakeholders actively contributing to and influencing the design. In contrast, the black bar represents the percentage of stakeholders participating in the study with limited influence on design outcomes.

Students exhibited the highest representation of passive involvement at 52%. For instance, students are often involved in the evaluation of the system (e.g., Ochoa and Dominguez, 2020; Lim et al., 2021), or being observed on their behaviours during a learning activity while using the system (e.g., Khosravi et al., 2019) but this does not necessarily means they are actively involved in the design of such systems. The active student involvement can be observed, for example, in the form of participating in ideas generation design activities (e.g., de Quincey et al., 2019; Wang et al., 2022). Conversely, despite students being the most involved stakeholders, their active involvement was comparatively lower than that of teachers (39%). In these studies, teachers were involved in co-design activities, from inception to evaluation, where the systems were designed to meet their teaching needs and preferences (e.g., Lawrence et al., 2023; Rodríguez-Triana et al., 2021; Olsen et al., 2021). An example of passive involvement of teachers (20%) is when they are invited to explore the system, thereby offering researchers valuable insights into the sensemaking process without clarifying whether such insights are further considered in the design or re-design of the system (e.g., Fernandez-Nieto et al., 2022; Alfredo et al., 2023).

Subject experts (e.g., Eradze et al., 2020) and educational designers (e.g., Tsai et al., 2022), 14% and 4% respectively, were more actively involved through participatory design. Administrators and others had almost equal proportions of active roles at 5% (e.g., De Silva et al., 2022; Bonnat and Sanchez, 2022). In summary, students and teachers were the most involved education stakeholders, with students having the highest representation of passive involvement and teachers having the highest representation of active involvement.

4.3 The levels of human control and computer automation (RQ3)

We examined the distribution of LA/AIED systems across the four quadrants of the HCAI framework (see Section 2.1.1). This distribution is summarised in Figure 5 and the details are in the Appendix (see Table 1).

Notably, a higher proportion of human-centred LA/AIED systems have already considered substantial human control features in their design, with a more significant percentage of systems categorised in Q4 at 47% (e.g., Pozdniakov et al., 2022; Alfredo et al., 2023) and in Q2 at 32% (e.g., Hu et al., 2022; Jeong et al., 2023). In contrast, a smaller proportion is found in Q1 at 12% (e.g., Kivimäki et al., 2019; Fernandez-Nieto et al., 2021) and in Q3 at 10% (e.g., Ochoa et al.,

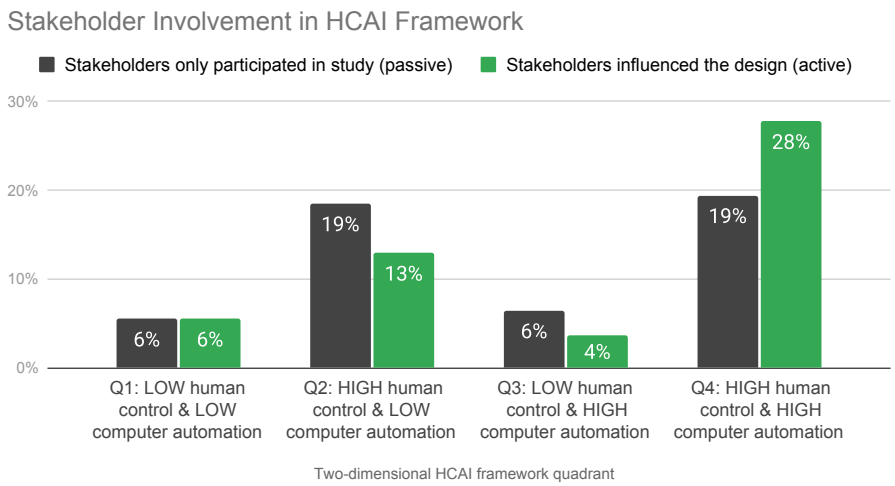


Figure 5: Distribution of human-centred LA/AIED systems categorised into the human control and computer automation quadrants of the HCAI framework (Q1-Q4, see definitions in Section 2.1.1). The green bars represent studies where the stakeholders actively engaged and contributed to the design of the tools. The black bars represent studies where stakeholders merely participated as test subjects, offering limited input during the design phase of LA/AIED systems.

2018; Ocumpaugh et al., 2017). Next, we present the findings of Q4 and Q2, as these were the most frequent categories, followed by Q1 and Q3, which were the least represented categories.

LA/AIED systems in **Q4** have had more active stakeholder involvement (28%) compared to systems in other quadrants, where stakeholders have often been passively involved. One example for Q4 is illustrated in Lawrence et al. (2023)'s work, in which teachers were actively involved in several design phases during a multi-year study to build an AI-powered classroom orchestration system that explicitly considers teachers' agency (i.e., giving the control to teachers to accept or reject recommendations coming from the AI). In contrast, passive stakeholder involvement in Q4 (19%) include studies where researchers documented observations and behavioural multimodal data (eye tracking data, interaction log files, and other physiological sensors) from students interacting with an AI-powered system (i.e., intelligent tutoring system) (e.g., Lallé et al., 2017).

Next, systems in **Q2** had less active stakeholder involvement in their design processes (13%) compared to a higher passive involvement (19%). In this quadrant, active stakeholder involvement includes participatory design activities to understand how stakeholders use systems that require customisation (i.e., manually selecting options to generate visualisations) on exploratory tasks, for example, how teachers use a data visualisation inquiry tool that requires them to control and customise visualisations (e.g., Shreiner and Guzdial, 2022). In contrast, passive involvement may be observed in the form of usability studies where participants are asked to evaluate systems through surveys or questionnaires, such as in the work by Muslim et al. (2016), where students and teachers were asked to rate the usefulness of a rule-based system to support flexible definition and dynamic generation of indicators to meet the needs of different stakeholders with diverse goals and questions (i.e., exploratory tasks using LA dashboards). This can be considered an indirect influence on design outcomes.

Regarding systems in **Q3** which feature high automation and low human control, we found less active stakeholder involvement in their design process (4%) compared to passive involvement (6%). For Q3, active involvement can be illustrated through co-design sessions aimed at identifying features that could be included in intelligent systems, for example, by including teachers' perspectives in the design of two modalities of a robot to support social interactions (as a social actor) or knowledge acquisition (as a didactic tool), and that would run fully automated during authentic classroom use (e.g., Ekstrom and Pareto, 2022). As for passive stakeholder involvement, these studies have commonly collected diverse log data and usability questionnaires to understand users' attitudes towards the use of automated intelligent support, such as the work presented by Wilson et al. (2021), in which teachers and students were asked to evaluate a fully-automated writing score system using usability and attitude questionnaires.

Lastly, systems in **Q1** that feature low human control and automation had an equal proportion of active and passive involvement, at 6%. An active involvement is exemplified in Garcia-Ruiz et al.'s (2022) study, in which participatory design activities were conducted with students (e.g., initial exploration, focus group discovery, and collaborative

prototyping with experts) to design a novel visualisation tool with a limited configuration setting. In contrast, Fernandez-Nieto et al.'s (2021) work reflects passive involvement, as teachers participated in interviews to share their sensemaking on a low-fidelity dashboard prototype, in which the prototype was manually generated, and the outcomes were not part of further studies. In summary, education stakeholders were more involved in Q4 than in other quadrants, indicating active stakeholder involvement in influencing design outcomes on systems considering end-user agency and using AI or advanced computer automation.

Concerning the different types of LA/AIED systems, as depicted in Figure 6, we found that the most prevalent type is visualisation or dashboards, accounting for 53% of the total of the included studies.

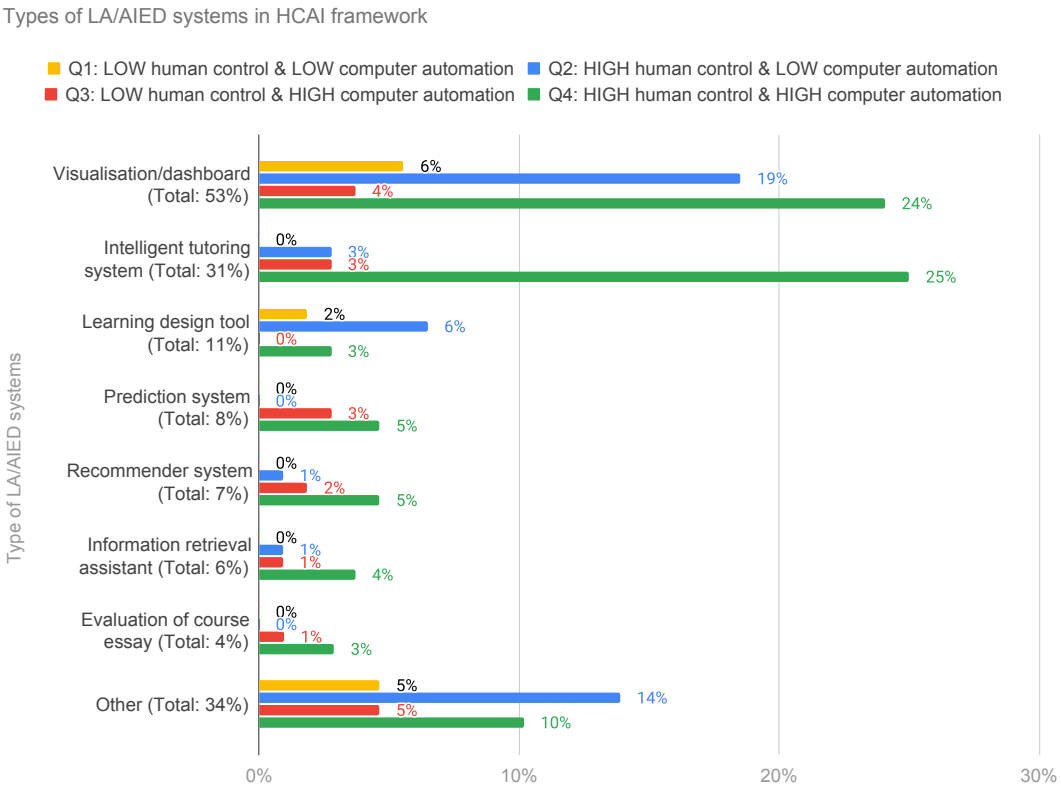


Figure 6: Distribution of human-centred LA/AIED systems on the two-dimensional HCAI framework by type of system. Each colour represents different quadrants in the HCAI framework (Q1-Q4, see Section 2.1.1).

The most prominent quadrant in this type is Q4, which comprises 24% of the total included studies. This type involves a high level of human control and computer automation, such as educators actively utilising and interpreting the visualised data generated by an AI-powered system which automatically make inferences by analysing complex and large amount of students' learning data (e.g., Alfredo et al., 2023; Nazaretsky et al., 2022b; Conijn et al., 2020b). The next most represented category is intelligent tutoring systems at a total of 31% (e.g., Holstein et al., 2019; Gerdes et al., 2017; Ngoon et al., 2023). Q4 also represents the largest quadrant for this type (25%), where high human control is given to users, such as overseeing learning activities (e.g., Tissenbaum and Slotta, 2019), accepting or rejecting automated interventions or recommendations from the system (e.g., Di Mitri et al., 2022), and making real-time adjustments (e.g., Lawrence et al., 2023).

Next, learning design support for teachers is represented by a total of 11%. The most prominent quadrant for this type is Q2 at 6% (e.g., Michos et al., 2020; Vezzoli et al., 2020; Pishtari et al., 2021), which favours high human control, and Q4 is the second highest at 3% (e.g., Rodríguez-Triana et al., 2021; Albó et al., 2022; Kaliisa et al., 2020), which may indicate learning design systems have lacked computer-automation. Learning design systems commonly consider high human control, as exemplified by the agency of learning designers in applying the inquiry process where they possess the autonomy to actively shape and refine learning experiences (e.g., Pishtari et al., 2021; Rodríguez-Triana et al., 2021).

The prediction system type accounts for a total of 8% and only has high computer automation (Q3 at 3% and Q4 at 5%). An example of Q3 of this type is in Ocumpaugh et al.'s (2017) study, where the system was trained on historical students' data to identify patterns, make predictions on students' engagement, and produce non-configurable early warning reports for educators to interpret, without having any further control on the system. On the other hand, an example in Q4 can be seen in Duan et al.'s (2023) study, where the teachers have more control over configuring the model and representation of prediction results to identify at-risk students, which ensures predictions are trustworthy and can support proper interventions. Evidently, no prediction systems are at low-level computer automation, as evidenced by no studies found in Q1 and Q2, since all prediction systems require a high level of computer automation (i.e., AI models) to operate.

The remaining three types—recommender system, information retrieval assistant, and evaluation of course essays—have Q4 as the most prominent quadrant. Recommender systems are represented by total of 7%, with 5% in Q4, that includes course recommender system (e.g., Chang et al., 2023), learning resources recommender (e.g., Ruiz-Calleja et al., 2019), and personalised learning recommendations (e.g., Khosravi et al., 2019). Information retrieval assistant comprises total of 6%, with 4% in Q4 (e.g., Bonnat and Sanchez, 2022) and evaluation of course essay 4%, with 3% in Q4 (e.g., Lee et al., 2023). Systems in Q4 across these three types have similarities where automation or an intelligent system relies on teachers' or students' oversight to interpret, make judgments, and contextualise the information from the provided interface to take actions in their teaching or learning environment close to real-time.

The other types of educational technologies accounted for smaller percentages, ranging from 1% to 5% of the total studies, adding up to 33%. These include game-based learning system (e.g., Tenório et al., 2022; Shute et al., 2021), online learning system (e.g., Martinez et al., 2020), mixed-reality systems such as augmented-reality and virtual-reality (Bonnat and Sanchez, 2022; Zwolinski et al., 2022; Kang et al., 2020), and social robots (e.g., Ekstrom and Pareto, 2022; Muldner et al., 2013). In studies about social robots, such as in Ekstrom and Pareto's (2022) study, Wizard of Oz tended to be employed in which the system was manually controlled by researchers in the background, but students still had full control of the interaction, such as having a flexible conversation with a robot to develop empathy and communication skills. Among these, the most notable quadrant when aggregated was Q2 (14%), in which students had full control when interacting with the system (i.e., playing learning games (Wang et al., 2019) or interacting with peers in online learning (Cukurova et al., 2017)), which was manually prepared with a pre-defined set of rules towards intended learning outcomes, without requiring advanced automation.

In summary, regardless of the level of computer automation, each type of LA/AIED system has already considered a high level of human control, indicated by a higher percentage of studies categorised in Q2 or Q4 for each type. These findings suggest that most reviewed human-centred LA/AIED systems have been designed to empower users with the support of human oversight.

4.4 Safety, reliability, and trustworthiness discussion (RQ4)

Of the total number of human-centred LA/AIED systems' studies reviewed, 54% contained a discussion or consideration of the HCAI principles of **safety**, **reliability**, and **trustworthiness** to some extent. The individual proportions were 35%, 37%, and 25%, respectively. Next, we describe the topics that emerged from each principle as a result of our thematic analysis.

4.4.1 Safety

Data privacy emerged as the most topic (18%), emphasising its significance when designing and implementing safe LA/AIED systems. Data privacy refers to the protection and control of personal and sensitive information of students, teachers, or any other individuals involved in the learning process. These studies involved discussions about safeguarding data related to personal identification (e.g., Kivimäki et al., 2019; Ngoon et al., 2023), academic performance (e.g., Echeverria et al., 2019), learning progress (e.g., Ma et al., 2022), and other sensitive details collected and processed by human-centred LA/AIED systems (e.g., Santos et al., 2012). Notably, discussion about privacy could influence stakeholders' willingness to adopt these systems in practice (e.g., Ma et al., 2022; Garcia-Ruiz et al., 2022).

The discussion of **data sharing** primarily focused on strategies to overcome privacy concerns (7%). Four studies advocated for anonymity features when sharing data as measures to protect participants' identities from others, promoting a more open discussion of learning activities (e.g., Echeverria et al., 2019; Conijn et al., 2020b; Sato et al., 2023; Barreiros et al., 2023). These anonymity features could also help students recall and discuss performed actions without feeling judged by their peers (i.e., Echeverria et al., 2019). Moreover, six studies discussed safety strategies for data sharing as part of their design methodologies. Two of them incorporated design principles such as "Privacy by Design" criteria (i.e., Bonnat and Sanchez, 2022), allowing privacy and design to co-exist, and "Risk Communication Principles" (i.e., Ocumpaugh et al., 2017), allowing communicating the risk involved and the certainty when presenting

data in a way that both experts and non-experts (i.e., students) can easily understand during design activity. Other studies leveraged co-design methods with teachers (e.g., Ngoon et al., 2023; Pishtari et al., 2021) and students (e.g., Lee et al., 2022; Garcia-Ruiz et al., 2022), to explore the type of data that can be shared with other users without breaching their privacy.

We found some discussions about **data collection** impact and procedures (10%). Two studies reported conflicting opinions from participants regarding the benefits and concerns associated with collecting and utilising student data (i.e., Wang et al., 2022; Kivimäki et al., 2019). For example, Wang et al. (2022) reported that several students expressed worries about potential privacy breaches and discomfort in asking questions in an online forum (due to being tracked and continuous data collection), but they wanted to have a more personalised learning experience from the system. It exemplifies how students may want something beneficial for their learning with an intelligent system but are unaware that such a system requires data collection. It might be necessary to establish clear communication between researchers and participants through informed consent. Five studies highlighted the importance of informed consent procedures to ensure transparent data collection (i.e., Conijn et al., 2020a; Khosravi et al., 2019; Wang et al., 2022; de Quincey et al., 2019; Ngoon et al., 2023). By acquiring explicit consent from students (e.g., opt-in/out process) and allowing users to control data access and explaining how the system operates (i.e., Bonnat and Sanchez, 2022; Wang et al., 2022), human-centred LA/AIED systems could potentially deliver a sense of safety to users. Yet, de Quincey et al. (2019) argued that some participants may overlook the privacy policy of the informed consent process because they may not have read the information thoroughly.

Moreover, **monitoring and surveillance** based on student data were discussed in four studies (i.e., Pozdniakov et al., 2022; Zhou et al., 2021; Alfredo et al., 2023; Tsai et al., 2020). For instance, Zhou et al. (2021) reported the experiences, privacy concerns, and impacts of providing students awareness for using monitoring tools in a collaborative learning setting (i.e., remote group meetings). They reported changes in students' behaviours, such as half of students feeling more motivated to engage and be more productive in the group conversations, while half of students still felt intense pressure and uncomfortable being monitored. In another study, Alfredo et al. (2023) further advised against using physiological data modelling and visualisation for surveillance purposes, including scenarios like exams and regular classrooms, as well as measuring students' performance. Another study by Ngoon et al. (2023) suggested that institutions that plan to adopt smart classroom systems with continuous monitoring features should conduct regular evaluations with both students and teachers, such as addressing privacy concerns through an iterative co-design process. Finally, a small number of studies (3%) proposed strategies to ensure secured data access (**data security**), such as two-factor authentication (i.e., Wang et al., 2022), using third-party software to establish privacy and security policies (Santos et al., 2012), and allowing limited data accessibility and visibility to relevant stakeholders (i.e., students) intended only for academic purposes (Martinez et al., 2020).

4.4.2 Reliability

In terms of reliability, we found that this principle is often addressed in terms of the accuracy of the system, the potential bias of these systems, and strategies to produce reliable data. We describe each of these topics below.

From the set of studies, we found **accuracy** (14%) as a key aspect to evaluate the system's reliability. These studies often consider two factors that could impact the accuracy and perception of the system's reliability: *machine algorithm* (3%) and *human interpretation* (11%). Accuracy in *machine algorithm* refers to the ability of the system to provide correct/accurate outcomes. Two studies discussed approaches to evaluate AI model accuracy that can lead to reliable outcomes (e.g., Chang et al., 2023; Khosravi et al., 2019). For example, Chang et al. (2023) used a mixed-method approach that integrated students' survey data to evaluate the accuracy of their proposed machine-learning model. Similarly, in a study on recommender systems, Khosravi et al. (2022) reported initial results about incorporating both students' subjective opinions and machine-learning algorithms, which can potentially improve the accuracy of the system in determining learning resource quality. On the other hand, accuracy in *human interpretation* refers to the significance of how end-users (teachers/students) correctly interpret and understand indicators presented by the LA/AIED systems.

Six studies reported stakeholders' responses about receiving **inaccurate** information or conclusions from a system that can introduce unreliability due to misinformation (i.e., Liaqat et al., 2021; Pozdniakov et al., 2022; Shreiner and Guzdial, 2022; Sato et al., 2023; Fernandez-Nieto et al., 2021; Kang et al., 2020). For example, in Shreiner and Guzdial's (2022) study, teachers critically analysed data visualisations to support students in learning data visualisations creation with a computer. Teachers discovered many different ways in which data visualisation generated by computers can be misleading. Specifically, teachers felt less confident in their own abilities to identify flaws and inaccuracies in computer-generated data visualisations. In their further exploration, authors reported teachers felt more confident creating data visualisations with paper and pencil over using computers. As a result, only a small number of the teachers who were supposed to use this system have actually incorporated them into their classrooms. Moreover, seven studies

reported how stakeholders questioned the accuracy of the results from LA/AIED systems (i.e., Ahn et al., 2021; Zhou et al., 2021; Shreiner and Guzdial, 2022; Barreiros et al., 2023), specifically in addressing social needs to promote social learning development (i.e., Wang et al., 2022), emotional support needs (i.e., Liaqat et al., 2021), and stress (i.e., Alfredo et al., 2023). For instance, in a couple of studies (i.e., Ahn et al., 2021; Zhou et al., 2021), study participants doubted whether the data they saw in the system's interface reflected what occurred in the learning activity. An example from Kang et al.'s (2020) study addresses this unreliability issue by providing students with control to correct the mistakes made by AI (i.e., incorrectly assigned labels to an interface). At the same time, it promotes students' agency over the tool to acknowledge that the system can be unreliable and corrections may be needed during use.

Another topic was **human bias** (4%), which refers to systematic and unfair preferences influencing the outcomes, decisions, or interpretations generated by systems (e.g. Fernandez-Nieto et al., 2022; Crain et al., 2022; Pozdniakov et al., 2022; Alfredo et al., 2023). Two studies highlighted the possibility of bias risk in LA/AIED systems due to the perceptions and preferences of users who have control over them (e.g. Fernandez-Nieto et al., 2022; Crain et al., 2022), arguing that excessive human control could lead to over-interpretation of data. The other three studies reported that insights formulated based on inaccurate visual analytics might be prone to interpretive bias (Pozdniakov et al., 2022; Alfredo et al., 2023; Fernandez Nieto et al., 2022). We also gathered several discussions about human bias in the design process, as examined by four studies (4%). Two studies explicitly discussed the risk of bias in the HCD process (i.e., Vinella et al., 2022; Di Mitri et al., 2022). For instance, Vinella et al. (2022) noted that the monetary incentive given to participants when employing HCD techniques (e.g., crowdsourcing in this study) could stir and bias stakeholders from the study's intended purpose. Participants who are more focused on immediate rewards may not necessarily invest the necessary time and effort to provide insights that contribute to the long-term reliability of the system. However, two studies discussed the importance of balancing researchers' bias with stakeholders' needs in the design process (i.e., Vannaprathip et al., 2022; Fernandez Nieto et al., 2022). For instance, Fernandez Nieto et al. (2022) highlighted that 'end-users' (teachers/students) should be actively involved in design and evaluation processes to perceive their real needs in order to minimise researchers' bias. When researchers or designers work in isolation, they may unintentionally introduce their own assumptions and biases into the system. Diverse perspectives are considered when involving end-users, leading to a better alignment with challenges found in authentic educational settings that can be more reliable for end-users use.

Furthermore, another topic that emerged is related to the **strategies to produce reliable data** (5%). Three studies discussed how they ensured the reliability of the data from their studies (i.e., Martinez et al., 2020; Lee et al., 2023; Ocumpaugh et al., 2017). For instance, Martinez et al. (2020) ensured the reliability of their results by applying criteria such as data security and confidentiality in data collection, triangulating data from multiple sources, and reviewing the analysis process for each design phase. Two studies reported that the methodological reliability for involving stakeholders in the design process could be influenced by the quality of evaluations from stakeholders (i.e., Ahn et al., 2019; Zhou et al., 2021). For instance, Ahn et al. (2019) highlighted that methodological reliability from a study might be influenced by how well the involved stakeholders understand the system's *validity* in an educational setting. Validity refers to how well the expectations of those involved match the intended purpose of the system, which promotes consistency and relevancy in evaluating the system to produce reliable data. Moreover, three studies addressed the issue of system validity, noting that while these systems may function effectively in laboratory settings, their sustainability in real-world scenarios remains uncertain (i.e., Zhang et al., 2022; Gibson et al., 2017; Hu et al., 2022). They suggested the need for *ecological validity* (the extent to which the systems and findings from studies can be generalised and applied to authentic educational settings) by incorporating intended stakeholders' perspectives to design a more reliable system.

4.4.3 Trustworthiness

Three emerging topics were identified in relation to the trustworthiness principle, such as *trust* (14%), *transparency* (9%), and *accountability* (3%) in LA/AIED systems.

Trust was discussed in several studies (14%). This concept was contextualised in how the stakeholders *perceive* trustworthiness when using or designing LA/AIED systems (e.g., Duan et al., 2023; Ngoon et al., 2023; Fernandez-Nieto et al., 2022; Ekstrom and Pareto, 2022; Barreiros et al., 2023; Ooge et al., 2023). In terms of evaluating trust of such LA/AIED systems, we have mixed findings on stakeholders' perceived sense of trust. Four studies reported that stakeholders trusted the outcomes of AI or computer automation (i.e., Liaqat et al., 2021; Gibson et al., 2017; Khosravi et al., 2019; Ma et al., 2022). For instance, Liaqat et al. (2021) reported that learners trusted automated feedback more than peer feedback, and Ma et al. (2022) noted that teachers were inclined to trust the system when the displayed data matched their expectations. On the other hand, four studies reported stakeholders were less trusting of outcomes from automation (i.e., Nazaretsky et al., 2022c; Wang et al., 2022; Ooge et al., 2023), especially when the systems utilised emotion or stress data (i.e., Alfredo et al., 2023). For instance, Nazaretsky et al. (2022c) reported that teachers were less trusting of the AI-powered system compared to receiving advice from peer teachers or experts. In another study, Alfredo et al. (2023) reported teachers doubted the system's ability to infer students' physiological

states accurately due the inherent complexity in modelling affective constructs. They suggested additional contextual information (e.g., explainable data) that could support making inferences and explaining this data back to students, especially if the intention is to use these systems in authentic settings. Interestingly, Ooge et al. (2023) argued that plainly giving learners a mechanism to control a recommender system, such as an ability to reveal the number and detail of recommendations during a learning activity, does not necessarily increase their trust in using the system. Instead, the authors highlighted this kind of human control could promote awareness of the algorithm behind computer automation, encouraging learners to reflect on their actions which may foster long-term trust in the system. Together, these results provide important insights into further investigating the nuanced dynamics of trust between stakeholders and the system, which may vary depending on who the end-user is (e.g., teachers or students).

Moreover, four studies discussed several approaches to cultivate stakeholders' trust in the system (i.e., Ahn et al., 2021; Wilson et al., 2021; Suleman et al., 2016; Gibson et al., 2017). For instance, Wilson et al. (2021) reported that balancing the presented information's accuracy with explainable information could foster learners' trust in the system. The authors ensured that the system is available to provide output whenever a learner requests feedback, but the output may be less accurate. This drawback was balanced by providing learners with an explanation that is more connected to the intended pedagogical component (i.e., formative feedback). It exemplifies an approach to improving trustworthiness when using human-centred LA/AIED systems by improving interpretability through explainability and system availability. Gibson et al. (2017) emphasised that stakeholders' trust in LA systems can be developed through reciprocity, meaning there exist continuous design processes that involve active evaluation and the ability to influence the programming code inside the system. The study highlights the importance of cultivating trust between stakeholders and systems to foster trust in using them in the authentic learning context.

Transparency (9%) was the second most discussed topic in the trustworthiness principle. Transparency refers to the clear and understandable way the system gathers and manages information to build trust among stakeholders who use or contribute to the LA/AIED system's design. Five studies discussed the transparency strategies that could foster trust, built around the user's perceptions and control (i.e., Shute et al., 2021; Ahn et al., 2021; Khosravi et al., 2022; Suleman et al., 2016; Dollinger et al., 2019). Shute et al. (2021) highlighted the importance of users' perceptions and their significant role in designing and testing the LA/AIED system impacting trust and engagement. Khosravi et al. (2019) reported that educating students on how their knowledge status is computed by AI and giving them access to the learning model (i.e., opening up the black-box AI models) could ultimately increase their trust and motivation to learn. Ahn et al. (2021) suggested that communicating the perception and intention of data use can foster trust through ongoing interpersonal interactions. Similarly, Dollinger et al. (2019) emphasised that transparent communication and clear research documentation are fundamental for building trust between stakeholders and establishing trustworthiness. This finding highlights that trust and engagement in LA/AIED systems can be positively influenced when users perceive transparency in knowledge computation and transparent communication between stakeholders and researchers is encouraged.

The concept of **accountability** refers to individuals or systems being responsible for their decisions and the consequences of their actions. This is discussed in only four studies, taking a perspective from system usage (i.e., Zhou et al., 2021; Duan et al., 2023; Lawrence et al., 2023) and design (i.e., Gibson et al., 2017). The study by Zhou et al. (2021) sees accountability from a system usage perspective. For them, accountability is not just about the teachers overseeing the system; it also involves students taking responsibility for adjusting their learning strategies based on the feedback they receive from the AI-powered system. This suggests that the accountability of the AI-powered system usage is not solely dependent on teachers but also on the students' willingness to use the information provided by the system in their learning. The importance of accountable use of an AI-powered system is also investigated in Duan et al.' (2023) study, where authors highlighted the need for trustworthy and transparent AI algorithms that align with the needs of stakeholders involvement in real educational settings. The authors recommended exploring explainable AI technologies capable of dynamically incorporating subject-matter experts' (i.e., teachers') contextual insights during the learning process for generating more accountable interventions aimed at helping students accomplish their learning goals. This kind of teachers' interventions are discussed in Lawrence et al.'s (2023) study. Authors highlighted that teachers felt responsible for classroom orchestration decisions made with the assistance of the AI-powered system, thus suggesting that teachers should be encouraged to trust their own judgement and decision-making ability over the AI. Teachers explained they tended to override the system's recommendations with their own judgement, indicating teachers may still hesitate to trust and follow AI suggestions completely. This study may suggest that when teachers have control over the system, the control features force them to be responsible for the system's effectiveness since they are accountable for their decisions, which may give them an extra burden to use in practice.

In contrast, Gibson et al. (2017) define accountability from a design standpoint study. The expectation was for teachers to exhibit a higher level of accountability, implying that they should be more critical during the evaluation phase of the design process. However, it was observed that the teachers did not meet this expectation and their evaluations were not as critical as anticipated. This highlights a need to minimise assumptions about stakeholders' accountability

in the design process. In summary, effective exploration of accountable usage in human-centred LA/AIED systems requires active participation from both teachers and students. However, the limited literature from a design perspective shows a gap in our understanding of the accountability of stakeholders when involved in the design process. Each study interprets its meaning differently, primarily focusing on accountability in system usage.

5 Discussion

5.1 Current state of Human-centred LA/AIED systems

The results from ***RQ1*** indicate that a high number of university/college studies align with other LA/AIED reviews (Sarmiento and Wise, 2022; Granić, 2022). Yet, there are still opportunities to explore human-centred LA/AIED systems in supporting informal learning (e.g. Zhang et al., 2022; Head et al., 2014), as noted by Buckingham Shum et al. (2019). Moreover, the HCD techniques most commonly used in designing human-centred LA/AIED systems that have considered stakeholders' voices include interviews, co-design sessions, and prototype validation, with few studies using other techniques such as surveys, workshops, and personas. Despite traditional techniques (interviews and questionnaires) being helpful, this area requires more innovative HCD techniques to leverage AI's potential challenges and benefits. For example, other techniques and methods that mimic human-AI interaction (i.e., Wizard of Oz (Vinella et al., 2022) or technology probes (Sato et al., 2023)) could be helpful to investigate how users interact with systems features that require complex automation without the burden of full implementation (Lawrence et al., 2023; Echeverria et al., 2023). However, researchers should consider the trade-off between high-fidelity interaction and resources needed, as these techniques and methods may require more resources and expertise than low-fidelity prototypes (Muldner et al., 2013).

The lack of human-centred LA/AIED systems in Phase 1 (18%) and Phase 5 (9%) highlights a need for greater attention to the inception and monitoring phases. Engaging stakeholders at the earliest and latest design phase can provide valuable perspectives that ensure the designed system aligns with real-world learning needs and concerns (Buckingham Shum et al., 2019) and becomes sustainable solutions (Yan et al., 2022). A closer look at the planning, scoping, and definition design phase (Phase 1) indicates quantitative methodologies are scarce. Integrating quantitative methodologies, such as surveys or crowdsourcing, could offer an objective, systematic approach to gathering and interpreting data that could capture ideas at a large scale for subsequent design phases. For example, researchers can conduct surveys on end-users (teachers/students) to assess potential risks or feasibility, which can be cost-effective and scalable; otherwise, it may require more effort and resources when stakeholders are involved in subsequent design phases (Lang and Davis, 2023). Researchers may also consider analysing historical or public data to develop relevant AI-powered features or models depending on the research objectives (i.e., by following data-driven design approaches, Gorkovenko et al., 2020). Moreover, the limited literature in the launch and monitoring design phase (Phase 5) could suggest a gap in reporting practices about LA/AIED systems post-deployment in authentic learning environments (Martinez-Maldonado, 2023) or a lack of longitudinal studies that manage to deploy human-centred LA/AIED systems sustainably.

Therefore, evidenced by our findings, we recommend:

- **Research Scope:** Expand human-centred LA and AI in Education research beyond the university and K-12 levels to include vocational education, workplace training, and informal learning.
- **Inception & Deployment Phases:** Pay greater attention to Phases 1 and 5 of the design process to balance stakeholder involvement, pedagogical focus, and research objectives at both inception (phase 1) and monitoring (phase 5) stages.
- **Quantitative Approaches:** Employ quantitative methods in the planning phase to help reach stakeholder consensus and be resource-effective. Consider large-scale surveys, crowdsourcing, quantified risk analysis, or leveraging public datasets on AI-powered features.
- **In-the-Wild Studies:** Report more in-the-wild studies during Phase 5 to understand the impact of human-centred LA/AIED systems on teaching or learning experiences in real-world practices. Combine this with existing human-centred approaches and design methodologies (e.g., design-based research by Reimann, 2016).
- **Innovative HCD Techniques:** Use innovative Human-Centered Design techniques beyond interviews and questionnaires to leverage AI's potential challenges and benefits. Consider techniques that mimic human-AI interaction, such as Wizard of Oz.

5.2 Stakeholder involvement in Design Phases

Regarding ***RQ2***, lower active stakeholder involvement in evaluating and refining human-centred LA/AIED systems can overlook design flaws that may persist in authentic educational scenarios (see Phase 4). Including their voices in the refinement process, such as conducting post-hoc reflection interviews to capture their experiences for using a system after pilot studies (Martinez-Maldonado et al., 2015), can provide an opportunity for end-users to reflect and re-align systems' features with their values, preferences, and needs (Chen and Zhu, 2019) before the system is finalised and deployed in real-world settings. This active stakeholder involvement should be considered in all design phases, which have currently been considered in Phases 1, 2, 3, and 5. Incorporating a "human-in-the-loop" practice in all design phases can be an approach to tap into education stakeholders' expertise, ensuring a nuanced grasp of context, end-user needs, and ethical considerations (Duan et al., 2023). Allowing end-users to share their voices can also foster a sense of ownership, which aligns with the main goal of human-centred LA/AIED systems to empower end-users (Usmani et al., 2023; Barreiros et al., 2023). Ultimately, this active involvement will foster adaptive and human-centric designs that align more effectively with real-world complexities and requirements (Buckingham Shum et al., 2019; Martinez-Maldonado, 2023). Indeed, it is crucial to acknowledge that a balanced combination of HCD techniques and design phases for involving stakeholders in educational settings, such as availabilities, policies, and resources, remains an open question that requires further exploration. Considering these benefits, this finding highlights the need for stakeholder involvement in all design phases of human-centred LA/AIED systems to co-create systems that can meet real-world educational needs (Dollinger et al., 2019).

The extent of stakeholder involvement indicates that students and teachers play a pivotal role in the design process as the most frequently engaged stakeholders (see Fig. 4). Despite students having the highest representation in the overall participant count (71%), the data highlighted their active involvement in the design process was relatively low (19%). Teachers may have expertise and pedagogy knowledge (e.g., how students learn, effective instructional strategies, and a supportive learning environment). In contrast, students may be just consumers of these systems or partake in another role in the design, such as being observed when exploring the system or participating in experiments to test the newly developed system without giving input or voice to the design. It could be caused by several challenges, such as communication, students' narrowed perspectives or knowledge, monetary incentives (Vinella et al., 2022) and privacy concerns, especially in K-12 education (Bond et al., 2023) (see Section 4.4). This may lead to a heavy reliance on teachers and other experts, which invites further investigation. Same teachers or experts may be more available to contribute to the design process from the beginning until the end of the project than students, which commonly occurs in the longitudinal study (e.g., Hutt et al., 2021).

The challenges above show that students' contributions tend to be punctual (occurring as one time-off) and sporadic (happening sometimes and inconsistently). These uncertainties signal a need for a more inclusive approach to the design process due to the underutilisation of students' expertise and voices in the design process of human-centred LA/AIED systems. To break this pattern, efforts should focus on recognising the value of students' unique insights (Dollinger et al., 2019; Alfredo et al., 2024), empowering them through agency (Hooshyar et al., 2023), and addressing communication problems, such as by clearly stating the research outcomes and benefits of participating in the design activities (Slade and Prinsloo, 2013). It is also essential to encourage collaboration among students, teachers, researchers, and other stakeholders (e.g., developers, designers, and administrators) while establishing transparent communication about the purpose and benefits of student involvement (Martinez-Maldonado, 2023). This multifaceted approach aims to collaboratively create a more inclusive and effective system (Dollinger et al., 2019).

Overall, we found a gap in which students' active involvement is still limited, implying students can be considered as underrepresented stakeholders (Martinez-Maldonado, 2023). Martinez-Maldonado recommended building robust relationships with these underrepresented stakeholders, offering compensation for their time in contributing to design activities and employing inclusive design toolkits that promote inclusivity and diversity. It urges researchers to be more proactive in including students' voices because their learning experiences will be the most impacted by these technologies (Kitto et al., 2018). We further argue that ensuring meaningful student engagement in the design process can enhance the relevance and effectiveness of adopting human-centred LA/AIED systems, promoting a more student-centred approach (de Quincey et al., 2019).

Therefore, we recommend:

- **Active Stakeholder Involvement**: Employ co-creation practices such as co-design, participatory design, and value-sensitive design. Emphasise active collaboration in all design phases and the central role of end-users and directly affected stakeholders (including students) in shaping outcomes.

- **End-User Needs and Preferences**: Ensure the system's outcomes meet the needs and preferences of the end-users, particularly teachers and students.

- **Student Involvement**: Address students' passive involvement in the design process. Recognise that human-centred LA/AIED systems can significantly impact students' learning experiences and outcomes.
- **Clear Communication**: Ensure all stakeholders are well-informed about the benefits and challenges of involving students in the design process.

5.3 Human Control and Computer Automation

Regarding ***RQ3***, by employing the two-dimensional HCAI framework (Shneiderman, 2022) as an analysis lens, the current SLR revealed that current human-centred LA/AIED systems have already prioritised human control with a significant percentage falling in this category (Q2=32% and Q4=47%). Human control is prioritised so students and teachers can be responsible for their learning and teaching practices. With AI-powered systems' help and high human control (Q4), students can have personalised learning experiences, and teachers can provide guidance and support (Dogan et al., 2023). This finding also indicates the need for further exploration of frameworks for collaboration between humans and AI in the LA/AIED systems (Holstein et al., 2020). This approach stands in line with the objectives of HCAI (Shneiderman, 2020b; Usmani et al., 2023; Renz and Vladova, 2021).

Regarding stakeholder involvement in the HCAI framework, notably, Q3 (low human control and high computer automation) has shown more passive stakeholder involvement (6%). The possible reason for this passive involvement, as highlighted by Nazaretsky et al. (2022a), could be a lack of technical expertise among the stakeholders. Systems in Q3 often have simpler interfaces, resulting in lower human control but it can be more complex behind the interface. It may require a certain understanding of technical knowledge, such as computer science or AI literacy, to understand its automation. If teachers and students lack this expertise, they might find it challenging to provide meaningful input in the design process, resulting in them taking a more passive role. Yet, stakeholders' involvement should not only be considered for human-centred LA/AIED systems that offer some sort of end-user interface but also in the design of systems that fully automate educational actions. For systems with low human control and high computer automation, it is important to ensure they are transparent, trustworthy, and user-friendly, as explored in (Duan et al., 2023)'s study (Q4) to balance it with giving more human control. Another example is a ZoomSense system with high computer automation in Zhou et al.'s (2021) and Pozdniakov et al.'s (2022) studies. Initially, Pozdniakov et al.'s (2022) study focused on computer automation with AI but had limited control (Q3). It was later updated by creating a query-driven dashboard that allows teachers to control the online classroom in Pozdniakov et al. (2022)'s study (Q4).

Nonetheless, we also found that systems in quadrant Q2 had more passive stakeholder involvement (19%). Since systems in Q2 are intended to provide an end-user interface that allows teachers or students to take control and make meaningful decisions about their teaching or learning process, it can be seen as short-sighted to neglect active stakeholder involvement just because AI or computer automation is unavailable. Buckingham Shum and Luckin (2019) and Dollinger et al. (2019) argued that designing human-centred LA/AIED systems requires a well-thought-out stakeholder engagement strategy that considers the diverse needs and values regardless of their level of computer automation.

In intelligent tutoring systems, the control is usually given to end-users to a limited extent since these systems commonly provide a set of problems to students, adapting to their prior and current knowledge (Lawrence et al., 2023; Schulz et al., 2022). While teachers often do not have control to choose problems for students, they are likely to take action based on those problems, such as providing additional instruction, offering feedback, or adjusting the course material in response to the problems and how students are handling them (Holstein et al., 2019). Hence, it could be that the system's control may often be attributed to teachers rather than students. This requires further exploration to determine how much the teacher's agency has been balanced with the student's agency (Echeverria et al., 2023; Lawrence et al., 2023). Overall, most existing human-centred LA/AIED systems have been designed with a high degree of human control, as indicated by the larger proportion of systems categorised into Q2 or Q4. It further suggests that these systems are already designed to empower end-users, supported by human oversight. This aligns with the notion of striking a balance between human control and computer automation, as too much of either can lead to users being overwhelmed with options to control or mistrust since it may operate in a 'black-box' manner, respectively (Ozmen Garibay et al., 2023).

Therefore, we recommend:

- **Define End-Users**: Clearly define the intended end-users to avoid ambiguity in agency between teachers and students in highly automated systems. Develop distinct interfaces for teachers and students to enhance system usability and effectiveness.
- **Involve Stakeholders**: Actively engage stakeholders in the development process regardless of the system's level of automation or AI. Consider the influence of stakeholders' technical expertise on their preference for human control and computer automation.

- **Maintain Control**: In high computer automation systems (Q3 and Q4), allow teachers and students to adjust system parameters to suit dynamic changes in use. This ensures system flexibility and adaptability according to the changing dynamics of the classrooms or learning environments.
- **HCAI Framework as a Design Tool**: Utilise the HCAI two-dimensional framework as a design tool to assess and balance features that offer human control and computer automation in the system (update sequentially from Q1 to Q4), empowering end-users and supporting self-efficacy.

5.4 Exploring Safety, Reliability, and Trustworthiness in Human-Centred LA/AIED Systems

Lastly, regarding ***RQ4***, we summarised how researchers address safety, reliability, and trustworthiness princples in human-centred LA/AIED systems. Finding reveals that reliability (37%) emerges as the most prominent principle, closely followed by safety (35%), with trustworthiness trailing at 25%. This distribution of principles is reflected in the design and evaluation of human-centred LA/AIED systems. While these principles are addressed to a certain degree, our review underscores the need for more robust evidence and methodological mechanisms to thoroughly understand and evaluate them. This is particularly true for trustworthiness, which is less represented in the current research. However, the importance of trustworthiness in the context of LA/AIED systems cannot be overstated. As highlighted by Shneiderman (2020b), the perceived trustworthiness of our systems can directly impact their adoption in real-world practices. If end-users (i.e., teachers or students) deem these smart systems untrustworthy, it could significantly hinder their widespread acceptance and use. Investigating further on trustworthiness, interestingly, we identified various perspectives on accountability; on one hand, there are several discussions about the accountability of teachers and students when using the system (i.e., Zhou et al., 2021; Duan et al., 2023; Lawrence et al., 2023), and on the other hand, one study refers to the accountability of teachers in providing design inputs during system evaluation to be trusted stakeholders. These findings underscore the complexity of accountability in the context of human-centred LA/AIED systems (i.e., Gibson et al., 2017). It is a shared responsibility involving teachers, students, and the system itself. Each plays a crucial role in ensuring the system's effectiveness and in building trust among stakeholders. It extends beyond the traditional notion of individuals or systems being responsible for their actions and their consequences in use. Further investigation is required as this insight could have significant implications for the design of AI-powered systems in education.

From our findings, the principles of safety, reliability, and trustworthiness are not isolated; rather, they are interconnected and mutually reinforcing in the context of human-centred LA/AIED systems. Safety is a fundamental aspect that can significantly contribute to building trust. When users, especially students, feel that their data is protected and that the system operates within safe parameters, their trust in the system will likely increase (Drachsler and Greller, 2016). It can empower users by giving them the confidence to use the system without fear of data breaches or misuse. For instance, robust data privacy measures (e.g., Kivimäki et al., 2019; Ngoon et al., 2023) and transparent data collection procedures (e.g., Conijn et al., 2020a; Wang et al., 2022) can enhance the perceived safety of the system, thereby fostering trust among users. Reliability, on the other hand, refers to the consistent performance of the system. Interestingly, a system that is reliable and consistent, even if less accurate, can often be perceived as more trustworthy (Duan et al., 2023). It may happen because users (i.e., teachers) can predict the system's behaviour and adjust their expectations accordingly, increasing trust. This reliable system is important in empowering users by providing consistent and predictable results, enabling them to make informed decisions. Moreover, trustworthiness is closely linked to both safety and reliability. A system that consistently protects user data (safety) and performs as expected (reliability) is likely to be deemed trustworthy. A trustworthy system empowers users by fostering confidence in the system's outputs, which in turn encourages its adoption and use (Khosravi et al., 2022).

In conclusion, these principles are intertwined, and improvements in one area can positively impact the others (Ozmen Garibay et al., 2023; Shneiderman, 2020b). The interconnected principles of safety, reliability, and trustworthiness, coupled with a balance of human control and computer automation, can significantly empower stakeholders in the context of human-centred LA/AIED systems. Therefore, future research and development efforts should enhance these principles collectively to foster user trust and promote the acceptance and effectiveness of human-centred LA/AIED systems in real-world practices, accompanied by the aforementioned human-centred design practices above. Notably, beyond stakeholders' perspectives, we could not find studies exploring external mechanisms to ensure a human-centred LA/AIED system is safe, trustworthy or reliable. These mechanisms can involve, for example, subjecting the systems to independent audits and assessments by third parties to verify their actual trustworthiness, to move beyond perceived trust, or ensure adherence to relevant legal and regulatory requirements, as well as industry standards (Shneiderman, 2020b).

Therefore, we recommend:

- **Safety Procedures**: Report safety procedures, including risk assessment, stakeholders' perspectives, and informed consent procedures.

- **Data Privacy**: Ensure data privacy of stakeholders when involving them in the design process. Include secure data storage procedures and risk assessments for safety in the educational environment.
- **Reliability & Collaboration**: Foster collaboration with end-users and employ system explainability. Educate end-users on system capabilities and limitations to improve privacy awareness and effective technology use.
- **Trustworthiness**: Report stakeholders' trust perceptions, ensure transparency of data outcomes and algorithms, and measure trustworthiness using human control and computer automation dimensions.
- **Accountability**: Explore stakeholder accountability in the design process, focusing on how teachers or students can be held accountable for their inputs in human-centred LA/AIED systems.

5.5 Limitations

This SLR is subject to limitations that should be considered in interpreting results and discussions. First, despite the comprehensive search strategy employed, there is always a possibility of missing relevant studies due to limitations in the search terms, databases searched, or the exclusion of grey literature (e.g., unpublished works) and groundbreaking results in short articles or posters. Although we took many studies to reduce bias, it is possible that the study selection process could be influenced by the reviewers' judgment or subjective criteria, which could introduce some selection bias. However, this is a known possibility and has been addressed through transparency and open communication between researchers. We further acknowledge that LA and AIED come from two different research communities and potentially different ways of reporting their research (Rienties et al., 2020). Nonetheless, we found a clear overlap between these two communities. This SLR attempted to comprehensively view these overlaps through a human-centred perspective, allowing future work to explore and advance this avenue.

6 Concluding remarks

The findings of this systematic literature review indicate a growing interest in human-centred LA and AIED research. The review highlights the importance of safety, reliability, and trust in the design and implementation of these data-intensive systems, as well as the need for transparent, effective communication and user control. The review also identifies gaps in the existing research and methodological challenges that need to be addressed for human-centred LA and AIED to remain relevant and potentially become part of mainstream practices in the foreseeable future. Overall, this review provides valuable insights into the current state of human-centredness in LA and AIED studies with support HCAI framework as a lens, and underscores the importance of ongoing research and development.

Acknowledgements

Riordan Alfredo gratefully acknowledges Monash University for his PhD scholarship. This research was funded partially by the Australian Government through the Australian Research Council (project number DP210100060).

References

Ahn, J., Campos, F., Hays, M., Digiacomo, D., 2019. Designing in context: Reaching beyond usability in learning analytics dashboard design. Journal of Learning Analytics 6, 70–85. doi:doi:10.18608/jla.2019.62.5.

Ahn, J., Campos, F., Nguyen, H., Hays, M., Morrison, J., 2021. Co-Designing for Privacy, Transparency, and Trust in K-12 Learning Analytics, in: LAK21: 11th International Learning Analytics and Knowledge Conference. ACM, pp. 55–65. doi:doi:10.1145/3448139.3448145.

Al-Doulat, A., Nur, N., Karduni, A., Benedict, A., Al-Hossami, E., Maher, M.L., Dou, W., Dorodchi, M., Niu, X., 2020. Making Sense of Student Success and Risk Through Unsupervised Machine Learning and Interactive Storytelling, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 12163, pp. 3–15.

Albó, L., Barria-Pineda, J., Brusilovsky, P., Hernández-Leo, D., 2022. Knowledge-Based Design Analytics for Authoring Courses with Smart Learning Content. Int. J. Artif. Intell. Educ. 32, 4–27. doi:doi:10.1007/s40593-021-00253-3.

Aleven, V., Blankestijn, J., Lawrence, L., Nagashima, T., Taatgen, N., 2022. A Dashboard to Support Teachers During Students' Self-paced AI-Supported Problem-Solving Practice, Springer International Publishing. pp. 16–30.

Alfredo, R., Echeverria, V., Jin, Y., Swiecki, Z., Gašević, D., Martinez-Maldonado, R., 2024. SLADE: A Method for Designing Human-Centred Learning Analytics Systems, in: LAK24: 14th International Learning Analytics and Knowledge Conference, ACM. doi:doi:10.1145/3636555.3636847.

Alfredo, R.D., Nie, L., Kennedy, P., Power, T., Hayes, C., Chen, H., McGregor, C., Swiecki, Z., Gašević, D., Martinez-Maldonado, R., 2023. "That Student Should Be a Lion Tamer!" StressViz: Designing a stress analytics dashboard for teachers. LAK23: 13th international learning analytics and knowledge conference , 57–67doi:doi:10.1145/3576050.3576058.

Alzahrani, A.S., Tsai, Y.S., Aljohani, N., Whitelock-Wainwright, E., Gasevic, D., 2023. Do teaching staff trust stakeholders and tools in learning analytics? a mixed methods study. Educational technology research and development 71, 1471–1501.

Assim, M., Al-Bahri, H., Zafar, Q., Shahada, M., Al-Ammary, J., 2021. Remote Teaching during COVID-19 pandemic in Higher Education Institutions in the Kingdom of Bahrain: Challenges and Innovative Solutions, in: 2021 Sustainable Leadership and Academic Excellence International Conference, SLAE 2021. doi:doi:10.1109/SLAE54202.2021.9788084.

Baker, R.S., 2016. Stupid tutoring systems, intelligent humans. Int. J. AI in Educ. 26, 600–614. doi:doi:10.1007/s40593-016-0105-0.

Barreiros, C., Leitner, P., Ebner, M., Veas, E., Lindstaedt, S., 2023. Students in Focus – Moving Towards Human-Centred Learning Analytics, in: Viberg, O., Grönlund, Å. (Eds.), Practicable Learning Analytics. Springer International Publishing, Cham. Advances in Analytics for Learning and Teaching, pp. 77–94. doi:doi:10.1007/978-3-031-27646-0_5.

van Berkel, N., Tag, B., Goncalves, J., Hosio, S., 2022. Human-centred artificial intelligence: a contextual morality perspective. Behaviour & Information Technology 41, 502–518.

Biesta, G., Priestley, M., Robinson, S., 2015. The role of beliefs in teacher agency. Teachers and Teaching 21, 624–640. doi:doi:10.1080/13540602.2015.1044325.

Bishop, E., Allington, D., Ringrose, T., Martin, C., Aldea, A., Garcia-Jaramillo, M., Leon-Vargas, F., Leal, Y., Henao, D., Gomez, A.M., 2022. Design and Usability of an Avatar-Based Learning Program to Support Diabetes Education: Quality Improvement Study in Colombia. Journal of diabetes science and technology doi:doi:10.1177/19322968221136141.

Boateng, S., Alex, J., Adelabu, F., Sihele, T., Momoti, V., 2022. Pre-Service Teachers' Perspectives towards the Use of GammaTutor in Teaching Physical Sciences in South African Secondary Schools. International Journal of Learning, Teaching and Educational Research 21, 304–323. doi:doi:10.26803/ijlter.21.6.18.

Bodily, R., Verbert, K., 2017. Trends and issues in student-facing learning analytics reporting systems research, in: Proceedings of the Seventh International Learning Analytics & Knowledge Conference, Association for Computing Machinery, New York, NY, USA. pp. 309–318. doi:doi:10.1145/3027385.3027403.

Bond, M., Viberg, O., Bergdahl, N., 2023. The current state of using learning analytics to measure and support K-12 student engagement: A scoping review, in: LAK2023: LAK23: 13th International Learning Analytics and Knowledge Conference. Association for Computing Machinery, New York, NY, USA, pp. 240–249. doi:doi:10.1145/3576050.3576085.

Bonnat, C., Sanchez, E., 2022. Toward a Digital Companion to Monitor a Mixed Reality Game. International Journal of Serious Games 9, 5–21. doi:doi:10.17083/ijsg.v9i3.504.

Borjigin, A., Miao, C., Lim, S.F., Li, S., Shen, Z., 2015. Teachable Agents with Intrinsic Motivation, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 9112, pp. 34–43.

Braun, V., Clarke, V., 2012. Thematic analysis., APA, Washington, DC, pp. 57–71. doi:doi:10.1037/13620-004.

Brossi, L., Castillo, A.M., Cortesi, S., 2022. Student-centred requirements for the ethics of ai in education, in: Holmes, W., Porayska-Pomsta, K. (Eds.), The Ethics of Artificial Intelligence in Education. Routledge, pp. 91–112.

Buckingham Shum, S., Ferguson, R., Martinez-Maldonado, R., 2019. Human-centred learning analytics. Journal of Learning Analytics 6, 1–9.

Buckingham Shum, S., Luckin, R., 2019. Learning analytics and AI: Politics, pedagogy and practices. British Journal of Educational Technology 50, 2785–2793. doi:doi:10.1111/bjet.12880.

Camara Olim, S.M., Nisi, V., Rubegni, E., 2023. "Periodic fable discovery" using tangible interactions and augmented reality to promote STEM subjects. Proceedings of the seventeenth international conference on tangible, embedded, and embodied interaction doi:doi:10.1145/3569009.3572804.

Carter, M., Egliston, B., 2023. What are the risks of virtual reality data? learning analytics, algorithmic bias and a fantasy of perfect data. New Media & Society 23, 485–504.

Cavalcante Siebert, L., Lupetti, M.L., Aizenberg, E., Beckers, N., Zgonnikov, A., Veluwenkamp, H., Abbink, D., Giaccardi, E., Houben, G.J., Jonker, C.M., van den Hoven, J., Forster, D., Lagendijk, R.L., 2023. Meaningful human control: Actionable properties for AI system development. AI and Ethics 3, 241–255. doi:doi:10.1007/s43681-022-00167-3.

Cavalcanti, A.P., Barbosa, A., Carvalho, R., Freitas, F., Tsai, Y.S., Gašević, D., Mello, R.F., 2021. Automatic feedback in online learning environments: A systematic literature review. Computers and Education: Artificial Intelligence 2, 100027.

Chakraborty, U., Banerjee, A., Saha, J.K., Sarkar, N., Chakraborty, C., 2022. Artificial intelligence and the fourth industrial revolution. CRC Press.

Challco, G.C., Bittencourt, I.I., Isotani, S., 2020. Can Ontologies Support the Gamification of Scripted Collaborative Learning Sessions?, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 12163, pp. 79–91.

Chang, H.T., Lin, C.Y., Jheng, W.B., Chen, S.H., Wu, H.H., Tseng, F.C., Wang, L.C., 2023. AI, Please Help Me Choose a Course: Building a Personalized Hybrid Course Recommendation System to Assist Students in Choosing Courses Adaptively 26, 203–217. doi:doi:10.30191/ETS.202301_26(1).0015.

Chatti, M., Muslim, A., Guesmi, M., Richtscheid, F., Nasimi, D., Shahin, A., Damera, R., 2020. How to Design Effective Learning Analytics Indicators? A Human-Centered Design Approach. volume 12315 LNCS. doi:doi:10.1007/978-3-030-57717-9_22.

Chavan, P., Mitra, R., 2022. Tcherly: A Teacher-Facing Dashboard for Online Video Lectures. J. Learn. Anal. 9, 125–151. doi:doi:10.18608/jla.2022.7555.

Chen, BD., Zhu, HY., 2019. Towards Value-Sensitive Learning Analytics Design, in: PROCEEDINGS OF THE 9TH INTERNATIONAL CONFERENCE ON LEARNING ANALYTICS & KNOWLEDGE (LAK'19), pp. 343–352. doi:doi:10.1145/3303772.3303798.

Chignell, M., Wang, L., Zare, A., Li, J., 2022. The evolution of hci and human factors: Integrating human and artificial intelligence. ACM Transactions on Computer-Human Interaction doi:doi:10.1145/3557891.

Conijn, R., Martinez-Maldonado, R., Knight, S., Shum, SB., Van Waes, L., van Zaanen, M., 2020a. How to provide automated feedback on the writing process? A participatory approach to design writing analytics tools. Computer assisted language learning doi:doi:10.1080/09588221.2020.1839503.

Conijn, R., Van Waes, L., van Zaanen, M., 2020b. Human-Centered Design of a Dashboard on Students' Revisions during Writing. volume 12315 LNCS. doi:doi:10.1007/978-3-030-57717-9_3.

Crain, P., Lee, J., Yen, Y.C.G., Kim, J., Aiello, A., Bailey, B., 2022. Visualizing topics and opinions helps students interpret large collections of peer feedback for creative projects. ACM Trans. Comput.-Hum. Interact. doi:doi:10.1145/3571817.

Cukurova, M., Mavrikis, M., Luckin, R., Clark, J., Crawford, C., 2017. Interaction Analysis in Online Maths Human Tutoring: The Case of Third Space Learning, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 10331, pp. 636–643.

Darvishi, A., Khosravi, H., Sadiq, S., Gašević, D., Siemens, G., 2024. Impact of ai assistance on student agency. Computers & Education 210, 104967.

Davenport, T.H., 2018. From analytics to artificial intelligence. Journal of Business Analytics 1, 73–80.

de Quincey, E., Briggs, C., Kyriacou, T., Waller, R., 2019. Student Centred Design of a Learning Analytics System, Association for Computing Machinery. pp. 353–362. doi:doi:10.1145/3303772.3303793.

De Silva, L., Chounta, I.A., Rodríguez-Triana, M., Roa, E., Gramberg, A., Valk, A., 2022. Toward an Institutional Analytics Agenda for Addressing Student Dropout in Higher Education: An Academic Stakeholders' Perspective. J. Learn. Anal. 9, 179–201. doi:doi:10.18608/jla.2022.7507.

Di Mitri, D., Schneider, J., Drachsler, H., 2022. Keep Me in the Loop: Real-Time Feedback with Multimodal Data. Int J Artif Intell Educ 32, 1093–1118. doi:doi:10.1007/s40593-021-00281-z.

Dogan, M.E., Goru Dogan, T., Bozkurt, A., 2023. The Use of Artificial Intelligence (AI) in Online Learning and Distance Education Processes: A Systematic Review of Empirical Studies. Applied Sciences 13, 3056. doi:doi:10.3390/app13053056.

Dollinger, M., Liu, D., Arthars, N., Lodge, J., 2019. Working together in learning analytics towards the co-creation of value. Journal of Learning Analytics 6, 10–26. doi:doi:10.18608/jla.2019.62.2.

Drachsler, H., Greller, W., 2016. Privacy and analytics: it's a DELICATE issue a checklist for trusted learning analytics, in: LAK '16: Proc. of the 6th Intl. Conf. on Learning Analytics & Knowledge. ACM, New York, NY, USA, pp. 89–98. doi:doi:10.1145/2883851.2883893.

Duan, X., Pei, B., Ambrose, G.A., Hershkovitz, A., Cheng, Y., Wang, C., 2023. Towards transparent and trustworthy prediction of student learning achievement by including instructors as co-designers: A case study doi:doi:10.1007/s10639-023-11954-8.

Easterday, M., Rees Lewis, D., Gerber, E., 2017. Designing Crowdcritique Systems for Formative Feedback. Int. J. Artif. Intell. Educ. 27, 623–663. doi:doi:10.1007/s40593-016-0125-9.

Echeverria, V., Martinez-Maldonado, R., Buckingham Shum, S., 2019. Towards collaboration translucence: Giving meaning to multimodal group data. Proceedings of the 2019 CHI conference on human factors in computing systems , 1–16doi:doi:10.1145/3290605.3300269.

Echeverria, V., Martinez-Maldonado, R., Shum, S.B., Chiluiza, K., Granda, R., Conati, C., 2018. Exploratory versus Explanatory Visual Learning Analytics: Driving Teachers' Attention through Educational Data Storytelling. Journal of Learning Analytics 5, 73—97–73—97. doi:doi:10.18608/jla.2018.53.6.

Echeverria, V., Yan, L., Zhao, L., Abel, S., Alfredo, R., Dix, S., Jaggard, H., Wotherspoon, R., Osborne, A., Buckingham Shum, S., Gašević, D., Martinez-Maldonado, R., 2024. Teamslides: a multimodal teamwork analytics dashboard for teacher-guided reflection in a physical learning space, in: LAK24: 14th International Learning Analytics and Knowledge Conference, ACM. doi:doi:10.1145/3636555.3636857.

Echeverria, V., Yang, K., Lawrence, L., Rummel, N., Aleven, V., 2023. Designing hybrid human–ai orchestration tools for individual and collaborative activities: A technology probe study. IEEE Transactions on Learning Technologies 16, 191–205. doi:doi:10.1109/TLT.2023.3248155.

Edelenbos, J., Klijn, E.H., 2006. Managing Stakeholder Involvement in Decision Making: A Comparative Analysis of Six Interactive Processes in the Netherlands. J. Public Adm. Res. Theory 16, 417–446. doi:doi:10.1093/jopart/mui049.

Ekstrom, S., Pareto, L., 2022. The dual role of humanoid robots in education: As didactic tools and social actors. Education and information technologies doi:doi:10.1007/s10639-022-11132-2.

Engelbart, D.C., 1962. Augmenting human intellect: A conceptual framework. Summary Report, Contract AF 49–1024 21.

Eradze, M., Rodríguez-Triana, M., Milikic, N., Laanpere, M., Tammets, K., 2020. Contextualising Learning Analytics with Classroom Observations: A Case Study. Interaction Design and Architecture(s) 44, 71–95.

Fernandez-Nieto, G., An, P., Zhao, J., Buckingham Shum, S., Martinez-Maldonado, R., 2022. Classroom Dandelions: Visualising Participant Position, Trajectory and Body Orientation Augments Teachers' Sensemaking, in: CHI '22: CHI Conference on Human Factors in Computing Systems, ACM. pp. 1–17. doi:doi:10.1145/3491102.3517736.

Fernandez Nieto, G.M., Kitto, K., Buckingham Shum, S., Martinez-Maldonado, R., 2022. Beyond the learning analytics dashboard: Alternative ways to communicate student data insights combining visualisation, narrative and storytelling, in: LAK22: 12th International Learning Analytics and Knowledge Conference, pp. 219–229.

Fernandez-Nieto, G.M., Martinez-Maldonado, R., Echeverria, V., Kitto, K., Gašević, D., Buckingham Shum, S., 2024. Data storytelling editor: A teacher-centred tool for customising learning analytics dashboard narratives, in: LAK24: 14th International Learning Analytics and Knowledge Conference.

Fernandez-Nieto, G.M., Martinez-Maldonado, R., Kitto, K., Buckingham Shum, S., 2021. Modelling Spatial Behaviours in Clinical Team Simulations using Epistemic Network Analysis: Methodology and Teacher Evaluation, in: LAK21: 11th International Learning Analytics and Knowledge Conference, ACM. pp. 386–396. doi:doi:10.1145/3448139.3448176.

Garcia-Ruiz, M., Santana-Mancilla, P., Gaytan-Lugo, L., Iniguez-Carrillo, A., 2022. Participatory Design of Sonification Development for Learning about Molecular Structures in Virtual Reality. Multimodal Technologies and Interaction 6. doi:doi:10.3390/mti6100089.

Gerdes, A., Heeren, B., Jeuring, J., van Binsbergen, L., 2017. Ask-Elle: An Adaptable Programming Tutor for Haskell Giving Automated Feedback. Int. J. Artif. Intell. Educ. 27, 65–100. doi:doi:10.1007/s40593-015-0080-x.

Giacomin, J., 2014. What is human centred design? The design journal 17, 606–623.

Gibson, A., Aitken, A., Sándor, Á., Buckingham Shum, S., Tsingos-Lucas, C., Knight, S., 2017. Reflective Writing Analytics for Actionable Feedback, Association for Computing Machinery. pp. 153–162. doi:doi:10.1145/3027385.3027436.

Gorkovenko, K., Burnett, D.J., Thorp, J.K., Richards, D., Murray-Rust, D., 2020. Exploring the future of data-driven product design, in: Proc. of the 2020 CHI Conference on Human Factors in Computing Systems, ACM, New York, NY, USA. p. 1–14. URL: https://doi.org/10.1145/3313831.3376560, doi:doi:10.1145/3313831.3376560.

Granić, A., 2022. Educational Technology Adoption: A systematic review. Educ. Inf. Technol. 27, 9725–9744. doi:doi:10.1007/s10639-022-10951-7.

Grudin, J., 2009. Ai and hci: Two fields divided by a common focus. AI magazine 30, 48–48.

Han, S., Liu, M., Pan, Z., Cai, Y., Shao, P., 2022. Making FAQ Chatbots More Inclusive: An Examination of Non-Native English Users' Interactions with New Technology in Massive Open Online Courses. Int. J. Artif. Intell. Educ. doi:doi:10.1007/s40593-022-00311-4.

Hanington, B., Martin, B., 2012. Universal Methods of Design: 100 Ways To Research Complex Problems, develop innovative ideas, and Design Effective Solutions. Rockport Publishers.

Head, A., Xu, Y., Wang, J., 2014. ToneWars: Connecting Language Learners and Native Speakers through Collaborative Mobile Games, in: Intelligent Tutoring Systems. Springer International Publishing, Cham. volume 8474, pp. 368–377.

Holmes, W., Porayska-Pomsta, K., Holstein, K., Sutherland, E., Baker, T., Shum, S.B., Santos, O.C., Rodrigo, M.T., Cukurova, M., Bittencourt, I.I., et al., 2022. Ethics of AI in education: Towards a community-wide framework. Int. J. AI in Educ. 32, 504–526.

Holstein, K., Aleven, V., Rummel, N., 2020. A Conceptual Framework for Human–AI Hybrid Adaptivity in Education, in: Bittencourt, I.I., Cukurova, M., Muldner, K., Luckin, R., Millán, E. (Eds.), Artificial Intelligence in Education. Springer International Publishing, Cham. volume 12163, pp. 240–254. doi:doi:10.1007/978-3-030-52237-7_20.

Holstein, K., McLaren, B.M., Aleven, V., 2019. Co-Designing a Real-Time Classroom Orchestration Tool to Support Teacher–AI Complementarity. Learning Analytics 6, 27–52. doi:doi:10.18608/jla.2019.62.3.

Hooshyar, D., Tammets, K., Ley, T., Aus, K., Kollom, K., 2023. Learning Analytics in Supporting Student Agency: A Systematic Review. Sustainability 15, 13662. doi:doi:10.3390/su151813662.

Hosseini, R., Akhuseyinoglu, K., Brusilovsky, P., Malmi, L., Pollari-Malmi, K., Schunn, C., Sirkiä, T., 2020. Improving Engagement in Program Construction Examples for Learning Python Programming. Int. J. Artif. Intell. Educ. 30, 299–336. doi:doi:10.1007/s40593-020-00197-0.

Hu, L., Wu, J., Chen, G., 2022. iTalk–iSee: A participatory visual learning analytical tool for productive peer talk. International Journal of Computer-Supported Collaborative Learning 17, 397–425. doi:doi:10.1007/s11412-022-09374-w.

Hutchins, N., Biswas, G., 2023. Using Teacher Dashboards to Customize Lesson Plans for a Problem-Based, Middle School STEM Curriculum, in: ACM International Conference Proceeding Series, pp. 324–332. doi:doi:10.1145/3576050.3576100.

Hutt, S., Krasich, K., Brockmole, JR., D'Mello, SK., 2021. Breaking out of the Lab: Mitigating Mind Wandering with Gaze-Based Attention-Aware Technology in Classrooms, in: CHI '21: PROCEEDINGS OF THE 2021 CHI CONFERENCE ON HUMAN FACTORS IN COMPUTING SYSTEMS. doi:doi:10.1145/3411764.3445269.

Jääskelä, P., Heilala, V., Kärkkäinen, T., Häkkinen, P., 2021. Student agency analytics: learning analytics as a tool for analysing student agency in higher education. Behaviour & Information Technology 40, 790–808. doi:doi:10.1080/0144929X.2020.1725130.

Jamalova, G., Aymatova, F., Ikromov, S., 2022. The state-of-the-art applications of artificial intelligence in distance education: A systematic mapping study, in: Proceedings of the 6th International Conference on Future Networks & Distributed Systems, ACM, Tashkent TAS Uzbekistan. pp. 600–606. doi:doi:10.1145/3584202.3584292.

Jelinek, T., Wallach, W., Kerimi, D., 2021. Policy brief: the creation of a g20 coordinating committee for the governance of artificial intelligence. AI and Ethics 1, 141–150.

Jeong, Y., Cho, H., Kim, T., Nam, T.J., 2023. AutomataStage: An AR-Mediated creativity support tool for hands-on multidisciplinary learning. Proceedings of the 2023 CHI conference on human factors in computing systems doi:doi:10.1145/3544548.3581408.

Jobin, A., Ienca, M., Vayena, E., 2019. The global landscape of ai ethics guidelines. Nature Machine Intelligence 1, 389–399.

Kaliisa, R., Jivet, I., Prinsloo, P., 2023. A checklist to guide the planning, designing, implementation, and evaluation of learning analytics dashboards. International Journal of Educational Technology in Higher Education 20, 28. doi:doi:10.1186/s41239-023-00394-6.

Kaliisa, R., Kluge, A., Mørch, A., 2020. Combining checkpoint and process learning analytics to support learning design decisions in blended learning environments. J. Learn. Anal. 7, 33–47. doi:doi:10.18608/JLA.2020.73.4.

Kang, S., Shokeen, E., Byrne, VL., Norooz, L., Bonsignore, E., Williams-Pierce, C., Froehlich, JE., 2020. ARMath: Augmenting Everyday Life with Math Learning, in: PROCEEDINGS OF THE 2020 CHI CONFERENCE ON HUMAN FACTORS IN COMPUTING SYSTEMS (CHI'20). doi:doi:10.1145/3313831.3376252.

Khachatryan, G., Romashov, A., Khachatryan, A., Gaudino, S., Khachatryan, J., Guarian, K., Yufa, N., 2014. Reasoning mind genie 2: An intelligent tutoring system as a vehicle for international transfer of instructional methods in mathematics. Int. J. Artif. Intell. Educ. 24, 333–382. doi:doi:10.1007/s40593-014-0019-7.

Khosravi, H., Kitto, K., Williams, J., 2019. RiPPLE: A crowdsourced adaptive platform for recommendation of learning activities. J. Learn. Anal. 6, 91–105. doi:doi:10.18608/jla.2019.63.12.

Khosravi, H., Shum, S.B., Chen, G., Conati, C., Tsai, Y.S., Kay, J., Knight, S., Martinez-Maldonado, R., Sadiq, S., Gašević, D., 2022. Explainable artificial intelligence in education. Computers and Education: Artificial Intelligence 3, 100074.

Kim, G., Kim, J., Kim, Y.S., 2023. "Explain what a treemap is": Exploratory investigation of strategies for explaining unfamiliar chart to blind and low vision users. Proceedings of the 2023 CHI conference on human factors in computing systems doi:doi:10.1145/3544548.3581139.

Kitto, K., Buckingham Shum, S., Gibson, A., 2018. Embracing imperfection in learning analytics, in: LAK '18: Proc. of the 8th Int. Conf. on Learning Analytics and Knowledge, ACM. pp. 451–460. doi:doi:10.1145/3170358.3170413.

Kivimäki, V., Pesonen, J., Romanoff, J., Remes, H., Ihantola, P., 2019. Curricular concept maps as structured learning diaries: Collecting data on self-regulated learning and conceptual thinking for learning analytics applications. J. Learn. Anal. 6, 106–121. doi:doi:10.18608/jla.2019.63.13.

Kloos, C.D., Dimitriadis, Y., Hernández-Leo, D., Alario-Hoyos, C., Martínez-Monés, A., Santos, P., Muñoz-Merino, P.J., Asensio-Pérez, J.I., Safont, L.V., 2022. H2O Learn - Hybrid and Human-Oriented Learning: Trustworthy and Human-Centered Learning Analytics (TaHCLA) for Hybrid Education, in: 2022 IEEE Global Engineering Education Conference (EDUCON), pp. 94–101. doi:doi:10.1109/EDUCON52537.2022.9766770.

Labonte-LeMoyne, E., Leger, PM., Robert, J., Babin, G., Charland, P., Michon, JF., 2017. Business intelligence serious game participatory development: Lessons from ERPsim for big data. BUSINESS PROCESS MANAGEMENT JOURNAL 23, 493–505. doi:doi:10.1108/BPMJ-12-2015-0177.

Lallé, S., Taub, M., Mudrick, N.V., Conati, C., Azevedo, R., 2017. The Impact of Student Individual Differences and Visual Attention to Pedagogical Agents During Learning with MetaTutor, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 10331, pp. 149–161. doi:doi:10.1007/978-3-319-61425-0_13.

Lang, C., Davis, L., 2023. Learning Analytics and Stakeholder Inclusion: What do We Mean When We Say "Human-Centered"?, in: LAK23: 13th International Learning Analytics and Knowledge Conference, ACM, Arlington TX USA. pp. 411–417. doi:doi:10.1145/3576050.3576110.

Lawrence, L., Echeverria, V., Yang, K., Aleven, V., Rummel, N., 2023. How teachers conceptualise shared control with an AI co-orchestration tool: A multiyear teacher-centred design process. British Journal of Educational Technology 00. doi:doi:10.1111/bjet.13372.

Lee, A., Luco, A., Tan, S., 2023. A Human-Centric Automated Essay Scoring and Feedback System for the Development of Ethical Reasoning. Educational Technology and Society 26, 147–159. doi:doi:10.30191/ETS.202301_26(1).0011.

Lee, E., Kamat, M., Temor, L., Schiafone, C., Fan, L., Liu, J., Coppin, P., Uribe-Quevedo, A., Ingino, R., Syed, A., Rojas, D., Lee, T., Perera, S., Dubrowski, A., Sukhai, M., 2022. Prototyping a Spatial Skills AR Authoring Tool for Partially Sighted, Blind, and Sighted Individuals, in: 2022 IEEE Games, Entertainment, Media Conference, GEM 2022. doi:doi:10.1109/GEM56474.2022.10017454.

Leitner, P., Khalil, M., Ebner, M., . Learning Analytics in Higher Education—A Literature Review, in: Peña-Ayala, A. (Ed.), Learning Analytics: Fundaments, Applications, and Trends: A View of the Current State of the Art to Enhance e-Learning. Springer International Publishing. Studies in Systems, Decision and Control, pp. 1–23. URL: https://doi.org/10.1007/978-3-319-52977-6_1, doi:doi:10.1007/978-3-319-52977-6_1.

Li, S., Gu, X., 2023. A Risk Framework for Human-centered Artificial Intelligence in Education. Educational Technology & Society 26, 187–202. URL: https://www.jstor.org/stable/48707976.

Liaqat, A., Munteanu, C., Demmans Epp, C., 2021. Collaborating with Mature English Language Learners to Combine Peer and Automated Feedback: A User-Centered Approach to Designing Writing Support. International Journal of Artificial Intelligence in Education 31, 638–679.

Lim, L.A., Gasevic, D., Matcha, W., Ahmad Uzir, N., Dawson, S., 2021. Impact of Learning Analytics Feedback on Self-Regulated Learning: Triangulating Behavioural Logs with Students' Recall, Association for Computing Machinery. pp. 364–374. doi:doi:10.1145/3448139.3448174.

Lin, C.C., Huang, A.Y.Q., Lu, O.H.T., 2023. Artificial intelligence in intelligent tutoring systems toward sustainable education: A systematic review. Smart Learning Environments 10, 41. doi:doi:10.1186/s40561-023-00260-y.

Long, Y., Aman, Z., Aleven, V., 2015. Motivational Design in an Intelligent Tutoring System that Helps Students Make Good Task Selection Decisions, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 9112, pp. 226–236.

Luckin, R., Underwood, J., du Boulay, B., Holmberg, J., Kerawalla, L., O'Connor, J., Smith, H., Tunley, H., 2006. Designing educational systems fit for use: A case study in the application of human centred design for aied. International Journal of Artificial Intelligence in Education 16, 353–380.

Luo, J., Zhang, S., 2022. Research on the design of project-based teaching & learning mode assisted by educational robot. Proceedings of the 4th world symposium on software engineering , 148–154doi:doi:10.1145/3568364.3568387.

Ma, S., Zhou, T., Nie, F., Ma, X., 2022. Glancee: An adaptable system for instructors to grasp student learning status in synchronous online classes, in: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems, pp. 1–25.

Macfadyen, L.P., 2022. Institutional implementation of learning analytics: Current state, challenges, and guiding frameworks, in: Lang, C., Siemens, G., Wise, A.F., Gašević, D., Merceron, A. (Eds.), The Handbook of Learning Analytics. 2 ed.. SoLAR, pp. 173–186. Section: 17.

Maguire, MARTIN., 2001. Methods to support human-centred design. International Journal of Human-Computer Studies 55, 587–634. doi:doi:10.1006/ijhc.2001.0503.

Maier, U., Klotz, C., 2022. Personalized feedback in digital learning environments: Classification framework and literature review. Computers and Education: Artificial Intelligence 3, 100080. doi:doi:https://doi.org/10.1016/j.caeai.2022.100080.

Mangaroska, K., Sharma, K., Giannakos, M., Trætteberg, H., Dillenbourg, P., 2018. Gaze Insights into Debugging Behavior Using Learner-Centred Analysis, Association for Computing Machinery. pp. 350–359. doi:doi:10.1145/3170358.3170386.

Markauskaite, L., Marrone, R., Poquet, O., Knight, S., Martinez-Maldonado, R., Howard, S., Tondeur, J., De Laat, M., Shum, S.B., Gašević, D., et al., 2022. Rethinking the entwinement between artificial intelligence and human learning: What capabilities do learners need for a world with ai? Computers and Education: Artificial Intelligence 3, 100056.

Martinez, JPC., Catasus, MG., Fontanillas, TR., 2020. Impact of using learning analytics in asynchronous online discussions in higher education. International Journal of Educational Technology in Higher Education 17. doi:doi:10.1186/s41239-020-00217-y.

Martinez-Maldonado, R., 2023. Human-centred learning analytics: Four challenges in realising the potential 1, 6–6. URL: https://learningletters.org/index.php/learn/article/view/6, doi:doi:10.59453/FIZJ7007.

Martinez-Maldonado, R., Pardo, A., Mirriahi, N., Yacef, K., Kay, J., Clayphan, A., 2015. The LATUX Workflow: Designing and Deploying Awareness Tools in Technology-Enabled Learning Settings, Association for Computing Machinery. pp. 1–10. doi:doi:10.1145/2723576.2723583.

Mavrikis, M., Cukurova, M., 2021. Same, same but different: The fading boundaries between LA and AIED.

Mcneill, G., Sondag, M., Powell, S., Asplin, P., Turkay, C., Moller, F., Archambault, D., 2023. From asymptomatics to zombies: Visualization-based education of disease modeling for children. Proceedings of the 2023 CHI conference on human factors in computing systems doi:doi:10.1145/3544548.3581573.

Michos, K., Lang, C., Hernandez-Leo, D., Price-Dennis, D., 2020. Involving teachers in learning analytics design: Lessons learned from two case studies, in: Proceedings of the 10th International Conference on Learning Analytics & Knowledge, pp. 94–99. doi:doi:10.1145/3375462.3375507.

Moore, J.W., 2016. What Is the Sense of Agency and Why Does it Matter? Frontiers in Psychology 7, 1272. doi:doi:10.3389/fpsyg.2016.01272.

Moya, B., Eaton, S.E., Pethrick, H., Hayden, K.A., Brennan, R., Wiens, J., McDermott, B., Lesage, J., 2023. Academic integrity and artificial intelligence in higher education contexts: A rapid scoping review protocol. Canadian Perspectives on Academic Integrity 5, 59–75.

Muldner, K., Lozano, C., Girotto, V., Burleson, W., Walker, E., 2013. Designing a Tangible Learning Environment with a Teachable Agent, in: Artificial Intelligence in Education. Springer Berlin Heidelberg, Berlin, Heidelberg. volume 7926, pp. 299–308.

Muslim, A., Chatti, M.A., Mahapatra, T., Schroeder, U., 2016. A Rule-Based Indicator Definition Tool for Personalized Learning Analytics, Association for Computing Machinery. pp. 264–273. doi:doi:10.1145/2883851.2883921.

Nazaretsky, T., Ariely, M., Cukurova, M., Alexandron, G., 2022a. Teachers' trust in AI-powered educational technology and a professional development program to improve it. Br. J. Educ. Technol. 53, 914–931. doi:doi:10.1111/bjet.13232.

Nazaretsky, T., Bar, C., Walter, M., Alexandron, G., 2022b. Empowering Teachers with AI: Co-Designing a Learning Analytics Tool for Personalized Instruction in the Science Classroom, in: ACM International Conference Proceeding Series, pp. 1–12. doi:doi:10.1145/3506860.3506861.

Nazaretsky, T., Cukurova, M., Alexandron, G., 2022c. An Instrument for Measuring Teachers' Trust in AI-Based Educational Technology, in: LAK22: 12th International Learning Analytics and Knowledge Conference, ACM, Online USA. pp. 56–66. doi:doi:10.1145/3506860.3506866.

Ngoon, T.J., Kovalev, D., Patidar, P., Harrison, C., Agarwal, Y., Zimmerman, J., Ogan, A., 2023. "An Instructor is [already] able to keep track of 30 students": Students' Perceptions of Smart Classrooms for Improving Teaching & Their Emergent Understandings of Teaching and Learning, in: Proceedings of the 2023 ACM Designing Interactive Systems Conference, ACM, Pittsburgh PA USA. pp. 1277–1292. doi:doi:10.1145/3563657.3596079.

Nguyen, Q., Rienties, B., Whitelock, D., 2020. A mixed-method study of how instructors design for learning in online and distance education. J. Learn. Anal. 7, 64–78. doi:doi:10.18608/JLA.2020.73.6.

Ochoa, X., Dominguez, F., 2020. Controlled evaluation of a multimodal system to improve oral presentation skills in a real learning setting. Br J Educ Technol 51, 1615–1630. doi:doi:10.1111/bjet.12987.

Ochoa, X., Domínguez, F., Guamán, B., Maya, R., Falcones, G., Castells, J., 2018. The RAP system: Automatic feedback of oral presentation skills using multimodal analysis and low-cost sensors, in: LAK '18: International Conference on Learning Analytics and Knowledge, ACM. pp. 360–364. doi:doi:10.1145/3170358.3170406.

Ocumpaugh, J., Baker, RS., Pedro, MOCZS., Hawn, MA., Heffernan, C., Heffernan, N., Slater, SA., 2017. Guidance Counselor Reports of the ASSISTments College Prediction Model (ACPM), in: Proceedings of the 7th International Conference on Learning Analytics & Knowledge, pp. 479–488. doi:doi:10.1145/3027385.3027435.

Oliver-Quelennec, K., Bouchet, F., Carron, T., Fronton Casalino, K., Pinçon, C., 2022. Adapting Learning Analytics Dashboards by and for University Students. volume 13450 LNCS. doi:doi:10.1007/978-3-031-16290-9_22.

Olsen, J.K., Rummel, N., Aleven, V., 2021. Designing for the Co-Orchestration of Social Transitions between Individual, Small-Group and Whole-Class Learning in the Classroom. Inter. Jour. of Artificial Intelligence in Education 31, 24–56. doi:doi:10.1007/s40593-020-00228-w.

Onnasch, L., Wickens, C.D., Li, H., Manzey, D., 2014. Human Performance Consequences of Stages and Levels of Automation: An Integrated Meta-Analysis. Human Factors 56, 476–488. doi:doi:10.1177/0018720813501549.

Ooge, J., Dereu, L., Verbert, K., 2023. Steering recommendations and visualising its impact: Effects on adolescents' trust in E-learning platforms. Proceedings of the 28th international conference on intelligent user interfaces , 156–170doi:doi:10.1145/3581641.3584046.

Ozmen Garibay, O., Winslow, B., Andolina, S., Antona, M., Bodenschatz, A., Coursaris, C., Falco, G., Fiore, S.M., Garibay, I., Grieman, K., et al., 2023. Six human-centered artificial intelligence grand challenges. International Journal of Human–Computer Interaction , 1–47.

Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372, n71. doi:doi:10.1136/bmj.n71.

Parasuraman, R., Sheridan, T.B., Wickens, C.D., 2000. A model for types and levels of human interaction with automation. IEEE Trans. Syst. Man Cybern. Part A Syst. Humans 30, 286–297. doi:doi:10.1109/3468.844354.

Pardo, A., Siemens, G., 2014. Ethical and privacy principles for learning analytics. Br. J. Educ. Technol. 45, 438–450. doi:doi:10.1111/bjet.12152.

Pinski, M., Adam, M., Benlian, A., 2023. AI Knowledge: Improving AI Delegation through Human Enablement, in: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, ACM, Hamburg Germany. pp. 1–17. doi:doi:10.1145/3544548.3580794.

Pishtari, G., Rodríguez-Triana, M., Väljataga, T., 2021. A multi-stakeholder perspective of analytics for learning design in location-based learning. International Journal of Mobile and Blended Learning 13, 1–17. doi:doi:10.4018/IJMBL.2021010101.

Pozdniakov, S., Martinez-Maldonado, R., Tsai, Y.S., Cukurova, M., Bartindale, T., Chen, P., Marshall, H., Richardson, D., Gasevic, D., 2022. The Question-driven Dashboard: How Can We Design Analytics Interfaces Aligned to Teachers' Inquiry?, in: ACM International Conference Proceeding Series, pp. 175–185. doi:doi:10.1145/3506860.3506885.

Prestigiacomo, R., Hadgraft, R., Hunter, J., Locker, L., Knight, S., Van Den Hoven, E., Martinez-Maldonado, R., 2020. Learning-centred translucence: An approach to understand how teachers talk about classroom data, in: Proceedings of the tenth international conference on learning analytics & knowledge, pp. 100–105.

Prieto, L.P., Rodríguez-Triana, M.J., Martínez-Maldonado, R., Dimitriadis, Y., Gašević, D., 2019. Orchestrating learning analytics (orla): Supporting inter-stakeholder communication about adoption of learning analytics at the classroom level. Australasian Journal of Educational Technology 35.

Prieto-Alvarez, C.G., Martinez-Maldonado, R., Anderson, T.D., 2018. Co-designing learning analytics tools with learners, in: Lodge, J., Thompson, K., Horvath, J., De Barba, P., Blumenstein, M. (Eds.), Learning analytics in the classroom. Routledge, pp. 93–110.

Rapp, C., Kauf, P., 2018. Scaling Academic Writing Instruction: Evaluation of a Scaffolding Tool (Thesis Writer). Int. J. Artif. Intell. Educ. 28, 590–615. doi:doi:10.1007/s40593-017-0162-z.

Read, S., DelaMerced, A., Zachry, M., 2012. Participatory Design in the Development of a Web-based Technology for Visualizing Writing Activity as Knowledge Work, in: Proceedings of the 30th ACM International Conference on Design of Communication, pp. 333–340.

Reimann, P., 2016. Connecting learning analytics with learning research: the role of design-based research, Routledge. volume 2, pp. 130–142. doi:doi:10.1080/23735082.2016.1210198.

Renz, A., Vladova, G., 2021. Reinvigorating the Discourse on Human-Centered Artificial Intelligence in Educational Technologies. Technology Innovation Management Review 11, 5–16. doi:doi:10.22215/timreview/1438.

Rienties, B., Køhler Simonsen, H., Herodotou, C., 2020. Defining the Boundaries Between Artificial Intelligence in Education, Computer-Supported Collaborative Learning, Educational Data Mining, and Learning Analytics: A Need for Coherence. Frontiers in Education 5.

Rodrigues, L., Palomino, P., Toda, A., Klock, A., Pessoa, M., Pereira, F., Oliveira, E., Oliveira, D., Cristea, A., Gasparini, I., Isotani, S., 2023. How Personalization Affects Motivation in Gamified Review Assessments. Int. J. Artif. Intell. Educ. doi:doi:10.1007/s40593-022-00326-x.

Rodríguez-Triana, M., Prieto, L., Dimitriadis, Y., de Jong, T., Gillet, D., 2021. Ada for ibl: Lessons learned in aligning learning design and analytics for inquiry-based learning orchestration. J. Learn. Anal. 8, 22–50. doi:doi:10.18608/JLA.2021.7357.

Ruiz-Calleja, A., Dennerlein, S., Kowald, D., Theiler, D., Lex, E., Ley, T., 2019. An infrastructure for workplace learning analytics: Tracing knowledge creation with the social semantic server. J. Learn. Anal. 6, 120–139. doi:doi:10.18608/jla.2019.62.9.

Santos, J.L., Govaerts, S., Verbert, K., Duval, E., 2012. Goal-Oriented Visualizations of Activity Tracking: A Case Study with Engineering Students, Association for Computing Machinery. pp. 143–152. doi:doi:10.1145/2330601.2330639.

Sarmiento, J.P., Campos, F., Wise, A., 2020. Engaging students as co-designers of learning analytics, in: Companion proceedings of the 10th international learning analytics and knowledge conference, SoLAR Frankfurt. pp. 29–32.

Sarmiento, J.P., Wise, A.F., 2022. Participatory and co-design of learning analytics: an initial review of the literature, in: LAK22: 12th International Learning Analytics and Knowledge Conference, pp. 535–541.

Sato, A.J., Sramek, Z., Yatani, K., 2023. Groupnamics: Designing an interface for overviewing and managing parallel group discussions in an online classroom. Proceedings of the 2023 CHI conference on human factors in computing systems doi:doi:10.1145/3544548.3581322.

Schmitz, M., Scheffel, M., Bemelmans, R., Drachsler, H., 2022. Fola 2–a method for co-creating learning analytics-supported learning design. Journal of Learning Analytics 9, 265–281.

Schulz, S., McLaren, B., Pinkwart, N., 2022. Towards a Tutoring System to Support Robotics Activities in Classrooms – Two Wizard-of-Oz Studies. Int. J. Artif. Intell. Educ. doi:doi:10.1007/s40593-022-00305-2.

Shardlow, M., Sellar, S., Rousell, D., 2022. Collaborative Augmentation and Simplification of Text (CoAST): Pedagogical Applications of Natural Language Processing in Digital Learning Environments. Learning Environments Research 25, 399–421.

Shetty, S.V., Guruvyas, K.R., Patil, P.P., Acharya, J.J., 2022. Essay Scoring Systems Using AI and Feature Extraction: A Review, in: Proceedings of Third International Conference on Communication, Computing and Electronics Systems. Springer, Singapore, pp. 45–57. doi:doi:10.1007/978-981-16-8862-1_4.

Shibani, A., Knight, S., Buckingham Shum, S., 2022. Questioning learning analytics? cultivating critical engagement as student automated feedback literacy, in: LAK22: 12th international learning analytics and knowledge conference, pp. 326–335.

Shneiderman, B., 2020a. Bridging the gap between ethics and practice: guidelines for reliable, safe, and trustworthy human-centered ai systems. ACM Transactions on Interactive Intelligent Systems (TiiS) 10, 1–31.

Shneiderman, B., 2020b. Human-Centered Artificial Intelligence: Three Fresh Ideas. AIS Transactions on Human-Computer Interaction , 109–124doi:doi:10.17705/1thci.00131.

Shneiderman, B., 2021. Human-Centered AI. Issues Sci. Technol. 37, 56–61.

Shneiderman, B., 2022. Human-Centered AI. Oxford University Press, Oxford, New York.

Shreiner, T., Guzdial, M., 2022. The information won't just sink in: Helping teachers provide technology-assisted data literacy instruction in social studies. British Journal of Educational Technology 53, 1134–1158. doi:doi:10.1111/bjet.13255.

Shute, V., Smith, G., Kuba, R., Dai, C.P., Rahimi, S., Liu, Z., Almond, R., 2021. The Design, Development, and Testing of Learning Supports for the Physics Playground Game. Int. J. Artif. Intell. Educ. 31, 357–379. doi:doi:10.1007/s40593-020-00196-1.

da Silva, F.L., Slodkowski, B.K., da Silva, K.K.A., Cazella, S.C., 2023. A systematic literature review on educational recommender systems for teaching and learning: research trends, limitations and opportunities. Educ. Inf. Technol. 28, 3289–3328. doi:doi:10.1007/s10639-022-11341-9.

Slade, S., Prinsloo, P., 2013. Learning Analytics: Ethical Issues and Dilemmas 57, 1510–1529. URL: https://doi.org/10.1177/0002764213479366, doi:doi:10.1177/0002764213479366.

Soreny, C., Takhar, P., Vickers, E., Bartlett, R., Coates, E., Stavroulakis, T., Fox, E., Davidson, H., Harrison, A., McDermott, C., 2022. Co-design of digital learning resources for care workers: Reflections on the neurocare knowhow project. Journal of Medical Engineering and Technology 46, 518–526. doi:doi:10.1080/03091902.2022.2089253.

Suleman, R., Mizoguchi, R., Ikeda, M., 2016. A New Perspective of Negotiation-Based Dialog to Enhance Metacognitive Skills in the Context of Open Learner Models. Int. J. Artif. Intell. Educ. 26, 1069–1115. doi:doi:10.1007/s40593-016-0118-8.

Swiecki, Z., Khosravi, H., Chen, G., Martinez-Maldonado, R., Lodge, J.M., Milligan, S., Selwyn, N., Gašević, D., 2022. Assessment in the age of artificial intelligence. Computers and Education: Artificial Intelligence 3, 100075.

Tegos, S., Demetriadis, S., Tsiatsos, T., 2014. A configurable conversational agent to trigger students' productive dialogue: A pilot study in the CALL domain. Int. J. Artif. Intell. Educ. 24, 62–91. doi:doi:10.1007/s40593-013-0007-3.

Tenório, K., Dermeval, D., Monteiro, M., Peixoto, A., Silva, A., 2022. Exploring Design Concepts to Enable Teachers to Monitor and Adapt Gamification in Adaptive Learning Systems: A Qualitative Research Approach. Int. J. Artif. Intell. Educ. 32, 867–891. doi:doi:10.1007/s40593-021-00274-y.

Ternblad, E.M., Haake, M., Anderberg, E., Gulz, A., 2018. Do Preschoolers 'Game the System'? A Case Study of Children's Intelligent (Mis)Use of a Teachable Agent Based Play-&-Learn Game in Mathematics, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 10947, pp. 557–569.

Tissenbaum, M., Slotta, J., 2019. Supporting classroom orchestration with real-time feedback: A role for teacher dashboards and real-time agents. International Journal of Computer-Supported Collaborative Learning 14, 325–351. doi:doi:10.1007/s11412-019-09306-1.

Topali, P., Ortega-Arranz, A., Chounta, IA., Asensio-Perez, JI., Martinez-Mones, A., Villagra-Sobrino, SL., 2022. Supporting instructors in the design of actionable feedback for MOOCs, in: Proceedings of the 2022 IEEE Global Engineering Education Conference, pp. 1881–1887. doi:doi:10.1109/EDUCON52537.2022.9766546.

Treasure-Jones, T., Dent-Spargo, R., Dharmaratne, S., 2018. How do students want their workplace-based feedback visualized in order to support self-regulated learning? Initial results & reflections from a co-design study in medical education, in: CEUR Workshop Proceedings.

Tsai, Y.S., Perrotta, C., Gašević, D., 2020. Empowering learners with personalised learning approaches? agency, equity and transparency in the context of learning analytics. Assessment & Evaluation in Higher Education 45, 554–567.

Tsai, Y.S., Singh, S., Rakovic, M., Lim, L.A., Roychoudhury, A., Gasevic, D., 2022. Charting Design Needs and Strategic Approaches for Academic Analytics Systems through Co-Design, in: ACM International Conference Proceeding Series, pp. 381–391. doi:doi:10.1145/3506860.3506939.

Usmani, U.A., Happonen, A., Watada, J., 2023. Human-Centered Artificial Intelligence: Designing for User Empowerment and Ethical Considerations, in: 2023 5th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA), IEEE, Istanbul, Turkiye. pp. 01–05. doi:doi:10.1109/HORA58378.2023.10156761.

Uttamchandani, S., Quick, J., 2022. An introduction to fairness, absence of bias, and equity in learning analytics, in: Lang, C., Siemens, G., Wise, A.F., Gašević, D., Merceron, A. (Eds.), The Handbook of Learning Analytics. 2 ed.. SoLAR, pp. 205–212. URL: https://www.solaresearch.org/publications/hla-22/hal22-chapter20/. section: 20.

Vannaprathip, N., Haddawy, P., Schultheis, H., Suebnukarn, S., 2022. Intelligent Tutoring for Surgical Decision Making: A Planning-Based Approach. Int. J. Artif. Intell. Educ. 32, 350–381. doi:doi:10.1007/s40593-021-00261-3.

Vezzoli, Y., Mavrikis, M., Vasalou, A., 2020. Inspiration Cards Workshops with Primary Teachers in the Early Co-Design Stages of Learning Analytics, in: Proceedings of the 10th International Conference on Learning Analytics & Knowledge, pp. 73–82. doi:doi:10.1145/3375462.3375537.

Viberg, O., Jivet, I., Scheffel, M., 2023. Designing Culturally Aware Learning Analytics: A Value Sensitive Perspective. Springer International Publishing, Cham. pp. 177–192. doi:doi:10.1007/978-3-031-27646-0_10.

Viberg, O., Mutimukwe, C., Grönlund, Å., 2022. Privacy in la research: Understanding the field to improve the practice. Journal of Learning Analytics 9, 169–182.

Vinella, F., Koppelaar, S., Masthoff, J., 2022. Forming Teams of Learners Online in a User as Wizard Study with Openness, Conscientiousness, and cognitive Ability, in: UMAP2022 - Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization, pp. 283–292. doi:doi:10.1145/3511047.3537660.

Wang, Q., Jing, S., Goel, A., 2022. Co-Designing AI Agents to Support Social Connectedness Among Online Learners: Functionalities, Social Characteristics, and Ethical Challenges, in: DIS 2022 - Proceedings of the 2022 ACM Designing Interactive Systems Conference: Digital Wellbeing, pp. 541–556. doi:doi:10.1145/3532106.3533534.

Wang, Y., Nguyen, H., Harpstead, E., Stamper, J., McLaren, B.M., 2019. How Does Order of Gameplay Impact Learning and Enjoyment in a Digital Learning Game?, in: Artificial Intelligence in Education. Springer International Publishing, Cham. volume 11625, pp. 518–531.

Weigand, A., Kindsmüller, M., 2021. HCD3A: An HCD Model to Design Data-Driven Apps. volume 12797 LNAI. doi:doi:10.1007/978-3-030-77772-2_19.

Wiley, K., Dimitriadis, Y., Linn, M., 2023. A human-centred learning analytics approach for developing contextually scalable K-12 teacher dashboards doi:doi:10.1111/bjet.13383.

Williamson, K., Kizilcec, R., 2022. A review of learning analytics dashboard research in higher education: implications for justice, equity, diversity, and inclusion, in: LAK22: 12th International Learning Analytics and Knowledge Conference, pp. 260–270.

Williamson, K., Kizilcec, R.F., 2021. Learning analytics dashboard research has neglected diversity, equity and inclusion, in: Proceedings of the Eighth ACM Conference on Learning@ Scale, pp. 287–290.

Wilson, J., Huang, Y., Palermo, C., Beard, G., MacArthur, C., 2021. Automated Feedback and Automated Scoring in the Elementary Grades: Usage, Attitudes, and Associations with Writing Outcomes in a Districtwide Implementation of MI Write. Int. J. Artif. Intell. Educ. 31, 234–276. doi:doi:10.1007/s40593-020-00236-w.

Wise, A.F., Sarmiento, J.P., Boothe Jr, M., 2021. Subversive learning analytics, in: LAK21: 11th international learning analytics and knowledge conference, pp. 639–645.

Xu, W., Dainoff, M.J., Ge, L., Gao, Z., 2023. Transitioning to human interaction with ai systems: New challenges and opportunities for hci professionals to enable human-centered ai. International Journal of Human–Computer Interaction 39, 494–518.

Yan, L., Zhao, L., Gasevic, D., Martinez-Maldonado, R., 2022. Scalability, Sustainability, and Ethicality of Multimodal Learning Analytics, in: LAK22: LAK22: 12th International Learning Analytics and Knowledge Conference. Association for Computing Machinery, New York, NY, USA, pp. 13–23. doi:doi:10.1145/3506860.3506862.

Yang, S.J.H., Ogata, H., Matsui, T., Chen, N.S., 2021. Human-centered artificial intelligence in education: Seeing the invisible through the visible. Computers and Education: Artificial Intelligence 2. doi:doi:10.1016/j.caeai.2021.100008.

Zhang, K., Aslan, A.B., 2021. AI technologies for education: Recent research & future directions 2, 100025. URL: https://www.sciencedirect.com/science/article/pii/S2666920X21000199, doi:doi:10.1016/j.caeai.2021.100025.

Zhang, Z., Xu, Y., Wang, Y., Yao, B., Ritchie, D., Wu, T., Yu, M., Wang, D., Li, T.J., 2022. StoryBuddy: A Human-AI Collaborative Chatbot for Parent-Child Interactive Storytelling with Flexible Parental Involvement, in: Conference on Human Factors in Computing Systems - Proceedings. doi:doi:10.1145/3491102.3517479.

Zhao, F., Liu, G.Z., Zhou, J., Yin, C., . A Learning Analytics Framework Based on Human-Centered Artificial Intelligence for Identifying the Optimal Learning Strategy to Intervene in Learning Behavior 26, 132–146. URL: https://www.jstor.org/stable/48707972, arXiv:48707972.

Zhou, Q., Suraworachet, W., Pozdniakov, S., Martinez-Maldonado, R., Bartindale, T., Chen, P., Richardson, D., Cukurova, M., 2021. Investigating Students' Experiences with Collaboration Analytics for Remote Group Meetings. volume 12748 LNAI. doi:doi:10.1007/978-3-030-78292-4_38.

Zwolinski, G., Kaminska, D., Laska-Lesniewicz, A., Haamer, R.E., Vairinhos, M., Raposo, R., Urem, F., Reisinho, P., 2022. Extended Reality in Education and Training: Case Studies in Management Education. ELECTRONICS 11. doi:doi:10.3390/electronics11030336.

A HCAI two-dimensional quadrants in LA/AIED references

No.	Quadrant	Reference
1	Q2	Ahn, J., Campos, F., Hays, M., & Digiacomo, D. (2019). Designing in context: Reaching beyond usability in learning analytics dashboard design. Journal of Learning Analytics, 6(2), 70–85. https://doi.org/10.18608/jla.2019.62.5
3	Q2	Al-Doulat, A., Nur, N., Karduni, A., Benedict, A., Al-Hossami, E., Maher, M. L., Dou, W., Dorodchi, M., & Niu, X. (2020). Making Sense of Student Success and Risk Through Unsupervised Machine Learning and Interactive Storytelling. In Artificial Intelligence in Education (Vol. 12163, pp. 3–15). Springer International Publishing. http://link.springer.com/10.1007/978-3-030-52237-7_1
4	Q4	Albó, L., Barria-Pineda, J., Brusilovsky, P., & Hernández-Leo, D. (2022). Knowledge-Based Design Analytics for Authoring Courses with Smart Learning Content. Int. J. Artif. Intell. Educ., 32(1), 4–27. https://doi.org/10.1007/s40593-021-00253-3
5	Q4	Aleven, V., Blankestijn, J., Lawrence, L., Nagashima, T., & Taatgen, N. (2022). A Dashboard to Support Teachers During Students' Self-paced AI-Supported Problem-Solving Practice. 16–30.
7	Q4	Alfredo, R. D., Nie, L., Kennedy, P., Power, T., Hayes, C., Chen, H., McGregor, C., Swiecki, Z., Gašević, D., & Martinez-Maldonado, R. (2023). "That Student Should Be a Lion Tamer!" StressViz: Designing a stress analytics dashboard for teachers. LAK23: 13th International Learning Analytics and Knowledge Conference, 57–67. https://doi.org/10.1145/3576050.3576058
9	Q3	Assim, M., Al-Bahri, H., Zafar, Q., Shahada, M., & Al-Ammary, J. (2021). Remote Teaching during COVID-19 pandemic in Higher Education Institutions in the Kingdom of Bahrain: Challenges and Innovative Solutions. 2021-January. https://doi.org/10.1109/SLAE54202.2021.9788084
11	Q2	Barreiros, C., Leitner, P., Ebner, M., Veas, E., & Lindstaedt, S. (2023). Students in Focus – Moving Towards Human-Centred Learning Analytics. In O. Viberg & Å. Grönlund (Eds.), Practicable Learning Analytics (pp. 77–94). Springer International Publishing. https://doi.org/10.1007/978-3-031-27646-0_5
14	Q1	Bishop, E., Allington, D., Ringrose, T., Martin, C., Aldea, A., Garcia-Jaramillo, M., Leon-Vargas, F., Leal, Y., Henao, D., & Gomez, A. M. (2022). Design and Usability of an Avatar-Based Learning Program to Support Diabetes Education: Quality Improvement Study in Colombia. Journal of Diabetes Science and Technology, 19322968221136140–19322968221136140. https://doi.org/10.1177/19322968221136141
15	Q1	Boateng, S., Alex, J. K., Adelabu, F. M., Sihele, T., & Momoti, V. (2022). Pre-Service Teachers' Perspectives towards the Use of GammaTutor in Teaching Physical Sciences in South African Secondary Schools. International Journal of Learning, Teaching and Educational Research, 21(6), 304–323. https://doi.org/10.26803/ijlter.21.6.18
18	Q4	Bonnat, C., & Sanchez, E. (2022). Toward a Digital Companion to Monitor a Mixed Reality Game. International Journal of Serious Games, 9(3), 5–21. https://doi.org/10.17083/ijsg.v9i3.504
19	Q4	Borjigin, A., Miao, C., Lim, S. F., Li, S., & Shen, Z. (2015). Teachable Agents with Intrinsic Motivation. In Artificial Intelligence in Education (Vol. 9112, pp. 34–43). Springer International Publishing. http://link.springer.com/10.1007/978-3-319-19773-9_4
24	Q2	Camara Olim, S. M., Nisi, V., & Rubegni, E. (2023). "Periodic fable discovery" using tangible interactions and augmented reality to promote STEM subjects. Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction. https://doi.org/10.1145/3569009.3572804
29	Q1	Challco, G. C., Bittencourt, I. I., & Isotani, S. (2020). Can Ontologies Support the Gamification of Scripted Collaborative Learning Sessions? In Artificial Intelligence in Education (Vol. 12163, pp. 79–91). Springer International Publishing. http://link.springer.com/10.1007/978-3-030-52237-7_7
30	Q4	Chang, H.-T., Lin, C.-Y., Jheng, W.-B., Chen, S.-H., Wu, H.-H., Tseng, F.-C., & Wang, L.-C. (2023). AI, Please Help Me Choose a Course: Building a Personalized Hybrid Course Recommendation System to Assist Students in Choosing Courses Adaptively. EDUCATIONAL TECHNOLOGY & SOCIETY, 26(1), 203–217. https://doi.org/10.30191/ETS.202301_26(1).0015

31	Q2	Chatti, M. A., Muslim, A., Guesmi, M., Richtscheid, F., Nasimi, D., Shahin, A., & Damera, R. (2020). How to design effective learning analytics indicators? A human-centered design approach. In Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics): Vol. 12315 LNCS (p. 317). https://doi.org/10.1007/978-3-030-57717-9_22
32	Q4	Chavan, P., & Mitra, R. (2022). Tcherly: A Teacher-Facing Dashboard for Online Video Lectures. J. Learn. Anal., 9(3), 125–151. https://doi.org/10.18608/jla.2022.7555
35	Q4	Conijn, R., Martinez-Maldonado, R., Knight, S., Shum, S., Van Waes, L., & van Zaanen, M. (2022). How to provide automated feedback on the writing process? A participatory approach to design writing analytics tools. COMPUTER ASSISTED LANGUAGE LEARNING. https://doi.org/10.1080/09588221.2020.1839503
36	Q4	Conijn, R., Van Waes, L., & van Zaanen, M. (2020). Human-centered design of a dashboard on students' revisions during writing. In Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics): Vol. 12315 LNCS (p. 44). https://doi.org/10.1007/978-3-030-57717-9_3
37	Q2	Crain, P., Lee, J., Yen, Y.-C. (Grace), Kim, J., Aiello, A., & Bailey, B. (2022). Visualizing topics and opinions helps students interpret large collections of peer feedback for creative projects. ACM Trans. Comput.-Hum. Interact. https://doi.org/10.1145/3571817
38	Q2	Cukurova, M., Mavrikis, M., Luckin, R., Clark, J., & Crawford, C. (2017). Interaction Analysis in Online Maths Human Tutoring: The Case of Third Space Learning. In Artificial Intelligence in Education (Vol. 10331, pp. 636–643). Springer International Publishing. http://link.springer.com/10.1007/978-3-319-61425-0_80
41	Q4	de Quincey, E., Briggs, C., Kyriacou, T., & Waller, R. (2019). Student Centred Design of a Learning Analytics System. 353–362. https://doi.org/10.1145/3303772.3303793
42	Q2	De Silva, L. M. H., Chounta, I.-A., Rodríguez-Triana, M. J., Roa, E. R., Gramberg, A., & Valk, A. (2022). Toward an Institutional Analytics Agenda for Addressing Student Dropout in Higher Education: An Academic Stakeholders' Perspective. J. Learn. Anal., 9(2), 179–201. https://doi.org/10.18608/jla.2022.7507
43	Q4	Di Mitri, D., Schneider, J., & Drachsler, H. (2022). Keep Me in the Loop: Real-Time Feedback with Multimodal Data. Int J Artif Intell Educ, 32(4), 1093–1118. https://doi.org/10.1007/s40593-021-00281-z
45	Q4	Dollinger, M., Liu, D., Arthars, N., & Lodge, J. M. (2019). Working together in learning analytics towards the co-creation of value. Journal of Learning Analytics, 6(2), 10–26. https://doi.org/10.18608/jla.2019.62.2
47	Q4	Duan, X., Pei, B., Ambrose, G. A., Hershkovitz, A., Cheng, Y., & Wang, C. (2023). Towards transparent and trustworthy prediction of student learning achievement by including instructors as co-designers: A case study. EDUCATION AND INFORMATION TECHNOLOGIES. https://doi.org/10.1007/s10639-023-11954-8
48	Q2	Easterday, M. W., Rees Lewis, D., & Gerber, E. M. (2017). Designing Crowdcritique Systems for Formative Feedback. Int. J. Artif. Intell. Educ., 27(3), 623–663. https://doi.org/10.1007/s40593-016-0125-9
49	Q1	Echeverria, V., Martinez-Maldonado, R., & Buckingham Shum, S. (2019). Towards collaboration translucence: Giving meaning to multimodal group data. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, 1–16. https://doi.org/10.1145/3290605.3300269
54	Q3	Ekstrom, S., & Pareto, L. (2022). The dual role of humanoid robots in education: As didactic tools and social actors. EDUCATION AND INFORMATION TECHNOLOGIES. https://doi.org/10.1007/s10639-022-11132-2
56	Q1	Eradze, M., Rodríguez-Triana, M. J., Milikic, N., Laanpere, M., & Tammets, K. (2020). Contextualising Learning Analytics with Classroom Observations: A Case Study. Interaction Design and Architecture(s), 44, 71–95.
58	Q1	Fernandez Nieto, G. M., Kitto, K., Buckingham Shum, S., & Martinez-Maldonado, R. (2022). Beyond the Learning Analytics Dashboard: Alternative Ways to Communicate Student Data Insights Combining Visualisation, Narrative and Storytelling. 219–229. https://doi.org/10.1145/3506860.3506895
57	Q2	Fernandez-Nieto, G., An, P., Zhao, J., Buckingham Shum, S., & Martinez-Maldonado, R. (2022). Classroom Dandelions: Visualising Participant Position, Trajectory and Body Orientation Augments Teachers' Sensemaking. 1–17. https://doi.org/10.1145/3491102.3517736

60	Q4	Fernandez-Nieto, G. M., Martinez-Maldonado, R., Kitto, K., & Buckingham Shum, S. (2021). Modelling Spatial Behaviours in Clinical Team Simulations using Epistemic Network Analysis: Methodology and Teacher Evaluation. 386–396. https://doi.org/10.1145/3448139.3448176
61	Q1	Garcia-Ruiz, M., Santana-Mancilla, P. C., Gaytan-Lugo, L. S., & Iniguez-Carrillo, A. (2022). Participatory Design of Sonification Development for Learning about Molecular Structures in Virtual Reality. Multimodal Technologies and Interaction, 6(10). https://doi.org/10.3390/mti6100089
62	Q4	Gerdes, A., Heeren, B., Jeuring, J., & van Binsbergen, L. T. (2017). Ask-Elle: An Adaptable Programming Tutor for Haskell Giving Automated Feedback. Int. J. Artif. Intell. Educ., 27(1), 65–100. https://doi.org/10.1007/s40593-015-0080-x
64	Q4	Gibson, A., Aitken, A., Sándor, Á., Buckingham Shum, S., Tsingos-Lucas, C., & Knight, S. (2017). Reflective Writing Analytics for Actionable Feedback. 153–162. https://doi.org/10.1145/3027385.3027436
68	Q4	Han, S., Liu, M., Pan, Z., Cai, Y., & Shao, P. (2022). Making FAQ Chatbots More Inclusive: An Examination of Non-Native English Users' Interactions with New Technology in Massive Open Online Courses. Int. J. Artif. Intell. Educ. https://doi.org/10.1007/s40593-022-00311-4
70	Q4	Head, A., Xu, Y., & Wang, J. (2014). ToneWars: Connecting Language Learners and Native Speakers through Collaborative Mobile Games. In Intelligent Tutoring Systems (Vol. 8474, pp. 368–377). Springer International Publishing. http://link.springer.com/10.1007/978-3-319-07221-0_46
73	Q4	Holstein, K., McLaren, B. M., & Aleven, V. (2019). Co-Designing a Real-Time Classroom Orchestration Tool to Support Teacher-AI Complementarity. Journal of Learning Analytics, 6(2), 27_52.
75	Q2	Hosseini, R., Akhuseyinoglu, K., Brusilovsky, P., Malmi, L., Pollari-Malmi, K., Schunn, C., & Sirkiä, T. (2020). Improving Engagement in Program Construction Examples for Learning Python Programming. Int. J. Artif. Intell. Educ., 30(2), 299–336. https://doi.org/10.1007/s40593-020-00197-0
76	Q2	Hu, L., Wu, J., & Chen, G. (2022). iTalk–iSee: A participatory visual learning analytical tool for productive peer talk. International Journal of Computer-Supported Collaborative Learning, 17(3), 397–425. https://doi.org/10.1007/s11412-022-09374-w
77	Q4	Hutchins, N., & Biswas, G. (2023). Using Teacher Dashboards to Customize Lesson Plans for a Problem-Based, Middle School STEM Curriculum. 324–332. https://doi.org/10.1145/3576050.3576100
78	Q3	Hutt, S., Krasich, K., Brockmole, J., & D'Mello, S. (2021). Breaking out of the Lab: Mitigating Mind Wandering with Gaze-Based Attention-Aware Technology in Classrooms (WOS:000758168003023). CHI '21: PROCEEDINGS OF THE 2021 CHI CONFERENCE ON HUMAN FACTORS IN COMPUTING SYSTEMS. https://doi.org/10.1145/3411764.3445269
82	Q2	Jeong, Y., Cho, H., Kim, T., & Nam, T.-J. (2023). AutomataStage: An AR-Mediated creativity support tool for hands-on multidisciplinary learning. Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. https://doi.org/10.1145/3544548.3581408
85	Q4	Kaliisa, R., Kluge, A., & Mørch, A. I. (2020). Combining checkpoint and process learning analytics to support learning design decisions in blended learning environments. J. Learn. Anal., 7(3), 33–47. https://doi.org/10.18608/JLA.2020.73.4
86	Q4	Kang, S., Shokeen, E., Byrne, V., Norooz, L., Bonsignore, E., Williams-Pierce, C., & Froehlich, J. (2020). ARMath: Augmenting Everyday Life with Math Learning (WOS:000695432500125). PROCEEDINGS OF THE 2020 CHI CONFERENCE ON HUMAN FACTORS IN COMPUTING SYSTEMS (CHI'20). https://doi.org/10.1145/3313831.3376252
87	Q4	Khachatryan, G. A., Romashov, A. V., Khachatryan, A. R., Gaudino, S. J., Khachatryan, J. M., Guarian, K. R., & Yufa, N. V. (2014). Reasoning mind genie 2: An intelligent tutoring system as a vehicle for international transfer of instructional methods in mathematics. Int. J. Artif. Intell. Educ., 24(3), 333–382. https://doi.org/10.1007/s40593-014-0019-7
88	Q4	Khosravi, H., Kitto, K., & Williams, J. J. (2019). RiPPLE: A crowdsourced adaptive platform for recommendation of learning activities. J. Learn. Anal., 6(3), 91–105. https://doi.org/10.18608/jla.2019.63.12

90	Q3	Kim, G., Kim, J., & Kim, Y.-S. (2023). "Explain what a treemap is": Exploratory investigation of strategies for explaining unfamiliar chart to blind and low vision users. Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. https://doi.org/10.1145/3544548.3581139
92	Q1	Kivimäki, V., Pesonen, J., Romanoff, J., Remes, H., & Ihantola, P. (2019). Curricular concept maps as structured learning diaries: Collecting data on self-regulated learning and conceptual thinking for learning analytics applications. J. Learn. Anal., 6(3), 106–121. https://doi.org/10.18608/jla.2019.63.13
94	Q4	Labonte-LeMoyne, E., Leger, P., Robert, J., Babin, G., Charland, P., & Michon, J. (2017). Business intelligence serious game participatory development: Lessons from ERP-sim for big data. BUSINESS PROCESS MANAGEMENT JOURNAL, 23(3), 493–505. https://doi.org/10.1108/BPMJ-12-2015-0177
95	Q4	Lallé, S., Taub, M., Mudrick, N. V., Conati, C., & Azevedo, R. (2017). The Impact of Student Individual Differences and Visual Attention to Pedagogical Agents During Learning with MetaTutor. In Artificial Intelligence in Education (Vol. 10331, pp. 149–161). Springer International Publishing. http://link.springer.com/10.1007/978-3-319-61425-0_13
97	Q4	Lawrence, L., Echeverria, V., Yang, K., Aleven, V., & Rummel, N. (2023). How teachers conceptualise shared control with an AI co-orchestration tool: A multiyear teacher-centred design process. British Journal of Educational Technology. https://doi.org/10.1111/bjet.13372
98	Q4	Lee, A. V. Y., Luco, A. C., & Tan, S. C. (2023). A Human-Centric Automated Essay Scoring and Feedback System for the Development of Ethical Reasoning. Educational Technology and Society, 26(1), 147–159. https://doi.org/10.30191/ETS.202301_26(1).0011
99	Q4	Lee, E., Kamat, M., Temor, L., Schiafone, C., Fan, L., Liu, J., Coppin, P., Uribe-Quevedo, A., Ingino, R., Syed, A. R., Rojas, D., Lee, T., Perera, S., Dubrowski, A., & Sukhai, M. (2022). Prototyping a Spatial Skills AR Authoring Tool for Partially Sighted, Blind, and Sighted Individuals. 2022 IEEE Games, Entertainment, Media Conference, GEM 2022. https://doi.org/10.1109/GEM56474.2022.10017454
102	Q4	Liaqat, A., Munteanu, C., & Demmans Epp, C. (2021). Collaborating with Mature English Language Learners to Combine Peer and Automated Feedback: A User-Centered Approach to Designing Writing Support. International Journal of Artificial Intelligence in Education, 31(4), 638–679.
103	Q4	Lim, L.-A., Gasevic, D., Matcha, W., Ahmad Uzir, N., & Dawson, S. (2021). Impact of Learning Analytics Feedback on Self-Regulated Learning: Triangulating Behavioural Logs with Students' Recall. 364–374. https://doi.org/10.1145/3448139.3448174
105	Q2	Long, Y., Aman, Z., & Aleven, V. (2015). Motivational Design in an Intelligent Tutoring System that Helps Students Make Good Task Selection Decisions. In Artificial Intelligence in Education (Vol. 9112, pp. 226–236). Springer International Publishing. http://link.springer.com/10.1007/978-3-319-19773-9_23
107	Q4	Luo, J., & Zhang, S. (2022). Research on the design of project-based teaching & learning mode assisted by educational robot. Proceedings of the 4th World Symposium on Software Engineering, 148–154. https://doi.org/10.1145/3568364.3568387
108	Q4	Ma, S., Zhou, T., Nie, F., & Ma, X. (2022). Glancee: An adaptable system for instructors to grasp student learning status in synchronous online classes. 1–25.
112	Q2	Mangaroska, K., Sharma, K., Giannakos, M., Trætteberg, H., & Dillenbourg, P. (2018). Gaze Insights into Debugging Behavior Using Learner-Centred Analysis. 350–359. https://doi.org/10.1145/3170358.3170386
114	Q2	Martinez, J., Catasus, M., & Fontanillas, T. (2020). Impact of using learning analytics in asynchronous online discussions in higher education. INTERNATIONAL JOURNAL OF EDUCATIONAL TECHNOLOGY IN HIGHER EDUCATION, 17(1). https://doi.org/10.1186/s41239-020-00217-y
116	Q4	Martinez-Maldonado, R., Pardo, A., Mirriahi, N., Yacef, K., Kay, J., & Clayphan, A. (2015). The LATUX Workflow: Designing and Deploying Awareness Tools in Technology-Enabled Learning Settings. 1–10. https://doi.org/10.1145/2723576.2723583
118	Q2	Mcneill, G., Sondag, M., Powell, S., Asplin, P., Turkay, C., Moller, F., & Archambault, D. (2023). From asymptomatics to zombies: Visualization-based education of disease modeling for children. Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. https://doi.org/10.1145/3544548.3581573

119	Q2	Michos, K., Lang, C., Hernandez-Leo, D., & Price-Dennis, D. (2020). Involving teachers in learning analytics design: Lessons learned from two case studies (WOS:000558753800013). 94–99. https://doi.org/10.1145/3375462.3375507
122	Q2	Muldner, K., Lozano, C., Girotto, V., Burleson, W., & Walker, E. (2013). Designing a Tangible Learning Environment with a Teachable Agent. In Artificial Intelligence in Education (Vol. 7926, pp. 299–308). Springer Berlin Heidelberg. http://link.springer.com/10.1007/978-3-642-39112-5_31
123	Q2	Muslim, A., Chatti, M. A., Mahapatra, T., & Schroeder, U. (2016). A Rule-Based Indicator Definition Tool for Personalized Learning Analytics. 264–273. https://doi.org/10.1145/2883851.2883921
125	Q4	Nazaretsky, T., Bar, C., Walter, M., & Alexandron, G. (2022). Empowering Teachers with AI: Co-Designing a Learning Analytics Tool for Personalized Instruction in the Science Classroom. 1–12. https://doi.org/10.1145/3506860.3506861
127	Q4	Ngoon, T. J., Kovalev, D., Patidar, P., Harrison, C., Agarwal, Y., Zimmerman, J., & Ogan, A. (2023). 'An Instructor is [already] able to keep track of 30 students': Students' Perceptions of Smart Classrooms for Improving Teaching & Their Emergent Understandings of Teaching and Learning. Proceedings of the 2023 ACM Designing Interactive Systems Conference, 1277–1292. https://doi.org/10.1145/3563657.3596079
128	Q1	Nguyen, Q., Rienties, B., & Whitelock, D. (2020). A mixed-method study of how instructors design for learning in online and distance education. J. Learn. Anal., 7(3), 64–78. https://doi.org/10.18608/JLA.2020.73.6
129	Q3	Ochoa, X., & Dominguez, F. (2020). Controlled evaluation of a multimodal system to improve oral presentation skills in a real learning setting. Br J Educ Technol, 51(5), 1615–1630. https://doi.org/10.1111/bjet.12987
131	Q3	Ocumpaugh, J., Baker, R., Pedro, M., Hawn, M., Heffernan, C., Heffernan, N., & Slater, S. (2017). Guidance Counselor Reports of the ASSISTments College Prediction Model (ACPM) (WOS:000570180700063). 479–488. https://doi.org/10.1145/3027385.3027435
132	Q2	Oliver-Quelennec, K., Bouchet, F., Carron, T., Fronton Casalino, K., & Pinçon, C. (2022). Adapting Learning Analytics Dashboards by and for University Students. In Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics): Vol. 13450 LNCS (p. 309). https://doi.org/10.1007/978-3-031-16290-9_22
133	Q4	Olsen, J. K., Rummel, N., & Aleven, V. (2021). Designing for the Co-Orchestration of Social Transitions between Individual, Small-Group and Whole-Class Learning in the Classroom. International Journal of Artificial Intelligence in Education, 31(1), 24–56.
135	Q4	Ooge, J., Dereu, L., & Verbert, K. (2023). Steering recommendations and visualising its impact: Effects on adolescents' trust in E-learning platforms. Proceedings of the 28th International Conference on Intelligent User Interfaces, 156–170. https://doi.org/10.1145/3581641.3584046
141	Q2	Pishtari, G., Rodríguez-Triana, M. J., & Väljataga, T. (2021). A multi-stakeholder perspective of analytics for learning design in location-based learning. International Journal of Mobile and Blended Learning, 13(1), 1–17. https://doi.org/10.4018/IJMBL.2021010101
142	Q4	Pozdniakov, S., Martinez-Maldonado, R., Tsai, Y.-S., Cukurova, M., Bartindale, T., Chen, P., Marshall, H., Richardson, D., & Gasevic, D. (2022). The Question-driven Dashboard: How Can We Design Analytics Interfaces Aligned to Teachers' Inquiry? 175–185. https://doi.org/10.1145/3506860.3506885
146	Q2	Rapp, C., & Kauf, P. (2018). Scaling Academic Writing Instruction: Evaluation of a Scaffolding Tool (Thesis Writer). Int. J. Artif. Intell. Educ., 28(4), 590–615. https://doi.org/10.1007/s40593-017-0162-z
147	Q2	Read, S., DelaMerced, A., & Zachry, M. (2012). Participatory Design in the Development of a Web-based Technology for Visualizing Writing Activity as Knowledge Work (WOS:000324841800045). 333–340.
151	Q2	Rodrigues, L., Palomino, P. T., Toda, A. M., Klock, A. C. T., Pessoa, M., Pereira, F. D., Oliveira, E. H. T., Oliveira, D. F., Cristea, A. I., Gasparini, I., & Isotani, S. (2023). How Personalization Affects Motivation in Gamified Review Assessments. Int. J. Artif. Intell. Educ. https://doi.org/10.1007/s40593-022-00326-x
152	Q4	Rodríguez-Triana, M. J., Prieto, L. P., Dimitriadis, Y., de Jong, T., & Gillet, D. (2021). Ada for ibl: Lessons learned in aligning learning design and analytics for inquiry-based learning orchestration. J. Learn. Anal., 8(2), 22–50. https://doi.org/10.18608/JLA.2021.7357

153	Q4	Ruiz-Calleja, A., Dennerlein, S., Kowald, D., Theiler, D., Lex, E., & Ley, T. (2019). An infrastructure for workplace learning analytics: Tracing knowledge creation with the social semantic server. J. Learn. Anal., 6(2), 120–139. https://doi.org/10.18608/jla.2019.62.9
154	Q3	Santos, J. L., Govaerts, S., Verbert, K., & Duval, E. (2012). Goal-Oriented Visualizations of Activity Tracking: A Case Study with Engineering Students. 143–152. https://doi.org/10.1145/2330601.2330639
157	Q4	Sato, A. J., Sramek, Z., & Yatani, K. (2023). Groupnamics: Designing an interface for overviewing and managing parallel group discussions in an online classroom. Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. https://doi.org/10.1145/3544548.3581322
159	Q4	Schulz, S., McLaren, B. M., & Pinkwart, N. (2022). Towards a Tutoring System to Support Robotics Activities in Classrooms – Two Wizard-of-Oz Studies. Int. J. Artif. Intell. Educ. https://doi.org/10.1007/s40593-022-00305-2
160	Q3	Shardlow, M., Sellar, S., & Rousell, D. (2022). Collaborative Augmentation and Simplification of Text (CoAST): Pedagogical Applications of Natural Language Processing in Digital Learning Environments. Learning Environments Research, 25(2), 399–421.
162	Q4	Shibani, A., Knight, S., & Buckingham Shum, S. (2022). Questioning learning analytics? Cultivating critical engagement as student automated feedback literacy. 326–335. https://doi.org/10.1145/3506860.3506912
167	Q2	Shreiner, T. L., & Guzdial, M. (2022). The information won't just sink in: Helping teachers provide technology-assisted data literacy instruction in social studies. British Journal of Educational Technology, 53(5), 1134–1158. https://doi.org/10.1111/bjet.13255
168	Q2	Shute, V. J., Smith, G., Kuba, R., Dai, C.-P., Rahimi, S., Liu, Z., & Almond, R. (2021). The Design, Development, and Testing of Learning Supports for the Physics Playground Game. Int. J. Artif. Intell. Educ., 31(3), 357–379. https://doi.org/10.1007/s40593-020-00196-1
171	Q1	Soreny, C., Takhar, P., Vickers, E., Bartlett, R., Coates, E., Stavroulakis, T., Fox, E., Davidson, H., Harrison, A., & McDermott, C. (2022). Co-design of digital learning resources for care workers: Reflections on the neurocare knowhow project. Journal of Medical Engineering and Technology, 46(6), 518–526. https://doi.org/10.1080/03091902.2022.2089253
172	Q4	Suleman, R. M., Mizoguchi, R., & Ikeda, M. (2016). A New Perspective of Negotiation-Based Dialog to Enhance Metacognitive Skills in the Context of Open Learner Models. Int. J. Artif. Intell. Educ., 26(4), 1069–1115. https://doi.org/10.1007/s40593-016-0118-8
174	Q4	Tegos, S., Demetriadis, S., & Tsiatsos, T. (2014). A configurable conversational agent to trigger students' productive dialogue: A pilot study in the CALL domain. Int. J. Artif. Intell. Educ., 24(1), 62–91. https://doi.org/10.1007/s40593-013-0007-3
175	Q2	Tenório, K., Dermeval, D., Monteiro, M., Peixoto, A., & Silva, A. P. (2022). Exploring Design Concepts to Enable Teachers to Monitor and Adapt Gamification in Adaptive Learning Systems: A Qualitative Research Approach. Int. J. Artif. Intell. Educ., 32(4), 867–891. https://doi.org/10.1007/s40593-021-00274-y
176	Q2	Ternblad, E.-M., Haake, M., Anderberg, E., & Gulz, A. (2018). Do Preschoolers 'Game the System'? A Case Study of Children's Intelligent (Mis)Use of a Teachable Agent Based Play-&-Learn Game in Mathematics. In Artificial Intelligence in Education (Vol. 10947, pp. 557–569). Springer International Publishing. http://link.springer.com/10.1007/978-3-319-93843-1_41
177	Q4	Tissenbaum, M., & Slotta, J. (2019). Supporting classroom orchestration with real-time feedback: A role for teacher dashboards and real-time agents. INTERNATIONAL JOURNAL OF COMPUTER-SUPPORTED COLLABORATIVE LEARNING, 14(3), 325–351. https://doi.org/10.1007/s11412-019-09306-1
178	Q2	Topali, P., Ortega-Arranz, A., Chounta, I., Asensio-Perez, J., Martinez-Mones, A., & Villagra-Sobrino, S. (2022). Supporting instructors in the design of actionable feedback for MOOCs (WOS:000836390500274). 1881–1887. https://doi.org/10.1109/EDUCON52537.2022.9766546
179	Q2	Treasure-Jones, T., Dent-Spargo, R., & Dharmaratne, S. (2018). How do students want their workplace-based feedback visualized in order to support self-regulated learning? Initial results & reflections from a co-design study in medical education. 2193. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85053678222&partnerID=40&md5=0e13f7113188599442308fba21e0cc57
181	Q2	Tsai, Y.-S., Singh, S., Rakovic, M., Lim, L.-A., Roychoudhury, A., & Gasevic, D. (2022). Charting Design Needs and Strategic Approaches for Academic Analytics Systems through Co-Design. 381–391. https://doi.org/10.1145/3506860.3506939

184	Q4	Vannaprathip, N., Haddawy, P., Schultheis, H., & Suebnukarn, S. (2022). Intelligent Tutoring for Surgical Decision Making: A Planning-Based Approach. Int. J. Artif. Intell. Educ., 32(2), 350–381. https://doi.org/10.1007/s40593-021-00261-3
185	Q1	Vezzoli, Y., Mavrikis, M., & Vasalou, A. (2020). Inspiration Cards Workshops with Primary Teachers in the Early Co-Design Stages of Learning Analytics (WOS:000558753800010). 73–82. https://doi.org/10.1145/3375462.3375537
188	Q2	Vinella, F. L., Koppelaar, S., & Masthoff, J. (2022). Forming Teams of Learners Online in a User as Wizard Study with Openness, Conscientiousness, and cognitive Ability. 283–292. https://doi.org/10.1145/3511047.3537660
189	Q2	Wang, Q., Jing, S., & Goel, A. K. (2022). Co-Designing AI Agents to Support Social Connectedness Among Online Learners: Functionalities, Social Characteristics, and Ethical Challenges. 541–556. https://doi.org/10.1145/3532106.3533534
190	Q4	Wang, Y., Nguyen, H., Harpstead, E., Stamper, J., & McLaren, B. M. (2019). How Does Order of Gameplay Impact Learning and Enjoyment in a Digital Learning Game? In Artificial Intelligence in Education (Vol. 11625, pp. 518–531). Springer International Publishing. http://link.springer.com/10.1007/978-3-030-23204-7_43
191	Q3	Weigand, A. C., & Kindsmüller, M. C. (2021). HCD3A: An HCD Model to Design Data-Driven Apps. In Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics): Vol. 12797 LNAI (p. 297). https://doi.org/10.1007/978-3-030-77772-2_19
192	Q4	Wiley, K., Dimitriadis, Y., & Linn, M. (2023). A human-centred learning analytics approach for developing contextually scalable K-12 teacher dashboards. BRITISH JOURNAL OF EDUCATIONAL TECHNOLOGY. https://doi.org/10.1111/bjet.13383
195	Q3	Wilson, J., Huang, Y., Palermo, C., Beard, G., & MacArthur, C. A. (2021). Automated Feedback and Automated Scoring in the Elementary Grades: Usage, Attitudes, and Associations with Writing Outcomes in a Districtwide Implementation of MI Write. Int. J. Artif. Intell. Educ., 31(2), 234–276. https://doi.org/10.1007/s40593-020-00236-w
201	Q4	Zhang, Z., Xu, Y., Wang, Y., Yao, B., Ritchie, D., Wu, T., Yu, M., Wang, D., & Li, T. J.-J. (2022). StoryBuddy: A Human-AI Collaborative Chatbot for Parent-Child Interactive Storytelling with Flexible Parental Involvement. Conference on Human Factors in Computing Systems - Proceedings. https://doi.org/10.1145/3491102.3517479
203	Q3	Zhou, Q., Suraworachet, W., Pozdniakov, S., Martinez-Maldonado, R., Bartindale, T., Chen, P., Richardson, D., & Cukurova, M. (2021). Investigating Students' Experiences with Collaboration Analytics for Remote Group Meetings. In Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics): Vol. 12748 LNAI (p. 485). https://doi.org/10.1007/978-3-030-78292-4_38
204	Q1	Zwolinski, G., Kaminska, D., Laska-Lesniewicz, A., Haamer, R. E., Vairinhos, M., Raposo, R., Urem, F., & Reisinho, P. (2022). Extended Reality in Education and Training: Case Studies in Management Education. ELECTRONICS, 11(3). https://doi.org/10.3390/electronics11030336

Table 1: A comprehensive reference list of included articles in the two-dimensional HCAI framework quadrants (Q1–Q4).

Videnovik *et al.*
International Journal of STEM Education (2023) 10:54
https://doi.org/10.1186/s40594-023-00447-2

International Journal of STEM Education

REVIEW

Open Access

Game-based learning in computer science education: a scoping literature review

Maja Videnovik[1], Tone Vold[2*], Linda Kiønig[2], Ana Madevska Bogdanova[3] and Vladimir Trajkovik[3]

Abstract

Using games in education has the potential to increase students' motivation and engagement in the learning process, gathering long-lasting practical knowledge. Expanding interest in implementing a game-based approach in computer science education highlights the need for a comprehensive overview of the literature research. This scoping review aims to provide insight into current trends and identify research gaps and potential research topics concerning game-based learning in computer science. Using standard methodology for scoping review, we identified 113 articles from four digital libraries published between 2017 and 2021. Those articles were analyzed concerning the educational level, type of the game, computer science topic covered by the game, pedagogical strategies, and purpose for implementing this approach in different educational levels. The results show that the number of research articles has increased through the years, confirming the importance of implementing a game-based approach in computer science. Different kinds of games, using different technology, concerning different computer science topics are presented in the research. The obtained results indicate that there is no standardized game or standardized methodology that can be used for the creation of an educational game for computer science education. Analyzed articles mainly implement a game-based approach using learning by playing, and no significant focus is given to the effectiveness of learning by designing a game as a pedagogical strategy. Moreover, the approach is mainly implemented for developing computational thinking or programming skills, highlighting the need for its implementation in other topics beyond programming.

Keywords Scoping review, Game-based learning, Educational games, Computer science, Computer science education

Introduction

The world is changing very fast due to the emergence of technology in our everyday lives. This tremendous change can be noticed in different areas, including education. Students are influenced by the digital era, surrounded by technology and working with a massive amount of digital information on an everyday base. They are used to interactive environments and fast communication and prefer learning by doing (Unger & Meiran, 2020). Traditional learning environments, where students should sit and listen to the information provided by the teachers are unacceptable for them (Campbell, 2020). Students require active learning environments, using the possibilities of various technology applications to gain knowledge. They seek more interesting, fun, motivating and engaging learning experiences (Anastasiadis et al., 2018).

Creating engaging learning environments can develop students' critical thinking, problem-solving skills, creativity and cooperation, preparing students for living in

*Correspondence:
Tone Vold
tone.vold@inn.no
[1] Center for Innovations and Digital Education Dig-Ed, Solunska glava br.3, 1000 Skopje, North Macedonia
[2] Department of Business Administration and Organizational Studies, Inland Norway University of Applied Sciences, Holsetgata 31, 2318 Hamar, Norway
[3] Faculty of Computer Science and Engineering, Ss Cyril and Methodius University in Skopje, Rugjer Boshkovikj 16, 1000 Skopje, North Macedonia

Springer Open

© The Author(s) 2023. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

a constantly changing world (Joshi et al., 2022; Lapek, 2018; Tang et al., 2020). Education needs to shift toward active learning approaches that will encourage students to engage on a deeper level than traditional lecture-based methods (Boyer et al., 2014). To achieve this, teachers must find an approach tied to digital tools that students use daily (Videnovik et al., 2020).

Implementation of a game-based learning approach for creating engaging learning environments

Game-based learning is considered one of the most innovative learning approaches for increasing students' interest in education by playing games (Priyaadharshini et al., 2020). It refers to using games as an educational tool or strategy to facilitate learning and engagement (Li et al., 2021). Game-based learning involves designing and incorporating educational content within a game format, where players actively participate and interact with the game mechanics to acquire knowledge or develop skills. Many approaches tackle the umbrella of application of game-based learning in different educational fields. Different playful experiences can enable children to construct knowledge by playing and exploring a real-world problem often driven by students' interest in inquiry (Hirsh-Pasek, 2020). Gamification is a process that uses game elements, such as points, rewards, badges and competition during the learning process, establishing interactive and engaging learning environments (Turan et al., 2016). Gamification aims to enhance motivation, engagement, and participation using the inherent appeal of games. Designing interactive and entertaining games, primarily for education, is a step forward in implementing game-based learning. Serious games enable players to cultivate their knowledge and practice their skills by overcoming numerous interruptions during gaming (Yu, 2019). Effectively designed serious games facilitate learning by stimulating creativity, igniting interest, promoting discourse, and cultivating a competitive drive for exploration in diverse fields. Different mobile and location-based technologies provide opportunities to embed learning in authentic environments and thereby enhance engagement and learning outside traditional formal educational settings (Huizenga et al., 2009). Those games can simulate various aspects of reality, such as driving a vehicle, managing a city, or piloting an aircraft, allowing players to experiment and make decisions in a safe space without real-world consequences (Toh & Kirschner, 2020).

Games enable the integration of intrinsic and extrinsic motivational components to create an environment, where players feel more motivated to engage in the activities (Hartt et al., 2020). When digital game-based learning is implemented, including key game design elements (collaboration, choice, feedback), there is typically a positive impact on student engagement (Serrano, 2019; Wang et al., 2022). Students approach gameplay with interest and dedication and are persistent in progressing it. Therefore, teachers must find different ways to implement a game-based approach in the classroom, utilizing students' engagement, persistence and motivation during gameplay for classroom activities. During game-based learning, students have fun and enjoy themselves with increased imagination and natural curiosity, which can lead to high levels of participation and the student's involvement in the learning process. In this way, students can be more successfully engaged in meaningful learning than traditional teaching methods (Hamari et al., 2016; Huizenga et al., 2009; Karram, 2021).

Research on using a game-based learning approach in education

In the last decade, the game-based approach is receiving increasing attention in the research community due to its potential to increase students' motivation and engagement, promoting a student-centred learning environment. Many researchers show that digital game-based learning is becoming a powerful tool in education, making learning more enjoyable, easier and efficient (Boyle et al., 2016; Hafeez, 2022). Implementation of a game-based learning approach can provide students with an engaging, motivating and stimulating environment (Ghergulescu & Muntean, 2012; Hwang et al., 2014), supporting them to focus on the task and increasing overall learning experiences (Hamari et al., 2016). Moreover, game-based learning has the potential to improve students' competencies and academic performance (Clark et al., 2016; López-Fernández et al., 2021a, 2021b; Mezentseva et al., 2021; Noroozi et al., 2020; Sanchez Mena & Martí-Parreño, 2017; Vu & Feinstein, 2017). It presents the learners with rich, immersive environments and experiences that are not just about learning facts but enables the development of problem-solving, decision-making, and strategic planning (Lymbery, 2012; Sung & Hwang, 2013) skills. In addition, the student's academic achievement using a game-based approach is better than those learning through the traditional method (Arcagök, 2021; Partovi & Razavi, 2019; Roodt & Ryklief, 2022; Wang et al., 2022). Educational games promote active and self-directed learning, enabling students to learn from authentic situations and receive immediate feedback (Pellas & Mystakidis, 2020; Zhao et al., 2021). It can be highly personalized, allowing students to learn at their own pace and in a way best suited to their individual needs and learning styles, engaging them in the self-assessment process (Videnovik et al., 2022). In a gaming environment, students can explore different scenarios,

make choices, and learn from the consequences of their actions without fear of making a mistake.

Despite the great potential of the game-based approach for learning, it must be noted that developing educational games can be very complex and costly, and faces significant challenges (Boyle et al., 2016). The process of designing an educational game needs a lot of planning and requires a lot of skills (Hussein et al., 2019). Teachers do not have necessary skills to develop a game that combines entertainment and educational elements to increase student's interest and motivation during learning (Qian & Clarck, 2016). On the other side, game developers have problem to align educational goals within the game. In addition, the games must be well-designed and with the right level of complexity so the learners should not be bored or frustrated during the play (Liu et al., 2020; Vlahu-Gjorgievska et al., 2018), taking into account both educational and entertainment elements. That is why educators cannot depend solely on professional game designers and must take on the responsibility of creating these immersive learning experiences themselves or by engaging their students in the design process.

Game-based learning approach in computer science education

The game-based approach provides a dynamic and effective way for students to learn and apply their knowledge in a variety of subjects, such as math (Vankúš, 2021), physics (Cardinot & Fairfield, 2019), languages (Lee, 2019), and history (Kusuma et al., 2021). This approach allows students to learn complex concepts and skills in a fun and interactive way while also fostering critical thinking and collaboration. It is particularly effective in computer science, where students can learn about algorithms, data structures, networks, software testing and programming languages by designing and testing their games and simulations (Kalderova et al., 2023). In addition, game-based learning can help to bridge the gap between theory and practice, allowing students to apply their knowledge in a real-world context (Barz et al., 2023).

The importance of computer science has been emphasized in the last decade through different campaigns and online platforms. Their main aim is to develop students' computational thinking skills and attract students to coding, mainly through a game-based approach (code.org, codeweek.org). They offer teachers access to materials and learning scenarios covering different unplugged activities and block-based programming. Students have an opportunity to play games and learn basic programming concepts through fun and interactive activities, developing collaboration and competitiveness at the same time. Game narratives, collecting points, and immediate feedback through these games increase students' engagement. These platforms are a valid option for developing computational thinking at an early age and a good way for students to develop creativity, critical thinking and problem-solving skills (Barradas et al., 2020).

Various block-based programming languages, which are also accessible online (Scratch,[1] Snap,[2] Blockly[3]), are used to develop students' computational thinking and block-based programming skills, especially in primary education. In addition, they support the development of interactive projects that students can use afterward (Tsur & Rusk, 2018). Moreover, students can develop animations, interactive stories, and games, which allow them to engage in the coding process, learn programming concepts and even learn about other computer science topics during game design.

Topics connected with programming are the most common in computer science, but learning how to program is often recognized as a frustrating activity (Yassine et al., 2018). Learning object-oriented programming languages is especially difficult for students, because programming concepts are complex, cognitively demanding, require algorithmic thinking and problem-solving skills, and is a long-term process (Zapušek & Rugelj, 2013). Game-based learning stimulates active learning and enables students to learn about programming concepts in fun and engaging ways through visual interfaces and engaging environments (CodeCombat,[4] Alice,[5] Greenfoot[6]). Those engaging and motivating environments enable simplifying complex programming concepts, such as inheritance, nested loops, and recursion (Karram, 2021).

Different pedagogical strategies can be used to implement game-based learning in computer science, empowering students' skills and increasing their active engagement in learning. For example, students can deepen their knowledge and skills on a given topic by playing the game (Hooshyar et al., 2021; Shabalina et al., 2017) or through the process of game design (Denner et al., 2012; Zhang et al., 2014). In both cases, the game-based approach can increase students' motivation and engagement in learning (Chandel et al., 2015; Park et al., 2020).

[1] https://scratch.mit.edu/

[2] https://snap.berkeley.edu/

[3] https://blockly.games/

[4] https://codecombat.com/

[5] https://www.alice.org/

[6] https://greenfoot.org/door

Table 1 Review articles concerning the implementation of game-based approach in computer science

Title	References	Outcomes
A systematic review of learning object-oriented programming through serious games and programming approaches	Abbasi et al. (2017)	Systematic literature review identifying serious games developed or incorporated for learning object-oriented programming, object-oriented programming concepts covered in those games, and programming approaches applied
Serious games to prevent and detect bullying and cyberbullying: A systematic serious games and literature review	Calvo-Morata et al. (2020)	The systematic literature review focused on the use of video games as tools for the prevention and detection of bullying and cyberbullying
Fostering computational thinking through unplugged activities: A systematic literature review and meta-analysis	Cheng et al. (2023)	Systematic review and meta-analyses summarizing the effect of unplugged activities on enhancing students' computational thinking skills
Evaluating Aspects of Usability in Video Game-Based Programming Learning Platforms	Diaz et al. (2021)	Scoping review of video game platforms that can be used for developing of programming skills
A Systematic Mapping Study on Game Elements and Serious Games for Learning Programming	Dos Santos et al. (2019)	Systematic review about serious games for learning programming, game elements and methods for their evaluation
Game-Based Information Security/Privacy Education and Awareness: Theory and Practice	Karagiannis et al. (2020)	Reviewing of methods and tools for deploying a game-based approach for security/privacy learning and awareness, and their assessment
A comparative analysis of programming games, looking through the lens of an instructional design model and a game attributes taxonomy	Laporte and Zaman (2018)	Qualitative, comparative analysis of 19 programming games from an instructional and game attributes perspective
A Review of Gamification for Learning Programming Fundamental	Shahid et al. (2019)	Scoping review of the existing literature of serious programming games, identifying game elements that should be included to ensure students' active participation
Improving girls' perception of computer science as a viable career option through game playing and design: Lessons from a systematic literature review	Sharma et al. (2021)	Systematic literature review about the relation between the various games playing or designing activities and their impact on girls' perception of computer science as a career choice
Cyber security training a survey of serious games in cyber security	Tioh et al. (2017)	A survey about background as well as the current state of serious games dealing specifically with the topic of cybersecurity

Existing reviews of game-based approach in computer science

Existing reviews of game-based approach in computer science provide valuable information about the latest trends in the implementation of game-based approach in the last few years. Table 1 presents latest trends in the implementation of game-based learning in computer science education.

Most of the review articles analyze publications that describe the implementation of game-based approach for learning programming (Abbasi et al., 2017; Diaz et al., 2021; Dos Santos et al., 2019; Laporte & Zaman, 2018; Shahid et al., 2019), from different aspects: game design, game elements, or their evaluation. However, there are some of them tackling other topics, such as cybersecurity (Karagiannis et al., 2020; Tioh et al., 2017) or cyberbullying (Calvo-Morata et al., 2020). Sharma et al. (2021) analyzes the impact of game-based learning on girls' perception toward computer science. There are review articles that focus on just one aspect of computer science. For example, Chen et al. (2023) provides meta-analyses to investigate potential of unplugged activities on computational thinking skills.

In our review, we aim to perform the broader analysis of the research articles referring to the game-based approach in various computer science topics, different educational levels and different types of games. For that purpose, instead of systematic review, we have opted to perform the scoping review on significantly larger set of articles.

Valuable insight regarding the game-based approach in computer science has been provided in research concerning different educational levels, computer science topics, and used games. However, computer science is a field that is changing very fast, and the number of games that can be used for developing students' knowledge and skills is increasing all the time. As a result, continuous research in this field should be done.

This research aims to elaborate on current trends concerning the game-based approach in computer science. It focuses on the educational level, covered computer science topic, type of the game, purpose for its use, and pedagogical strategies for the implementation of this approach. Moreover, possible gaps and potential research topics concerning game-based learning in computer science in primary education are identified.

Current review

This research represents scoping review that identifies the educational context and the type of games used for implementing a game-based learning approach in computer science. The scoping review method was selected over systematic literature review, because we wanted to determine the scope of the literature in the field of game-based learning in computer science education, to examine how research is done on this topic and to identify and analyze research gaps in the literature (Munn et al., 2018).

Following Arksey and O'Malley (2005) five-step framework, which adopts a rigorous process of transparency, enabling replication of the search method and increasing the reliability of the results, the steps of the applied review process are: to (1) identify research questions (2) identify relevant studies, (3) study selection of papers, (4) charting the data, (5) summarizing and reporting the results.

Research questions

The focus of our research was to analyze what type of games were used in computer science, the subject's topics that were covered by the game and pedagogical strategies for implementing game-based learning, comparing all these in different educational levels. Starting from this, our research questions are:

RQ1: What kind of educational games are usually used during the implementation of the game-based approach in computer science?

Various games are used to cover topics from computer science, from block-based serious games (Vahldick et al., 2020) to educational escape rooms (López-Pernas et al., 2019). Using different games influences the learning process differently (Chang et al., 2020). The RQ1 seeks to identify and understand the types of educational games that are commonly utilized in the context of teaching computer science. Exploration of the variety of used games provides insights into the different approaches, mechanics, and formats used to enhance learning outcomes.

RQ2: Which pedagogical strategy is mostly used in the published research?

There are various strategies for implementing game-based learning in computer science education. The implementation strategies refer to whether students should learn by playing the game (Malliarakis et al., 2014) or by designing a game (Denner et al., 2012). The strategies can differ based on the gender of students (Harteveld et al., 2014), students' age (Bers, 2019), or the adopted approach by policymakers (Lindberg et al., 2019). RQ2 aims to identify the predominant pedagogical strategy employed in the published research on game-based approaches in computer science education. By examining the pedagogical strategies, researchers can gain insights into the most effective instructional methods that facilitate learning through game-based approaches. Furthermore, the findings can inform educators and researchers in designing and implementing effective instructional

strategies that align with the goals of computer science education.

RQ3: Which computer science topics are covered by the game-based approach?

Game-based learning can be used to teach different computer science topics, from introduction topics (Fagerlund et al., 2021; Mathew et al., 2019), to core topics (Karram, 2021). RQ3 aims to provide value in exploring the specific computer science topics addressed through game-based approaches. In addition, it helps identify the range of topics that have been integrated into educational games. By understanding the computer science topics covered, researchers can assess the breadth and depth of the game-based approach and identify potential gaps or areas for further exploration in the curriculum.

RQ4: What are the potential research topics concerning the implementation of a game-based approach in computer science?

RQ4 is essential as it seeks to identify potential areas for future research in the implementation of game-based approaches in computer science education. It might include specific computer science topics (Calvo-Morata et al., 2020), strategies to implement game-based learning in computer science (Hooshyar et al., 2021), or ways to analyze the effects of game-based learning (Scherer et al., 2020). By exploring research topics that have not been extensively studied or require further investigation, researchers can identify new directions and opportunities for advancing the field. This can contribute to the ongoing development and improvement of game-based approaches in computer science education, fostering innovation and addressing emerging challenges.

Methodology

To answer research questions, we analyzed the contents of articles published from 2017 to 2021. Due to the rapid development of technology and change in the learnt computer science topics as well as designed game with new technology and tools, we have decided to research the articles that refer just to the interval of 5 years. As technology progresses swiftly, studying 5 year interval of the published literature ensures that scoping review results analyze the most current tools, approaches, and methodologies being utilized in the field of computer science education.

The research was done according to the PRISMA-ScR (Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews) guidelines (Peters et al., 2020). The PRISMA-ScR methodology is a structured approach used to conduct comprehensive and transparent scoping reviews. It involves identifying a research question, performing a systematic search of relevant literature, applying inclusion and exclusion criteria to select studies, extracting data from the included studies, analyzing and synthesizing the data to identify key themes or patterns, and reporting the findings. It aims to map the existing literature on a particular topic, identify key concepts, and examine the extent, range, and nature of research available. It is particularly useful for exploring complex and diverse research questions.

There is a large number of articles regarding the topic, so performing this kind of research manually seemed like labor-intensive work. Therefore, we have identified the opportunity to use the Natural Language Processing (NLP) toolkit (Zdravevski et al., 2019) to automate the literature search, scanning, and eligibility assessment. We have used this toolkit for article identification and selection (i.e., scanning procedures and eligibility criteria assessment). The search considered articles indexed in four digital libraries: IEEE, PubMed, Springer and Elsevier. The NLP toolkit requires structured data input comprising keywords, properties, property groups, required relevance, included sources, and start and end years.

The provided keywords serve as search criteria within available libraries, acting as the primary filter to determine which articles will be gathered for further analysis. At the beginning of setting up the NLP toolkit for the research, to address different games that can be used in education, we have identified the main keywords to be "Serious Games", "Educational Games", "Games in education" or "Games for learning". The NLP toolkit used these keywords to identify the potentially relevant articles in the mentioned digital libraries.

Furthermore, the NLP toolkit was adjusted to search specific properties (words or phrases) within the title, abstract, or keywords of already identified articles to select relevant articles in more detail, according to the features (properties groups) of the game-based learning approach that we are interested in: subject, educational level, educational context, purpose and used technology. Properties groups address synonyms and various versions of the phrase (e.g., educational games and serious games). To be included in the results, at least one representative from each property group must appear in the title or abstract of the article, thereby functioning as a secondary filter for identifying relevant articles.

The property group "subject" was set as mandatory during the search, because we were interested in analyzing articles that refer to game-based learning just in computer science. Since the name of this subject is different in different countries, we have used synonyms, such as "programming", "coding", and "informatics". The property group "age" or educational level included different synonyms for primary and secondary education, as well as higher education, although we did not make this

Table 2 NLP toolkit input parameters: keywords, properties, and property groups

Input parameter	Natural language processing toolkit input parameters	
Keyword	Serious Games OR Educational Games OR Games in education OR Games for learning	
Property Group	Subject	Computer Science, programming, coding, informatics
	Age	primary education, K12, primary school, pupils, elementary school, secondary education, high school, adolescents, teenagers, secondary school, university
	Purpose	engagement, Quality of Experience, educational evaluation, learning achievement, interest, motivation, game evaluation
	Technology	Web, online, mobile, augmented reality, virtual reality

property mandatory. To search about the used technology (web, online, mobile, augmented reality, virtual reality), we have set one property group to include a different kind of used technology, and we also set a property group that refers to the aim of using these educational games (to achieve students' engagement, increase motivation, evaluation of educational results, etc.). A more detailed description of the properties groups is given in Table 2.

The following input parameter for the NLP toolkit setup is the minimum relevant properties. In this research, it was set that each article has to contain a minimum of two of the previously defined properties to be considered relevant. The quality analysis of the relevant articles followed in the next step of the methodology.

Study selection

The initial search in four digital libraries: IEEE, PubMed, Springer and Elsevier, has identified 43,885 articles concerning using game-based learning in computer science. After articles had been identified based on the specified keywords and retrieved from the publishers, the duplicates were identified according to the article DOI as their unique identifier and removed, which has decreased the number of articles to 21,002. In the next step, the articles selection (screening and eligibility assessment) procedures followed, discarding articles not published in the required period or for which the title or abstract could not be analyzed because of parsing errors, unavailability, or other reasons. The screening process eliminated 11,129 articles and the remaining 9873 articles underwent an automated eligibility assessment using the advanced NLP toolkit functionalities. The automated eligibility analysis involved the following processing: tokenization of sentences (Manning et al., 2014; Webster et al., 1992) and English stop words removal, stemming, and lemmatization using the Natural Language Toolkit library (Bird, 2006). Furthermore, articles containing less than two properties were removed, which left 1209 articles eligible for further manual analysis and inclusion in identifying the research trends and summarizing the results.

For each of the articles from the collection of relevant articles, the toolkit automatically generated a bibliographic file (as defined by BibTeX reference management software). This file was manually analyzed in more detail to identify the most relevant articles for the purpose of our study. First, the abstract was read to see whether the article was relevant, and if that did not provide enough information, the whole article was read. For each of the research questions we used the same approach, but with different focuses. For the first research question, we looked for any specific game name. For the second research question, we were looking for any mentioning of the pedagogical approaches or strategies. For the third research question, we looked for different computer science topics used in computer science curricula. In that way, the most relevant articles concerning first three research questions were identified. The last research question is related to future potential research topics in the field of game-based learning in computer science education, so it was not used during this phase of selection of relevant articles.

As a result of the manual analysis of articles' titles, articles that did not refer to computer science subjects were excluded, which left just 206 articles. We could not obtain the full text for some of articles, so they were excluded from further analyses. Some articles did not refer to using games to teach computer science topics, so they were also removed. The same was the case with a few articles not written in English. Finally, we had 125 relevant articles.

Nine relevant articles were review papers that referred to different game-based learning approaches at different educational levels. Among identified articles is a book describing different teaching methods in computer science education, including game-based learning (Hazzan et al., 2020). Two book chapters refer to different approaches of using game-based learning in education (Bellas et al., 2018; Zaw & Hlaing, 2020). These articles were also excluded from the list.

Finally, we finished the selection process and got 113 relevant articles using educational games in computer science that were the subject of further analysis.

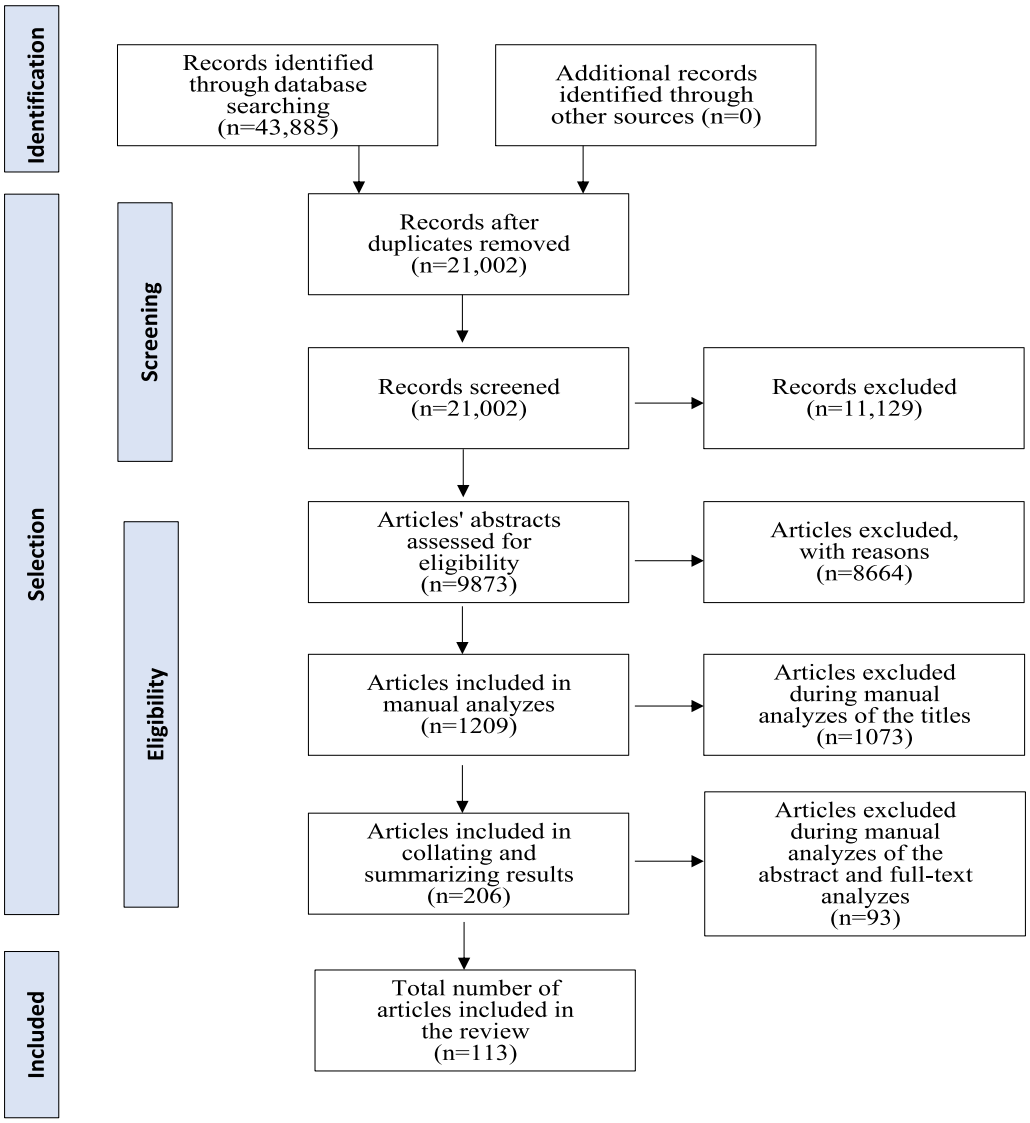


Fig. 1 Flowchart of the PRISMA-SCR-based selection process

The information flowchart presenting the numbers of identified, screened, processed, and removed articles in the automated NLP procedure and articles removed during the manual analysis is presented in Fig. 1.

After the final identification of the most relevant studies concerning game-based learning in computer science, summaries were developed for each article. Information about their correspondence to education, educational level, used game, type of the game, covered computer science topic, educational context and general usefulness of the article was provided.

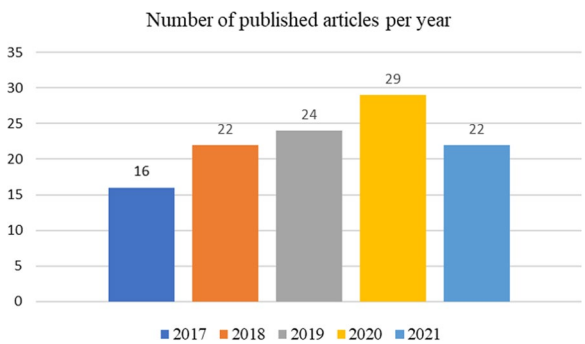


Fig. 2 Distribution of the published articles through the years

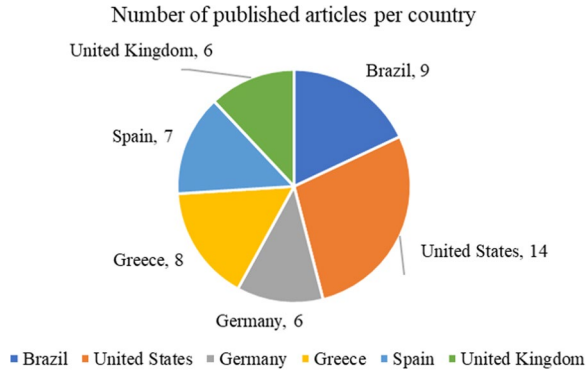


Fig. 3 Distribution of the published articles per country, showing countries with more than five published articles

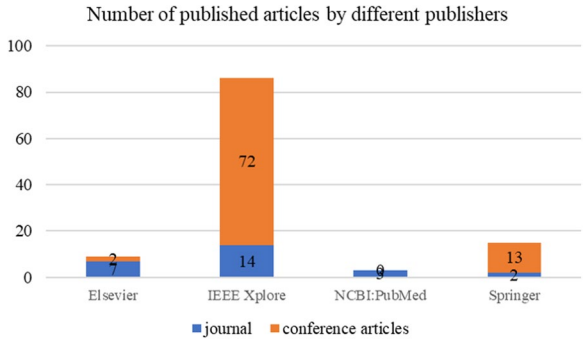


Fig. 4 Distribution of the published articles by different publishers

Results

Distribution of published articles through the years

The distribution of the articles concerning the game-based approach in computer science through the years is presented in Fig. 2. It can be noticed that the number of articles was increasing through the years, but then suddenly, in 2021, that number decreased. The reason might be found in the situation with the pandemic, because in 2020 and 2021, most of the schools were closed. In some of them, the teaching was transferred online, which resulted in a huge change in the way of teaching and learning, and it was a period of adaptation for teachers and students at the same time, which might lead to a decrease of the research articles.

Distribution of published articles per country

The distribution of the published articles per country differs from country to country. Figure 3 presents the distribution of published articles per country, showing only the countries that have more than five published articles concerning game-based learning between 2017 and 2021. Most articles are published in the United States, followed by Brazil and Greece.

Further analysis of the relevant articles depending on the country, where the research was conducted, shows that just 17 (of 113) articles are joint work of researchers from different countries. Moreover, just two present joint research on game-based learning from three countries. The first one describes the methodology implemented within the European initiative Coding4girls, which proposes to teach coding through a game design based on a design thinking methodological approach linked to creativity and human-centred solutions (De Carvalho et al., 2020). The second joint research (Agbo et al., 2021) describes the students' online co-creation of mini-games to develop their computational thinking skills. Interestingly, all other published articles describe implementing a game-based learning approach in computer science in the local context, making it difficult to generalize the conclusions and the research outcomes.

Distribution of published articles by publisher

Most of the relevant researched articles are published by IEEE Xplore (86 of 113) but mostly published as part of the proceedings at different conferences. This might explain why the number of published articles from IEEE Xplore differs from other publishing companies. Figure 4 presents the distribution of the articles by each of the publishers in detail, comparing published articles in journals and at conferences.

Distribution of published articles by educational level

Identifying the number of articles according to the educational level was more complicated due to the different educational systems in different countries, resulting in a different understanding of the terms "primary", and "secondary" education. In some countries, the same educational level is entitled as "primary", and in others as "lower secondary" or even "middle school". For example, in some countries, the primary school includes 6–14-year-old students; in others, it is divided, so there are primary (from 6 to 10 years), middle (11–13 years) and high schools (14–18 years); and in some, there are even lower secondary school (12–16 years). Therefore, we have tried to combine different categories according to the student's age and to gather three levels: primary, secondary and university, according to the local context (primary education includes 6–14 years, secondary education includes 15–18 years). The situation with the distribution of the relevant articles is presented in Fig. 5.

It can be noticed that most of the articles concern universities, although the number of articles that concern using games in computer science in primary and secondary schools is not small. It can be expected, because most of the articles refer to using games for developing

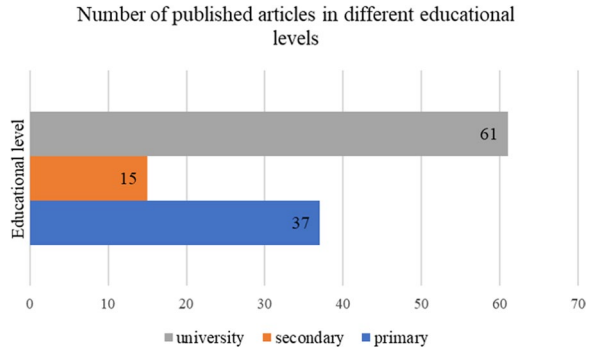


Fig. 5 Distribution of the published articles in different educational levels

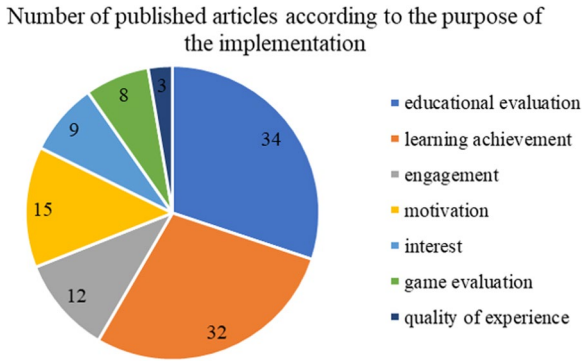


Fig. 6 Distribution of the published articles according to the purpose of the implementation

programming skills, which is present mainly at the university level. However, in some countries, primary school students learn fundamental programming concepts.

Distribution of published articles by the purpose of implementation

The purpose of the research concerning game-based learning in computer science is different and mostly depends on the type of the game as well as the topic that is covered by the game. The distribution of the published articles according to the purpose of the implementation of the research is presented in Fig. 6. However, it must be mentioned that it was difficult to distinguish the purposes of implementing the game-based approach in computer science, because the purpose was not clearly stated in the articles or there was overlapping among different categories.

In the most articles (66 of 113), the research is done to measure students' learning achievement or to evaluate the benefits of the game-based approach by comparing students' knowledge and skills before and after implementing this approach. In addition, some articles are interested in students' engagement and raising students' interest and motivation for the learning process by implementing a game-based approach. However, just a few articles refer to using this approach for measuring students' overall satisfaction with the whole experience (3 of 113).

Distribution of published articles by implemented pedagogical strategy and used technology

Manual analyses of the included articles gave us insight into additional aspects of implementing a game-based approach in computer science. When we talk about the game-based approach, there are two main pedagogical strategies for implementation: students can learn by playing the game, and students can learn while creating the game. The distribution of those two approaches in the published articles indicates that learning by playing games is more frequently used than learning by creating games. Only 19 of 113 relevant articles refer to the implementation of a game-based approach, where students learn during the process of game design or are involved themselves in the creation of the game. In most of the articles, students just use the created game (previously created or designed for the purpose of the research) to develop their competencies on a given topic. Regarding the technology used for the creation of the games in the published articles, it can be noticed that most of the games are web-based (although they have a mobile version, too), and there are just a few articles concerning the use of the unplugged activities as a game-based approach for learning computer science.

Distribution of published articles by covered computer science topic

Most of the articles concerning computer science topics covered during the implementation of the game-based approach refer to using to develop students' programming skills in object-oriented programming, followed by the articles concerning block-based programming and the development of computational thinking skills. The number of articles that utilize the game-based approach in all other computer science topics is significantly smaller (in total, 14 from 113 articles). Figure 7 contains more detailed information about this distribution.

Discussion

Types of educational games used for implementation of the game-based approach in computer science

Our research aims to provide information about the latest research trends concerning game-based learning in computer science education. Table 3 gives information about the implemented game, the type of the game, the computer science topic covered by the game, and the

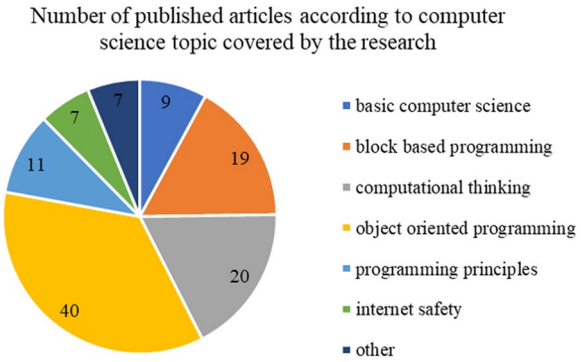


Fig. 7 Distribution of the published articles according to the covered computer science topics

educational level, where the research concerning the game-based approach in computer science was carried out. The type of the game refers to the origin of the game creation, whether the game was already created and can be used or is created for the research by the author or by the students (they are learning during the game design process).

Detailed analysis of these relevant articles shows that different educational games are used to implement game-based learning in computer science, implementing different technologies for their design. Articles refer to using different platforms, environments or engines for creating games using different technology. In primary education, most implemented approaches include block-based environments, such as Blocky, Snap!, and Scratch. Those platforms give access to the already created game (De Carvallho et al., 2020; Sáiz Manzanares et al., 2020; Vourletsis & Politis, 2022) but also offer possibilities a game to be created by a teacher (Bevčič & Rugelj, 2020; Holenko Dlab & Hoic-Bozic, 2021; Wong & Jiang, 2018) or by the students during the learning process (Funke et al., 2017; Zeevaarders & Aivaloglouor, 2021). Even more, their use as a platform to code Arduino boards is presented in two of the articles (Sharma et al., 2019; Yongqiang et al., 2018). Block-based environments are used in the research in secondary education, too. For example, Araujo et al. (2018) measured students' motivation for learning block-based programming by involving students in creating games in Scratch. Schatten and Schatten (2020) involve students in creating different games using CodeCombat during the CodeWeek initiative to increase their interest in programming, and Chang and Tsai (2018) are implementing an approach for learning programming in pairs while coding Kinnect with Scratch.

However, in the research articles concerning secondary education, it can be noticed that some specified games are created by the researcher (or teacher) to develop some concrete computer science skills. In these cases, the articles focus on the evaluation of the effectiveness of the game as an approach. For example, the chatbot's serious game "PrivaCity" (Berger et al., 2019) is designed to raise students' privacy awareness, as a very important topic among teenagers.

Similarly, "Capture the flag" is a game designed for learning about network security in a vocational school (Prabawa et al., 2017). The effectiveness of using the educational game "Degraf" in a vocational high school as supplementary material for learning graphic design subjects is measured by Elmunsyah et al. (2021). Furthermore, Hananto and Panjaburee (2019) developed the semi-puzzle game "Key and Chest" to develop algorithm thinking skills and concluded that this digital game could lead to better achievement than if the physical game is used for the same purpose. The number of games developed at the university level on a specific topic by the researchers is even more significant. However, there is still no standardized game, and the games differ among themselves depending on the topic covered by the game and the country, where the game is implemented.

Only a few games are mentioned more than once in the list of relevant articles. The implementation of "Code defenders" to enable students to learn about software testing in a fun and competitive way is researched by Clegg et al. (2017) and Fraser et al. (2020). However, the studies continue each other, presenting improvements in the game. Different block-based programming languages and online platforms such as Scratch, Snap!, and Code Combat are mentioned in several articles, too. Implementation of a game-based approach during the assessment process through the creation of quizzes in Kahoot is presented by Abidin and Zaman (2017) and Videnovik et al. (2018). Finally, several articles refer to the use of Escape room as a popular game implemented in an educational context (Giang et al., 2020; López-Pernas et al., 2019, 2021; Seebauer et al., 2020; Towler et al., 2020). However, all these Escape room-style games are created on different platforms and cover different topics. Therefore, it can be concluded that no standardized type of game is implemented at a certain educational level or concerning a specific topic.

Further analyses were done concerning the type of the game, referring to the origin of the game: already created and just used for the research, created by the researcher for the purpose of the research or created by the students during the learning process. The distribution of the number of articles according to the type of the game in different educational levels is presented in Fig. 8.

Most of the articles describe the implementation of a game-based approach when the author creates the game to test the game's efficiency and make improvements

Table 3 Analyzed articles concerning the implementation of game-based learning in computer science

References	Educational level	Implemented game	Type of the game	Computer science topics covered by the game
Abdellatif et al. (2018)	University	Robocode	Already created	Object-oriented programming
Abidin & Zaman (2017)	University	Kahoot	Author	Object-oriented programming
Agalbato and Loiacono (2018)	University	Robo	Already created	Block-based programming
Agbo et al. (2021)	University	Online platform	Students	Computational thinking
Alatrista-Salas and Nunez-Del-Prado (2018)	University	CodeCombat	Already created	Object-oriented programming
Araujo et al. (2018)	Secondary	Scratch	Students	Block-based programming
Baek and Oh (2019)	Primary	Code Planet	Author	Object-oriented programming
Barriga & Besoaín (2020)	University	No specific game	Students	Programming principles
Berger et al. (2019)	Secondary	Privacy	Author	Internet safety
Bevcic and Rugelj (2020)	Primary	Snap!	Author	Block-based programming
Borna and Rad (2018)	University	Unity 3D Engine, Kahoot, Pulse	Author	Programming principles
Chang and Tsai (2018)	Secondary	Kinnect + Scratch	Students	Object-oriented programming
Clegg et al. (2017)	University	Code Defenders	Author	Other
Corda et al. (2019)	University	BashDungeon	Author	Basic computer science
Daungcharone et al. (2017)	University	PC game	Author	Object-oriented programming
De Carvalho et al. (2020)	Primary	Snap!, Run Marco, CodeCombat, Human Resource Machine, Light Bot, May's journey	Already created	Block-based programming
De Kereki and Adorjan (2018)	University	Kulami. Morelli, Pentago, FlipFlop, Quinamid	Already created	Computational thinking
De Troyer et al. (2019)	University	TrueBiters	Author	Basic computer science
Dočkalová Burská et al. (2021)	University	Training Analysis Tool	Author	Internet safety
Dos Santos et al. (2018)	University	Code Combat, Code Hunt	Already created	Object-oriented programming
Duch and Jaworski (2018)	University	Arduino-based educational Board	Students	Object-oriented programming
Eleftheriadis and Xinogalos (2020)	University	Office Madness	Author	Object-oriented programming
Elmunsyah et al. (2018)	Secondary	Mobile-based educational game	Author	Object-oriented programming
Elmunsyah et al. (2021)	Secondary	Degraff	Author	Other
Emembolu et al. (2019)	Primary	Gamefoot engine	Students	Block-based programming
Evripidou et al. (2021)	Primary	Bee-Bot robot	Students	Computational thinking
Fraser et al. (2020)	University	Code Defenders	Author	Other
Funke et al. (2017)	Primary	Scratch	Students	Computational thinking
Gaborik et al. (2019)	Secondary	Board Nanu game	Already created	Object-oriented programming
Garcia-Ruiz et al. (2021)	Primary	Micro:bit projects	Students	Object-oriented programming
Gardeli and Vosinakis (2019)	Primary	Request	Author	Computational thinking
Giang et al. (2020)	University	Escape room	Author	Object-oriented programming
Gossen et al. (2018)	University	Games with Cinco Adventure Game Tool	Students	Computational thinking
Groza et al. (2020)	University	MineFOL game	Author	Basic computer science
Gulec et al. (2019)	University	CONGO	Author	Object-oriented programming
Hananto and Panjaburee (2019)	Secondary	Key and Chest	Author	Computational thinking
Holanda et al. (2020)	Secondary	Kodu, MIT Inventor, Arduino,	Students	Object-oriented programming
Holenko Dlab and Hoic-Bozic (2021)	Primary	Snap!	Author	Block-based programming
Hong and Chu (2017)	University	Situated 3D game	Author	Object-oriented programming
Horst et al. (2019)	University	FunPlogs	Author	Programming principles
Huang et al. (2019)	Secondary	Set of board games	Author	Object-oriented programming
Huang et al. (2021)	University	pyMaze	Author	Object-oriented programming
Jagušt et al. (2018)	Primary	Unplugged activities	Already created	Computational thinking
Jovanov et al. (2017)	University	Unity 3D platform	Students	Other
Kanellopoulou et al. (2021)	Primary	Code	Already created	Block-based programming

Table 3 (continued)

References	Educational level	Implemented game	Type of the game	Computer science topics covered by the game
Kannappan (2019)	University	La Petite Fee Cosmo	Author	Programming principles
Kantharaju et al. (2020)	University	Parallel	Already created	Object-oriented programming
Kazimoglu (2020)	University	Program Your Robot	Author	Computational thinking
Kintsakis and Rangoussi (2017)	Primary	Scratch	Author	Block-based programming
Klimová et al. (2021)	Primary	Minecraft	Students	Object-oriented programming
Kučera et al. (2020)	University	Interactive application in Unity	Author	Object-oriented programming
Kurniawati et al. (2018)	Primary	2D Maze, 3D adventures	Author	Block-based programming
López-Fernández et al. (2021a, 2021b)	University	Flappy Bird	Author	Programming principles
López-Fernández et al. (2021a, 2021b)	University	LEGO® Serious Play	Students	Object-oriented programming
López-Pernas et al. (2019)	University	Escape Room	Author	Object-oriented programming
López-Pernas et al. (2021)	University	Escape room	Author	Object-oriented programming
Lotfi et al. (2019)	University	OOP SG	Already created	Object-oriented programming
Malizia et al. (2017)	University	TAPASPlay	Author	Computational thinking
Martins et al. (2019)	Secondary	Kahoot	Author	Other
Meftah et al. (2019)	University	Marco Run	Already created	Object-oriented programming
Miljanovic and Bradbury (2020)	University	GidgetML	Author	Programming principles
Min et al. (2020)	Primary	Engage	Author	Block-based programming
Montes et al. (2021)	Secondary	DFD-C	Author	Object-oriented programming
Mosquera et al. (2020)	Primary	sCool platform	Author	Object-oriented programming
Nche et al. (2019)	Primary	CodeTracesure	Author	Computational thinking
Nche et al. (2020)	primary	Codetracesure	Author	Computational thinking
Noval et al. (2019)	University	Robocode, Battlebot	Author	Basic computer science
Paiva et al. (2020)	University	Asura environment	Students	Object-oriented programming
Pila et al. (2019)	Primary	Daisy the Dinosaur, Kodable	Already created	Block-based programming
Popović et al. (2017)	Primary	Angry Birds	Already created	Computational thinking
Prabawa et al. (2017)	Secondary	Capture the flag	Already created	Internet safety
Priyadarshini et al. (2020)	University	The game-based learning mobile app	Author	Object-oriented programming
Rajeev and Sharma (2018)	University	Vizard platform	Author	Object-oriented programming
Riera et al. (2019)	University	HOME I/O, Scratch 2.0	Already created	Basic computer science
Roussou and Rangoussi (2020)	Primary	Robot Code & Go Robot Mouse	Students	Computational thinking
Rozali and Zaid (2017)	University	Mobile game	Author	Other
Sáiz Manzanares et al. (2020)	Primary	Blockly Games	Already created	Block-based programming
Schatten and Schatten (2020)	Secondary	CodeCombat	Students	Block-based programming
Seebauer et al. (2020)	University	Escape Room	Author	Basic computer science
Seralidou and Douligeris (2020)	Primary	Kodu Game Lab, Light Bot, Kahoot, Scratch	Students	Programming principles
Sharaf et al. (2020)	Primary	Treasure hunt style game	Author	Object-oriented programming
Sharma et al. (2019)	Primary	Scratch + Arduino	Students	Block-based programming
Shim et al. (2017)	Primary	Robot game	Students	Object-oriented programming
Siakavaras et al. (2018)	University	Platforms for LBG—Tale Blazer, ARIS, 7scenes, Wherigo	Author	Internet Safety
Silva et al. (2020)	University	The Turing Project	Author	Basic computer science
Simões Gomes et al. (2018)	Primary	Code Baymax; LightBot	Already created	Block-based programming
Sookhanaphibarn and Choensawat (2020)	University	Laptop Security, Social Network, Cyber Defender, Quiz Tank, Cyber Runner	Author	Internet safety
Stigall and Sharma (2017)	University	Second Life	Author	Object-oriented programming
Tabuti et al. (2020)	University	Online migration of traditional game	Students	programming principles
Tacouri and Nagowah (2021)	University	Code Saga	Author	Object-oriented programming

Table 3 (continued)

References	Educational level	Implemented game	Type of the game	Computer science topics covered by the game
Taylor et al. (2019)	Secondary	IntelliBlox toolkit	Students	Block-based programming
Towler et al. (2020)	University	Logic Descent—Escape room	Author	Basic computer science
Tretinjak (2019)	Secondary	Unplugged games	Author	Computational thinking
Vahldick et al. (2020)	University	NoBug's, SnackBar	Author	Computational thinking
Valle et al. (2017)	University	Testing Game	Author	Other
Venkatesh et al. (2021)	Primary	Unplugged activities	Author	Computational thinking
Ventura et al. (2017)	University	Video games using Unity3D	Students	Object-oriented programming
Videnovik et al. (2018)	Primary	Kahoot	Author	Basic computer science
Visoottiviseth et al. (2018)	University	Lord of Secure	Author	Internet safety
Voštinár (2021)	Primary	MakeCode Arcade	Author	Block-based programming
Vourletsis and Politis (2022)	Primary	Games in Scratch	Already created	Computational thinking
Wong and Jiang (2018)	Primary	Games in Scratch	Author	Computational thinking
Wong and Yatim (2018)	University	Greenfoot, Darwin CodeCombat	Author	Object-oriented programming
Wong et al. (2017)	University	The Odyssey of Phoenix	Author	Object-oriented programming
Workman et al. (2021)	University	Secure Code Warrior	Already created	Internet safety
Xian (2021)	University	Sandbox game LE	Author	Object-oriented programming
Yallihep and Kutlu (2020)	Primary	Lightbot	Already created	Object-oriented programming
Yokoyama et al. (2020)	University	RPA	Author	Object-oriented programming
Yongqiang et al. (2018)	Primary	Scratch + Arduino	Students	Block-based programming
Yücel and Rızvanoğlu (2019)	Primary	Code Combat	Already created	Block-based programming
Zapata-Cáceres & Martín-Barroso (2021)	Primary	Visual environment Blue Ant Code	Author	Computational thinking
Zeevaarders and Aivaloglou (2021)	Primary	Scratch	Students	Object-oriented programming
Zhao et al. (2021)	University	Three games on NEWTELP platform	Author	Object-oriented programming

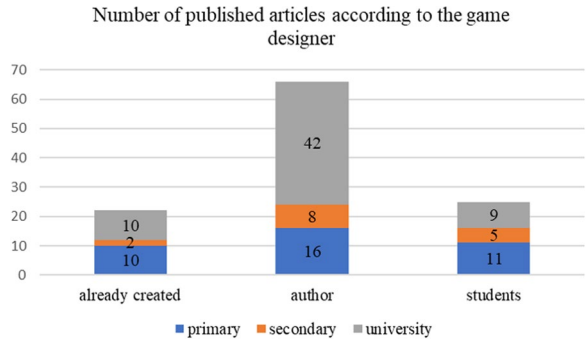


Fig. 8 Distribution of the published articles according to the game designer in different educational levels

based on the feedback received by the students. The number of games created by the author is the biggest at the university level, and the most balanced distribution of different kinds of games (created by the author, students or already created) is present in primary education. Interestingly, the most significant number of articles that concern using games created by students is in primary education. It shows that students in primary education have been the most involved in the process of game design, although they are young and have less knowledge and skills than students at other educational levels. This could be result of the fact that the articles that refer to primary education present a game's design only in a block-based environment and using basic programming concepts. However, research articles do not refer to a standardized methodology of a framework for the creation of a game, and each game is designed individually depending on the used technology, topic and educational level.

Pedagogical strategies for implementation of the game-based approach in computer science

A detailed analysis of the pedagogical strategies for implementing a game-based approach shows that most relevant articles use games as a tool for learning the content. This trend continues in the recent period as well (Kaldarova et al., 2023). Hence, students play the game (already created or created by an author) to gather knowledge or develop their skills. Detail distribution of the research articles regarding pedagogical strategies for implementing a game-based approach is presented in

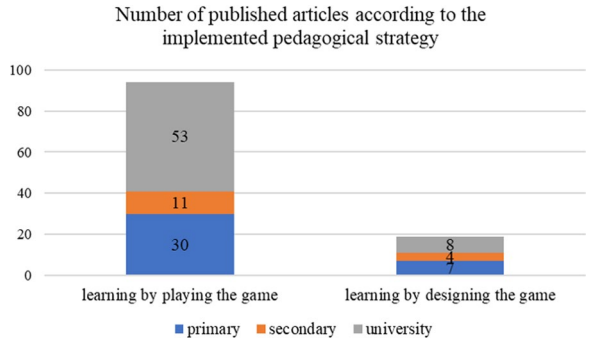


Fig. 9 Distribution of the published articles according to the implemented pedagogical strategy

mainly taught by a game-based approach at university is object-oriented programming. The situation is similar in secondary schools. Game-based approach is suitable classroom strategy for fostering higher order thinking skills, such as problem solving, group collaboration, and critical thinking, that are developed during learning object-oriented programming, which is consistent with previous research conducted by Chen et al. (2021).

This can be expected, because the topic is complex for the students, and teachers must find different approaches and strategies to make it more understandable. In addition, in those educational levels, there is a distribution of the articles in different mentioned computer science topics (although it is not equally distributed).

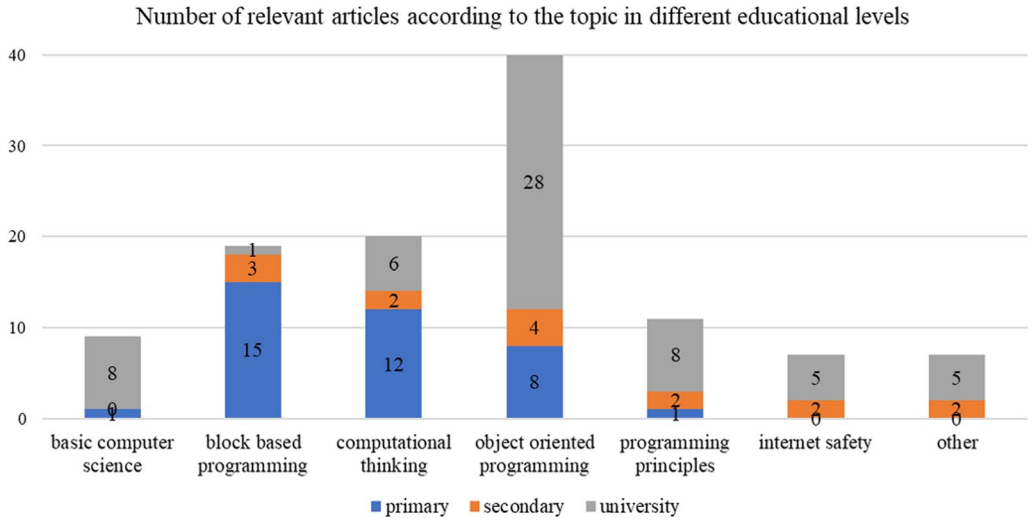


Fig. 10 Distribution of the published articles concerning the covered computer science topics

Fig. 9 and more detailed data can be found in Table 3. Some articles explain how students learn during the process of the creation of a game. Those are different games at different educational levels, but they all concern the process of designing a game on some platform that will develop their programming skills. Unfortunately, no article describes the process of developing students' knowledge and skills on different computer science topics than programming while designing a game. It is a critical gap that should be considered as a topic in future research: to see whether students can learn about other computer science topics during the game creation process (while they develop their programming skills).

Computer science topics covered by game-based approach in computer science

Figure 10 gives insight into the distribution of the relevant articles concerning the computer science topic covered by the game-based approach. The topic that is

However, if we analyze the topics covered by the game-based approach in primary education, it can be noticed that this approach is implemented in several topics only, mainly connected with the development of students' computational thinking skills and fundaments of programming languages (see Table 3 for detailed overview). This trend continues in the recent years (Cheng et al., 2023; Mozelius & Humble, 2023).

Students in primary education mostly learn block-based programming languages, so it is expected that this will be the most frequent topic covered by the game-based approach. However, some articles also refer to object-oriented programming taught in upper grades. The interesting finding is that there are no articles about using educational games to learn other computer science topics, such as hardware, some applications, networks, and cybersecurity, in primary education, as there are in other educational levels. For example, there are

two articles that elaborate on learning about internet safety using games in secondary education (Berger et al., 2019; Prabawa et al., 2017), and no article on game-based learning for internet safety in primary education. This lack of research articles concerning using the game-based approach for learning other topics in computer science in primary education can help identify potential future research topics.

Potential research topics concerning the game-based approach in computer science

While the lack of research articles concerning using the game-based approach for learning other topics in computer science in primary education is a good starting point for identifying potential future research topics, it is important to consider it in combination with practical constraints such are lack of knowledge, access to technology or teacher training on a specific subject. In that context, "Identifying the challenges, opportunities and solutions for integrating game-based learning methods in primary schools for specific computer science topics" can be a future research topic. It should be noted, that although some articles on specific topics can be found in the recent literature (Alam, 2022), there is a huge pool of topics, such are internet safety and digital citizenship that can be explored in this context.

There is an evident lack of articles on the use of game-based learning in primary and secondary schools. The findings in the existing literature that elaborate on how specific game design elements influence the learning process are minimal (Baek & Oh, 2019; Dos Santos et al., 2019; Emembolu et al., 2019; Kanellopoulou et al., 2021). These findings, combined with the finding of a limited number of articles that use existing games in the process of learning, define the potential future research topic "Assessing the role of game design elements in enhancing engagement and understanding of computer science concepts among primary and/or secondary school students". This research topic can use conceptual framework that investigates how specific elements of game design can contribute to increased engagement and improved understanding of computer science concepts in primary or/and education.

This research topic includes various specific research questions and theoretical frameworks. One possible set of research questions can investigate the specific elements of game design that can be incorporated into educational games or learning activities to enhance the learning experience. These elements may include interactive interfaces, engaging narratives, immersive environments, feedback mechanisms, competition or collaboration features, levels of difficulty, rewards, and progression systems. Different theories such are social cognitive theory (Lim et al., 2020) and self-determination theory (Ryan et al., 2006) can be used to better understand the motivational factors of different game design elements (interactivity, challenges, and rewards), and how they influence student engagement and sustain student interest and active participation in computer science learning.

All mentioned research questions can be investigated by conducting experiments, surveys, observations, or interviews to gather quantitative and qualitative data on student experiences and perceptions. Combined with data from learning outcomes, these potential findings can provide the information about overall effectiveness of using the elements of a game-based approach to learning computer science in primary schools.

Limitations

This scoping review focuses on the articles in four digital libraries, potentially leaving a significant number of articles out of the analyzing process.

Using the NLP toolkit automates searching for relevant articles. Undoubtedly, a human reader might better understand the context and better assess the relevance of an article and potentially include some articles that NLP toolkit classified as irrelevant. In addition, after the initial selection by NLP toolkit, we performed the quality assessment of the identified articles, for each of the research questions. In that way, we ensured that only relevant articles are included in the study, but it might happen that, due to the phase of selection some relevant articles were omitted from the study.

Detailed meta-analyses within the selected group of articles concerning a particular research feature can further contribute to the existing body of knowledge. Similar analyses exist, but not on learning computer science (Gui et al., 2023). For example, in our manuscript, we did not consider the size of the student population, existence of the control group of students, or replicability of the studies.

Conclusion

This scoping review discusses implementation of game-based approach in computer science by analyzing research articles in four digital libraries published between 2017 and 2021. In total, 113 research articles were analyzed concerning the educational level, where the game-based approach is implemented, the type of the game, covered computer science topic, pedagogical strategy and purpose of the implementation. The results show that the number of research articles is increasing through the years, confirming the importance of implementing a game-based approach in computer science. Most of these articles refer to the research in just one country, in the local context, making it difficult to generalize the

research outcomes and conclusions on the international level.

The article presents various games using various technologies concerning several computer science topics. However, there is no standardized game or methodology that can be used for designing an educational game. Implemented game in each of the researched articles depends on the educational level, covered topic and game type. From our findings, it is evident that most articles refer to the implementation of the game-based approach, where students gather the necessary knowledge and skills while playing a game. Just a few of them incorporate the process of learning by designing educational games, and this learning is connected to developing computational thinking or programming skills.

Potential future research might be focused on identifying the challenges, opportunities, and solutions for integrating game-based learning methods for a specific computer science topic. Example topics might be internet safety and digital citizenship.

The lack of research articles on game-based learning in primary and secondary schools, along with limited findings on the influence of game design elements, highlights the need to assess how different elements enhance engagement and understanding of computer science concepts.

Acknowledgements
Not applicable.

Author contributions
VT and AMB had the idea for the article. TV defined the methodological steps. MV and AMB performed the literature search and data analyses. Data analyses were supported by VT and LK. MV drafted the article, while TV critically revised the work.

Funding
No funding was received for conducting this study.

Availability of data and materials
All data generated and analyzed during this study are included in this article.

Declarations

Competing interests
The authors declare that they have no competing interests.

Received: 27 March 2023 Accepted: 18 August 2023
Published online: 06 September 2023

References

Abbasi, S., Kazi, H., & Khowaja, K. (2017). A systematic review of learning object-oriented programming through serious games and programming approaches. In *2017 4th IEEE International Conference on Engineering Technologies and Applied Sciences (ICETAS)* (pp. 1–6). IEEE. https://doi.org/10.1109/ICETAS.2017.8277894

Abdellatif, A. J., McCollum, B., & McMullan, P. (2018). Serious games quality characteristics evaluation: The case study of optimizing Robocode. In *2018 International Symposium on Computers in Education (SIIE)* (pp. 1–4). IEEE. https://doi.org/10.1109/SIIE.2018.8586730

Abidin, H. Z., & Kamaru Zaman, F. H. (2017). Students' perceptions on game-based classroom response system in a computer programming course. In *2017 IEEE 9th International Conference on Engineering Education (ICEED)* (pp. 254–259). IEEE. https://doi.org/10.1109/ICEED.2017.8251203

Agalbato, F., & Loiacono, D. (2018). Robo^3: A puzzle game to learn coding. In *2018 IEEE Games, Entertainment, Media Conference (GEM)* (pp. 359–366). IEEE. https://doi.org/10.1109/GEM.2018.8516515

Agbo, F. J., Oyelere, S. S., Suhonen, J., & Laine, T. H. (2021). Co-design of mini games for learning computational thinking in an online environment. *Education and Information Technologies, 26*(5), 5815–5849. https://doi.org/10.1007/s10639-021-10515-1

Alam, A. (2022). A digital game-based learning approach for effective curriculum transaction for teaching-learning of artificial intelligence and machine learning. In *2022 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS)* (pp. 69–74). IEEE. https://doi.org/10.1109/icscds53736.2022.9760932

Alatrista-Salas, H., & Nunez-Del-Prado, M. (2018). Teaching software engineering through computer games. In *2018 IEEE World Engineering Education Conference (EDUNINE)* (pp. 1–4). IEEE. https://doi.org/10.1109/EDUNINE.2018.8450996

Anastasiadis, T., Lampropoulos, G., & Siakas, K. (2018). Digital game-based learning and serious games in education. *International Journal of Advances in Scientific Research and Engineering., 4*(12), 139–144. https://doi.org/10.31695/IJASRE.2018.33016

Araujo, L. G. J., Bittencourt, R. A., & Santos, D. M. B. (2018). Contextualized spiral learning of computer programming in Brazilian vocational secondary education. In *2018 IEEE Frontiers in Education Conference (FIE)* (pp. 1–9). IEEE. https://doi.org/10.1109/FIE.2018.8658456

Arcagök, S. (2021). The impact of game-based teaching practices in different curricula on academic achievement. *International Online Journal of Education and Teaching, 8*(2), 778–796.

Arksey, H., & O'Malley, L. (2005). Scoping studies: towards a methodological framework. *International journal of social research methodology*, *8*(1), 19–32. https://doi.org/10.1080/1364557032000119616.

Baek, J., & Oh, G. (2019). Development of a puzzle game to learn coding for elementary students. In C. Stephanidis (Ed.), *HCI International 2019 – Late Breaking Papers* (pp. 267–279). Springer International Publishing. https://doi.org/10.1007/978-3-030-30033-3_21

Barradas, R., Lencastre, J., Soares, S., & Valente, A. (2020). Developing computational thinking in early ages: A review of the code.org platform. In *Proceedings of the 12th International Conference on Computer Supported Education (CSEDU2020)* (pp. 157–168). SCITEPRESS – Science and Technology Publications https://doi.org/10.5220/0009576801570168

Barriga, N. A., & Besoain, F. (2020). Artificial intelligence and mobile programming courses for a video game development program in Chile. *Computing in Science & Engineering, 22*(4), 17–25. https://doi.org/10.1109/MCSE.2020.2986758

Barz, N., Benick, M., Dörrenbächer-Ulrich, L., & Perels, F. (2023). The effect of digital game-based learning interventions on cognitive, metacognitive, and affective-motivational learning outcomes in school: A meta-analysis. *Review of Educational Research*. https://doi.org/10.3102/00346543231167795

Bellas, F., Naya, M., Varela, G., Llamas, L., Prieto, A., Becerra, J. C., Bautista, M., Faiña, A., & Duro, R. (2018). The Robobo project: Bringing educational robotics closer to real-world applications. In W. Lepuschitz, M. Merdan, G. Koppensteiner, R. Balogh, & D. Obdržálek (Eds.), *Robotics in education: Latest results and developments* (pp. 226–237). Springer. https://doi.org/10.1007/978-3-319-62875-2_20

Berger, E., Sæthre, T. H., & Divitini, M. (2019). PrivaCity: A chatbot game to raise privacy awareness among teenagers. In S. N. Pozdniakov & V. Dagienė (Eds.), *Informatics in schools. New ideas in school informatics* (pp. 293–304). Springer. https://doi.org/10.1007/978-3-030-33759-9_23

Bers, M. U. (2019). Coding as another language: A pedagogical approach for teaching computer science in early childhood. *Journal of Computers in Education, 6*(4), 499–528. https://doi.org/10.1007/s40692-019-00147-3

Bevcic, M., & Rugelj, J. (2020). Game design-based learning of programming for girls. In *2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)* (pp. 576–580). IEEE. https://doi.org/10.23919/MIPRO48935.2020.9245175

Bird, S. (2006). NLTK: the natural language toolkit. In *Proceedings of the COLING/ACL 2006 Interactive Presentation Sessions* (pp. 69–72). https://doi.org/10.3115/1225403.1225421

Borna, K., & Rad, H. M. (2018). Serious games in computer science learning goals. In *2018 2nd National and 1st International Digital Games Research Conference: Trends, Technologies, and Applications (DGRC)* (pp. 161–166). IEEE. https://doi.org/10.1109/DGRC.2018.8712030

Boyer, S. L., Edmondson, D. R., Artis, A. B., & Fleming, D. (2014). Self-directed learning: A tool for lifelong learning. *Journal of Marketing Education, 36*(1), 20–32. https://doi.org/10.1177/0273475313494010

Boyle, E. A., Hainey, T., Connolly, T. M., Gray, G., Earp, J., Ott, M., Lim, T., Ninaus, M., Ribeiro, C., & Pereira, J. (2016). An update to the systematic literature review of empirical evidence of the impacts and outcomes of computer games and serious games. *Computers & Education, 94*, 178–192. https://doi.org/10.1016/j.compedu.2015.11.003

Calvo-Morata, A., Alonso-Fernández, C., Freire, M., Martínez-Ortiz, I., & Fernández-Manjón, B. (2020). Serious games to prevent and detect bullying and cyberbullying: A systematic serious games and literature review. *Computers & Education, 157*, 103958. https://doi.org/10.1016/j.compedu.2020.103958

Campbell, L. (2020). Teaching in an inspiring learning space: An investigation of the extent to which one school's innovative learning environment has impacted on teachers' pedagogy and practice. *Research Papers in Education, 35*(2), 185–204. https://doi.org/10.1080/02671522.2019.1568526

Cardinot, A., & Fairfield, J. A. (2019). Game-based learning to engage students with physics and astronomy using a board game. *International Journal of Game-Based Learning, 9*(1), 42–57. https://doi.org/10.4018/IJGBL.2019010104

Chandel, P., Dutta, D., Tekta, P., Dutta, K., & Gupta, V. (2015). Digital game-based learning in computer science education. *CPUH-Research Journal, 1*(2), 33–37.

Chang, C. S., Chung, C. H., & Chang, J. A. (2020). Influence of problem-based learning games on effective computer programming learning in higher education. *Educational Technology Research and Development, 68*, 2615–2634. https://doi.org/10.1007/s11423-020-09784-3

Chang, C. K., & Tsai, Y. T. (2018). Pair-programming curriculum development of motion-based game for enhancing computational thinking skills. In *2018 7th International Congress on Advanced Applied Informatics (IIAI-AAI)* (pp. 284–287). IEEE. https://doi.org/10.1109/IIAI-AAI.2018.00061

Chen, P. Y., Hwang, G. J., Yeh, S. Y., Chen, Y. T., Chen, T. W., & Chien, C. H. (2021). Three decades of game-based learning in science and mathematics education: An integrated bibliometric analysis and systematic review. *Journal of Computers in Education, 9*(3), 455–476. https://doi.org/10.1007/s40692-021-00210-y

Chen, P., Yang, D., Metwally, A. H. S., Lavonen, J., & Wang, X. (2023). Fostering computational thinking through unplugged activities: A systematic literature review and meta-analysis. *International Journal of STEM Education, 10*(1), 1–25. https://doi.org/10.1186/s40594-023-00434-7

Cheng, Y. P., Lai, C. F., Chen, Y. T., Wang, W. S., Huang, Y. M., & Wu, T. T. (2023). Enhancing student's computational thinking skills with student-generated questions strategy in a game-based learning platform. *Computers & Education, 200*, 104794. https://doi.org/10.1016/j.compedu.2023.104794

Clark, D. B., Tanner-Smith, E. E., & Killingsworth, S. S. (2016). Digital games, design, and learning: A systematic review and meta-analysis. *Review of Educational Research, 86*, 79–122. https://doi.org/10.3102/0034654315582065

Clegg, B. S., Rojas, J. M., & Fraser, G. (2017). Teaching software testing concepts using a mutation testing game. In *2017 IEEE/ACM 39th International Conference on Software Engineering: Software Engineering Education and Training Track (ICSE-SEET)* (pp. 33–36). IEEE. https://doi.org/10.1109/ICSE-SEET.2017.1

Corda, F., Onnis, M., Pes, M., Spano, L. D., & Scateni, R. (2019). BashDungeon: Learning UNIX with a video-game. *Multimedia Tools and Applications, 78*(10), 13731–13746. https://doi.org/10.1007/s11042-019-7230-3

Daungcharone, K., Panjaburee, P., & Thongkoo, K. (2017). Using digital game as compiler to motivate C programming language learning in higher education. In *2017 6th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI)* (pp. 533–538). IEEE. https://doi.org/10.1109/IIAI-AAI.2017.77

De Carvalho, C. V., Cerar, Š, Rugelj, J., Tsalapatas, H., & Heidmann, O. (2020). Addressing the gender gap in computer programming through the design and development of serious games. *IEEE Revista Iberoamericana De Tecnologias Del Aprendizaje, 15*(3), 242–251. https://doi.org/10.1109/RITA.2020.3008127

De Kereki, I. F., & Adorjan, A. (2018). Serious games: Using abstract strategy games in computer science 2: An experience report and lessons learned. In *2018 IEEE Global Engineering Education Conference (EDUCON)* (pp. 169–174). IEEE. https://doi.org/10.1109/EDUCON.2018.8363224

De Troyer, O., Lindberg, R., Maushagen, J., & Sajjadi, P. (2019). Development and evaluation of an educational game to practice the truth tables of logic. In *2019 IEEE 19th International Conference on Advanced Learning Technologies (ICALT)* (pp. 92–96). IEEE. https://doi.org/10.1109/ICALT.2019.00032

Denner, J., Werner, L., & Ortiz, E. (2012). Computer games created by middle school girls: Can they be used to measure understanding of computer science concepts? *Computers & Education, 58*(1), 240–249. https://doi.org/10.1016/j.compedu.2011.08.006

Díaz, J., López, J. A., Sepúlveda, S., Ramírez Villegas, G. M., Ahumada, D., & Moreira, F. (2021). Evaluating aspects of usability in video game-based programming learning platforms. *Procedia Computer Science, 181*, 247–254. https://doi.org/10.1016/j.procs.2021.01.141

Dočkalová Burská, K., Rusňák, V., & Ošlejšek, R. (2022). Data-driven insight into the puzzle-based cybersecurity training. *Computers & Graphics, 102*, 441–451. https://doi.org/10.1016/j.cag.2021.09.011

Dos Santos, A. L., Souza, M. R. D. A., Dayrell, M., & Figueiredo, E. (2018). Exploring game elements in learning programming: An empirical evaluation. In *2018 IEEE Frontiers in Education Conference (FIE)* (pp. 1–9). IEEE. https://doi.org/10.1109/FIE.2018.8658505

Dos Santos, A. L., Souza, M. R. D. A., Dayrell, M., & Figueiredo, E. (2019). A systematic mapping study on game elements and serious games for learning programming. In B. M. McLaren, R. Reilly, S. Zvacek, & J. Uhomoibhi (Eds.), *Computer supported education* (Vol. 1022, pp. 328–356). Springer. https://doi.org/10.1007/978-3-030-21151-6_17

Duch, P., & Jaworski, T. (2018). Enriching computer science programming classes with Arduino game development. In *2018 11th International Conference on Human System Interaction (HIS)* (pp. 148–154). IEEE. https://doi.org/10.1109/HSI.2018.8430994

Eleftheriadis, S., & Xinogalos, S. (2020). Office Madness: Design and pilot evaluation of a serious game for learning the C++ programming language. In I. Marfisi-Schottman, F. Bellotti, L. Hamon, & R. Klemke (Eds.), *Games and learning alliance* (pp. 389–394). Springer. https://doi.org/10.1007/978-3-030-63464-3_36

Elmunsyah, H., Kusumo, G. R., Pujianto, U., & Prasetya, D. D. (2018). Development of mobile based educational game as a learning media for basic programming in VHS. In *2018 5th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI)* (pp. 416–420). IEEE. https://doi.org/10.1109/EECSI.2018.8752658

Elmunsyah, H., Herwanto, H. W., Smaragdina, A. A., Anggraini, N. S., & Utomo, W. M. (2021). DEGRAF educational game as a supplement for basic graphic design subjects for vocational high school students. In *2021 7th International Conference on Electrical, Electronics and Information Engineering (ICEEIE)* (pp. 128–132). IEEE. https://doi.org/10.1109/ICEEIE52663.2021.9616829

Emembolu, I., Strachan, R., Davenport, C., Dele-Ajayi, O., & Shimwell, J. (2019). Encouraging diversity in computer science among young people: Using a games design intervention based on an integrated pedagogical framework. In *2019 IEEE Frontiers in Education Conference (FIE)* (pp. 1–8). IEEE. https://doi.org/10.1109/FIE43999.2019.9028436

Evripidou, S., Amanatiadis, A., Christodoulou, K., & Chatzichristofis, S. A. (2021). Introducing algorithmic thinking and sequencing using tangible robots. *IEEE Transactions on Learning Technologies, 14*(1), 93–105. https://doi.org/10.1109/TLT.2021.3058060

Fagerlund, J., Häkkinen, P., Vesisenaho, M., & Viiri, J. (2021). Computational thinking in programming with Scratch in primary schools: A systematic review. *Computer Applications in Engineering Education, 29*(1), 12–28. https://doi.org/10.1002/cae.22255

Fraser, G., Gambi, A., & Rojas, J. M. (2020). Teaching software testing with the Code Defenders testing game: Experiences and improvements. In *2020 IEEE International Conference on Software Testing, Verification and*

Validation Workshops (ICSTW) (pp. 461–464). IEEE. https://doi.org/10.1109/ICSTW50294.2020.00082

Funke, A., Geldreich, K., & Hubwieser, P. (2017). Analysis of Scratch projects of an introductory programming course for primary school students. In *2017 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1229–1236). IEEE. https://doi.org/10.1109/EDUCON.2017.7943005

Gabaruk, J., Logofatu, D., Groskreutz, D., & Andersson, C. (2019). On teaching Java and object oriented programming by using children board games. In *2019 IEEE Global Engineering Education Conference (EDUCON)* (pp. 601–606). IEEE. https://doi.org/10.1109/EDUCON.2019.8725264

Garcia-Ruiz, M. A., Alvarez-Cardenas, O., & Iniguez-Carrillo, A. L. (2021). Experiences in developing and testing BBC Micro: Bit games in a K-12 Coding Club during the COVID-19 Pandemic. In *2021 IEEE/ACIS 20th International Fall Conference on Computer and Information Science (ICIS Fall)* (pp. 161–164). IEEE. https://doi.org/10.1109/ICISFall51598.2021.9627364

Gardeli, A., & Vosinakis, S. (2019). ARQuest: A tangible augmented reality approach to developing computational thinking skills. In *2019 11th International Conference on Virtual Worlds and Games for Serious Applications (VS-Games)* (pp. 1–8). IEEE. https://doi.org/10.1109/VS-Games.2019.8864603

Ghergulescu, I., & Muntean, C. H., et al. (2012). Measurement and analysis of learner's motivation in game-based e-learning. In D. Ifenthaler (Ed.), *Assessment in game-based learning: Foundations, innovations, and perspectives* (pp. 355–378). Cham: Springer. https://doi.org/10.1007/978-1-4614-3546-4_18

Giang, C., Chevalier, M., Negrini, L., Peleg, R., Bonnet, E., Piatti, A., & Mondada, F. (2020). Exploring Escape games as a teaching tool in educational robotics. In M. Moro, D. Alimisis, & L. Iocchi (Eds.), *Educational robotics in the context of the maker movement* (pp. 95–106). Springer. https://doi.org/10.1007/978-3-030-18141-3_8

Gossen, F., Kuhn, D., Margaria, T., & Lamprecht, A. (2018). Computational thinking: Learning by doing with the Cinco adventure game tool. In *2018 IEEE 42nd Annual Computer Software and Applications Conference (COMPSAC)* (pp. 990–999). IEEE. https://doi.org/10.1109/COMPSAC.2018.00175

Groza, A., Baltatescu, M. M., & Pomarlan, M. (2020). MineFOL: A game for learning first order logic. In *2020 IEEE 16th International Conference on Intelligent Computer Communication and Processing (ICCP)* (pp. 153–160). IEEE. https://doi.org/10.1109/ICCP51029.2020.9266174

Gui, Y., Cai, Z., Yang, Y., Kong, L., Fan, X., & Tai, R. H. (2023). Effectiveness of digital educational game and game design in STEM learning: A meta-analytic review. *International Journal of STEM Education, 10*(1), 1–25. https://doi.org/10.1186/s40594-023-00424-9

Gulec, U., Yilmaz, M., Yalcin, A. D., O'Connor, R. V., & Clarke, P. M. (2019). CENGO: A web-based serious game to increase the programming knowledge levels of computer engineering students. In A. Walker, R. V. O'Connor, & R. Messnarz (Eds.), *Systems, software and services process improvement* (pp. 237–248). Springer. https://doi.org/10.1007/978-3-030-28005-5_18

Hafeez, M. (2022). Effects of game-based learning in comparison to traditional learning to provide an effective learning environment: A comparative review. *Contemporary Educational Researches Journal, 12*(2), 89–105. https://doi.org/10.18844/cerj.v12i2.6374

Hamari, J., Shernoff, D. J., Rowe, E., Coller, B., Asbell-Clarke, J., & Edwards, T. (2016). Challenging games help students learn: An empirical study on engagement, flow and immersion in game-based learning. *Computers in Human Behavior, 54*, 170–179. https://doi.org/10.1016/j.chb.2015.07.045

Hananto, A. A. F., & Panjaburee, P. (2019). Proposing an online peer-feedback approach in digital game from a semi-puzzle game-based learning perspective. In *2019 8th International Congress on Advanced Applied Informatics (IIAI-AAI)* (pp. 254–259). IEEE. https://doi.org/10.1109/IIAI-AAI.2019.00058

Harteveld, C., Smith, G., Carmichael, G., Gee, E., & Stewart-Gardiner, C. (2014). A design-focused analysis of games teaching computer science. *Proceedings of Games+ Learning_ Society, 10*, 1–8.

Hartt, M., Hosseini, H., & Mostafapour, M. (2020). Game on: Exploring the effectiveness of game-based learning. *Planning Practice & Research, 35*(5), 589–604. https://doi.org/10.1080/02697459.2020.1778859

Hazzan, O., Ragonis, N., & Lapidot, T. (2020). Teaching methods in computer science education. In O. Hazzan, N. Ragonis, & T. Lapidot (Eds.), *Guide to teaching computer science: An activity-based approach* (pp. 181–220). Springer. https://doi.org/10.1007/978-3-030-39360-1_10

Hirsh-Pasek, K., Hadani, H. S., Blinkoff, E., & Golinkoff, R. M. (2020). A new path to education reform: playful learning promotes 21st century skills in school and beyond. *Policy Brief*. https://www.brookings.edu/wp-content/uploads/2020/10/Big-Ideas_Hirsh-Pasek_PlayfulLearning.pdf

Holanda, M., Araujo, A. P. F., & Walter, M. E. (2020). Meninas.comp project: Programming for girls in high school in Brazil. In *2020 Research on Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT)* (pp. 1–2). IEEE. https://doi.org/10.1109/RESPECT49803.2020.9272419

Holenko Dlab, M., & Hoic-Bozic, N. (2021). Effectiveness of game development-based learning for acquiring programming skills in lower secondary education in Croatia. *Education and Information Technologies, 26*(4), 4433–4456. https://doi.org/10.1007/s10639-021-10471-w

Hong, T. Y., & Chu, H. C. (2017). Effects of a situated 3D computational problem-solving and programming game-based learning model on students' learning perception and cognitive loads. In *2017 6th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI)* (pp. 596–600). IEEE https://doi.org/10.1109/IIAI-AAI.2017.96

Hooshyar, D., Pedaste, M., Yang, Y., Malva, L., Hwang, G.-J., Wang, M., Lim, H., & Delev, D. (2021). From gaming to computational thinking: An adaptive educational computer game-based learning approach. *Journal of Educational Computing Research, 59*(3), 383–409. https://doi.org/10.1177/0735633120965919

Horst, R., Naraghi-Taghi-Off, R., Diez, S., Uhmann, T., Müller, A., & Dörner, R. (2019). FunPlogs – a serious puzzle mini-game for learning fundamental programming principles using visual scripting. In G. Bebis, R. Boyle, B. Parvin, D. Koracin, D. Ushizima, S. Chai, S. Sueda, X. Lin, A. Lu, D. Thalmann, C. Wang, & P. Xu (Eds.), *Advances in Visual Computing* (pp. 494–504). Springer International Publishing. https://doi.org/10.1007/978-3-030-33720-9_38

Huang, L. W., Chen, L. W., & Cheng, P. H. (2019). Board game design for Python programming education. In *2019 IEEE International Conference on Engineering, Technology and Education (TALE)* (pp. 1–6). IEEE. https://doi.org/10.1109/TALE48000.2019.9225988

Huang, L. W., Cheng, P. H., & Chen, L. W. (2021). Web-based board game for learning Python. In *2021 IEEE World Conference on Engineering Education (EDUNINE)* (pp. 1–6). IEEE. https://doi.org/10.1109/EDUNINE51952.2021.9429144

Huizenga, J., Admiraal, W., Akkerman, S., & Dam, G. T. (2009). Mobile game-based learning in secondary education: Engagement motivation and learning in a mobile city game. *Journal of Computer Assisted Learning., 25*(4), 332–344. https://doi.org/10.1111/j.1365-2729.2009.00316.x

Hussein, M. H., Ow, S. H., Cheong, L. S., & Thong, M.-K. (2019). A digital game-based learning method to improve students' critical thinking skills in elementary science. *IEEE Access, 7*, 96309–96318. https://doi.org/10.1109/ACCESS.2019.2929089

Hwang, G. J., Hung, C. M., & Chen, N. S. (2014). Improving learning achievements, motivations and problem-solving skills through a peer assessment-based game development approach. *Educational Technology Research and Development, 62*(2), 129–145. https://doi.org/10.1007/s11423-013-9320-7

Jagust, T., Krzic, A. S., Gledec, G., Grgic, M., & Bojic, I. (2018). Exploring different unplugged game-like activities for teaching computational thinking. In *2018 IEEE Frontiers in Education Conference (FIE)* (pp. 1–5). IEEE. https://doi.org/10.1109/FIE.2018.8659077

Joshi, D. R., Adhikari, K. P., Khanal, B., Khadka, J., & Belbase, S. (2022). Behavioral, cognitive, emotional and social engagement in mathematics learning during COVID-19 pandemic. *PLoS ONE, 17*(11), e0278052. https://doi.org/10.1371/journal.pone.0278052

Jovanov, M., Ilijoski, B., Stankov, E., & Armenski, G. (2017). Creation of educational games—project based learning in e-learning systems course. In *2017 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1274–1281). IEEE. https://doi.org/10.1109/EDUCON.2017.7943012

Kaldarova, B., Omarov, B., Zhaidakbayeva, L., Tursynbayev, A., Beissenova, G., Kurmanbayev, B., & Anarbayev, A. (2023). Applying game-based learning to a primary school class in computer science terminology learning. *Frontiers in Education*. https://doi.org/10.3389/feduc.2023.1100275

Kanellopoulou, I., Garaizar, P., & Guenaga, M. (2021). First steps towards automatically defining the difficulty of maze-based programming challenges. *IEEE Access, 9*, 64211–64223. https://doi.org/10.1109/ACCESS.2021.3075027

Kannappan, V. T., Fernando, O. N. N., Chattopadhyay, A., Tan, X., Hong, J. Y. J., Seah, H. S., & Lye, H. E. (2019). La Petite Fee Cosmo: Learning data structures through game-based learning. In *2019 International Conference on Cyberworlds (CW)* (pp. 207–210). IEEE. https://doi.org/10.1109/CW.2019.00041

Kantharaju, P., Alderfer, K., Zhu, J., Char, B., Smith, B., & Ontanon, S. (2020). Modeling player knowledge in a parallel programming educational game. *IEEE Transactions on Games, 14*(1), 64–75. https://doi.org/10.1109/TG.2020.3037505

Karagiannis, S., Papaioannou, T., Magkos, E., & Tsohou, A. (2020). Game-based information security/privacy education and awareness: Theory and practice. In M. Themistocleous, M. Papadaki, & M. M. Kamal (Eds.), *Information Systems* (Vol. 402, pp. 509–525). Cham: Springer. https://doi.org/10.1007/978-3-030-63396-7_34

Karram, O. (2021). The role of computer games in teaching object-oriented programming in high schools—Code Combat as a game approach. *WSEAS Transactions on Advances in Engineering Education, 18*, 37–46. https://doi.org/10.37394/232010.2021.18.4

Kazimoglu, C. (2020). Enhancing confidence in using computational thinking skills via playing a serious game: A case study to increase motivation in learning computer programming. *IEEE Access, 8*, 221831–221851. https://doi.org/10.1109/ACCESS.2020.3043278

Kintsakis, D., & Rangoussi, M. (2017). An early introduction to STEM education: Teaching computer programming principles to 5th graders through an e-learning platform: A game-based approach. In *2017 IEEE Global Engineering Education Conference (EDUCON)* (pp. 17–23). IEEE. https://doi.org/10.1109/EDUCON.2017.7942816

Klimová, N., Šajben, J., & Lovászová, G. (2021). Online game-based learning through Minecraft: Education edition programming contest. In *2021 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1660–1668). IEEE. https://doi.org/10.1109/EDUCON46332.2021.9453953

Kucera, E., Haffner, O., & Leskovsky, R. (2020). Multimedia application for object-oriented programming education developed by Unity engine. In *2020 Cybernetics & Informatics (K&I)* (pp. 1–8). IEEE. https://doi.org/10.1109/KI48306.2020.9039853

Kurniawati, A., Akbar, N. H., & Prasetyo, D. (2018). Visual learning on mobile phone for introduction basic programming in vocational high school. In *2018 International Conference on Computer Engineering, Network and Intelligent Multimedia (CENIM)* (pp. 186–191). IEEE. https://doi.org/10.1109/CENIM.2018.8710873

Kusuma, G. P., Putera Suryapranata, L. K., Wigati, E. K., & Utomo, Y. (2021). Enhancing historical learning using role-playing game on mobile platform. *Procedia Computer Science, 179*, 886–893. https://doi.org/10.1016/j.procs.2021.01.078

Lapek, J. (2018). Promoting 21st century skills in problem-based learning environments. *CTETE-Research Monograph Series, 1*(1), 66–85. https://doi.org/10.21061/ctete-rms.v1.c.4

Laporte, L., & Zaman, B. (2018). A comparative analysis of programming games, looking through the lens of an instructional design model and a game attributes taxonomy. *Entertainment Computing, 25*, 48–61. https://doi.org/10.1016/j.entcom.2017.12.005

Lee, S. M. (2019). *Her Story* or their own stories? Digital game-based learning, student creativity, and creative writing. *ReCALL, 31*(3), 238–254. https://doi.org/10.1017/S0958344019000028

Li, F. Y., Hwang, G. J., Chen, P. Y., & Lin, Y. J. (2021). Effects of a concept mapping-based two-tier test strategy on students' digital game-based learning performances and behavioral patterns. *Computers & Education, 173*, 104293. https://doi.org/10.1016/j.compedu.2021.104293

Lim, J. S., Choe, M. J., Zhang, J., & Noh, G. Y. (2020). The role of wishful identification, emotional engagement, and parasocial relationships in repeated viewing of live-streaming games: A social cognitive theory perspective. *Computers in Human Behavior, 108*, 106327. https://doi.org/10.1016/j.chb.2020.106327

Lindberg, R. S., Laine, T. H., & Haaranen, L. (2019). Gamifying programming education in K-12: A review of programming curricula in seven countries and programming games. *British Journal of Educational Technology, 50*(4), 1979–1995. https://doi.org/10.1111/bjet.12685

Liu, Z. Y., Shaikh, Z., & Gazizova, F. (2020). Using the concept of game-based learning in education. *International Journal of Emerging Technologies in Learning (iJET), 15*(14), 53–64. https://doi.org/10.3991/ijet.v15i14.14675

Lopez-Fernandez, D., Gordillo, A., Alarcon, P. P., & Tovar, E. (2021a). Comparing traditional teaching and game-based learning using teacher-authored games on computer science education. *IEEE Transactions on Education, 64*(4), 367–373. https://doi.org/10.1109/TE.2021.3057849

Lopez-Fernandez, D., Gordillo, A., Ortega, F., Yague, A., & Tovar, E. (2021b). LEGO® serious play in software engineering education. *IEEE Access, 9*, 103120–103131. https://doi.org/10.1109/ACCESS.2021.3095552

López-Pernas, S., Gordillo, A., Barra, E., & Quemada, J. (2019). Analyzing learning effectiveness and students' perceptions of an educational Escape room in a programming course in higher education. *IEEE Access, 7*, 184221–184234. https://doi.org/10.1109/ACCESS.2019.2960312

López-Pernas, S., Gordillo, A., Barra, E., & Quemada, J. (2021). Comparing face-to-face and remote educational Escape rooms for learning programming. *IEEE Access, 9*, 59270–59285. https://doi.org/10.1109/ACCESS.2021.3073601

Lotfi, E., Othman, B. Y., & Mohammed, B. (2019). Towards a mobile serious game for learning object oriented programming paradigms. In M. BenAhmed, A. A. Boudhir, & A. Younes (Eds.), *Innovations in smart cities applications Edition 2* (pp. 450–462). Springer. https://doi.org/10.1007/978-3-030-11196-0_38

Lymbery, J. (2012). The potential of a game-based learning approach to improve learner outcomes. *Computers in New Zealand Schools, 24*(1), 21–39.

Malizia, A., Fogli, D., Danesi, F., Turchi, T., & Bell, D. (2017). TAPASPlay: A game-based learning approach to foster computation thinking skills. In *2017 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)* (pp. 345–346). IEEE. https://doi.org/10.1109/VLHCC.2017.8103502

Malliarakis, C., Satratzemi, M., & Xinogalos, S. (2014). Educational games for teaching computer programming. *Research on e-Learning and ICT in Education Technological, Pedagogical and Instructional Perspectives.* https://doi.org/10.1007/978-1-4614-6501-0_7

Manning, C., Surdeanu, M., Bauer, J., Finkel, J., Bethard, S., & McClosky, D. (2014). The Stanford CoreNLP natural language processing toolkit. In *Proceedings of 52nd Annual Meeting of the Association for Computational Linguistics: System Demonstrations* (pp. 55–60). https://doi.org/10.3115/v1/P14-5010

Martins, E. R., Geraldes, W. B., Afonseca, U. R., & Gouveia, L. M. B. (2019). Using Kahoot as a learning tool. In I. Ramos, R. Quaresma, P. Silva, & T. Oliveira (Eds.), *Information systems for industry 4.0* (Vol. 31, pp. 161–169). Springer. https://doi.org/10.1007/978-3-030-14850-8_11

Mathew, R., Malik, S. I., & Tawafak, R. M. (2019). Teaching problem solving skills using an educational game in a computer programming course. *Informatics in Education, 18*(2), 359–373. https://doi.org/10.15388/infedu.2019.17

Meftah, C., Retbi, A., Bennani, S., & Idrissi, M. K. (2019). Evaluation of user experience in the context of mobile serious game. In *2019 International Conference on Intelligent Systems and Advanced Computing Sciences (ISACS)* (pp. 1–5). IEEE. https://doi.org/10.1109/ISACS48493.2019.9068872

Mezentseva, L. V., Kolganov, S. V., Dvoryankin, O. A., & Tuganova, S. V. (2021). Game-driven learning in the digital age: A systematic review and meta-analysis. *Turkish Journal of Computer and Mathematics Education (TURCOMAT)., 12*(10), 2244–2253. https://doi.org/10.17762/turcomat.v12i10.4747

Miljanovic, M. A., & Bradbury, J. S. (2020). GidgetML: An adaptive serious game for enhancing first year programming labs. In *2020 IEEE/ACM 42nd International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)* (pp. 184–192). IEEE. https://doi.org/10.1145/3377814.3381716

Min, W., Mott, B., Park, K., Taylor, S., Akram, B., Wiebe, E., Boyer, K. E., & Lester, J. (2020). Promoting computer science learning with block-based programming and narrative-centered gameplay. In *2020 IEEE Conference on Games (CoG)* (pp. 654–657). IEEE. https://doi.org/10.1109/CoG47356.2020.9231881

Montes, H., Hijon-Neira, R., Perez-Marin, D., & Montes, S. (2021). Using an online serious game to teach basic programming concepts and facilitate gameful experiences for high school students. *IEEE Access, 9*, 12567–12578. https://doi.org/10.1109/ACCESS.2021.3049690

Mosquera, C. K., Steinmaurer, A., Eckhardt, C., & Guetl, C. (2020). Immersively learning object oriented programming concepts with sCool. In *2020 6th International Conference of the Immersive Learning Research Network*

(ILRN) (pp. 124–131). IEEE. https://doi.org/10.23919/iLRN47897.2020.9155144

Mozelius, P., & Humble, N. (2023). Design factors for an educational game where girls and boys play together to learn fundamental programming. In *International Conference on ArtsIT, Interactivity and Game Creation* (pp. 134–148). Springer, Cham. https://doi.org/10.1007/978-3-031-28993-4_10

Munn, Z., Peters, M. D. J., Stern, C., Tufanaru, C., McArthur, A., & Aromataris, E. (2018). Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Medical Research Methodology, 18*(1), 143. https://doi.org/10.1186/s12874-018-0611-x

Nche, O. M., Welter, J., Che, M., Kraemer, E. T., Sitaraman, M., & Zordan, V. B. (2019). CodeTracesure—combining gaming, CS concepts, and pedagogy. In *2019 Research on Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT)* (pp. 1–2). IEEE. https://doi.org/10.1109/RESPECT46404.2019.8985865

Nche, O. M., Boulware, R., Che, S. M., Kraemer, E. T., Sitaraman, M., & Zordan, V. B. (2020). Basic code understanding challenges for elementary school children. In *2020 Research on Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT)* (pp. 1–2). IEEE. https://doi.org/10.1109/RESPECT49803.2020.9272421

Noroozi, O., Dehghanzadeh, H., & Talaee, E. (2020). A systematic review on the impacts of game-based learning on argumentation skills. *Entertainment Computing, 35*, 100369. https://doi.org/10.1016/j.entcom.2020.100369

Noval, B. A., Safrodin, M., & Hakkun, R. Y. (2019). Battlebot: Logic learning based on visual programming implementation in multiplayer game online. In *2019 International Electronics Symposium (IES)* (pp. 138–142). IEEE. https://doi.org/10.1109/ELECSYM.2019.8901628

Paiva, J. C., Leal, J. P., & Queirós, R. (2020). Authoring game-based programming challenges to improve students' motivation. In M. E. Auer & T. Tsiatsos (Eds.), *The challenges of the digital transformation in education* (Vol. 916, pp. 602–613). Springer. https://doi.org/10.1007/978-3-030-11932-4_57

Park, K., Mott, B., Min, W., Wiebe, E., Boyer, K. E., & Lester, J. (2020). Generating game levels to develop computer science competencies in game-based learning environments. In I. I. Bittencourt, M. Cukurova, K. Muldner, R. Luckin, & E. Millán (Eds.), *Artificial Intelligence in Education* (Vol. 12164, pp. 240–245). Cham: Springer. https://doi.org/10.1007/978-3-030-52240-7_44

Partovi, T., & Razavi, M. R. (2019). The effect of game-based learning on academic achievement motivation of elementary school students. *Learning and Motivation, 68*, 101592. https://doi.org/10.1016/j.lmot.2019.101592

Pellas, N., & Mystakidis, S. (2020). A systematic review of research about game-based learning in virtual worlds. *JUCS - Journal of Universal Computer Science, 26*(8), 1017–1042. https://doi.org/10.3897/jucs.2020.054

Peters, M. D. J., Marnie, C., Tricco, A. C., Pollock, D., Munn, Z., Alexander, L., McInerney, P., Godfrey, C. M., & Khalil, H. (2020). Updated methodological guidance for the conduct of scoping reviews. *JBI Evidence Synthesis, 18*(10), 2119–2126. https://doi.org/10.11124/JBIES-20-00167

Pila, S., Aladé, F., Sheehan, K. J., Lauricella, A. R., & Wartella, E. A. (2019). Learning to code via tablet applications: An evaluation of Daisy the Dinosaur and Kodable as learning tools for young children. *Computers & Education, 128*, 52–62. https://doi.org/10.1016/j.compedu.2018.09.006

Popovic, J., Korolija, N., Markovic, Z., & Bojic, D. (2017). Developing algorithmic skills of pupils in Serbian schools using code.org materials. In *2017 25th Telecommunication Forum (TELFOR)* (pp. 1–4). IEEE. https://doi.org/10.1109/TELFOR.2017.8249483

Prabawa, H. W., Junaeti, E., & Permana, Y. (2017, October). Using capture the flag in classroom: Game-based implementation in network security learning. In *2017 3rd International Conference on Science in Information Technology (ICSITech)* (pp. 690–695). IEEE. https://doi.org/10.1109/icsitech.2017.8257201

Priyaadharshini, M., Dakshina, R., & Sandhya, S. (2020). Learning analytics: Game-based learning for programming course in higher education. *Procedia Computer Science, 172*, 468–472. https://doi.org/10.1016/j.procs.2020.05.143

Qian, M., & Clark, K. R. (2016). Game-based Learning and 21st century skills: A review of recent research. *Computers in Human Behavior, 63*, 50–58. https://doi.org/10.1016/j.chb.2016.05.023

Rajeev, S., & Sharma, S. (2018). Educational game-theme based instructional module for teaching introductory programming. In *IECON 2018 - 44th Annual Conference of the IEEE Industrial Electronics Society* (pp. 3039–3044). IEEE. https://doi.org/10.1109/IECON.2018.8592835

Riera, B., Philippot, A., & Annebicque, D. (2019). Teaching the first and only logic control course with HOME I/O and Scratch 2.0. *IFAC-PapersOnLine, 52*(9), 109–114. https://doi.org/10.1016/j.ifacol.2019.08.133

Roodt, S., & Ryklief, Y. (2022). Using digital game-based learning to improve the academic efficiency of vocational education students. In *Research Anthology on Vocational Education and Preparing Future Workers* (pp. 643–671). IGI Global. https://doi.org/10.4018/978-1-6684-5696-5.ch037

Roussou, E., & Rangoussi, M. (2020). On the use of robotics for the development of computational thinking in kindergarten: Educational intervention and evaluation. In M. Merdan, W. Lepuschitz, G. Koppensteiner, R. Balogh, & D. Obdržálek (Eds.), *Robotics in Education* (pp. 31–44). Springer. https://doi.org/10.1007/978-3-030-26945-6_3

Rozali, N. F., & Zaid, N. M. (2017). Code puzzle: ActionScript 2.0 learning application based on problem-based learning approach. In *2017 6th ICT International Student Project Conference (ICT-ISPC)* (pp. 1–4). IEEE. https://doi.org/10.1109/ICT-ISPC.2017.8075329

Ryan, R. M., Rigby, C. S., & Przybylski, A. (2006). The motivational pull of video games: A self-determination theory approach. *Motivation and Emotion, 30*, 344–360. https://doi.org/10.1007/s11031-006-9051-8

Sáiz Manzanares, M. C., Rodríguez Arribas, S., Pardo Aguilar, C., & Queiruga Dios, M. A. (2020). Effectiveness of self-regulation and serious games for learning STEM knowledge in primary education. *Psicothema, 32*(4), 516–524. https://doi.org/10.7334/psicothema2020.30

Sánchez Mena, A., & Martí Parreño, J. (2017). Teachers´ acceptance of educational video games: A comprehensive literature review. *Journal of E-Learning and Knowledge Society, 13*(2), 47–63. https://doi.org/10.20368/1971-8829/1319

Schatten, M., & Schatten, M. (2020). A comparative study of gamification in progamming education in a Croatian high school. In *2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)* (pp. 700–704). IEEE. https://doi.org/10.23919/MIPRO48935.2020.9245237

Scherer, R., Siddiq, F., & Viveros, B. S. (2020). A meta-analysis of teaching and learning computer programming: Effective instructional approaches and conditions. *Computers in Human Behavior, 109*, 106349. https://doi.org/10.1016/j.chb.2020.106349

Seebauer, S., Jahn, S., & Mottok, J. (2020). Learning from Escape rooms? A study design concept measuring the effect of a cryptography educational Escape room. In *2020 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1684–1685). IEEE. https://doi.org/10.1109/EDUCON45650.2020.9125333

Seralidou, E., & Douligeris, C. (2020). Creating and using digital games for learning in elementary and secondary education. In *2020 5th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)* (pp. 1–8). IEEE. https://doi.org/10.1109/SEEDA-CECNSM49515.2020.9221799

Serrano, K. (2019). The effect of digital game-based learning on student learning: A literature review. *Graduate Research Papers*, 943. https://scholarworks.uni.edu/grp/943

Shabalina, O., Malliarakis, C., Tomos, F., & Mozelius, P. (2017). Game-based learning for learning to program: from learning through play to learning through game development. In *11th European Conference on Games Based Learning 2017, Graz, Austria, 5–6 October 2017* (Vol. 11, pp. 571–576). Academic Conferences and Publishing International Limited.

Shahid, M., Wajid, A., Haq, K. U., Saleem, I., & Shujja, A. H. (2019). A Review of gamification for learning programming fundamental. In *2019 International Conference on Innovative Computing (ICIC)* (pp. 1–8). IEEE. https://doi.org/10.1109/ICIC48496.2019.8966685

Sharaf, N., Ahmed, G., & Ihab, S. (2020). Virtual/mixed reality control of a game through Scratch. In *2020 24th International Conference Information Visualisation (IV)* (pp. 689–693). IEEE. https://doi.org/10.1109/IV51561.2020.00119

Sharma, K., Papavlasopoulou, S., & Giannakos, M. (2019). Coding games and robots to enhance computational thinking: How collaboration and engagement moderate children's attitudes? *International Journal of Child-Computer Interaction, 21*, 65–76. https://doi.org/10.1016/j.ijcci.2019.04.004

Sharma, K., Torrado, J. C., Gómez, J., & Jaccheri, L. (2021). Improving girls' perception of computer science as a viable career option through game playing and design: Lessons from a systematic literature review. *Entertainment Computing, 36*, 100387. https://doi.org/10.1016/j.entcom.2020.100387

Shim, J., Kwon, D., & Lee, W. (2017). The effects of a robot game environment on computer programming education for elementary school students. *IEEE Transactions on Education, 60*(2), 164–172. https://doi.org/10.1109/TE.2016.2622227

Siakavaras, I., Papastergiou, M., & Comoutos, N. (2018). Mobile games in computer science education: Current state and proposal of a mobile game design that incorporates physical activity. In T. A. Mikropoulos (Ed.), *Research on e-Learning and ICT in Education* (pp. 243–255). Springer. https://doi.org/10.1007/978-3-319-95059-4_15

Silva, J. P., Silveira, I. F., Kamimura, L., & Barboza, A. T. (2020). Turing project: An open educational game to teach and learn programming logic. In *2020 15th Iberian Conference on Information Systems and Technologies (CISTI)* (pp. 1–6). IEEE. https://doi.org/10.23919/CISTI49556.2020.9141122

Simões Gomes, T. C., Pontual Falcão, T., de Azevedo, C., & Restelli Tedesco, P. (2018). Exploring an approach based on digital games for teaching programming concepts to young children. *International Journal of Child-Computer Interaction, 16*, 77–84. https://doi.org/10.1016/j.ijcci.2017.12.005

Sookhanaphibarn, K., & Choensawat, W. (2020). Educational games for cybersecurity awareness. In *2020 IEEE 9th Global Conference on Consumer Electronics (GCCE)* (pp. 424–428). IEEE. https://doi.org/10.1109/GCCE50665.2020.9291723

Stigall, J., & Sharma, S. (2017). Virtual reality instructional modules for introductory programming courses. In *2017 IEEE Integrated STEM Education Conference (ISEC)* (pp. 34–42). IEEE. https://doi.org/10.1109/ISECon.2017.7910245

Sung, H. Y., & Hwang, G. J. (2013). A collaborative game-based learning approach to improving students' learning performance in science courses. *Computers & Education, 63*, 43–51. https://doi.org/10.1016/j.compedu.2012.11.019

Tabuti, L. M., de Azevedo da Rocha, R. L., & Nakamura, R. (2020). Proposal of method for converting a physical card game to digital for logical reasoning competencies on the data structure subject. In *2020 IEEE Frontiers in Education Conference (FIE)* (pp. 1–9). IEEE. https://doi.org/10.1109/FIE44824.2020.9274001

Tacouri, H., & Nagowah, L. (2021). Code Saga – A mobile serious game for learning programming. In *2021 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS)* (pp. 190–195). IEEE. https://doi.org/10.1109/IoTaIS53735.2021.9628484

Tang, T., Vezzani, V., & Eriksson, V. (2020). Developing critical thinking, collective creativity skills and problem solving through playful design jams. *Thinking Skills and Creativity, 37*, 100696. https://doi.org/10.1016/j.tsc.2020.100696

Taylor, S., Min, W., Mott, B., Emerson, A., Smith, A., Wiebe, E., & Lester, J. (2019). Position: IntelliBlox: A toolkit for integrating block-based programming into game-based learning environments. In *2019 IEEE Blocks and Beyond Workshop (B&B)* (pp. 55–58). IEEE. https://doi.org/10.1109/BB48857.2019.8941222

Tioh, J. N., Mina, M., & Jacobson, D. W. (2017). Cyber security training a survey of serious games in cyber security. In *2017 IEEE Frontiers in Education Conference (FIE)* (pp. 1–5). IEEE. https://doi.org/10.1109/FIE.2017.8190712

Toh, W., & Kirschner, D. (2020). Self-directed learning in video games, affordances and pedagogical implications for teaching and learning. *Computers & Education, 154*, 103912. https://doi.org/10.1016/j.compedu.2020.103912

Towler, A., Aranda, D., Ramyaa, R., & Kuo, R. (2020). Using educational game for engaging students in learning foundational concepts of propositional logic. In *2020 IEEE 20th International Conference on Advanced Learning Technologies (ICALT)* (pp. 208–209). IEEE. https://doi.org/10.1109/ICALT49669.2020.00067

Tretinjak, M. F. (2019). Methodology for using games as an educational tool. In *2019 42nd International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)* (pp. 547–551). IEEE. https://doi.org/10.23919/MIPRO.2019.8757046

Tsur, M., & Rusk, N. (2018). Scratch microworlds: designing project-based introductions to coding. In *Proceedings of the 49th ACM Technical Symposium on Computer Science Education* (pp. 894–899). https://doi.org/10.1145/3159450.3159559

Turan, Z., Avinc, Z., Kara, K., & Goktas, Y. (2016). Gamification and education: Achievements, cognitive loads, and views of students. *International Journal of Emerging Technologies in Learning (IJET), 11*(07), 64. https://doi.org/10.3991/ijet.v11i07.5455

Unger, S., & Meiran, W. R. (2020). Student attitudes towards online education during the COVID-19 viral outbreak of 2020: Distance learning in a time of social distance. *International Journal of Technology in Education and Science, 4*(4), 256–266. https://doi.org/10.46328/ijtes.v4i4.107

Vahldick, A., Farah, P. R., Marcelino, M. J., & Mendes, A. J. (2020). A blocks-based serious game to support introductory computer programming in undergraduate education. *Computers in Human Behavior Reports, 2*, 100037. https://doi.org/10.1016/j.chbr.2020.100037

Valle, P. H. D., Toda, A. M., Barbosa, E. F., & Maldonado, J. C. (2017). Educational games: A contribution to software testing education. In *2017 IEEE Frontiers in education Conference (FIE)* (pp. 1–8). IEEE. https://doi.org/10.1109/FIE.2017.8190470

Vankúš, P. (2021). Influence of game-based learning in mathematics education on students' affective domain: A systematic review. *Mathematics, 9*(9), 986. https://doi.org/10.3390/math9090986

Venkatesh, P., Das, S., & Das, A. K. (2021). Design and development of low-cost unplugged activities for teaching computational thinking at K-5 level. In *Design for Tomorrow—Volume 3* (pp. 523–534). Springer, Singapore. https://doi.org/10.1007/978-981-16-0084-5_42

Ventura, J., Qualls, J., Ventura, M., McGinnis, R., Baker, C., & Nikaido, B. (2017). Development of a video game design program in the college of engineering. In *SoutheastCon 2017* (pp. 1–7). IEEE. https://doi.org/10.1109/SECON.2017.7925397

Videnovik, M., Kionig, L., Vold, T., & Trajkovik, V. (2018). Testing framework for investigating learning outcome from quiz game: A Study from Macedonia and Norway. In *2018 17th International Conference on Information Technology Based Higher Education and Training (ITHET)* (pp. 1–5). IEEE. https://doi.org/10.1109/ITHET.2018.8424777

Videnovik, M., Trajkovik, V., Kiønig, L. V., & Vold, T. (2020). Increasing quality of learning experience using augmented reality educational games. *Multimedia Tools and Applications, 79*(33–34), 23861–23885. https://doi.org/10.1007/s11042-020-09046-7

Videnovik, M., Vold, T., Dimova, G., Kiønig, L. V., & Trajkovik, V. (2022). Migration of an Escape room–style educational game to an online environment: Design Thinking methodology. *JMIR Serious Games, 10*(3), e32095. https://doi.org/10.2196/32095

Visoottiviseth, V., Phungphat, A., Puttawong, N., Chantaraumporn, P., & Haga, J. (2018). Lord of Secure: The virtual reality game for educating network security. In *2018 Seventh ICT International Student Project Conference (ICT-ISPC)* (pp. 1–6). IEEE. https://doi.org/10.1109/ICT-ISPC.2018.8523947

Vlahu-Gjorgievska, E., Videnovik, M., & Trajkovik, V. (2018). Computational thinking and coding subject in primary schools: Methodological approach based on alternative cooperative and individual learning cycles. In *2018 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)* (pp. 77–83). IEEE. https://doi.org/10.1109/tale.2018.8615334

Vostinar, P. (2021). MakeCode Arcade: Interesting environment for programming 2D games. In *2021 IEEE World Conference on Engineering Education (EDUNINE)* (pp. 1–6). IEEE. https://doi.org/10.1109/EDUNINE51952.2021.9429132

Vourletsis, I., & Politis, P. (2022). Exploring the effect of remixing stories and games on the development of students' computational thinking. *Computers and Education Open, 3*, 100069. https://doi.org/10.1016/j.caeo.2021.100069

Vu, P., & Feinstein, S. (2017). An exploratory multiple case study about using game-based Learning in STEM classrooms. *International Journal of Research in Education and Science*. https://doi.org/10.21890/ijres.328087

Wang, L. H., Chen, B., Hwang, G. J., Guan, J. Q., & Wang, Y. Q. (2022). Effects of digital game-based STEM education on students' learning achievement: A meta-analysis. *International Journal of STEM Education, 9*(1), 1–13. https://doi.org/10.1186/s40594-022-00344-0

Webster, J. J., & Kit, C. (1992). Tokenization as the initial phase in NLP. In *Proceedings of the 14th Conference on Computational Linguistics*, (Vol. 4, pp. 1106–1110). https://doi.org/10.3115/992424.992434

Wong, Y. S., Hayati, I. M., Yatim, M., & Hoe, T. W. (2017). A propriety game-based learning mobile game to learn object-oriented programming—Odyssey of Phoenix. In *2017 IEEE 6th International Conference on Teaching, Assessment, and Learning for Engineering (TALE)* (pp. 426–431). IEEE. https://doi.org/10.1109/TALE.2017.8252373

Wong, G. K. W., & Jiang, S. (2018). Computational thinking education for children: Algorithmic thinking and debugging. In *2018 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)* (pp. 328–334). https://doi.org/10.1109/TALE.2018.8615232

Wong, Y. S., & Yatim, M. H. M. (2018). A propriety multiplatform game-based learning game to learn object-oriented programming. In *2018 7th International Congress on Advanced Applied Informatics (IIAI-AAI)* (pp. 278–283). IEEE. https://doi.org/10.1109/IIAI-AAI.2018.00060

Workman, M. D., Luevanos, J. A., & Mai, B. (2021). A study of cybersecurity education using a Present-Test-Practice-Assess model. *IEEE Transactions on Education, 65*(1), 40–45. https://doi.org/10.1109/TE.2021.3086025

Xian, T. (2021). Exploring the effectiveness of Sandbox game-based learning environment for game design course in higher education. In T. Ahram, R. Taiar, & F. Groff (Eds.), *Human interaction, emerging technologies and future applications IV* (pp. 262–272). Cham: Springer. https://doi.org/10.1007/978-3-030-74009-2_33

Yallihep, M., & Kutlu, B. (2020). Mobile serious games: Effects on students' understanding of programming concepts and attitudes towards information technology. *Education and Information Technologies, 25*(2), 1237–1254. https://doi.org/10.1007/s10639-019-10008-2

Yassine, A., Berrada, M., Tahiri, A., & Chenouni, D. (2018). A cross-platform mobile application for learning programming basics. *International Journal of Interactive Mobile Technologies (IJIM), 12*(7), 139. https://doi.org/10.3991/ijim.v12i7.9442

Yokoyama, H., Fukui, M., Matsumoto, T., Mori, Y., & Takami, T. (2020). Use gameplay RPA as programming materials. In *2020 Nicograph International (NicoInt)* (pp. 92–92). IEEE. https://doi.org/10.1109/NicoInt50878.2020.00028

Yongqiang, C., Xiaojun, W., & Chengbin, Q. (2018). Computer programming education for primary school students. In *2018 13th International Conference on Computer Science & Education (ICCSE)* (pp. 1–5). IEEE. https://doi.org/10.1109/ICCSE.2018.8468824

Yu, Z. (2019). A meta-analysis of use of serious games in education over a decade. *International Journal of Computer Games Technology, 2019*, 1–8. https://doi.org/10.1155/2019/4797032

Yücel, Y., & Rızvanoğlu, K. (2019). Battling gender stereotypes: A user study of a code-learning game, "Code Combat" with middle school children. *Computers in Human Behavior, 99*, 352–365. https://doi.org/10.1016/j.chb.2019.05.029

Zapata-Caceres, M., & Martin-Barroso, E. (2021). Applying game learning analytics to a voluntary video game: Intrinsic motivation, persistence, and rewards in learning to program at an early ge. *IEEE Access, 9*, 123588–123602. https://doi.org/10.1109/ACCESS.2021.3110475

Zapušek, M., & Rugelj, J. (2013). Learning programming with serious games. *EAI Endorsed Transactions on Game-Based Learning, 1*(1), e6. https://doi.org/10.4108/trans.gbl.01-06.2013.e6

Zaw, H. H., & Hlaing, S. Z. (2020). Verifying the gaming strategy of self-learning game by using PRISM-games. In P. Vasant, I. Zelinka, & G.-W. Weber (Eds.), *Intelligent computing and optimization* (pp. 148–159). Springer. https://doi.org/10.1007/978-3-030-33585-4_15

Zdravevski, E., Lameski, P., Trajkovik, V., Chorbev, I., Goleva, R., Pombo, N., & Garcia, N. M. (2019). Automation in systematic, scoping and rapid reviews by an NLP toolkit: A case study in enhanced living environments. In I. Ganchev, N. M. Garcia, C. Dobre, C. X. Mavromoustakis, & R. Goleva (Eds.), *Enhanced living environments* (Vol. 11369, pp. 1–18). Springer. https://doi.org/10.1007/978-3-030-10752-9_1

Zeevaarders, A., & Aivaloglou, E. (2021). Exploring the programming concepts practiced by Scratch users: An analysis of project repositories. In *2021 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1287–1295). IEEE. https://doi.org/10.1109/EDUCON46332.2021.9453973

Zhang, F., Kaufman, D., & Fraser, S. (2014). Using video games in computer science education. *European Scientific Journal, 10*(22), 37–52. https://doi.org/10.19044/esj.2014.v10n22p%25p

Zhao, D., Muntean, C. H., Chis, A. E., & Muntean, G.-M. (2021). Learner attitude, educational background, and gender influence on knowledge gain in a serious games-enhanced programming course. *IEEE Transactions on Education, 64*(3), 308–316. https://doi.org/10.1109/TE.2020.3044174

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Submit your manuscript to a SpringerOpen journal and benefit from:

- ► Convenient online submission
- ► Rigorous peer review
- ► Open access: articles freely available online
- ► High visibility within the field
- ► Retaining the copyright to your article

Submit your next manuscript at ► springeropen.com

arXiv:2207.07369v2 [cs.CY] 2 Dec 2022

A Systematic Literature Review of Game-based Assessment Studies: Trends and Challenges

Manuel J. Gomez, José A. Ruipérez-Valiente, Félix J. García Clemente
Faculty of Computer Science
University of Murcia
Murcia (Spain)
{manueljesus.gomezm, jruiperez, fgarcia}@um.es

Abstract

Technology has become an essential part of our everyday life, and its use in educational environments keeps growing. In addition, games are one of the most popular activities across cultures and ages, and there is ample evidence that supports the benefits of using games for assessment. This field is commonly known as game-based assessment (GBA), which refers to the use of games to assess learners' competencies, skills, or knowledge. This paper analyzes the current status of the GBA field by performing the first systematic literature review on empirical GBA studies. It is based on 65 research papers that used digital GBAs to determine: (1) the context where the study has been applied; (2) the primary purpose; (3) the domain of the game used; (4) game/tool availability; (5) the size of the data sample; (6) the computational methods and algorithms applied; (7) the targeted stakeholders of the study; and (8) what limitations and challenges are reported by authors. Based on the categories established and our analysis, the findings suggest that GBAs are mainly used in K-16 education and for assessment purposes, and that most GBAs focus on assessing STEM content, and cognitive and soft skills. Furthermore, the current limitations indicate that future GBA research would benefit from the use of bigger data samples and more specialized algorithms. Based on our results, we discuss current trends in the field and open challenges (including replication and validation problems), providing recommendations for the future research agenda of the GBA field.

Keywords Game-based assessment · educational technology · game-based learning · learning analytics

1 Introduction

Technology is progressively changing the world in which we live. During the last decade, it has started to make a significant impact on educational environments, and increasing evidence has been accumulated showing the positive impact of technology in education [1]. One of the most prominent examples of technology in education is the use of digital games [2]. This type of games has become a significant part of families and, especially, among young people around the world. In fact, three-quarters of all U.S. households have at least one person who plays video games [3], while in Europe, 51% of the population aged 6-61 years play video games (an average of 8.6 hours/week) [4]. Moreover, many educators see digital games as powerfully motivating digital environments because of their potential to enhance student engagement and motivation in learning [5]. This increasing interest provides an opportunity to use video games as a tool to improve learning and education. Specifically, there is much enthusiasm in the field of education about game-based assessment (GBA) because conventional assessment methods do not seem to fully have the power to measure all aspects of students' knowledge, skills, and attributes [6].

Accompanying this explosion in technology use is the quantity, range and scale of data that can be collected, which have increased exponentially over the last decade [7]. In education, the increase in e-learning resources, educational software like Google Classroom or Kahoot, and the use of the Internet have created large repositories that provide a goldmine of educational data that can be explored and used to understand how students learn [8]. Regarding games, they allow recreating more authentic situations compared to traditional classroom activities, such as lectures or written exercises. From these situations, we can collect a vast amount of detailed data on students' interaction with the game,

which provides a great opportunity to make game-based assessments (GBAs) in ways that are not possible in traditional testing [9].

In the past 10 years, numerous studies (see the work in [10] for a meta-analysis) have reported that games can be more effective for learning than other traditional teaching methods. In addition, when measuring the competencies acquired, most traditional tests present individual and decontextualized items to learners, while 21st-century competencies benefit from being applied in context for more accurate measurements. Furthermore, classic assessment often interrupts the learning process, and it does little to motivate learners [11]. Since digital games often employ challenging, interesting, and complex problems, they can be used to generate evidence of 21st-century competencies, which are traditionally difficult to measure using conventional forms of assessment [12]. The advantages of using games as a form of assessment are manifold [11, 13, 14]: they are engaging and motivating (which provides more valid assessments), and they allow us to create more complex and authentic scenarios required to assess the application of knowledge and skills. Moreover, immediate feedback based on learners' activity can reveal teachers' specific areas of difficulty to make learners keep up with the pace of the class, and such assessment would result in an adaptive game environment, which changes with learners' activity.

The implementation of assessment features into game environments is only in its early stages because it adds a very time-consuming step to the design process [15]. This situation calls for a review of the current state of the art in the GBA field for effective implementations. In this respect, we found some previous works that performed meta-reviews of the existing research on the different applications of games in learning and education. For example, the authors in [16] reviewed 137 papers to determine what empirical evidence existed concerning the effects of Game-based Learning (GBL) on 21st-century competencies and identified successful game-design elements that aligned well with established learning theories. Moreover, Alonso-Fernandez et al.[17] carried out a review focused on data science applications to game learning analytics data, showing that the primary purpose when analyzing data from serious games was assessment. Furthermore, Gris and Bengtson [18] aimed to answer how learning, engagement, and usability of games are evaluated in GBL research. To this aim, they conducted a systematic review of 91 empirical studies and categorized their measures and instruments. The researchers concluded that future research in GBL studies should add direct assessments and indirect measures to assess engagement and usability. Guan et al. [19] provided a systematic review of 35 experimental studies that substantially integrated gaming elements in primary school lessons and they noted that gamification was the most frequently used game genre. Finally, Chen et al. [20] conducted a systematic review of 146 articles related to GBL in science and mathematics education. These researchers concluded that GBL is mainly used to increase learner motivation and engagement and reduce learning anxiety. They also revealed that analyzing higher-order thinking skills (e.g., problem-solving, group collaboration) is one of the main hot topics in the community.

Despite the previous reviews of the use of games in learning in education, we have not found any specific study reviewing literature about GBA. For this reason, the current paper aims to conduct the first systematic literature review on the applications of empirical GBA studies and answer some research questions based on the analysis performed to discover current trends and open challenges in this area. The results obtained will provide an overall view of the GBA field, defining its current status and potential future steps in the research in this area.

The rest of the paper is organized as follows. Section 2 describes the methods, including some terminology clarifications, the research questions, databases and search terms, research selection as well as review process. Section 3 presents the analysis and synthesis of our results. Then, we end the paper with a discussion in Section 4 and conclusions in Section 5.

2 Methods

We followed a standard systematic literature review methodology, using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [21] as a basis for conducting our systematic review. First of all, (1) we formulated each research question (RQ). Then, we used (2) a fixed set of queries on a pre-identified bibliographical database, and (3) a set of inclusion and exclusion criteria. Next, (4) we made a full paper review and coding process of the RQs, and, finally, we carried out (5) a synthesis and analysis. No time restrictions were set. We can see a flow diagram representing the different stages of our systematic review (following the PRISMA template [22]) in Figure 1.

2.1 Terminology clarifications

In this Section, we present a set of definitions that aim to clarify the concept of GBA, which is the focus of this systematic literature review. Firstly, we can define a **game** as "a system in which players engage in an artificial conflict, defined by rules, that results in a quantifiable outcome" [23, p. 80]. In addition, games also have clearly-defined goals

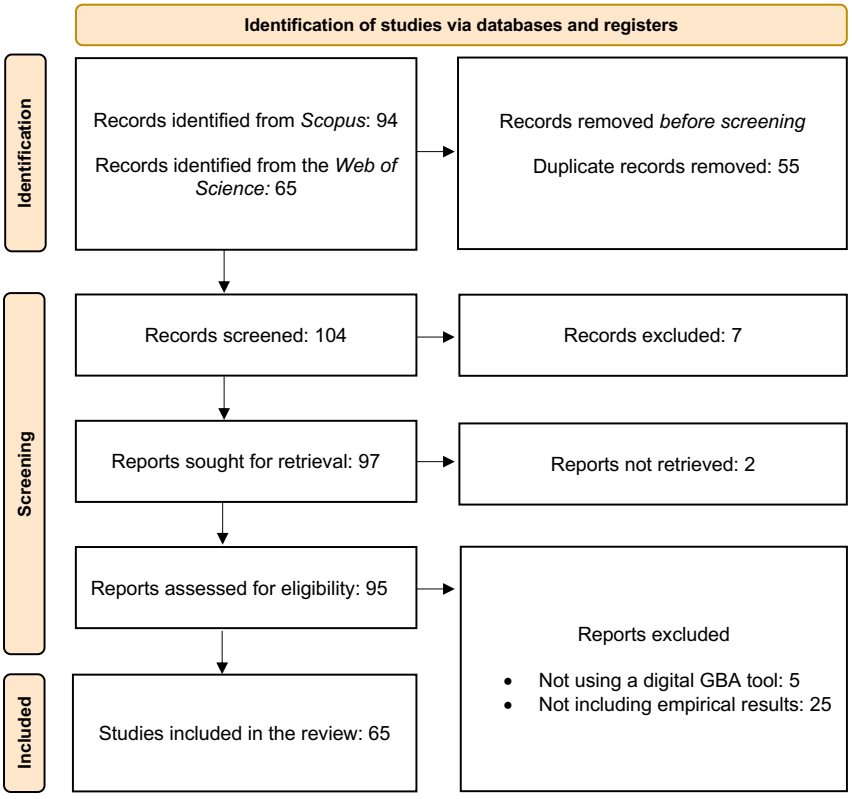


Figure 1: Flow diagram representing the different phases of the systematic review.

and obstacles for the player to overcome, providing only intrinsic rewards (satisfaction for getting the right answer) [24]. Secondly, **GBL** can be regarded as an innovative learning approach where a game is developed to deliver immersive and attractive learning experiences aiming at particular learning goals, experiences and results [25]. Thus, GBL uses a game containing learning content derived from school curricula or essential life skills to improve the learning experience. Moreover, **GBA** is a specific application of games, referring to a type of assessment that uses players' interactions with the game, both digital and non-digital, as a source of evidence to make meaningful inferences about what players know and can do (i.e., knowledge, skills), and how individual players interact with the game as a problem-solving process [15, 26]. Finally, we have the concept of **gamification**, which is usually defined as "the use of game design elements in non-game contexts" [27, p. 9].

Although GBL and GBA are often confused with gamification and gamified assessment, it is undeniable that some differences exist between them. While GBL implies the use of a game developed for learning purposes, gamification utilizes game elements in non-game contexts, not necessarily using full games inside the activities [28]. Thus, GBA also implies the use of a game developed for assessment purposes, using players' interaction with the game as a way to obtain evidence and use this evidence as a form of assessment. Therefore, tools that use gamified activities to assess students' knowledge (e.g., Kahoot, Duolingo) use gamified assessments, and cannot be considered as GBAs. We can also make a clear distinction between GBA and a simple measurement using games since GBA is intended for evaluating players' skills or knowledge based on their interaction with the game. As Ghergulescu and Muntean state, "measurement represents the process of collecting the information needed for assessment" [29, p. 357]. In other words, measurements are used as evidence to make meaningful inferences about what players know and can do, while measurements using games do not perform that evaluation. These are the definitions that we applied as part of the systematic review screening process to consider a given paper within the GBA field or not, including or discarding that study.

2.2 Research questions

To state each one of the RQs, we analyzed and simplified the steps in empirical GBA research [30], which can be seen in Figure 2.

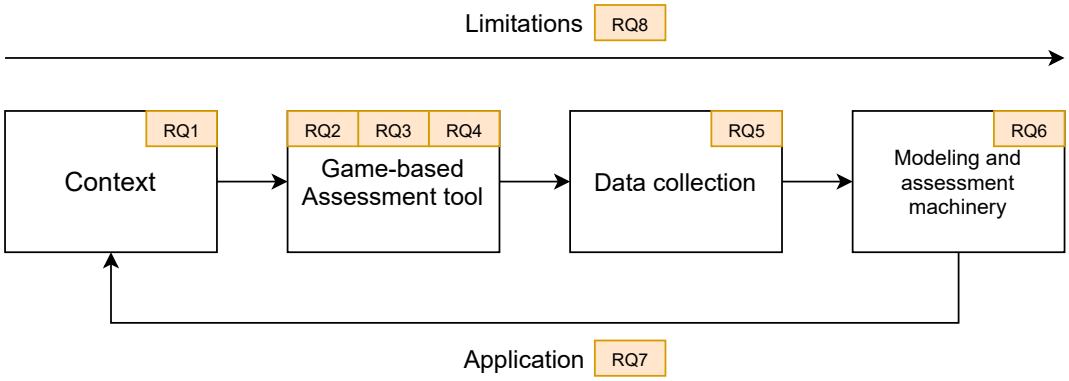


Figure 2: A simplified view of the steps in game-based assessment empirical research.

In this process, we can identify five different stages: (1) Learning environments, with the context and learners; (2) The GBA tool that is going to be used in the research; (3) Data collection, to identify which data has to be collected and how to store them; (4) Modeling and assessment machinery; and (5) Educational application, to identify the final objective and target users. From these stages, we identified the following RQs, which allow us to understand the open challenges and current trends in the area:

RQ1. In what context or environment has GBA been applied?
RQ2. What is the primary purpose of GBA?
RQ3. What is the domain of GBA?
RQ4. Is the game/tool used available to the public?
RQ5. What is the size of the data sample used in the study?
RQ6. What computational methods and algorithms have been applied in the research?
RQ7. What stakeholder is the intended recipient of the research results?
RQ8. What limitations and challenges do the authors address?

In addition, Figure 2 also shows the mapping between the different stages and the RQs identified: RQ1 is based on the first stage, related to the learning context. Then, RQ2, RQ3, and RQ4 are based on the second stage, which refers to the GBA tool used, its primary purpose, the domain and availability. Next, we wanted to investigate the sample size (RQ5) which is situated in the data collection stage. Regarding the modeling and assessment machinery, our objective was to investigate the computational methods and algorithms applied in the research (RQ6). RQ7 refers to the application of the research in the desired context, identifying the main stakeholder of the research. Finally, RQ8 aims to identify the research limitations at any stage.

2.3 Databases and search terms

We have queried two databases: Scopus and the Web of Science since they are the most widely used databases in different scientific fields and are often used for surveying the literature [31]. Scopus is the world's largest citation database of peer-reviewed research literature, with over 22,000 titles (including journals, conferences and book series) from more than 5,000 international publishers, of which 20,000 are peer-reviewed journals in the scientific, technical, medical, and social sciences [32]. Moreover, the Web of Science, the second biggest bibliographic database, can be used to track ideas going back several decades from almost 1.9 billion cited references from over 171 million records [33].

To perform the search on both databases, we restricted the query to title and keywords: 1) we included the term "game-based assessment" and searched for it within the paper titles; 2) we included the term "game-based assessment" and searched for it within the paper keywords. Thus, we used the following final search query:

(TITLE("game-based assessment") OR *KEY*("game-based assessment")).

The initial selection of studies was retrieved in January 2021, and this query generated 159 initial studies (94 from Scopus and 65 from the Web of Science).

2.4 Inclusion/Exclusion criteria

After obtaining the initial collection, we excluded the duplicated records from the two databases (55 studies). Then, we made a first brief review of all papers, comparing them against the inclusion and exclusion criteria. This first review was conducted by one of the authors. After the first analyses, we classified studies as *included* or *excluded*, and the coding results were discussed collaboratively by the three authors in order to obtain the final set of included and excluded studies and avoid possible errors. The inclusion/exclusion criteria followed are described below. Given these criteria, the paper was included if all of the following conditions were met (i.e., if one condition was not met, the paper was excluded). Furthermore, the conditions were applied sequentially, so that a paper not matching a condition was excluded immediately from the collection:

- The paper was written in English or Spanish (languages in which the authors have high proficiency): 0% of the papers were excluded.
- The paper was fully accessible: 1.9% of the papers were excluded (2 studies).
- The paper was published in conference proceedings, journals or edited books/volumes (i.e., book chapter): 0% of the papers were excluded.
- The paper was not extended at a later time (i.e., a conference paper that was later on extended in a journal paper): 6.9% of the papers were excluded (7 studies).
- The paper used a digital GBA tool: 5.3% of the papers were excluded (5 studies). See Section 2.1 above for relevant definitions.
- The paper included empirical evidence related to the outcomes of applying the GBA tool: 27.8% of the papers were excluded (25 studies).

2.5 Final paper collection

After the first brief review to ensure that every paper met our inclusion/exclusion criteria, we excluded a total of 39 papers. Thus, the final paper collection consists of 65 studies.

Figure 3 shows the distribution of papers within the final collection by publication year. We see an increasing interest in this particular topic: between the years 2013 and 2016, we only have 21 (32.3%) published papers that matched our criteria, while between 2017 and 2020 there are 44 (67.7%) of them.

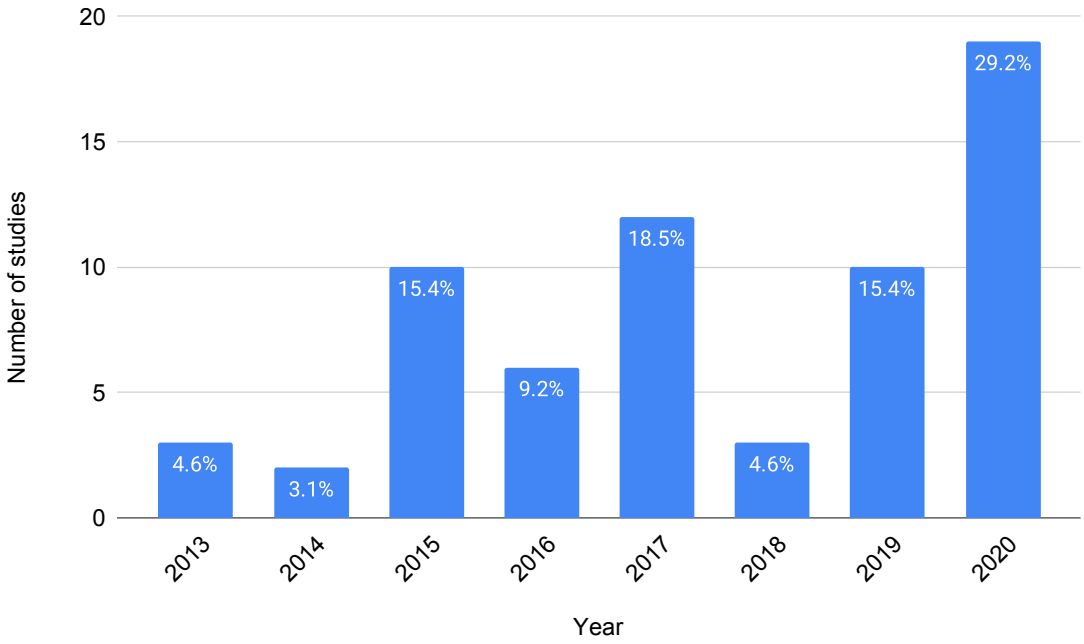


Figure 3: Number of selected studies per year of publication.

We also collected each paper's keywords and made a brief analysis to describe our paper collection. For our analysis, we excluded the "game-based assessment" keyword since it was the most common one. The most frequent keywords

are presented in Figure 4. The total sum of keywords is 299 while there are 201 unique keywords. The average keyword was found 1.48 times. As we can see, the predominant keywords strongly focused on games, assessment, and analytics.

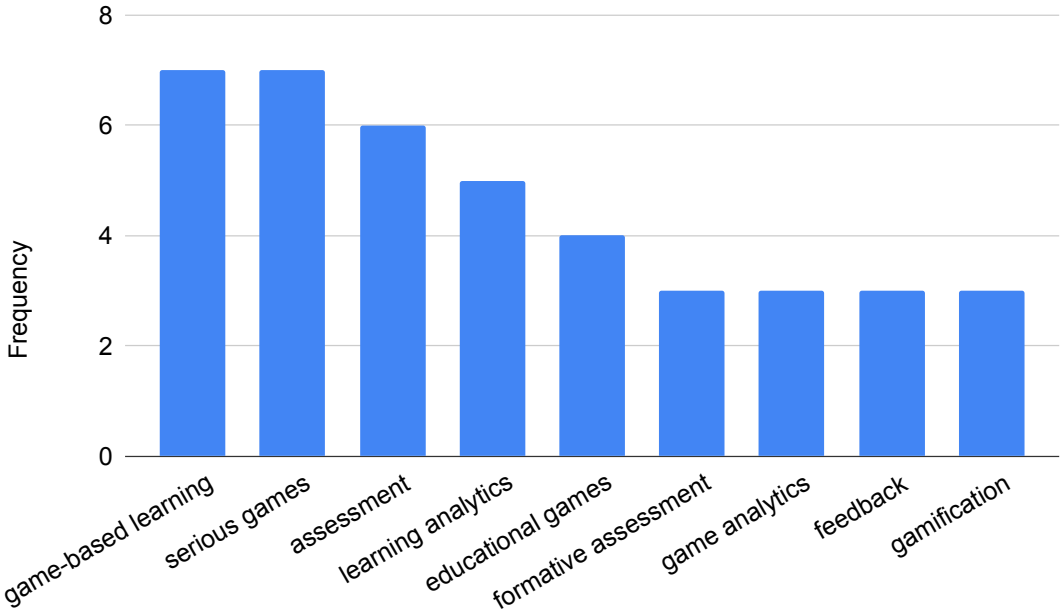


Figure 4: Distribution of keywords across articles in the final collection.

2.6 Review and coding process

In the coding stage, we collected the data of the selected studies that we consider to be the most valuable to address the RQs in Section 2.2. Based on the aim of the review, we followed an inductive coding scheme (also called open coding). This means that the codes created were based on the qualitative data itself [34]. This is an iterative process since researchers can add new codes, split an existing code into two, or compress two existing codes into one as they continue reviewing data. Specifically, in our analysis, we first made a brief review of each paper (conducted by one author), collecting all the necessary information to code each RQ at once. After that, we followed an iterative process whereby we continued reviewing the information corresponding to each RQ sequentially, and unclear results were discussed and contrasted by the three authors. The full results of the coding process per paper are available in [35]. In addition, it should be noted that each paper can fit into more than one of the codes created for each RQ.

3 Results

3.1 In what context or environment has the GBA been applied? (RQ1)

GBAs can be used in very different environments. Our analysis reveals that there are three main contexts where GBAs have been used:

1. `K-16 education`: some papers use GBAs in K-16 education (e.g., school, university) to support teaching and learning. More specifically, games are most commonly used in middle school and high school (23.1%). However, games are also used in other K-16 education environments such as primary school (15.4%) and university (10.8%). For example, Di Cerbo et al.[36] used game data from 751 US middle school players.
2. `Medical`: games can also be used in medical environments for different purposes (e.g., rehabilitation). For example, the authors in [37] examined the feasibility of administering the GBA in a sample of inpatients with chronic schizophrenia with low levels of functioning. Moreover, the authors in [38] aimed to present data on construct validity, test–retest reliability and feasibility, measuring motor-cognitive functions in multimorbid patients with mild-to-moderate dementia. Regarding construct validity, the authors tested eight hypotheses and confirmed seven of them (87.5%), thus indicating excellent construct validity. Moreover, the authors found moderate-to-high test–retest reliability (ICC=0.47-0.92).

3. `Workforce`: another option is the use of games to assess in professional environments. In this context, enterprises can use games to evaluate their employees or provide them with additional feedback. Even now, companies can include the use of GBA for the recruitment of staff and the selection process [39]. This idea is supported by the fact that in-game constructs show similar relationships with in-game performance to what the workforce constructs do with job performance [40].

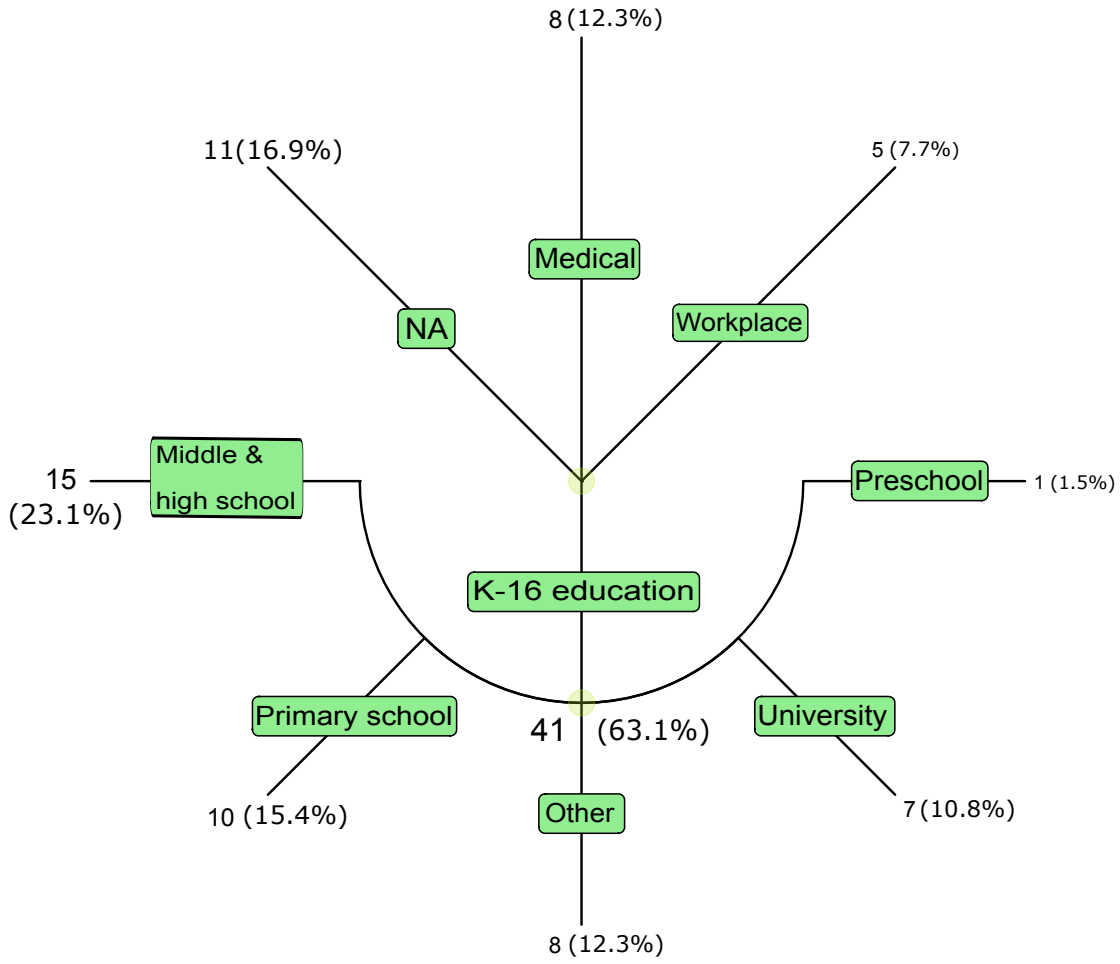


Figure 5: Papers' category distribution based on RQ1.

There are also some studies such as [41] that did not specify in what context their games were used (11 studies, 16.9%). We can see the number of papers fitting in each category in Figure 5. As the figure shows, GBAs are mostly used in `K-16 education` (41 studies, 63.1%), followed by `medical` (8 studies, 12.3%) and `workforce` contexts (5 studies, 7.7%).

3.2 What is the primary purpose of the GBA research? (RQ2)

In this RQ, we wanted to know what was the main purpose of each GBA in each study. We coded the papers' main purpose into six different categories: `GBA evaluation`, `study of in-game behaviors`, `assessment`, `interventions`, `framework proposal` and `game design proposal`. Next, we describe in detail each one of these categories.

1. `GBA evaluation`: in these studies, authors evaluate the game by checking if it achieves its initial objectives using some measure to prove that the game or tool is suitable for an educational environment. In [42], the authors showed how they applied the methodology for an assessment game for ICT managers in secondary vocational education, checking if this assessment was content-valid compared to a face-to-face assessment. Moreover, the authors in [43] aimed to investigate whether it is possible to perform an in-Basket test (which

is widely used by companies and organizations in order to map employees' soft skills) online with the same effect as that of the onsite one.

2. `In-game behaviors`: in these studies, authors investigate in-game players' behaviors (e.g., persistence, engagement). By identifying these behaviors, we can group players according to different behaviors or simply check if a student shows a specific one. For example, Dicerbo [11] used evidence extracted from log files to create a measure of persistence. Similarly, Ventura & Shute [44] also created a measure of persistence, validating it against another existing measure and concluding that the GBA predicted students' learning.
3. `Assessment`: in these studies, games are used to report measures that aim to evaluate students. This allows for improvements in the learning process using this evaluation measure instead of classic evaluation methods or providing personalized feedback. In their work, Weiner & Sanchez [45] created an alternative measure using a virtual reality game that calculated scores to indicate specific cognitive abilities.
4. `Interventions`: games can also be used to investigate the effect of some interventions while playing. For example, we can use feedback messages to notify the learner with positive (or negative) feedback to observe how this intervention influences its performance and behavior. Another typical example is switching the order of in-game elements or testing different game features. In [46], the authors used a psychophysiological methodology to investigate attention allocation to different feedback valences (i.e., positive and negative feedback). With that purpose in mind, they used an eye tracker to collect accurate information about individuals' locus of attention when they process feedback.
5. `Framework proposal`: in these papers, the authors propose the design of a new framework to be used within the context of GBA. We can see an example in [47], where the authors examined the process of creating a Bayesian network framework through different techniques (e.g., using correlation matrixes, IRT) to create scoring models for assessing students.
6. `Game design proposal`: authors provide a game design that can be used for assessment purposes. For example, the authors in [48] show the design of an online GBA to help students improve their learning outcomes and promote the development of general and transferable skills, such as the ability to solve problems in complex situations, and working under pressure.

Some studies focused on more than one of the categories described above. For example, Weiner & Sanchez [45] used a virtual reality game to calculate a score measure for each student (`assessment`) and they proved that these calculated scores are best used by comparing them to classic measures (`GBA evaluation`).

We can see the number of papers fitting each category in Figure 6. `GBA evaluation` is the most common category (38 studies, 58.5%), followed by `assessment` (34 studies, 52.3%) and `framework proposal` (12 studies, 18.5%). The less common category is `game design proposal`, with only three papers fitting (4.6%). We can conclude that most papers focused on using games to assess learning, but they also tried to prove that this assessment was a valid measure to be used in real educational contexts.

3.3 What is the domain of the GBA? (RQ3)

From reviewing the selected papers, we identify four major domain categories: `STEM`, `humanities and social sciences`, `cognitive and soft skills`, and `physiological capacities`. As some of the categories have more than one related area, we also consider some sub-categories fitting into them. We describe each domain category below in detail:

1. `STEM`: in this category, we include papers that are related to `science`, `technology`, `engineering` and `mathematics`. For example, Chiu & Hsieh [49] showed the different teaching methods of second-grade elementary students in fraction concepts (`mathematics`), while Kim et al. [26] aimed to assess the understanding of Newton's three laws of physics using a two-dimensional physics game.
2. `Humanities and social sciences`: papers related to humanities and social science areas (e.g., art, music, language) fit in this category. As this is a wide area, we have also defined some sub-categories to better categorize the papers. These sub-categories are `language`, `art` and `history`. Studies that do not fit into one of those three categories are categorized as `other`. As an example of the `art` category, we highlight the work in [50], where the authors used a game in which players collect data about the musical interests of an in-game character and use these data to make decisions about which artists to sign and what songs to record. We can see another example (related to `language`) in [51], where the researchers described the design of an argumentative reasoning task within a scenario-based assessment enhanced with game elements.
3. `Cognitive and soft skills`: cognitive skills are the core skills your brain uses to think, read, learn, remember, reason, and pay attention [52]. Cognitive skills help to process new information by taking that

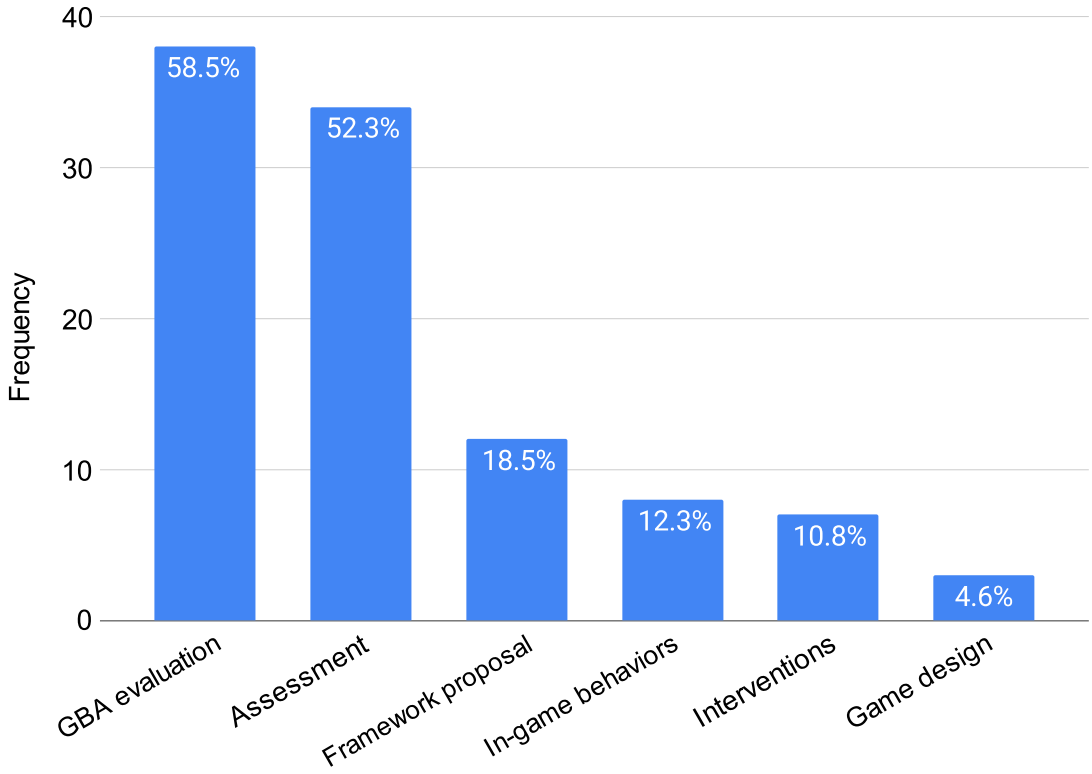


Figure 6: Main purpose of each research. More than one purpose is possible for each paper.

information and distributing it into the appropriate areas in the brain. Developing cognitive skills helps to complete this process more quickly and efficiently, helping people to understand and effectively process new information [53]. Moreover, soft skills are described as a combination of interpersonal and social skills, including the ability to communicate, coordinate, work under pressure, and solve problems [54]. In this category, we consider `attention`, `memory`, `logic and reasoning`, `visual processing and speed`, and `soft skills`. We find papers that have measured interesting skills, such as [55], which included a series of reasoning activities to measure argumentation skills (which is related to `logic and reasoning`), or [56, 57], which used GBAs to assess candidates' soft skills.

4. `Physiological capacities`: physiological functional capacity is the ability to perform the physical tasks of daily life and the ease with which these tasks can be performed. We could assess daily physical tasks, like Rodríguez de Pablo et al. [58], who used a set of games to provide a fast, quantitative and automatic evaluation of the arm movement function. Furthermore, other works focused on assessing mental abilities, such as motivating children with autism to make more eye contact [59].

There are also papers fitting more than one category at once. For example, in [60, 61], researchers used a GBA for measuring argumentation and pragmatic skills. This research measured language competencies (which is part of `humanities and social sciences`), but it also measured `cognitive and soft skills`. We can see the full tree showing the distribution of studies into categories and sub-categories in Figure 7.

The three predominant categories are `cognitive and soft skills` (28 studies, 43.1%), `STEM` (19 studies, 29.2%), and `humanities and social sciences` (17 studies, 26.2%). Taking a look at each sub-category, we note that the main area in `STEM` is `science` (10 studies, 15.4%). The main field in `cognitive and soft skills` is `logic and reasoning`, with 17 papers (26.2%), while in `humanities and social sciences`, the predominant sub-category is `language`, with nine papers (13.8%).

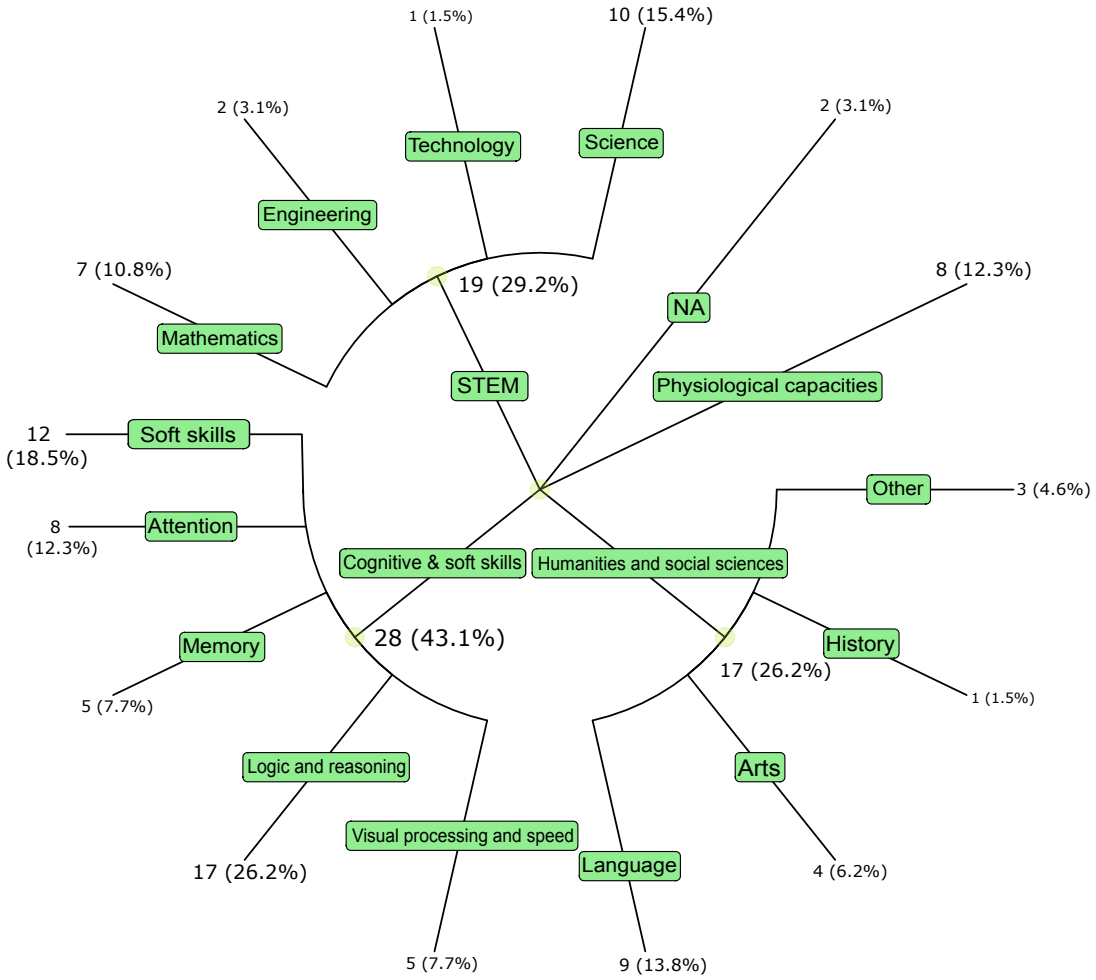


Figure 7: Category tree for RQ3.

3.4 Is the game/tool used available to the public? (RQ4)

A critical aspect of research is the availability of the results obtained to be used by the general public. It is essential to make tools accessible so that researchers can replicate experiments and practitioners can use them as part of their teaching. From our analysis, we find three primary categories: `Currently available`, `Not available (NA)` and `Not specified`.

1. `Currently available`: the game/tool used in the corresponding research is currently available (using the web portal specified by the authors) for public use (e.g., [62, 63]).

2. `Not available`: the game/tool used in the research was presented as initially available in the paper, but currently, it is no longer accessible based on our attempt to access the site (e.g., [64, 65, 66]).

3. `Not specified`: researchers did not specify the tool's availability; it is more than likely that it is not accessible (e.g., [67, 68, 69]).

Figure 8 shows that most papers (51 studies, 78.5%) did not specify if the tool is accessible or not. Another minority of papers (9 studies, 13.8%) offered their tool publicly. The rest of the studies (five papers, 7.7%) initially offered their games, but they are currently unavailable. In addition, we did not find any open-source game across the studies included in the review.

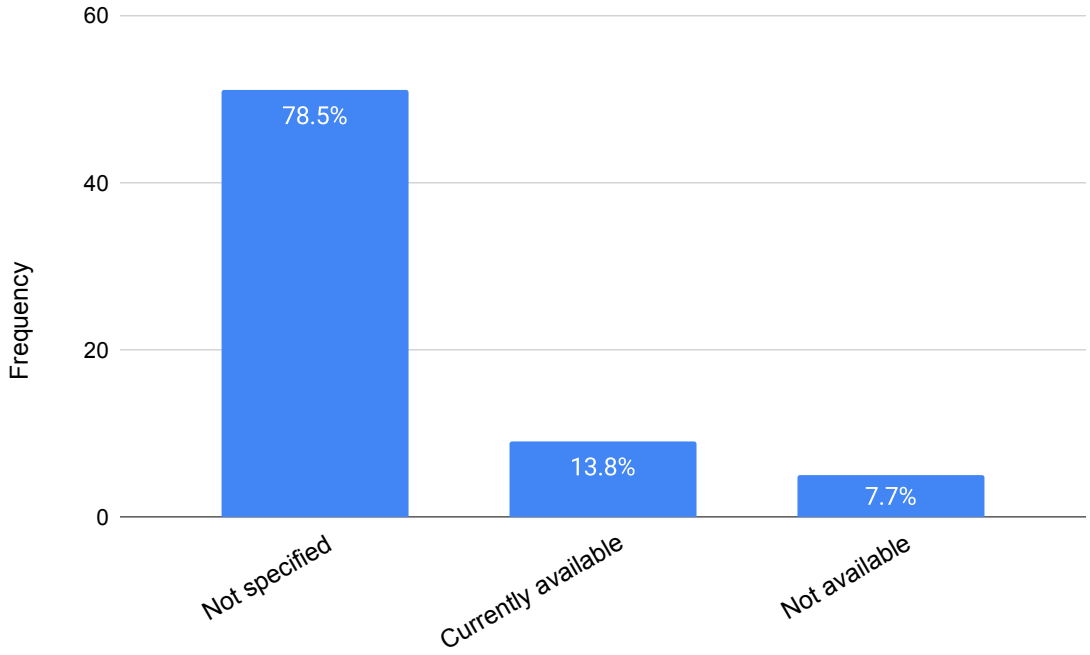


Figure 8: Number of papers based on their availability.

3.5 What is the size of the data sample used in the study? (RQ5)

In this research question, we classify the different data collections used in the studies based on their data sample size. Investigating the sample size is relevant since the use of larger data samples will allow better generalization of the research results, as well as the possibility of applying more complex algorithms (e.g., neural networks), which often require large amounts of data to outperform other models [17]. Although the sample size can be relevant for some aspects, such as preventing overfitting in some methods, it is not related to the study's rigor (i.e., using a larger sample does not make a study more rigorous). From the coding process, we present four categories:

1. `Fewer than 50 participants`: these papers involved fewer than 50 participants in their empirical studies. We find studies with small data samples, such as [70], using data from 30 postgraduate students, or [71], which used a sample of 20 healthy controls patients and 18 patients with Alzheimer's disease to evaluate the usability of a tool created to assess cognitive functions.
2. `Between 50 and 250 participants`: these papers involved between 50 and 250 participants in their studies. For example, Leonardou et al. [72] used data from 77 primary school pupils for assessing and improving multiplication skills. We see another example in [73], which used data from 95 children from the final year in preschool to measure psychoacoustic thresholds.
3. `Between 250 and 500 participants`: these papers involved between 250 and 500 participants in their studies. For example, Gjicali et al. [74] used data from 433 students who played a game simulating an artificial culture with norms embodying two cultural concepts: hierarchy and collectivism.
4. `More than 500 participants`: these papers used data from more than 500 participants in their research. Hautala et al. [75] used data from 723 students to investigate reading difficulties, concluding that the GBA could be successfully used to identify students with reading difficulties with acceptable reliability (Cronbach's alpha 0.93 and 0.87). Some other studies used a huge sample, such as [12], which used data from 5,545 students to measure engagement and cluster students to finally report four different engagement profiles.

Other papers (e.g., [76, 77]) did not specify the data sample size of the study and we categorize them as `NA`. Figure 9 summarizes the results of the data sizes across papers. We can see that only 16 papers (24.6%) used more than 250 participants in their students, and only nine papers (13.8%) used data from more than 500 students, meanwhile most of the papers (58.5%) used data from fewer than 250 participants. We also see that a significant amount of papers (11 studies, 16.9%) did not specify the data sample size in their studies.

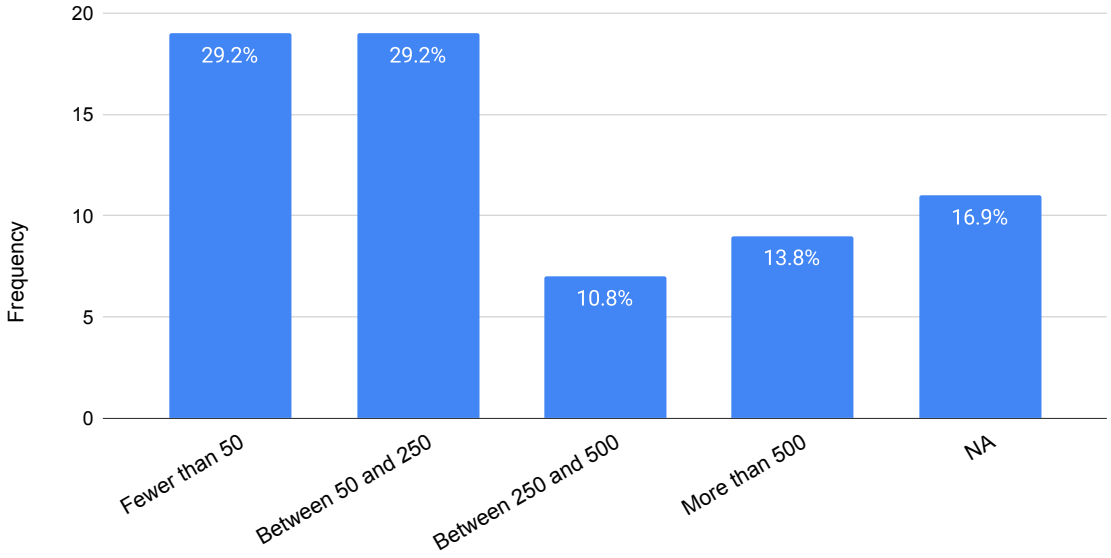


Figure 9: Paper distribution based on the number of participants involved in empirical studies.

3.6 What computational methods and algorithms have been applied in the research? (RQ6)

After exploring the data samples that were retrieved across papers, our goal was to examine the methods that were applied for its analysis. We believe that the methods being applied are crucial, since they are the link between the evidence generated by learners and the assessment. Accordingly, we identified five different groups of methods for analyzing the data: `Descriptive statistics`, `Machine learning`, `Knowledge inference`, `Deep learning`, and `Sequence mining`. Below is a description of each group in detail:

1. `Descriptive statistics`: these encompass further mathematical analyses covering various methods, tests, and visualizations. We identified several papers that applied `summary statistics` (e.g., mean, variances) [78], `correlations` [79] and `visualizations` [80].
2. `Machine learning`: it is a part of AI and covers a set of methods that allow systems to learn and improve from historical data automatically. We noted that the authors used two significant families of machine learning methods: `supervised learning` and `unsupervised learning`. `Supervised learning` includes techniques such as regression [81] while `unsupervised learning` uses other methods, such as clustering techniques like *k*-means [12] or dimensionality reduction techniques like Principal Component Analysis (PCA) [82].

   For example, the authors in [83] developed a game for evaluating the logic abilities of first-year university students. They tried to compare the measures obtained by paper-based tests with those obtained using the game by conducting a linear regression (which is a supervised method). The authors concluded that the measures obtained from both methods were not significantly different.
3. `Knowledge inference`: it refers to the acquisition of new knowledge from existing facts based on certain rules and constraints. One way of representing these rules and constraints is through the use of logic rules, formally known as knowledge representation [84]. Common knowledge inference methods that several studies have used are `Bayesian networks` [85] and `fuzzy cognitive maps` [67].

   In [86], the researchers proposed a dynamic Bayesian network modeling approach for measuring student performance from an educational video game. The results supported the usefulness of Bayesian networks to characterize and accumulate evidence regarding students in games and related assessment environments.
4. `Deep learning`: an artificial intelligence function that imitates the workings of the human brain in processing data and creating patterns for decision-making [87]. An example is the work of Chen et al. [88], who used Long Short-Term Memory (LSTM), an artificial recurrent neural network architecture.
5. `Sequence mining`: the objective of sequence mining is to unlock useful knowledge hidden in sequence data [89]. Specifically, Gomez et al. used [90] sequence mining to identify sequences and errors by transforming raw data into meaningful sequences that are interpretable and actionable for teachers.

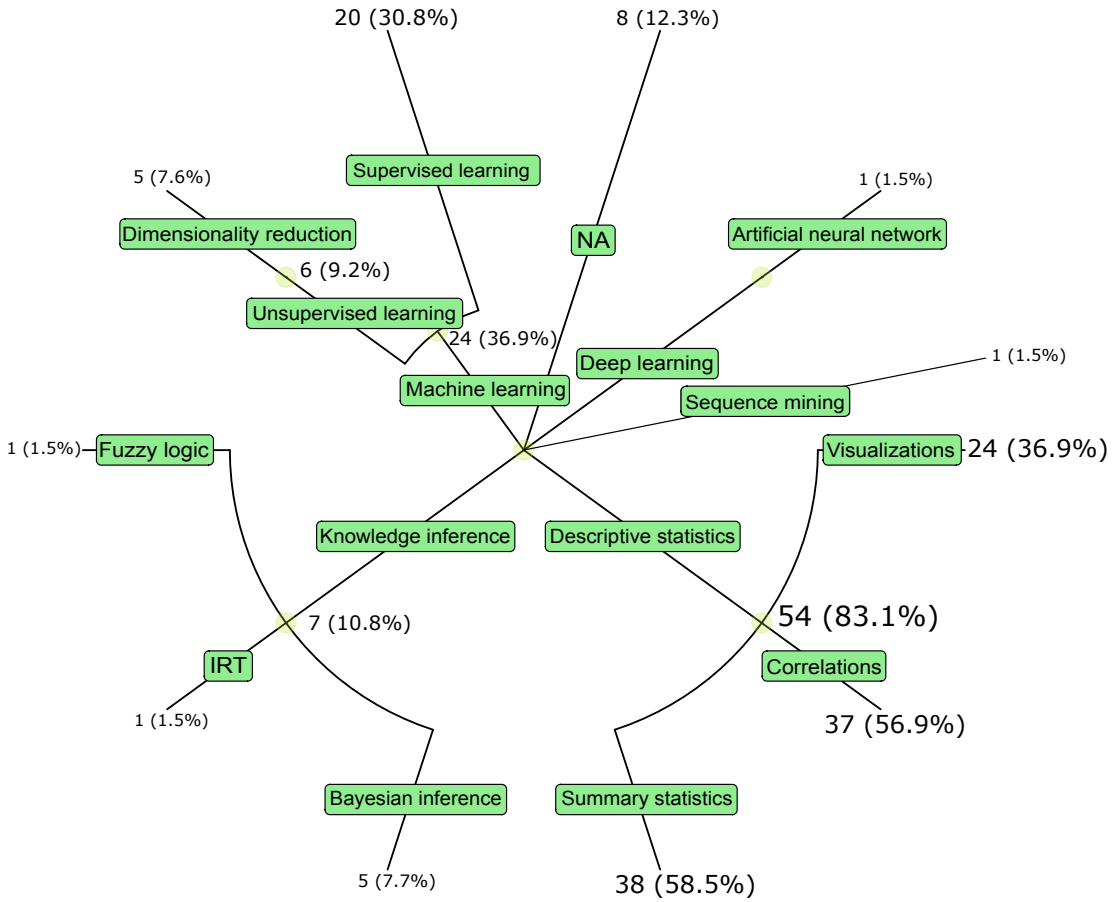


Figure 10: Paper distribution based on methods used.

In Figure 10 we can see the different families of techniques and the number of papers that used them in their research. We can see that most papers (83.1%) used `descriptive statistics`, and almost none of them used `deep learning` (only one paper, 1.5%). We also noted that 83.3% of the papers that used `machine learning` techniques used `supervised learning` too, specifically, most of them used different types of regressions.

3.7 What stakeholder is the intended recipient of the research results? (RQ7)

A stakeholder is defined as a person with an interest or concern in something, especially a business [91]. In our study, we consider the paper's stakeholder as the person to whom the results are directed, even though the paper's contribution might have other secondary stakeholders. Specifically, we have two main groups of stakeholders: `researchers` and the `final user`.

1. `Researchers`: if the paper's contribution is methodological, we expect that the paper's main stakeholders will be researchers. For example, Lonergan et al. [92] created a Paper-based Assessment (PBA) and a GBA in order to measure students' performance, cognitive states and satisfaction related to both assessment methods. The authors concluded that smaller versatile GBAs may have a greater impact on the student's cognitive capabilities, and could enhance student performances during, for example, a final exam or short formative assessments. Moreover, Tsai et al. [93] proposed an online learning system using different gaming modes of classic tic-tac-toe games to explore how different gaming modes and feedback types in this game-based formative assessment affect knowledge acquisition effectiveness and perceptions of participation.
2. `Final user`: if the paper's results are to be used by final users or are validating the GBA, we consider that the main stakeholder will be the final user in that context (e.g., teachers and students). In their work, Ciman et al. [94] designed a game to support children with cerebral visual impairment, developing a mobile version of the game to be used by children easily at home on any platform. Delcker & Ifenthaler [95] also developed a

mobile app that makes an automated analysis of the data and provides information about children's language skills. Other papers focused on teachers, such as [96], where the authors used a GBA to develop a set of visualizations to support teachers in classrooms.

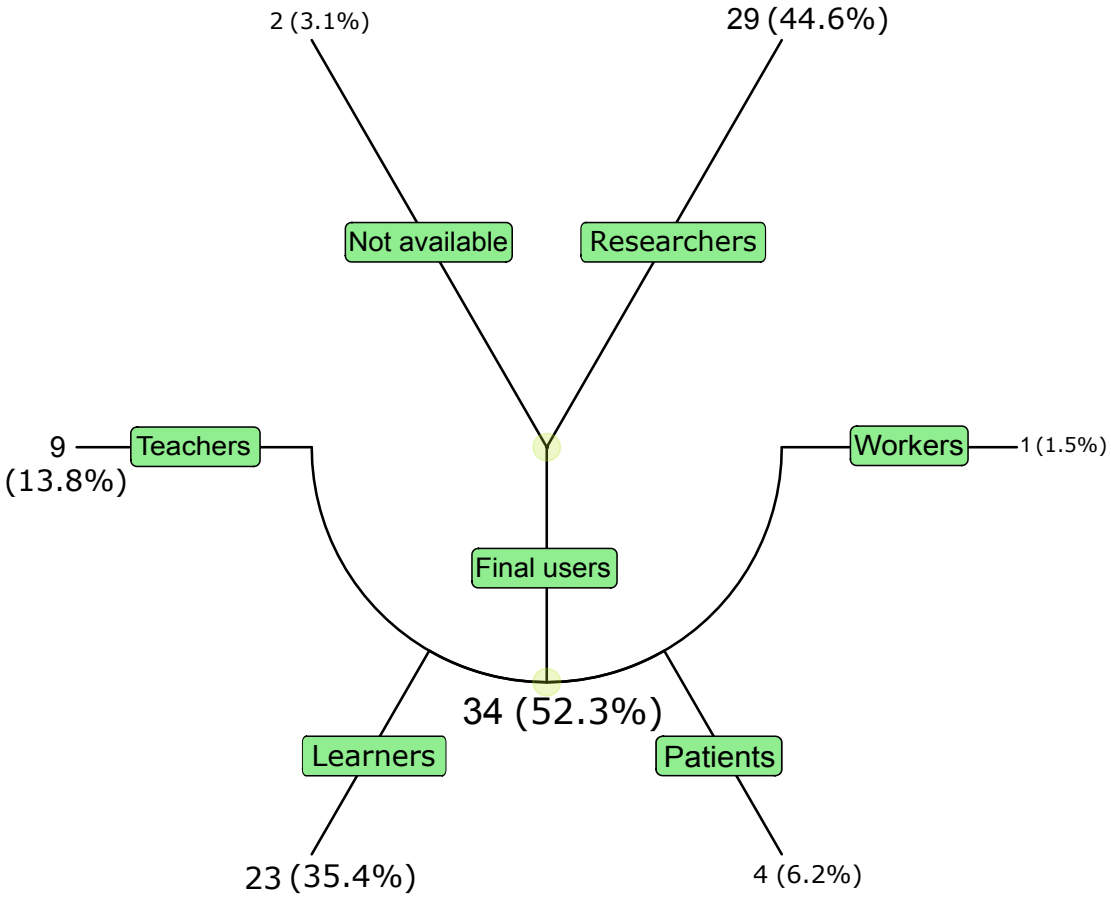


Figure 11: Paper distribution tree based on the main stakeholder.

Figure 11 shows the number of studies that focused their work on each of the stakeholders. We see that 34 studies (52.3%) were directed to `final users`, mainly students. Moreover, 29 studies (44.6%) focused on `researchers` as the main result recipients. Two further studies (e.g., [97]) did not provide results and we categorized them as `NA`. Focusing on studies directed to the final user, we see that the majority of papers are directed to `learners` (35.4%) and `teachers` (13.8%).

3.8 What limitations and challenges do the authors address? (RQ8)

Limitations show potential weak points of the study that researchers usually highlight regarding their work such as constraints in research design or methodology. We can group the limitations that the authors faced in the six following categories: `game design`, `data sample`, `methodological`, `technical`, `integration` and `validation`.

1. `Game design`: an appropriate design of a game is crucial for learners' assessment since the GBA design must be adapted based on the constructs that will be evaluated. It requires a great effort to design a good GBA, aligning the evidence collected with the final purpose of the assessment. Designers might have different goals when developing a GBA [14]: "for game design, engagement; for instructional design, developing key concepts and capabilities in the target domain; for assessment design, evoking evidence of those capabilities for the intended use case(s)." Moreover, many game design decisions play a role in what kind of game performance is achieved and its meaning [98]. Future designers should consider these concerns to achieve better designs, thus, creating more engagement and facilitating the development of the key concepts and capabilities intended for learning.

2. `Data sample`: data were crucial for our review because GBA is based on the evidence, stored as data, generated by the students' interaction with the games. We examined each paper and found several limitations related to data. Jackson et al. [99] reported that they had a small sample size and that larger sample sizes would be necessary to detect smaller effects. We see a similar example in [100], where the authors had a sample collection of 67 students, but only four of those 67 student samples were used in their empirical study. The work in [101] described the difficulty of designing a good data model, as there are usually conflicts between programmers and assessment designers, usually complicated by constraints related to budgets and schedules.

3. `Methodological`: this category includes challenges and limitations related to the methods, algorithms, or techniques used. For example, Yu et al. [102] wanted to collect additional data to explore learners' behavioral patterns during gameplay. We see another example in [103] since the authors reported that the assessment developed in this study only includes a part of number sense (this term refers to a group of key math abilities), and, in order to complete the number-sense battery, the assessment tools for the other components of number sense are needed to be developed.

4. `Technical`: it is defined as a challenge involving how a machine or system works. This could include storage limitations, computing power, or even limitations related to sensors used in the study. In [104], the authors pointed out the necessity of a database (to store information about students' achievements), since they could not store that information, as well as the necessity of an administrator module to facilitate developing and modifying game elements.

5. `Integration`: incorporating game activities as part of the curriculum in schools remains limited due to certain factors such as the schools' budget or the rigidity of subjects' classic curriculum. Halverson & Owen [64] claimed that if GBAs can show that games can serve as assessments that generate reliable evidence, we could then legitimize the potential of games and then break the social conventions that limit the potential of learning and assessment technologies in schools.

6. `Validation`: one of the most significant parts of the research is the validation of the results. Validation is intended to ensure that the proposed methods and the accomplished results proved satisfactory by conducting empirical experiments. Sanchez & Langer [105] suggested that the games used in their study were entertainment games, and further research could be oriented to validate their results with games designed for assessment purposes.

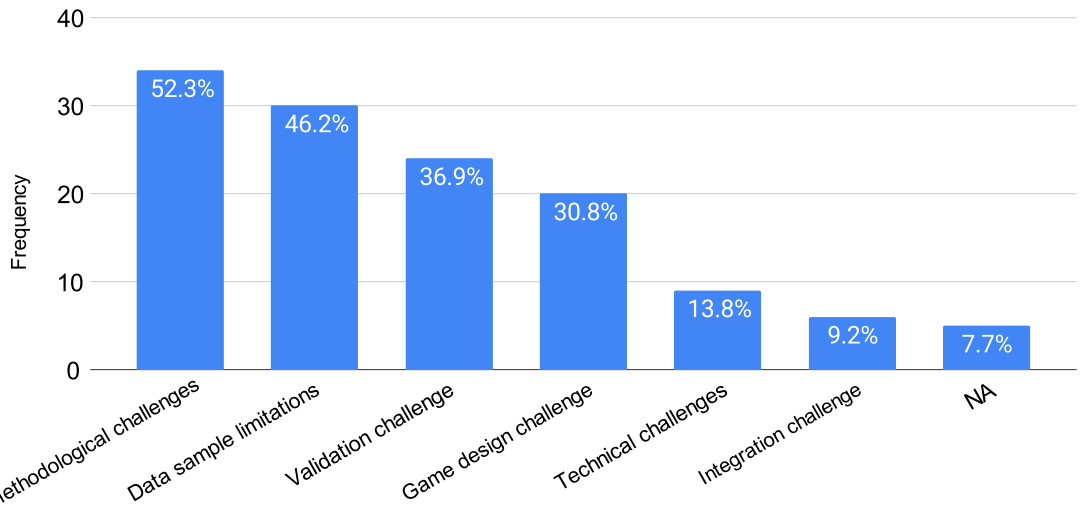


Figure 12: Number of papers based on their limitations and challenges.

Some other studies did not report any challenges or limitations (e.g., [106]). Figure 12 shows that `methodological` challenges are the most common ones (34 studies, 52.3%), followed by `data sample` limitations (30 studies, 46.2%). On the other hand, the least frequent limitations are related to `integration` (6 studies, 9.2%) and `technical challenges` (9 studies, 13.8%).

Research question	Categories	Count	%
Context (RQ1)	Formal education	41	63.1%
	Medical	8	12.3%
	Workplace	5	7.7%
	Not Available	11	16.9%
Main Purpose (RQ2)	GBA evaluation	38	58.5%
	In-game behaviors	8	12.3%
	Assessment	34	52.3%
	Interventions	7	10.8%
	Framework proposal	12	18.5%
	Game design proposal	3	4.6%
Domain (RQ3)	STEM	19	29.2%
	Cognitive and soft skills	28	43.1%
	Humanities and social sciences	17	26.2%
	Physiological capacities	8	12.3%
	Not Available	2	3.1%
Availability (RQ4)	Not specified	51	78.5%
	Currently available	9	13.8%
	Not available	5	7.7%
Sample Size (RQ5)	Fewer than 50 participants	19	29.2%
	Between 50 and 250 participants	19	29.2%
	Between 250 and 500 participants	7	10.8%
	More than 500 participants	9	13.8%
	Not available	11	16.9%
Algorithms/ techniques (RQ6)	Descriptive statistics	54	83.1%
	Machine learning	24	36.9%
	Deep learning	1	1.5%
	Sequence mining	1	1.5%
	Knowledge inference	7	10.8%
	Not available	8	12.3%
Stakeholder (RQ7)	Researchers	29	44.6%
	Final user	34	52.3%
	Not available	2	3.1%
Limitations (RQ8)	Technical	9	13.8%
	Game design	20	30.8%
	Data sample	30	46.2%
	Methodological	34	52.3%
	Integration	6	9.2%
	Validation	24	36.9%
	Not available	5	7.7%

Table 1: Summary of the main findings.

4 Discussion

In this section, we first present a summary and discussion of our main findings. We can also see a summary of these main findings in Table 1. Next, we present a discussion of past and future challenges regarding games for assessment. Finally, we address existing limitations in this study and the implications of our research.

4.1 Current trends

First of all, we analyzed the contexts where the studies were applied (RQ1), finding that most of them took place in `K-16 education`, especially in `high school and middle school`. This is an exciting finding because young kids and teenagers represent the major force whose 21st-century competencies development will be heavily impacted by technology [16, 107]. Moreover, children and adolescents are an ideal target since the familiarity of these users with gaming environments and game mechanics facilitates their interactions with games [17].

Regarding RQ2, we found that the majority of GBA studies focused on students' `assessment` and the `validation` of the game or the tool used. This suggests that having established that games are helpful for other purposes beyond

entertainment, there is an increasing interest in using games as a natural alternative to classic evaluation methods, validating and comparing them against those traditional alternatives. Moreover, the fact that researchers also focused on the validation of the GBA used is a promising finding. Specifically, Gris and Bengtson [18] pointed out the lack of evidence about engagement and usability needs, especially with well-assessed reliability and validity. We also noticed that few studies had the main purpose of proposing or validating a `game design` for assessment. Although many studies proved that GBA could improve students' learning outcomes, we should not forget game design. The literature reveals that game design is essential, and several distinctive design elements, such as narrative context, rules, goals, rewards, multi-sensory cues, and interactivity, seem necessary to stimulate the desired outcomes [108, 109].

We also extracted four predominant domains (RQ3) across studies. A large proportion of the analyzed studies aimed at practicing and assessing content related to `STEM` as well as `humanities and social sciences`. This is not surprising since many of the studies took place in schools and high schools, and the use of games in these contexts is an ideal opportunity to teach content related to the main subjects at those ages. Another large number of papers also focused on developing and measuring `cognitive and soft skills`. Using game design as a context to teach higher-order thinking skills has drawn attention from researchers since schools usually place heavy emphasis on covering and delivering content knowledge [110]. Moreover, this could be useful not only in educational contexts, as we have seen some studies that measure cognitive skills for `medical` purposes (e.g., rehabilitation) [71, 68]. However, the researchers in [111] pointed out the lack of research on 21st-century skills such as creativity and critical thinking.

We discovered that many of the studies had small data samples (RQ5). Furthermore, a significant part of the studies did not specify the data sample size used in the experiment. This is also noticed by the researchers themselves, as nearly half of the studies reported data sample limitations. Moreover, apart from collecting the sample size, we also tried to collect information about the type of data collected. However, almost no study included information related to the type or format of the data used.

Across papers, researchers used many different algorithms and techniques (RQ6) to analyze the data. We classified them into five categories and found that the most common ones are `descriptive statistics` and `machine learning`. With that said, we note that the majority of papers used statistical analyses or basic machine learning algorithms, and few studies used more complex or advanced methods, which might be more adequate to model students' knowledge properly. However, those techniques that are easier to implement are also the ones chosen more frequently by researchers. Therefore, more work is needed to develop specialized GBA methods, that are also affordable to implement. This could perhaps be done through open-source libraries and more reproducible research. Moreover, we consider that making results interpretable is an essential part of the assessment, and one way to reach this interpretability is by using `visualizations`. Visualizations are essential components of research presentation and communication because of their ability to represent large amounts of data [112] and because it is easier for the brain to comprehend an image as opposed to words or numbers [113]. We think this is a promising way to integrate games in schools, and we realized that studies now tend to use visualizations to communicate their results (e.g., [62, 80]).

Finally, we wish to report the scarce information regarding games and tools availability (RQ4). The majority of studies did not provide any information on how to access or use their tools. In addition, some studies made tools public but expired, being inaccessible at present. This underscores the low transference of this research to practice and, thus, we encourage authors to make their products and results publicly accessible since we consider that this is an essential part of this type of research.

4.2 Open challenges

From our results and previous related reviews, we find some open challenges in the area that authors usually report. A description of each of these challenges is found below:

In [114], the authors address the challenge of how to make appropriate assessments. They noted that pre-test and post-test measures are a good manner to make an assessment, and they also recommended unobtrusive ways to collect data, such as another person taking notes during game-play. In our review, we noted that, at present, most studies found an appropriate method to make good assessments using Evidence-centered Design (ECD) and stealth assessment. ECD framework views assessment as an evidentiary argument, that is, an argument from which we observe what students say, do, or make in a few particular circumstances [115]. Moreover, stealth assessment represents a unobtrusive, yet powerful process by which learner performance data are continuously gathered during playing and learning, and inferences are made about the level of relevant competencies, maintaining the learners' flow and engagement [116]. Since ECD and stealth assessment are two common practices in current research, we could claim that the objective of making appropriate assessments using unobtrusive methods has been accomplished.

What data are going to be collected is as important as how to collect these data, and another present challenge is the design of games for specific assessment purposes. Akcaoglu & Koehler [110] indicated that games that present a hidden

questionnaire format do not engage learners, while well-designed games can engage learners in reflective thinking [117]. Although we identified a few papers with the main objective of providing a good game design, many of them have developed an excellent game for other purposes. Some examples are [12, 71], and [73]. Future research should focus on complex game designs rather than the typical simple quiz design, employing multiple game-design elements such as collaboration, role-playing, narrative, exploration and complexity [16].

An important open challenge at present is replication and transferring the research to practice. In addition to the findings in our literature review about the game or tool being unavailable in most cases, All et al. [118] mentioned replication issues with certain studies due to missing information in multiple areas of the study. It is crucial to provide a detailed description of the procedure followed to conduct the study. While the community is currently demanding more standardized open science practices, this problem is still present currently. Besides, Alonso-Fernandez et al. [17] noted that most papers did not describe the format in which they collected the data, so we cannot know if they used a standard or relied on their data formats, which represent even more replication and reusability issues. In addition, having open-source games or tools would be especially helpful for researchers. Unfortunately, we did not find any available open-source games across the studies. This problem of missing information is a familiar issue in multiple research fields (nearly every field is affected), and it leads to other problems such as low reproducibility. In fact, the terms "reproducibility crisis" and "replication crisis" have gained significant popularity over the last decade [119]. To fix this issue, the community is demanding more pre-registered studies, open data, open analyses, and open access publications [120], and this can be systematized by the guidelines of the publishers, governments and research communities [121].

Regarding the methods and techniques, we identify the challenge of implementing learner modeling algorithms. As we mentioned above, researchers usually use simple techniques to conduct their studies. In addition to Alonso-Fernandez et al. [17] noting that limitation, we confirmed it in our results. In our review, 52.3% of the papers reported methodological challenges to be addressed in future research, most of them related to the use of more complex metrics and techniques to infer new information. It is important to benefit from more advanced techniques (e.g., knowledge inference techniques, deep learning techniques) that can allow us to infer more complex and valuable information from the data collected. However, an important limitation of many of those advanced techniques is their low interpretability. Even if visualizations are a promising way to improve the presentation of results and communication, they cannot improve the model's interpretability themselves. According to the researchers in [122], with machine learning models being increasingly used, there has been an interest in developing interpretable models. However, there have been relatively few experimental studies investigating whether these models achieve their intended effects. Thus, the development of new models to provide better interpretability in GBA environments and their validation is still an open challenge.

We found several studies that described data sample and validation challenges. Since most evaluations are conducted with small samples, typically corresponding to one classroom's size, these studies present low statistical power, having a reduced chance of detecting actual effects [123]. Thus, studies must use larger data samples to improve the results' generalization and validity. However, collecting large samples of in-context data is also a cumbersome task. Finally, a few empirical studies discussed the challenge of implementing GBA in the classroom, but this is a significant problem. Many teachers are still unsure about how to integrate game activities with the regular curriculum, and it is crucial to provide guidelines that can facilitate teachers to deploy games in the classroom more easily and flexibly [124].

4.3 Limitations and implications

This review is mainly limited by the paper selection. First of all, we have only used the key term "game-based assessment" to perform our document search, based on the papers' keywords and titles. However, other communities could also be working on games for assessment purposes, but they might be using slightly different terms to describe their work. Therefore, those studies might not be included in our review. Nevertheless, we purposely opted for this term to analyze the core of GBA while also having a manageable selection of papers for this study. Furthermore, we focused our attention on Scopus and the Web of Science, the two primary academic databases. However, there could be other peer-reviewed academic papers indexed in different databases, as well as non-peer-reviewed publications including pre-prints, technical or white reports that could be missing in our review, and also non-academic work being conducted in industrial companies and by practitioners. Regarding the computational methods and algorithms used, we have identified a set of categories based on the qualitative review of each selected paper. However, there might be studies using less quantitative approaches that might be missing in this review due to the review methodology itself. Finally, we have based our RQ generation on a simplified process that involves the general steps required in GBA projects, but there might be other potential and valuable RQs about the GBA field missing in this review.

We found that most studies emphasized GBA implementation and comparisons between games and classic assessment methods. More studies are needed to systematically develop and improve game design, adopting design-based research methods, as mentioned in [125]. The potential of GBA is now emerging, coinciding with the rise of big data. Data mining and visualization techniques on player interaction logs can provide different stakeholders with valuable insights

into how players interact with the game [126]. The increasing interest in games as a learning tool also indicates their potential as actual assessment tools. In our review, we found that GBAs are not only being used in K-16 education but also in medical and professional areas, among others. As expected, the most frequent area where GBAs are being applied is `K-16 education` since children and adolescents are the leading groups whose development will be affected by technology.

Despite this dominating use in education, we can see the great potential that GBAs have in many other contexts. Concerning the professional environment, companies have begun to include assessment games for the recruitment of staff and the selection process. This is a relatively new trend due to certain limitations, such as the cross-domain generalizability of behaviors between game and workforce contexts, which needs further research [40]. In medical environments, the use of GBAs can also be helpful for multiple purposes. Some examples are the possibility to recreate a virtual environment with daily life activities, allowing a precise and complete cognitive evaluation, which can be useful to treat certain diseases such as Alzheimer's [71] or using games to rehabilitate children with cerebral visual impairment using an eye-tracker [94]. Due to the above, we firmly believe that the future of games for assessment is promising; however, further research is needed to overcome the existing problems, and increase the still limited application of games in real-life environments, in order to start building the classrooms of the future.

5 Conclusions

Technology is changing and improving every day, and this is also making a significant impact on educational areas. Moreover, playing games is one of the most popular activities in the world, and the technological revolution that we are experiencing allows the implementation of games as alternative assessment tools in educational environments. However, previous studies suggest that the use of games also presents some challenges, such as finding the time for both the presenter/instructor and student to learn the systems employed, the financial impact on both parties, and technical limitations [1, 6]. We can tackle all these challenges by facing current limitations and revealing the great potential games have for assessment. This study represents a novel analysis and the first literature review of the emerging research field of GBA. Its main purpose was to review empirical studies of digital GBAs published until 2020. Based on a detailed systematic review of the 65 selected papers, we concluded that games are mainly used in K-16 education for assessment and validation purposes. The domain of the games used is usually related to STEM and cognitive skills, but other domains emerged from our analysis, such as social sciences and physiological capacities. Moreover, we note that, although few GBA studies had the purpose of proposing an adequate game design for assessment, most studies used games designed specifically for assessment purposes, employing complex game-design elements such as collaboration, narrative, or role-playing. In addition, we found that most of the studies used small data samples and simple techniques to process these data and assess students. Finally, we found that most of the studies do not provide public access to their tools, or they overlook links and let them expire over time, which makes it impossible to reproduce the results or even try their game.

Future work should address the current challenges emerging from our review, as those are the main barriers to actual systematic adoption of games for assessment. For example, the next generation of GBA studies should ensure that enough data is collected to have meaningful and reliable results since one of the main limitations of the current research was the size of the data sample collected. Moreover, they should also address the game design that will be used for assessment, as many studies use games designed for other purposes (e.g., entertainment) and overlook the vital link between the design of a game and collecting the necessary evidence for the assessment. In that sense, it would be good to work on conceptual GBA pieces or frameworks that can provide a set of guidelines for the design. Moreover, classic performance indicators such as completion times or scores could still be included in future studies, but GBA also needs to apply more specific and complex algorithms (e.g. knowledge inference or deep learning techniques) specifically designed for learner modeling and assessment purposes. The use of more complex techniques, along with larger data samples, could substantially improve the reliability and generalization of the results. We also believe that future studies should continue exploring the use of visualizations and dashboards to integrate games in schools, adopting a more intuitive approach rather than providing teachers with raw numerical outputs or metrics, which are usually harder to understand. Teachers should also have a more important role in future work to address digital and assessment literacy issues, as well as the potential interpretability and actionability of GBAs. Finally, there are no theoretical frameworks within the GBA area (a related one regarding serious games could be [127]). Considering this lack of theoretical papers focused on describing GBA foundations, we believe that future work should address publications with additional content on the theoretical side.

Therefore, further research is needed to overcome current limitations and to continue exploring the possibilities of games as assessment tools in other contexts and environments. Finally, we encourage authors to document their research in a reproducible and verifiable way, using beneficial open science practices by pre-registering their study, sharing data

and code for replication purposes, and if possible open sourcing the GBA tools with clear descriptions so that they can be used by interested stakeholders and researchers.

References

[1] Lea S Eiland and Timothy J Todd. Considerations when incorporating technology into classroom and experiential teaching. *The Journal of Pediatric Pharmacology and Therapeutics*, 24(4):270–275, 2019.

[2] Sara De Freitas. Are games effective learning tools? a review of educational games. *Journal of Educational Technology & Society*, 21(2):74–84, 2018.

[3] ESA. 2020 essential facts about the computer and video game industry. Technical report, Entertainment Software Association, 2020.

[4] ISFE. Isfe key facts 2020. Technical report, ISFE, 2020.

[5] Stamatios Papadakis. The use of computer games in classroom environment. *International Journal of Teaching and Case Studies*, 9(1):1–25, 2018.

[6] Sebastiaan de Klerk and Pamela M Kato. The future value of serious games for assessment: Where do we go now? *Journal of Applied Testing Technology*, 18(S1):32–37, 2017.

[7] Doug Clow. An overview of learning analytics. *Teaching in Higher Education*, 18(6):683–695, 2013.

[8] Cristobal Romero and Sebastian Ventura. Data mining in education. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 3(1):12–27, 2013.

[9] Yoon J Kim and Valerie J Shute. Opportunities and challenges in assessing and supporting creativity in video games. In *Video games and creativity*, pages 99–117. Elsevier, 2015.

[10] Douglas B Clark, Emily E Tanner-Smith, and Stephen S Killingsworth. Digital games, design, and learning: A systematic review and meta-analysis. *Review of educational research*, 86(1):79–122, 2016.

[11] K.E. Dicerbo. Game-based assessment of persistence. *Educational Technology and Society*, 17(1):17–28, 2013.

[12] J.A. Ruiperez-Valiente, M. Gaydos, L. Rosenheck, Y.J. Kim, and E. Klopfer. Patterns of engagement in an educational massively multiplayer online game: A multidimensional view. *IEEE Transactions on Learning Technologies*, 13(4):648–661, 2020.

[13] Dirk Ifenthaler, Deniz Eseryel, and Xun Ge. Assessment for game-based learning. In *Assessment in game-based learning*, pages 1–8. Springer, 2012.

[14] R.J. Mislevy, S. Corrigan, A. Oranje, K. DiCerbo, M.I. Bauer, A. Von Davier, and M. John. *Psychometrics and game-based assessment*. 2015.

[15] Yoon Jeon Kim and Dirk Ifenthaler. Game-based assessment: The past ten years and moving forward. In *Game-Based Assessment Revisited*, pages 3–11. Springer, 2019.

[16] Meihua Qian and Karen R Clark. Game-based learning and 21st century skills: A review of recent research. *Computers in human behavior*, 63:50–58, 2016.

[17] Cristina Alonso-Fernandez, Antonio Calvo-Morata, Manuel Freire, Ivan Martinez-Ortiz, and Baltasar Fernández-Manjón. Applications of data science to game learning analytics data: A systematic literature review. *Computers & Education*, 141:103612, 2019.

[18] Gabriele Gris and Clarissa Bengtson. Assessment measures in game-based learning research: a systematic review. *International Journal of Serious Games*, 8(1):3–26, 2021.

[19] Xin Guan, Chunmei Sun, Gwo-jen Hwang, Kegan Xue, and Zhuo Wang. Applying game-based learning in primary education: a systematic review of journal publications from 2010 to 2020. *Interactive Learning Environments*, pages 1–23, 2022.

[20] Pei-Ying Chen, Gwo-Jen Hwang, Ssu-Yin Yeh, Yi-Ting Chen, Ting-Wei Chen, and Chih-Hsuan Chien. Three decades of game-based learning in science and mathematics education: an integrated bibliometric analysis and systematic review. *Journal of Computers in Education*, pages 1–22, 2021.

[21] Matthew J Page, David Moher, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, Roger Chou, Julie Glanville, Jeremy M Grimshaw, Asbjørn Hróbjartsson, Manoj M Lalu, Tianjing Li, Elizabeth W Loder, Evan Mayo-Wilson, Steve McDonald, Luke A McGuinness, Lesley A Stewart, James Thomas, Andrea C Tricco, Vivian A Welch, Penny Whiting, and Joanne E McKenzie. Prisma 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. *BMJ*, 372, 2021.

[22] Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, et al. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *Bmj*, 372, 2021.

[23] Katie Salen Tekinbas and Eric Zimmerman. *Rules of play: Game design fundamentals*. MIT press, 2003.

[24] Bomi Kang, Sophia H Tan, et al. Interactive games: Intrinsic and extrinsic motivation, achievement, and satisfaction. *Journal of Management and Strategy*, 5(4):110–116, 2014.

[25] Sara De Freitas. *Learning in immersive worlds: A review of game-based learning*. Jisc, 2006.

[26] Y.J. Kim, R.G. Almond, and V.J. Shute. Applying evidence-centered design for the development of game-based assessments in physics playground. *International Journal of Testing*, 16(2):142–163, 2016.

[27] Sebastian Deterding, Dan Dixon, Rilla Khaled, and Lennart Nacke. From game design elements to gamefulness: defining" gamification". In *Proceedings of the 15th international academic MindTrek conference: Envisioning future media environments*, pages 9–15, 2011.

[28] Hanif Al Fatta, Zulisman Maksom, and Mohd Hafiz Zakaria. Game-based learning and gamification: Searching for definitions. *International Journal of Simulation: Systems, Science and Technology*, 19(6):41–1, 2018.

[29] Ioana Ghergulescu and Cristina Hava Muntean. Measurement and analysis of learner's motivation in game-based e-learning. In *Assessment in game-based learning*, pages 355–378. Springer, 2012.

[30] Jose A. Ruipérez-Valiente. Unveiling the potential of learning analytics in game-based learning: Case studies with a geometry game. In *Handbook of Research on Promoting Economic and Social Development Through Serious Games*, pages 524–544. IGI Global, 2022.

[31] Arezoo Aghaei Chadegani, Hadi Salehi, Melor Yunus, Hadi Farhadi, Masood Fooladi, Maryam Farhadi, and Nader Ale Ebrahim. A comparison between two main academic literature collections: Web of science and scopus databases. *Asian social science*, 9(5):18–26, 2013.

[32] Elsevier. About scopus, 2021.

[33] Clarivate. Web of science, 2021.

[34] Alyona Medelyan. Coding qualitative data: How to code qualitative research, 2021.

[35] Manuel J. Gomez, José Ruipérez-Valiente, and Félix J. García Clemente. Supplementary materials: A systematic literature review of game-based assessment empirical studies: Current trends and open challenges, December 2021. https://osf.io/34jk9/?view_only=865f94046fa84c45a013d986bb1c4f87, Accessed November 28, 2022.

[36] K.E. Di Cerbo, M. Bertling, S. Stephenson, Y. Jia, R.J. Mislevy, M. Bauer, and G.T. Jackson. *An application of exploratory data analysis in the development of game-based assessments*. 2015.

[37] J.-P. Lindenmayer, A. Goldring, S. Borne, A. Khan, R.S.E. Keefe, B.J. Insel, A. Thanju, I. Ljuri, and B. Foreman. Assessing instrumental activities of daily living (iadl) with a game-based assessment for individuals with schizophrenia. *Schizophrenia Research*, 223:166–172, 2020.

[38] Stefanie Wiloth, Nele Lemke, Christian Werner, and Klaus Hauer. Validation of a computerized, game-based assessment strategy to measure training effects on motor-cognitive functions in people with dementia. *JMIR Serious Games*, 4(2):e5696, 2016.

[39] Andrew B Collmus, Michael B Armstrong, and Richard N Landers. Game-thinking within social media to recruit and select job candidates. In *Social media in employee selection and recruitment*, pages 103–124. Springer, 2016.

[40] E. Short and N. Weidner. Gamers at work: Predicting workplace-relevant behaviours across domains. *Journal of Gaming and Virtual Worlds*, 11(2):161–177, 2019.

[41] Cătălin Gabriel Ioniță Alina Stanciu and Adrian Toșcă Dan Florin Stănescu. Development of an integrated game based assessment approach–the next generation of psychometric testing. *European Journal of Sustainable Development*, 8(5):270–270, 2019.

[42] H.G.K. Hummel, D. Joosten-ten Brinke, R.J. Nadolski, and L.K.J. Baartman. Content validity of game-based assessment: case study of a serious game for ict managers in training. *Technology, Pedagogy and Education*, 26(2):225–240, 2017.

[43] Agostino Marengo and Alessandro Pagano. Innovative ways to assess soft-skills: the in-basket game online experience. In *European Conference on e-Learning*, pages 325–XVII. Academic Conferences International Limited, 2020.

[44] M. Ventura and V. Shute. The validity of a game-based assessment of persistence. *Computers in Human Behavior*, 29(6):2568–2572, 2013.

[45] E.J. Weiner and D.R. Sanchez. Cognitive ability in virtual reality: Validity evidence for vr game-based assessments. *International Journal of Selection and Assessment*, 28(3):215–235, 2020.

[46] M. Cutumisu, K.-L. Turgeon, T. Saiyera, S. Chuong, L.M. González Esparza, R. MacDonald, and V. Kokhan. Eye tracking the feedback assigned to undergraduate students in a digital assessment game. *Frontiers in Psychology*, 10, 2019.

[47] R.G. Almond. Tips and tricks for building bayesian networks for scoring game-based assessments. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 9505:250–263, 2015.

[48] L.F.Z. Rivera and C.A. Suescún. Game-based assessment for radiofrequency circuits courses in engineering. In *Proceedings - Frontiers in Education Conference, FIE 2015*, volume 2014, 2015.

[49] F.-Y. Chiu and M.-L. Hsieh. Role-playing game based assessment to fractional concept in second grade mathematics. *Eurasia Journal of Mathematics, Science and Technology Education*, 13(4):1075–1083, 2017.

[50] S. Basu, B. Disalvo, D. Rutstein, Y. Xu, J. Roschelle, and N. Holbert. The role of evidence centered design and participatory design in a playful assessment for computational thinking about data. In *Annual Conference on Innovation and Technology in Computer Science Education, ITiCSE 2020*, pages 985–991, 2020.

[51] Y. Song and J.R. Sparks. Measuring argumentation skills through a game-enhanced scenario-based assessment. *Journal of Educational Computing Research*, 56(8):1324–1344, 2019.

[52] Learningrx. What are cognitive skills?, 2021.

[53] Indeed Editorial Team. Cognitive skills: What they are and how to improve them, 2021.

[54] Jami Dixon, Cody Belnap, Chad Albrecht, and Konrad Lee. The importance of soft skills. *Corporate finance review*, 14(6):35, 2010.

[55] Y. Song and J.R. Sparks. Building a game-enhanced formative assessment to gather evidence about middle school students' argumentation skills. *Educational Technology Research and Development*, 67(5):1175–1196, 2019.

[56] I. Nikolaou, K. Georgiou, and V. Kotsasarlidou. Exploring the relationship of a gamified assessment with performance. *Spanish Journal of Psychology*, 2019.

[57] Esraa M Mosalam, Ghada A El Khayat, Shaimaa Lazem, Lilia Cheniti-Belcadhi, and Bilal Said. Assessing modelling readiness in a games environment. In *2019 7th International conference on ICT & Accessibility (ICTA)*, pages 1–6. IEEE, 2019.

[58] C. Rodríguez-de Pablo, A. Savić, and T. Keller. Game-based assessment in upper-limb post-stroke telerehabilitation. *Biosystems and Biorobotics*, 15:413–417, 2017.

[59] V. Korhonen, H. Räty, and E. Kärnä. A pilot study: a computer game-based assessment of visual perspective taking of four children with autism with high support needs. *Scandinavian Journal of Disability Research*, 19(4):281–294, 2017.

[60] G. Tanner Jackson, B. Lehman, and L.D. Grace. Awkward annie: Impacts of playing on the edge of social norms. In *ACM International Conference Proceeding Series 2020*, 2020.

[61] G.T. Jackson, L. Grace, P. Inglese, J. Wain, and R. Hone. Awkward annie: Game-based assessment of english pragmatic skills. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 10714 LNCS:795–808, 2018.

[62] H. Song, D.-J. Yi, and H.-J. Park. Validation of a mobile game-based assessment of cognitive control among children and adolescents. *PLoS ONE*, 15(3), 2020.

[63] H. Ketamo and K. Devlin. Replacing pisa with global game based assessment. In *Proceedings of the European Conference on Games-based Learning 2014*, volume 1, pages 258–264, 2014.

[64] R. Halverson and V.E. Owen. Game-based assessment: An integrated model for capturing evidence of learning in play. *International Journal of Learning Technology*, 9(2):111–138, 2014.

[65] O. Gaggi, T.M. Sgaramella, L. Nota, M. Bortoluzzi, and S. Santilli. A serious games system for the analysis and the development of visual skills in children with cvi: A pilot study with kindergarten children. *Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICST*, 195 LNICST:155–165, 2017.

[66] M. Bertling, G. Tanner Jackson, A. Oranje, and V.E. Owen. Measuring argumentation skills with game-based assessments: Evidence for incremental validity and learning. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 9112:545–549, 2015.

[67] H.B. Barón, R.G. Crespo, J. Pascual Espada, and O.S. Martínez. Assessment of learning in environments interactive through fuzzy cognitive maps. *Soft Computing*, 19(4):1037–1050, 2015.

[68] M. Loachamin-Valencia, M.-C. Juan, M. Mendez-Lopez, and E. Perez-Hernandez. Auditory and spatial assessment in inattentive children using smart devices and gesture interaction. In *Proceedings - IEEE 17th International Conference on Advanced Learning Technologies, ICALT 2017*, pages 106–110, 2017.

[69] S Pouezevara, S Powers, G Moore, C Strigel, and K McKnight. Assessing soft skills in youth through digital games. In *ICERI2019 Proceedings*, pages 3057–3066. IATED, 2019.

[70] A. Mavridis and T. Tsiatsos. Game-based assessment: investigating the impact on test anxiety and exam performance. *Journal of Computer Assisted Learning*, 33(2):137–150, 2017.

[71] V. Vallejo, P. Wyss, L. Rampa, A.V. Mitache, R.M. Müri, U.P. Mosimann, and T. Nef. Evaluation of a novel serious game based assessment tool for patients with alzheimer's disease. *PLoS ONE*, 12(5), 2017.

[72] A. Leonardou, M. Rigou, and J. Garofalakis. Techniques to motivate learner improvement in game-based assessment. *Information (Switzerland)*, 11(4), 2020.

[73] V.V. Abeele, J. Wouters, P. Ghesquière, A. Goeleven, and L. Geurts. Game-based assessment of psychoacoustic thresholds: Not all games are equal! In *CHI PLAY 2015 - Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*, pages 331–342, 2015.

[74] K. Gjicali, B.M. Finn, and D. Hebert. Effects of belief generation on social exploration, culturally-appropriate actions, and cross-cultural concept learning in a game-based social simulation. *Computers and Education*, 156, 2020.

[75] J. Hautala, R. Heikkilä, L. Nieminen, V. Rantanen, J.-M. Latvala, and U. Richardson. Identification of reading difficulties by a digital game-based assessment technology. *Journal of Educational Computing Research*, 58(5):1003–1028, 2020.

[76] Y.J. Kim and V.J. Shute. The interplay of game elements with psychometric qualities, learning, and enjoyment in game-based assessment. *Computers and Education*, 87:340–356, 2015.

[77] J.C. Perry, S. Balasubramanian, C. Rodriguez-De-Pablo, and T. Keller. Improving the match between ability and challenge: Toward a framework for automatic level adaptation in game-based assessment and training. In *IEEE International Conference on Rehabilitation Robotics*, 2013.

[78] M.C. Gómez-Álvarez, J.A. Echeverri, and L. González-Palacio. Games-based assessment strategy: Case systems engineer of universidad de medellín [estrategia de evaluación basada en juegos: Caso ingeniería de sistemas universidad de medellín]. *Ingeniare*, 25(4):633–642, 2017.

[79] Y. Jaffal and D. Wloka. Employing game analytics techniques in the psychometric measurement of game-based assessments with dynamic content. *Journal of E-Learning and Knowledge Society*, 11(3):101–115, 2015.

[80] M. Cutumisu, D.B. Chin, and D.L. Schwartz. A digital game-based assessment of middle-school and college students' choices to seek critical feedback and to revise. *British Journal of Educational Technology*, 50(6):2977–3003, 2019.

[81] D.B. Chin, K.P. Blair, and D.L. Schwartz. Got game? a choice-based learning assessment of data literacy and visualization skills. *Technology, Knowledge and Learning*, 21(2):195–210, 2016.

[82] C.M. Forsyth, T. Jackson, D. Hebert, B. Lehman, P. Inglese, and L. Grace. Striking a balance: user-experience and performance in computerized game-based assessment. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 10331 LNAI:502–505, 2017.

[83] C. Arce-Lopera and A. Perea. Logic evaluation through game-based assessment. *Advances in Intelligent Systems and Computing*, 973:243–250, 2020.

[84] Luis Tari. *Knowledge Inference*, pages 1074–1078. Springer New York, New York, NY, 2013.

[85] V. Shute and L. Wang. *Assessing and Supporting Hard-to-Measure Constructs in Video Games*. 2016.

[86] R. Levy. Dynamic bayesian network modeling of game-based diagnostic assessments. *Multivariate Behavioral Research*, 54(6):771–794, 2019.

[87] Marshall Hargrave. Deep learning, 2020.

[88] F. Chen, Y. Cui, and M.-W. Chu. Utilizing game analytics to inform and validate digital game-based assessment with evidence-centered game design: A case study. *International Journal of Artificial Intelligence in Education*, 30(3):481–503, 2020.

[89] Guozhu Dong and Jian Pei. *Sequence data mining*, volume 33. Springer Science & Business Media, 2007.

[90] Manuel J Gomez, José A Ruipérez-Valiente, Pedro A Martinez, and Yoon Jeon Kim. Exploring the affordances of sequence mining in educational games. In *Eighth International Conference on Technological Ecosystems for Enhancing Multiculturality*, pages 648–654, 2020.

[91] Oxford. Oxford languages and google, 2021.

[92] M.A. Lonergan, L. De Wet, and A.J. Burger. Technology as a tool to improve student understanding of assessment questions. *Advances in Intelligent Systems and Computing*, 1217 AISC:250–256, 2020.

[93] F.-H. Tsai, C.-C. Tsai, and K.-Y. Lin. The evaluation of different gaming modes and feedback types on game-based formative assessment in an online learning environment. *Computers and Education*, 81:259–269, 2015.

[94] M. Ciman, O. Gaggi, T.M. Sgaramella, L. Nota, M. Bortoluzzi, and L. Pinello. Serious games to support cognitive development in children with cerebral visual impairment. *Mobile Networks and Applications*, 23(6):1703–1714, 2018.

[95] J. Delcker and D. Ifenthaler. Mobile game-based language assessment. *International Journal of Emerging Technologies in Learning*, 15(3):195–206, 2020.

[96] P.A. Martínez, M.J. Gómez, J.A. Ruipérez-Valiente, G.M. Pérez, and Y.J. Kim. Visualizing educational game data: A case study of visualizations to support teachers. In *CEUR Workshop Proceedings*, volume 2671, pages 97–111, 2020.

[97] J.C. Paiva and J.P. Leal. Asura: A game-based assessment environment for mooshak. In *OpenAccess Series in Informatics*, volume 62, 2018.

[98] C. Harteveld and S.C. Sutherland. The goal of scoring: Exploring the role of game performance in educational games. In *Conference on Human Factors in Computing Systems 2015 - Proceedings*, volume 2015-April, pages 2235–2244, 2015.

[99] D.J.R. Jackson, S. Kim, C. Lee, Y. Choi, and J. Song. Simulating déjà vu: What happens to game performance when controlling for situational features? *Computers in Human Behavior*, 55:796–803, 2016.

[100] B. Lehman, D. Hebert, G.T. Jackson, and L. Grace. Affect and experience: Case studies in games and test-taking. In *Conference on Human Factors in Computing Systems 2017 - Proceedings*, volume Part F127655, pages 917–924, 2017.

[101] J. Hao and R.J. Mislevy. The evidence trace file: A data structure for virtual performance assessments informed by data analytics and evidence-centered design. *ETS Research Report Series*, 2018(1), 2018.

[102] H.-H. Yu, J.-K. Yang, H.-W. Chen, T.-F. Yu, and H.-T. Hou. Jingnan campaign© - using game-based assessment with the mechanism of strategy games for history teaching: System development and learning evaluation. In *Proceedings - 2015 IIAI 4th International Congress on Advanced Applied Informatics, IIAI-AAI 2015*, pages 727–728, 2016.

[103] S.-C. Shih, B.-C. Kuo, and S.-J. Lee. An online game-based computational estimation assessment combining cognitive diagnostic model and strategy analysis. *Educational Psychology*, 39(10):1255–1277, 2019.

[104] E. Ibryamova and G. Stefanov. Developing and implementing a labyrinth game for self-assessment. In *ACM International Conference Proceeding Series*, pages 106–110, 2020.

[105] D.R. Sanchez and M. Langer. Video game pursuit (vgpu) scale development: Designing and validating a scale with implications for game-based learning and assessment. *Simulation and Gaming*, 51(1):55–86, 2020.

[106] M. Ponticorvo, F. Ferrara, R. Di Fuccio, A. Di Ferdinando, and O. Miglino. Sniff: A game-based assessment and training tool for the sense of smell. *Advances in Intelligent Systems and Computing*, 617:126–133, 2017.

[107] Anna S Robberts and Linda Van Ryneveld. Design principles for introducing 21st century skills by means of game-based learning. *Industry and Higher Education*, page 09504222221079210, 2022.

[108] Mary Jo Dondlinger. Educational video game design: A review of the literature. *Journal of applied educational technology*, 4(1):21–31, 2007.

[109] James Paul Gee. Are video games good for learning? *Nordic Journal of Digital Literacy*, 1(03):172–183, 2006.

[110] Mete Akcaoglu and Matthew J Koehler. Cognitive outcomes from the game-design and learning (gdl) after-school program. *Computers & Education*, 75:72–81, 2014.

[111] Mahmood H Hussein, Siew Hock Ow, Monther M Elaish, and Erik O Jensen. Digital game-based learning in k-12 mathematics education: a systematic literature review. *Education and Information Technologies*, pages 1–33, 2021.

[112] Colin Ware. *Information visualization: perception for design*. Morgan Kaufmann, 2019.

[113] Kenneth Cukier. A special report on managing information. *The Economist*, 394(8671):3–18, 2010.

[114] Jeffrey Chin, Richard Dukes, and William Gamson. Assessment in simulation and gaming: A review of the last 40 years. *Simulation & Gaming*, 40(4):553–568, 2009.

[115] Robert J Mislevy and Geneva D Haertel. Implications of evidence-centered design for educational testing. *Educational measurement: issues and practice*, 25(4):6–20, 2006.

[116] Valerie J Shute. Stealth assessment in computer-based games to support learning. *Computer games and instruction*, 55(2):503–524, 2011.

[117] Cheryl I. Johnson and Richard E. Mayer. Applying the self-explanation principle to multimedia learning in a computer-based game-like environment. *Computers in Human Behavior*, 26(6):1246–1252, 2010. Online Interactivity: Role of Technology in Behavior Change.

[118] Anissa All, Elena Patricia Nunez Castellar, and Jan Van Looy. Measuring effectiveness in digital game-based learning: A methodological review. *International Journal of Serious Games*, 1(2), 2014.

[119] Fiona Fidler and John Wilcox. Reproducibility of scientific results. *The Stanford Encyclopedia of Philosophy*, 2018.

[120] Tim van der Zee and Justin Reich. Open education science. *AERA Open*, 4(3):2332858418787466, 2018.

[121] Stuart Buck. Solving reproducibility, 2015.

[122] Forough Poursabzi-Sangdeh, Daniel G Goldstein, Jake M Hofman, Jennifer Wortman Wortman Vaughan, and Hanna Wallach. Manipulating and measuring model interpretability. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pages 1–52, 2021.

[123] Giani Petri and Christiane Gresse von Wangenheim. How games for computing education are evaluated? a systematic literature review. *Computers & education*, 107:68–90, 2017.

[124] Manuel J Gomez, José A Ruipérez-Valiente, Pedro A Martínez, and Yoon Jeon Kim. Applying learning analytics to detect sequences of actions and common errors in a geometry game. *Sensors*, 21(4):1025, 2021.

[125] Ming-Chaun Li and Chin-Chung Tsai. Game-based learning in science education: A review of relevant research. *Journal of Science Education and Technology*, 22(6):877–898, 2013.

[126] Manuel Freire, Ángel Serrano-Laguna, Borja Manero, Iván Martínez-Ortiz, Pablo Moreno-Ger, and Baltasar Fernández-Manjón. Game learning analytics: learning analytics for serious games. In *Learning, design, and technology*, pages 1–29. Springer Nature Switzerland AG, 2016.

[127] Christian Sebastian Loh, Yanyan Sheng, and Dirk Ifenthaler. Serious games analytics: Theoretical framework. In *Serious games analytics*, pages 3–29. Springer, 2015.

ActiveAI: Introducing AI Literacy for Middle School Learners with Goal-based Scenario Learning

Ying Jui Tseng, Gautam Yadav
Carnegie Mellon University
yingjuit@andrew.cmu.edu

Brief Abstract

The ActiveAI project addresses key challenges in AI education for grades 7-9 students by providing an engaging AI literacy learning experience based on the AI4K12 knowledge framework. Utilizing learning science mechanisms such as goal-based scenarios, immediate feedback, project-based learning, and intelligent agents, the app incorporates a variety of learner inputs like sliders, steppers, and collectors to enhance understanding. In these courses, students work on real-world scenarios like analyzing sentiment in social media comments. This helps them learn to effectively engage with AI systems and develop their ability to evaluate AI-generated output. The Learning Engineering Process (LEP) guided the project's creation and data instrumentation, focusing on design and impact. The project is currently in the implementation stage, leveraging the intelligent tutor design principles for app development. The extended abstract presents the foundational design and development, with further evaluation and research to be conducted in the future.

The Learning Engineering Process Model

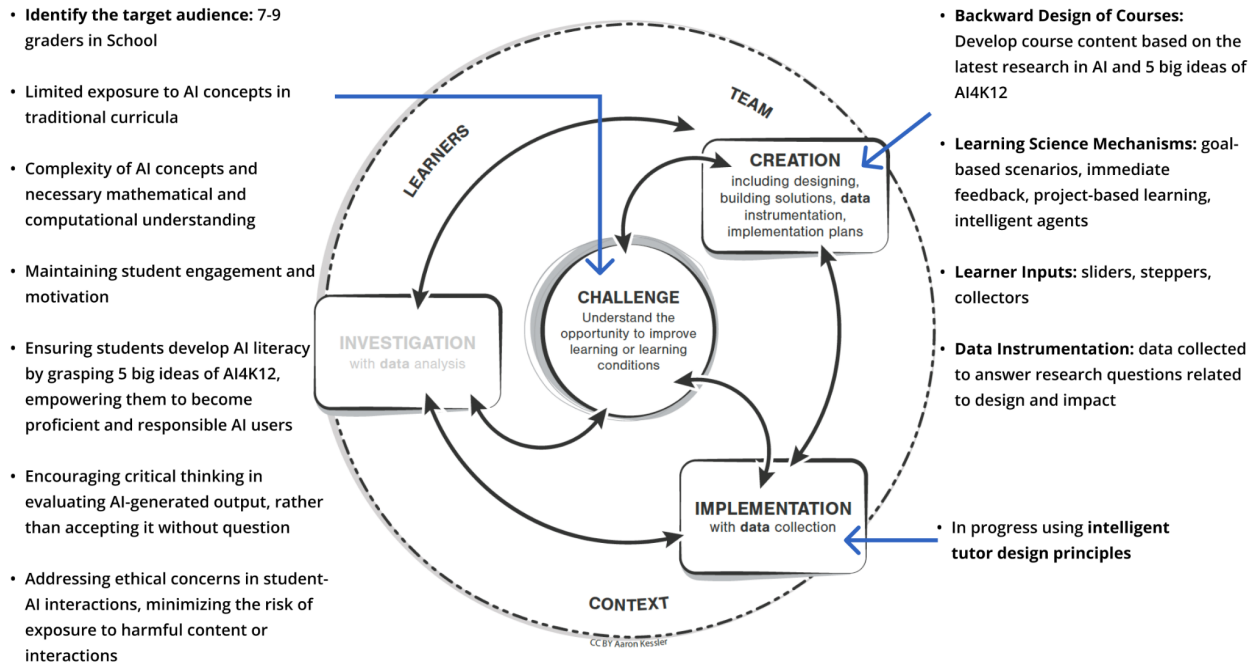


Figure 1. Visualizing the Learning Engineering Process: A Comprehensive Model for Designing and Implementing ActiveAI (Kessler et al., 2022)

Extended Abstract

ActiveAI project offers 7-9 grade students an engaging and accessible AI learning experience, grounded in AI4K12's five big ideas (Touretzky et al., 2022). These five big ideas include "Perception," "Representation & Reasoning," "Learning," "Natural Interaction," and "Societal Impact." The objective is to foster an understanding of AI concepts and applications in an ever-changing technological landscape.

Challenge Stage: The challenges the project seeks to address include:

1. Limited exposure to AI concepts in traditional curricula.
2. Complexity of AI concepts and necessary mathematical and computational understanding.
3. Maintaining student engagement and motivation.
4. Ensuring students develop AI literacy by grasping 5 big ideas of AI4K12, empowering them to become proficient and responsible AI users.
5. Encouraging critical thinking in evaluating AI-generated output, rather than accepting it without question.
6. Addressing ethical concerns in student-AI interactions, minimizing the risk of exposure to harmful content or interactions.

Creation Stage: ActiveAI Creation Step is designed to teach 7-9 grade school students the five big ideas of AI4K12 effectively. The design of ActiveAI relies on proven learning science mechanisms and a range of learner inputs to engage students and promote in-depth understanding.

1. Goal-based scenarios: ActiveAI adopts a goal-oriented approach, offering a range of scenarios in which students tackle real-world challenges that call for AI principles and methodologies (Schank et al., 1994). By immersing learners in simulated situations that present them with a clear objective, they are incentivized to invest more effort into grasping fundamental concepts and honing their skills.
   As an example, students take on the role of content creators seeking to gauge the sentiment of comments on their TikTok videos. They leverage AI techniques to develop a sentiment analysis model, which they utilize to gain insights into their audience's reactions and produce more engaging content.
   In another module, students engage in a task of classifying dog images, much like an automated image tagging system that makes search results more precise in platforms like Instagram. They utilize supervised image classification techniques to label images as "dog sleeping" or "dog playing". During this process, they face the challenge of a skewed training dataset where most images show dogs sleeping indoors and playing outdoors. This could lead to the learning of irrelevant features and biases in the results. Through this interactive task, students not only delve into the workings of machine learning models, but they also learn about potential biases that can emerge from unbalanced datasets, and the importance of addressing these issues to ensure accurate and fair outcomes.
2. Immediate feedback: Throughout the learning experience, students receive immediate feedback on their performance through the app's intelligent agents and the learner inputs.
3. Project-based learning: ActiveAI encourages students to work on authentic hands-on projects that require the application of AI concepts and skills.

4. Intelligent Agents: The app utilizes intelligent agents that enable students to engage with real AI algorithms without the need for coding skills. This approach effectively reduces the learning barrier for students and enables them to access more advanced concepts that were previously beyond their reach (Ng et al., 2022). Studies have demonstrated that incorporating intelligent agents in educational settings can effectively engage middle school learners and promote the development of AI literacy skills (Rodríguez-García et al., 2020).
5. Learner input: ActiveAI strategically employs a select set of learner input interactions – sliders, steppers, and collectors – to facilitate student engagement and comprehension of complex AI concepts. By limiting the interactions to these three types, the app ensures a consistent and familiar interaction across the system, reducing the learning curve and allowing students to focus on mastering AI Literacy concepts without the need to constantly adapt to new interaction styles. The following descriptions detail each interaction type:
    a. Collector: Collector lets learners capture their own datasets or label provided dataset into different classes to train the intelligent agent. For example, students can use their tablet's camera to take pictures for a basic classification model. By participating actively in data collection, learners gain a deeper understanding of the process and a stronger connection to the problem. This hands-on experience enhances student engagement and reinforces the importance of data collection for real-world problem-solving.
    b. Slider: With sliders, learners can adjust variables and receive immediate feedback on the screen. Sliders can be employed to modify various quantitative variables, such as the threshold for a classification model or the amount of training data used. This real-time feedback fosters deeper understanding and experimentation as students explore AI literacy concepts.
    c. Stepper: Steppers are designed to control time-related variables or provide learners with step-by-step explanations. By offering a guided approach to navigating complex AI processes, steppers help students grasp concepts sequentially and methodically.

    Figure 2 describes how the collector and slider works in the dog classification scenario.
    Figure 3 describes how the stepper works in the TikTok sentiment analysis scenario.

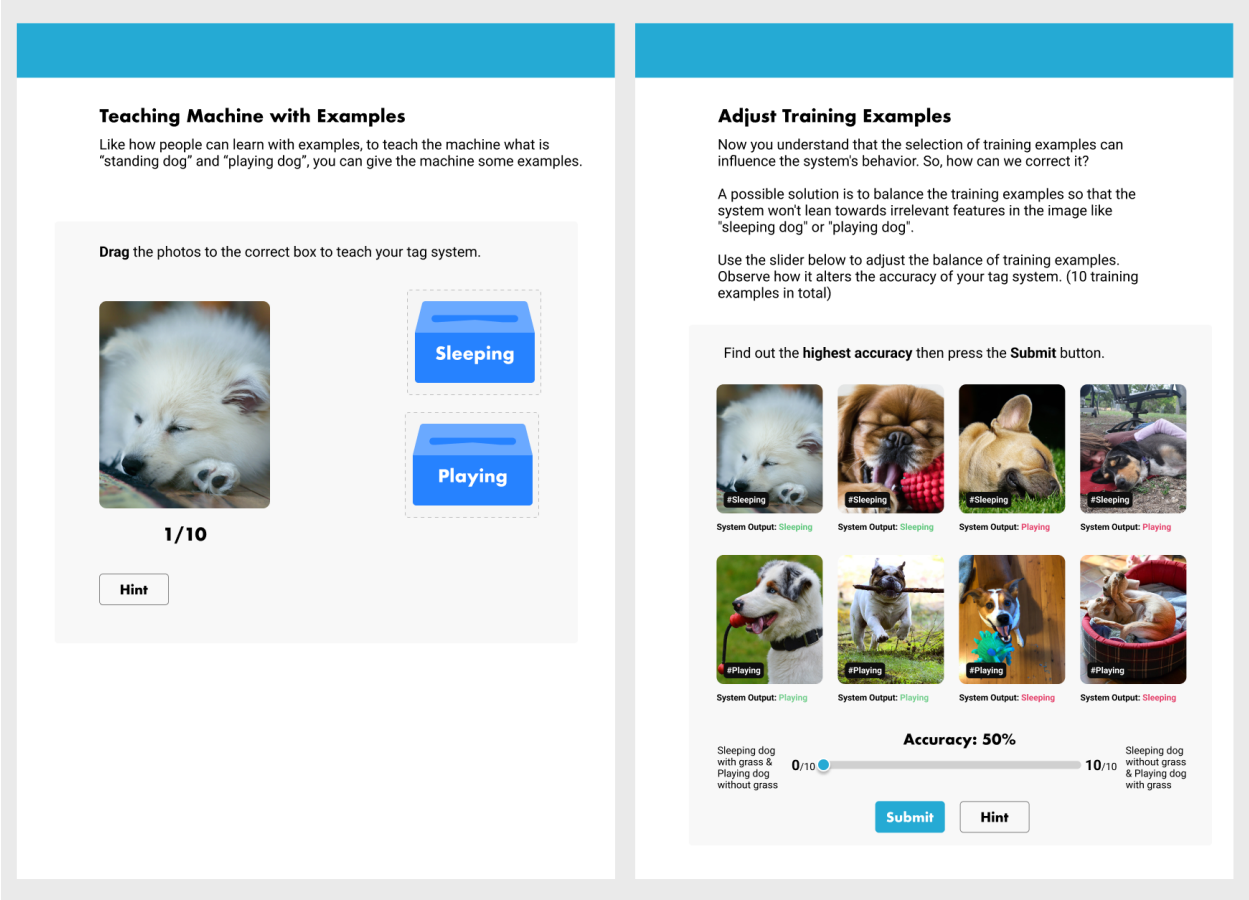


Figure 2. Students employ a collector to categorize 10 images into two classes - "sleeping" and "playing" to create a training set for a classifier (left side). In another learning activity, they will manipulate a slider to adjust the number of training examples, thereby observing how this modification influences the accuracy of their automated tag system on a given test dataset of 8 images (right side).

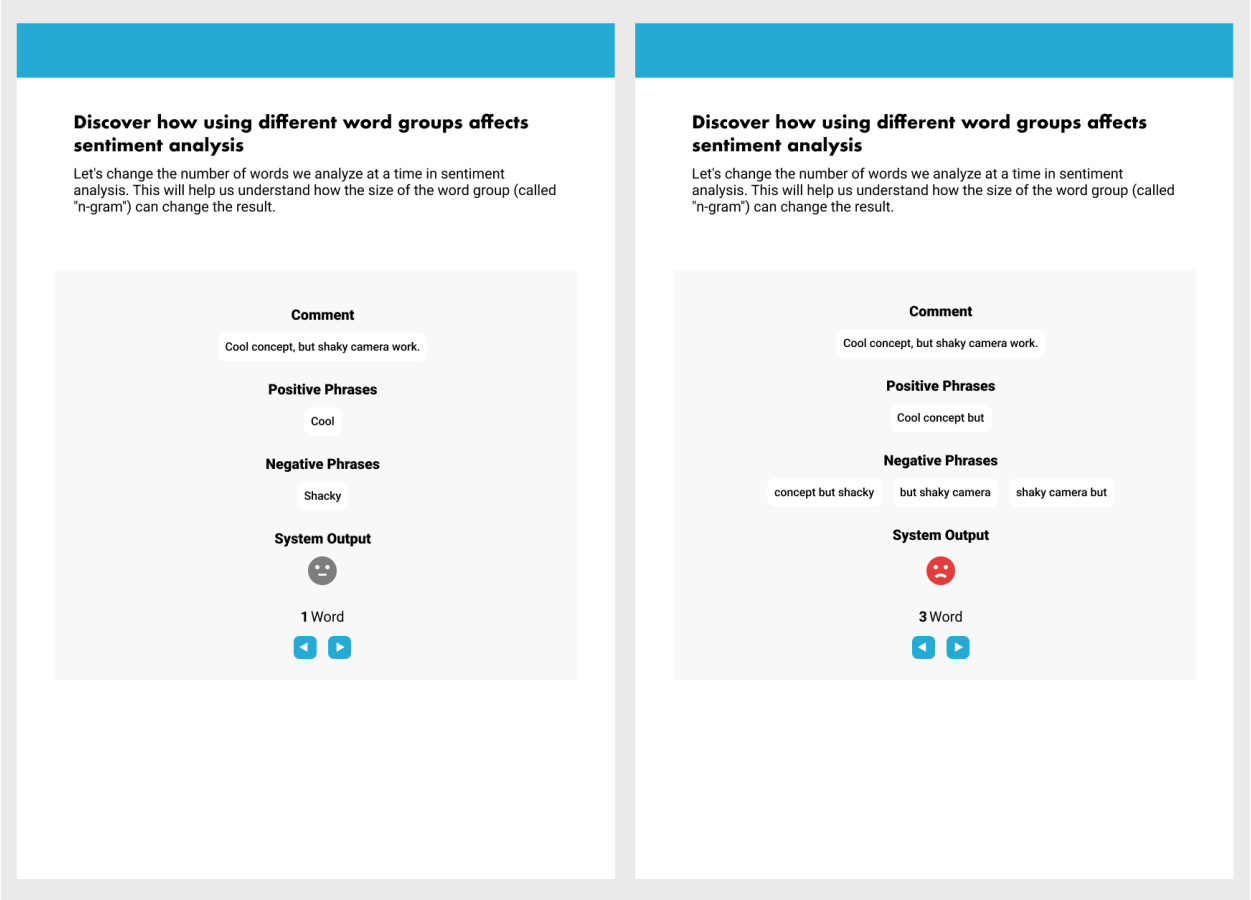


Figure 3. Students use a stepper to modify the number of words considered in the sentiment analysis, enabling them to explore how using different word groups affects sentiment analysis results sequentially and gain insight into how machines “understand” sentiment.

The data collected from students' app interactions will be analyzed to answer research questions related to design and impact (Torrence, 2022), providing insights into the learning experience's effectiveness and continuous improvement.

Table 1. Design Focus Data Instrumentation

Research Questions	**Hypothesis**	**Method**	**Metrics**
How effectively are students engaging with the app and its different interactions?	Increased use of interaction types leads to better understanding and problem-solving.	Analyzing frequency and type of interactions	Number of interactions, interaction types, time spent on each interaction
How do the hints provided by the tutor affect student performance?	Hints offered by tutor will positively impact student performance by	Analyzing hint usage and impact on performance	Number of hints used, performance before and after hint usage

	providing appropriate guidance.		
How does the level of assistance provided by the tutor correlate with student performance?	A tailored level of assistance will result in improved student performance compared to generic hints.	Correlating assistance level with student performance	Assistance level, student performance

Table 2. Impact Focus Data Instrumentation

Research Questions	**Hypothesis**	**Method**	**Metrics**
How quickly are students progressing through the app's modules and activities?	Students who spend more time on the app will demonstrate faster progress through modules and activities.	Tracking student progress through modules and activities	Time spent on app, number of modules completed, time taken to complete activities
What factors contribute to students' time taken to complete each activity or module?	Factors such as prior knowledge, level of tutor assistance, and engagement will impact completion times.	Analyzing time taken and factors influencing completion times	Time taken to complete activities, prior knowledge, engagement, tutor assistance
How does student performance on assessments change after using the app?	Student performance on assessments will improve after using the AI literacy app.	Comparing pre- and post-app assessment scores	Pre- and post-app assessment scores
Which components of the app have the most significant impact on assessment scores?	Hands-on activities and personalized feedback will have the most significant impact on improving scores.	Analyzing correlation between app components and assessment scores	App component usage, Assessment scores
How does student performance on assessments related to the same KC change over multiple opportunities?	As students have more opportunities to practice the same KC, their performance on related assessments will improve.	Analyzing assessment performance over multiple opportunities	Number of opportunities, assessment scores for the same KC, student progress
How does the performance of students using the intelligent agent compare to those using pre-defined practice questions?	Students using the intelligent agent will show better performance compared to those using pre-defined practice.	Comparing performance between intelligent agent users and pre-defined practice question users	Assessment scores, usage of intelligent agent, usage of pre-defined practice questions

Implementation Stage: The ActiveAI project is currently in the implementation stage, utilizing the intelligent tutor design principles (Corbett et al., 1997) to develop an interactive learning environment that incorporates the learning science mechanisms and learner inputs discussed in the Creation Step (Aleven et al., 2006). By leveraging intelligent tutor design principles, we can build an adaptive and personalized learning experience, ensuring students receive appropriate support and guidance throughout their AI literacy journey.

This extended abstract presents the foundational design and development of the ActiveAI program, with further evaluation and research to be conducted in the future.

References

Aleven, V., McLaren, B. M., Sewall, J., & Koedinger, K. R. (2006). The cognitive tutor authoring tools (CTAT): Preliminary evaluation of efficiency gains. *In Intelligent Tutoring Systems: 8th International Conference, ITS 2006, Jhongli, Taiwan, June 26-30, 2006. Proceedings 8* (pp. 61-70). Springer Berlin Heidelberg.

Corbett, A. T., Koedinger, K. R., & Anderson, J. R. (1997). Intelligent tutoring systems. *In Handbook of human-computer interaction* (pp. 849-874). North-Holland.

Kessler, A., Craig, S.D., Goodell, J., Kurzweil, D., & Greenwald, S.W. (2022). Learning Engineering is a Process. In Goodell & Kolodner (Eds.), *Learning Engineering Toolkit: Evidence-based Practices from the Learning Sciences, Instructional Design, and Beyond*. Routledge: New York and London.

Ng, D.T.K., Leung, J.K.L., Su, M.J., Yim, I.H.Y., Qiao, M.S., Chu, S.K.W. (2022). AI Literacy Education in Secondary Schools. In: *AI Literacy in K-16 Classrooms*. Springer, Cham.

Rodríguez-García, J. D., Moreno-León, J., Román-González, M., & Robles, G. (2021). Introducing Artificial Intelligence Fundamentals with LearningML: Artificial Intelligence made easy. In TEEM'20: *Eighth International Conference on Technological Ecosystems for Enhancing Multiculturality* (pp. 18-20).

Schank, R. C., Fano, A., Bell, B., & Jona, M. (1994). The design of goal-based scenarios. *The journal of the learning sciences, 3*(4), 305-345.

Torrence, K., (2022, October 27) Building Better VR Solutions with Learning Engineering Practices (Conference presentation). *DevLearn 2022 Conference & Expo*. Las Vegas, NV, USA

Touretzky, D., Gardner-McCune, C., & Seehorn, D. (2022). Machine Learning and the Five Big Ideas in AI. *International Journal of Artificial Intelligence in Education*, 1-34.

arXiv:2006.10214v1 [cs.CV] 18 Jun 2020

MediaPipe Hands: On-device Real-time Hand Tracking

Fan Zhang Valentin Bazarevsky Andrey Vakunov
Andrei Tkachenka George Sung Chuo-Ling Chang Matthias Grundmann
Google Research
1600 Amphitheatre Pkwy, Mountain View, CA 94043, USA
{zhafang, valik, vakunov, atkach, gsung, chuoling, grundman}@google.com

Abstract

We present a real-time on-device hand tracking solution that predicts a hand skeleton of a human from a single RGB camera for AR/VR applications. Our pipeline consists of two models: 1) a palm detector, that is providing a bounding box of a hand to, 2) a hand landmark model, that is predicting the hand skeleton. It is implemented via MediaPipe[12], a framework for building cross-platform ML solutions. The proposed model and pipeline architecture demonstrate real-time inference speed on mobile GPUs with high prediction quality. MediaPipe Hands is open sourced at https://mediapipe.dev.

1. Introduction

Hand tracking is a vital component to provide a natural way for interaction and communication in AR/VR, and has been an active research topic in the industry [2] [15]. Vision-based hand pose estimation has been studied for many years. A large portion of previous work requires specialized hardware, *e.g.* depth sensors [13][16][17][3][4]. Other solutions are not lightweight enough to run real-time on commodity mobile devices[5] and thus are limited to platforms equipped with powerful processors. In this paper, we propose a novel solution that does not require any additional hardware and performs in real-time on mobile devices. Our main contributions are:

- An efficient two-stage hand tracking pipeline that can track multiple hands in real-time on mobile devices.
- A hand pose estimation model that is capable of predicting 2.5D hand pose with only RGB input.
- And open source hand tracking pipeline as a ready-to-go solution on a variety of platforms, including Android, iOS, Web (Tensorflow.js[7]) and desktop PCs.

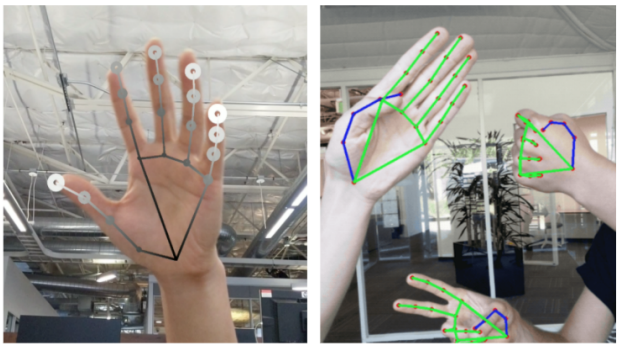


Figure 1: Rendered hand tracking result. (Left): Hand landmarks with relative depth presented in different shades. The lighter and larger the circle, the closer the landmark is towards the camera. (Right): Real-time multi-hand tracking on Pixel 3.

2. Architecture

Our hand tracking solution utilizes an ML pipeline consisting of two models working together:

- A palm detector that operates on a full input image and locates palms via an oriented hand bounding box.
- A hand landmark model that operates on the cropped hand bounding box provided by the palm detector and returns high-fidelity 2.5D landmarks.

Providing the accurately cropped palm image to the hand landmark model drastically reduces the need for data augmentation (*e.g.* rotations, translation and scale) and allows the network to dedicate most of its capacity towards landmark localization accuracy. In a real-time tracking scenario, we derive a bounding box from the landmark prediction of the previous frame as input for the current frame, thus avoiding applying the detector on every frame. Instead, the detector is only applied on the first frame or when the hand prediction indicates that the hand is lost.

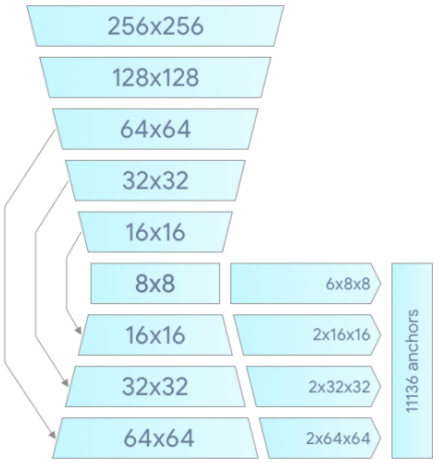


Figure 2: Palm detector model architecture.

2.1. BlazePalm Detector

To detect initial hand locations, we employ a single-shot detector model optimized for mobile real-time application similar to BlazeFace[1], which is also available in MediaPipe[12]. Detecting hands is a decidedly complex task: our model has to work across a variety of hand sizes with a large scale span (∼20x) and be able to detect occluded and self-occluded hands. Whereas faces have high contrast patterns, *e.g.*, around the eye and mouth region, the lack of such features in hands makes it comparatively difficult to detect them reliably from their visual features alone.

Our solution addresses the above challenges using different strategies.

First, we train a palm detector instead of a hand detector, since estimating bounding boxes of rigid objects like palms and fists is significantly simpler than detecting hands with articulated fingers. In addition, as palms are smaller objects, the non-maximum suppression algorithm works well even for the two-hand self-occlusion cases, like handshakes. Moreover, palms can be modelled using only square bounding boxes [11], ignoring other aspect ratios, and therefore reducing the number of anchors by a factor of 3∼5.

Second, we use an encoder-decoder feature extractor similar to FPN[9] for a larger scene-context awareness even for small objects.

Lastly, we minimize the focal loss[10] during training to support a large amount of anchors resulting from the high scale variance. High-level palm detector architecture is shown in Figure 2. We present an ablation study of our design elements in Table 1.

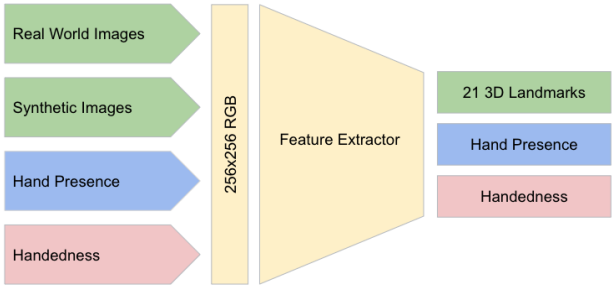


Figure 3: Architecture of our hand landmark model. The model has three outputs sharing a feature extractor. Each head is trained by correspondent datasets marked in the same color. See Section 2.2 for more detail.

2.2. Hand Landmark Model

After running palm detection over the whole image, our subsequent hand landmark model performs precise landmark localization of 21 2.5D coordinates inside the detected hand regions via regression. The model learns a consistent internal hand pose representation and is robust even to partially visible hands and self-occlusions. The model has three outputs (see Figure 3):

1. 21 hand landmarks consisting of x, y, and relative depth.
2. A hand flag indicating the probability of hand presence in the input image.
3. A binary classification of handedness, *e.g.* left or right hand.

We use the same topology as [14] for the 21 landmarks. The 2D coordinates are learned from both real-world images as well as synthetic datasets as discussed below, with the relative depth w.r.t. the wrist point being learned only from synthetic images. To recover from tracking failure, we developed another output of the model similar to [8] for producing the probability of the event that a reasonably aligned hand is indeed present in the provided crop. If the score is lower than a threshold then the detector is triggered to reset tracking. Handedness is another important attribute for effective interaction using hands in AR/VR. This is especially useful for some applications where each hand is associated with a unique functionality. Thus we developed a binary classification head to predict whether the input hand is the left or right hand. Our setup targets real-time mobile GPU inference, but we have also designed lighter and heavier versions of the model to address CPU inference on the mobile devices lacking proper GPU support and higher accuracy requirements of accuracy to run on desktop, respectively.

3. Dataset and Annotation

To obtain ground truth data, we created the following datasets addressing different aspects of the problem:

- In-the-wild dataset: This dataset contains 6K images of large variety, *e.g*. geographical diversity, various lighting conditions and hand appearance. The limitation of this dataset is that it doesn't contain complex articulation of hands.
- In-house collected gesture dataset: This dataset contains 10K images that cover various angles of all physically possible hand gestures. The limitation of this dataset is that it's collected from only 30 people with limited variation in background. The in-the-wild and in-house dataset are great complements to each other to improve robustness.
- Synthetic dataset: To even better cover the possible hand poses and provide additional supervision for depth, we render a high-quality synthetic hand model over various backgrounds and map it to the corresponding 3D coordinates. We use a commercial 3D hand model that is rigged with 24 bones and includes 36 blendshapes, which control fingers and palm thickness. The model also provides 5 textures with different skin tones. We created video sequences of transformation between hand poses and sampled 100K images from the videos. We rendered each pose with a random high-dynamic-range lighting environment and three different cameras. See Figure 4 for examples.

For the palm detector, we only use in-the-wild dataset, which is sufficient for localizing hands and offers the highest variety in appearance. However, all datasets are used for training the hand landmark model. We annotate the real-world images with 21 landmarks and use projected ground-truth 3D joints for synthetic images. For hand presence, we select a subset of real-world images as positive examples and sample on the region excluding annotated hand regions as negative examples. For handedness, we annotate a subset of real-world images with handedness to provide such data.

4. Results

For the hand landmark model, our experiments show that the combination of real-world and synthetic datasets provides the best results. See Table 2 for details. We evaluate only on real-world images. Beyond the quality improvement, training with a large synthetic dataset leads to less jitter visually across frames. This observation leads us to believe that our real-world dataset can be enlarged for better generalization.

Our target is to achieve real-time performance on mobile devices. We experimented with different model sizes and found that the "Full" model (see Table 3) provides a good trade-off between quality and speed. Increasing model capacity further introduces only minor improvements in quality but decreases significantly in speed (see Table 3 for details). We use the TensorFlow Lite GPU backend for on-device inference[6].

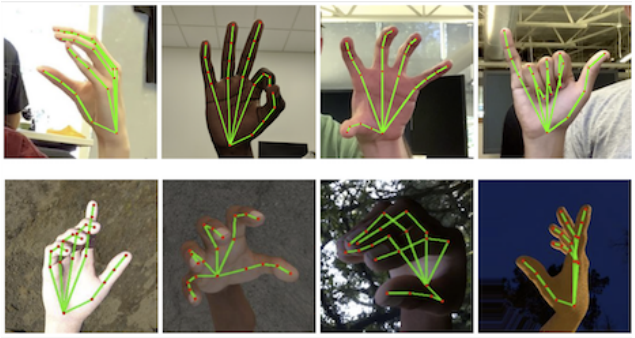


Figure 4: Examples of our datasets. (Top): Annotated real-world images. (Bottom): Rendered synthetic hand images with ground truth annotation. See Section 3 for details.

Model Variation	Average Precision
No decoder + cross entropy loss	86.22%
Decoder + cross entropy loss	94.07%
Decoder + focal loss	95.7%

Table 1: Ablation study of palm detector design elements of palm detector.

Dataset	MSE normalized by palm size
Only real-world	16.1%
Only synthetic	25.7%
Combined	13.4%

Table 2: Results of our model trained from different datasets.

Model	Params (M)	MSE	Time(ms) Pixel 3	Time(ms) Samsung S20	Time(ms) iPhone11
Light	1	11.83	6.6	5.6	1.1
Full	1.98	10.05	16.1	11.1	5.3
Heavy	4.02	9.817	36.9	25.8	7.5

Table 3: Hand landmark model performance characteristics.

5. Implementation in MediaPipe

With MediaPipe[12], our hand tracking pipeline can be built as a directed graph of modular components, called Calculators. Mediapipe comes with an extensible set of Calculators to solve tasks like model inference, media processing, and data transformations across a wide variety of devices and platforms. Individual Calculators like cropping, rendering and neural network computations are further optimized to utilize GPU acceleration. For example, we employ TFLite GPU inference on most modern phones.

Our MediaPipe graph for hand tracking is shown in Figure 5. The graph consists of two subgraphs one for hand detection and another for landmarks computation. One key optimization MediaPipe provides is that the palm detector only runs as needed (fairly infrequently), saving significant computation. We achieve this by deriving the hand location in the current video frames from the computed hand landmarks in the previous frame, eliminating the need to apply the palm detector on every frame. For robustness, the hand tracker model also outputs an additional scalar capturing the confidence that a hand is present and reasonably aligned in the input crop. Only when the confidence falls below a certain threshold is the hand detection model reapplied to the next frame.

6. Application examples

Our hand tracking solution can readily be used in many applications such as gesture recognition and AR effects. On top of the predicted hand skeleton, we employ a simple algorithm to compute gestures, see Figure 6. First, the state of each finger, *e.g.* bent or straight, is determined via the accumulated angles of joints. Then, we map the set of finger states to a set of predefined gestures. This straightforward, yet effective technique allows us to estimate basic static gestures with reasonable quality. Beyond static gesture recognition, it is also possible to use a sequence of landmarks to predict dynamic gestures. Another application is to apply AR effects on top of the skeleton. Hand based AR effects currently enjoy high popularity. In Figure 7, we show an example AR rendering of the hand skeleton in neon light style.

7. Conclusion

In this paper, we proposed MediaPipe Hands, an end-to-end hand tracking solution that achieves real-time performance on multiple platforms. Our pipeline predicts 2.5D landmarks without any specialized hardware and thus, can be easily deployed to commodity devices. We open sourced the pipeline to encourage researchers and engineers to build gesture control and creative AR/VR applications with our pipeline.

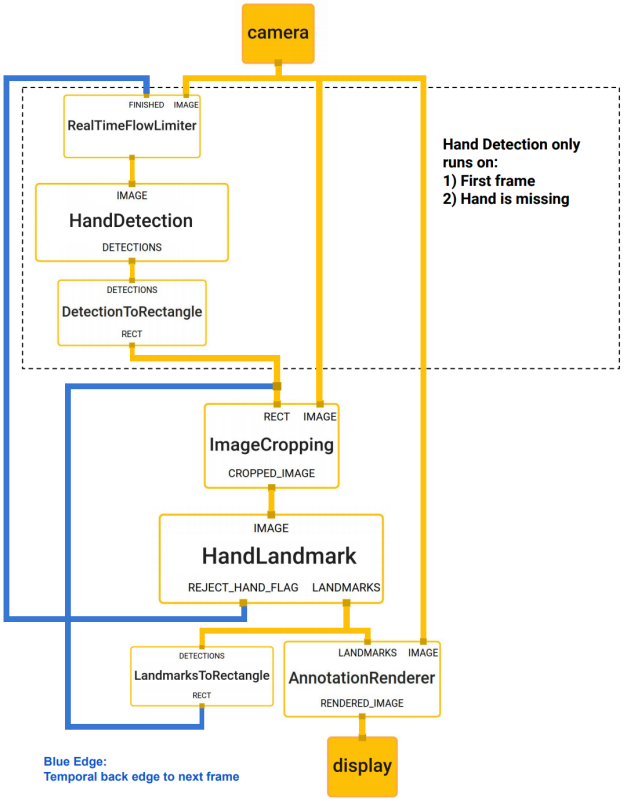


Figure 5: The hand landmark models output controls when the hand detection model is triggered. This behavior is achieved by MediaPipes powerful synchronization building blocks, resulting in high performance and optimal throughput of the ML pipeline.

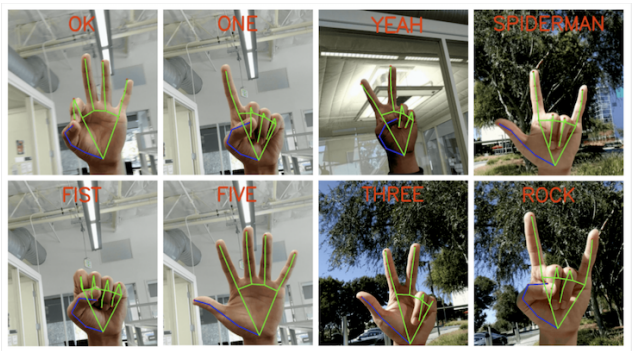


Figure 6: Screenshots of real-time gesture recognition. Semantics of gestures are rendered at top of the images.

References

[1] Valentin Bazarevsky, Yury Kartynnik, Andrey Vakunov, Karthik Raveendran, and Matthias Grundmann. Blazeface: Sub-millisecond neural face detection on mobile gpus.

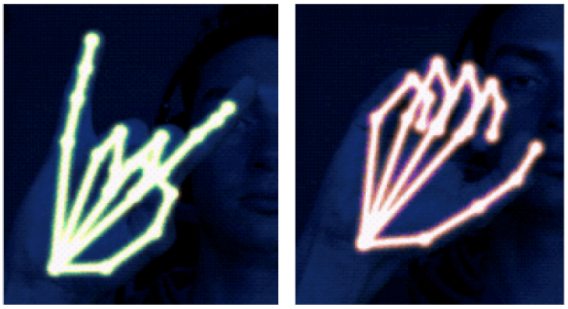


Figure 7: Example of real-time AR effects based on our predicted hand skeleton.

CoRR, abs/1907.05047, 2019. 2

[2] Facebook. Oculus Quest Hand Tracking. https://www.oculus.com/blog/oculus-connect-6-introducing-hand-tracking-on-oculus-quest-facebook-horizon-and-more/. 1

[3] Liuhao Ge, Hui Liang, Junsong Yuan, and Daniel Thalmann. Robust 3d hand pose estimation in single depth images: from single-view cnn to multi-view cnns. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3593–3601, 2016. 1

[4] Liuhao Ge, Hui Liang, Junsong Yuan, and Daniel Thalmann. Robust 3d hand pose estimation from single depth images using multi-view cnns. *IEEE Transactions on Image Processing*, 27(9):4422–4436, 2018. 1

[5] Liuhao Ge, Zhou Ren, Yuncheng Li, Zehao Xue, Yingying Wang, Jianfei Cai, and Junsong Yuan. 3d hand shape and pose estimation from a single rgb image. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 10833–10842, 2019. 1

[6] Google. Tensorflow lite on GPU. https://www.tensorflow.org/lite/performance/gpu_advanced. 3

[7] Google. Tensorflow.js Handpose. https://blog.tensorflow.org/2020/03/face-and-hand-tracking-in-browser-with-mediapipe-and-tensorflowjs.html. 1

[8] Yury Kartynnik, Artsiom Ablavatski, Ivan Grishchenko, and Matthias Grundmann. Real-time facial surface geometry from monocular video on mobile gpus. *CoRR*, abs/1907.06724, 2019. 2

[9] Tsung-Yi Lin, Piotr Dollár, Ross B. Girshick, Kaiming He, Bharath Hariharan, and Serge J. Belongie. Feature pyramid networks for object detection. *CoRR*, abs/1612.03144, 2016. 2

[10] Tsung-Yi Lin, Priya Goyal, Ross B. Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. *CoRR*, abs/1708.02002, 2017. 2

[11] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott E. Reed, Cheng-Yang Fu, and Alexander C. Berg. SSD: single shot multibox detector. *CoRR*, abs/1512.02325, 2015. 2

[12] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. Mediapipe: A framework for building perception pipelines. volume abs/1906.08172, 2019. 1, 2, 4

[13] Iason Oikonomidis, Nikolaos Kyriazis, and Antonis A Argyros. Efficient model-based 3d tracking of hand articulations using kinect. 1

[14] Tomas Simon, Hanbyul Joo, Iain A. Matthews, and Yaser Sheikh. Hand keypoint detection in single images using multiview bootstrapping. *CoRR*, abs/1704.07809, 2017. 2

[15] Snapchat. Lens Studio by Snap Inc. https://lensstudio.snapchat.com/templates/object/hand/. 1

[16] Andrea Tagliasacchi, Matthias Schröder, Anastasia Tkach, Sofien Bouaziz, Mario Botsch, and Mark Pauly. Robust articulated-icp for real-time hand tracking. In *Computer Graphics Forum*, volume 34, pages 101–114. Wiley Online Library, 2015. 1

[17] Chengde Wan, Thomas Probst, Luc Van Gool, and Angela Yao. Self-supervised 3d hand pose estimation through training by fitting. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 10853–10862, 2019. 1

arXiv:2111.00038v1 [cs.CV] 29 Oct 2021

On-device Real-time Hand Gesture Recognition

George Sung Kanstantsin Sokal Esha Uboweja Valentin Bazarevsky Jonathan Baccash
Eduard Gabriel Bazavan Chuo-Ling Chang Matthias Grundmann
Google Research
1600 Amphitheatre Pkwy, Mountain View, CA 94043, USA
{gsung, kanstantsin, eshauboweja, bazarevsky, jbaccash, egbazavan, chuoling, grundman}@google.com

Abstract

We present an on-device real-time hand gesture recognition (HGR) system, which detects a set of predefined static gestures from a single RGB camera. The system consists of two parts: a hand skeleton tracker and a gesture classifier. We use MediaPipe Hands [14, 2] as the basis of the hand skeleton tracker, improve the keypoint accuracy, and add the estimation of 3D keypoints in a world metric space. We create two different gesture classifiers, one based on heuristics and the other using neural networks (NN).

1. Introduction

Hand gesture recognition (HGR) is a natural and intuitive method for human-computer interaction (HCI), and has been an active research area [11, 10]. A wide variety of input devices and techniques have been investigated, and skeleton-based HGR is a popular choice due to its robustness to background and light variations [7, 6]. Many skeleton-based HGR systems rely on depth sensors, such as RGBD cameras, which are not nearly as common as RGB cameras on mobile devices. Our HGR, on the other hand, requires only a single RGB camera. It does so by first predicting 3D skeleton keypoints from a camera image, then running a gesture classifier on the keypoints.

We design two gesture classifiers with different use cases in mind. The heuristics-based classifier is easier to create and extend, without the need for training data, and more intuitive to develop and troubleshoot. The NN-based classifier is more accurate and precise, especially for borderline cases. It's also more forgiving of errors in skeleton keypoints.

Our HGR runs real-time at 30fps on mainstream mobile devices.

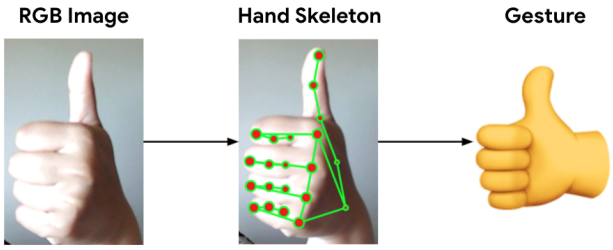


Figure 1. Our hand gesture recognition system

2. Architecture

Our HGR consists of two parts: a hand skeleton tracker improved from MediaPipe Hands and a gesture classifier, as shown on Figure 1. The two-step approach has a few advantages:

- Reduced engineering effort by leveraging the hand tracker which is already real-time, robust, and fair [4].
- Simpler gesture classifier design which processes skeleton keypoints instead of raw pixels.
- Optimized complexity by running the gesture classifier only when hands are tracked.

2.1. Hand Skeleton Tracker

Both our gesture classifiers operate on a keypoint level (not with RGB data), therefore accurate hand keypoint estimation is a key prerequisite for gesture classification. As a basis for our gesture recognition pipeline, we improve MediaPipe Hands [2] by training with a new sophisticated hand poses data (like American Sign Language). Analysis of various gestures indicates that robust estimation of hand rotation angle and normalization distance is key to an accurate hand tracker. The original hand rotation and scale estimation is based on the 2D vector from the wrist to the middle finger knuckle. For various cases (like frontal view)

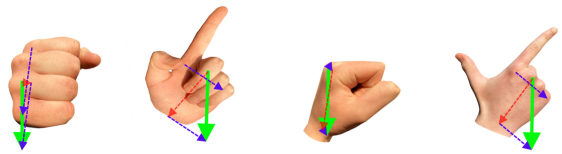


Figure 2. Hands rotation angle derived from the sum of two vectors: index to pinky base knuckle (in green) and middle base knuckle to wrist (in red).

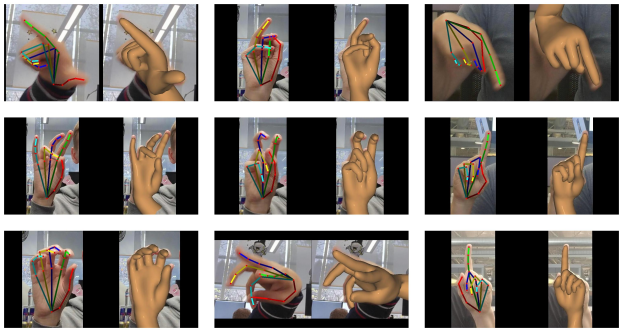


Figure 3. Sample images with overlaid 2D annotations and the corresponding GHUM [12] hand models fitted and rendered on top.

such normalization becomes very unstable as the normalization distance approaches zero, which results in a significant degradation in tracking quality. To overcome this problem, we introduce a new algorithm similar to the approach taken by BlazePose [5].

For our case, we define two virtual keypoints to describe hand center, scale and rotation angle: a center keypoint and an alignment keypoint. The center keypoint is estimated as the average of the index, middle and pinky knuckles. The alignment keypoint location is estimated so it forms the rotation/scale vector with the center keypoint. The rotation angle is estimated from a sum of two vectors: from the middle base knuckle to the wrist, and from the index to the pinky base knuckle. As the component vectors tend to be orthogonal in the majority of cases; the resulting sum vector changes smoothly for any hand pose, and never degrades to zero, as shown on Figure 2. This increases overall hand tracking quality for frontal hand cases. The scalar value of the alignment vector is estimated as the distance from the center keypoint to the farthest knuckle of the same hand. The new rotation and scale normalization results in a significant quality boost for the whole hand pose estimation pipeline: 71.3 mAP vs 66.5 mAP (for the original MediaPipe Hands [2] pipeline) on our validation dataset with complex ASL hand poses.

Accurate hand pose estimation in the 3D space is a vital component for both angle based and few-shot learning gesture classification. It minimizes the ambiguity among projections of the same hand pose from different observer positions in space, and allows the gesture classification to be invariant to rotation. Therefore, in addition to predicting the hand pose in the screen "pixel" space, we also estimate the pose in a world metric space relatively to the hand wrist. To obtain a 3D hand pose ground truth in a metric space, we fit our 2D hand annotation with a statistical and highly realistic GHUM [12, 13] hand model as shown in Figure 3. Due to the nature of perspective projection, objects of different sizes may have the same projection on the image plane: two objects with the same shape (bigger and smaller) will have the same projection if placed respectively further and closer to the camera. Therefore we have to make the following assumptions when we perform the hand fitting from 2D keypoints: human hands have minor variations in size, those variations are always reflected in hand shapes, and thus covered by the statistical GHUM hand model. For images with unknown camera intrinsic parameters, we assume the focal length is the maximum of the image width and height, and the optical center is the center of the image. Since both the model training and inference operate on a cropped image, we normalize the absolute world coordinates such that the origin is at the middle finger knuckle, while the scale remains the same. We also normalize 3D coordinates in a roll plane using virtual keypoints to be consistent with input image transformations.

Overall, our hand tracking model achieves a mean average prediction error of 1.5cm, and provides 21 hand keypoints in a metric 3D space to the gesture classfier.

2.2. Heuristics Gesture Classifier

Based on the hand skeleton tracker, we build a single-shot, heuristics-based classifier for a small set of static gestures. We start with the simple gesture classification approach described in [14], which first derives a set of angles between various 2D hand keypoints, then applies thresholds to the derived angles to define a discrete state for each finger (e.g. bent or straight), and finally defines a static gesture as a logic expression based on the finger states. To get a more accurate set of underlying angles, we replace features based on the 2D hand keypoints with features based on the 3D world metric hand keypoints. In order to de-correlate the gesture classifier features and make manual threshold picking easier, we distinguish between *extrinsic* and *intrinsic* features with respect to the palm pose in the 3D world metric space during the preprocessing stage:

- The *extrinsic* features are composed of palm pose components such as rotation, scale and translation. For classification purposes, we only use the rotation pose component of the palm represented by its three Euler angles.

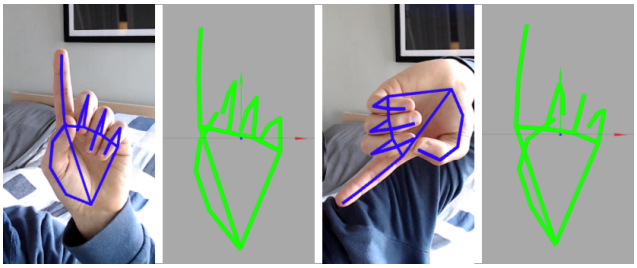


Figure 4. 3D hand keypoints decoupled from the palm pose during the preprocessing stage. The blue hand skeleton is based on the 2D hand keypoints. The green hand skeleton is based on the preprocessed 3D hand keypoints.

- The *intrinsic* features are angles between various hand keypoints. For classification purposes, we derive a single feature angle for each finger with a goal of thresholding them to define a discrete state (e.g. fully bent, fully straight or neither). In accordance with the underlying hand skeleton topology, each finger is represented by a base joint keypoint, two intermediate joint keypoints and a tip keypoint. To define an individual finger feature angle, we first introduce a 3D polygonal chain: starting from the wrist keypoint, to the finger's base joint keypoint, the first intermediate joint keypoint, the second intermediate joint keypoint, and finishing at the tip keypoint. Then, we define the feature angle as the maximum angle between the first chain segment and each of the remaining chain segments. Additionally, we derive a feature angle for each adjacent pair of fingers with the same goal of thresholding them to define a discrete state (e.g. fingers crossed, apart or neither).

As shown on Figure 4, removing the influence of the extrinsic features from the the intrinsic features produces a consistent 3D hand keypoint set for a fixed hand shape configuration, regardless of its position on the input frames. In turn, this significantly de-correlates features and makes it manageable to manually pick thresholds on a larger feature set in order to define a more complex static gesture.

As the final stage of the heuristics-based classifier, we establish a system of 6 gesture definitions based on the features derived from the pre-processed 3D world metric hand keypoints. Figure 5 showcases the supported gestures. This particular gesture set is chosen so that it covers some of the most recognizable and common static hand gestures.

To evaluate our approach, we collect and annotate an in-house gesture dataset. The dataset contains 1882 short video clips that cover various angles and lighting conditions for 21 static hand gestures (for the full list of gestures, please see Appendix A). The dataset contains all 6 gestures supported by the heuristics-based classifier (see Figure 5)

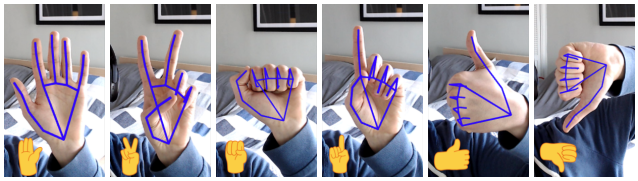


Figure 5. Visualization of gestures supported by the heuristic-based classifier. Left-to-right: *OpenPalm*, *Victory*, *ClosedFist*, *PointingUp*, *ThumbUp*, *ThumbDown*.

and those are used as positive samples. The remaining 15 gestures are used as negative samples. The presence of a diverse negative sample collection allows us to evaluate how often the classifier recognizes an unknown gesture as a known one. The limitation of this dataset is that it's collected from only 18 users with limited variation in background. The classifier achieves 0.86% false positive rate and 44.4% recall rate on the dataset.

2.3. NN Gesture Classifier

The dataset used in Section 2.2 is for evaluation only and too small for training NN models. We collect and mine another in-house dataset containing 7307 images from 6478 users, representing a rich variety of hand shapes of both gestures and non-gestures in the wild. In addition to regular positive samples we collect easy and hard *Negative* samples. Please see Figure 6 for some example images.

We train an NN classifier to distinguish among six static hand gestures, namely, *OpenPalm*, *ClosedFist*, *PointingUp*, *Victory*, *ThumbUp*, *ThumbDown* and a background *Negative* class. The NN model consists of 3 fully connected layers of 50 neurons each. The model inputs are the intrinsic and extrinsic features computed in Section 2.2. We use focal loss [8] to deal with the class imbalance, where there are a lot more negative samples than positive samples in our dataset.

The NN classifier achieves an average recall rate of 87.9% across the 6 static gesture classes at a false positive rate of 1%.

3. MediaPipe Implementation

The proposed HGR system is implemented using the open source MediaPipe framework [1]. The system adds to MediaPipe Hands, primarily consisting of a hand-detection and a hand-keypoint component [14, 2], an additional gesture-classification component as discussed in Section 2.2 and 2.3.

In many applications, such as remote control, the user gestures only once in a while but the HGR is always running in the background. In order to reduce average computation requirement and maintain real-time performance across a wide range of devices, the HGR system hand de-

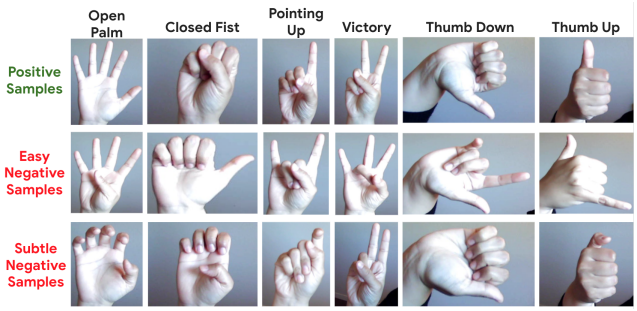


Figure 6. Some examples of true positive samples for gesture classes, easy samples for *Negative* hand shapes and subtle variations of hand shapes that should not be confused with the gesture class.

tection is configured to run only as needed, capped at a maximum frequency lower than the frequency of hand keypoint generation and gesture classification, by utilizing the flow-control and stream-synchronization support in MediaPipe [9] similar to [14]. When there's no hand in the camera view, the HGR runs hand detection at a lower frequency to save computation and power. As soon as a hand is detected, it's tracked at a higher frequency for better accuracy and temporal resolution. Furthermore, GPU acceleration is heavily exploited end-to-end, covering tasks like ML model inference as well as image and tensor processing, via OpenGL/OpenCL/Metal on mobile devices and WebGL locally in Web browsers (similar to the Web ML effort enabling background blur/replace in Google Meet [3]).

4. Applications

Our HGR can be used as an HCI mechanism for various applications, such as a virtual touchscreen for desktop computers, sending visual commands for robots, a controller for virtual reality gaming systems, and a remote control for large screen displays [11].

References

[1] MediaPipe. https://mediapipe.dev/, 2019. 3

[2] MediaPipe Hands. https://solutions.mediapipe.dev/hands, 2019. 1, 2, 3

[3] Background Features in Google Meet, Powered by Web ML. https://ai.googleblog.com/2020/10/background-features-in-google-meet.html, 2020. 4

[4] MediaPipe Hands Model Card. https://mediapipe.page.link/handmc, 2020. 1

[5] Valentin Bazarevsky, Ivan Grishchenko, Karthik Raveendran, Tyler Zhu, Fan Zhang, and Matthias Grundmann. BlazePose: On-device Real-time Body Pose Tracking, 2020. 2

[6] Quentin De Smedt, Hazem Wannous, and Jean-Philippe Vandeborre. Skeleton-based dynamic hand gesture recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pages 1–9, 2016. 1

[7] Guillaume Devineau, Fabien Moutarde, Wang Xi, and Jie Yang. Deep learning for hand gesture recognition on skeletal data. In *2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018)*, pages 106–113. IEEE, 2018. 1

[8] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pages 2980–2988, 2017. 3

[9] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. MediaPipe: A Framework for Building Perception Pipelines. In *CVPR Workshop on Computer Vision for AR/VR*, 2019. 4

[10] Munir Oudah, Ali Al-Naji, and Javaan Chahl. Hand gesture recognition based on computer vision: a review of techniques. *journal of Imaging*, 6(8):73, 2020. 1

[11] Siddharth S Rautaray and Anupam Agrawal. Vision based hand gesture recognition for human computer interaction: a survey. *Artificial intelligence review*, 43(1):1–54, 2015. 1, 4

[12] Hongyi Xu, Eduard Gabriel Bazavan, Andrei Zanfir, William T Freeman, Rahul Sukthankar, and Cristian Sminchisescu. GHUM & GHUML: Generative 3D Human Shape and Articulated Pose Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6184–6193, 2020. 2

[13] Andrei Zanfir, Eduard Gabriel Bazavan, Hongyi Xu, William T. Freeman, Rahul Sukthankar, and Cristian Sminchisescu. Weakly Supervised 3D Human Pose and Shape Reconstruction with Normalizing Flows. In *Computer Vision – ECCV 2020*, pages 465–481, 2020. 2

[14] Fan Zhang, Valentin Bazarevsky, Andrey Vakunov, George Sung, Chuo-Ling Chang, Matthias Grundmann, and Andrei Tkachenka. MediaPipe Hands: On-device Real-time Hand Tracking. In *CVPR Workshop on Computer Vision for Augmented and Virtual Reality, Seattle, WA*, 2020. 1, 2, 3, 4

Appendix A. In-house static gesture dataset: gesture code names

1. OpenPalm
2. Victory
3. ClosedFist
4. PointingUp
5. ThumbUp
6. ThumbDown
7. OK
8. CallMe
9. IndexMiddlePointingUp
10. Three
11. Four
12. ILoveYou
13. FingerHeart
14. HandHeart
15. IndexMiddlePointingUpWithClosedThumb
16. IndexMiddlePointingUpWithOpenThumb
17. IndexPointingToCamera
18. Loser
19. PinchedFingers
20. VulcanSalute
21. SignOfTheHorns

arXiv:2309.10858v1 [cs.CV] 19 Sep 2023

On-device Real-time Custom Hand Gesture Recognition

Esha Uboweja David Tian Qifei Wang Yi-Chun Kuo Joe Zou Lu Wang
George Sung Matthias Grundmann
Google LLC
1600 Amphitheatre Pkway, Mountain View, CA 94043, USA
{eshauboweja, dctian, qfwang, yichunkuo, zouj, luwa, gsung, grundman}@google.com

Abstract

Most existing hand gesture recognition (HGR) systems are limited to a predefined set of gestures. However, users and developers often want to recognize new, unseen gestures. This is challenging due to the vast diversity of all plausible hand shapes, e.g. *it is impossible for developers to include all hand gestures in a predefined list.*

In this paper, we present a user-friendly framework that lets users easily customize and deploy their own gesture recognition pipeline. Our framework provides a pre-trained single-hand embedding model that can be fine-tuned for custom gesture recognition. Users can perform gestures in front of a webcam to collect a small amount of images per gesture. We also offer a low-code solution to train and deploy the custom gesture recognition model. This makes it easy for users with limited ML expertise to use our framework. We further provide a no-code web front-end for users without any ML expertise. This makes it even easier to build and test the end-to-end pipeline. The resulting custom HGR is then ready to be run on-device for real-time scenarios. This can be done by calling a simple function in our open-sourced model inference API, MediaPipe Tasks. *This entire process only takes a few minutes.*

1. Introduction

Hand gesture recognition (HGR) plays a pivotal role in enabling natural and intuitive human-computer interactions, such as in augmented reality (AR), virtual reality (VR), video conferencing and remote control applications. As these technologies evolve, the ability to accurately detect, interpret and respond to hand gestures is key to creating immersive user experiences without disruption.

We present an innovative approach to train accurate and robust HGR models with limited training data. Our approach uses a pre-trained model that has been trained on a large dataset of videos of people fingerspelling words in sign language. We then fine-tune the weights of this pre-

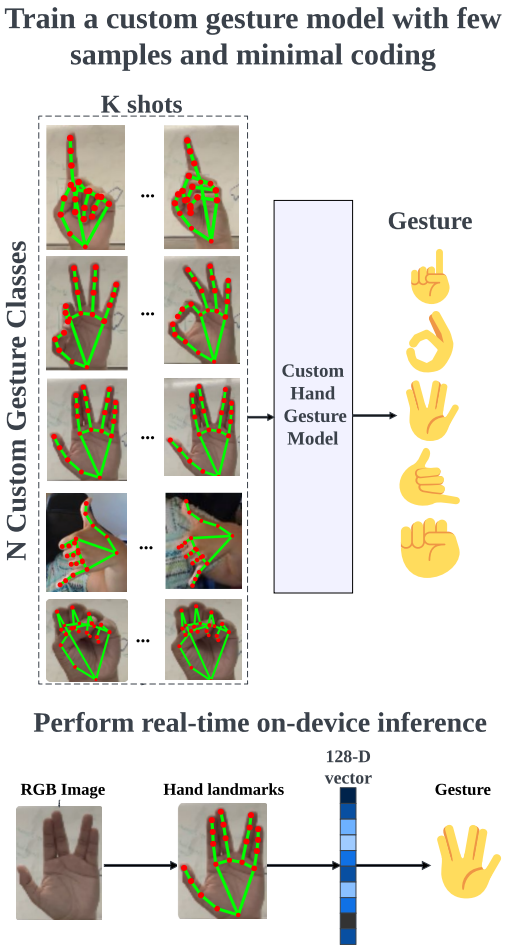


Figure 1. Our custom hand gesture recognition system enables any user without ML expertise to use a small number of images per gesture class for training and immediately use the model for real-time on-device inference. Here we show how our solution extracts the hand landmarks of each hand to compute a 128 dimensional embedding vector which is used for custom gesture classification. (The landmarks in this figure are best viewed digitally).

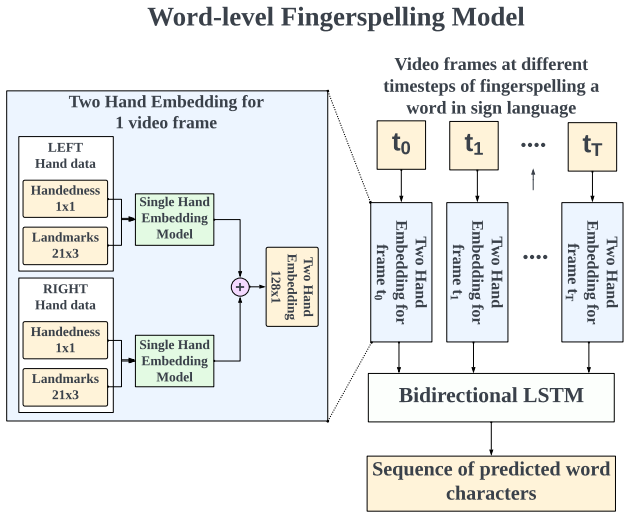


Figure 2. Model architecture used for training the *word-level fingerspelling* model and the *single-hand embedding* sub-model.

trained model for custom gesture classification (see Figure 1). This approach has two main benefits:

1. We are able to train an accurate model with a relatively small amount of training data, as few as 50 images per gesture.
2. The pre-trained model captures information of a wider range of hand shapes and movements, including transition states that are harder to capture with still images.

Our HGR inference pipeline works as follows:

1. An RGB camera captures an image.
2. The HGR extracts the 3D skeletal key points (or landmarks) and the handedness (left, right) of each hand from the input image.
3. The landmarks and handedness information are supplied to the newly trained custom gesture recognition model for inference.

Our HGR runs real-time at 30+ FPS (frames per second) on mainstream mobile devices.

2. Architecture

We use the work presented in "On-device Real-Time Hand Gesture Recognition" [1] as the starting point for building a system for custom hand gesture recognition. As shown in Figure 1, our solution uses a model that extracts hand landmarks and runs in real-time [2].

To train our *word-level fingerspelling* model, we use an in-house collected dataset of $79K$ videos of $21K$ unique fingerspelled words. In each video, a subject fingerspells a word using either the left or the right hand. During training we discard frames that don't contain any hands. We use normalized hand landmarks after processing the input videos using the hand landmark model [2].

As shown in Figure 2, the *word-level fingerspelling* model extracts embedding vectors from each hand's landmarks in each video frame. Since each frame contains only one of *left*, *right* hands, the model extracts a *single-hand embedding* for each hand and adds the two embedding vectors. Since addition is commutative, the model is invariable to the order of the two embedding vectors. The embedding vector of a single video frame contains structural piece-wise skeletal information. All per-frame embedding vectors along with hand location information are sent to a lightweight bidirectional LSTM [3] [4] to predict character level sequences of the fingerspelled word.

Using a Connectionist Temporal Classification (CTC) loss [5] for training the *word-level fingerspelling* model in Figure 2, we are able to guide the *single-hand embedding* sub-model to extract discriminative features that capture the subtle differences in a wide range of real-world hand configurations. We are thus able to use weights of the *single-hand embedding* model for training a custom gesture recognition model with minimal training data via transfer learning [6].

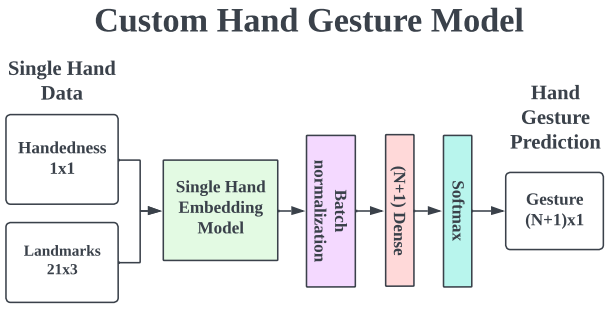


Figure 3. Custom hand gesture model. This model classifies the input hand data into one of $N+1$ classes,(N gestures and 1 *background* class).

We propose that the weights of the pre-trained *single-hand embedding* model represent essential features that are useful for custom gesture recognition. By fine-tuning the weights of the pre-trained embedding model and the custom hand gesture model head, we observe that our model can recognize gestures accurately. This approach significantly reduces number of images required for training. Figure 3 shows the model architecture of the custom hand gesture recognition model with the *single-hand embedding* model as its feature extractor.

3. Results

In Figure 4, we report the results of training a custom gesture recognition model by fine-tuning the weights of the *single-hand embedding* model (shown in Figure 3).

We used an in-house dataset of 8 classes, with 7 gesture classes and 1 *background* class. Samples that could not be labeled as any of the 7 gesture classes were labeled as the *background* class. To explore how much data is required to train the custom gesture recognition model, we conducted trials with varying values of the average number of training samples per gesture, K. We used the following values of K : $10, 20, 50, 100, 200, 500$. For example, when $K = 20$, we train a model with 20 positive and negative samples of each of the N gesture classes, *i.e.* the total number of samples used for training were $N \times 20$ (140 for 7 gesture classes). The negative samples are labeled as the *background* class.

During inference, an input hand shape can be labeled as one of the 8 classes. We report the performance on the 7 gesture classes. To account for the performance of the *background* class in our results, we use *specificity* and *sensitivity*:

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

True negatives account for samples that are correctly labeled as the *background* class. Similarly, *false positives* account for samples that belong to the *background* class but are incorrectly labeled as one of the gesture classes. To concisely represent our results, we combine *sensitivity* and *specificity* into one metric, namely the SS F_1score which is the harmonic mean of these two metrics:

$$\text{SS F}_1\text{score} = \frac{2 \times \text{Sensitivity} \times \text{Specificity}}{\text{Sensitivity} + \text{Specificity}}$$

Most of our models achieve SS F_1score values close to 1.0. So we present results in Figure 4 as

$$\text{complementary SS F}_1\text{score} = 1 - \text{SS F}_1\text{score}$$

This allows us to measure the model's performance by focusing on misclassification errors.

To explore the effectiveness of the *fine-tuned embedding* model for custom gesture classification, we conducted an ablation study on the model's weights. The models we trained for the study have the same architecture as the *fine-tuned embedding* custom gesture model. We defined two experiments:

1. *Random initial weights*: The initial weights of all layers are randomized, so the model trains on raw hand landmark data from scratch.
2. *Frozen embedding*: The weights of the *single-hand embedding* model layers are frozen. Only the weights of the classification head are updated during training.

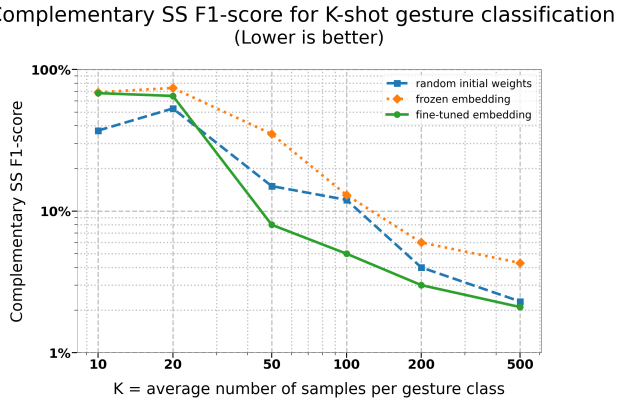


Figure 4. *Complementary SS F_1score* for K-shot gesture classification.

We report the results of K-shot gesture classification for these models in Figure 4. All models perform reasonably well when the value of K is high, *i.e.* $K = 500$. For very small values of K, *i.e.* $K = 10$ and $K = 20$, all models perform poorly. Note that the model with random initial weights performs well for these values of K but the *complementary SS F_1score* is still unacceptably much higher than 10%.

For values of $K = 50$ and above, we observe that the *complementary SS F_1score* is lower than 10%, steadily decreasing for higher values of K. The *fine-tuned embedding* model outperforms the other two models at $K = 50$, $K = 100$ and $K = 200$. These results demonstrate the advantage of fine-tuning a pre-trained *single-hand embedding* model instead of training a model with random initial weights to recognize hand gestures from raw hand landmarks.

4. Hand Landmark Detection Improvements

When two hands are very close to each other or occlude each other, the landmark model fails to accurately extract all hand landmarks for both hands. This failure cascades to the gesture recognition system that relies on accurate landmark detection to correctly infer the gesture depicted by a hand shape. In Figure 5 for example, we can see that the baseline hand landmark model is unable to extract landmarks of the right hand in panels (a) and (b).

To improve landmark accuracy when two hands are near each other or are overlapping with each other, we experimented with providing a handedness hint to the hand landmark model during training and inference. This guides the model to extract the landmarks of the hand with the same handedness as the input handedness hint.

In Figure 5, we can see that the new model extracts both the left and right hand's landmarks respectively with the correct handedness hint.

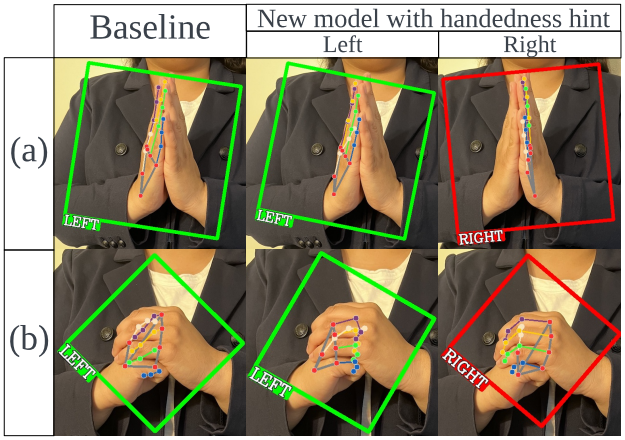


Figure 5. Results of using a hand landmark model with a handedness hint input compared to using the baseline hand landmark model without a handedness hint input.

Quantitatively, on an in-house dataset of $3,310$ images where hands are near or overlapping each other, the new model has a Mean Normalized Absolute Error (MNAE) of 13.09 compared to the baseline model's MNAE of 13.89. This improvement enables the custom hand gesture recognition pipeline to perform well when multiple hands are present in the input image or video.

5. Implementation

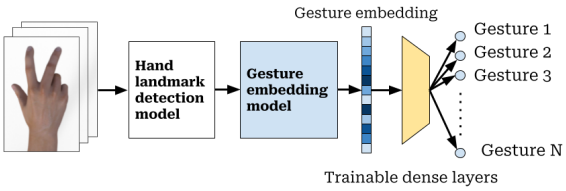


Figure 6. Training and Inference Pipelines.

5.1. Training Pipeline

We developed a low-code training pipeline called *MediaPipe Model Maker* [7], that enables users to effortlessly train new hand gesture recognition models. In the pipeline, the custom gesture recognition model is defined as a set of dense layers as shown in Figure 6. This model maps the gesture embedding vectors generated by the pre-trained *single-hand embedding* model to the target labels of the input images.

To train the custom gesture recognition model, users need to supply a small set of images. Each image should be annotated with a hand gesture label. All input images are pre-processed by the hand landmark model to generate hand landmarks on the fly during model training.

Our training pipeline allows the users to customize the neural network attributes such as the dense layer shapes and the training hyperparameters such as learning rate, batch size, and training epochs, etc. Because of low training data requirements, each training session only takes a few minutes on most local computers and on Google's public Colab [8] runtime to produce accurate gesture recognition models. The trained custom gesture model is then converted to a TFLite [9] model format for our end-to-end inference pipeline introduced below.

5.2. Inference Pipeline

The gesture recognition inference pipeline has been implemented as a modular structure, as shown in Figure 6.

The pipeline consumes a raw hand image sequence as input and processes all images sequentially. The hand landmark detection module converts the input images into landmark vectors. The gesture embedding module further maps the landmark vectors to 128-dimensional gesture embedding vectors. The gesture recognition module outputs the probabilities of each label. This modular graph structure allows users to control or replace any module as desired.

Our benchmarks show that this end-to-end pipeline achieves real time performance (16.76 ms per frame) on Pixel 6 devices.

Our inference pipeline *MediaPipe Tasks* [10] offers a user-friendly API that supports multiple platforms, including Java, Python, and WebJs. This API allows users to easily integrate their customized gesture recognition model into the pipeline.

Both the training and inference pipeline have been open-sourced via *MediaPipe Model Maker* [7] and *Gesture Recognizer API* in *MediaPipe Tasks* [10].

6. Conclusion

In conclusion, our research presents an easy-to-use approach to train accurate custom hand gesture recognition models with just a small set of training examples by fine-tuning pre-trained embeddings of hand landmarks. We also present our improvements to the hand landmark model which enhance the effectiveness of our hand gesture recognition system. These findings underscore the practicality of our custom hand gesture recognition system in real world scenarios and pave the way for better human-computer interactions in various domains, such as virtual reality, augmented reality, video conferencing and remote control applications.

References

[1] George Sung, Kanstantsin Sokal, Esha Uboweja, Valentin Bazarevsky, Eduard Gabriel Bazavan Jonathan Baccash, Chuo-Ling Chang, and Matthias Grundmann. On-device

Real-time Hand Gesture Recognition . In *ICCV Workshop on Computer Vision for Augmented and Virtual Reality*, 2021. 2

[2] Fan Zhang, Valentin Bazarevsky, Andrey Vakunov, George Sung, Chuo-Ling Chang, Matthias Grundmann, and Andrei Tkachenka. MediaPipe Hands: On-device Real-time Hand Tracking. In *CVPR Workshop on Computer Vision for Augmented and Virtual Reality, Seattle, WA*, 2020. 2

[3] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997. 2

[4] Alex Graves, Santiago Fernández, and Jürgen Schmidhuber. Bidirectional recurrent neural networks. *Computational Linguistics*, 31(2):343–365, 2005. 2

[5] Alex Graves, Santiago Fernández, and Jürgen Schmidhuber. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 369–376. ACM, 2006. 2

[6] Mohammadreza Iman, Hamid Reza Arabnia, and Khaled Rasheed. A review of deep transfer learning and recent advancements. *Technologies*, 11(2):40, mar 2023. 2

[7] Custom Gesture Training by MediaPipe Model Maker. https://developers.google.com/mediapipe/solutions/customization/gesture_recognizer, 2022. 4

[8] Google Colaboratory. https://colab.research.google.com, 2023. 4

[9] TensorFlow Lite. https://www.tensorflow.org/lite, 2023. 4

[10] Custom Gesture Inference by MediaPipe Tasks. https://developers.google.com/mediapipe/solutions/vision/gesture_recognizer, 2022. 4

[11] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift, 2015.

Questioning the AI: Informing Design Practices for Explainable AI User Experiences

Q. Vera Liao
IBM Research AI
Yorktown Heights, NY, USA
vera.liao@ibm.com

Daniel Gruen
IBM Research
Cambridge, MA, USA
daniel_gruen@us.ibm.com

Sarah Miller
IBM Research
Cambridge, MA, USA
millers@us.ibm.com

arXiv:2001.02478v3 [cs.HC] 3 Sep 2021

ABSTRACT

A surge of interest in explainable AI (XAI) has led to a vast collection of algorithmic work on the topic. While many recognize the necessity to incorporate explainability features in AI systems, how to address real-world user needs for understanding AI remains an open question. By interviewing 20 UX and design practitioners working on various AI products, we seek to identify gaps between the current XAI algorithmic work and practices to create explainable AI products. To do so, we develop an algorithm-informed *XAI question bank* in which user needs for explainability are represented as prototypical questions users might ask about the AI, and use it as a study probe. Our work contributes insights into the design space of XAI, informs efforts to support design practices in this space, and identifies opportunities for future XAI work. We also provide an extended XAI question bank and discuss how it can be used for creating user-centered XAI.

Author Keywords

Explainable AI; human-AI interaction; User experience

INTRODUCTION

The rapidly growing adoption of Artificial Intelligence (AI), and Machine Learning (ML) technologies using opaque deep neural networks in particular, has spurred great academic and public interest in explainability to make AI algorithms understandable by people. This issue appears in popular press, industry practices [2, 10], regulations [24], as well as hundreds of recent papers published in AI and related disciplines. These XAI works often express an algorithm-centric view, relying on "*researchers' intuition of what constitutes a 'good' explanation*" [63]. This is problematic because AI explanations are often demanded by lay users, who may not have deep technical understanding of AI, but hold preconception of what constitutes useful explanations for decisions made in a familiar domain. As an example, one of the most popular approaches to explain a prediction made by a ML classifier, as dozens of XAI algorithms strive to do [40], is by listing the features with the highest weights contributing to a model's prediction. For example, a model predicting a patient having the flu may explain by saying "the symptoms of sneeze and headache are contributing to this prediction" [74]. However, it is questionable whether such an explanation satisfies a doctor's needs to understand the AI, or adds significant value to a clinical decision-support tool.

To close the gap between XAI algorithms and user needs for effective transparency, the HCI community has called for interdisciplinary collaboration [4] and user-centered approaches to explainability [92]. This emerging area of work tends to either build on frameworks of human explanations from social science, or empirically study how explanation features impact user interaction with AI. In this paper, we take a complementary approach by investigating challenges faced by industry practitioners to create explainable AI products, with the goal of identifying gaps between the algorithmic work of XAI and what is needed to address real-world user needs.

Recently, an increasing number of open-source toolkits (e.g. [1, 2, 3, 10]) are making XAI techniques, which produce various forms of explanation for "black-box" ML models, accessible to practitioners. However, little is known about how to put these techniques from research literature into practice. As we will show, it is challenging work to bridge user needs and technical capabilities to create effective explainabilty features in AI products. This kind of work often falls to those with a bridging role in product teams–the design and user experience (UX) practitioners, whose job involves identifying user needs, communicating with developers and stakeholders, and creating design solutions based on demands and constraints on both sides. We study, therefore, how AI explainability is approached by design and UX practitioners, explore together with them how XAI techniques can be applied in various products, and identify opportunities to better support their work and thus the creation of user-centered explainable AI applications.

Given the early status of XAI in industry practices, we anticipate a lack of established means to uncover user needs or a shared technical understanding. Therefore, we develop a novel probe to ground our investigation, namely an XAI algorithm informed *question bank*. As an explanation can be seen as an answer to a question [63], we represent user needs for explainability in terms of the questions a user might ask about the AI. Drawn on relevant ML literature and prior work on question-driven explanations in various domains, we create a list of prototypical user questions that can be addressed by current XAI algorithms. These questions thus represent the

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
CHI '20, April 25–30, 2020, Honolulu, HI, USA.
© 2020 Association for Computing Machinery.
ACM ISBN 978-1-4503-6708-0/20/04 ...$15.00.
http://dx.doi.org/10.1145/3313831.3376590

current availability of algorithmic methods for AI explainability, allowing us to explore how they can be applied in various AI products, and identify their limitations for addressing real-world user needs. Our contributions are threefold:

- We provide insights into how user needs for different types of explainability are presented in various AI products. We suggest how these user needs should be understood, prioritized and addressed. We also identify opportunities for future XAI work to better satisfy these user needs.
- We summarize current challenges faced by design practitioners to create explainable AI products, including variability of user needs for explainability, discrepancies between algorithmic explanations and human explanations and a lack of support for design practices.
- We present an extended XAI question bank (Figure 1) by combining algorithm-informed questions and user questions identified in the study. We discuss how it can be used as guidance and tool to support the needs specification work to create user-centered XAI applications.

BACKGROUND

Explainable artificial intelligence (XAI)

Although XAI first appeared in expert systems almost four decades ago [22, 87], it is gaining widespread visibility as a field focusing on ML interpretability [19]. The term explainability is used by the research community with varying scope. In much of the ML literature, XAI aims to make the reasons behind a ML model's decisions comprehensible to humans [40, 59, 74]. In a broader view, explainability encompasses everything that makes ML models transparent and understandable, also including information about the data, performance, etc. [10, 45]. Our view aligns with the latter.

Recent papers surveyed this rapidly growing field and identified its key research threads [5, 19, 34, 40, 64, 73]. We will discuss taxonomies of XAI techniques in the next section. Another core thread is the evaluation of explanations, which answers whether an explanation is good enough, and how to compare different explanations. These questions are not only critical for choosing appropriate XAI techniques, but also underlie the development of intelligent systems that optimize the choice of explanation, such as interactive or personalized explanations [4, 81, 94]. Toward this goal, many sought to define the desiderata of XAI [19, 40, 45, 77], including fidelity, completeness, robustness, etc. Despite the conceptual discussions, there are few established means of quantifying explainability. Partly, the reason is that the effectiveness of an explanation is relative to the recipient, and on a philosophical ground, the question asked [17]. So the same explanation may be seen as more or less comprehensible to different users, or even to the same user engaged in a different understanding. Many therefore advocate that the evaluation of XAI needs to involve real users within the targeted application [30, 44].

Given its recipient-dependent nature, it is clear that work on XAI must take a human-centered approach. By conducting a literature review in social science on how humans give and receive explanations, Miller identified a list of human-friendly characteristics of explanation that are not given sufficient attention in the algorithmic work of XAI, including contrastiveness (to a counterfactual case), selectivity, social process, focusing on the abnormal, etc. Wang et al. [92] proposed a conceptual framework to connect XAI techniques and cognitive patterns in human-decision making to guide the design of XAI systems.

With a fundamental interest in creating user-centered technologies, the HCI community is seeing burgeoning efforts around designing and studying user interactions with explainable AI [14, 18, 21, 29, 45, 49, 54, 71]. As a literature analysis performed by Abdul et al. [4] shows, before this wave of work on ML systems, the HCI community have studied explainable systems in various contexts, most notably context-aware applications [12, 56, 57], recommender systems [42], debugging tools [53] and algorithmic transparency [28, 80]. Specific to XAI, recent studies largely focused on empirically understanding the effect of explanation features on users' interaction with and perception of ML systems, usually through controlled lab or field experiments. Notably, although explanations were found to improve user understanding of the AI systems, conclusions about its benefits for user trust and acceptance were mixed [21, 49, 54, 70], suggesting potential gaps between algorithmic explanations and end user needs.

Our work shares the goal of bridging between the algorithm-centric XAI and user-centered explanations. In contrast to prior work centered around end users, we focus on the people that engage in this bridging work, namely UX and design practitioners. By studying their current practices, we explore the largely undefined design space of XAI and identify challenges for creating explainable AI products.

Supporting AI practitioners

We join a growing group of scholars studying the work of industry practitioners who create AI products [7, 15, 45, 46, 67, 78]. By better supporting their work, we can ameliorate downstream usability, ethical and societal problems associated with AI. For example, Boukhelifa et al. [15] interviewed 12 data scientists to understand their coping strategies around uncertainty in data science work, and proposed a process model for uncertainty-aware data analytics. Holstein et al. interviewed 35 ML practitioners to conduct the first investigation of commercial product teams' challenges for developing fairer ML systems, and identified the disconnect between their needs and the solutions proposed in the fair ML research literature.

Most studies of AI practitioners focused on data scientists. As creating explainable AI products requires a user-centered approach, design practitioners should also perform an indispensable role. Despite a growing body of HCI work on AI technologies, there is a lack of design guidelines for AI systems. One notable exception is a recent paper by Amershi et al. [7], which synthesized a set of 18 usability guidelines for AI systems. Several of these guidelines (e.g., make clear what the system can do, how well it can do, why it did what it did) are relevant to explainability, but they do not provide actionable guidance on how to actualize these capabilities. Meanwhile, recent papers explored design methods supporting the creation of explainable AI systems. Wolf [96] proposed a scenario-based approach to identify user needs for explainability ("*what people might need to understand about AI systems*") early on in system development. Eiband et al. [31]

proposed a stage-based participatory design process, which guides product-specific needs specification–*what* to explain, followed by iterative design of solutions–*how* to explain.

Our work is motivated by a similar pragmatic goal of supporting design practices of XAI. More specifically, in the face of increasingly available XAI techniques, we are interested in the design work that connects user needs and these technical capabilities. In particular, we probe the challenges to identify the suitability of XAI techniques. A recent stream of guidance in the public domain (e.g. [1, 2, 10]) on how to select among XAI algorithms suggest their suitability can be difficult to determine. More problematically, these guidelines are targeting data scientists, using criteria grounded in the development process (e.g., explaining data or features, pre- or post-training). They do not directly address end user needs for understanding AI, nor support the navigation of the design space of XAI.

Question-driven explanations

Outside the ML field, many explored the space of user needs for explanation using a question-driven framework. Fundamentally, an explanation is "*an answer to a (why-) question* [63]." These questions are also user and context dependent, described as "triggers" by Hoffman et al. [44] representing "*tacitly an expression of a need for a certain kind of explanation...to satisfy certain user purposes of user goals*. "

In the early generation of AI work, question-driven frameworks were used to generate explanations for knowledge-based systems [20, 38, 88]. AQUA [72] is a reasoning model that uses questions to generate explanations and identify knowledge gaps for learning. AQUA was built upon a taxonomy of questions, including anomaly detection questions, hypothesis verification questions, etc. Silveira et al. provided a taxonomy of user questions about software to drive the design of help systems [82]. Building on it, Glass et al. [35] investigated users' explanation requirements in using an adaptive agent and showed that user needs for different types of explanation vary. In context-aware computing, Lim and Dey [56] developed a taxonomy of user needs for intelligibility by crowdsourcing user questions in scenarios of context-aware applications. These questions were coded into *intelligibility types*, including input, output, conceptual model (why, how, why not, what else, what if) and non-functional types (certainty, control). This taxonomy enabled a toolkit that supports the generation of explanations for context-aware applications [57].

Inspired by the prior work, we use an XAI question bank, containing prototypical questions that users may ask for understanding AI systems, as a study probe representing user needs for AI explanability. Instead of using question taxonomies that are not specific to ML, we start by performing a literature review to arrive at a taxonomy of existing XAI techniques, and use it to guide the creation of user questions. Thereby we constrain the probe to reflect the current availability of XAI techniques to understand how user needs for such explainability are presented in real-world AI products.

XAI QUESTION BANK

We now describe how we developed the XAI question bank by first identifying a list of explanation methods supported by current XAI algorithms, for which we focus on those generating post-hoc explanations for opaque ML models [9, 40]. For the scope of this paper, we will leave out the technical details of the algorithms but provide references for interested readers. There have been many efforts to create taxonomies of XAI methods [5, 9, 19, 34, 40, 64, 65, 73, 79]. Commonly, they differentiate between an explanation *method*–a pattern or mechanism to explain ML models–and specific XAI algorithms. One type of explanation method can be generated by multiple algorithms, which may vary in performance or applicability to specific ML models. Common dimensions to categorize explanation methods include: 1) The scope of the explanation, i.e. whether to support understanding the entire model (*global*) versus a single prediction (*local*); 2) The complexity of the model to be explained; 3) The dependency on the model used, i.e., whether the technique applies to any ML model or to only one type [5]; and 4) The stage of model development to apply the explanation [19].

Except for the first one, these dimensions are data scientist centric as they are concerned with the characteristics of the underlying model. For our purpose of mapping user questions, we seek a taxonomy that lists the forms of explanation as presented to users. For example, we disregard the complexity of the model or the explanation's applicability to specific models, but instead differentiate between methods that describe the model logic as *rules*, *decision trees* or *feature importance*. Also, to identify user questions an explanation addresses, we believe it is sufficient to stay at the general mechanism, and ignore the specificity of the presentation such as textual or by visualization [74]. Guided by these principles, we found the taxonomy of *explanators* in Guidotti et al. [40] closest to our purpose. Using it as a starting point, we consulted other survey papers and iteratively consolidated a taxonomy of explanation methods. In addition to the three categories in [40]–methods that explain the entire model (*global*), an individual outcome (*local*), and inspect how the output changes with instance changes (*inspect counterfactual*), we added *example based* explanations [45, 65], since they represent a distinct mechanism to explain. Finally, we arrived at the taxonomy presented in the second column of Table 1.

To map the explanation methods to user questions they can answer, we consulted prior work that provided taxonomies of questions for explanations [56, 57, 72, 82]. The closest to our purpose is the *intelligibility types* by Lim et al. [56, 57], developed by eliciting user questions in scenarios of context-aware computing. In particular, the intelligibility types of *How* (system logic), *Why* (a prediction), *Why not*, *What if* are directly applicable to ML systems. By mapping these questions to explanation methods listed in Table 1, we identified two additional types of question that can be addressed by existing XAI techniques: 1) *How to be that*: what are the changes required, often implying minimum changes, for an instance to get a different target prediction; 2) *How to still be this*: what are the permitted changes, often implying maximum changes, for an instance to still get the same prediction. We note that the questions of *What if*, *How to be...* are considered *counterfactual questions* and best answered by inspection or example based explanations, which allow users to understand the deci-

Category of Methods	Explanation Method	Definition	Algorithm Examples	Question Type
Explain the model (**Global**)	Global feature importance	Describe the weights of features used by the model (including visualization that shows the weights of features)	[41, 60, 69, 90]	**How**
	Decision tree approximation	Approximate the model to an interpretable decision-tree	[11, 47, 52]	**How**, Why, Why not, What if
	Rule extraction	Approximate the model to a set of rules, e.g., if-then rules	[26, 93, 102]	**How**, Why, Why not, What if
Explain a prediction (**Local**)	Local feature importance and saliency method	Show how features of the instance contribute to the model's prediction (including causes in parts of an image or text)	[61, 74, 83, 85, 101]	**Why**
	Local rules or trees	Describe the rules or a decision-tree path that the instance fits to guarantee the prediction	[39, 75, 99]	**Why**, **How to still be this**
Inspect counterfactual	Feature influence or relevance method	Show how the prediction changes corresponding to changes of a feature (often in a visualization format)	[8, 33, 36, 51]	**What if**, How to be that, How to still be this
	Contrastive or counterfactual features	Describe the feature(s) that will change the prediction if perturbed, absent or present	[27, 91, 100]	**Why**, **Why not**, **How to be that**
Example based	Prototypical or representative examples	Provide example(s) similar to the instance and with the same record as the prediction	[13, 48, 50]	**Why**, How to still be this
	Counterfactual example	Provide example(s) with small differences from the instance but with a different record from the prediction	[37, 55, 66]	**Why**, **Why not**, How to be that

Table 1. Taxonomy of XAI methods mapping to user question types. Questions in bold are the primary ones that the XAI method addresses. Questions in regular font are ones that only a subset of cases the XAI method can address. For example, while a global decision tree approximation can potentially answer *Why, Why not, and What if* questions for individual instances [58], the approximation may not cover certain instances.

sion boundaries of a ML model. Table 1 was reviewed by 4 additional experts working in the field of XAI.

Taking a broad view on explainability, we also consider descriptive information that could make a ML model more transparent. We added three more types based on [45, 56, 57]– questions regarding model *Input* (training data), *Output*, and *Performance*. In the rest of the paper, we refer these 9 types of questions as 9 *explainability needs categories* as they represent categories of prototypical questions users may ask to understand AI. For each category, we created a leading question (e.g.,"*Why is this instance given this prediction*" for the *Why* category[1]), and supplemented 2-3 additional example questions, inquiring about features and examples whenever applies. The list of questions developed in this step are shown in Figure 1 without an asterisk. We do not claim the exhaustiveness of this list, but deem it to be sufficient as a study probe.

STUDY DESIGN

We conducted semi-structured interviews with 20 UX and design practitioners recruited from multiple product lines at IBM. All but two (I-6 and I-20) informants worked on different products without shared AI models. Three informants were design team leads overseeing multiple products. The AI products included mature commercial platforms, systems in the testing phase, and internal platforms used by IBM employees. 50% of informants were female. All but two were based in the United States, in 7 different locations. Table 2 summarizes the primary areas of the products and informants' job titles. Our samples focused on AI systems that support utility tasks such as analytics and decision-support, as explainability is critical for high-stakes tasks where people would want to understand the AI's decisions [19, 30]. Informants were recruited from internal chat groups relevant to design and UX of AI products. The recruiting criteria indicated that one should have worked on the design of an AI product and had a good understanding

[1]We instructed that 'prediction' is used to refer the AI output for an instance. In the context of a product, it can mean a score/ recommendation/ classification/ answer, etc.

Technology area	Job title	Informant IDs
Business decision support	HCI researcher, Designer, Designer, Data scientist, User researcher	I-4, I-5, I-12, I-17, I-19
Medical analytic or decision support	Product lead, Design researcher, Design researcher, Designer, Design researcher	I-1, I-6, I-7, I-11, I-20
AI model training or customization tools	Designer, Project manager, Designer	I-10, I-14, I-15
Human resource support	Designer	I-3
Enterprise social	User researcher	I-9
Natural resource analytic	UX researcher	I-2
Customer care chatbot	UX researcher	I-13
Multiple areas	Design team leads	I-8, I-16, I-18

Table 2. Informants information

of its users, and mentioned that the interview would focus on user needs around understanding the AI.

We noticed that the current status of explainability in commercial AI products vary–about two thirds of the products (68.8%) have descriptive explanations about the data or algorithm, only a subset (37.5%) provide explanations for individual decisions, and certain products (e.g., chatbot) have neither. To explore the design space of XAI, we were interested in user needs for explainability uncovered by the design practitioners instead of the current system capabilities. The XAI question bank was able to scaffold the discussions, both to enumerate on the explainability needs categories, and to ground the discussion on user questions instead of venturing into the technical details.

Using MURAL–a visual collaboration tool, we created a card for each question category listed in Figure 1, with the leading and example questions (without an asterisk). Informant went through each card and discussed whether they encountered these questions from users; If not, we asked whether they saw the questions would apply and in what situations. After pilot testing, for efficiency, we combined the *Why* and *Why not* into one card to represent user needs to understand a prediction; and *What if*, *How to be that*, *How to still be this* into one card to

represent user needs to understand counterfactual cases. Thus there were 6 cards plus a blank card if one wanted to create an additional category. If time permitted, we asked informants to sort the cards according to their priority to address, and elicited the reasons for the ordering.

Interviews lasted 45-60 minutes, conducted remotely using a video conferencing system and MURAL. We started by asking informants to pick an AI product they worked on and had good knowledge of the users, in which they saw user needs for understanding the AI. We asked them to describe the system and the AI components. They could either use screen sharing or send screenshots to show us the system. We then asked whether the users had *needs to understand the AI*, and probed on why, when and where they had such needs (or lack thereof), and how the needs could be addressed, currently or speculatively. We then asked informants to reflect on what questions users would ask about the AI and listed as many as they could. User questions were also added to MURAL by the researchers if they appeared in other parts of the discussion. Thereby, we gathered user questions in a bottom-up fashion that allowed us to identify gaps in the algorithm-informed XAI question bank. It also prepared informants to move to discussions around the question cards. We closed the interview by asking informants to reflect on common challenges to build explainability features in AI products, and what kind of support they wished to have. For the three informants on lead roles, we focused on discussing the general status of explainability in AI products.

Analysis

Around 1000 minutes of interviews were recorded and transcribed, from which we extracted 607 passages broadly relevant to explainability. We performed open coding and axial coding on these passages as informed by Grounded Theory research [25]. The iterative coding was conducted by one researcher, with frequent discussions with the other researchers. We returned to the passages, interview video and the AI products repeatedly as necessary. The iterative coding process resulted in a set of 24 axial codes. We combined them into selective codes to be discussed as the main themes in the results section, where the axial codes are presented in bold.

Two additional sets of code were applied: 1)We identified 170 *user questions* appeared in the question-listing activity and the rest of the interviews. 2)We coded these questions and other passages, wherever applied, with the *explainability needs category*. The intersection of the two sets of code was 124 *covered questions*, as covered by the categories of the question bank, and the remaining 46 *uncovered questions*.

To perform gap analysis on the XAI question bank, we followed two steps. For the covered questions in each needs category, we identified new forms of questions that were not covered by the original example questions, as shown with asterisks in Figure 1. By *forms*, we grouped together questions with the same intent but phrased differently. For example, "*how was the data created*", and "*where did the data come from*" were both regarding the source of the training data, and covered by an original question in the *Input* category, while "*what is the sample size*" would be regarding a different characteristic of the input. In the second step, we examined the 46 uncovered questions. We first excluded 22 questions not generalizable to AI products, such as "*what is the summary of the article?*". We then iteratively grouped and coded the intent of the remaining 24 questions and identified 5 additional forms of question in the *Others* category in Figure 1. Insights from the analysis will be discussed the results section.

RESULTS

The results are divided into two parts. We start by discussing the general themes emerged in the interviews around the design work to create explainable AI products, which highlight some of the gaps between the algorithmic perspective of XAI and the practices to address user needs to understand AI. We then discuss how each category of user needs for explainability is presented in real-world AI products and based on that, reflect on the opportunities and limitations of XAI work.

From XAI algorithms to design practices

The diverse motivations for and utility of explainability

The historical context for the surge of XAI can be attributed to a fear of lacking understanding and control on increasingly complex ML models. Explanation is often embraced as a cure for "black box" models to gain user trust and adoption. So a common pursuit is to produce interpretable, often simplified, descriptions of model logic to make an opaque ML model seen as transparent. This is a necessary effort, but insufficient to deliver a satisfying user experience if we ignore users' motivation for explanations. As I-8 put: "*Explainability isn't just telling me how you get there, but also, can you expand on what you just told me...explanation has its own utility*".

We identified several utility goals driving user demands for explanations of AI. In the context of AI-assisted decision-making, explanations are most frequently sought to **gain further insights or evidence**, as users are not satisfied by merely seeing a recommendation or score given by the AI. There are several ways people use these insights. When seeing disagreeable, unexpected or unfamiliar output, explanations are critical for people to assess the AI's judgment to **make an informed decision**. Even when users' decision aligns with the AI's, explanations could help **enhance decision confidence** or **generate hypothesis about the causality** for follow-up actions, as illustrated by I-5's comment, who worked on a tool supporting supply chain management: "*users need to know why the system is saying this will be late because the reason is going to determine what their next action is...If it's because of a weather event, so no matter what you do you're not going to improve this number, versus something small,if you just make a quick call, you can get that number down*." In some cases, users also deem explanations of the AI's decision as potential **mitigation of their own decision biases**.

To **appropriately evaluate the capability of** the AI system is identified as the second theme of motivation, both to **determine the overall system adoption** (e.g., evaluating data quality, transferability to a new context, whether the model logic aligns with domain knowledge), and at the operational level to **beware of the system's limitations**. I-6 commented on why explanations matter for users of a medical imaging system: "*There is a calibration of trust, whether people will use it over time. But also saying hey, we know this fails in this*

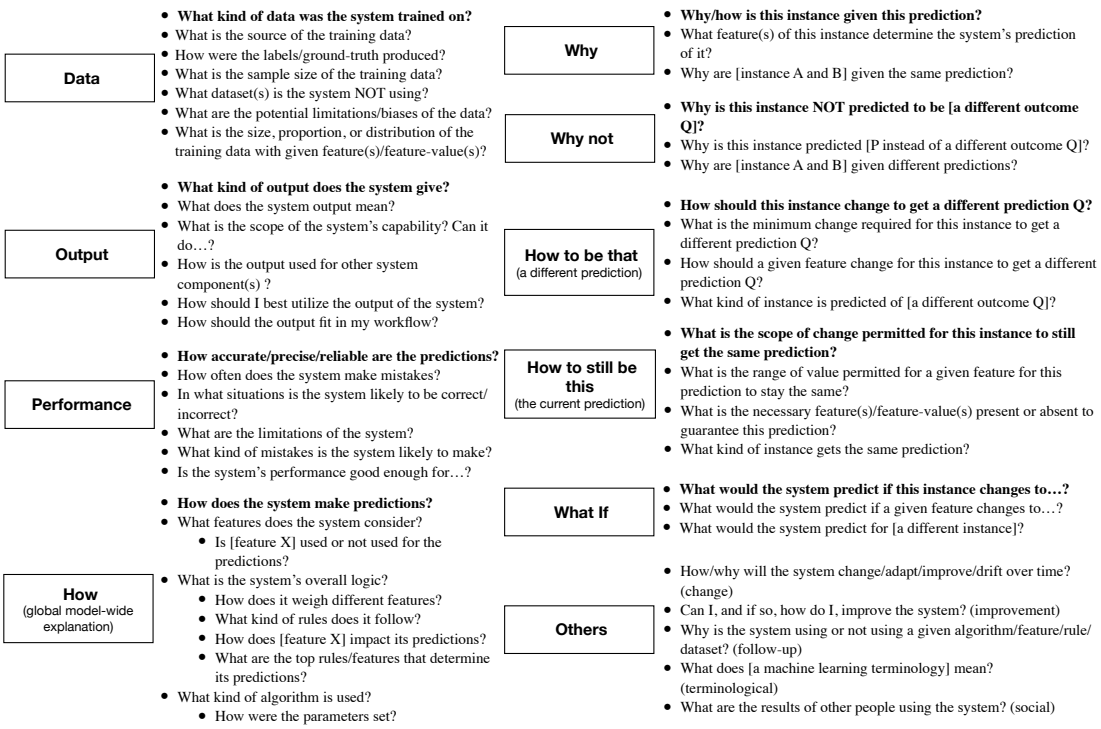


Figure 1. XAI Question Bank, with leading questions in bold, and new questions identified from the interviews with * (updated September 2021)

way." We note that appropriating trust should be distinguished from *enhancing trust*. Though from a product team's perspective, the concern is often on users' under-trusting of the AI system and explanations are sought to improve adoption.

The third theme of motivation for explainability is to **adapt usage or interaction behaviors to better utilize the AI**. I-7 described users' desire to understand how the AI extracted information from clinic notes so they could adapt their notes-taking practices. I-17 mentioned users of a sales inventory management tool would want to focus on cases where the AI prediction was likely to err. I-13 commented that explanation could suggest to chatbot users what kind of things they could ask. Furthermore, explanations could also **convince users to invest in the system** "*if they know how the system will improve*"(I-11) (e.g., access to personal information, feedback).

Several informants working on AI systems supporting analysts' work or model training tools considered explanations as an integral part of a "*feedback loop*" (I-11) to **improve AI performance**. Such needs are not only seen in debugging tools [45], but also in cases where the user could manipulate the data or correct the instance: "*[Explaining] why it thinks we are where we are and the opportunity to say, 'no, I need you to just understand that we're in Phase 2"'* (I-5).

Last but not least, informants reflected on their **ethical responsibilities** to provide explanations:" *What are we responsible for as creators of tools... whether it's out of the kindness of our hearts or whether it's because there's a true risk to others or society... We have to provide that level of explainability*"(I-8). While some of these motivations have been discussed conceptually in prior work [5, 9, 19, 30, 40], our study provides concrete examples in real-world AI products. It is worth noting that the motivation for explainability is grounded in users' system goals such as improving decision-making, so explanation is not merely to provide transparency but support downstream user actions such as adapting interactions or acting on the decisions. Unpacking the motivation and downstream user actions should ultimately guide the selection of explanation methods. For example, if the goal is to gain further insights, example-based explanations could be more useful than feature-based explanations that describe the algorithm's logic, as I-2 described: "*[Users of natural resource analytic tools] already rely a lot on the analogy... [Similar examples] are very good to their study, e.g. give clues on which year this was formed.*"

Moreover, unpacking the motivation may help foresee the limitations of algorithmic explanations and fill the gaps in designing user experiences. For example, if the motivation is to mitigate biases, then users may desire to see "*both positive and negative evidence*" (I-1). If it is to support adaption of interaction, the system could supplement information of "*this is what other people do*" (I-13). Some informants criticized designing for the mental model of AI as a decision-maker explaining its rationale, and argued to focus on what utility explanations could provide to support users' end goals. As I-1, who worked on a clinical decision-support tool, put:"*[explanations by system rationale] are essentially 'this is how I do it, take it or leave it'. But doctors don't like this approach...Thinking that [AI is] giving treatment recommendations is the wrong place*

to start, because doctors know how to do it. It's everything that happens around that decision they need help with... more discussions about the output, rather than how you get there."

In quest for human-like explanations

Explanation is an integral part of human communication, and invites preconception of what constitutes effective and natural ways to explain. Informants were constantly dealing with discrepancies between algorithmic and human explanations. Some of the discrepancies are inherent to the mechanisms of AI algorithms, such as using features or learned patterns that are not aligned with how people make decisions. Others are in the forms constrained by technical capabilities and foreign to human interactions. For example: "*People have this unspoken norm, I trust you and if you are not sure you would let me know. But nobody goes around saying how confident they are in the thing that they're saying. It may be implied in the language they're using. So a system has high precision but 67% confidence...it is a stupid and hard to use metric* " (I-5).

Several informants attempted to **mimic how people, especially domain experts, explain** in their design work. By aligning with how humans explain, it aligns user perception of the AI with existing mental model of decision-making in the domain, suggested by prior work as critical to build trust [84]. This is best exemplified by I-1's work in designing explanations for a clinical-decision support system that performs information extraction from medical literature: "*We mirror the way a doctor would do. So if a doctor was asked, how would you go and find the evidence? ... You went to PubMed you found a paper, the paper matches my patient... you're showing me the statements in the paper [on whether] it was a good or bad idea, and putting all that together...So when you manage to reflect with AI literally how the doctor thinks about the problem, the black box kind of disappears.*"

We identified several themes on the desirable properties of human explanation that echoed Miller's arguments on informing XAI with how humans explain [63]. First, explanations are **selected**, often focusing on one or two causes from a sometimes infinite number of causes [63]. Informants discussed the importance of selectivity as "*a balance of providing enough information that is trustworthy and compelling without overwhelming the user*" (I-11) and acknowledging that "*AI will have a degree of [randomness] and may not be 100% explainable*" (I-8). Second, explanations are **social**, as part of a conversation and presented based on the recipient's current beliefs [63]. This social aspect is not only seen in tailoring explanations "*for people with different backgrounds*" (I-4), but also in accommodating the evolving needs for explanation as one builds understanding and trust during the interaction process, "*once we trust it, it's about going deeper into it, the kind of questions goes from broad to ultra-specific* " (I-8).

The selective and social nature of explanation has made many to argue that XAI has to be *interactive* or even *conversational* [62, 63, 94], tailoring explanation to different questions asked by different users, who would also ask follow-up questions to keep closing the gap of understanding, a process known as grounding in human communication [23]. Following prior work [56, 57, 72, 94], we postulate that a question-driven framework provides a viable path to interactive explanations.

XAI: challenges and needs in design practices

It is challenging work to create design solutions that bridge user needs for explainability and technical capabilities. The satisfaction of user needs is frequently **hampered by the current availability of XAI techniques**, which we will discuss in detail in the next section. Informants also had to work with other **product goals that are at odds with explainability**, such as protecting proprietary data, avoiding legal or marketing concerns from exposing the AI algorithm. Sometimes, explainability presents challenges to other aspects of user experience –"*any opportunities we have to give them more explainability comes at the cost of the seamless integration. And [doctors] are just so clear that not breaking their workflow is the most important factor to their happiness*"(I-6). Or, it might expose corner cases or rationales that some individuals found "*wrong*" (I-2), "*over-simplified*" (I-9), or "*outdated*" (I-5), resulting in unnecessary user aversion and making the product "*victim of trying to be too transparent*" (I-2).

In addition, unlike XAI algorithmic work's focus on one or a class of AI algorithm, creating explainable AI products requires **a holistic approach** that links multiple algorithms or system components to accommodate users' goal of better understanding and interacting with the system, as described by I-4: "*There is the traditional what we think about XAI, explaining what the model is doing. But there is this huge wrapper or the situation around it that people are really uncertain... what do I need to do with this output, how do I incorporate it into other processes, other tools? So it is thinking about it as part of complex systems rather than one model and one output.*"

In short, inherent tension often exits between explainability and other system and business goals. Design practitioners are eager to advocate for explainability, but the realization requires teamwork with data scientists, developers and other stakeholders. Their advocacy is often hindered by **skill gaps** to engage themselves and the team in "*finding the right pairing to put the ideas of what's right for the user together with what's doable given the tools or the algorithms*"(I-8), and the **cost of time and resource** that a product team is reluctant to invest with a release schedule. These challenges can potentially be overcome by having resources that help sensitizing designers to the design opportunities in algorithmic explanations [97] and enable conversations with the rest of the team, as expressed by many informants. We summarize informants' desirable support for XAI design in two areas:

1. Guidance for **explainability needs specification**, for which we saw requests for both: 1) general principles of what types of explainability should be provided, as heuristic guidelines that a product can be developed or evaluated with; 2) guidance to identify product, user, and context specific needs to help the product team prioritize the effort.
2. Guidance for creating **explainability solutions** to address user needs, paired with example artifacts (e.g., UI elements, design patterns), to support the exploration of tangible solutions and communication with developers and stakeholders.

The two areas correspond to the *what to explain* and *how to explain* stages in Eiband et al.'s design process for transparent

interfaces [31]. We argue that the question bank could potentially support needs specification work, as it essentially lays out the space of users' prototypical questions to understand AI systems. The above requests suggest the needs to further understand the key factors that may lead to variability of user questions, and how these questions should be appropriately answered. We work towards these goals in the next section.

Understanding user needs for explainability

We use the *explainability needs category* codes to guide our analysis on each category. We focus on two questions: 1) The variability of explainability needs, i.e., what factors make a category of user questions more or less likely to be asked. 2) The potential gaps between algorithmic explanations and user needs, by examining passages coded as design challenge, and the additional questions identified in the gap analysis (Figure 1). To help answer the former, we first discuss key factors that may lead to the variability of explainability needs, which we identified by coding informants' reasons to include, exclude or prioritize a needs category.

- **Motivation for explainability**:The diverse motivations discussed in the last section for demanding explainability could lead to wanting different kinds of explanation.
- **Usage point**: Informants mentioned common points during the usage of AI systems where certain type of explianabiltiy was of particular interest, including on-boarding, reliance or delegation to AI, encountering abnormal results, system breakdown, and seeing changes in the system.
- **Algorithm or data type**: Different algorithms invoke different questions. For lay users, it might be more relevant to consider the type of data the AI is used with rather than specific algorithms, e.g., tabular data, text, images or video.
- **Decision context**: We identified codes describing the nature of the decision context that led to prominent needs for certain type of explainability, including outcome criticality, time-sensitivity, and decision complexity.
- **User type**: Codes describing the characteristics of users include AI knowledge, domain knowledge, attitude towards AI, roles or responsibilities.

In prior work, the variability of user needs for explaianability has been discussed regarding the roles of the users [9, 10, 43, 79, 95], e.g., regulators, model developers, decision-makers, consumers. The diverse criteria used by our informants suggest many other factors to consider for the suitability of XAI techniques. This paper does not conclude on how these factors vary user needs. Rather, they should be seen as *sensitizing concepts* by Bowen's [16] and Ribes's [76] definitions–"*tell where to look but not what to see.*" The sheer number of these factors highlight the challenge to pre-define users' explainability needs, vindicating the recent effort [31, 96] to provide structured guidance to support empirically identifying application-specific user needs. Below we present informants' discussions on each category of explainability needs and highlight how these factors heighten the needs (in italic).

Input/data

Understanding training data for the AI model was most frequently seen to serve the motivation to *appropriately evaluate AI capabilities* for use. It was considered a prominent need during the *on-boarding* stage, and by both the *decision-makers* and people in *quality-control roles*. Explanations of data were also important in cases where the users could directly manipulate the data to either *adapt the usage to better utilize the AI* or to *improve the AI performance*.

Additional questions identified from the gap analysis indicate a desire to gauge the AI's limitations by inquiring about the sample size, potential biases, sampling of critical sub-groups, missing data, etc. Additional codes include to understand the system's compliance with regulations regarding data sampling, and transferability of the AI model: "*Not necessarily source, but more conceptual like...[are we] making the solutions based on what occurred yesterday*" (I-4). These patterns imply that users demand **comprehensive transparency of training data, especially the limitations**.

Output

While understanding the output is often an neglected aspect in algorithmic work of XAI, we saw frequent questions on it, indicating users' desire to understand the value of the AI system to *appropriately evaluate the capability* and to *better utilize the AI*, often in the *on-boarding stage* or dealing with *complex decisions*. Explaining output and explaining input/data were considered as "*static explanations*" that more likely come up in the early stage of system usage, instead of frequent "*day-to-day, or transaction-to-transaction interactions*" (I-8).

The most frequently asked questions were not regarding descriptive information of the algorithmic output, but at a high level, inquiring how to best utilize the output. We also identified two additional questions–"*the scope of the capability*", and "*how the output impacts other system components*." To address such user needs requires **contextualizing explaining the system's output in downstream system tasks and the users' overall workflow**.

Performance

To our surprise, the performance category was repeatedly ranked at the bottom, especially for users *without AI background* and in decision contexts considered *less critical*. There was a common hesitation among informants to present ML performance metrics such as accuracy, not only because a numerical value could be hard to interpret by lay users, but also there may be discrepancy between performances on the test data and the actual data, creating different "experienced accuracy" [98] that might deter users. Some also believed that small differences in these metrics would not change how users interact: "*Technically that's great, but, it's still not a hundred... there's always going to be work that the users have to do to verify or double check*" (I-4).

As many informants pointed out, and suggested by the additional questions, the goal of explaining performance should be to **help users understand the limitations of the AI, and make it actionable** as to answer "*Is the performance good enough for...*" There are constraints of technical capabilities. For example, confidence scores were repeatedly dismissed as not providing enough actionability –"*[users] struggle to really understand, does it mean it's going to do what I want it to do, or, can I trust it?* " (I-15). Regarding the additional

question on "*What kind of mistake*", informants mentioned the precision-recall trade-off is a deliberately decided limitation that should be explained as it might change users' course of actions [49] :"*It's use case dependent... for the [doctors] if they miss a tumor, that's a life changing. So they have a very high tolerance for false positives* "(I-7).

How–global model

Informants recognized the importance of providing global explanations on how the AI made decisions, both to help users *appropriately evaluate the system capabilities*, and build a mental model to *better interact with* or *improve the system*. Such needs were prominent in cases where users were in a *quality-control* role, or in a position able to *adjust the model or the data-collection process*: "*The company really care about which of these attributes are the most important... then they will forward the manufacturer to include those in the data"* (I-17). Informants also agreed that users with *AI or analytic background* were more likely to seek global explanations.

As Table 1 shows, to answer the *How* question, XAI algorithms commonly employ ranked features, decision trees or rules. However, some informants were referring to high-level descriptions, such as "*I would just say keywords matching, it is intuitive, and it's been around*" (I-3). Some were also concerned about fitting a complete *How* explanation into the users' workflow: "*I can't imagine [doctors are] going into their workflow and be like, I'm so busy, let me read more about this AI. But, they would probably want some kind of confirmation about how it makes decisions*" (I-11). So the design challenge is to identify the **appropriate level of details to explain the model globally**. This challenge is reflected in the question bank as well. While most XAI methods focus on answering "*What is the overall logic*", we discovered that many questions were simply asking about the top features or whether certain feature was used, meanwhile a small set of questions by users with AI background were regarding the technical details of the model.

Why, Why not–local prediction

Understanding a particular decision was often ranked at the top, and in user questions mentioned in all products. These questions were naturally raised after a surprising or abnormal event: "*For everyday interactions, most likely it's how did the system give me this answer? Not just any answer, but all of a sudden, here's this thing that I'm [not expecting] seeing*" (I-8). This pattern is pointed out by Miller [63] as the *contrastive* nature of human explanations, which are often implying *Why not* the expected event. We observed a shared **struggle with available technical solutions answering *Why* but not *Why not***. Several informants working on text-based ML commented on the inadequacy of the common approach by highlighting keywords that contribute to the prediction:"*even though we explained conceptually how it's working, it wouldn't be able to explain that error. So it would actually be counter-intuitive why it should make that error*" (I-4). I-17 discussed the limitation of a state-of-the-art explanation algorithm, LIME [77], which generates feature importance for "black-box" ML models. She found the static explanation to be unsatisfying: "*LIME would say 'it is boot cut which is why [it's not going to sell]', but would it be different if it was a skinny cut?*"

Many current XAI algorithms focus on the *Why* question. We note that a challenge for algorithmic explanations is that the contrastive outcome is often not explicitly available to the model. These observations again suggest the benefit of interactive explanations, allowing users to explicitly reference the contrastive outcome and asking follow-up *What if* questions.

What if, How to be–inspecting counterfactual

This category of explainability needs was not ranked high, and informants mentioned only 3 related user questions. Currently, these kinds of explanation are not widely adopted in commercial AI products. As prior work suggested, awareness of new types of explainability could change user demand [56]. In fact, informants recognized its **potential utility as system features to test different scenarios** for users to *gain further insights* for the decision, and to **understand the boundary of system capabilities** to enable *adapting interaction behaviors*. Informants also identified that such features align with how data scientists currently debug to *improve ML models*. For example, I-4 was excited to consider how *What if* explanations might support supply chain managers make decisions–"*you can run different scenarios... the system can make an initial recommendation and then they can tweak it to see, the impact on the cost after that.*" I-13 speculated that *How to be that* explanations (how the chatbot would understand differently) could help chatbot users better phrase their queries. I-15 working on a tool for customizing entity recognition models commented that seeing how instance changes impact the output could help users debug the training data.

As seen in Table 1, there is a growing collection of XAI techniques addressing the counterfactual questions. However, currently the feature influence methods are mostly used in data science tools [45]. Contrastive feature and example based methods are relatively new areas of XAI work [27, 91, 100]. Our results suggest their potentials as utility features in a broad spectrum of AI products. Future work should explore these potentials and sensitize practitioners to these possibilities.

Additional explainability needs

We also identified a set of questions that were not covered by the algorithm-informed needs categories. They point to additional areas of interest that users have for understanding AI. A critical area is to **understand the changes and adaption of AI**, both in explaining changes in the system, and how users can change the system. Other areas are **follow-up questions** by further inquiring why a certain feature or data is used, and **terminological questions** such as "*what do you mean by...*", both of which may naturally emerge in an interactive explanation paradigm. Lastly, some users might be interested in knowing other people's experience with the system, suggesting a new mechanism for an AI system to provide **social explanations** with regard to other users' outcomes and actions.

DISCUSSION

With widespread calls for transparent and responsible AI, industry practitioners are eager to take up the ideas and solutions from the XAI literature. However, despite recent effort toward a scientific understanding of human-AI interaction [29, 68, 103], XAI research still struggles with a lack of understanding of real-world user needs for AI transparency, and by far little

consideration of what practitioners need to create explainable AI products. Our study suggests the following directions both for algorithmic work to close the gaps addressing user needs, and design support to reduce technical and practical barriers to create user-friendly XAI products.

- XAI research should direct its attention to techniques that address user needs, and we suggest a question-driven framework to embody these needs. Our results point to a few common questions and their desired answers that future work of XAI should explore, for example, *How* question answered by multi-level details describing the algorithm, *Why not* question referencing an expected alternative outcome, and *How/Why will the system change*. Considering the coverage of user questions, especially common and new questions identified, could help the community move toward more human-centric effort. The question bank presented in this paper is just a starting point. Future work could continue building the repository by directly eliciting questions from end users of different types of AI systems [56].
- Practitioners struggle with the gaps between algorithmic output and creating human-consumable explanations. To close the gap requires inter-disciplinary work that studies how humans explain, and formalizes the patterns in algorithmic forms. Such a practice has already been engaged in interactive ML [6, 86] and "socially guided machine learning" [89]. Prior work repeatedly pointed out that a prerequisite for explanations to be truly human-like is to be *interactive* [62, 63, 94], because explanation is a grounding process where people incrementally close the belief gaps. Indeed, our study found that some user questions are closely connected with or followed by other questions. Future work could explore interaction protocols, for example through statistical modeling of how humans ask different explanation-seeking questions [62], to drive the flow of interactive or conversational explanations.
- Our study revealed the variability of user questions and its complex mechanisms, highlighting the challenge to identify product-specific user needs. While prior work attempted at top-down descriptions of needs of users in different roles [9, 10, 43, 79, 95], it may not be sufficient for design work that has to consider specific actions, usage points, models, etc. Recent HCI work on XAI encourages empirically identifying user needs with structured procedures [31, 96]. We suggest several ways the XAI question bank can be used for needs specification. First as heuristic guidance, a product team could enumerate on whether each question category has been addressed and which should be prioritized. Second, it can be used in user research to scaffold the elicitation of user needs. For example, card-sorting exercises of the questions can be performed with users (adaptation may be required for specific AI applications). We invite practitioners to use, revise and expand the XAI question bank.
- The technical barriers for designers and practitioners in general to navigate the space of XAI remains a primary challenge for product teams to optimize XAI user experiences. To support design work for ML, Yang suggested research opportunities to "sensitize designers to the breadth of ML capabilities" [97]. Informants also expressed strong desire for support of technical discussions with data scientists and stakeholders, as mitigating the friction is critical for the success of their advocacy for explainability. An opportunity for sensitizing support is to create concrete mapping between user questions and algorithmic capabilities, serving as a shared cognitive artifact between the designers and data scientists. One example, perhaps over-simplified, is the taxonomy of XAI methods we presented in Table 1. We may envision a *question-driven design process*: by user research, a design practitioner identifies the primary type of user question as *what to explain* (e.g. How to be that), and also the requirements to address the question as *How to explain*. Table 1 then suggests candidate explanation method(s) to answer the question (e.g. contrastive features). Together with a data scientist, the team find the most suitable solution to implement from the list of suggested algorithms, then work toward closing the gaps between the algorithmic output and user requirements to answer the question. By suggesting conceptually this question-driven design process, we invite the research community to develop more fine-grained frameworks of XAI features (e.g., considering UI patterns) that connect user questions and technical capabilities.

Limitations

First of all, the user questions were explored through design practitioners instead of end users, so we cannot claim this is a complete analysis of user needs for explainability. The results only reflect design practitioners' views. Future work could study other roles involved in AI product development (e.g., data scientists) to better understand the challenges to create XAI products. Our product samples focus on ones supporting high-stakes tasks, where needs for explainability might have been greater, and the current status of XAI more advanced. We do not claim the completeness of the XAI methods discussed, especially as this is a fast advancing research field. Practitioners' increasing accessibility to XAI techniques may also change the demands and concerns expressed in the study. Finally, our informants worked for the same organization. Although this is not uncommon for studies of practitioners [7, 32, 67] and we recruited informants from diverse product lines and locations, we acknowledge that design practices may be different in other companies or organizations.

CONCLUSION

Although the research field of XAI is experiencing exponential growth, there is little shared practices of designing user-friendly explainable AI applications. We take the position that the suitability of explanations is *question* dependent and requires an understanding of user questions for a specific AI application. We develop an XAI question bank to bridge the spaces of user needs for AI explainability and technical capabilities provided by XAI work. Using it as a study probe, we explored together with industry design practitioners the opportunities and challenges in putting XAI techniques into practice. We illustrated the great variability of user questions that may subject to many motivational, contextual and individual factors. We also identified gaps between current algorithmic solutions of XAI and what's needed to deliver satisfying user experiences, in the types of user questions to address and how they are addressed. We join many others in this field advocating

a user-centered approach to XAI [4, 30, 63, 92]. Our work suggests opportunities for the HCI and AI communities, as well as industry practitioners and academics, to work together to advance the field of XAI through translational work and shared knowledge repository that maps between user needs for explainability and XAI technical solutions.

ACKNOWLEDGEMENTS

We thank all our anonymous participants. We also thank Zahra Ashktorab, Rachel Bellamy, Amit Dhurandhar, Werner Geyer, Michael Hind, Stephanie Houde, David Millen, Michael Muller, Chenhao Tan, Richard Tomsett, Kush Varshney, Justin Weisz, Yunfeng Zhang, and anonymous CHI 2020 reviewers for their helpful feedback.

REFERENCES

[1] 2018. DALEX: Descriptive Machine Learning EXplanations. (2018). Accessed September 18, 2019 from `http://uc-r.github.io/dalex`.

[2] 2018. H2O Driverless AI. (2018). Accessed September 18, 2019 from `https://www.h2o.ai/products/h2o-driverless-ai/`.

[3] 2019. Alibi. (2019). Accessed September 18, 2019 from `https://github.com/SeldonIO/alibi`.

[4] Ashraf Abdul, Jo Vermeulen, Danding Wang, Brian Y Lim, and Mohan Kankanhalli. 2018. Trends and trajectories for explainable, accountable and intelligible systems: An hci research agenda. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. ACM, 582.

[5] Amina Adadi and Mohammed Berrada. 2018. Peeking inside the black-box: A survey on Explainable Artificial Intelligence (XAI). *IEEE Access* 6 (2018), 52138–52160.

[6] Saleema Amershi, Maya Cakmak, William Bradley Knox, and Todd Kulesza. 2014. Power to the people: The role of humans in interactive machine learning. *AI Magazine* 35, 4 (2014), 105–120.

[7] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, and others. 2019. Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 3.

[8] Daniel W Apley. 2016. Visualizing the effects of predictor variables in black box supervised learning models. *arXiv preprint arXiv:1612.08468* (2016).

[9] Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Bennetot, Siham Tabik, Alberto Barbado, Salvador García, Sergio Gil-López, Daniel Molina, Richard Benjamins, and others. 2019. Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI. *Information Fusion* (2019).

[10] Vijay Arya, Rachel KE Bellamy, Pin-Yu Chen, Amit Dhurandhar, Michael Hind, Samuel C Hoffman, Stephanie Houde, Q Vera Liao, Ronny Luss, Aleksandra Mojsilović, and others. 2019. One Explanation Does Not Fit All: A Toolkit and Taxonomy of AI Explainability Techniques. *arXiv preprint arXiv:1909.03012* (2019).

[11] Osbert Bastani, Carolyn Kim, and Hamsa Bastani. 2017. Interpretability via model extraction. *arXiv preprint arXiv:1706.09773* (2017).

[12] Victoria Bellotti and Keith Edwards. 2001. Intelligibility and accountability: human considerations in context-aware systems. *Human–Computer Interaction* 16, 2-4 (2001), 193–212.

[13] Jacob Bien, Robert Tibshirani, and others. 2011. Prototype selection for interpretable classification. *The Annals of Applied Statistics* 5, 4 (2011), 2403–2424.

[14] Reuben Binns, Max Van Kleek, Michael Veale, Ulrik Lyngs, Jun Zhao, and Nigel Shadbolt. 2018. 'It's Reducing a Human Being to a Percentage': Perceptions of Justice in Algorithmic Decisions. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, 377.

[15] Nadia Boukhelifa, Marc-Emmanuel Perrin, Samuel Huron, and James Eagan. 2017. How data workers cope with uncertainty: A task characterisation study. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, 3645–3656.

[16] Glenn A Bowen. 2006. Grounded theory and sensitizing concepts. *International journal of qualitative methods* 5, 3 (2006), 12–23.

[17] Sylvain Bromberger. 1992. *On what we know we don't know: Explanation, theory, linguistics, and how questions shape them*. University of Chicago Press.

[18] Carrie J Cai, Jonas Jongejan, and Jess Holbrook. 2019. The effects of example-based explanations in a machine learning interface. In *Proceedings of the 24th International Conference on Intelligent User Interfaces*. ACM, 258–262.

[19] Diogo V Carvalho, Eduardo M Pereira, and Jaime S Cardoso. 2019. Machine Learning Interpretability: A Survey on Methods and Metrics. *Electronics* 8, 8 (2019), 832.

[20] Bruce Chandrasekaran, Michael C Tanner, and John R Josephson. 1989. Explaining control strategies in problem solving. *IEEE Intelligent Systems* 1 (1989), 9–15.

[21] Hao-Fei Cheng, Ruotong Wang, Zheng Zhang, Fiona O'Connell, Terrance Gray, F Maxwell Harper, and Haiyi Zhu. 2019. Explaining Decision-Making Algorithms through UI: Strategies to Help Non-Expert Stakeholders. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 559.

[22] William J Clancey. 1983. The epistemology of a rule-based expert system—a framework for explanation. *Artificial intelligence* 20, 3 (1983), 215–251.

[23] Herbert H Clark, Susan E Brennan, and others. 1991. Grounding in communication. *Perspectives on socially shared cognition* 13, 1991 (1991), 127–149.

[24] European Commission. 2016. General Data Protection Regulation. (2016). Accessed September 18, 2019 from `https://eugdpr.org/`.

[25] Juliet Corbin, Anselm L Strauss, and Anselm Strauss. 2015. *Basics of qualitative research*. sage.

[26] Sanjeeb Dash, Oktay Gunluk, and Dennis Wei. 2018. Boolean decision rules via column generation. In *Advances in Neural Information Processing Systems*. 4655–4665.

[27] Amit Dhurandhar, Pin-Yu Chen, Ronny Luss, Chun-Chen Tu, Paishun Ting, Karthikeyan Shanmugam, and Payel Das. 2018. Explanations based on the missing: Towards contrastive explanations with pertinent negatives. In *Advances in Neural Information Processing Systems*. 592–603.

[28] Nicholas Diakopoulos. 2015. Algorithmic accountability: Journalistic investigation of computational power structures. *Digital journalism* 3, 3 (2015), 398–415.

[29] Jonathan Dodge, Q Vera Liao, Yunfeng Zhang, Rachel KE Bellamy, and Casey Dugan. 2019. Explaining models: an empirical study of how explanations impact fairness judgment. In *Proceedings of the 24th International Conference on Intelligent User Interfaces*. ACM, 275–285.

[30] Finale Doshi-Velez and Been Kim. 2017. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608* (2017).

[31] Malin Eiband, Hanna Schneider, Mark Bilandzic, Julian Fazekas-Con, Mareike Haug, and Heinrich Hussmann. 2018. Bringing transparency design into practice. In *23rd International Conference on Intelligent User Interfaces*. ACM, 211–223.

[32] Thomas Erickson, Catalina M Danis, Wendy A Kellogg, and Mary E Helander. 2008. Assistance: the work practices of human administrative assistants and their implications for it and organizations. In *Proceedings of the 2008 ACM conference on Computer supported cooperative work*. ACM, 609–618.

[33] Jerome H Friedman. 2001. Greedy function approximation: a gradient boosting machine. *Annals of statistics* (2001), 1189–1232.

[34] Leilani H Gilpin, David Bau, Ben Z Yuan, Ayesha Bajwa, Michael Specter, and Lalana Kagal. 2018. Explaining explanations: An overview of interpretability of machine learning. In *2018 IEEE 5th International Conference on data science and advanced analytics (DSAA)*. IEEE, 80–89.

[35] Alyssa Glass, Deborah L McGuinness, and Michael Wolverton. 2008. Toward establishing trust in adaptive agents. In *Proceedings of the 13th international conference on Intelligent user interfaces*. ACM, 227–236.

[36] Alex Goldstein, Adam Kapelner, Justin Bleich, and Emil Pitkin. 2015. Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation. *Journal of Computational and Graphical Statistics* 24, 1 (2015), 44–65.

[37] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. 2014. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572* (2014).

[38] Shirley Gregor and Izak Benbasat. 1999. Explanations from intelligent systems: Theoretical foundations and implications for practice. *MIS quarterly* (1999), 497–530.

[39] Riccardo Guidotti, Anna Monreale, Salvatore Ruggieri, Dino Pedreschi, Franco Turini, and Fosca Giannotti. 2018. Local rule-based explanations of black box decision systems. *arXiv preprint arXiv:1805.10820* (2018).

[40] Riccardo Guidotti, Anna Monreale, Salvatore Ruggieri, Franco Turini, Fosca Giannotti, and Dino Pedreschi. 2019. A survey of methods for explaining black box models. *ACM computing surveys (CSUR)* 51, 5 (2019), 93.

[41] Andreas Henelius, Kai Puolamäki, Henrik Boström, Lars Asker, and Panagiotis Papapetrou. 2014. A peek into the black box: exploring classifiers by randomization. *Data mining and knowledge discovery* 28, 5-6 (2014), 1503–1529.

[42] Jonathan L Herlocker, Joseph A Konstan, and John Riedl. 2000. Explaining collaborative filtering recommendations. In *Proceedings of the 2000 ACM conference on Computer supported cooperative work*. ACM, 241–250.

[43] Michael Hind. 2019. Explaining explainable AI. *XRDS: Crossroads, The ACM Magazine for Students* 25, 3 (2019), 16–19.

[44] Robert R Hoffman, Shane T Mueller, Gary Klein, and Jordan Litman. 2018. Metrics for explainable AI: Challenges and prospects. *arXiv preprint arXiv:1812.04608* (2018).

[45] Fred Hohman, Andrew Head, Rich Caruana, Robert DeLine, and Steven M Drucker. 2019. Gamut: A design probe to understand how data scientists understand machine learning models. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 579.

[46] Kenneth Holstein, Jennifer Wortman Vaughan, Hal Daumé III, Miro Dudik, and Hanna Wallach. 2019. Improving fairness in machine learning systems: What do industry practitioners need?. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 600.

[47] Ulf Johansson and Lars Niklasson. 2009. Evolving decision trees using oracle guides. In *2009 IEEE Symposium on Computational Intelligence and Data Mining*. IEEE, 238–244.

[48] Been Kim, Cynthia Rudin, and Julie A Shah. 2014. The bayesian case model: A generative approach for case-based reasoning and prototype classification. In *Advances in Neural Information Processing Systems*. 1952–1960.

[49] Rafal Kocielnik, Saleema Amershi, and Paul N Bennett. 2019. Will You Accept an Imperfect AI?: Exploring Designs for Adjusting End-user Expectations of AI Systems. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 411.

[50] Pang Wei Koh and Percy Liang. 2017. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning-Volume 70*. JMLR. org, 1885–1894.

[51] Josua Krause, Adam Perer, and Kenney Ng. 2016. Interacting with predictions: Visual inspection of black-box machine learning models. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. ACM, 5686–5697.

[52] R Krishnan, G Sivakumar, and P Bhattacharya. 1999. Extracting decision trees from trained neural networks. *Pattern recognition* 32, 12 (1999).

[53] Todd Kulesza, Margaret Burnett, Weng-Keen Wong, and Simone Stumpf. 2015. Principles of explanatory debugging to personalize interactive machine learning. In *Proceedings of the 20th international conference on intelligent user interfaces*. ACM, 126–137.

[54] Vivian Lai and Chenhao Tan. 2018. On Human Predictions with Explanations and Predictions of Machine Learning Models: A Case Study on Deception Detection. *arXiv preprint arXiv:1811.07901* (2018).

[55] Thibault Laugel, Marie-Jeanne Lesot, Christophe Marsala, Xavier Renard, and Marcin Detyniecki. 2017. Inverse Classification for Comparison-based Interpretability in Machine Learning. *arXiv preprint arXiv:1712.08443* (2017).

[56] Brian Y Lim and Anind K Dey. 2009. Assessing demand for intelligibility in context-aware applications. In *Proceedings of the 11th international conference on Ubiquitous computing*. ACM, 195–204.

[57] Brian Y Lim and Anind K Dey. 2010. Toolkit to support intelligibility in context-aware applications. In *Proceedings of the 12th ACM international conference on Ubiquitous computing*. ACM, 13–22.

[58] Brian Y Lim, Anind K Dey, and Daniel Avrahami. 2009. Why and why not explanations improve the intelligibility of context-aware intelligent systems. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 2119–2128.

[59] Zachary C Lipton. 2016. The mythos of model interpretability. *arXiv preprint arXiv:1606.03490* (2016).

[60] Yin Lou, Rich Caruana, Johannes Gehrke, and Giles Hooker. 2013. Accurate intelligible models with pairwise interactions. In *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*. ACM, 623–631.

[61] Scott M Lundberg and Su-In Lee. 2017. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*. 4765–4774.

[62] Prashan Madumal, Tim Miller, Liz Sonenberg, and Frank Vetere. 2019. A Grounded Interaction Protocol for Explainable Artificial Intelligence. In *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems*. International Foundation for Autonomous Agents and Multiagent Systems, 1033–1041.

[63] Tim Miller. 2018. Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence* (2018).

[64] Sina Mohseni, Niloofar Zarei, and Eric D Ragan. 2018. A survey of evaluation methods and measures for interpretable machine learning. *arXiv preprint arXiv:1811.11839* (2018).

[65] Christoph Molnar and others. 2018. Interpretable machine learning: A guide for making black box models explainable. *E-book at< https://christophm. github. io/interpretable-ml-book/>, version dated* 10 (2018).

[66] Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. 2020. Explaining machine learning classifiers through diverse counterfactual explanations. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. 607–617.

[67] Michael Muller, Ingrid Lange, Dakuo Wang, David Piorkowski, Jason Tsay, Q Vera Liao, Casey Dugan, and Thomas Erickson. 2019. How Data Science Workers Work with Data: Discovery, Capture, Curation, Design, Creation. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 126.

[68] Menaka Narayanan, Emily Chen, Jeffrey He, Been Kim, Sam Gershman, and Finale Doshi-Velez. 2018. How do humans understand explanations from machine learning systems? an evaluation of the human-interpretability of explanation. *arXiv preprint arXiv:1802.00682* (2018).

[69] Anh Nguyen, Jason Yosinski, and Jeff Clune. 2016. Multifaceted feature visualization: Uncovering the different types of features learned by each neuron in deep neural networks. *arXiv preprint arXiv:1602.03616* (2016).

[70] Forough Poursabzi-Sangdeh, Daniel G Goldstein, Jake M Hofman, Jennifer Wortman Vaughan, and Hanna Wallach. 2018. Manipulating and measuring model interpretability. *arXiv preprint arXiv:1802.07810* (2018).

[71] Emilee Rader, Kelley Cotter, and Janghee Cho. 2018. Explanations as mechanisms for supporting algorithmic transparency. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, 103.

[72] Ashwin Ram. 1989. Question-driven understanding: An integrated theory of story understanding, memory and learning. (1989).

[73] Gabriëlle Ras, Marcel van Gerven, and Pim Haselager. 2018. Explanation methods in deep learning: Users, values, concerns and challenges. In *Explainable and Interpretable Models in Computer Vision and Machine Learning*. Springer, 19–36.

[74] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. Why should i trust you?: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. ACM, 1135–1144.

[75] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2018. Anchors: High-precision model-agnostic explanations. In *Thirty-Second AAAI Conference on Artificial Intelligence*.

[76] David Ribes. 2017. Notes on the concept of data interoperability: Cases from an ecology of AIDS research infrastructures. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. ACM, 1514–1526.

[77] Marko Robnik-Šikonja and Marko Bohanec. 2018. Perturbation-Based Explanations of Prediction Models. In *Human and Machine Learning*. Springer, 159–175.

[78] Adam Rule, Aurélien Tabard, and James D Hollan. 2018. Exploration and explanation in computational notebooks. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, 32.

[79] Wojciech Samek and Klaus-Robert Müller. 2019. Towards explainable artificial intelligence. In *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*. Springer, 5–22.

[80] Christian Sandvig, Kevin Hamilton, Karrie Karahalios, and Cedric Langbort. 2014. Auditing algorithms: Research methods for detecting discrimination on internet platforms. *Data and discrimination: converting critical concerns into productive inquiry* 22 (2014).

[81] Johanes Schneider and Joshua Handali. 2019. Personalized explanation in machine learning. *arXiv preprint arXiv:1901.00770* (2019).

[82] Milene Selbach Silveira, Clarisse Sieckenius de Souza, and Simone DJ Barbosa. 2001. Semiotic engineering contributions for designing online help systems. In *Proceedings of the 19th annual international conference on Computer documentation*. ACM, 31–38.

[83] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. 2013. Deep inside convolutional networks: Visualising image classification models and saliency maps. *arXiv preprint arXiv:1312.6034* (2013).

[84] Aaron Springer and Steve Whittaker. 2019. Progressive disclosure: empirically motivated approaches to designing effective transparency. In *Proceedings of the 24th International Conference on Intelligent User Interfaces*. ACM, 107–120.

[85] Erik Štrumbelj and Igor Kononenko. 2014. Explaining prediction models and individual predictions with feature contributions. *Knowledge and information systems* 41, 3 (2014), 647–665.

[86] Simone Stumpf, Vidya Rajaram, Lida Li, Margaret Burnett, Thomas Dietterich, Erin Sullivan, Russell Drummond, and Jonathan Herlocker. 2007. Toward harnessing user feedback for machine learning. In *Proceedings of the 12th international conference on Intelligent user interfaces*. ACM, 82–91.

[87] William R Swartout. 1983. XPLAIN: A system for creating and explaining expert consulting programs. *Artificial intelligence* 21, 3 (1983), 285–325.

[88] William R Swartout and Stephen W Smoliar. 1987. On making expert systems more like experts. *Expert Systems* 4, 3 (1987), 196–208.

[89] Andrea L Thomaz and Cynthia Breazeal. 2006. Transparency and socially guided machine learning. In *5th Intl. Conf. on Development and Learning (ICDL)*.

[90] Gabriele Tolomei, Fabrizio Silvestri, Andrew Haines, and Mounia Lalmas. 2017. Interpretable predictions of tree-based ensembles via actionable feature tweaking. In *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining*. ACM, 465–474.

[91] Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2017. Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GPDR. *Harv. JL & Tech.* 31 (2017), 841.

[92] Danding Wang, Qian Yang, Ashraf Abdul, and Brian Y Lim. 2019. Designing Theory-Driven User-Centric Explainable AI. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 601.

[93] Dennis Wei, Sanjeeb Dash, Tian Gao, and Oktay Gunluk. 2019. Generalized Linear Rule Models. In *International Conference on Machine Learning*. 6687–6696.

[94] Daniel S Weld and Gagan Bansal. 2018. The challenge of crafting intelligible intelligence. *arXiv preprint arXiv:1803.04263* (2018).

[95] Adrian Weller. 2017. Challenges for transparency. *arXiv preprint arXiv:1708.01870* (2017).

[96] Christine T Wolf. 2019. Explainability scenarios: towards scenario-based XAI design. In *Proceedings of the 24th International Conference on Intelligent User Interfaces*. ACM, 252–257.

[97] Qian Yang. 2018. Machine Learning as a UX Design Material: How Can We Imagine Beyond Automation, Recommenders, and Reminders?. In *2018 AAAI Spring Symposium Series*.

[98] Ming Yin, Jennifer Wortman Vaughan, and Hanna Wallach. 2019. Understanding the Effect of Accuracy on Trust in Machine Learning Models. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, 279.

[99] Quanshi Zhang, Yu Yang, Haotian Ma, and Ying Nian Wu. 2019. Interpreting cnns via decision trees. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 6261–6270.

[100] Xin Zhang, Armando Solar-Lezama, and Rishabh Singh. 2018. Interpreting neural network judgments via minimal, stable, and symbolic corrections. In *Advances in Neural Information Processing Systems*. 4874–4885.

[101] Bolei Zhou, Aditya Khosla, Agata Lapedriza, Aude Oliva, and Antonio Torralba. 2016. Learning deep features for discriminative localization. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2921–2929.

[102] Zhi-Hua Zhou, Yuan Jiang, and Shi-Fu Chen. 2003. Extracting symbolic rules from trained neural network ensembles. *Ai Communications* 16, 1 (2003), 3–15.

[103] Jichen Zhu, Antonios Liapis, Sebastian Risi, Rafael Bidarra, and G Michael Youngblood. 2018. Explainable AI for designers: A human-centered perspective on mixed-initiative co-creation. In *2018 IEEE Conference on Computational Intelligence and Games (CIG)*. IEEE, 1–8.

frontiers | Frontiers in Neuroscience

ORIGINAL RESEARCH
published: 24 June 2022
doi: 10.3389/fnins.2022.883385

Designing for Confidence: The Impact of Visualizing Artificial Intelligence Decisions

Alexander John Karran, Théophile Demazure, Antoine Hudon, Sylvain Senecal and Pierre-Majorique Léger*

HEC Montréal, Université de Montréal, Montreal, QC, Canada

Explainable artificial intelligence aims to bring transparency to artificial intelligence (AI) systems by translating, simplifying, and visualizing its decisions. While society remains skeptical about AI systems, studies show that transparent and explainable AI systems can help improve the Human-AI trust relationship. This manuscript presents two studies that assess three AI decision visualization attribution models that manipulate morphological clarity (MC) and two information presentation-order methods to determine each visualization's impact on the Human-AI trust relationship through increased confidence and cognitive fit (CF). The first study, N = 206 (Avg. age = 37.87 ± 10.51, Male = 123), utilized information presentation methods and visualizations delivered through an online experiment to explore trust in AI by asking participants to complete a visual decision-making task. The second study, N = 19 (24.9 ± 8.3 years old, Male = 10), utilized eye-tracking technology and the same stimuli presentation methods to investigate if cognitive load, inferred through pupillometry measures, mediated the confidence-trust relationship. The results indicate that low MC positively impacts Human-AI trust and that the presentation order of information within an interface in terms of adjacency further influences user trust in AI. We conclude that while adjacency and MC significantly affect cognitive load, cognitive load alone does not mediate the confidence-trust relationship. Our findings interpreted through a combination of CF, situation awareness, and ecological interface design have implications for the design of future AI systems, which may facilitate better collaboration between humans and AI-based decision agents.

Keywords: explainability, confidence, trust, HCAI, cognitive fit, decision support

OPEN ACCESS

Edited by:
Waldemar Karwowski, University of Central Florida, United States

Reviewed by:
Andreas Holzinger, Medical University of Graz, Austria
Rob Gleasure, Copenhagen Business School, Denmark

****Correspondence:***
Alexander John Karran
alexander-john.karran@hec.ca

Specialty section:
This article was submitted to Decision Neuroscience, a section of the journal Frontiers in Neuroscience

Received: *25 February 2022*
Accepted: *30 May 2022*
Published: *24 June 2022*

Citation:
Karran AJ, Demazure T, Hudon A, Senecal S and Léger P-M (2022) Designing for Confidence: The Impact of Visualizing Artificial Intelligence Decisions. Front. Neurosci. 16:883385. doi: 10.3389/fnins.2022.883385

INTRODUCTION

Artificial intelligence (AI) and machine learning algorithms are increasing in sophistication and accuracy to automate an ever-increasing array of tasks. However, these algorithms' rapid growth in complexity and performance makes them more opaque and less interpretable. Thus, the decisions taken and processes used by these AI systems to make said decisions are more challenging to understand and explain for its end users (Rudin, 2019).

Within the field of AI, interpretability can be defined as "the ability to explain or to present in understandable terms to a human" (Doshi-Velez and Kim, 2017). It represents the degree to which a human can understand the basis of a decision or the extent to which a human can predict a machine learning model's result with a high degree of consistency. The more interpretable the model, the easier it is to understand why certain decisions or predictions have been made. This relationship

also engenders confidence between the user of a decision support engine and the machine learning algorithm that forms and then provides those decisions.

In this work, we report on a study that aims to assess the effect of interpretability techniques and visualizations on user confidence in the AI system providing decisions. Current research in the field focuses more on creating mathematically interpretable models, neglecting the human who uses these explanations (Abdul et al., 2018). Taking a user-centric approach to interpretability and designing to foster confidence may better facilitate collaboration between humans and AI decision systems and assuage societal concerns around the use of AI (Bryson and Theodorou, 2019). This study attempts to bridge this gap in interpretability by providing human-centered explanations using various techniques without compromising the faithfulness of the AI visualization.

Research in explainable AI (XAI) seeks to create interpretability models and methods to address the problems associated with a lack of transparency in AI systems, therefore making them more explainable and fairer (Barredo Arrieta et al., 2020). Some methods developed for image recognition systems attempt to highlight which components of the AI model or system are perturbed to create decisions. Visualizations or descriptions of discriminative mechanisms are then used to represent perturbed components of the system for end-users. These methods are essential to identify sources of potential bias in the training data and ensure that algorithms perform as expected (Gilpin et al., 2018). Providing explanations of system behaviors as a form of transparency has been shown to have a considerable positive impact on developing confidence in new technology (Lee and See, 2004; Cofta, 2009; Eiband et al., 2019; Glikson and Woolley, 2020; Meske and Bunde, 2020). Implementing XAI methods within HCI design for systems that include AI as decision support will help users become more aware of a system's behavior and support a richer collaboration between humans and AI (Gilpin et al., 2018). Similarly, XAI research is also critical in domains such as transportation, finance, security, legal, and medicine, where AI decisions can potentially impact human lives.

Improving congruence between algorithmic decisions and human perceptions concerning those decisions is a serious challenge within advanced and critical machine learning applications (Doshi-Velez and Kim, 2017; Rudin, 2019). However, the complexity, recursivity, and high degree of nonlinearity of current machine learning systems make it arduous to dissect the decision process and, thus, provide straightforward and understandable explanations for a human to process. To tackle this challenge, current approaches in XAI create post-hoc algorithmic and mathematical methods to help explain initially opaque models. Moreover, researchers in the field have recently begun to investigate how to visually represent the AI decision process to produce intelligible explanations for humans (Sundararajan et al., 2019), highlighting the importance of visualization during the interpretation process.

According to a recent call for research (Abdul et al., 2018), a growing societal need is driving the rise in research interest investigating XAI. Furthermore, this call states that there exists a perception of bias in AI decision-making systems that affects both users of those systems and those for whom decisions are being made more generally. To address this need for research, we present a study investigating the effects of design choices of an AI model's explanation visualization (EV) on the level of confidence between a human and an AI system. As a basis for the study presented here, we formulated the following research questions: "To what extent will the EV of an AI system's decision affect a user's confidence?". And furthermore, to highlight if aspects of information presentation further affect user confidence, "To what extent does presentation order and visualization technique promote or degrade user confidence in the AI system and its decisions?".

We present two studies designed to address these questions using Cognitive Fit (CF) as a theoretical basis and utilizing several EV methods of AI decisions output to manipulate morphological clarity (MC) and two presentation-order methods to investigate adjacency. The first study was administered online and assessed each EV's impact while participants completed a simple decision-making task using an AI system's outputs. The second study was performed using the same stimulus material and methodology in a laboratory. However, in this study, we recorded measures of pupillometry to infer cognitive load. Both studies investigate how user confidence in future system predictions is affected by each type of EV.

PREVIOUS WORK

Trust and Confidence in Artificial Intelligence

It has been proposed that control or perception of control moderates confidence, whereby increasing confidence acts cumulatively to build trust in alliances and partnerships (Das and Teng, 1998). Trust in a specific technology, such as an AI system, affects the value-added proposition of using the technology after its adoption (Mcknight et al., 2011). Moreover, a user who has high confidence in a technologies capabilities and thus places trust in a specific technology is more likely to explore and use its features (Mcknight et al., 2011). Therefore, it has become essential to consider the trust relationship between user and technology when developing and implementing an AI system in order to accelerate its acceptance within the workplace. This relationship has a few proven antecedents, such as navigational structure and visual appeal, ease of use, and the national culture of the user (Vance et al., 2008). Furthermore, trust in an AI system is also dependent on more specific factors such as privacy, security, reliability, stated and perceived accuracy, and transparency (Cofta, 2009; Fairclough et al., 2015; Yin et al., 2019; Glikson and Woolley, 2020).

As described in Lee and See (2004), trust within a human-AI partnership is the attitude that an agent, such as an AI system, "will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability." In this form of partnership, insufficient trust placed in a system may result in distrust and disuse of the system, whereas too much trust may result in over-reliance in the system, whereby the AI takes away from human

agency and the ability to make decisions. Depending on the scale, importance, and impact of the tasks and decisions taken by the AI system, misuses or disuses of the system by its users can lead to safety and profitability problems (Parasuraman and Riley, 1997; Lee and See, 2004). While trust mediates how much reliance humans are willing to place on AI systems, trust also mediates how much humans rely on each other (Lankton et al., 2015). However, humans lose confidence in AI systems more rapidly than other humans, even if both parties make the same mistake (Dietvorst et al., 2015). Thus, it can be stated that trust in AI systems tends to decrease rapidly as a function of the number of errors the system makes over the duration of interaction and that the restoration of trust in AI systems or tools requires an undetermined but more significant amount of time (Glikson and Woolley, 2020; Nourani et al., 2020). Moreover, studies have shown that humans display greater trust toward other human agents than AI agents, even though both perform the same tasks and make identical mistakes (Jakesch et al., 2019; Glikson and Woolley, 2020).

However, according to (DeCamp and Tilburt, 2019), it is erroneous to speak of a user-AI trust relationship. Referring to trust in AI as a concrete psycho-affective relationship would imply that the system belongs to the same category of human agents that can be trusted, such as a physician or member of law enforcement. Currently, for an AI system, the thoughts, motives, and actions, which comprise the human psycho-affective framework involved in developing and allocating trust, go beyond its technical and mechanical capacities. Furthermore, moving forward, the development of AI will have a significant moral impact in the long term as the performance of AI improves, given that human capacities, in terms of technical accuracy, speed, and judgment, may well prove inferior to that of AI systems, resulting in a blanket of distrust regardless of evidence proving their accuracy and worth.

Moreover, some researchers suggest a game-theoretic approach where it is only possible to talk about trust when the trustor is in a situation of vulnerability and uncertainty with the trustee and when the consequences of a betrayal of trust are more significant than the benefits sought (Luhmann, 2001; Vance et al., 2008). In the context of the current research study, it would therefore be more appropriate to speak of developing confidence rather than trust in AI systems. Confidence in AI can be defined as a measure of risk as to how sure users are that they received the correct suggestions by the AI system and if they consider the system to be reliable, i.e., the system consistently operates properly, functional, i.e., the system does what it is supposed to do, and helpful, i.e., the system provides adequate help for the users (Lankton et al., 2015; Wanner et al., 2021).

Cognitive Fit Theory

Cognitive Fit theory (Vessey, 1991) offers a means to understand how design choices of AI decision-making visualizations affect human cognitive performance and potentially toward building confidence in the system. This theory proposes that congruence between the task and the structure of the representation of a problem in an individual's mental model results in reduced cognitive effort and superior performance. The complexity and usefulness of this mental model are dependent on the user's working memory capacity (Vessey and Galletta, 1991; Goodhue and Thompson, 1995; Adipat et al., 2011). Therefore, additional cognitive effort is required when there is a lack of fit between the task at hand and the information format used to complete the task (Vessey, 1991). The individual must mentally transform that information into a format that enables them to accomplish the task, resulting in reduced performance (Adipat et al., 2011). This relationship between CF and cognitive effort was demonstrated using neuroscientific methods (Nuamah et al., 2020).

Cognitive Fit theory was first proposed to assess the effect of numerical data presented in a tabular versus function format paired with a symbolic and spatial task on cognitive performance. This theory has been applied in several studies over the years, within other research domains, and using multiple other information presentation formats (Adipat et al., 2011; van der Land et al., 2013; Chen, 2017; Gillespie et al., 2018). CF theory can also be used to support the theoretical basis of this study, where numerical data are translated into visualizations using colors and shapes for human usage. In this respect, the visualization of the AI decision process can potentially help provide a stronger CF between the task and the information required to complete that task in human-AI collaboration contexts, where understanding the AI decision process is of importance.

The result of designing tasks using this process is greater efficiency and effectiveness, manifested as lower cognitive load, increased accuracy, and speed in problem-solving. Therefore, applying this process to the presentation and visualization of AI decision reasoning may significantly impact the level of confidence engendered between a human and an AI system. Moreover, a visualization method with a stronger CF would help users perceive the reasoning behind the AI system's prediction, consequently helping to avoid over or lack of confidence resulting in a misuse or disuse of the AI system (Lee and See, 2004). Furthermore, results from a recent study using the same stimuli and AI algorithm (Hudon et al., 2021) reported a negative correlation between cognitive load and the level of confidence in the AI system resulting from the use of an EV. These results further support the assertion that improving the CF between the task and the EV type may reduce cognitive load and positively influence a user's confidence in the system.

Progress in Explainable Artificial Intelligence

Recent advances in XAI have made it possible to produce explainable models and learning methods, especially in image recognition systems that have developed multiple techniques (Barredo Arrieta et al., 2020). Techniques such as Grad-CAM (Selvaraju et al., 2020) and integrated gradient (Sundararajan et al., 2017) aim to show relationships between inputs and outputs, focusing on processing information in the models. Studies have investigated the potential effect of interpretability and explanations on humans. Moreover, it has been shown that transparency in a model helped users simulate the model's prediction (Poursabzi-Sangdeh et al., 2018). Others indicated that visualizations have a considerable influence on the effectiveness

of the explanations (Sundararajan et al., 2019), that they can help discriminate between classes more accurately, help reveal a model's trustworthiness, and help identify biases in datasets or models (Gilpin et al., 2018; Selvaraju et al., 2020). However, it is unknown what impact these visualization techniques have on developing confidence between human and AI systems to our knowledge.

HYPOTHESIS DEVELOPMENT AND RESEARCH MODEL

The research model (**Figure 1**) posits that the design of an EV will affect a user's confidence in the system. More precisely, we hypothesize that the effect of EV adjacency and MC, in addition to the interaction between both, will affect cognitive load influencing perceived confidence. The rationale behind these relationships is developed below.

Adjacency is a form of presentation order in which an adjacent EV is presented with its explanation data displayed directly upon the original image by coloring the areas in different colors to indicate their data values. In contrast, a non-adjacent EV is presented with the same explanation data but separated from the image (Dennis and Carte, 1998). However, non-adjacent EV requires greater cognitive effort to process since there is a loss in correspondence between the explanation and the visualization (Sundararajan et al., 2019). For spatial tasks, similar to the one used in the study presented here, (Dennis and Carte, 1998) showed that adjacent representations led to faster and more accurate decisions. They state that adjacent visualizations should help decision-makers associate areas of a visualization image with its data, simplifying the complex relationship between them, leading to faster, more accurate decisions. We posit that this adjacency relationship will hold for newer spatial tasks in which the visualization and data presented represent decisions already made by an AI and where the user of the AI is asked if he has confidence in future AI classification abilities.

Furthermore, we theorize that MC, which represents the degree to which a visualization displays clear delimited features by adjusting the appearance or removing specific data (e.g., noise) to help make the delimitation clearer for the user (Sundararajan et al., 2019), will also affect user confidence in the AI system. This study uses three levels of MC (i.e., low, medium, and high). First, low MC EVs are faithful to the model's behavior as they allow the user to precisely identify at a pixel level what areas of the image are relevant. However, this precision makes the EV more cluttered, preventing the user from having a clear overarching view of the image. Alternatively, High MC EVs are more faithful to the MC definition by having clear form features while reducing the noise due to excess information. High MC EV provides greater clarity for the user and tends to be easier to comprehend by reducing his cognitive load but at the cost of faithfully depicting the model's behavior. On the other hand, low MC EV might cause humans to misuse or ignore the explanation altogether by giving them too much information to process (Sundararajan et al., 2019; Müller et al., 2021). This concept of information overload (i.e., too much available information or too much high-quality information) has been shown to decrease the user's decision effectiveness and can even make the user less confident about their decision (Keller and Staelin, 1987; Müller et al., 2021).

As discussed previously, CF theory (Vessey, 1991) proposes that the level of congruence between task and information presentation mediates task performance. Such that, while solving a problem, an individual creates a mental representation of the problem based on the information presented. This mental representation's complexity and usefulness are dependent on the user's working memory capacity (Vessey and Galletta, 1991; Goodhue and Thompson, 1995; Adipat et al., 2011). Thus, additional cognitive effort is required in the presence of incongruence between task and the format of the information presented to help complete the task (Nuamah et al., 2020), requiring the individual to mentally transform that information into a format suitable for accomplishing the task, resulting in reduced performance (Adipat et al., 2011).

We hypothesize that specific EVs will result in a better CF, reducing cognitive load, defined as the demand imposed by a task on the user's working memory (Wickens, 2008). Therefore, a task

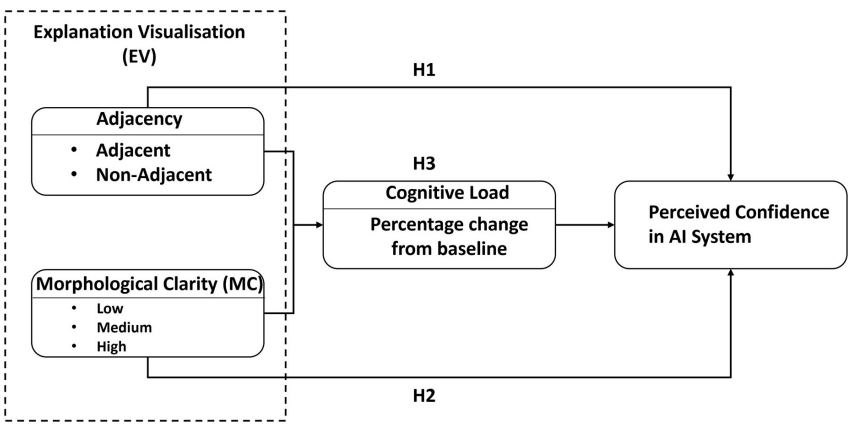


FIGURE 1 | Research model.

requiring significant mental resources is more likely to prompt more user errors than a task requiring less cognitive resources, resulting in less perceived effort and greater CF (Palinko et al., 2010b; Nuamah et al., 2020).

Thus, we seek to determine whether *Adjacent EVs will result in higher confidence in the AI system* (H1) or if *High* MC *EVs will result in higher confidence in the AI system* (H2) or whether *the combination of adjacent and high* MC *EVs mediated by cognitive load will result in higher confidence in the AI system* (H3).

MATERIALS AND METHODS

Study One

Experimental Design

We designed an online experiment to test our hypotheses. Specifically, we used a 2 × 3 within-subject factorial design to examine the effects of adjacency and MC of AI EV on confidence between the user and the AI system. The first factor considers the adjacency of the representation with two levels: EV with (adjacent) or without (non-adjacent) image background, the second factor considers the MC of the EV with three levels: low-cloud of points (CP), medium-heatmap (HM), and high-outline (ON). We considered CP EV to be low MC since it faithfully depicts the model's attributions by highlighting the image's pixels that positively impact the model's classification. However, CP also displays much information that can be useless to the user. We considered HM EV to be a low MC since it is less precise than CP, showing only the model's focal point on the stimuli image. HM also does not delimit its shapes enough to be considered a High MC visualization. Finally, ON EV draws only the most essential zones of the image used in the classification, therefore providing a high MC, but at the cost of pixel-level precision. See **Table 1** for EV examples and the Generation of Explanations section for detailed information about these factors' implementation.

Participants

Amazon Mechanical Turk (MTurk) was used to recruit participants. Based on (Jia et al., 2017) recommendations the participant pool was screened for North American residency to prevent linguistic difficulty. In the study, we restricted ourselves to local explanations of the model as they are deemed to be accessible for novices. This selection criterion avoids confounding factors due to participants' potential machine learning expertise. A total of 350 North American participants took part in the survey, of which 206 (Avg. age = 37.87 ± 10.51, 123 Male) provided usable data. Sixty-one participants considered themselves AI or Data experts based on criteria posited by Mohseni et al. (2018). To improve data quality, data were removed for participants who failed any of the two attention checks, did the survey multiple times, or did not correctly submit their survey. The attention checks verified participant attention to the stimuli by displaying black squares instead of the images. Our institution's ethics committee approved the study, and each participant provided informed consent and was compensated $CAD 20 upon completion.

Experimental Procedure

Once recruited, participants were directed to an online survey on Qualtrics. Participants were provided instructions and an example of a trial task upon beginning the study. For each trial, several elements were presented on screen (**Figure 2**): (1) The image analyzed by the system, (2) the system's EV, (3) the system's output, and (4) a related but unclassified image. In this case, the system's output is the classification of the image predicted by the system; classifications were presented irrespective of correctness. Classification mistakes were evenly distributed among all EV types. For our 206 participants, we found on average per EV type 350 classification mistakes (STD = 10.61) compared to 1,259 correct classification (STD = 14,15). Participants were then asked to rate their agreement with the following statement using a 7-point Likert scale ranging from "Strongly disagree" to "Strongly agree": "Given the information above, I am confident that the system will correctly identify the next picture." The decision-making task consisted of 50 trials per participant, with two trials consisting of attention checks.

Stimulus material was presented randomly, with one label associated with a pair of similar images. In this context, similar images belong to the same class, therefore having the same label (e.g., two images of a car). A participant could not see the same pair of images more than one time. EV types were also presented in a random order for each task trial. The combination label-explanation type was also randomized amongst participants to avoid confounding effects due to the image label, object, or prediction. Therefore, a participant could see the image of an elephant paired with an adjacent-medium MC EV, and another could see the same image paired with a non-adjacent-low MC EV. After 50 trials, all six different EV types were seen and judged eight times by the participant. After completing the main task, participants were asked to answer demographic questions regarding their age, gender, and education level.

Study Two

Experimental Design

For the second study, we designed a 2 × 3 within-subject factorial design to investigate the effects of adjacency and MC of AI-EVs on the user's cognitive load and perceived confidence. The first factor considers the representation's adjacency with two levels: EV with (adjacent) or without (non-adjacent) image background. The second factor considers the MC of the EV with three levels: low-CP, medium-heatmap (HM), and high-outline (ON). CP faithfully depicts the model's attributions by highlighting pixels of the image that positively impact the model's classification but also displays a high amount of superfluous information. HM is less precise than CP in terms of classification feature granularity. However, it shows the stimuli image's prime focus used to make a classification but does not have precisely delimited features. The ON visualization draws only the most essential zones of the image used in the classification but at the cost of pixel-level precision. CP and ON-EVs were both implemented using the Integrated Gradients method (Sundararajan et al., 2017, 2019), and the HM-EV using the Grad-CAM class activation function (Selvaraju et al., 2020).

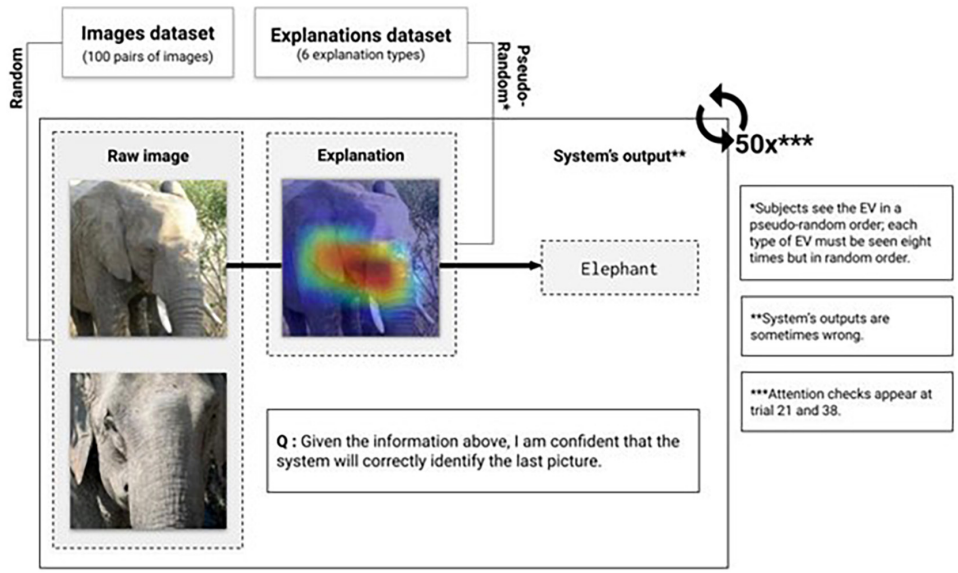


FIGURE 2 | Task's design. Images selected from the ImageNet dataset (Deng et al., 2009).

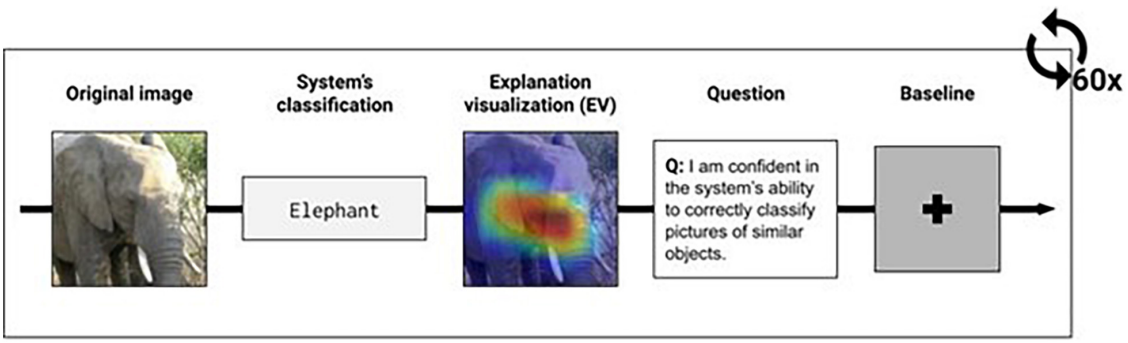


FIGURE 3 | Task design. Images selected from the ImageNet dataset.

Experimental Procedure

Nineteen participants (24.9 ± 8.3 years old, 10 Male) took part in the study, all signed consent and were compensated $CAD 30 upon completion. The study was approved by the ethics committee of our institution, the task consisted of a spatial task repeated over 60 randomized trials. Each trial involves a series of elements displayed on screen (see **Figure 3**) in the following order: (1) original image (e.g., image of an elephant), (2) classification of the image given by the AI system[1] (e.g., "Elephant"), (3) the AI EV (e.g., an overlay IA explanation onto the original image), and (4) a perceived confidence question (e.g., confidence in the system).

Participants were asked to rate their agreement with the following statement using a 7-point Likert scale ranging from "Strongly disagree" to "Strongly agree" and "I am confident in the system's ability to classify pictures of similar objects correctly." A baseline image was finally shown at the end of each trial for 1 s. Each participant saw all six types of visualizations ten times. To measure pupil dilation, we used the Tobii ×60 eye tracker.

Illuminance Testing

We measured lux using an Arduino light-dependent resistor (photoresistor) housed in a custom 3-D printed 1.5 cm depth × 5 cm area housing. Lux is a standardized unit of measurement of light intensity where 1 lux is equal to the illumination of a one-meter square surface that is 1 m away from a single candle. The experiment took place in a light-controlled room with an ambient lux value of lx (19.97). Luminance measurements for the display screen and per condition images were taken from a 70 cm distance and a height of approximately 1 m (representing a seated participant of average height). The default screen brightness at idle was lx (1.59), with the baseline fixation image lx (1.59). The average lux values per experimental

[1]The Xception (extreme inception; Chollet, 2017) algorithm which comes with pre-trained weights on the ImageNet dataset was used to classify the images.

condition were as follows: CP Adjacent lx (1.612), CP Non-adjacent lx (1.607), HM Adjacent lx (1.609), HM Non-adjacent lx (1.61), ON Adjacent lx (1.611), and ON Non-adjacent lx (1.67). We performed a 1-way ANOVA to determine if a significant difference existed between lux values that may affect pupillometry measures, no significant difference in luminance values between conditions was found F_{Welch} (5, 24.637) = 0.710, $p = 0.621$.

Calculating Cognitive Load

Changes in Pupil dilation when a user faces a task requiring a high cognitive effort are referred to as the Task-evoked pupillary response (Beatty, 1982). We used pupil diameter to estimate the user's cognitive effort required to process each EV. We computed the average percentage change from a baseline taken from a neutral image (PcB) for each participant and each EV in this analysis. We used the percentage change of pupil diameter rather than the raw pupil size variation due to inter-participant variance (Attard-Johnson et al., 2019).

Selection of Stimuli

Subsets of stimulus images were selected from the ImageNet dataset (Deng et al., 2009), a publicly released image dataset with 1.2 million quality-controlled categorized images and associated human annotations. When building the stimulus image dataset, neutrality and unambiguity were used as the criterion. With these criteria in mind, straightforward categories of image and annotation pairs such as "Dog" instead of "Golden Retriever" or "Elephant" instead of "African elephant" were chosen to reduce confusion and present images as "platonic" classes. Image categories were chosen based on (Snodgrass and Vanderwart, 1980), who defined a standardized set of 260 illustrations of different concepts. These concepts were chosen based on three criteria: (1) They are unambiguously picturable, (2) they include exemplars from the widely used category norms of Battig and Montague (1969), and (3) which represent concepts at the basic level of categorization. As described by Battig and Montague (1969), unambiguity is the degree to which subjects will show consensus about the name to give the object. From these 260 concepts, 100 were selected to represent the categories of the images used within the stimulus dataset.

ImageNet organizes images according to the WordNet hierarchy (Miller, 1995), where each concept is described by one or multiple names called a "synonym set" or "synset." The following criteria were established to select the stimulus images: (1) The image's synset must contain the name of one of the 100 previously selected concepts, (2) No human subject is present in the image, and (3) the image must be neutral (no shocking or disturbing depiction). In total, 200 images were selected, giving two images per concept. Image ambiguity, familiarity, and complexity were measured and used as control variables in the final analysis.

Validation of Stimuli

To ensure that the selected images are an unambiguous representation of the concepts outlined by Snodgrass and Vanderwart (1980), we used a panel of 18 judges to validate the images used for stimulus presentation. To ensure that three judges independently coded each stimulus image, the validation process was split into two rounds, and judges were split into six groups of three, thus there were nine judges in both rounds, and each group of three judges validated the entire stimulus corpus. As part of the evaluation process, each judge was asked to complete an online survey that displayed images to a screen and then write a label composed of one or two words to describe the image. Each group of three judges would assess between 65 and 68 images. This ensured that three judges provided labels for all images. Furthermore, judges were prompted to write a spontaneous description to minimize subjective bias. Following guidelines presented by Snodgrass and Vanderwart (1980), for each picture, the judges could also specify if: (1) They did not know the object (DNO), (2) if they knew the object but did not know its name (DKN), (3) and if they knew the name of the object, but it was momentarily irretrievable [tip of the tongue (TOT)].

In order to assess if a judges description (label) could be accepted or rejected, we specified the following criteria: (1) It is the same as the concept name (expected: bee, label: bee), (2) it is a synonym of the concept name (expected: aeroplane, label: plane), (3) it is a more precise term than the concept's name (expected: bear, label: polar bear), and (4) it is listed in the nondominant list of name of the concept (Snodgrass and Vanderwart, 1980; expected: alligator, label: crocodile). An image was replaced if it matches one of these criteria: (1) At least two out of three judges put a rejected label or checked DKO, DKN, or TOT, or (2) at least two out of three judges use a nondominant name to describe the image.

The first round of stimuli verification was composed of three groups of three judges who produced an average Cohen's Kappa of 0.838 (**Table 1**), which is considered a strong agreement rate (Viera and Garrett, 2005). However, eight images were replaced that did not meet the level of agreement criteria. We conducted a second round of stimuli validation to include the new images and nine new judges, which produced an average Cohen's Kappa of 0.879 (**Table 1**). We did not replace any images after this second round of verification, given the high rate of inter-rater agreement.

Choice of the Algorithm

To classify the stimulus images and act as the automated system in the study, we chose the Xception (Chollet, 2017) algorithm, which is packaged with pre-trained weights trained on the ImageNet dataset. Developed by a team at Google, Xception is a deep learning algorithm that relies heavily on prior effort done in the area of Convolutional Neural Networks. Xception has proven to be a very effective, compact (88 MB), and accurate (0.790) algorithm for computer vision problems (Chollet, 2017). We applied the trained Xception algorithm to the validated stimulus image. Output classifications, in the format of a sysnet-ID and label, were then compared to the "platonic" class of image. For example, if we are looking for the platonic class of "Cat" and the algorithm returned the synset label of "Cat" or "Tiger Cat," we considered the classification acceptable as they both represent the class of a cat. After training, we obtained a reasonable

TABLE 1 | Overall inter-rater agreement for each group (G#) at each round (R#).

		Kappa		Raw agreement		Fleiss's K		Krippendorff's alpha		*N*	
		R1	R2	R1	R2	R1	R2	R1	R2	R1	R2
G1	J1 – J2	0.829	0.969	75.40%	78.80%	0.816	0.854	0.81	0.855	*N* = 65	*N* = 66
	J1 – J3	0.752	0.798								
	J2 – J3	0.858	0.798								
G2	J1 – J2	0.938	0.85	84.80%	82.10%	0.896	0.853	0.896	0.855	*N* = 66	*N* = 67
	J1 – J3	0.907	0.848								
	J2 – J3	0.845	0.863								
G3	J1 – J2	0.845	0.907	72.70%	89.40%	0.803	0.927	0.804	0.927	*N* = 66	*N* = 66
	J1 – J3	0.752	0.891								
	J2 – J3	0.814	0.984								
Average		0.838	**0.879**	77.63%	**83.43%**	0.838	**0.878**	0.837	**0.879**		

Values in bold represent the final average across rater groups.

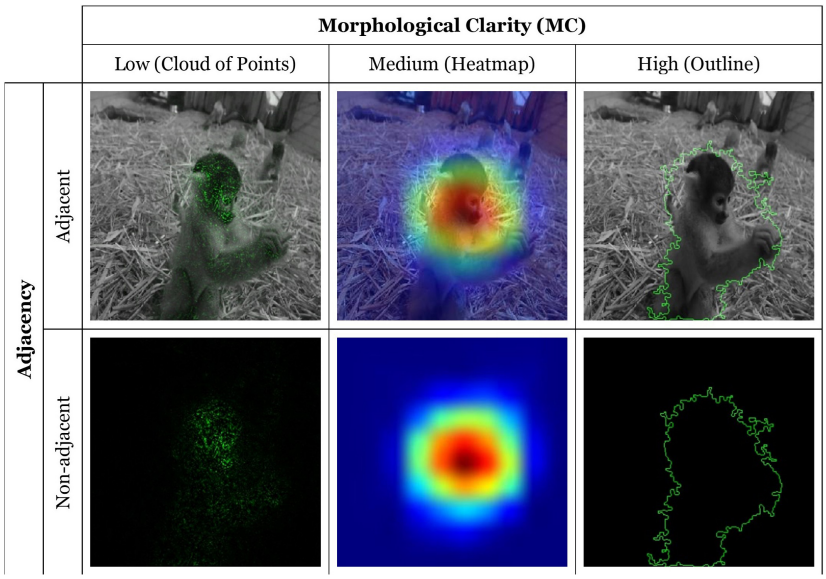


FIGURE 4 | Example of each type of EV for the classification "Monkey". Images selected from the ImageNet dataset (Deng et al., 2009).

classification rate of 0.76, close enough to the algorithm's current maximum accuracy rate.

Generation of Explanation Visualizations

In order to actualize the concepts of adjacency and MC during the experimental task, we utilized several visualization and presentation methods to provide the participant with an AI EV. The low and high MC visualizations were both implemented using the Integrated Gradients method (Sundararajan et al., 2017, 2019). These visualizations highlight in green the areas of the stimulus image that positively impact the model's classification. The medium MC visualization was implemented using the Grad-CAM class activation function (Selvaraju et al., 2020). This method displays a heatmap that highlights activated regions important for the classification of the image.

Adjacency representations were implemented using the Grad-CAM class activation visualization overlaid upon the original image and the Integrated Gradients visualization overlaid upon a grayscale version of the original image, ensuring that the attribution colors did not blend with those of the image as recommended by Sundararajan et al. (2019). Non-adjacent representations were implemented by showing a black background under each visualization. In total, six combinations of EV per image were generated (**Figure 4**).

Analysis

Using SAS 9.4, we performed a repeated-measures ANOVA (Bonferroni corrected) for the dependent variable Perceived Confidence (PConf) with both Adjacency, MC, and the interaction of those two as within-subject factors. PConf is a discrete variable taking values from 1 to 7, where 7 represents the highest value of confidence between the user and the system. Adjacency is a categorical binary variable (i.e., adjacent and non-adjacent), and MC is a categorical, ordinal variable (i.e.,

low, medium, and high). Since a classification's validity can significantly impact a user's perceived confidence in the system, we used the variable "Classification" as a covariate in the model. Classification is a Boolean variable indicating whether the model's classification is correct.

In addition to the analysis performed in Hudon et al. (2021), for study two, we computed a within-subject, repeated measures mediation analysis using Monte Carlo methods (Yzerbyt et al., 2018), in *R* (JSmediation) to estimate the confidence interval of the indirect effect. Mediation analysis tests whether an IV affects a DV through a third variable, the mediator, in this case, cognitive load.

RESULTS

Study One

The results show a statistically significant main effect of Adjacency [$F(1, 205) = 246.96$, $p < 0.001$], and MC [$F(2, 410) = 5.69$, $p = 0.004$]. Additionally, we observed a significant interaction between MC and Adjacency [$F(2, 410) = 6.10$, $p = 0.003$]. The covariate variable Classification also has a main effect on confidence [$F(1, 205) = 768.13$, $p < 0.001$].

Post hoc comparisons of simple effects (**Table 2**) reported that adjacent EV ($M = 5.01$, SD = 0.06) results in higher confidence than non-adjacent EV [$M = 4.55$, SD = 0.06; $t(205) = 15.71$, $p < 0.001$], providing support for H1. Concerning MC, *post hoc* comparisons showed that a low MC (Cloud of Points) EV ($M = 4.85$, SD = 0.06) resulted in higher confidence than both medium MC (Heatmap; $M = 4.77$, SD = 0.06; $t(410) = 2.36$, $p = 0.019$), and high MC (Outline; $M = 4.73$, SD = 0.06; $t(410) = 3.27$, $p = 0.001$), providing no support for H2, as the reported results highlight that the effect is the opposite of what we had hypothesized. The comparison between good classifications ($M = 5.27$, SD = 0.06) results in higher confidence than bad classification [$M = 4.29$, SD = 0.06; $t(205) = 27.72$, $p < 0.001$].

As shown in **Table 3**, adjacent-high MC EV did not significantly differ from the two other adjacent representations. Therefore, H3 is also not supported. Surprisingly, a non-adjacent-low MC EV ($M = -0.25$, SD = 0.09) resulted in a significantly greater positive impact on confidence than both non-adjacent-medium MC EV, and non-adjacent-high MC EV. **Figure 5** illustrates that all three adjacent EV types have a significantly greater positive impact on confidence than their non-adjacent equivalent. It also shows that low MC EV has a significantly greater positive impact on trust than medium and high MC EV.

TABLE 2 | *Post hoc* comparisons adjacency, MC, and classification on confidence.

Factor (1)	Factor (2)	Mean diff (1–2)	DF	*t*	*p*
Adjacent	Non-adjacent	0.46	205	15.71***	<0.001
Low MC (Cloud of points)	Medium MC (Heatmap)	0.08	410	2.36*	0.019
Low MC (Cloud of points))	High MC (Outline)	0.12	410	3.27**	0.001
Medium MC (Heatmap)	High MC (Outline)	0.03	410	0.91	0.363
Good classification	Bad classification	0.98	205	27.72***	<0.001

*$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$.

Study Two

Preliminary results from the second study were published in (Hudon et al., 2021). These results reported a significant main effect of adjacency ($p < 0.001$) and MC ($p < 0.001$) and a significant interaction between both factors ($p < 0.001$) upon perceived confidence. However, in this preliminary analysis, any cognitive load mediation effect was never considered. Shown in **Figure 6** are the results from a mediation analysis that follows the research model proposed above (**Figure 1**).

While the direct effect of adjacency on cognitive load was significant [$t(17) = 5.26$, $p < 0.001$], the indirect effect on perceived confidence [$t(18) = 2.13$, $p = 0.47$] *via* cognitive load is not, showing no indirect mediating effect of cognitive load on perceived confidence. Congruent with the above results, the Monte Carlo (5000 bootstrap samples) confidence interval for the indirect effect for adjacency reported no indirect effect upon perceived confidence and cognitive load (CI95% [–0.885; 0.408]).

Similarly, for MC, while the effect of MC on perceived confidence [$t(18) = 2.33$, $p < 0.032$] and cognitive load [$t(18) = 4.17$, $p < 0.001$] is significant, the effect of cognitive load on perceived confidence is not [$t(16) = 0.04$, $p = 0.969$], showing a lack of indirect effect. Monte Carlo confidence interval for MC reported no indirect effect upon perceived confidence and cognitive load (CI95% [–0.615; 0.655]).

These results indicate that the pathway to confidence *via* cognitive load as a mediator (H3) is not supported. However, the effects of both adjacency and MC have a very strong influence on the level of perceived confidence; and furthermore, each presentation method strongly influences cognitive load (see **Figures 7**, **8**).

DISCUSSION

Overall, the results from these studies indicate a clear difference when comparing EV types, such that adjacency strongly influences perceived user confidence. Looking at this significant difference in more detail, we found that adjacent EVs had a more significant positive effect on user confidence than non-adjacent EVs, confirming our hypothesis. However, contrary to our initial postulation in which we surmised that a high MC would provide more than sufficient information and accuracy of representation to influence user confidence positively, low MC EVs resulted in the highest confidence between all three levels of MC. Furthermore, looking at the results from the interaction of both adjacency and MC, the findings indicate no clear method of information presentation, in that all three adjacent EVs have an approximately equivalent impact on user confidence. Moreover, the results indicate that in situations of non-adjacency, there is a significant difference between low and medium MC in addition

TABLE 3 | *Post hoc* comparisons of the interaction between adjacency and MC on confidence.

Adjacency	MC level (1)	MC level (2)	Mean diff (1–2)	DF	*t*	*p*
Adjacent	Low (Cloud of points)	Medium (Heatmap)	–0.024	410	–0.48	0.632
	Low (Cloud of points)	High (Outline)	0.009	410	0.18	0.857
	Medium (Heatmap)	High (Outline)	0.033	410	0.66	0.511
Non-adjacent	Low (Cloud of points)	Medium (Heatmap)	0.191	410	3.82***	<0.001
	Low (Cloud of points)	High (Outline)	0.223	410	4.44***	<0.001
	Medium (Heatmap)	High (Outline)	0.032	410	0.63	0.530

****p < 0.001.*

to low and high MC, where the latter resulted in lower confidence in the system. Results from study 2 confirm these results. Finally, and perhaps counterintuitively, cognitive load decreases as MC increases with adjacent visualizations, with low MC showing the lowest cognitive load while this representation shows the highest density in terms of information and noise.

The Impact of Adjacency on User Confidence

We first hypothesized that adjacent visualizations would have a greater significant impact on user confidence than non-adjacent EVs, as adjacency is posited to provide a superior CF with the task at hand in terms of providing visual scan patterns for pertinent information, potentially leading to higher confidence through a reduction in epistemic uncertainty. The results indicate, at least in part, that our hypothesis holds true and that adjacency does indeed impact user confidence significantly. This effect can be explained by the lower cognitive effort required to identify the correspondence between the highlighted areas of the explanation and the original image. Users were able to promptly identify which areas of the image influenced the AI system in its decision, helping them compare AI classification reasoning with their own. Indeed, (Dennis and Carte, 1998) posited that adjacent visualizations led to more accurate decision-making in less time when the user is faced with a spatial task, resulting in a better CF. However, non-adjacent EVs have the advantage of aiding the user in inspecting the details of the explanation without the distraction of the underlying image or reviewing the original image to verify what the attributions highlight in order to form a fresh opinion

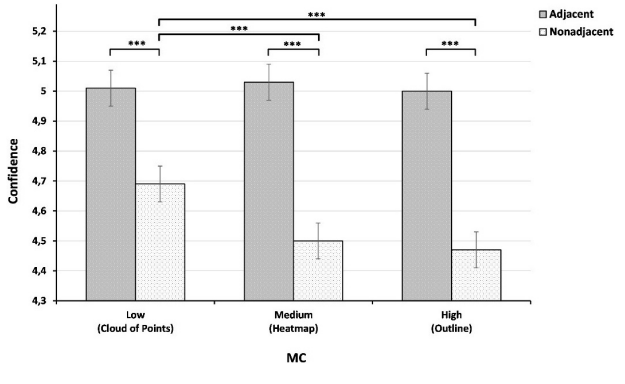


FIGURE 5 | Comparison of adjacent and non-adjacent visualizations based on their MC from low to high. ***$p < 0.001$.

(Sundararajan et al., 2019). Moreover, non-adjacent EVs ask the user to continuously gaze back and forth between the original image and the EV. Thus, in terms of CF, we suggest that users need to mentally associate the original image zones with the AI decision visualization, creating a more complex mental map requiring more cognitive resources.

The Conditional Effect of Morphological Clarity on User Confidence

We further hypothesized that high MC visualizations would have a more significant positive effect on user confidence. This visualization technique focuses and highlights important regions of the classified images, reducing the signal-to-noise ratio related to the EV representing the network's results, thus diminishing the information load required. However, our results surprisingly indicate the opposite to be true, showing in this case that low MC EV had a more significant positive effect on user's confidence than high and medium MC EVs. An alternate interpretation for this relationship could be that low MC EV's include more information, even if that additional information would ordinarily be considered noise. Furthermore, this supplementary information may implicitly increase the perception of transparency in the model and positively affect user confidence. This interpretation aligns with (Brunk et al., 2019), who showed that the interpretability of black-box algorithms is perceived to be more transparent and trustworthy if additional information is present.

However, the adjacency comparisons (**Figure 5**) indicate no significant difference between levels of MC when explanation data are presented adjacently. As can be seen in the table, in our case, the main significant variance between levels of MC is between non-adjacent low MC with high and medium MC EVs. These results imply that low MC EV has a greater positive effect on confidence but only in situations of non-adjacency. With these conditions in mind, and in light of our interpretations, the results strongly suggest that users require more information about the explanation when adjacency requirements are not met. We interpret this difference between non-adjacent MC levels to be a perceptual and processing disassociation between EV and the subject matter of the input image. Moreover, while participants could make a certain sense of the target objects form in the case of the non-adjacent-low MC EV, the non-adjacent-medium and high MC EVs were too abstract to identify clear forms and associate them with the input image. As shown in **Figure 4**, the non-adjacent-low MC EV allows the shape of the monkey's head to be distinguished, contrary to the two other non-adjacent EVs.

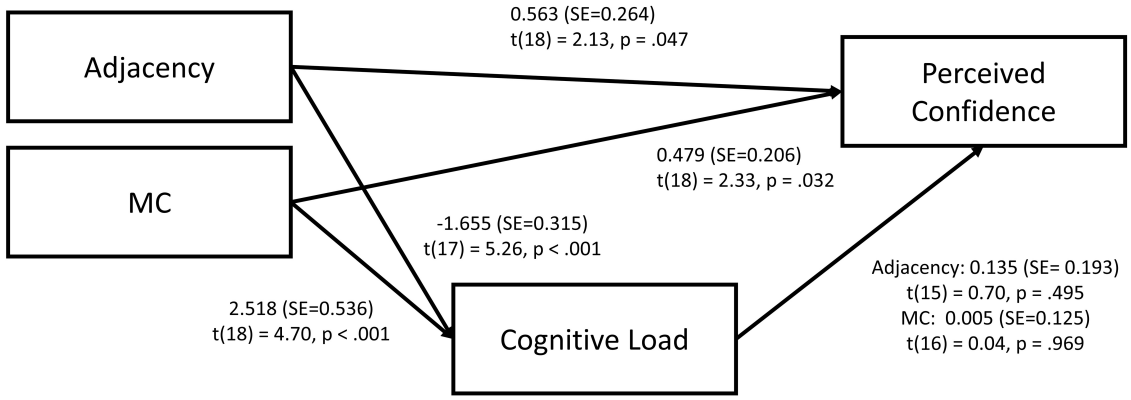


FIGURE 6 | Mediation model for adjacency and MC on perceived confidence *via* cognitive load.

Therefore, we cannot conclude overall that low MC EV has a greater effect on user confidence.

Overall, the results appear to indicate that in a task context requiring no precise knowledge (i.e., identifying a simple object in an image), confidence in an AI system can be improved with explanations, providing there is adjacency between the EV and the original image and furthermore, that the EV highlights areas of the original image that correspond to the user's perceptual understanding of the task, regardless of the EV's level of precision. In contrast, when users with greater knowledge in a particular subject are tasked with identifying specific patterns requiring precision (e.g., detecting diabetic retinopathy), they are much more critical about the type and characteristics of the EV used (Sundararajan et al., 2019). However, further research is required to investigate the relationship between the type of task and the explanations provided to complete that task in more detail.

The Mediating Effect of Cognitive Load on User Confidence

Our previous analysis (Hudon et al., 2021) indicated that adjacent EVs resulted in lower cognitive load. We posited that this implied that the cognitive effort required to process and understand the AI's EV through adjacent presentation is significantly lower than non-adjacent EV, given a reduced need to mentally associate the EV with the original image as the association is already made implicit within the EV itself. This effect was hypothesized by Dennis and Carte (1998), who stated that combining adjacent presentation order with a spatial task may lead to faster and more accurate decision making, resulting in a better CF. Furthermore, we reported low cognitive load for non-adjacent-medium MC EV. We posited that because this type of EV is very abstract, it potentially makes it difficult to process the original image's target object and allows for a snap judgment and lower cognitive workload through a disengagement effect.

However, as the current analysis indicates, it is primarily adjacency and MC that act as mediators toward increased user confidence in the AI decision visualizations and not the level of cognitive load directly. Overall, in terms of CF, the combination of low MC and non-adjacency results in the highest perceived confidence for non-adjacent visualizations even under conditions of high cognitive load. We previously proposed (Hudon et al., 2021) that the high density of information presented in this EV potentially helped users identify the target object forms by reducing epistemic uncertainty. In that, the extraneous but useful information allowed the user to swap from cognitive processing to perceptual processing to understand the model's behavior. Ongoing research in human factors is investigating the concept of epistemic uncertainty and how it affects man-machine teaming (Tomsett et al., 2020) in complex systems.

The results from the mediation analysis further indicate that the pathway to increased confidence *via* cognitive load as a mediator (H3) is not supported in our case. However, the effects of both adjacency and MC appear to have a strong direct influence on the level of perceived confidence, and that each presentation method significantly influences cognitive load. This result opens new questions for research concerning the mechanisms of action and the relationship between MC, adjacency, and perceived confidence in the results of AI decision making, which may impact how these systems are created and utilized by end-users.

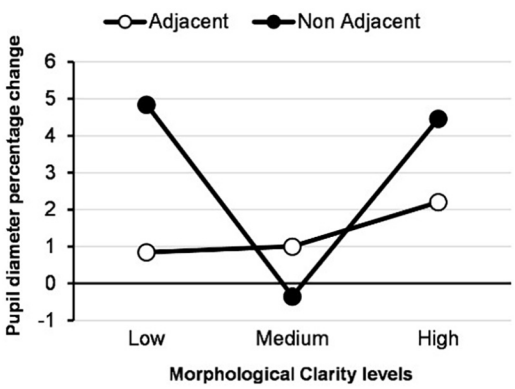


FIGURE 7 | Percentage change of pupil diameter by each type of EV.

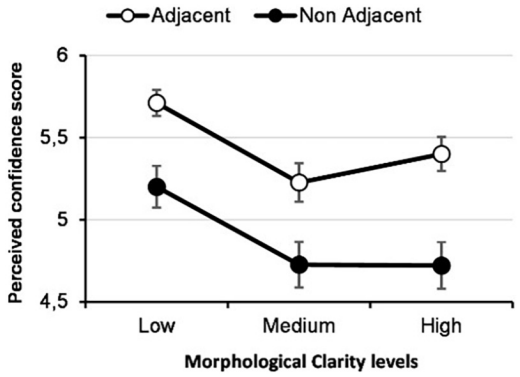


FIGURE 8 | Perceived confidence for each type of EV.

Decision-Making Dynamics: Situation Awareness, Ecological Interface Design, and Fast and Slow Thinking

Speaking in terms of decision-making dynamics, another complementary perspective from which to view these results may be through the combination of the model situation awareness (SA) proposed by Endsley (1995) ecological interface design (Bennett and Flach, 2019) and "fast and slow thinking" (Kahneman, 2011). When combined, these three approaches provide a dynamic model that blends design, perception and mechanisms that may explain the effects of MC and adjacency with regard to AI decision visualizations upon the user.

Situation awareness was defined by Endsley (1995; ref pp 1) as "the perception of the elements in the environment within a volume of time and space, the comprehension of their meaning, and the projection of their status in the near future," the model thus consists of three tiers: perception, comprehension, and projection, where each tier within the model details the human factors associated with a dynamic system's current and future state of information. The overarching goal of ecological interface design is to transform the behaviors required for workers to do their jobs: from activities requiring the worker to draw upon limited-capacity cognitive resources to activities that allow powerful perception and action capabilities to be leveraged. A critical process in achieving this goal is to understand the abstract, complex, semantic structure of a work domain and then design graphical displays that make it both concrete (i.e., easy to see) and meaningful (i.e., easy to interpret) to the worker.

Kahneman (2011) proposed that the human decision-making process is split into two parts a "fast" system and a "slow" system or system one and system two. System one includes two variants of intuitive thought, the expert and the heuristic, and operates at relatively fast speeds and automatically with providing no sense of voluntary control drawing upon the automatic neural systems involved with perception and memory. This system is the primary reactive system that consequently biases decisions *ex ante* based on a coherent interpretation of reality given current knowns and unknowns. System two by contrast, operates at a slower rate and at the subconscious level to allocate attentional resources to effortful mental activities and the conscious level to engage reasoning and volition such as when performing calculations or consciously attending to a problem. In terms of overall functionality and versatility, system two has the broader range in terms of decision making, however, it is system one that is dominant in the majority of situations as it is tied to many autonomic sensory and cognitive functions.

Thus, using this perspective to interpret the results reported here, the sense of SA creates task specific expectations within the user as it builds during task completion, these expectations are influenced by the elements of the interface design, in the current case the various visualizations of AI decisions. The developing sense of SA and the interactions with the interface dynamically cohere toward a more complete sense of SA related to the task, mediated by the adjacency and MC of information presentation within the interface. Therefore, the effects of MC and adjacency speak to methods of creating ecologically valid interfaces that positively affect SA. These further engage the decision-making process at various levels where the combination of both fast and slow processes mediated by adjacency and MC aggregate toward an increase in trust and confidence, i.e. if a task is repetitive requiring a quick "instinctual" decision, design interfaces with adjacency as a focus which engenders confidence and trust in the system until an error occurs. If, however, a more considered decision is required, such as a diagnosis from medical imagery for example, then design with non-adjacency, low MC and moderate cognitive load as a focus, to engage more rational processes through evidence gathering and consonance with expected outputs (building toward good SA) which engenders confidence building through perceived collaborative effort where errors are shared between system and user.

Contribution to Theory and Implications for Practice

The theory of CF has been used in previous research as a framework to explain the impact of different information presentation methods paired with various tasks on decision-making performance. Studies from various fields have provided validation of this theory, such as computer system development (Shaft and Vessey, 2006), geographic information systems (Dennis and Carte, 1998), and e-commerce (Brunelle, 2009; Chen, 2017). Our study incorporates CF theory into XAI by assessing which combination of presentation method and type of EV provided a better "fit" and positively influenced confidence using a spatial task. The results reported in this study provide evidence showing that adjacent EVs facilitate the CF of a problem and its representation for the user, in that they help users effectively utilize their working memory while performing the task by presenting the information in a more "understandable" and structured format than non-adjacent EVs.

Studies have shown that visualized explanations of a system or model positively impact confidence (Eiband et al., 2019; Meske and Bunde, 2020) and that the presence of transparent design further increases this impact (Kizilcec, 2016; Weitz et al., 2019). We add to this design template for confidence by showing that the choice of EV type in terms of its adjacency and MC may also

modulate the impact on user confidence. Furthermore, designs that impede the level of CF between interpretability and the task, such as in the case of non-adjacent EV, may have a null or even negative impact on confidence compared to cases where there is no explanation transparency. Implying that not all explanations of AI decision processes, be they visual or not, positively impact user trust in AI decision systems. Therefore, when designing AI decision tasks and interpretability interfaces, the aim should be to achieve a high level of congruence between the problem-solving task and the problem representation to align with the user's mental representation and knowledge of the task.

LIMITATIONS AND FUTURE WORK

The current study did not consider the cognitive distance or the degree of similarity between two images representing the same label. Consequently, some pairs of images may have appeared similar (e.g., two images of an elephant's face) and others rather different (e.g., dogs of different breeds). Users presented with an AI system's classification could potentially be more confident that the system will correctly classify a similar image than if they are presented with a dissimilar image. This difference in similarity between pairs of images could result in a bias in the results. Therefore, future work should control the degree of similarity between pairs of images. We see two opportunities to address this challenge. Firstly, one should adopt an algorithmic perspective and compute the similarity between pictures and secondly, one could task a new panel of judges to assess the perceived visual distance between images.

In this study, we only performed an analysis on the effect of Adjacency and MC on the user's confidence without including the sample's demographic data in the model. These data could have a moderating effect on the user-AI confidence (e.g., the confidence of an AI expert might be less influenced by the different EV representation than an AI novice). Moreover, we did not consider the potential impact on the confidence of the individual's propensity to trust AI. Some people are more likely to trust technology than others, and it would have been interesting to see the extent of this effect on confidence.

Finally, we did not control for algorithm type and design in this study. Further work should concentrate on replicating our results for different algorithms to examine if our observations generalize. It has been demonstrated that even if the explanation algorithm is model-agnostic, intrinsic differences in the image recognition system design can lead to differences in predictive features (Ribeiro et al., 2016). Thus, future research could investigate the impact of design choices in black-box models on the interpretability explanation and its impact on the human-algorithm interaction.

While most of the work on interpretability focuses on algorithmic approaches to create interpretable models, this study shows that human-centered EVs of such algorithms can provide an avenue for new research, where explanations are designed to fit the users' capabilities and the context of use. Future studies could investigate the impact of the highlighted areas of an image to determine if the highlighted areas that show a similar thinking process to humans have a greater impact on confidence than others.

Furthermore, explainability as currently recognized by the explainable AI community, highlights technically relevant parts of machine representations and machine models, i.e., those elements that have contributed to model accuracy (as both negative and positive attribution in the current study). However, this form of explainability does not make reference to a human-readable model which is potentially a limitation of the current approach. As a consequence, researchers introduced the term "causability," defined as cause identifiability, as an extension to "cause suitability" as a potential measurable model for human interpretability (Holzinger et al., 2020). Utilizing this model as a means through which to measure explainability would allow the measurement of how an explanation of a statement reaches a specified level of causal understanding for a user, in which the effectiveness, efficiency and satisfaction of an explanation can be determined for a given context of use. As previously discussed and further highlighted by researchers in the field (Holzinger and Müller, 2021), this will require new human-AI interfaces in the future that enable contextual understanding and also allow domain experts to ask questions and counterfactuals, i.e., "what if" questions. Investigating this new model and form of measurement is something to be considered moving forward toward explainable AI.

Concerns related to the lack of confidence between humans and AI are not only relevant to the field of HCI. They have far-reaching societal implications, as AI systems take critical decisions that can impact human lives. To continue developing better collaboration between human and AI agents, there is a need to focus on developing interpretable, transparent models that present information in a way that is compatible with human cognitive processes (European Commission, 2021). In this regard, research in the field could be expanded to include the concurrent use of neurophysiological assessment methods such as eye-fixation related potential (Palinko et al., 2010a). Studying the active brain while testing new interpretation methods would add an extra dimension that allows the testing of CF against the cognitive load of adjacent and non-adjacent low, medium, and high MC visualizations of new interpretation methods.

CONCLUSION

This study investigated the relationship between various types of EV of an AI system's output and user confidence in the system's decisions. The results show that visualization and information presentation design choices have the potential to positively impact a user's confidence in the AI system. By drawing upon CF theory (Vessey, 1991), we show that in this case, adjacency and the level of MC of the visualization have a conditional relationship upon user confidence. An EV overlaid upon the original image, no matter the level of precision of the visualization, results in a better fit between the decision task and the explanation resulting in higher confidence. On the other hand, users showed a preference for visualizations providing precision in situations of non-adjacency of the EV.

Furthermore, the relationships between various types of EVs used to explain an AI system's output and users' cognitive load cannot be explained simply as a function of cognitive load, while visualization presentation methods do contribute to both greater and lesser cognitive load, cognitive load in and of itself does not mediate the effect on a user's confidence in system decision output. The results indicate that design choices related to EVs can positively impact a user's confidence in AI systems by reducing epistemic uncertainty. Overall, our results strongly suggest that the careful consideration and application of CF theory, adjacency methods and EVs containing low MC to AI interface and task design may help improve the confidence relationship between a user and an AI decision support system. Most of the advances on interpretability and machine learning are still model-centric; this research proposes a human-centered approach to evaluate interpretability design choices.

DATA AVAILABILITY STATEMENT

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

ETHICS STATEMENT

The studies involving human participants were reviewed and approved by HEC Montreal Research Ethics Board (REB). The patients/participants provided their written informed consent to participate in this study.

AUTHOR CONTRIBUTIONS

AK: ideation, coordination, design and development, analysis, interpretation, and writing. TD: ideation, design and development, implementation, architecture, and analysis. AH: experimental design and procedures and analysis. P-ML: supervision, insight, and editing ideation. SS: ideation, insight, and supervision. All authors contributed to the article and approved the submitted version.

FUNDING

This research was financially supported by NSERC and Prompt (IRCPJ/514835-16, and 61_Léger 2016.12, respectively).

REFERENCES

Abdul, A., Vermeulen, J., Wang, D., Lim, B. Y., and Kankanhalli, M. (2018). "Trends and trajectories for explainable, accountable and intelligible systems," in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, (New York, NY), 1–18. doi: 10.1145/3173574.3174156

Adipat, B., Zhang, D., and Zhou, L. (2011). The effects of tree-view based presentation adaptation on mobile web browsing. *MIS Q. Manag. Inf. Syst.* 35, 99–121. doi: 10.2307/23043491

Attard-Johnson, J., Ó Ciardha, C., and Bindemann, M. (2019). Comparing methods for the analysis of pupillary response. *Behav. Res. Methods* 51, 83–95. doi: 10.3758/s13428-018-1108-6

Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., et al. (2020). Explainable Artificial Intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf. Fusion* 58, 82–115. doi: 10.1016/j.inffus.2019.12.012

Battig, W. F., and Montague, W. E. (1969). Category norms of verbal items in 56 categories a replication and extension of the Connecticut category norms. *J. Exp. Psychol.* 80:1.

Beatty, J. (1982). Task-evoked pupillary responses, processing load, and the structure of processing resources. *Psychol. Bull.* 91, 276–292. doi: 10.1037/0033-2909.91.2.276

Bennett, K. B., and Flach, J. (2019). Ecological interface design: thirty-plus years of refinement. *Progr. Potential. Hum. Factors* 61, 513–525. doi: 10.1177/0018720819835990

Brunelle, E. (2009). The moderating role of cognitive fit in consumer channel preference. *J. Electron. Commer. Res.* 10:8.

Brunk, J., Mattern, J., and Riehle, D. M. (2019). "Effect of transparency and trust on acceptance of automatic online comment moderation systems," in *Proceedings of the 2019 IEEE 21st Conference on Business Informatics (CBI)*, (Moscow), 429–435.

Bryson, J. J., and Theodorou, A. (2019). "How society can maintain human-centric artificial intelligence," in *Human-Centered Digitalization and Services. Translational Systems Sciences*, Vol. 19, eds M. Toivonen and E. Saari (Singapore: Springer), 305–323. doi: 10.1007/978-981-13-7725-9_16

Chen, C.-W. (2017). Five-star or thumbs-up? The influence of rating system types on users' perceptions of information quality, cognitive effort, enjoyment and continuance intention. *Intern. Res.* 27, 478–494. doi: 10.1108/IntR-08-2016-0243

Chollet, F. (2017). "Xception: deep learning with depthwise separable convolutions," in *Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, (Honolulu, HI), 1800–1807. doi: 10.1109/CVPR.2017.195

Cofta, P. (2009). "Designing for trust," in *Handbook of Research on Socio-Technical Design and Social Networking Systems*, (Hershey, PA: IGI Global), 388–401. doi: 10.4018/978-1-60566-264-0.ch026

Das, T. K., and Teng, B. (1998). Between trust and control?: developing confidence in partner cooperation in alliances. *Acad. Manag. Rev.* 23, 491–512.

DeCamp, M., and Tilburt, J. C. (2019). Why we cannot trust artificial intelligence in medicine. *Lancet Digit. Heal.* 1:e390. doi: 10.1016/S2589-7500(19)30197-9

Deng, J., Dong, W., Socher, R., Li, L.-J., Kai, Li, and Fei-Fei, L. (2009). "ImageNet: a large-scale hierarchical image database," in *Proceedings of the 2009 IEEE Conference on Computer Vision and Pattern Recognition*, (Miami, FL), 248–255. doi: 10.1109/cvprw.2009.5206848

Dennis, A. R., and Carte, T. A. (1998). Using geographical information systems for decision making: extending cognitive fit theory to map-based presentations. *Inf. Syst. Res.* 9, 194–203. doi: 10.1287/isre.9.2.194

Dietvorst, B. J., Simmons, J. P., and Massey, C. (2015). Algorithm aversion: people erroneously avoid algorithms after seeing them err. *J. Exp. Psychol. Gen.* 144, 114–126. doi: 10.1037/xge0000033

Doshi-Velez, F., and Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv* [Preprint] 1–13. doi: 10.48550/arXiv.1702.08608

Eiband, M., Buschek, D., Kremer, A., and Hussmann, H. (2019). "The impact of placebic explanations on trust in intelligent systems," in *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, (New York, NY), 1–6. doi: 10.1145/3290607.3312787

Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. *Hum. Factors* 37, 32–64. doi: 10.1518/001872095779049543

European Commission (2021). *Proposal for a Regulation of the European Parliament and of the Council Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence act) and Amending Certain Union Legislative Acts. Com 206 Final*. Brussels: European Commission.

Fairclough, S. H., Karran, A. J., and Gilleade, K. (2015). "Classification accuracy from the perspective of the user," in *Proceedings of the 33rd Annual ACM*

Conference on Human Factors in Computing Systems, (New York, NY), 3029–3038. doi: 10.1145/2702123.2702454

Gillespie, B., Muehling, D. D., and Kareklas, I. (2018). Fitting product placements: affective fit and cognitive fit as determinants of consumer evaluations of placed brands. *J. Bus. Res.* 82, 90–102. doi: 10.1016/j.jbusres.2017.09.002

Gilpin, L. H., Bau, D., Yuan, B. Z., Bajwa, A., Specter, M., and Kagal, L. (2018). "Explaining explanations: an overview of interpretability of machine learning," in *Proceedings of the 2018 IEEE 5th International Conference on Data Science and Advanced Analytics (DSAA*, (Turin), 80–89. doi: 10.1109/DSAA.2018.00018

Glikson, E., and Woolley, A. W. (2020). Human trust in artificial intelligence: review of empirical research. *Acad. Manag. Ann.* 14, 627–660. doi: 10.5465/annals.2018.0057

Goodhue, D. L., and Thompson, R. L. (1995). Task-Technology Fit and Individual Performance. *MIS Q.* 19:213. doi: 10.2307/249689

Holzinger, A., Carrington, A., and Müller, H. (2020). Measuring the quality of explanations: the system causability Scale (SCS): comparing human and machine explanations. *KI Kunstl. Intell.* 34, 193–198. doi: 10.1007/S13218-020-00636-Z/TABLES/1

Holzinger, A. T., and Müller, H. (2021). Toward human–AI interfaces to support explainability and causability in medical AI. *Computer* 54, 78–86.

Hudon, A., Demazure, T., Karran, A., Léger, P. M., and Sénécal, S. (2021). "Explainable artificial intelligence (XAI): how the visualization of AI predictions affects user cognitive load and confidence," in *Lecture Notes in Information Systems and Organisation*, eds F. D. Davis, R. Riedl, J. vom Brocke, P. M. Léger, A. B. Randolph, and G. Müller-Putz (Cham: Springer), 237–246. doi: 10.1007/978-3-030-88900-5_27

Jakesch, M., French, M., Ma, X., Hancock, J. T., and Naaman, M. (2019). "AI-Mediated communication," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, (New York, NY), 1–13. doi: 10.1145/3290605.3300469

Jia, R., Steelman, Z., and Reich, B. H. (2017). Using mechanical turk data in IS research: risks, rewards, and recommendations. *Commun. Assoc. Inf. Syst.* 41, 301–318. doi: 10.17705/1CAIS.04114

Kahneman, D. (2011). *Thinking, Fast and Slow*. London: Macmillan.

Keller, K. L., and Staelin, R. (1987). Effects of Quality and Quantity of Information on Decision Effectiveness. *J. Consum. Res.* 14:200. doi: 10.1086/209106

Kizilcec, R. F. (2016). "How much information?: effects of transparency on trust in an algorithmic interface," in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, (New York, NY), 2390–2395. doi: 10.1145/2858036.2858402

Lankton, N., McKnight, D. H., and Tripp, J. (2015). Technology, humanness, and trust: rethinking trust in technology. *J. Assoc. Inf. Syst.* 16, 880–918. doi: 10.17705/1jais.00411

Lee, J. D., and See, K. A. (2004). Trust in automation: designing for appropriate reliance. *Hum. Factors* 46, 50–80. doi: 10.1518/hfes.46.1.50_30392

Luhmann, N. (2001). Familiarity, confidence, trust: problems and alternatives. *Réseaux* 108:15. doi: 10.3917/res.108.0015

Mcknight, D. H., Carter, M., Thatcher, J. B., and Clay, P. F. (2011). Trust in a specific technology. *ACM Trans. Manag. Inf. Syst.* 2, 1–25. doi: 10.1145/1985347.1985353

Meske, C., and Bunde, E. (2020). "Transparency and trust in human-AI-interaction: the role of model-agnostic explanations in computer vision-based decision support," in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics*, eds H. Degen and L. Reinerman-Jones (Cham: Springer International Publishing), 54–69. doi: 10.1007/978-3-030-50334-5_4

Miller, G. A. (1995). WordNet: a lexical database for English. *Commun. ACM* 38, 39–41. doi: 10.1145/219717.219748

Mohseni, S., Zarei, N., and Ragan, E. D. (2018). A multidisciplinary survey and framework for design and evaluation of explainable AI systems. *ACM Trans. Interact. Intell. Syst.* 1:1. doi: 10.1145/3387166

Müller, R., Kessler, F., Humphrey, D. W., and Rahm, J. (2021). Data in context: how digital transformation can support human reasoning in cyber-physical production systems. *Future Intern.* 13:156.

Nourani, M., Honeycutt, D. R., Block, J. E., Roy, C., Rahman, T., Ragan, E. D., et al. (2020). "Investigating the importance of first impressions and explainable ai with interactive video analysis," in *Proceedings of the Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, (New York, NY), 1–8. doi: 10.1145/3334480.3382967

Nuamah, J. K., Seong, Y., Jiang, S., Park, E., and Mountjoy, D. (2020). Evaluating effectiveness of information visualizations using cognitive fit theory: a neuroergonomics approach. *Appl. Ergon.* 88:103173. doi: 10.1016/j.apergo.2020.103173

Palinko, O., Kun, A. L., Shyrokov, A., and Heeman, P. (2010b). "Estimating cognitive load using remote eye tracking in a driving simulator," in *Proceedings of the 2010 Symposium on Eye-Tracking Research & Applications, ETRA 2010*, (Austin, TX).

Palinko, O., Kun, A. L., Shyrokov, A., and Heeman, P. (2010a). "Estimating cognitive load using remote eye tracking in a driving simulator," in *Proceedings of the 2010 Symposium on Eye-Tracking Research & Applications - ETRA '10*, (New York, NY), 141. doi: 10.1145/1743666.1743701

Parasuraman, R., and Riley, V. (1997). Humans and automation: use, misuse, disuse, abuse. *Hum. Factors J. Hum. Factors Ergon. Soc.* 39, 230–253. doi: 10.1518/001872097778543886

Poursabzi-Sangdeh, F., Goldstein, D. G., Hofman, J. M., Vaughan, J. W., and Wallach, H. (2018). "Manipulating and measuring model interpretability," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, (New York, N).

Ribeiro, M. T., Singh, S., and Guestrin, C. (2016). ""Why should i trust you?" Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, (New York, NY), 1135–1144. doi: 10.1145/2939672.2939778

Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat. Mach. Intell.* 1, 206–215. doi: 10.1038/s42256-019-0048-x

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2020). Grad-CAM: visual explanations from deep networks via gradient-based localization. *Int. J. Comput. Vis.* 128, 336–359. doi: 10.1007/s11263-019-01228-7

Shaft, T. M., and Vessey, I. (2006). The role of cognitive fit in the relationship between software comprehension and modification. *MIS Q. Manag. Inf. Syst.* 30, 29–55. doi: 10.2307/25148716

Snodgrass, J. G., and Vanderwart, M. (1980). A standardized set of 260 pictures: norms for name agreement, image agreement, familiarity, and visual complexity. *J. Exp. Psychol. Hum. Learn. Mem.* 6, 174–215. doi: 10.1037/0278-7393.6.2.174

Sundararajan, M., Taly, A., and Yan, Q. (2017). Axiomatic attribution for deep networks. 34th Int. *Conf. Mach. Learn. ICML* 2017, 5109–5118.

Sundararajan, M., Xu, S., Taly, A., Sayres, R., and Najmi, A. (2019). Exploring principled visualizations for deep network attributions. Proceedings of the ACM IUI 2019 Workshops, Los Angeles, CA. 11.

Tomsett, R., Preece, A., Braines, D., Cerutti, F., Chakraborty, S., Srivastava, M., et al. (2020). Rapid trust calibration through interpretable and uncertainty-aware AI. *Patterns* 1:100049. doi: 10.1016/j.patter.2020.100049

van der Land, S., Schouten, A. P., Feldberg, F., van den Hooff, B., and Huysman, M. (2013). Lost in space? Cognitive fit and cognitive load in 3D virtual environments. *Comput. Human Behav.* 29, 1054–1064. doi: 10.1016/j.chb.2012.09.006

Vance, A., Elie-Dit-Cosaque, C., and Straub, D. W. (2008). Examining trust in information technology artifacts: the effects of system quality and culture. *J. Manag. Inf. Syst.* 24, 73–100. doi: 10.2753/MIS0742-1222240403

Vessey, I. (1991). cognitive fit: a theory-based analysis of the graphs versus tables literature. *Decis. Sci.* 22, 219–240. doi: 10.1111/j.1540-5915.1991.tb00344.x

Vessey, I., and Galletta, D. (1991). Cognitive fit: an empirical study of information acquisition. *Inf. Syst. Res.* 2, 63–84. doi: 10.1287/isre.2.1.63

Viera, A. J., and Garrett, J. M. (2005). Understanding interobserver agreement: the kappa statistic. *Fam. Med.* 37, 360–363.

Wanner, J., Herm, L.-V., Heinrich, K., Janiesch, C., and Zschech, P. (2021). "white, grey, black: effects of XAI augmentation on the confidence in AI-based decision support systems," in *Proceedings of the 41st International Conference*

on Information Systems, ICIS 2020, Making Digital Inclusive: Blending the Locak and the Global, (Hyderabad), 9.

Weitz, K., Schiller, D., Schlagowski, R., Huber, T., and André, E. (2019). "Do you trust me?: increasing user-trust by integrating virtual agents in explainable AI interaction design," in *Proceedings of the 19th ACM International Conference on Intelligent Virtual Agents*, (New York, NY), 7–9. doi: 10.1145/3308532.3329441

Wickens, C. D. (2008). Multiple resources and mental workload. *Hum. Factors* 50, 449–455. doi: 10.1518/001872008X288394

Yin, M., Wortman Vaughan, J., and Wallach, H. (2019). "Understanding the effect of accuracy on trust in machine learning models," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, (New York, NY), 1–12. doi: 10.1145/3290605.3300509

Yzerbyt, V., Muller, D., Batailler, C., and Judd, C. M. (2018). New recommendations for testing indirect effects in mediational models: the need to report and test component paths. *J. Pers. Soc. Psychol.* 115, 929–943. doi: 10.1037/PSPA0000132

Conflict of Interest: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Publisher's Note: All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Copyright © 2022 Karran, Demazure, Hudon, Senecal and Léger. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Check for updates

Article

Designing and Learning With Pedagogical Agents: An Umbrella Review

Journal of Educational Computing Research
2025, Vol. 62(8) 1907–1936
© The Author(s) 2024
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/07356331241288476
journals.sagepub.com/home/jec
Sage

Noah L. Schroeder[1], Robert O. Davis[2], and Eunbyul Yang[2]

Abstract

Pedagogical agents are virtual characters that instructional designers include in learning environments to help students learn. Research in the area has flourished for thirty years, yet there are still critical questions about the efficacy of pedagogical agents for influencing learning and affect. As such, we conducted an umbrella review to synthesize the field. We located 17 systematic reviews or meta-analyses focused on the use of pedagogical agents in educational settings. We found that agents can have small positive effects on learning, motivation, and other affective variables. However, we still cannot say how one should design a pedagogical agent for any given educational context. We highlight the limitations of existing theory in the area, as well as existing reviews from a practical and methodological perspective, and highlight productive areas for future research.

Keywords

pedagogical agent, virtual human, review, learning, motivation

Introduction

For the last three decades researchers have investigated the use of virtual characters, or pedagogical agents (PAs) within multimedia learning environments to help students

[1]Wright State University, OH, USA
[2]Chonnam National University, Republic of Korea

Corresponding Author:
Robert O. Davis, Department of Education, Chonnam National University, 77 Yongbong-ro, Buk-gu, Gwangju 61186, Republic of Korea.
Email: rdavis@jnu.ac.kr

learn (Siegle et al., 2023). Researchers have examined the use of PAs in various contexts. For example, Abdelghani et al. (2023) designed an agent that utilized a large-language model, Jaques et al. (2023) used a PA to support collaborative learning, and Liu et al. (2022) used a PA to support meditation. To date, it is largely believed that pedagogical agents can be effective learning tools, but the results often have been mixed due to contextual differences within studies (Castro-Alonso et al., 2021; Heidig & Clarebout, 2011; Schroeder et al., 2013).

While this ever-growing body of research has provided us with vast knowledge about PAs, we still lack the ability to provide evidence-based, generalizable design principles for pedagogical agents given any particular participant population, intervention type (e.g., game-based environment, lecture video, etc.), or instructional media (e.g., virtual reality, computer-based environment, etc.). This can be problematic as PAs are continually integrated into educational technologies. Despite this limitation, the use of PAs continues to be a common research topic and there have been a number of recent reviews attempting to synthesize the literature (Armando et al., 2022; Davis et al., 2023; Peng & Wang, 2022; Wang et al., 2023). Yet any systematic research synthesis is constrained by its inclusion and exclusion criteria,the result of which is that while we may have many reviews in the field of PAs, we have no meta-synthesis providing overarching insights into the effects and design of PAs. Accordingly, the purpose of this study is to conduct an umbrella review, also known as a review of reviews, to determine what research has been synthesized, what the findings of various reviews mean together, and extract practical implications for PA design. In addition, we document the limitations of the field that emerge through the umbrella review and highlight productive directions for future research.

Literature Review

As PA researchers, we are often asked a direct and rather blunt question: why might PAs facilitate learning? The literature provides no definitive answer. For example, some may suggest it is the pedagogy PAs can embody (Dinçer & Doganay, 2015), while others maintain it is the social processes within the learner that their presence in the learning environment initiates (Wang et al., 2008). Presumably the answer lies somewhere in between these two different conceptual positions – PAs can facilitate a variety of instruction techniques that are long established for aiding learning, such as visual signaling (Choi & Clark, 2006). However, what may provide advantages over other types of multimedia learning is their ability to initiate social processes while supporting evidence-based instructional techniques. As such, in this section we explore why social cues may be important in multimedia learning, how they can support the learning process, and how PAs can specifically embody social cues within multimedia learning. We then explore the various aspects researchers must consider when designing PAs and summarize the diverging research findings in the area.

The Significance of Social Cues in Multimedia Learning

A frequent use of PAs in the literature is as a lecturer or an instructor to facilitate the learning process (Schroeder & Gotch, 2015). According to social agency theory, the presence of a PA exhibiting human-like social cues allows users to adopt a social stance towards the computer-generated character. The auditory and visual representations of PAs enable users to engage within a social interaction framework, leading to deeper learning (Mayer et al., 2003). This deeper cognitive engagement is due to the PA being perceived as another entity in a communicative interaction (Atkinson et al., 2005; Lee & Nass, 2003). However, researchers have shown that the qualities of the PA may determine the extent to which learning is supported by PAs. For example, human-like functionalities, such as eye gaze, facial expressions, body movement, and gestures, contribute to anthropomorphizing the PA, thereby maintaining deep user engagement (Woo, 2009). As a result, designing PAs capable of eliciting social responses can influence the cognitive and affective processes considered crucial for learning.

While social agency theory has provided a theoretical foundation for PA research for twenty years, there are few meaningful extensions of the theory beyond its initial implication that social cues can support learning. Researchers have explored how social cues can support learning. For example, Domagk (2010) found that the valence of social cues can influence learning; however, there are few explorations based on social agency theory of how PAs influence measured social processes. This could be due to the relative lack of specific hypotheses as to which social processes influence which specific parts of the learning process. For example, from a cognitive theory of multimedia learning theory (Mayer, 2014b) perspective, how and to what extent would social processes influence the selection of relevant information, the organization of that information, or the integration of that information with prior knowledge? In an effort to address some of these theoretical questions, Schneider et al. (2022) proposed the cognitive-affective-social theory of learning in digital environments (CASTLE) that provides more specific hypotheses of how social processes may influence multimedia learning.

Cognitive-Affective-Social Theory of Learning in Digital Environments (CASTLE). The CASTLE argues that learning in digital environments inherently involves social processes that influence learning (Schneider et al., 2022). Building off a foundation of the cognitive theory of multimedia learning (Mayer, 2014b) and the cognitive-affective theory of learning with media (Moreno, 2006), CASTLE integrates social behaviors into a comprehensive theory that considers cognitive, affective, and social processes related to learning in digital environments. Based on the assumption that learners' long-term memory activates social processes when exposed to social cues, CASTLE proposes that learning can be enhanced through social facilitation in multimedia learning settings. Specifically, Schneider et al. (2022) proposed that integration of social entities into digital educational materials can boost learner motivation and foster positive shifts in attitudes towards these resources. Consequently, PAs can be viewed as instrumental elements that support learning outcomes.

While CASTLE provides explanations for some empirical observations and the theorized effects of social cues in various learning materials, it doesn't offer explicit predictions about the magnitude or nature of how these cues influence learning outcomes (Schneider et al., 2022). Moreover, the long-standing "computers as social actors" (CASA) paradigm (Lee & Nass, 2003) suggests that interactions between humans and technological platforms, such as computers, inherently have a social dimension, extending beyond just PAs.

Summary of Relevant Theories. In summary, social agency theory and CASTLE underscore the potential to boost learner engagement and performance through the meticulous design of PAs, presumably due to the initiation of social responses within the learner. In particular, the level of anthropomorphism embedded in the design of these agents can play a crucial role in shaping how learners perceive and engage with educational content. Research to date, to some extent, supports this hypothesis (Aljaroodi et al., 2019; Garau et al., 2003). As a result, a significant body of research has focused on refining the design of PAs to amplify the social cues they offer in multimedia learning settings, with the aim of potentially enhancing learning outcomes. However, designing pedagogical agents is not a simple process. Rather, designing an effective pedagogical agent requires an in-depth understanding of the conditions it will be used in (see the Pedagogical Agent - Conditions of Use model in Heidig & Clarebout, 2011). In the next section, we explore the multitude of factors designers must consider once they have an understanding of their contextual use case.

Designing Pedagogical Agents

The design of PAs can be critically examined and compared using the Pedagogical Agents—Levels of Design (PALD) framework proposed by Heidig and Clarebout (2011). PALD organizes PA design considerations into three hierarchical layers: (1) Global design, (2) Medium design, and (3) Detailed design.

Global Design Level. At this foundational tier, designers are presented with two primary decisions: the nature of the character (whether it's human or non-human) and its mode of representation (either animated or static). While human representations are common, there are also options to use animal or object-based entities. For instance, PAs have been designed as pets for emotional support (Chen & Chen, 2014) or even as parrots that instruct in math (Atkinson, 2005).

Medium Design Level. Heidig and Clarebout (2011) suggest that the medium level is made up of two categories of decisions: technical decisions and character choices.

Technical Decisions. On the technical front, decisions split into visual and auditory considerations (Heidig & Clarebout, 2011). Visually, designers must weigh factors like the degree of realism, intricacy of animation, and whether to opt for a partial or

complete portrayal of the PA. The visual spectrum can span from basic illustrations to lifelike animations where the PA's movements are fluid and natural. Emotional subtleties can be integrated to enhance the agent's lifelike appearance (Schneider et al., 2022), and animations can incorporate human-like social cues such as gestures and facial expressions (Mayer & DaPra, 2012; Mayer & Estrella, 2014). Auditory considerations encompass voice selection and modulation, with decisions such as for the PA to convey information via spoken word or text-based dialogue (Davis, 2018), or what type of voice the agent should use (Craig & Schroeder, 2017).

Character Choices. The second facet of the medium level revolves around character selection. Bendel (2003) proposes a tripartite framework for this process:

1. Determine the features or capabilities the PA should possess, as these will dictate its visual and behavioral traits.
2. Define the PA's functional role, be it as an expert, teacher, mentor, peer, or facilitator (Sikström et al., 2022).
3. Decide on a physical model that aligns with the PA's role, which can range from realistic to fictional archetypes inspired by societal, artistic, or scientific contexts.

Detailed Design Level. This most intricate level involves choices that descend from decisions made in the preceding tiers. For instance, if a human character is selected at the global level, detailed choices about age, gender, physique, clothing, and other features must be determined (Heidig & Clarebout, 2011). Furthermore, if spoken dialogue is the chosen mode of communication, phonetic nuances such as intonation, emphasis, and speech pace need to be defined at this level.

The Present Study

The academic discourse surrounding the effectiveness of PAs for influencing learning and affective outcomes is inconsistent. For example, if we examine a brief (and admittedly, incomplete) chronology, Heidig and Clarebout (2011) found largely non-significant results, while Schroeder et al. (2013) suggested a modest yet statistically significant improvement in learning outcomes due to PAs. A few years later, a review by Martha and Santoso (2019) found that only half of the studies they reviewed indicated a positive influence of PAs on learning outcomes. Similarly, Dai et al.'s (2022) analysis highlighted inconsistent empirical data on PAs in multimedia learning environments.

To summarize, reviews to date have arrived at different conclusions, ranging from mixed results (Dai et al., 2022), to small positive effects (Castro-Alonso et al., 2021; Schroeder et al., 2013), and the conclusion that we are examining too simplistic of a question that does not reflect the realities of learning environments (Heidig & Clarebout, 2011). The field has a plethora of review studies, yet all have slightly

different goals, inclusion criteria, and methodological approaches. The result has been that we have seen a variety of conclusions that are difficult to reconcile.

In an effort to synthesize the vast body of review studies in the field of PAs, we conducted an umbrella review. Umbrella reviews are reviews of systematic reviews focused on understanding the overall findings in a field around a phenomenon (Aromataris et al., 2015). They enable researchers to compare and contrast the results from various syntheses to try and draw synthesized conclusions (Aromataris et al., 2015). As such, an umbrella review is the ideal methodology to help us synthesize the considerable number of systematic reviews in the field of PA research. Using this methodology, we examine the following research questions:

1. What are the publication trends in regards to systematic reviews in the field of pedagogical agents?
2. What do we know about the impact of PAs?
3. What do we know about designing PAs?
4. How well have the existing systematic reviews in the fields been conducted and reported?

Methods

An umbrella review was conducted using reporting standards from the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021) and Aromataris et al.'s (2015) guidance for planning, conducting, and reporting umbrella reviews.

Literature Search

A comprehensive search of the literature was conducted to identify existing systematic reviews and meta-analyses focused on PAs. On July 15, 2023, the following nine databases were searched: Academic Search Complete, ACM Digital Library, APA PsychInfo, CINAHL, Education Research Complete, ERIC, Google Scholar, Medline with full text, and Web of Science Core Collection. This process identified 177 studies for consideration. The search string used for each database and the number of studies identified from each database are provided in Table 1. After removing 69 duplicates using Zotero (https://www.zotero.org/), 108 studies remained for consideration. An informal literature search produced an additional six studies for consideration.

Inclusion Criteria

In order to be included in this review, existing systematic reviews and meta-analyses in the field of PAs had to meet the following criteria:

Table 1. List of Databases Searched, the Search String Used for Each Database, and the Number of Studies Identified in Each Database.

Database	Studies located
Academic search complete	15
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
ACM digital Library	4
Search string (abstract): [[Abstract: "Embodied agent*"] OR [abstract: "Animated agent*"] OR [abstract: "Pedagogical agent*"] OR [abstract: "Virtual human*"]] AND [[abstract: Review] OR [abstract: Meta-ana*]]	
APA PsycInfo	19
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
CINAHL	5
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Education research complete	17
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
ERIC	11
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Google scholar	12
Search string: Allintitle: "Embodied agent" OR "animated agent" OR "pedagogical agent" OR "virtual human" AND (review OR meta-ana)	
Medline with full text	21
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Web of science core collection	73
Search string (abstract): ("Embodied agent"* OR "animated agent"* OR "pedagogical agent"* OR "virtual human"*) AND (review OR meta-ana*)	
Total before duplicate removal	177
Duplicates removed	69
Total after duplicate removal	108
Added via informal searches	6
For screening	114

- Studies must be systematic reviews and/or meta-analyses. Reviews that did not describe a systematic search process were excluded.
- The review must be focused on non-artificially intelligent PAs. We excluded studies reviewing artificially intelligent agents programmed to interact with users or the learning environment[1], as well as reviews of avatars that were representations of self.

- The review must provide an analysis of categorizable variables or basic descriptive statistics. In other words, novel frameworks (e.g., Morrow et al., 2021) or narrative literature reviews (e.g., Siegle et al., 2023) were excluded.
- Studies must be written in either English or Korean.

Phase I Screening

The first phase of screening was conducted by one author, who reviewed all the titles and, if not clear from the title, abstracts of each study to determine if the study met the inclusion criteria. If the title and abstract were unclear if it should be included in the analysis, it was retained for full-text examination in Phase II screening. As shown in Figure 1, 41 studies were retained for full-text review.

Phase II Screening

The second phase of screening was conducted by the same author who conducted the phase I screening, and involved obtaining the full-text copies of the retained 41 studies. The studies were read in full to see if they met the inclusion criteria. As shown in Figure 1, this process identified 17 studies that were retained for data extraction and analysis.

Data Extraction

A variety of data were extracted from each study in order to address each research question. Specifically, Table 2 specifies what information was extracted and how it was coded in respect to each research question. Our full dataset containing the codes extracted from each study is available in Supplementary Materials 1.

Inter-Rater Agreement

Four studies (24% of the sample) were randomly selected to be coded by a second author to establish inter-rater agreement. Inter-rater agreement was found to be lower than desired (75%) when the metric for study quality was included. With study quality removed from the calculations, inter-rater agreement was substantially higher (89%). The quality evaluation metric will be addressed in the discussion section. When disagreements were found, the team referred back to the original study and discussed discrepancies until consensus was reached.

Data Analysis

Data were analyzed based on the various approaches specified below.

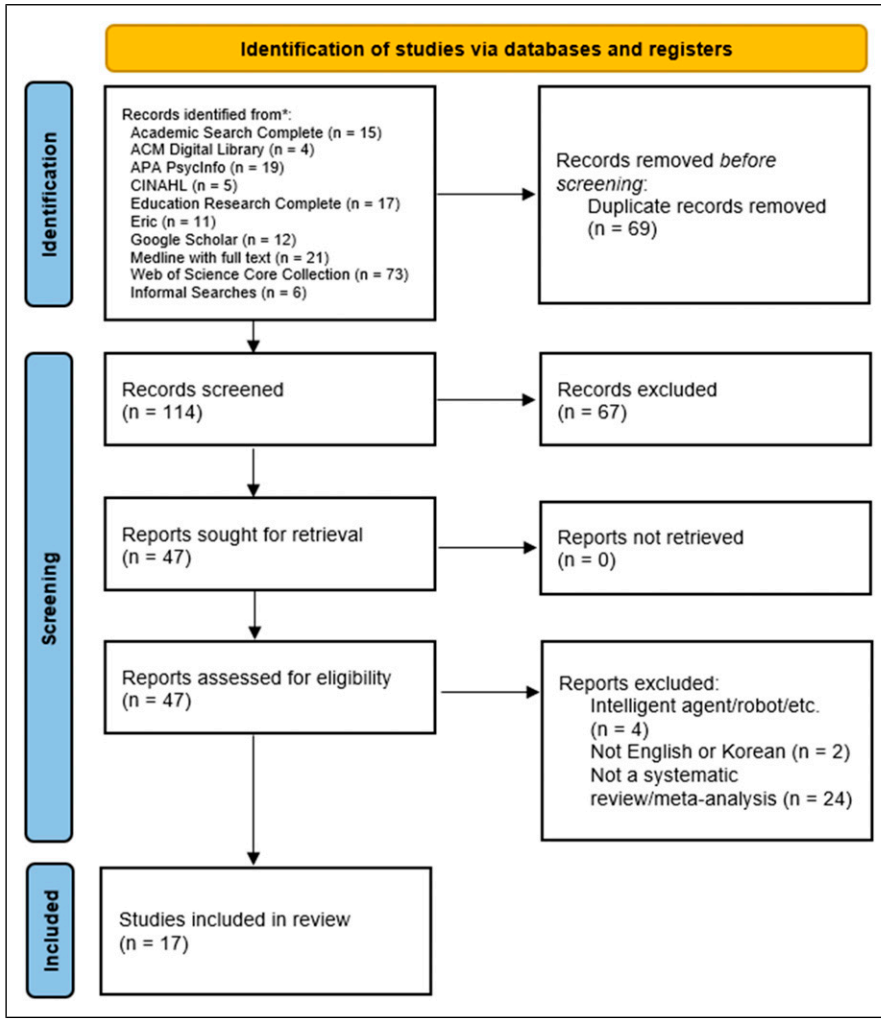


Figure 1. PRISMA flow diagram of our search and screening process (adapted from Page et al., 2021) created using Haddaway et al.'s (2022) online tool.

Research Question 1: What are the publication trends in regards to systematic reviews in the field of pedagogical agents?

This research question examines aspects such as how many reviews have been published in the area and the methods used in each study. To address this research question, basic descriptive statistics were examined.

Table 2. The Information That was Extracted and How It was Coded in Respect to Each Research Question.

Research question	Code category	Coding scheme
1	Study	APA style citation
1	Year	Year of publication
1	Type of review	Coded as systematic review or meta-analysis
1	Analysis software	Coded as N/A (systematic review), comprehensive meta-analysis (CMA), R, or SPSS.
1	Date of database search	As reported by authors
1	Number of databases searched	As reported by authors
1	Number of articles included	As reported by authors
1	Number of participants included	As reported by authors. We coded the total number of participants across all studies if the authors reported this information. If the authors provided the number of participants in each study but did not provide a summary across all studies, it was coded as 'not reported'
1	Countries/regions of origin studies	As reported by authors
1	Quality of studies examined	Coded as the authors evaluated quality or authors did not evaluate study quality
2, 3	Research question(s) and Findings(s)	As reported by the authors. We synthesized the findings of each review in alignment to each research question. We recorded only key findings. Significant results were recorded and what was not significant was recorded when quantitative data were reported in the review. However, we did not record every single effect size reported in meta-analyses or quantitative systematic reviews
4	"Is the review question clearly and explicitly stated?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the inclusion criteria appropriate for the review question?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A

(continued)

Table 2. (continued)

Research question	Code category	Coding scheme
4	"Was the search strategy appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the sources and resources used to search for studies adequate?" (Aromataris et al., 2015, p. 136)	Coded as yes (if more than one database was searched), no, unclear, N/A
4	"Were the criteria for appraising studies appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Was critical appraisal conducted by two or more reviewers independently?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the methods used to combine studies appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Was the likelihood of publication bias assessed?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were recommendations for policy/practice supported by reported data?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A
4	"Were the specific directives for new research appropriate?" (Aromataris et al., 2015, p. 136)	Coded as yes, no, unclear, N/A

Research Question 2: What do we know about the impact of PAs?

Researchers have examined the impacts of pedagogical agents in relation to a variety of outcome variables. As such, outcome variables from each review were categorized so that the evidence around each outcome could be synthesized.

Research Question 3: What do we know about designing PAs?

Due to the variety of details in pedagogical agent design, it was important to establish what areas reviews of pedagogical agents have been conducted. To address this research question, Heidig and Clarebout's (2011) PALD framework for pedagogical agent research was used. Study findings were organized by the levels of the PALD framework and the data were synthesized.

Research Question 4: How well have the existing systematic reviews in the fields been conducted and reported?

Examining study quality is a necessary component of the PRISMA statement (Page et al., 2021). It is also necessary to examine the quality of systematic reviews when conducting an umbrella review (Aromataris et al., 2015). However, study quality is not frequently examined in systematic reviews of educational technologies. To address this gap in the literature, we used items from Aromataris et al.'s (2015) quality analysis instrument and examined the results descriptively.

Results

In total, 17 systematic reviews and/or meta-analyses met the inclusion criteria. Below, the results are presented in relation to each research question.

Research Question 1: What are the publication trends in regards to systematic reviews in the field of pedagogical agents?

Using the coding categories specified by Aromataris et al. (2015), the sample of systematic reviews and meta-analyses that met the inclusion criteria can be objectively described. As shown in Figure 2, while systematic reviews or meta-analyses have been consistently published in the field, the last three years have seen a dramatic increase in their number.

Of the 17 reviews that met our inclusion criteria, seven of the 17 (41%) were meta-analyses. All of the meta-analyses were conducted using Comprehensive Meta-Analysis software (Biostat Inc, 2023). Nine of the reviews (53%) did not report

Figure 2. Number of systematic reviews or meta-analyses published about pedagogical agents by year. Note that our literature search was conducted in July, 2023.

when their literature searches were conducted, and authors searched between 1 and 10 databases in their literature search (mean = 5.88, median = 5). Systematic reviews and meta-analyses in the sample reviewed a minimum of 12 studies and a maximum of 115 studies (mean = 35.41, median = 28). In 10 of the 17 (59%) reviews included in our analyses the authors did not explicitly report the total number of participants in the studies they reviewed. However, those that did reported analyses with a minimum of 752 participants and a maximum of 4817 participants. Only four of the included studies discussed the countries or regions of the world in which their studies were published. However, there were varying approaches to this. For example, Armando et al. (2022) stated that they reviewed studies from "western culture", whereas *Yusoff et al. (2022) reported the number of studies by country. Finally, none of the reviews in our sample reported evaluating the quality of the studies they reviewed.

Research Question 2: What do we know about the impact of PAs?

This research question specifically focused on the use of pedagogical agents in comparison to non-agent systems (or in the case of Davis (2018) and Davis et al. (2023), the use of a static agent). Table A1 (see Appendix A) provides the summarized findings from each study in regards to learning outcomes, motivational outcomes, and other outcomes.

As shown in Table A1, ten of the reviews in our sample examined the impact of pedagogical agents on learning outcomes. Seven of the 10 studies were meta-analyses, and all showed significant, positive effects in favor of pedagogical agents. Of the three systematic reviews, one found mostly positive results (Papoutsi & Rangoussi, 2020), while two found mostly null or mixed results (Heidig & Clarebout, 2011; Martha & Santoso, 2019).

Four of the reviews in our sample examined the impact of pedagogical agents on motivational outcomes. While the two meta-analyses found statistically significant, positive effects showing that pedagogical agents can positively impact learner motivation, the two systematic reviews did not. Specifically, Heidig and Clarebout (2011) found no convincing evidence that agents significantly impact learner motivation, and Schroeder and Adesope (2014) found a few interesting trends regarding motivation, but these were largely not supported by statistically significant results.

Finally, three reviews examined the impacts of pedagogical agents on other types of outcomes. Specifically, Wang et al. (2023) found that affective pedagogical agents can lead to significant improvements in learners' positive emotions. Meanwhile, Schroeder and Adesope (2014) and Davis (2018) examined the influence of pedagogical agents on learners' perceptions of agent persona and cognitive load. Results indicated that learners may be positively influenced by pedagogical agents in relation to their ratings of the agent's persona. Regarding cognitive load, Davis (2018) found that gestures provided minimal benefits, but did not significantly reduce cognitive load. However, Schroeder and Adesope (2014) observed a positive influence on cognitive load when

the control group was a text-only learning condition, but the results were mixed when the control group was a voice-only learning condition.

Research Question 3: What do we know about designing PAs?

As shown in Table A2, 13 of the 17 reviews (76%) in our sample addressed the impacts of pedagogical agent design in their findings. We first organized these findings according to Heidig and Clarebout's (2011) PALD framework.

The global level of Heidig and Clarebout's (2011) addresses the question of if a PA should appear as a human or some other type of character. Four of the reviews in our analysis evaluated this question. However, the results were mixed. Two of the reviews found that the agent's appearance style did not make a significant difference (Peng & Wang, 2022; Schroeder et al., 2013). However, one of the other reviews found that humanoid agents that gestured decreased cognitive load, while non-humanoid characters significantly increased cognitive load (Davis, 2018), and the other found that virtual agents were more effective than human instructors for supporting positive emotions and learners' intrinsic motivation (Wang et al., 2023).

Twelve of the 17 reviews (71%) discussed findings in relation to the medium level of pedagogical agent design, which deals with the technical decisions and specific functions or roles of pedagogical agents within the learning environment (Heidig & Clarebout, 2011). While there are a variety of different design factors that can be considered at this level, the results show a few patterns in the literature. First, there seems to be some evidence among reviews that agents should be anthropomorphized in regards to their use of nonverbal social cues (Aljaroodi et al., 2019; Dai et al., 2022). However, this is difficult to reconcile with the findings of reviews that examined the impact of gestures. For example, while three reviews (Davis, 2018; Davis et al., 2021, 2023) found that gesturing agents improved learning, two other meta-analyses did not find gesturing to be a significant moderator with respect to learning (Castro-Alonso et al., 2021; *Yusoff et al., 2022). Similarly mixed results were found in regards to facial expression and emotions. Davis et al. (2021) found that facial expressions improved learnings ratings of the pedagogical agent's persona in most cases, yet Castro-Alonso et al. (2021) and Peng and Wang (2022) did not find that facial expressions significantly moderated agent's impacts on learning in their meta-analyses. As a whole, our analysis indicates that the data and findings from design decisions at the medium level are mixed and often contradictory. Yet, it is important to remember the context of the reviews. For example, Castro-Alonso et al. were comparing PA to non-PA systems. Their finding that facial expressions did not moderate learning outcomes is not directly comparing a PA with expressions to a PA without expressions, but rather a PA with expressions compared to a non-agent system, compared to a PA without expressions compared to a non-agent system.

Finally, four of the 17 reviews (24%) examined the detail level of the PALD framework, which focuses on the design of specific visual components of the agent such as its age or clothing (Heidig & Clarebout, 2011). From this data we also find

mixed results. For example, Castro-Alonso et al. (2021) found that the agent's gender did not significantly moderate learning outcomes. However, Armando et al. (2022) argued that gender stereotypes exist when learners interact with agents. That said, their results also showed mixed results in regards to the impact of an agent's gender on learning outcomes. There are other aspects at the detail level of pedagogical agent design that were also addressed by reviews in the sample. One review addressed the issue of an agent's age. *Yusoff et al. (2022) found that the agent's age did not make a significant impact on cognitive load. In addition, one review had implications for the detail level of pedagogical agent design from a broader perspective. Aljaroodi et al. (2019) argued that it may be important to have similarities between the agent and the learner, an argument reminiscent of the similarity-attraction hypothesis (Moreno & Flowerday, 2006).

Research Question 4: How well have the existing systematic reviews in the fields been conducted and reported?

The quality of the existing systematic reviews was examined using a metric by Aromataris et al. (2015). The metric examines 10 different aspects of systematic review and/or meta-analysis reporting.

First, it was examined if the reviews in the sample clearly and explicitly stated the research question(s) being investigated. The majority of the studies (14/17) explicitly stated the research question(s); however, three studies did not. Next, an assessment regarding the appropriateness of the inclusion criteria for answering the research questions was conducted. All reviews except one provided clear inclusion criteria to address the stated research questions.

In regards to the search strategies used; all the reviews used what could be considered an appropriate search strategy. However, two of the reviews appeared to use a limited database search, so the resources used for the search seemed inadequate.

The next examination focused on how the reviews appraised the studies they included, which can be interpreted as their data extraction method from the primary studies (i.e., the coding paradigm). Generally, these criteria were appropriate, with only three studies having unclear appraisal criteria.

However, these relatively positive evaluations do not extend further to the methods reporting of the reviews evaluated. For example, only five (29%) of the reviews had two or more independent coders extracting data. Moreover, of the seven meta-analyses, five did not explicitly address how they combined studies. Fortunately, all but one meta-analysis did report some form of publication bias assessment.

Finally, 14 of the 17 reviews provided recommendations for policy and/or practice that were supported by their data. This left three reviews in question. Two of them were coded as "unclear" because it was not certain if the recommendations were supported by the data, and one of the reviews was coded as offering recommendations not based on its reported data. In contrast to this, all reviews suggested specific research directions, with each providing productive directions for new research.

Discussion

Research Question 1: What are the publication trends in regards to systematic reviews in the field of pedagogical agents?

The number of systematic reviews and meta-analyses in the field of pedagogical agents has increased in recent years. Regarding systematic reviews, a variety of approaches were observed, ranging from investigating impacts on learning and motivation (Heidig & Clarebout, 2011; Schroeder & Adesope, 2014) to examining specific design components, such as how pedagogical agents communicate with learners (Sikström et al., 2022). While it is important to understand how pedagogical agents broadly affect learning and motivation at the global and medium levels of Heidig and Clarebout's (2011) design framework, future research and analyses should explore design elements (detail level) of pedagogical agents and the environment. Thorough, in-depth systematic reviews in these areas may provide ample insights into how the detail level of designs impact human experiences such as perception, reception, processing, and learning.

We found that 41% of the studies in our sample were meta-analyses. Yet, all meta-analyses were conducted using the same software program, Comprehensive Meta-Analysis (Biostat Inc, 2023) versions 2 (Guo & Goh, 2015; Schroeder et al., 2013) or 3 (Castro-Alonso et al., 2021; Davis, 2018; Davis et al., 2023; Peng & Wang, 2022; Wang et al., 2023). To the best of our knowledge, this software allows for the computation of conventional meta-analyses. However, a limitation of conventional (two-level) meta-analysis is that it does not account for dependent data, so each participant's scores can only be included in one effect size. This means that groups are often excluded or combined with another group, which can dilute the specificity of the results. There are now different approaches to meta-analyses that allow for the inclusion of dependent data, namely, three-level meta-analysis. Conducting a three-level meta-analysis can be done using Simple Meta-Analysis (Schroeder, 2024) or with statistical coding such as in (R Core Team, 2022) using packages such as metafor (Viechtbauer, 2010). Fortunately, there are walk-throughs available that simplify this process (Harrer et al., 2021) and recently a three-level meta-analysis has been published in the field of educational technology that provided commented R code that could be used as a template (Schroeder et al., 2023). Therefore, meta-analysts are encouraged to explore and utilize more modern approaches to meta-analysis over conventional meta-analysis when the data and research questions support it. For example, many of the existing meta-analyses around pedagogical agents might have been better suited as three-level analyses rather than two-level (conventional) analyses.

Research Question 2: What do we know about the impact of PAs?

The results of this umbrella review showed that pedagogical agents can clearly have quantifiable benefits to learners in regards to both learning (Castro-Alonso et al., 2021;

Davis, 2018; Davis et al., 2023; Guo & Goh, 2015; Peng & Wang, 2022; Schroeder et al., 2013; Wang et al., 2023) and motivational outcomes (Guo & Goh, 2015; Wang et al., 2023). While systematic review studies found mixed results (Heidig & Clarebout, 2011; Martha & Santoso, 2019; Papoutsi & Rangoussi, 2020; Schroeder & Adesope, 2014), meta-analytic results showed statistically significant, but frequently small, benefits from the incorporation of PAs.

What is known about the impact of PAs could be the results of methodology and inclusion criteria in existing reviews. Meta-analyses quantitatively pool studies into a single effect size, potentially overlooking nuanced outcomes. Systematic reviews qualitatively assess consistencies or discrepancies but may yield inconclusive overall effects. These analyses can lead to an oversimplification of complex data. In addition, inclusion criteria can further influence results, since overlapping studies between analyses can bias outcomes. Future studies might examine the need for a mixed-methods approach with adding the qualitative analysis of systematic reviews with the quantitative assessment of a meta-analysis. Likewise, future research should investigate study overlap in meta-analyses and systematic reviews to determine if results reflect true impact or inclusion bias. We encourage readers to pragmatically evaluate outcomes, scrutinize inclusion criteria, and consider nuances in the data, especially when studies report conflicting results.

Research Question 3: What do we know about designing PAs?

By analyzing the existing systematic reviews and meta-analyses using Heidig and Clarebout's (2011) frameworks, we hoped to establish what design components are more or less effective in different contexts. However, despite the availability of numerous reviews available investigating different layers of the framework, only two main ideas emerged from the analysis. First, in relation to the PALD framework levels of global, medium, and detail, most reviews focused on the medium level that is concerned more with technical decisions. Although some reviews measured multiple levels within a single review (Aljaroodi et al., 2019; Castro-Alonso et al., 2021; Davis, 2018; Peng & Wang, 2022; Schroeder et al., 2013; *Yusoff et al., 2022), 12 of the 17 reviews focused on the medium level. One possible explanation for the lack of review studies evaluating the global and detail levels could be the scarcity of research that evaluates those levels of design. Future experimental research should focus on the global and detailed levels of design so that future review studies can potentially provide more robust insights into those levels of design.

Secondly, a key idea explored in the medium level of pedagogical agent design was nonverbal social cues that are associated with the so-called embodiment principle. The embodiment principle states that highly embodied agents will be more effective at supporting learning than static agents (Mayer, 2014a). Since the embodiment principle does not refer to one specific aspect of PA design, but rather conflates multiple variables such as body movement, eye gaze, and gestures, we refer to these variables as nonverbal social cues. These non-verbal social cues were specifically examined in eight of

the 12 studies at the medium design level (Castro-Alonso et al., 2021; Dai et al., 2022; Davis, 2018; Davis et al., 2021; Davis et al., 2023; Peng & Wang, 2022; Wang et al., 2023; *Yusoff et al., 2022). Unfortunately, there is a lack of consistent evidence in relation to specific individual cues such as eye gaze or gestures. However, eight of the 12 review studies suggest that nonverbal social cues are overall more beneficial to measured outcomes (Aljaroodi et al., 2019; Dai et al., 2022; Davis, 2018; Davis et al., 2021, 2023; Heidig & Clarebout, 2011; Sikstrom et al., 2022; Wang et al., 2023). Only two reviews found no benefits to designing agents with nonverbal social cues (Castro-Alonso et al., 2021; Peng & Wang, 2022), and two produced mixed results (Schroeder et al., 2013; *Yusoff et al., 2022). Furthermore, when review studies are evaluated in terms of holistic nonverbal social cue strategies such as communication strategy (Sikstrom et al., 2022) and realism in behavior (Aljaroodi et al., 2019), there is a clear pattern of benefits for anthropomorphized designs that include social cues compared to disembodied conditions. In other words, there seems to be a clear benefit of including an anthropomorphized PA in the learning environment. We note however, as Heidig and Clarebout (2011) did, that there is nuance to this finding and the effectiveness of the PA will likely contextually differ and the design of the PA, including its nonverbal social cues, can influence its effectiveness (Domagk, 2010). We also note that it seems highly likely that multiple nonverbal social cues can interact with one another. For example, a facial expression may consist of mouth movements, eye brow movements, and eye movements. If any of these variables were missing or inconsistent, it may not be perceived as intended nor have the desired effect.

Research Question 4: How well have the existing systematic reviews in the fields been conducted and reported?

The findings suggested that quality of the review studies, according to Aromataris et al.'s (2015) metric, were appropriate in some respects but lacking in others. Reviews in our sample generally explicitly stated research questions, provided appropriate inclusion criteria, and used what could be considered as an appropriate database search. Moreover, all but one meta-analysis reported publication bias, and most reviews provided policy or practice recommendations. However, using Aromataris et al.'s (2015) measure highlighted that in some reviews it was not clear if the appropriate methodologies were used. Of course, it is quite possible appropriate methodologies were used but simply not reported in the published review. For example, only five of the 17 studies documented two or more coders extracting data and only two of the seven meta-analyses explained how data was combined. Both of these points are, however, critical to understanding the trustworthiness of a review. The inter-rater agreement or reliability speaks to the reliability of the data extraction process and the understandability of the coding criteria. Meanwhile, in meta-analysis, how studies were combined can highly influence the data analysis and reported results.

Although Aromataris et al. (2015) provided a metric to evaluate the quality of review studies, we found that the evaluation can be subjective and difficult to utilize. It was

challenging for our team to come to consensus on when to label something as "unclear" or "no" in some cases. For example, it is not easy to understand if inclusion criteria are appropriate or not because there are many ways to write inclusion criteria that would address the research question. Therefore, unless inclusion criteria are clearly inappropriate, a study is likely to be labeled as "yes" for this question. In addition, unless only one database was searched, what number of database searches should be conducted to be determined to be "yes" (appropriate) or "unclear?" Even though the metric lists some appropriate components that need to be considered for a review study, the subjectivity and many possibly appropriate approaches to systematic review and meta-analysis methodologies make it difficult to definitely determine what is acceptable except for when it is not addressed.

Unsurprisingly, we found that the inter-rater agreement was less than optimal when we included the ratings for each study on the quality evaluation metric. However, quality evaluation is a necessary component of umbrella reviews (Aromataris et al., 2015) and *should* be reported. In regards to systematic reviews and meta-analyses, one notable issue reviewers face is that what determines a "quality" study will likely vary by field, research question, etc. A result is that quality evaluations we have seen published in the education literature seem rather ad-hoc. This may be one reason why PRISMA moved away from using the terminology of "quality" to the terminology of "certainty" of evidence (Page et al., 2021). While we commend reviewers for attempting to create a paradigm for evaluating the quality of educational studies, it is becoming clear that the field needs some sort of guidance, similar to the PRISMA statement, for evaluating the certainty of educational studies. We therefore call for research synthesists to develop a protocol for evaluating the certainty of educational studies and reviews of educational research.

Implications for Theory

While this review was not designed to test specific theories, there are nonetheless implications for theory from this work. We found that PAs impact learning, but we still do not know how different design decisions may influence learning and why. At this point it is unclear if PAs support learning due to specific social processes during the learning process, the pedagogy they facilitate, or a combination of both factors. We postulate it is a combination of both the pedagogy and social processes, but there is limited evidence to support either perspective individually. Furthermore, it seems plausible that personal and cultural norms may also interact with how PAs are perceived by learners, but this too is yet to be broadly investigated in the literature.

Moreover, we see ample room for research in this space around related theories and outcomes outside of learning outcomes specifically. For example, social presence is an aspect of PAs that needs to be explored beyond the mere presence or absence of an image in the design. It should focus more on how users perceive PAs in technology-mediated environments as being 'real' and how designs influence that perception (Kreijns et al., 2022). Regarding emotional benefits, earlier research explored the

impact of empathetic responses and agent roles on agent design and user learning (Kim & Baylor, 2015). However, further research is needed to understand the mechanisms behind agent role design and how users recognize these roles. As a final example, consider motivational outcomes. Dai et al. (2022) provided evidence that agents can increase users' motivation, enhance agent perception, and improve learning for lower-level learning objectives. However, the lack of conclusive findings around specific design features requires a more detailed examination to understand how agent design increases motivation for educational purposes.

Taken together, our synthesis of the literature implies that social agency theory (Mayer et al., 2003) may not provide the mechanistic specificity that PA researchers require to move the field forward. However, the nonverbal social cues that make up the embodiment principle provide design elements that allow for deeper exploration in how social cues might influence user interaction with PAs. CASTLE outlines how specific social processes may influence learning, including learning from PAs. It is therefore suggested that researchers explore the use of CASTLE (Schneider et al., 2022) and the measurement of social processes to explore the effects of both verbal and nonverbal social cues embodied by PAs on learning.

Implications for Practice

Instructional designers should take from this study a few different key points. Namely, it is clear from quantitative syntheses that PAs can positively influence learning and motivation, although the effect may not be large. What is less clear is in what situations PAs may be especially useful, or how PAs should be optimally designed for any given instructional context. As a result, we suggest that PAs be employed in situations in which they can add *educational value*. We define educational value broadly, implying positive impacts on not only learning outcomes, but also the learner's social, emotional, and motivational development. For example, PAs have been used to add educational value in ways as varied as supporting career interest (Rosenberg-Kima et al., 2008) or guiding the learner's visual attention to different areas of the screen (Johnson et al., 2013). The possibilities are endless, however much more research is needed to understand the consequences of specific design considerations. Yet, we note that PAs are not the solution to all learning problems. They are simply one educational technology tool available, and should be used when appropriate.

For systematic reviewers and meta-analysts the results of this umbrella review have clear implications as well. First, it is important that reviews are reported in accordance with the PRISMA guidelines (Page et al., 2021). We encourage fellow research synthesists to review the full PRISMA statement and the specific PRISMA extensions for different types of reviews they may be interested in. Second, as we highlighted previously, our experience highlighted that it is quite clear that educational research synthesists would benefit from a certainty evaluation metric specifically designed for educational studies.

Limitations and Opportunities for Future Research

Our inclusion criteria required that the reviews in this analysis be of non-artificially intelligent agents. This was an intentional decision because artificial intelligence adds complexity to understanding the effects of PAs. For instance, artificially intelligent agents may provide different levels of scaffolding, switch between significantly different roles in the environment (e.g., motivator, peer, expert), or even be capable of answering or responding to students in very different ways depending on the underlying intelligence. One may be drawn to conclude that artificially intelligent agents share as much in common with PAs as they do educational chatbots. As such, it seems reasonable to view the conclusions from non-intelligent PA research, such as those presented here, as a foundation for research with intelligent PAs. For example, it seems reasonable that broad findings such as ensuring the PAs add educational value will carry over to intelligent systems, as may, for example, findings around the types of voice that the PA uses to communicate with learners. We acknowledge however, that it is possible that the results reported here may not be fully generalizable to PAs powered by artificial intelligence, so research is needed to better understand what, if any, differences there may exist.

Second, when analyzing the results of this umbrella review, we noted that relatively few reviews examined outcomes other than learning outcomes. Despite this, it is clear that PAs have the potential to significantly impact learners' motivation (Guo & Goh, 2015; Wang et al., 2023). Motivation is a complex construct and there are a variety of related variables PAs may be able to influence, such as persistence, resilience, or self-efficacy. Primary studies indicate that agents may be able to improve these variables in different contexts. For example, Lane et al. (2013) found that a PA could improve self-efficacy for programming and Kim et al. (2007) found that a PA could influence girls' math self-efficacy. Clearly, there is promise in this area for both primary research and research synthesis.

Conclusion

We conducted an umbrella review with the aim to better understand the state of the PA literature. The goal was to explore questions such as publication trends and quality of reviews, through more specific questions such as the impact of PAs and how PAs should be designed to support specific learning processes.

Overall, the results of meta-analyses indicate that pedagogical agents seem to have consistently produced small, positive effects on learning and motivation. However, the results around designing pedagogical agents are not as clear, providing only the high-level implication that PAs should be anthropomorphized. There is still no clear guidance on how pedagogical agents should be designed to facilitate any specific outcome in any specific context. From a methodological perspective, we found that there is room for improving the reporting of existing systematic reviews and meta-analyses in the field.

In closing, while pedagogical agents seem to have beneficial effects during learning, future research needs to explore why and how. Given that large language models and generative artificial intelligence are rapidly changing the educational technology landscape, it is expected to see these technologies combined with PAs more regularly into the future. However, researchers are encouraged to remember that while impressive technologies *can* be used for design, it is still important to understand *why* and *how* these technologies influence learning. Much more research is needed to understand why PAs have the effects they do and under what circumstances they may be most effective.

* indicates a study included in the analysis

Appendix A

Table A1. The impact of pedagogical agents on various outcomes compared to non-agent or static agent conditions.

	Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Learning outcomes		
	Castro-Alonso et al. (2021)	Overall effect: $g = 0.20$, $p = .003$
	Davis (2018)	Retention: $g = 0.39$, $p < .001$ Near transfer: $g = 0.28$, $p = .04$
	Davis et al. (2023)	Multiple choice: $g = 0.24$, $p < .001$ Transfer: $g = 0.35$, $p < .001$ Cued recall: $g = 0.29$, $p < .01$ Free recall: $g = 0.29$, $p = .04$
	Guo & Goh (2015)	Retention: $r = .29$, $p = .01$ Transfer: $r = .26$, $p < .001$
	Heidig & Clarebout (2011)	No notable effects on learning
	Martha & Santoso (2019)	Only 12 (of 50) papers found significant effects from agents. 76% of papers found learning impacts, and 50% found student behavior impacts
	Papoutsi & Rangoussi (2020)	72% report positive effects on learning, 14.7% neutral, and 6% negative impacts on learning
	Peng & Wang (2022)	Overall effect of pedagogical agents: $g = 0.42$, $p < .001$
	Schroeder et al. (2013)	$g = .19$, $p < .001$
	Wang et al. (2023)	Retention: $g = 0.26$, $p < .001$ Transfer: $g = 0.34$, $p < .001$

(continued)

Table A1. (continued)

	Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Motivation outcomes		
	Guo & Goh (2015)	Motivation: r = .35, p < .001
	Heidig & Clarebout (2011)	No notable effects on motivation
	Schroeder & Adesope (2014)	"Overall, our findings indicate that pedagogical agents tend to produce more positive than negative motivational effects." p. 242 "In sum, there is not clear evidence that pedagogical agents provide any advantages for persona, motivation, or cognitive-load measures in any particular domain" p. 246 "Overall, we can see that compared to a text-only control condition, pedagogical agents may facilitate learners' perception of the systems persona, increase learner motivation, and decrease cognitive load. However, when compared to a voice-only condition, the results are mixed." p. 247
	Wang et al. (2023)	Intrinsic motivation: g = 0.26, p < .001
Other outcomes		
	Davis (2018)	Agent persona: g = 0.44, p = .02 Cognitive load: g = 0.13, p = .16
	Schroeder & Adesope (2014)	"Overall, it appears as though the learners' evaluation of the pedagogical agent's persona is heavily dependent on the comparison group...Thus, learners appear to prefer pedagogical agents more than non-agent multimedia learning systems, yet they may not yield the same responses as true human interaction." p.238 Results were mixed in regards to cognitive load outcomes "In sum, there is not clear evidence that pedagogical agents provide any advantages for persona, motivation, or cognitive-load measures in any particular domain" p. 246 "Overall, we can see that compared to a text-only control condition, pedagogical agents may facilitate learners' perception of the systems persona, increase learner motivation, and decrease cognitive load. However, when compared to a voice-only condition, the results are mixed." p. 247
	Wang et al. (2023)	Positive emotions: g = 0.26, p = .002

Table A2. The design implications of pedagogical agents on various outcomes.

Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Global level (focusing on choosing between a human or non-human agent (Heidig & Clarebout, 2011))	
Davis (2018)	Humanoid gesturing agents decreased cognitive load while a non-humanoid character increased it
Peng & Wang (2022)	No significant moderation depending on the agent's appearance style
Schroeder et al. (2013)	No significant moderating effect depending on agent's appearance
Wang et al. (2023)	Positive emotions: Virtual agents were more effective than human instructors Intrinsic motivation: Virtual agents were more effective than human instructors
Medium level (focusing on technical decisions (e.g., communication techniques) and specific functions or roles of the agent (Heidig & Clarebout, 2011))	
Aljaroodi et al. (2019)	The agent should reflect human realism in behavior, appearance, and emotion. Similarity between agents and the human user may be important (not only physical appearance or gender, but also culture and ethnicity)
Castro-Alonso et al. (2021)	Appearance: 2D agents were more effective than 3D agents Agent gesturing, eye gaze, facial expressions, motion, and voice type were all non-significant moderators
Dai et al. (2022)	"Agents with more human-like modalities tend to perform better than less human-like ones" p. 10
Davis (2018)	Agents that use gestures can improve learning compared to non-gesturing agents: Near transfer: $g = 0.39$, $p < .001$. Gesturing agents using human voice were more effective than text-to-speech voiced agents. Retention: $g = 0.28$, $p = .04$ Agents that use gestures can improve learners' perceptions of agent persona compared to non-gesture conditions $g = 0.44$, $p = .02$. Gesturing was more effective in shorter videos (<6 minutes) than longer videos in regards to persona ratings Agents that use gestures can reduce cognitive load compared to non-gesture conditions $g = 0.13$, $p = .16$. Humanoid gesturing agents decreased cognitive load while a non-humanoid character increased it
Davis et al. (2021)	Gesturing agents were rated better on facilitating learning in 5/8 studies, on credibility in 2/7 studies, human-likeness in 3/7 studies, and engagement in 4/7 studies Facial expressions improved the ratings on the facilitated learning subscale in 5/8 studies, and for every study for the other subscales Voice yielded mixed results on facilitation of learning and credibility subscales. Human voice was rated higher on human-likeness and engagement scales
Davis et al. (2023)	Multiple choice results: Full body agents were more effective than talking heads. Agents with deictic gestures had higher scores than only facial gestures Transfer results: Fully embodied agents were more effective than taking heads Free recall results: Deictic, multiple, and facial gestures were more effective than conversational or iconic gestures
Heidig & Clarebout (2011)	Explanatory feedback was better than corrective feedback to facilitate transfer performance. Agents should speak rather than provide on-screen text
Peng & Wang (2022)	No significant moderating variables (additional support, facial expressions)
Schroeder et al. (2013)	On screen text was significantly more effective than narration. Animated agents were significantly more effective than static agents
Sikstrom et al. (2022)	Pedagogical agents communicate, including interpersonally, to support self-regulation, self-efficacy, motivation, and metacognition
Wang et al. (2023)	Retention: No significant differences based on PA characteristics Transfer: Pedagogical agents that moved led to more learning than static agents Positive emotions: Multiple emotional cues were more effective than PAs with only a single emotional cue Intrinsic motivation: Multiple emotional cues were more effective than PAs with only a single emotional cue
*Yusoff et al., 2022	Design elements (5/17 studies): a "pedagogical agent made up with good multimedia principles give a positive result towards learner's cognitive load" p.122. No effect of cues, anxiety coping strategies or message framing Social cues (11/17 studies): Voice types showed mixed results. Deictic gestures, types of speech did not show significant improvement on cognitive load

(continued)

Table A2. (continued)

Study	Summarized finding (positive effect sizes indicate pedagogical agents improved outcomes)
Detail level (focusing on specific visual design components like the agent's age and clothing (Heidig & Clarebout, 2011))	
Aljaroodi et al. (2019)	The agent should reflect human realism in behavior, appearance, and emotion. Similarity between agents and the human user may be important (not only physical appearance or gender, but also culture and ethnicity)
Armando et al. (2022)	Gender stereotypes exist in HCI. The results were contradictory in regards to if a pedagogical agent's gender influences students' academic performance and self-evaluations The discussion suggests that agents could be used to counter/reduce stereotype threats in STEM fields The discussion suggests that a female agent may improve women's learning, interest, and self-efficacy in math The discussion suggests that the agent's gender will depend on the agent's role The discussion suggests that researchers should be careful using androgynous agents to combat gender stereotypes because the opposite of the intended effect could occur
Castro-Alonso et al. (2021)	Agent gender was a non-significant moderator
*Yusoff et al., 2022	Visual design: (1/17 studies): Age of agent had no significant impact on cognitive load

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

This material is partially based upon work supported by the National Science Foundation and the Institute of Education Sciences under Grant #2229612.

ORCID iD

Noah L. Schroeder https://orcid.org/0000-0002-3281-2594

Note

1. We excluded artificially intelligent agents because they are often fundamentally different than the PAs investigated here in regards to their educational functions and capabilities. This is further examined in the "Limitations and Opportunities for Future Research" section.

References

*Yusoff, M. D. M., Yusof, A. M., & Jamaludin, N. S. A. (2022). Pedagogical agent element and design for learning: A systematic literature review. E-Journal of Economics. *Entrepreneurship and Management Sciences*, *1*(1), 116–126. http://hdl.handle.net/123456789/4130.

Abdelghani, R., Wang, Y. H., Yuan, X., Wang, T., Lucas, P., Sauzéon, H., & Oudeyer, P. Y. (2023). GPT-3-driven pedagogical agents to train children's curious question-asking skills. *International Journal of Artificial Intelligence in Education*, *34*(2), 483–518. https://doi.org/10.1007/s40593-023-00340-7

*Aljaroodi, H. M., Adam, M. T., Chiong, R., & Teubner, T. (2019). Avatars and embodied agents in experimental information systems research: A systematic review and conceptual framework. *Australasian Journal of Information Systems*, *23*(1), 1–38. https://doi.org/10.3127/ajis.v23i0.1841

*Armando, M., Ochs, M., & Régner, I. (2022). The impact of pedagogical agents' gender on academic learning: A systematic review. *Frontiers in Artificial Intelligence*, *5*(1), 862997. https://doi.org/10.3389/frai.2022.862997

Aromataris, E., Fernandez, R., Godfrey, C. M., Holly, C., Khalil, H., & Tungpunkom, P. (2015). Summarizing systematic reviews: Methodological development, conduct and reporting of an umbrella review approach. *JBI Evidence Implementation*, *13*(3), 132–140. https://doi.org/10.1097/XEB.0000000000000055

Atkinson, R. K., Mayer, R. E., & Merrill, M. M. (2005). Fostering social agency in multimedia learning: Examining the impact of an animated agent's voice. *Contemporary Educational Psychology*, *30*(1), 117–139. https://doi.org/10.1016/j.cedpsych.2004.07.001

Bendel, O. (2003). *Pädagogische Agenten im Corporate E-Learning*. Institut für Wirtschaftsinformatik. https://beat.doebe.li/bibliothek/b01472.html

*Castro-Alonso, J. C., Wong, R. M., Adesope, O. O., & Paas, F. (2021). Effectiveness of multimedia pedagogical agents predicted by diverse theories: A meta-analysis. *Educational Psychology Review*, *33*(3), 989–1015. https://doi.org/10.1007/s10648-020-09587-1

Chen, Z. H., & Chen, S. Y. (2014). When educational agents meet surrogate competition: Impacts of competitive educational agents on students' motivation and performance. *Computers & Education*, *75*(2), 274–281. https://doi.org/10.1016/j.compedu.2014.02.014

Choi, S., & Clark, R. E. (2006). Cognitive and affective benefits of an animated pedagogical agent for learning English as a second language. *Journal of Educational Computing Research*, *34*(4), 441–466. https://doi.org/10.2190/A064-U776-4208-N145

Craig, S. D., & Schroeder, N. L. (2017). Reconsidering the voice effect when learning from a virtual human. *Computers & Education*, *114*(1), 193–205. https://doi.org/10.1016/j.compedu.2017.07.003

*Dai, L., Jung, M. M., Postma, M., & Louwerse, M. M. (2022). A systematic review of pedagogical agent research: Similarities, differences and unexplored aspects. *Computers & Education*, *190*(1), 104607. https://doi.org/10.1016/j.compedu.2022.104607

*Davis, R. O. (2018). The impact of pedagogical agent gesturing in multimedia learning environments: A meta-analysis. *Educational Research Review*, *24*(1), 193–209. https://doi.org/10.1016/j.edurev.2018.05.002

*Davis, R. O., Park, T., & Vincent, J. (2021). A systematic narrative review of agent persona on learning outcomes and design variables to enhance personification. *Journal of Research on Technology in Education*, *53*(1), 89–106. https://doi.org/10.1080/15391523.2020.1830894

*Davis, R. O., Park, T., & Vincent, J. (2023). A meta-analytic review on embodied pedagogical agent design and testing formats. *Journal of Educational Computing Research*, *61*(1), 30–67. https://doi.org/10.1177/07356331221100556

Dinçer, S., & Doganay, A. (2015). The impact of pedagogical agent on learners' motivation and academic success. *Practice and Theory in Systems of Education*, *10*(4), 329–348. https://doi.org/10.1515/ptse-2015-0032

Domagk, S. (2010). Do pedagogical agents facilitate learner motivation and learning outcomes? *Journal of Media Psychology*, *22*(2), 84–97. https://doi.org/10.1027/1864-1105/a000011

Garau, M., Slater, M., Vinayagamoorthy, V., Brogni, A., Steed, A., & Sasse, M. A. (2003). The impact of avatar realism and eye gaze control on perceived quality of communication in a shared immersive virtual environment. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 529–536). https://doi.org/10.1145/642611.642703

*Guo, Y. R., & Goh, D. H. L. (2015). Affect in embodied pedagogical agents: Meta-analytic review. *Journal of Educational Computing Research*, *53*(1), 124–149. https://doi.org/10.1177/0735633115588774

Haddaway, N. R., Page, M. J., Pritchard, C. C., & McGuinness, L. A. (2022). *PRISMA2020*: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis. *Systematic Reviews*, *18*(2), Article e1230. https://doi.org/10.1002/cl2.1230

Harrer, M., Cuijpers, P., Furukawa, T.A., & Ebert, D.D. (2021). *Doing Meta-Analysis with R: A Hands-On Guide*. Boca Raton, FLLondon and Chapman & Hall/CRC Press. ISBN 978-0-367-61007-4.

*Heidig, S., & Clarebout, G. (2011). Do pedagogical agents make a difference to student motivation and learning? *Educational Research Review*, *6*(1), 27–54. https://doi.org/10.1016/j.edurev.2010.07.004

Jaques, P., Andrade, A., Jung, J., Bordini, R., & Vicari, R. (2023). Using pedagogical agents to support collaborative distance learning. In *Computer support for collaborative learning* (pp. 546–547). Routledge.

Johnson, A. M., Ozogul, G., Moreno, R., & Reisslein, M. (2013). Pedagogical agent signaling of multiple visual engineering representations: The case of the young female agent. *Journal of Engineering Education*, *102*(2), 319–337. https://doi.org/10.1002/jee.20009

Kim, Y., & Baylor, A. L. (2016). Research-based design of pedagogical agent roles: A review, progress, and recommendations. *International Journal of Artificial Intelligence in Education*, *26*(1), 160–169. https://doi.org/10.1007/s40593-015-0055-y

Kim, Y., Wei, Q., Xu, B., Ko, Y., & Ilieva, V. (2007). In R. Luckin (Ed.), *MathGirls: Toward developing girls' positive attitude and self-efficacy through pedagogical agents*.

Kreijns, K., Xu, K. M., & Weidlich, J. (2022). Social presence: Conceptualization and measurement. *Educational Psychology Review*, *34*(1), 139–170. https://doi.org/10.1007/s10648-021-09623-8

Lane, H. C., Cahill, C., Foutz, S., Auerbach, D., Noren, D., Lussenhop, C., & Swartout, W. (2013). The effects of a pedagogical agent for informal science education on learner behaviors and self-efficacy. In *Artificial intelligence in education: 16th international conference, AIED 2013, Memphis, TN, USA, july 9-13, 2013. Proceedings, 16* (pp. 309–318). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-39112-5_32

Lee, K. M., & Nass, C. (2003). Designing social presence of social actors in human computer interaction. In G. Cockton & P. Korhonen (Eds.), *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 289–296). https://doi.org/10.1145/642611.642662

Liu, C., Chen, H., Zhou, F., Chiang, C. H., Chen, Y. L., Wu, K., Huang, D. H., Liu, C. Y., Chiou, W. K., & Chiou, W. K. (2022). Effects of animated pedagogical agent-guided loving-

kindness meditation on flight attendants' spirituality, mindfulness, subjective wellbeing, and social presence. *Frontiers in Psychology*, *13*(1), 894220. https://doi.org/10.3389/fpsyg.2022.894220

*Martha, A. S. D., & Santoso, H. B. (2019). The design and impact of the pedagogical agent: A systematic literature review. *Journal of Educators Online*, *16*(1), 1–15. https://doi.org/10.9743/jeo.2019.16.1.8

Mayer, R. (2014a). Principles based on social cues in multimedia learning: Personalization, voice, image, and embodiment principles. In R. Mayer (Ed.), *The cambridge handbook of multimedia learning (cambridge handbooks in psychology)* (pp. 345–368). Cambridge University Press.

Mayer, R. E. (Ed.), (2014b). *The cambridge handbook of multimedia learning* (2nd ed.). Cambridge University Press.

Mayer, R. E., & DaPra, C. S. (2012). An embodiment effect in computer-based learning with animated pedagogical agents. *Journal of Experimental Psychology: Applied*, *18*(3), 239–252. https://doi.org/10.1037/a0028616

Mayer, R. E., Dow, G. T., & Mayer, S. (2003). Multimedia learning in an interactive selfexplaining environment: What works in the design of agent-based microworlds? *Journal of Educational Psychology*, *95*(4), 806–812. https://doi.org/10.1037/0022-0663.95.4.806

Mayer, R. E., & Estrella, G. (2014). Benefits of emotional design in multimedia instruction. *Learning and Instruction*, *33*(1), 12–18. https://doi.org/10.1016/j.learninstruc.2014.02.004

Moreno, R. (2006). Learning in high-tech and multimedia environments. *Current Directions in Psychological Science*, *15*(2), 63–67. https://doi.org/10.1111/j.0963-7214.2006.00408.x

Morrow, D. G., Lane, H. C., & Rogers, W. A. (2021). A framework for design of conversational agents to support health self-care for older adults. *Human Factors*, *63*(3), 369–378. https://doi.org/10.1177/0018720820964085

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., & Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *International Journal of Surgery*, *10*(1), 89. https://doi.org/10.1186/s13643-021-01626-4

*Papoutsi, F., & Rangoussi, M. (2020). Pedagogical agents in E-learning: A review of recent (2009–2019) research results. In *Proceedings of the 24th pan-hellenic conference on informatics* (pp. 316–321). https://doi.org/10.1145/3437120.3437332

*Peng, T. H., & Wang, T. H. (2022). Developing an analysis framework for studies on pedagogical agent in an e-learning environment. *Journal of Educational Computing Research*, *60*(3), 547–578. https://doi.org/10.1177/07356331211041701

R Core Team. (2022). *R: A language and environment for statistical computing.* [Computer software]. R Foundation for Statistical Computing. https://www.r-project.org/

Rosenberg-Kima, R. B., Baylor, A. L., Plant, E. A., & Doerr, C. E. (2008). Interface agents as social models for female students: The effects of agent visual presence and appearance on female students' attitudes and beliefs. *Computers in Human Behavior*, *24*(6), 2741–2756. https://doi.org/10.1016/j.chb.2008.03.017

Schneider, S., Beege, M., Nebel, S., Schnaubert, L., & Rey, G. D. (2022). The cognitive-affective-social theory of learning in digital environments (CASTLE). *Educational Psychology Review*, *34*(1), 1–38. https://doi.org/10.1007/s10648-021-09626-5

*Schroeder, N. L. (2024). Simple meta-analysis. https://github.com/noah-schroeder/simple_meta-analysis/

*Schroeder, N. L., & Adesope, O. O. (2014). A systematic review of pedagogical agents' persona, motivation, and cognitive load implications for learners. *Journal of Research on Technology in Education*, *46*(3), 229–251. https://doi.org/10.1080/15391523.2014.888265

Schroeder, N. L., Adesope, O. O., & Gilbert, R. B. (2013). How effective are pedagogical agents for learning? A meta-analytic review. *Journal of Educational Computing Research*, *49*(1), 1–39. https://doi.org/10.2190/EC.49.1.a

Schroeder, N. L., & Gotch, C. M. (2015). Persisting issues in pedagogical agent research. *Journal of Educational Computing Research*, *53*(2), 183–204. https://doi.org/10.1177/0735633115597625

Siegle, R. F., Schroeder, N. L., Lane, H. C., & Craig, S. D. (2023). Twenty-five years of learning with pedagogical agents: History, barriers, and opportunities. *TechTrends*, *67*(5), 851–864. https://doi.org/10.1007/s11528-023-00869-3

*Sikström, P., Valentini, C., Sivunen, A., & Kärkkäinen, T. (2022). How pedagogical agents communicate with students: A two-phase systematic review. *Computers & Education*, *188*(1), 104564. https://doi.org/10.1016/j.compedu.2022.104564

Wang, N., Johnson, W., Mayer, R., Rizzo, P., Shaw, E., & Collins, H. (2008). The politeness effect: Pedagogical agents and learning outcomes. *International Journal of Human-Computer Studies*, *66*(2), 98–112. https://doi.org/10.1016/j.ijhcs.2007.09.003

Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, *36*(3). https://doi.org/10.18637/jss.v036.i03

*Wang, Y., Gong, S., Cao, Y., Lang, Y., & Xu, X. (2023). The effects of affective pedagogical agent in multimedia learning environments: A meta-analysis. *Educational Research Review*, *38*(2), 100506. https://doi.org/10.1016/j.edurev.2022.100506

Woo, H. L. (2008). Designing multimedia learning environments using animated pedagogical agents: Factors and issues. *Journal of Computer Assisted Learning*, *25*(3), 203–218. https://doi.org/10.1111/j.1365-2729.2008.00299.x

Author Biographies

Noah L. Schroeder, Ph.D. is a Professor of Educational Technology and Instructional Design at Wright State University. His research focuses on pedagogical agents, the design of multimedia learning environments, and research synthesis.

Robert O. Davis is an Assistant Professor in the Department of Education at Chonnam National University. He is also a member of both the Convergence Research Center for

Immersive Learning Technology and the Institute of Educational Research. His current research interests include nonverbal communication with pedagogical agents, the social acceptance of virtual characters, human interaction with computer-based environments, and the use of AI in the classroom.

Eunbyul Yang is a postdoctoral researcher in the Department of Education at Chonnam National University, South Korea, specializing in instructional design and immersive learning technologies. She earned her Ph.D. in Education, focusing on instructional technology, where her research explored the intersection of technology-enhanced learning environments and self-regulated learning. Current research focuses on multimodal, multisensory data analysis to identify and assess cognition and emotion in educational settings.

AMERICAN PSYCHOLOGICAL ASSOCIATION

Psychological Bulletin

© 2021 American Psychological Association
ISSN: 0033-2909

2021, Vol. 147, No. 8, 806–827
https://doi.org/10.1037/bul0000337

Advancing Our Understanding of Psychological Flow: A Scoping Review of Conceptualizations, Measurements, and Applications

Cameron Norsworthy[1], Ben Jackson[1, 2], and James A. Dimmock[3]

[1] School of Human Sciences (Exercise and Sport Science), The University of Western Australia

[2] Telethon Kids Institute, Perth, Australia

[3] Department of Psychology, College of Healthcare Sciences, James Cook University

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Research on psychological flow is well established, although criticisms remain regarding conceptual and measurement issues associated with the construct. This scoping review maps flow-related research across scientific disciplines, examining the conceptualization, measurement instruments, and outcomes of flow between 2012 and 2019. Across 236 sources that met the review criteria, 108 different flow-related constructs were measured by 141 instruments, and 84 possible antecedents were identified. Despite the varied approaches, a common set of overarching antecedent constructs included "optimal challenge" and "high motivation," and recurring characteristics of the flow experience itself included "absorption," "effort-less control," and "intrinsic reward." Applied studies—albeit inconsistent in approach and largely correlational in nature—predominantly linked flow to "positive development" (i.e., well-being and health), "high functioning," and "further engagement." We contextualize the findings of the review relative to important work on flow that has recently emerged (following the review period)—in doing so, we hope this review offers a contemporary framework that can be used for the study of flow across scientific disciplines.

Public Significance Statement

This review scopes flow research across all scientific disciplines and identifies key commonalities and differences in how researchers have conceptualized the psychological state of flow—an optimal experience often referred to as being "in the zone." It addresses inconsistencies and concerns of practical application that characterize some contemporary flow-related models and studies. This review also provides a framework—derived from the findings of the review—for the study of flow antecedents and experiential dimensions.

Keywords: flow theory, performance, stress, motivation, health

Supplemental materials: https://doi.org/10.1037/bul0000337.supp

In his pursuit to understand what brings meaning to one's life, Csikszentmihalyi (1975) described a unique type of intrinsically rewarding, or "autotelic," experience, which later he defined as "flow." Deriving from the Greek words *auto* (self) and *telos* (goal), an autotelic experience referred to an experience so rewarding that participation alone was the goal. In Csikszentmihalyi's original interviews, participants consistently described an intrinsically rewarding, harmonious psychological state of total focus and absorption that facilitated a fluidity to their actions and a playfulness to their efforts as if the experience was "flowing" like water (Csikszentmihalyi, 1975). Csikszentmihalyi has since outlined nine dimensions associated with the flow experience: (a) challenge-skills balance (i.e., matching perceived skills and challenges in a particular situation); (b) merging of action and awareness (i.e., where deep involvement leads to automaticity and spontaneity); (c) clear goals (i.e., clear purpose and grasp of what to do next); (d) unambiguous feedback (i.e., direct and immediate feedback about one's effectiveness in the moment); (e) concentration on the task at hand (i.e., exclusion of any unnecessary distractions); (f) sense of control (i.e., a sense of personal control or agency over the activity); (g) loss of self-consciousness (i.e., when the self disappears, worrying about oneself and what others may think of him or herself cease to exist); (h) transformation of time (i.e., one's subjective experience of time is altered); and (i) autotelic experience (i.e., an intrinsically rewarding activity, an end in itself; adapted from Jackson & Csikszentmihalyi, 1999).

Further work has been undertaken to position these factors as antecedents, core experiences, or outcomes of flow. For instance, Nakamura and Csikszentmihalyi (2009, 2014) outlined that

Cameron Norsworthy https://orcid.org/0000-0002-6036-7053

No potential conflict of interest was reported by the authors.

This work was supported by a doctoral scholarship allocated by The University of Western Australia.

Correspondence concerning this article should be addressed to Cameron Norsworthy, School of Human Sciences (Exercise and Sport Science), The University of Western Australia, M408, Stirling Highway, Crawley, WA 6009, Australia. Email: cameron.norsworthy@research.uwa.edu.au

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

"challenge-skills balance," "clear goals," and "unambiguous feedback" dimensions may be considered structural task characteristics or proximal conditions to flow, whereas the other six dimensions describe the experience itself. Despite some support for the nine-dimensional model (Jackson & Eklund, 2002; Nakamura & Csikszentmihalyi, 2002), evidence of correlations between the other eight dimensions and the "challenge-skill balance" dimension (Beard & Hoy, 2010; Engeser & Rheinberg, 2008; Jackson & Marsh, 1996) has resulted in many researchers utilizing only the "challenge-skill balance" dimension when attempting to operationalize flow—therefore examining flow as a unidimensional construct (Fong et al., 2015).

Understanding, measuring, and describing flow is by no means straightforward. Flow is complex in nature—it encompasses, for example, both hedonic (i.e., intrinsic reward or pleasure) and Eudemonic (i.e., use of an individual's skills and abilities leading to individual and cultural progressive development) characteristics (Seligman, 2002; Vittersø & Søholt, 2011). It could also be argued that there is something paradoxical about the high concentration levels (and associated physiological symptoms of effort) that are proposed to co-occur alongside effortless attention and a fluidity of performance. Despite such complexities, flow research has become and remains popular across sport, educational, health, and organizational psychology domains. Importantly though, alongside empirical reports of flow antecedents, correlates, experiences, and interventions, work in these areas over the last decade has also been characterized by critiques of the conceptualization and measurement of the construct. Broadly, these critiques have highlighted four important considerations—specifically, issues relating to construct validity, theoretical compatibility, relational ambiguity, and definitional inconsistency.

With respect to *construct validity*, it has been recognized that flow dimensions may vary between activities—that is, that not all nine dimensions are necessary to experience flow in varying contexts (e.g., Boyd et al., 2018; Corbatto, 2018; Kawabata & Mallett, 2016). For example, in an educational setting, Heutte et al.'s (2016) factor analysis showed the dimensional use of "cognitive absorption," "time transformation," "loss of self-consciousness," and "autotelic experience" to measure flow (dimensions)—challenge/skill balance, clear goal, unambiguous feedback and high sense of control—were merged into one antecedent (factor). Further, Sinnamon et al. (2012) suggested that the dimensions of "merging of action and awareness" and "loss of self-consciousness" may not be valid dimensions for musicians, and Swann et al.'s (2012) systematic review in elite athletes highlighted that less than 30% of athletes in included studies reported the dimensions of "loss of self-consciousness" and "transformation of time." In addition to contextual variability, construct validity has also been questioned due to (a) some researchers not including any of the nine dimensions when measuring flow (for example, Zito et al., 2018); (b) certain dimensions (for example, "time transformation") exhibiting only modest relationships with the other dimensions (See, for example, see Jackson & Eklund, 2002); and (c) additional experiential dimensions beyond the nine "core" aspects being frequently reported (for example, "bodily sensations"; Chavez, 2008; Jackman et al., 2017), including other widely used psychological constructs (for example, arousal, motivation, confidence; see Swann et al., 2012) that do not naturally fit into one of the nine dimensions.

In terms of *theoretical compatibility*, Csikszentmihalyi's (1990) nine-dimensional model utilizes terms that are not commonly utilized in scientific research (for example, autotelic experience, merging of action and awareness), and in some respects, these constructs can be difficult to align with existing research or theory. For example, self-efficacy appears to be characterized in three different flow dimensions—"challenge-skill balance," "clear goals," and a "sense of control" (for example, Jackson & Csikszentmihalyi, 1999)—which indicates a degree of intradimensional overlap and poses problems for any theoretical integration efforts. Moreover, Swann et al. (2018) suggested that Csikszentmihalyi's (1990) nine-dimensional model should not be considered a "theory" as it lacks clear causal constructs and outcomes, making it difficult to examine experimentally or integrate with other theories.

Regarding *relational ambiguity*, Engeser and Schiepe-Tiska (2012) argued that although the nine dimensions of flow actually include antecedents, experiential constructs, and outcomes to flow, no interactional relationships are proposed or offered in the model (see also Guertin, 2016). Consequently, for example, some researchers regard concentration and enjoyment as elements of the flow experience (for example, Ghani, 1995), while others view concentration as an antecedent and enjoyment as an outcome of flow (for example, Chen, 2006). Finally, in terms of *definitional inconsistency*, it is recognized that definitions of flow vary immensely—even those presented by Csikszentmihalyi himself—and that many are not wholly compatible with the nine-dimensional model (Cheng, 2014; Hoffman & Novak,2009; Swann et al., 2018). To illustrate, working definitions of flow include:

- "Flow is a positive valenced state (affective component), resulting from an activity that has been appraised as an optimal challenge (cognitive component), characterized by optimized physiological activation (physiological component) for full concentration on coping with environmental/task demands (behavioral component)" (Peifer, 2012, p. 149).
- Flow incorporates the extent to which (a) the user perceives a sense of control over the computer interaction, (b) the user perceives that his or her attention is focused on the interaction, (c) the user's curiosity is aroused during the interaction, and (d) the user finds the interaction intrinsically interesting" (Webster et al., 1993, p. 413).
- "The state occurring during network navigation which is (1) characterized by a seamless sequence of responses facilitated by machine interactivity, (2) intrinsically enjoyable, (3) accompanied by a loss of self-consciousness, and (4) self-reinforcing" (Novak & Hoffman, 1997, p. 2).
- "The holistic sensation that people feel when they act with total involvement" (Csikszentmihalyi, 1975, p. 36).

These conceptual inconsistencies and (as some have argued) ambiguities have caused difficulties for researchers and practitioners when attempting to compare and synthesize flow research (Auld, 2014). Mahnke et al.'s (2012) flow review in information systems, for example, indicated that 75% of the authors of studies included in their review either modified the original constructs of flow (beyond adaptation to the research context), combined constructs from different concepts (for example, Huang, 2006; Novak et al., 2000; Shin, 2006), or customized terminology and interpretation of constructs to better suit their own context (for example,

cognitive absorption in education; Heutte et al., 2016)—highlighting the need for an overarching definition. Further, Swann et al. (2018) contended that the current (that is, most widely accepted and used nine-dimensional) flow model is too descriptive and not sufficiently mechanistic, and argued as a result that flow science may be at a "crisis point." More specifically, these authors also asserted that the flow model cannot be considered a "theory" as it is missing specific definitions and propositions of how interrelated constructs create clear causal relationships for a particular phenomenological occurrence (or nonoccurrence).

Despite the substantial number of flow-related studies conducted in the past decade, only four systematic reviews of psychological flow literature have been published—one with an emphasis on musical applications (Chirico et al., 2015), one on game-based learning (Perttula et al., 2017), one in computer-mediated learning environments (Auld, 2014), and one on elite athleticism (Swann et al., 2012). Although informative, these reviews have typically focused on a specific domain or specific questions, rather than examining flow science in its entirety (that is, across multiple conceptual issues and multiple contexts). Equally, informative books, chapter narratives, and conceptual reviews have helped to synthesize aspects of flow research (for example, Abuhamdeh, 2020; Engeser, 2012a; Harmat et al., 2016; Nakamura & Csikszentmihalyi, 2014), including the collected works of Mihaly Csikszentmihalyi (2014a); however, no formal review has been undertaken to chart psychological flow research across domains (for example, sport, education) and scientific disciplines (for example, physiology and neurobiology).

The broad purpose of this study, therefore, was to undertake and present a comprehensive scoping review that brought together —and charted, in one place—the varied conceptualizations of flow. Specifically, our aim in this review was to describe the consistency (and inconsistency) in constructs, dimensions, and working definitions utilized to describe flow, and to chart the varied antecedents, measurement instruments, and outcomes utilized when operationalizing flow. In line with recommendations for cross-disciplinary approaches, and calls to understand flow phenomenologically, cognitively, neurologically, physiologically, socially, and behaviorally (Beard, 2015; Csikszentmihalyi, 2014b; Kawabata & Mallett, 2016; Marr, 2001; Peifer, 2012; Rich, 2013; Stoll & Pithan, 2016), a secondary aim (if possible) was to highlight any common findings that could be used and operationalized across scientific disciplines.

Given the aims outlined above, we felt that a narrow search strategy using predetermined parameters and set domains (that is, a PRISMA-adherent systematic review: Moher et al., 2009) may limit the scope of our review. Our broad goal, however, did align closely with Arksey and O'Malley's (2005) and Tricco et al.'s (2016) guidelines for scoping reviews (that is, examining the extent, range, and nature of research activity; determining the value of undertaking a systematic review; summarizing and disseminating research findings; and identifying research gaps in the existing literature). As a result, a scoping review methodology was chosen for the study. The scoping review adhered to The Joanna Briggs Institute (JBI) guidelines (Peters et al., 2015) and the PRISMA extension for scoping reviews (PRISMA-ScR; Tricco et al., 2018), and was registered a priori with JBI on August 1, 2019 to ensure methodological rigor (Colquhoun et al., 2014). To adhere to scoping review procedures (Peters et al., 2015), the objectives, inclusion criteria, methods of analysis, and a priori search terms were specified in advance and documented in a protocol.

Method

Inclusion/Exclusion Criteria

There were no exclusion criteria for methodological approaches, participant characteristics, geography, or language. Literature from 2012 through to 2019 (the year at commencement of the review) that specifically analyzed "flow" conceptually was included to ensure theoretical rigor. The year 2012 was selected due to Engeser's (2012b) seminal theoretical update on flow within "Advances in Flow Research" (Engeser, 2012a)—a text that also included updates and commentaries from numerous flow researchers (including Csikszentmihalyi). Additionally, we selected 2012 because this year was when Peifer's (2012) article was published on the psychophysiological correlates of flow. This work was in part responsible for a subsequent proliferation in what might be termed "nonpsychological" (for example, physiological, psychophysiological, and neuroscientific research) flow-related research (for example, Harris et al., 2017). Finally, 2012 was also the year in which Baumann (2012) presented three different flow definitions (not descriptions) encompassing different research disciplines (general, cognitive, affective). Reviews of the flow literature have been published since 2012 (for example, Abuhamdeh, 2020; Chirico et al., 2015; Nakamura & Csikszentmihalyi, 2014); however, in comparison with those sources, our inclusion criteria were designed to offer broader coverage of flow conceptualization and measurement across multiple disciplines.

Search Strategy

A three-step search strategy was utilized (Peters et al., 2015). Initially, the keyword "flow" was used to begin an iterative search in Embase (1947) and PsycINFO (1806) to analyze common affiliated terms. In Step 2, the selected terms, and negative keywords, were utilized in title and abstract searches within the databases Web of Science (1900), PubMed (1997), Medline (1966), Scopus (1966), Embase (1947), PsycINFO (1806), SPORTDiscus (1930), Google Scholar, OSF, and PsyArXiv. Subsequently, a wider search, ("flow" in the title plus "optimal experience" in the article text) was carried out to ensure all relevant articles were captured and safeguard against any articles that might have been missed from previous, more reductive searches. In Step 3, the keyword "flow" was searched in additional related journals, and the reference list of all identified relevant reports and articles were searched for additional studies. (An online supplemental materials lists the specific terms, negative keyword, and journals.) Where necessary, study authors were contacted for access and clarification.

Data Charting

Search results were recorded in Microsoft Excel, Evernote, Zotero, and Bubbl. Data were charted into the following extraction categories: Document type (that is, book, journal), year of publication, author(s), study location (that is, location of associated institution), scientific discipline, scientific discipline - subdomain,

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

study population, methodology (experimental work was categorized using Campbell and Stanley's (1963) pre-experimental, quasi-experimental and experimental categories), definition of flow, described constructs, compounding factors, measures, antecedents, intervention, linked outcomes, reported outcomes, quality rating, individual or group orientation, and key findings. Methodological quality was assessed using Kmet et al.'s (2004) standard quality assessment criteria for primary quantitative and qualitative studies; literature reviews were assessed using Oxman and Guyatt's (1988) guidelines, and systematic reviews were assessed using AMSTAR (Shea et al., 2007)—all common assessment tools for reviews. A preliminary meeting was held between all authors to ensure a consistent interpretation of descriptors within the quality assessment tools. The lead author then recorded quality scores, and this recording was verified by the other two authors for accuracy (with discussion in cases where any disagreements arose). Data extraction followed the PRISMA-ScR (Tricco et al., 2018) and JBI scoping review guidelines (Peters et al., 2015).

To ensure high interrater agreement during data extraction (which followed the PRISMA and JBI scoping review guidelines), sample data were extracted by the lead author and cross-checked by the two coauthors for reliability (Brooker et al., 2015) to test the process and ensure accuracy of data extraction. Data from each source were then mined into a predefined data sheet by the lead author. Additionally, longer narrative and sentence extraction was recorded from each source within bubbl.us—a live online mapping tool designed to help group large amounts of data into themes and ensure original data accuracy. Both sets of data were continuously cross-checked by (a) additional researchers acting as "critical friends" (to aid integrity and reflexivity; Appleton, 2011), and (b) the two coauthors—for accuracy of extraction, interpretation, and categorization—on a fortnightly basis over a period of several months. Certain core extraction categories (for example, conceptual dimensions) yielded over 300 different individual items. Consequently, a thematic content analysis (TCA) approach was conducted by the authors in order to identify common concepts/themes across the antecedents, experiential dimensions, and outcomes of flow. Anderson's (2007) TCA protocol was adhered to when synthesizing terms, including identifying distinct units, grouping and regrouping similar and dissimilar units, retaining source language where possible, iteratively relabeling categories based on new source data, and rereading source data postanalysis without looking at final categorizations. Any disagreements were resolved through open discussion between the three authors while revisiting source data to ensure integrity of the source's meaning.

Results

The State of Flow Conceptualization, Measurement, and Application

Literature Search

The literature search resulted in 2,633 citations (see Figure 1). After eligibility screening, 2,391 articles were removed. Three articles were added from reference screening, three books were removed because all the book chapters were already listed, and six articles were not accessible even after contacting the authors. A final 236 records were included in the review (see supplementary material S1, tab Review Data for full list).

Source Characteristics

Eligible sources included 58 books or book chapters and 178 journal articles, the majority of which were published in 2016, 2017, and 2018 (see supplementary material S2, Figure 1). More than half of all the records came from authors affiliated with European institutions (k = 133), followed by North America (k = 66), Asia (k = 18), Australasia (k = 14), South America (k = 3), and Africa (k = 2). Work reported in journal articles predominantly came from the United States (k = 60), followed by Germany (k = 31), England (k = 19), Italy (k = 17), Australia (k = 14), Spain (k = 11), Hungary (k = 6), Taiwan (k = 5), and the Netherlands (k = 5; see S2, Figures 2 and 3). The most prevalent disciplines included psychology (k = 173), psychophysiology (k = 21), neuroscience (k = 19), and computing (k = 13; see S2, Figure 4). Research was most prevalent within the subdisciplines of positive psychology (k = 42), sport and exercise psychology (k = 24), occupational psychology (k = 18), educational psychology (k = 15), computing psychology (k = 12), personality psychology (k = 9), neurobiology (k = 8), human computer interaction (HCI; k = 8), positive psychophysiology (k = 7), cognitive psychology (k = 7), and cognitive neuroscience (k = 7; see S2, Table 1).

Methodology and Quality Assessment

Pre-experimental studies (k = 80), literature reviews (k = 63), theorizing literature (k = 27), qualitative works (k = 22), and confirmatory factor analyses (k = 21) were the most widely-used research methods (see S2, Figure 5). The literature was given a high (k = 27), medium (k = 84), and low (k = 125) "quality" score; however, the practical value of these scores was limited—as can be the case when conducting scoping reviews (Tricco et al., 2018)—because much of the theoretical analysis was carried out in narrative reviews or expert commentaries (rather than, for example, experimental studies).

Sample Population

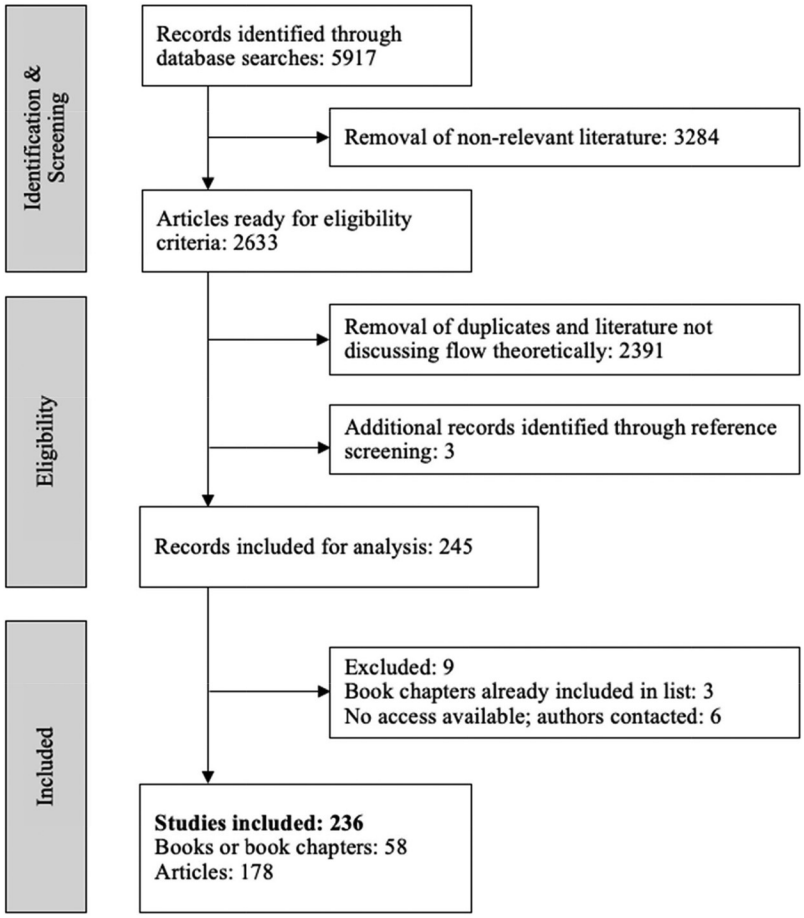
Students (k = 57), general population (k = 24), athletes (k = 20), musicians (k = 10), gamers (k = 10), workers (k = 8), and teachers (k = 3) were the samples most utilized (see S2, Table 2). Studies were focused on individuals (k = 222), groups (k = 10), or both (k = 5).

Definitions and Dimensions of Flow

The majority (80%) of authors created their own phrase (that is, flow definition) or cited one of Csikszentmihalyi's statements to describe flow. Only three exact descriptive phrases were utilized by multiple sources (namely, "holistic sensation that people feel when they act with total involvement," k = 14; "being fully absorbed in an activity," k = 4; "in which people are so intensely involved in an activity that nothing else seems to matter; the experience itself is so enjoyable that people will do it even at great cost, for the sheer sake of doing it" k = 3). Twenty-seven records (11%) did not list a clear definition or description of the focal construct. In total, 270 psychological and physiological markers, dimensions, or terms were associated with the conceptualization of flow. A total of 150 different constructs, terms, or dimensions were utilized to *describe* the flow experience itself. Csikszentmihalyi's

Figure 1

PRISMA Flow Diagram Identifying the Included Records (Peters et al., 2015)



This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

nine dimensions were referred to together in 109 (46%) of the records. Of those nine dimensions, the most frequently described were "sense of control" (k = 166) and "total concentration/focus" (k = 158), followed by "immersion" or "absorption" (k = 115), "joy" (k = 66), "optimal experience" (k = 66), "enjoyment" (k = 66), "total engagement" (k = 66), "motivation" (k = 31), "intrinsically rewarding" (k = 28), "nothing else matters" (k = 25), "full attention" (k = 25), "effortless attention" (k = 23), and "effortless involvement" (k = 23). See S2, Tables 3 and 4 for a full listing of these terms.

Antecedents

Eligible sources identified 84 different antecedents to flow, described 158 compounding factors, and measured 30 variants in applied studies (see S2, Tables 5 and 6). Csikszentmihalyi's (Nakamura & Csikszentmihalyi, 2009) three theorized "preconditions" to flow—namely challenge-skill balance (described, k = 149/measured, k = 52), clear-goals (described, k = 105/measured, k = 34), and unambiguous feedback (described, k = 102/measured, k = 32)—were the most widely utilized, and were often reported together (k = 96). "Skills-demand fit" was utilized in work from European researchers (e.g., Engeser & Schiepe-Tiska, 2012), while "concentration," "attention," "sense of control," and "positive affect" were frequently described across sources (see S2, Tables 5 and 6). Additionally, "autonomy," "choice," "motivational competence," and "interest" were repeatedly observed as proposed or measured antecedents, indicating that there is a motivational antecedent additional to the challenge-skill balance dimension. In the two applied experimental studies (Colzato et al., 2018; Peifer et al., 2015), challenge and arousal levels were targeted through manipulating the level of task difficulty, vagus nerve stimulation, and cortisol levels (see also S1, tab Applied Studies).

Outcomes

Flow was described to have 133 different outcomes in all studies, while studies targeting outcomes of flow (k = 75) measured 45 varying outcomes (see S2, Tables 7 and 8) across a range of designs; 3% experimental, 5% quasi-experimental, and 75% pre-experimental designs—other designs included reviews, confirmatory factor analyses and qualitative approaches. Flow was (or proposed dimensions of flow were) most frequently positively correlated with performance (k = 6), behavioral engagement (k = 4), learning (k = 3), joy (k = 2), further flow (k = 2), enjoyment (k = 2), well-being (k = 2), and mental health (k = 2). Measured outcomes were spread across emotional and motivational (k = 18; 27%), behavioral (k = 17; 25%), cognitive (k = 17; 25%), physiological (k = 14; 21%) and trait-based (k = 1; 1%) indices (note:

studies reported more than one outcome). Findings were largely correlational in nature and researchers utilized inconsistent conceptual and measurement approaches; therefore, any causal conclusions are premature. In the few experimental studies, reported outcomes included further flow, high levels of arousal, and inverted U-shape heart-rate-variability (HRV)–though arousal levels and HRV may better represent measurement methods associated with experiential components of flow (rather than outcomes).

Measurement

Overall, questionnaires assessing flow (or dimensions thereof) appeared in 210 sources—including 39 experience sampling method (ESM) studies—followed by electrocardiogram (ECG) measures (k = 36), electroencephalography (EEG) measures (k = 17), and functional MRI (fMRI) measures (k = 16). The most commonly utilized instruments were multicomponent measures within ESM studies (k = 33), Flow State Scales (k = 29; FSS-2; Jackson & Eklund, 2002), Dispositional Flow Scales (k = 22; DFS-2; Jackson & Eklund, 2002), Flow Short Scale (k = 20; Engeser & Rheinberg, 2008; Rheinberg et al., 2002), The WOrk-reLated Flow inventory (k = 10; WOLF; Bakker, 2008), and Swedish Flow Proneness Questionnaire (k = 9; SFPQ; Ullén et al., 2012). See S2, Table 9 and 10. In these instruments, scored responses offered insight into flow "intensity" or "proneness" levels, but no instrument clearly identified whether or when a person was "in" flow or not. These instruments also did not account for multiple flow entries (and exits) in a given experience.

In terms of psychological assessments, 108 constructs or concepts were *measured* through 141 different instruments (see S2, Tables 4 and 9). The nine dimensions were the most frequently assessed constructs, with "level of challenge" (k = 58) and "level of skill" (k = 33) also being measured independently from "challenge-skill balance" (k = 107). In addition, "immersion" or "absorption" (k = 54), "positive affect" (k = 34), "reasons for participation" (k = 33), "satisfaction" (k = 33), "enjoyment" (k = 32), "fluency" (k = 25), and "importance" (k = 23) were also commonly measured.

In terms of physiological assessments, sources examined markers of flow associated with the brain (k = 40), heart (k = 38), facial expressions/movement (k = 16), skin (k = 12), hormones (k = 13), breathing (k = 7), and muscle movement (k = 4). Physiological and neuroscientific instruments predominantly targeted arousal levels (k = 66) and the autonomic nervous system (k = 4; for example, heart-rate-variability; HRV), active geographical areas in the brain (k = 35; for example, blood flow), valence (k = 14; for example, corrugator supercilii muscle movement), attention (k = 9; eye pupil movement), affect and reward processing (k = 7; ventral striatum or putamen activity), fluency of movement (k = 4; for example, cortico-muscular coherence), and mental effort (k = 3; for example, heartbeat stroke volume). See S1, tab Measured Terms, for a detailed listing. Arousal levels and HRV (that is, the variability and coherence of the length of inter heartbeat intervals) were targeted to examine optimal levels of challenge, concentration, stress, sense of control, and activity within the autonomic nervous system (ANS). Brain activity was used to measure cognitive and biological functions and associations with attention, self-consciousness, and time monitoring. Valence and affect were targeted to examine the attractiveness of the experience and the positive or negative affect that flow yielded, as well as indicating the likelihood for subsequent focus. Attention and mental effort were assessed in order to examine high attentional activity and paradoxical reports of effortless attention. Similarly, fluency of movement was examined to understand effortless action, reduced pain, and feeling of smoothness in actions when exerting high levels of effort. Cardiac output and respiratory rates were measured to analyze mental effort and autonomic nervous system (ANS) activity. In summary, preliminary physiological findings reveal a somewhat coherent set of physiological markers that suggest flow may differ from alternative physiological modes of operating.

Physiological findings appeared to confirm an optimal facilitative level of arousal that corresponds with a down regulation of neuroanatomical components of the brain that are associated with effortful activity (Peifer, 2012). For example, higher flow intensity levels (as measured by Flow Short Scale; Rheinberg et al., 2002) have been linked in an inverted u-shaped relationship with sympathetic arousal and cortisol production, while parasympathetic activity has been linearly and positively correlated with higher flow intensity levels during coactivation of sympathetic activity (e.g., Peifer et al., 2014). Further, higher intensities of flow have been characterized by a faster respiratory rate, deeper respiration, moderate heart rate, and moderate HRV, thereby suggesting that flow occurs with highly focused attention that is facilitated through an increased parasympathetic modulation of sympathetic activity (Bian et al., 2016; Tian et al., 2017). A higher sense of control during high levels of flow intensity is physiologically evident from eye movement and pupil dilation; specifically, flow has been linked with higher attention signified by increased pupil size and reduced blinking (for example, Kim et al., 2013). Further, through the measure of functional magnetic resonance perfusion imaging (fMRPI), increases in neural activity in the left anterior inferior frontal gyrus (IFG) and the left putamen suggest a deeper sense of cognitive control and coding of increased outcome probability (see Ulrich et al., 2014). Further, this level of control appears to have a physiological basis of ease, fluidity, or effortlessness as indicated by EMG and EEG reports of high levels of cortico-muscular coherence (that is, synchronization of neural activity between the brain and body) during higher levels of flow intensity (Cheron, 2016). The above physiological findings help to suggest how an individual can be challenged and subsequently highly aroused and alert, yet also able to act calmly and with ease as psychologically evident during flow (see Effortless Control section below for further discussion).

Physiological explanations were also offered for the phenomenological characteristics of absorption, loss of self-consciousness, and time transformation. Studies examining electrical activity in the brain (often measured with EEG) linked states of flow with reduced beta brain wave frequencies, suggesting a reduction in conscious thinking and self-reflexive activity such as self-consciousness (for example, Pluta et al., 2018). Similarly, neuroanatomy studies (often using fMRI to understand which areas of the brain, and associated faculties, are activated and inhibited) revealed that the brain's attentional resources are highly engaged during high flow intensities, while brain activity associated with worry and self-concern about one's performance is inhibited (for example, de Sampaio Barros, 2018; Sadlo, 2016). Additionally, researchers have linked flow with a moderate (that is, not high or

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

low) default mode network (DMN: a large-scale network of brain regions that is responsible for our normal modes of conscious thinking and waking consciousness) activation (for example, Esterman et al., 2013) and transient hypofrontality (for example, Nakamura & Roberts, 2016). Such work physiologically indicates the loss or reduction of normal modes of self-consciousness, time monitoring, and awareness that are often considered to be absent during flow. Lastly, skin conductance tests (that is, testing the ability of the skin to resist electrical currents) have been used to measure an individual's stress, arousal, and valence in order to examine the intensity of flow (e.g., Bian et al., 2016).

Moving Toward Conceptual Clarity and Parsimony

Despite the large number of terms (n = 270) utilized within the review sources, many of the terms were used across studies and common themes bringing together multiple terms were apparent. For example, only 15 (described)/16 (measured) terms were utilized in more than 10 sources, and Csikszentmihalyi's original nine dimensions accounted for 60% of all described terms and 65% of all measured terms. Accordingly, we sought to synthesize the 270 terms into coherent (and discrete) antecedents, experiential constructs and outcomes through Anderson's (2007) protocol of TCA. Our efforts to do so yielded two broad antecedent dimensions ("optimal challenge" and "high motivation") and three broad experiential dimensions ("absorption," effort-less control," "intrinsic reward"), which are presented below and supported throughout by citations to relevant literature and review evidence (see Figures 2 and 3). See also S1, tabs Thematic Grouping of Terms for a full list of synthesized terms. Robust conclusions about the outcomes of flow experiences were limited due to methodological issues associated with research in the area. Only 9% of applied studies on flow-related outcomes included a control group, with the majority of studies characterized by correlational and cross-sectional designs. Additionally, various flow instruments have been utilized to measure possible outcomes of the construct, leading to some ambiguity in the identification of key outcomes. Nevertheless, on the basis of research examined in this review, below we also provide preliminary thematic insight into three outcomes of flow: positive development, high functioning, and further engagement. See S1, tab, Thematic Groupings of Outcomes for the full synthesis.

Flow Antecedent Dimension 1: Optimal Challenge

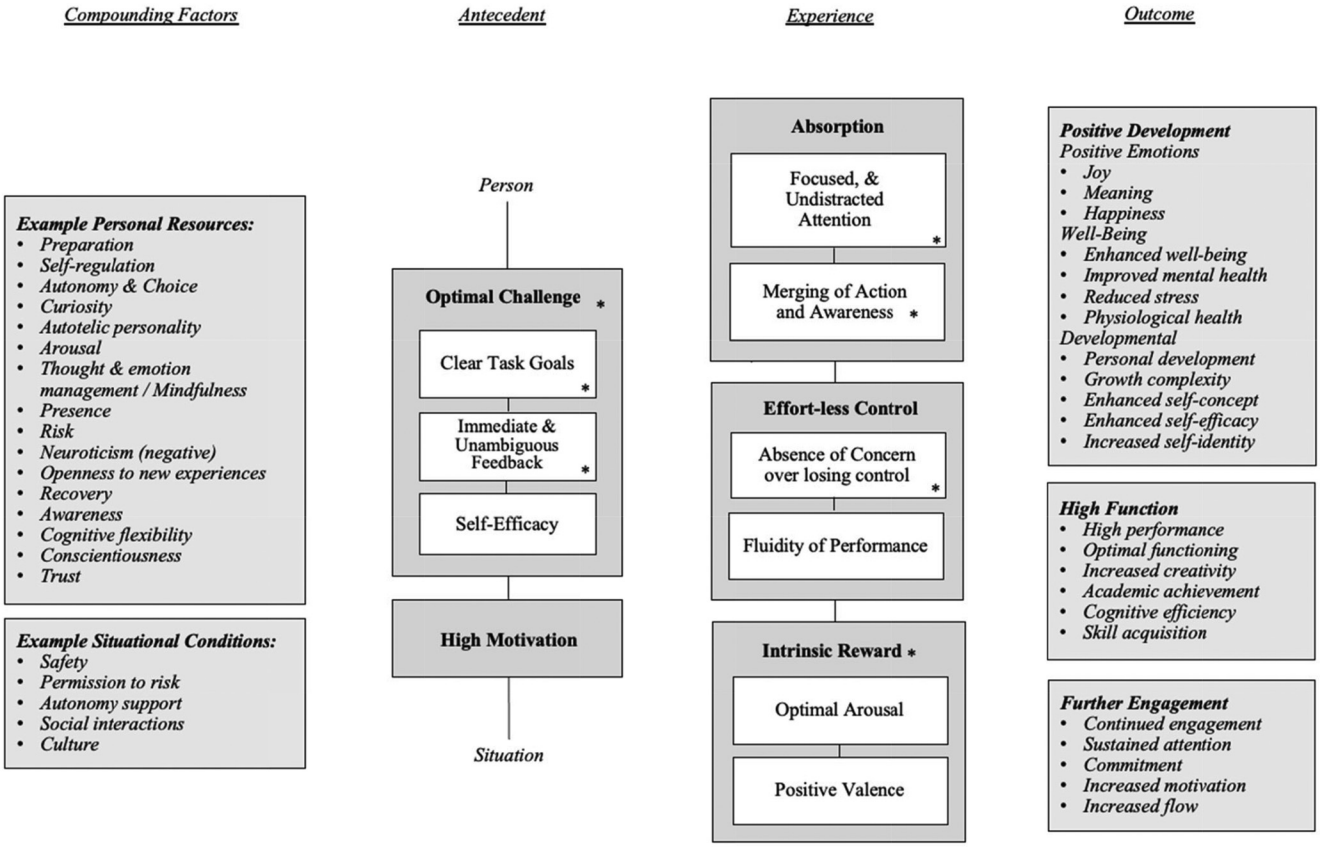
Sources most commonly identified Csikszentmihalyi's (1990) "challenge-skill balance" dimension as the primary (and frequently the only) antecedent (described: k = 148/measured: k = 52) to flow. This dimension of the original model was frequently operationalized as an antecedent to flow in both correlational (for example, Bian et al., 2016; de Sampaio Barros et al., 2018; Flett, 2015) and experimental (for example, Tozman et al., 2017; Yousef, 2013) studies. Originally, Csikszentmihalyi (1975) contended that flow depends on establishing a balance between "action capabilities" [skills] and "action opportunities" [challenges] (p. 49), with the premise being that flow occurs when one is engaged in a challenge that stretches one's capability (Nakamura & Csikszentmihalyi, 2014). This "balance" or "stretching" between the *perceived* level of skill and challenge is proposed to be neither overmatching nor underutilizing (that is, matching at a 50:50 ratio above average levels; see Moneta, 2012; Nakamura & Csikszentmihalyi, 2014). If the level of challenge exceeds the level of skill, then feelings of arousal and concern dominate one's experience. If the level of challenge is too low, however, then the individual's capability is not challenged and underarousal occurs (see Csikszentmihalyi's octant model of flow; Engeser, 2012a). Across studies, this optimal level of challenge was measured and described through a number of related terms (n = 29) such as "anxiety," "arousal," and "skills-demand fit," accounting for 31.29% of all utilized terms.

Psychophysiological evidence within the reviewed sources supported the notion that flow occurs when there is an optimal level of challenge and corresponding physiological arousal. Specifically, by examining ANS responses during flow (with participants engaged in tasks that challenged their ability but were not overly demanding), researchers reported an optimal (that is, moderate) level of activation in sympathetic activity (for example, Peifer et al., 2014; Tian et al., 2017), hypothalamic pituitary adrenal (HPA) axis levels (Bian et al., 2016; Peifer et al., 2014; Tozman & Peifer, 2016), high-frequency HRV (HF-HRV) and low-frequency HRV (LF-HRV), cortisol levels (Peifer et al., 2015), and respiratory depth (that is, larger breaths indicative of increased parasympathetic activity; Harmat et al., 2015). In other words, flow appeared to occur during increased parasympathetic modulation of sympathetic activity (Tian et al., 2017) and in an inverted U-shaped arousal (Peifer et al., 2014). This optimal (moderate) level reflects an inverted U-shaped relationship between challenge and flow (e.g., Tozman et al., 2017), suggesting that an individual first needs to be challenged or aroused, and that flow then occurs at an optimal level, around the peak of the inverted-U (when challenged but not overly so).

The literature examined in this review indicated that it may be most appropriate to term this antecedent dimension *optimal challenge:* a perceived capability to meet the challenging demands of the situation. The term optimal challenge subsumes constructs labeled by others as "challenge-skill balance," "level of challenge," "skills-demand fit," "skill level," "optimally challenged," and "skill integration," for example, The reasons for this relabeling are as follows. First, researchers have criticized the use of (original) terms such as "balance" or "50:50 ratio" because flow has been observed during instances of a slight imbalance between these concepts (for example, Engeser & Rheinberg, 2008; Lovoll & Vitterso, 2014; Vaughn, 2016). Similarly, anticipating flow in response to "above average" levels of these concepts may be conceptually ambiguous because what is considered "average" (skill or challenge) may differ between and within individuals and contexts (Keller & Landhausser, 2012; Yoshida et al., 2013; Mahnke et al., 2012)— the term optimal challenge instead suggests direction toward a positive high degree of challenge without limiting this degree to an absolute balance or discounting intra- or interdifferences. Second, sources within this review often measured the level of "challenge" and "skill" separately, and researchers subsequently made inferences about the degree of balance between those concepts. This is potentially problematic insofar as the terms challenge and skill may be partly conflated. Assessing the level of challenge may already acknowledge, to a degree, one's level of skill (for example, the level of challenge one faces is dependent, in part, upon the resources, or skills, one possesses). This separate assessment approach may result, therefore, in skill being measured twice, and may contribute to measurement inaccuracy (Keller &

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Figure 2
Constructs, Compounding Factors, and Outcomes of Flow



Note. 'Optimal Challenge' represents a more unidimensional construct than 'challenge/skill balance', which often measures challenge/difficulty and skill separately. 'Total concentration on the task at hand' has been updated to 'focused and undistracted attention'. 'Autotelic experience' has been updated to 'intrinsic reward'. 'Sense of control' has been more aptly defined by Csikszentmihalyi as an absence of concern or anxiety over losing control – summarized as 'absence of doubt'. 'Time transformation' and 'loss of reflective self-consciousness' have been considered as variable experiences within 'absorption'. * Csikszentmihalyi's nine dimensions.

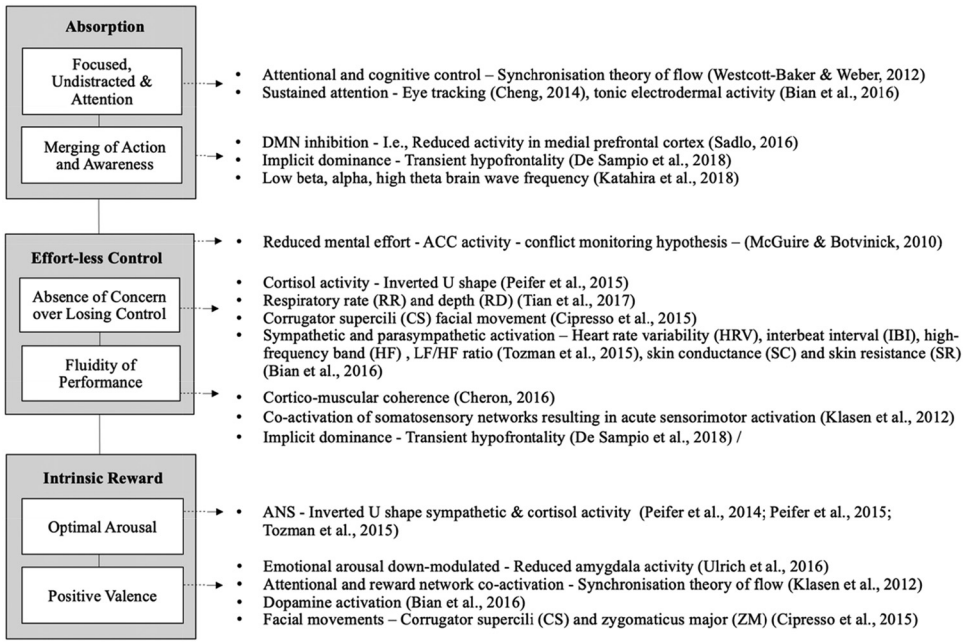
Landhasser, 2012; Mahnke et al., 2012). In addition to issues of conflation, the independent measurement of challenge has, at times, also proved difficult, such as those instances when researchers have used performance-related markers of success (e.g., Klasen et al., 2012).

The evidence in this review indicated that this antecedent dimension may incorporate three subthemes that are commonly referred to in flow research, and that were often used to help operationalize this dimension—namely *clear task goals*, *immediate and unambiguous feedback*, and *self-efficacy*. The construct clear task goals subsumes the example terms (utilized by review sources) including "specified objectives," "clarity of purpose," "confidence"; "immediate and unambiguous feedback" subsumes the example constructs "coherent, noncontradictory demands" and "direct feedback"; and "self-efficacy" subsumes the example constructs "self-confidence" and "self-trust." See Figure 2 and S1, tab Thematic Grouping of Terms for further details. The premise behind goal- and feedback-related concepts being included in this theme appears to be that if goal (that is, clear goals) and diagnostic information are not clear and immediate (that is, immediate and unambiguous feedback), then the experience may be confusing or overdemanding (and no longer optimally challenging; Cseh, 2016; Drengner et al., 2018; Guertin, 2016; Keller & Landhausser, 2012; Moneta, 2017; Simlesa et al., 2018; Valenzuela et al., 2018). When examining goal type, sources reported that goal congruency (for example, Keller & Landhausser, 2012), personal growth and exploration goals (for example, "seeing how well I can do"; Swann et al., 2016), "open" (that is, nonoutcome orientated) as opposed to "fixed" (that is, outcome orientated) goals (Swann et al., 2017), and task-orientated goals (Csikszentmihalyi, 2014c) result in increased flow. Finally, with respect to the self-efficacy subtheme, it has been recognized for some time that "perceived ability," and more recently, self-efficacy (for example, Cheng, 2014; Chen & Sun, 2016; Salanova et al., 2014), acts as a facilitator of flow. Waples and Knight (2017), for example, described that self-efficacy and flow may be related in a way that mimics an "upward spiral" effect. In terms of its positioning within this antecedent theme, self-efficacy is considered a personal ("capacity" or "control") resource that buffers against stress and has resilient properties (Bandura, 1986), thereby supporting individuals' capabilities to meet challenging situations more effectively (Keeler et al., 2015).

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Figure 3

Physiological and Neuroscientific Measures and Markers to Flow



Note. Certain markers may better reflect the experience (as a whole) rather than specific dimensions or subdimensions.

Flow Antecedent Dimension 2: High Motivation

Review sources also brought to light a second broad antecedent dimension—capturing concepts relating to high motivational forces)—which was especially valuable when examining the occurrence of flow in nonachievement settings (e.g., Engeser & Schiepe-Tiska, 2012). The construct "high motivation" subsumes the example terms (utilized by review sources) "importance," "intrinsic motivation," "and "interest" (e.g., Bricteux et al., 2017; Csikszentmihalyi, 2014c; Rheinberg & Engeser, 2018). See Figure 2 and S1, tab Thematic Grouping of Terms for further details. Authors of reviewed sources posited that flow was more likely to occur when individuals reported or displayed a high level of motivation for the activity (e.g., Boffi et al., 2016; Geyser et al., 2015; Llorens et al., 2013; Schüler & Brandstätter, 2013). For example, de Sampaio et al. (2018) concluded that "flow feelings are highly connected to conditions that promote individuals' choice and autonomy" (p. 810). Baumann (2012) also proposed that the achievement flow motive operationalizes an autotelic personality toward flow; and, Engeser and Schiepe-Tiska (2012) proposed that flow emerges from the interaction of one's motives and the motive-specific incentives available in the situation. Accordingly, instances where flow was described as a function of "importance," "motivation," "reasons for participation," "interest," "curiosity," and "autonomy" were all captured within this dimension. Together, these terms accounted for 4.59% of all terms and 34% of the total measured antecedents.

The occurrence of flow has long been associated with intrinsic motivation (Bakker et al., 2017) due in part to the intrinsically rewarding nature of the experience and Csikszentmihalyi's discussion of "autotelic experience" (see Csikszentmihalyi, 2014b). Accordingly, sources posited that task interest (Bricteux et al., 2017; Cheng, 2014; Culbertson et al., 2015; Llorens & Salanova, 2017; Shernoff et al., 2014), "outcome importance" (Abuhamdeh & Engeser, 2012), "task importance," "goal congruency" (e.g., Keller, 2016), "motivational regulatory orientations" (e.g., Keller & Landhausser, 2012), and "subjective value" predict the likelihood and intensity of flow (also see Csikszentmihalyi, 2014b,' 2014c; Keller & Landhausser, 2012). For example, Auld's (2014), review concluded that *interest* in content and subject was an integral antecedent for finding flow in learning, and Keller and Landhausser (2012) propose that the *subjective value* predicts and explains the variations between micro and macro (i.e., deep and mild) experiences of flow—which to date, have only been explained through the escalating balance between challenge and skill (i.e., the higher the challenge and skill level the higher the flow intensity). However, reviewed sources also highlighted that extrinsic motivation concepts may play a pivotal role in the occurrence of flow (e.g., Schüler & Brandstätter, 2013; MacNeill & Cavanagh, 2013; Valenzuela et al., 2018; Vaughn, 2016; Vuorre & Metcalfe, 2016). For example, Bassi and delle Fave (2012) reported work-related flow to be associated with controlled regulation, and Vaughn (2016) reported that NFL players reported high levels of extrinsic (as well as intrinsic) motives in their most memorable flow experiences. In fact, extrinsic factors leading toward flow are so common in work settings that in some flow literature this is known as the "paradox of work"' (e.g., Bassi & delle Fave, 2012).

In summary, although research has consistently linked flow experiences with a high degree of motivation (e.g., Boffi et al.,

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

2016; Geyser et al., 2015; Llorens et al., 2013), further research is needed in order to better understand the specific types of motivation that are most conducive to flow experiences and how they relate, or not, to an "optimal challenge."

The themes we described above were those that, on the balance of evidence, appeared to be considered (and act) as antecedents, or important preconditions, for flow experiences. These themes mirror, in some respects, elements of the most widely-used (nine-dimensional) flow model (i.e., challenge-skill balance, clear goals, unambiguous feedback), but in this instance it is important to note that they were positioned as precursors to (rather than components of) flow experiences. We did, however, also observe some consistency in terms of the characteristics and components of flow experiences, and in the sections that follow we overview the three themes that we observed in this respect.

The Flow Experience Dimension 1: Absorption

The evidence we reviewed demonstrated that an absorption in the task—characterized by focused and undistracted attention (i.e., total concentration; see Moran, 2012; Ulrich et al., 2016a) and a merging of action and awareness—is a key experiential component of the flow experience (Csikszentmihalyi, 2014a). For example, Nakamura and Csikszentmihalyi (2014) explained that flow involves an intense and focused concentration on what one is doing in the present moment, resulting in a merging of action and awareness. Scholars have argued that during this absorption, attentional resources that ordinarily facilitate reflective self-consciousness (i.e., the hyper egoic self or Mead's "Me") and temporal experience are instead allocated to present attentional demands, resulting in a "loss of *reflective* self-consciousness" and "time distortion" (e.g., Nakamura & Roberts, 2016). However, scholars have also pointed out inconsistencies with the existing dimensions "loss of reflective self-consciousness" and "time transformation" (see Boyd et al., 2018; Swann et al., 2012; Vaughn, 2016). For example, an individual in flow may also be acutely aware of time if the activity requires, such as running down the clock in a basketball match. Correspondingly, 72 terms accounting for 34.44% of the total described and measured terms were synthesized into this theme. Commonly-used terms in the flow literature—and that are recognizable from the nine-dimensional model—that were coded in this theme included "total concentration," "loss of self-consciousness," "merging of action and awareness," and (in some instances) "time transformation." Parenthetically, it is also worth noting that "absorption/immersion" is often referred to using different labels across subdomains, such as "telepresence" in human computer interface or gaming (e.g., Guertin, 2016; Klasen, 2012; Lazoc & Luiza, 2012; Miles, 2012). Additionally, we found sources frequently associated this dimension (absorption) with a deeper sense of "connection" (e.g., Csikszentmihalyi, 2014a; Garces-Bacsal, 2016) to equipment, people, or other components relevant to the task (e.g., "being at one with the activity," "self-transcendence," "connection," "sense of unity," "shared identity"). Reports are preliminary, however, and further research is required to identify reliable characteristics of such a phenomenon.

Many researchers highlighted that absorption is critical to the flow experience (e.g., Drengner et al., 2018; Fullagar & Kelloway, 2012; Harris et al., 2017), and this notion is reflected in multiple flow instruments (e.g., FSS, Rheinberg et al., 2002; WOLF, Bakker, 2008; EGameFlow, Fu, et al., 2009). In addition, authors also noted that failing to account for this dimension (i.e., absorption) in measurement efforts may produce misleading conclusions (Schiefele, 2013), because it is this concept that differentiates flow, at least in part, from other states of concentrated action (e.g., Csikszentmihalyi, 2014a; Sheldon et al., 2015). This differentiation is perhaps best illustrated by those sources in which brain wave frequencies were examined (through EEG measures). Specifically, whereas high attentional states of control are associated with high beta frequencies, flow experiences have been shown to display varying theta, moderate alpha (Katahira et al., 2018), and low beta (Pluta et al., 2018) activity, which is typically more closely representative of trance-like immersive states.

The sources in this review widely supported the notion that absorption is a defining characteristic of flow; however, researchers differed, at times, in how they positioned this theme relative to (or within) Csikszentmihalyi's nine dimensions. For example, some authors grouped Csikszentmihalyi's dimensions "merging of action of awareness," "loss of (reflective) self-consciousness," and "time transformation" as related but independent subcomponents of "immersion" (e.g., Fang et al., 2013; Sillaots & Jesmin, 2016; Simlesa et al., 2018; Yoshida et al., 2014). Alternatively, others identified absorption through measures of time transformation-only (e.g., Grigorieva, 2015), or categorized "loss of reflective self-consciousness" and "time transformation" as outcomes of the flow experience (e.g., Herrando et al., 2018). Overall, the sources we reviewed revealed a trend toward measuring absorption as a single construct, accounting for "established" flow concepts "merging of action of awareness," "loss of reflective self-consciousness," and "transformation of time" (for example, Bakker et al., 2017; Fullagar & Kelloway, 2012; Llorens & Salanova, 2017; and, although not included in this review, see also Engeser & Rheinberg's, 2008; conceptualization of flow as "absorption" and "fluency of performance"). The premise behind this simplification appears to be that when attention is wholly invested in and directed toward a task, action and awareness merge in the absence of spare attention, and self-related thoughts and monitoring of experiential factors (such as reflective self-consciousness) fade from awareness (Csikszentmihalyi, 2014b; Nakamura & Csikszentmihalyi, 2002; Pelet et al., 2017). Interestingly, part of the support for this absorption theme came from a branch of neuroscientific research, in which it was suggested that flow occurrence results in a depleting "onion-peeling" effect of higher cognitive processes (e.g., Stoll & Pithan, 2016; Ulrich et al., 2016b). During flow, it is argued by some that the highest layers of consciousness —those that include higher cognitive functions (i.e., self-reflective thinking and time monitoring)—are most susceptible to change when alterations to mental states occur and resources are reallocated (Stoll & Pithan, 2016). Accordingly, the more intense the flow experience, the greater number of higher cognitive functions that are down-regulated. Evidence indicates that flow is associated with (a) reduced activity in the medial prefrontal cortex (a neuroanatomical area for reflexive conscious thinking; e.g., Ulrich et al., 2014); (b) inhibited activity in a group of brain regions called the default mode network (also known as self-awareness network; Sadlo, 2016; Ulrich et al., 2014; Ulrich et al., 2016b); and (c) a down-regulation of the explicit cognitive system (i.e., volitional and conscious skills) as outlined in transient hypofrontality theory (see Dietrich, 2004; Nakamura & Roberts, 2016). This depleting

"onion-peeling" effect may shed light on the neuroscientific mechanisms that explain the reliability inconsistencies of certain measured constructs from Csikszentmihalyi's nine dimensions (e.g., "loss of self-consciousness" and "time transformation").

The Flow Experience Dimension 2: Effort-Less Control

In addition to the notion of absorption, there was evidence that flow experiences are also characterized by a degree of "effortless control"—a high sense of control in which the task feels less effortful than is typical for that person, characterized by fluidity of performance and an absence of concern over losing control. For example, In Beard's (2015) interview, Csikszentmihalyi describes the effortless behavior in flow as "the kind you do not have to think about it, you're just doing it" (p. 358); or, more recently, Csikszentmihalyi et al. (2018) described flow as "an almost automatic, effortless, yet highly focused state of consciousness" (p. 216). To illustrate, 66 terms accounting for 12.81% of the total described and measured terms were synthesized into this dimension. Commonly-used terms in the flow literature that were coded in this theme included "sense of control," "fluency," and "effortless involvement."

A sense of control, irrespective of how it was defined, was the most commonly used descriptor of flow in this review. This sense of control—or, as Csikszentmihalyi (2014a) more recently described, "a lack of anxiety about losing control" (p. 231) and not a strained conscious control that may otherwise occur from an *increase* in anxiety over losing control—is considered integral to the unusual fluidity of action and felt effortlessness (e.g., Bian et al., 2016; Järvilehto, 2016) that identifies flow from other high-performance acts (i.e., clutch states; Swann et al., 2017). For example, in a meta-analytic review within computer-mediated learning, Auld (2014) stated that "flow enables the individual to enjoy the activity in a seemingly effortless manner. . . feeling in control of one's actions during a task, despite a seeming automaticity and effortlessness to one's behaviors; hence participants have described this feeling as if they were "flowing" on a current" (p. 6). Equally, in neuroscientific or psychophysiological studies, flow is frequently described as a subjective state of effortlessness (e.g., Bian et al., 2016) in which attention is effortless (e.g., de Manzano et al., 2013). In the literature on neurophysiology, subjective effort-less control has been aligned with fast and efficient physiological processes that are internally coherent and harmonious. Specifically, when experiencing effortless control, there appears cortico-muscular coherence (e.g., Cheron, 2016), coactivation of brain regions (e.g., Weber et al., 2016), balanced ANS activity (e.g., Tozman et al., 2017), and synchronized changes in neural activity (i.e., inferior frontal gyrus, left putamen, and amygdala) that permit the psychophysiological processes of a deeper sense of cognitive control, increased coding of outcome probability (putamen), and decreased negative arousal (e.g., Ulrich et al., 2014; Ulrich et al., 2016b).

Flow has also been associated with a *reduction* in conflict-monitoring cognitive processes as evident by reduced activity in the anterior cingulate cortex—an area of the brain designed to assign effortful resources when conflict is assumed (see conflict-monitoring theory: McGuire & Botvinick, 2010). Additionally, an inhibition of the posterior cingulate cortex (a core region of the default mode network; Huskey et al., 2018) and explicit cognitive system (i.e., transient hypofrontality; see Nakamura & Roberts, 2016), which results in a reduction of concern about self-ability and allows for more efficient and fluid processing and movement, is highlighted to be apparent during flow (Bian et al., 2016). In EMG and EEG studies, high levels of cortico-muscular coherence suggest that flow is linked with a synchronization of neural activity between the brain and body, which is responsible for the high level of control and, more precisely, the smoothness perception or fluidity of action that is described in flow (see Cheron, 2016). In further examination of the high degree of felt control in flow, flow is linked with synchronized activity in somatosensory networks and acute sensorimotor activation, which facilitates heightened feedback and a subsequent increased sense of control (Klasen et al., 2012).

In regard to flow feeling less effortful than typical, the synchronization theory of flow outlines how a highly efficient "switching on" of networks associated with goal-directed activity, and a corresponding "switching off" (due to limited capacity) of areas related to the self and perception of effort reduces the effort felt, even during high physiological effort (see Westcott-Baker & Weber, 2012; Huskey et al., 2018). In adjacent research, authors have highlighted an inhibition in debilitative emotional responses that may require effort to overcome. For example, some scholars have highlighted that the heightened concentration during flow is posited to temporarily exclude all irrelevant thoughts and feelings from consciousness (Csikszentmihalyi, 2014a; Guertin, 2016; Swann et al., 2017). Concurrently, neuroscientific and psychophysiological researchers indicate that a defining feature of flow is the reduced emotional activity that is apparent in (a) the down-modulation of emotional arousal mediated by the amygdala (Ulrich et al., 2016b); (b) a down-regulation of medial prefrontal cortex, which plays an important role in emotional biasing during decision-making (Nah et al., 2017); and (c) reduced activity in the dorsomedial prefrontal cortex, which is responsible for activity surrounding self-related emotions (Nah et al., 2017). Further research is required to understand the psychological ramifications of these neuroscientific findings, though this preliminary research helps to shed light on how flow can feel effort-less even during objectively effortful acts.

In summary, this effort-less control, or reduced felt effort deriving from a reduction of mental conflict and increased fluidity of movement, is seen as a principal characteristic of flow because traditional models of focused engagement otherwise require high degrees of felt effort (i.e., cognitive/mental effort and arousal) to override external distractors (Gaggioli et al., 2013; Harris et al., 2017; Peifer et al., 2014; Romero & Calvillo-Gamez, 2014; Tozman et al., 2015). Järvilehto (2016) argues that it is precisely this reduced effort and inhibition of the explicit system that allows intuitive rather than self-reflective action, and posits that it is this intuitiveness of the behavior that allows such effortless control to be felt.

The Flow Experience Dimension 3: Intrinsic Reward

This review found a diverse range of constructs representing flow as an intrinsically rewarding (i.e., "autotelic") experience that is characterized in a psychophysiological way by positive valence and optimal levels of arousal. For example, the majority of sources pointed toward a sense of enjoyment or intrinsic reward during

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

flow as a fundamental characteristic of flow (e.g., Nakamura & Csikszentmihalyi, 2014; Valenzuela et al., 2018; Yoshida et al., 2013). Correspondingly, 28 terms accounting for 9.77% of the total described and measured constructs were synthesized into this construct, including the common terms employed in flow literature: "autotelic" (e.g., Csikszentmihalyi, 2014a), "joy" (e.g., Chilton, 2013), "enjoyment" (Llorens et al., 2013), "positive affect" (e.g., Omori et al., 2014), "intrinsic reward" (e.g., Romero & Calvillo-Gamez, 2014), "positive valence" (e.g., Peifer et al., 2014), "pleasure" (e.g., Lazoc & Luiza, 2012), and "satisfaction" (Wolfigiel & Czerw, 2017).

An autotelic experience was the seventh most described (k = 141) and ninth most measured (k = 70) term of flow, likely due to its inclusion in the nine-dimensional model. It is unclear, however, whether Csikszentmihalyi meant for an autotelic experience to represent a holistic description of the entire flow experience (flow was originally labeled autotelic experience; e.g., Abuhamdeh & Engeser, 2012), a part of the experience (e.g., Araújo & Hein, 2016; Kaur et al., 2016), a consequence of flow (e.g., Perttula et al., 2017; Schiefele, 2013; Swann et al., 2018), or a reason for participation (e.g., Corbatto, 2018; Monson, 2013). Accordingly, authors utilized this autotelic component differently. The terms intrinsic reward (e.g., Nakamura & Csikszentmihalyi, 2014) and enjoyment (e.g., Miles, 2012) were instead used more consistently, with most scholars positioning them as a component of the flow experience. From a neurobiological perspective, there is a strong link to attention and reward such that the sustained attention in flow requires the experience to be somewhat rewarding, as is evident in the activation of midbrain reward structures (Nah et al., 2017) and increased dopamine production during flow (Bian et al., 2016). Dopamine is considered to be an essential neurochemical in the brain's reward system, making the flow experience feel highly rewarding. In fact, the correlation between flow and dopamine is considered so strong that flow proneness has been related to a polymorphism of the dopamine D2 receptor coding gene (DRD2 C957T rs6277) which affects striatal D2 receptor availability and thus an individual's ability for flow proneness (Gyurkovics et al., 2016). For example, Huskey et al. (2018) posited in the synchronization theory of flow (i.e., synchronization process between structures within cognitive control and reward networks) that intrinsic reward is biologically inevitable during flow, and that fMRI studies show that optimal levels of challenge corresponded to the highest levels of intrinsic reward (when compared with problems that are too easy or difficult).

Lastly, of note, is the importance of differentiating the *real-time* intrinsic reward felt during the experience and emotions (see also Abuhamdeh, 2020) that accompany one's reflective assessment of the event—the latter may be a better reflection of the participant's immediate *response* to the experience (e.g., enjoyment, joy, surprise, happiness; Llorens et al., 2013); therefore, the term intrinsic reward rather than enjoyment was deemed a more accurate reflection of this dimension. Concurrently, experimental studies have shown that flow experiences are characterized by optimal levels of arousal, and high values for positive valence that then lead to positive affective states (see Cheng, 2014; Peifer et al., 2014; Ulrich et al., 2016b). Equally, emotion theorists (e.g., Barrett et al., 2007; Moors,2009) often describe emotion as a composite of two dimensions, valence and arousal, that combine to produce an "affective quality" or emotion (Moors,2009) such as enjoyment (subsumed in this dimension). Given this dimension encapsulates the emotional term "enjoyment," we thought it best to position arousal and valence indicators under the dimension "intrinsic reward." Further research is required, however, to understand whether arousal markers best represent this dimension, another dimension, or all three dimensions (i.e., the experience as a whole).

In terms of understanding those constructs that arise as a result of flow, rather than those that describe the flow experience itself, this review revealed three common outcome themes (which we present below).

Flow Outcome Theme 1: Positive Development

Positive development accounted for 45.6% of total reported outcomes of flow and was the most commonly studied consequence of the construct. Emotional, well-being, health, and developmental outcomes were all synthesized into this theme, with specific experiences such as joy, meaning, satisfaction, happiness, personal development, growth complexity, enhanced self-concept, increased self-identity, physiological health, mental health, and reduced stress all linked to flow. In the event of further causal research shedding light on flow outcomes, researchers may find it more appropriate to position developmental and well-being (i.e., health and emotion) outcomes as distinguishable subcategories, or as separate outcome themes.

One rationale for an association between flow and positive development was offered by Nakamura and Roberts (2016); these authors argued that self-reflective attention, which is absent during flow, is chiefly responsible for many self-defeating processes such as anxiety, stress, negative affect, negative health, and the causation of depressive tendencies. Even at a basic level, self-focused attention can impair the quality of one's experience and limit attentional capacity causing moment-to-moment attentional conflict, effort, and subsequent stress. Because flow facilitates an absorption that inhibits this self-reflective activity, attentional conflict and stress is reduced, which subsequently allows room for positive outcomes (e.g., Nakamura & Roberts, 2016). For example, Fullagar et al. (2013) reported that flow—as operationalized through challenge-skill balance and measured through an adaptation of the FSS-2 scale—and task-specific anxiety were antithetical experiences such that when flow was highest, performance anxiety was lowest (and vice versa). These intrinsically rewarding experiences of effort-less control result in a more ordered self, stronger self-efficacy (e.g., Waples & Knight, 2017), personal satisfaction (e.g., Miles, 2012) and ultimately improved well-being and health (e.g., Magyarodi & Oláh, 2015). For example, Riva et al.'s (2016) review indicated that flow in everyday life was connected with individuals' well-being, reduced symptomatology, and positive and meaningful changes both inside and outside the therapeutic and rehabilitation processes. Developmentally, Freire et al. (2016) posited that these flow experiences can also be referred to as "optimal leisure," which facilitate several developmental functions and positive personal and social skills and achievements. Overall, flow has been viewed as an effective and rewarding coping mechanism for anxiety- and stress-reduction when facing daily-life challenges (e.g., Llorens & Salanova, 2017; Sadlo, 2016), associated with moderate cortisol levels (i.e., not associated with high levels of cortisol that might ordinarily be associated

with distress) during challenging task engagements (e.g., Peifer et al., 2014), and for deriving potential health benefits.

Flow Outcome Theme 2: High Functioning

A theme relating to "high functioning" (28.44%) brought together terms such as "high performance," "optimal functioning," "increased creativity," "academic achievement," "optimal learning," "increased productivity," "cognitive efficiency," "skill acquisition," and "better judgments." For example, Andersen (2016) associated flow, as measured through ESM in combination with observations and interviews, with better STEM learning. Further, Flett (2015) found that challenge-skill balance was correlated with positive functioning and led to higher performance ratings while playing tennis, and Wang and Hsu (2013) reported that flow improved learning performance and learning satisfaction. Although the majority of sources reported perceived functioning or performance as outcomes of flow (e.g., Swann et al., 2017), some authors considered flow to be synonymous with objective performance success (e.g., Klasen et al., 2012). Given that peak performance (e.g., winning a sporting event) is dependent in many cases on external influences outside of an individual's control (Privette, 1983), however, performance success may be best considered as a possible (rather than expected) outcome of the construct.

Flow Outcome Theme 3: Further Engagement

Finally, a theme emerged reflecting "further engagement" (19.94%), which accounted for (a) continued engagement, adherence, and ongoing attentional recourses allocated to the relevant task in real time (e.g., Nakamura & Roberts, 2016), and (b) a temporally delayed reengagement (e.g., Csikszentmihalyi, 2014a). Terms such as "continued engagement," "sustained attention," "commitment," "exercise adherence," "optimal engagement," "further flow," and "increased motivation" were included in this theme. Sources considered flow to be a highly rewarding experience that led individuals to be naturally drawn toward reengagement even at great risk (Csikszentmihalyi, 2014b). For example, Nakamura and Csikszentmihalyi (2009) described emergent motivation, meaning that once flow is achieved, the emergence of new intrinsic motives facilitates further engagement. In an application of this principle, Rodriguez-Ardura and Meseguer-Artola (2017) reported that flow, as measured through a self-report questionnaire (examining challenge level, control, focused attention, presence, and positive affect), enhanced e-learners' academic performance and contributed to students' effective continuance in e-learning. Further, Harris et al. (2017) reported that flow resulted in continued attentional effort and allocation of attentional resources—which has both biological and sociological ramifications.

Discussion

Given the variation in conceptualizations of flow and the divergence in methods used to operationalize the construct, flow research is considered by some to have reached "crisis point" (e.g., Swann et al., 2018). Guided by scoping review recommendations, the broad purpose of this study was to bring together—and chart, in one place—the varied conceptualizations of flow that have appeared in the literature since 2012. We also sought, if possible, to identify commonalities in flow conceptualization that may provide a more coherent platform for future research in the area. Our review of over 200 sources demonstrated that the study of flow remains a popular research topic—flow research has been undertaken (since 2012) by researchers from 37 countries (predominately in Europe and the United States) and has appeared within 48 different scientific disciplines, including cognitive neuroscience, art therapy, sociology, clinical psychology, electrophysiology, artificial intelligence, human computer interface, and neuropsychophysiology. Unsurprisingly, flow research has been most frequently published in the broad field of psychology (accounting for 73% of studies in this review), especially in positive psychology (18%), sport and exercise psychology (10%), occupational psychology (7%), educational psychology (6%), and personality psychology (4%) fields. Importantly, although the review in part highlighted the disparate nature of flow research (e.g., 150 different constructs, terms, or concepts were used to describe the flow experience, and 84 different terms were positioned as flow antecedents), some commonalities also emerged with respect to flow antecedents, characteristics, and (to a lesser extent) outcomes. In the material that follows, we consider key implications of these findings.

The Conceptualization and Measurement of Flow

This review identified two commonly reported or studied antecedent themes (optimal challenge and high motivation) and three experiential themes (absorption, effort-less control, and intrinsic reward), with the experiential constructs associated with psychological and physiological markers (see Figure 3 and the Results section for a full analysis). With respect to the experience of flow itself, review sources indicated that flow is characterized by focused and undistracted attention, a merging of action and awareness, fluidity of performance, an absence of concern over losing control, positive valence, and optimal levels of arousal (see Table 1). It appears, therefore, that (according to the sources reviewed in this study at least) an appropriate way to define flow may be as follows: *flow is an intrinsically rewarding state of absorption in a task in which control feels effort-less*. In addition to evidence for common antecedents and experiential concepts, three broad outcomes of flow were highlighted or studied—namely, positive development, high functioning, and further engagement.

This review is important not least because it addresses recent calls to reduce (unnecessary) theoretical complexity, eliminate redundancy, and strive for conceptual parsimony in psychological research (e.g., Hagger, 2014; Le et al., 2010). Specifically, compared with the somewhat inconsistent application (and in some cases, conceptualization) of the nine-dimensional model, the two antecedent and three "flow experience" themes that emerged in this review may provide greater clarity for those seeking to understand, measure, and facilitate flow (and associated mechanisms or processes). In addition, and with those calls in mind, researchers have also been encouraged to consider the nature of flow relative to other related constructs, such as "peak experience," "ecstasy," "work engagement," "immersion," "peak performance," "presence," "intrinsic motivation," "mindfulness," "cognitive absorption," or "deep play." The results of this review indicate that there is clearly a continued research interest in, and attraction to, the

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Table 1
Flow Dimensions and Descriptions

Proposed definition of Flow	Flow is an intrinsically rewarding state of absorption in a task in which control feels effort-less
Antecedents to flow	
Optimal challenge	A perceived capability to meet the challenging demands of the situation
High motivation	A high motivational force
Experiential dimensions to flow	
Absorption	A state of absorption in the task characterized by focused, undistracted attention, and a merging of action and awareness
Effort-less control	A high sense of control in which the task feels less effortful than is typical for that person, characterized by fluidity of performance and an absence of concern over losing control
Intrinsic reward	An intrinsically rewarding experience characterized by positive valence and optimal levels of arousal

Note. Bold terms indicate categories.

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

flow construct—an interest and attraction that cuts across research methodologies and disciplines. This continued focus on flow may be in part due to the nature (i.e., the quality) of the experience itself. That is, although other related constructs may play a role in (and share elements of) the experience of flow, it appears that the combination of absorption, effort-less control, and intrinsic reward makes flow appealing from both a practical and a scientific perspective. It is interesting to note, for example, that individuals rarely focus on singular elements or components when describing flow. Most often, and as was observed in many of the sources reviewed in this study, it is the combination of (i.e., absorption, effort-less control, and intrinsic reward) elements that characterize individuals' descriptions of a latent flow experience. In addition to providing greater clarity on the psychological dimensions of flow, this review also helped identify the "nonpsychological" markers associated with flow experiences (see Figure 3). By integrating recent neuroscientific and physiological studies with those focused on people's experiences of flow, these findings provide a foundation for future research that intersects neurological, psychological, and physiological perspectives on the construct.

With an emphasis on the measurement of flow experiences (and antecedents), we observed substantial variation and a lack of consistency in the instruments used within the studies in this review. In addition to identifying a plethora of measurement instruments, it was apparent that the most commonly used instruments have attracted at least some criticism (see Moneta, 2012; Swann et al., 2018). It was also apparent that no existing instrument (a) adequately assesses all of the dimensions of flow outlined in this review; (b) is able to identify when a person is "in" or "out" of flow (Engeser, 2012b); or (c) offers a clearly superior way of measuring the intensity of the flow experience (Jackman et al., 2017; Moneta, 2012). Regarding the issue of identifying the presence or absence of flow, researchers are encouraged to move beyond identifying flow when summed mean dimensional scores are above a midpoint (often labeled, *neither agree or disagree*) as low dimensional scores may better reflect nonflow states (Jackman et al., 2017). Further, accounting for multiple occurrences of flow within a single activity or defined time period may be more accurate. One potential solution in this respect may be provided by Ellis et al.'s (2019) suggestion to integrate Hsiang et al.'s (2000) binary method of measurement with Csikszentmihalyi's original flow questionnaire methodology. Such a method could be administered postactivity (or through ESM), whereby participants are asked to respond "yes" or "no" (e.g., binary response instead of, or as well as, a continuous response scale) to a description of flow and its dimensions. This method may provide insight into whether focal individuals consider themselves to be experiencing (or to have experienced) flow. Meanwhile, a more detailed assessment of flow intensity and frequency could be achieved through questions focused directly on soliciting participant ratings of these descriptors. Respondents might be asked, for example, "Can you relate to experiencing this description?" (yes/no), and if yes, to identify the strength or intensity of the experience, and whether it reflected one continual experience or was a case of entering and exiting the experience multiple times. If indicating the latter, participants may also be asked to indicate the proportion of the total time they were in flow within the activity and add commentary on flow restoration or other temporal factors. Such a solution could be compared against the common approach of measuring flow through a Likert scale instrument (continuous measure) to ascertain possible thresholds of flow entry.

Integration of Postreview Research

Flow research has continued to accumulate since the period during which this review was conducted. It is important, therefore, to highlight certain important contemporary works that offer insight into the fundamental conclusions of this review. Most notably, Peifer and Engeser (2021a) edited a recent update to the 2012 collection of articles on flow (Engeser, 2012a), and in support of our review findings concluded that (a) "a gold standard for the modeling and measurement of flow is not at close reach" (p. 61)—highlighting that measuring flow is in need of continuous improvements in line with theoretical advancements; and (b) there was support for synthesizing flow descriptors into three core experiential dimensions (absorption, perceived demand-skill balance, enjoyment). Peifer and Engeser's (2021b) work also synthesized a number of flow characteristics into a core dimension, labeled "absorption," to represent the high level of control felt in flow. In addition, Peifer and Engeser (2021b) labeled a secondary core dimension as "perceived demand-skill balance." An important differentiating approach between the findings of this review and Peifer and Engeser's (2021b) labeling of "perceived demand-skill balance" is that sources within this review highlighted (both psychologically and neuroscientifically) that it is the sense of "effortlessness" (i.e., a subjective sense of the act being less effortful or more fluid than usual) toward the sense of control that differentiates flow from other forms of high control (also see

Peifer & Tan, 2021). Additionally, we observed that review sources in our work recognized the optimal level of challenge dimension (i.e., perceived demand-skill balance) as an antecedent to flow (e.g., Barthelmäs & Keller, 2021)—one that should be separated, therefore, from the experience of a high effort-less sense of control. Another important differentiating approach between this review and Peifer and Engeser's (2021b) suggested core dimensions, is that although both the findings of this review and Peifer and Engeser's (2021b) work aimed to encapsulate the autotelic experience of flow, we labeled that dimension as "intrinsic reward," differing from Peifer and Engeser's (2021b) utilization of the term "enjoyment." In this review, sources highlighted that to acknowledge an experience as "enjoyable" (or not) involves reflective cognitive processes that are deemed open to bias from outcomes and contextual or social factors. To illustrate, one may experience an intrinsically rewarding hit of dopamine during an optimally aroused state, for example, yet still reflect on the experience as not enjoyable if they lost the sporting contest or received negative feedback from a colleague (see Abuhamdeh's, 2021; review on the relationship between flow and enjoyment). The term intrinsic reward, therefore, was deemed as a more appropriate depiction of a flow-like experience.

In terms of understanding flow intensity, frequency, and duration, there is also a long-standing debate about whether flow is a discrete (i.e., in-or-out) or continuous (i.e., mild-to-intense) state. Although most flow researchers measure flow with Likert-type scales (thereby assuming flow is a continuous state), Abuhamdeh (2020) suggested that, if flow is conceptualized as an "optimal" experience, it should be measured as a discrete state and not in terms of varying levels of intensity. That is, Abuhamdeh argued that lower intensities are unlikely to be optimal, and that flow can only be experienced once a high threshold is met. Peifer and Engeser (2021b), on the other hand, suggest that flow can be both a discrete *and* continuous state; an experience can be considered "flow" when certain markers above a threshold are met (i.e., discrete), then "the intensity of the experience becomes even more intense (i.e., continuous) the more the components are present and pronounced" (p. 425). Further clarity in this area is clearly required in order to develop more effective forms of measuring (and promoting) flow.

Postreview neuroscience and psychophysiology studies continue to support the flow components of absorption, effortless control, and intrinsic reward. Three publications, however, offer additional insight beyond the review findings. First, van der Linden et al. (2021) positioned that the higher dopaminergic activity that makes flow intrinsically rewarding (a) inhibits mood states such as fatigue, hunger, or/and low expectations, and (b) activates moods and cognitive states such as optimism and energy, helping to explain certain descriptive and behavioral characteristics of flow (e.g., vitality, long durations without eating). In addition, van der Linden et al. (2021) suggested that the characteristics of flow can be partially explained by the interaction between three large networks in the brain. Van der Linden et al. argued that during flow, the default mode network (DMN; as explained earlier) activity is inhibited causing absorption effects (e.g., loss of self-consciousness), the central executive network becomes activated supporting the high degree of focused attention in flow, and the salience network (that continuously evaluates feedback and affirms that one is still "in control") activating may account for the high degree of control felt during flow and the continual inhibition of negative stress reactions. Second, Peifer and Tan (2021) published a revised and updated version of the article, "Psychophysiological Correlates of Flow Experience" (Peifer, 2012), offering an integrative account on physiological activation during flow. Peifer and Tan described findings that mirror those presented in this review (moderate sympathetic, cortisol, and parasympathetic activation, reduced DMN activity, and synchronized brain regions involving reward and cognitive control), though in addition, they also suggested a high activation of the brain's cognitive "multiple-demand system." And third, Leroy and Cheron (2020) highlighted the neural generators and rapid changes in cortical or subcortical activity that characterize flow. Specifically, Leroy and Cheron experimentally analyzed a tight-rope walker under ecological conditions and reported that the flow state was accompanied by the recruitment of additional EEG generators (globus pallidus) and basal ganglia activity (suggesting transient hypofrontality to momentarily suppress the analytical metaconscious abilities of the explicit system) not found in other high challenging (but not flow) situations.

Unanswered Questions and Future Directions

Despite providing insight into important conceptual and methodological issues, this review also highlighted key areas that require greater scrutiny by those who study flow. First, in examining the causal processes involved in flow experiences, constructs are often interchangeably referred to as antecedents, preconditions, mediators, compounding factors, or influencing factors. Moving forward, we recommend that constructs recognized or targeted as causing flow experiences to be labeled as "antecedents" so that research can be more easily amalgamated. Second, it is important that researchers' attention be directed toward understanding the nature of flow within groups. Recent research regarding flow in groups is sparse—only 6.33% of studies in this review fully or partially focused on the concept of group flow. Nevertheless, there is preliminary evidence that flow may have a contagion effect in groups (Culbertson et al., 2015; Kim, 2016). There was inconsistency among sources in this review as to whether "group flow" or "collective flow" may reflect a process of individuals being "drawn into" flow by one another in a contagion-like manner (i.e., an amalgamation of individual flow experiences; e.g., Culbertson et al., 2015), or whether a higher-order (group-level) flow construct exists (e.g., Corbatto, 2018; Kim, 2016). To better understand the nature of any group-level flow construct—such as in work or sport teams—it would be valuable to draw from Chan's (1998) classic text in which he outlines the ways through which constructs (e.g., flow) may operate at different levels of analysis. The careful development and testing of composition models in the flow literature (i.e., specifying how lower- and higher-order flow constructs are related) may enable interesting research into potential group-level correlates of any higher-order construct, such as heightened social awareness (Lucas, 2017), a sense of unity or joint progress, and shared identity (van den Hout et al., 2018). In addition, such work would enable researchers to test proposals relating to theorized antecedents of flow in team settings, such as institutional, cognitive, and affective trust (Corbatto, 2018), collective ambition, common goals, open communication, skill

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

integration, safety, mutual commitment (van den Hout et al., 2018), and collective efficacy beliefs (Salanova et al., 2014).

Aside from considering group-level, team-related issues, it is also important that researchers attend to integrating flow "theory" with other established theoretical models. It was apparent in these review sources, that there are overlaps between flow components and those in other prominent theoretical frameworks. If research efforts are directed toward studying these "overlaps" in greater detail, it is possible that such work will lead to a more nuanced understanding of (what appear to be) the common flow characteristics and antecedents outlined previously. For instance, the evidence relating to optimal challenge and arousal within flow appears to have connections to research on stress and optimal functioning (Tozman & Peifer, 2016). Specifically, the notions of "challenge" within the transactional-stress model (Lazarus, 1993; Lazarus & Folkman, 1984), "eustress" within Selye's (1983) general adaptation syndrome, and "optimal arousal" within Hanin's (2000) individual zones of optimal functioning model, all emphasize deep concentration and optimal functioning occurring at the top of an inverted-U bell curve (Tozman et al., 2015; Tozman & Peifer, 2016). Similarly, a closer consideration of self-efficacy theory (Bandura, 1997) may inform flow researchers' understanding of the "optimal challenge" antecedent, and aspects of self-determination theory (SDT; Ryan & Deci, 2000) and motivational competence (see Engeser & Schiepe-Tiska, 2012) may offer insight into the "high motivation" antecedent theme. As one example, Schüler and Brandstätter (2013) reported, for instance, that subjects whose psychological (competence and relatedness) needs were not satisfied reported less flow. Lastly, the two antecedents (i.e., optimal challenge and high motivation), as posited in this review, appear to be consistent with Petty and Cacioppo's (1986) delineation between ability and motivation as unique attributes to changing behavior.

There are also a range of important measurement and methodological considerations that stem from the findings of this review. We have already considered measurement issues relating to the assessment of flow experiences, but with respect to antecedents, it appears it would be valuable to assess optimal levels of challenge using instruments in which "demands" are not separated from "ability" (Keller & Landhausser, 2012). It would also be valuable to determine the extent to which optimal challenge is operationally distinguishable and independent from one's high motivation, and the combinations between these factors that are most likely to give rise to intense and sustained flow experiences. Research is also needed to better understand the types of motivation that lead to flow—perhaps most notably, it would be interesting to test in more detail whether autonomous motives are necessary for flow experiences, or whether (and if so, when, and in what relation to intrinsic motives) flow may occur in the presence of more controlled motives. Finally, researchers are encouraged to continue to chart the physiological markers that characterize flow experiences, partly with an emphasis on understanding whether these processes help reveal a graduated process for entering flow (vs. a more discrete "turning on" of flow).

Practical Implications

Before considering limitations of this study and offering conclusions, it is important to note the practical implications that emerged from this review. For those who seek to experience, train, or facilitate flow, review sources indicated that there may be promise in the use of self-regulatory training (Riva et al., 2016), hypoegoic self-regulation (Nakamura & Roberts, 2016), metacognition of flow training (Moneta, 2017; Wilson & Moneta, 2016), and flow mindset educational interventions (Norsworthy et al., 2017). The two broad antecedent themes outlined in this review provide direction that may help individuals achieve flow (e.g., generating high levels of motivation, and seeking challenging opportunities while maintaining a sense of capability to meet these demands and foster an optimal level of challenge). Further, Swann et al. (2017) suggested that researchers report on, or address directly, the restoration and temporal characteristics of flow experiences. Recommendations with this goal in mind might include attention training or meditation that targets focused and undistracted attention, and relaxation techniques that help manipulate individuals' arousal levels and parasympathetic activity during arousing activities (see MacNeill & Cavanagh, 2013; Peifer et al., 2014). We additionally suggest that noninvasive electrocardiographic biofeedback devices incorporating HRV training may also prove useful in this respect, especially when technology in this space becomes mobile.

Limitations

It is worth reiterating that although our aims were suited to the conduct of a scoping review, it remains that systematic reviews and meta-analyses within more specific domains (or focused on more specific questions) of application may help further refine our knowledge of applied flow science. It may also be valuable in the future to apply a Delphi method (see Landeta, 2006) to the themes presented and conclusions offered in this review, and to solicit expert commentary from those representing the subdisciplines from which review sources were drawn. A specific limitation of our Integration of Postreview Research is that included studies were not a comprehensive representation of all flow research that has appeared in the literature between 2019 and the time of publication. Rather, in that section, we simply hoped to highlight a *selection* of what we consider to be important contemporary articles published outside the window of our review period.

In relation to the limitations of review sources, it became apparent during our analysis that more experimental studies utilizing the same conceptualization of flow (see Moving Toward Conceptual Clarity and Parsimony) are required to draw more conclusive findings. Further, authors from review sources varied in their framing of constructs as flow antecedents, preconditions, mediators, compounding factors, influencing factors, experiential descriptors, and even outcomes—making the interpretation and synthesis of research difficult. The rise of nonpsychological (i.e., neuroscientific, physiological) research is helping to further our appreciation about the nature of flow (as evident by this review), although many of the review sources (a) assumed an "optimal challenge" was the only flow antecedent, and (b) assumed that flow was apparent when difficulty levels were *objectively* balanced with skill levels. It was unclear, therefore, in certain instances, what exactly was being measured (e.g., flow or a high level of challenge nonflow state). Given that flow is a subjective phenomenon, objective measures should be analyzed alongside (rather than replacing) subjective experiential measures.

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Conclusion

The work of Csikszentmihalyi and of others who have devoted sustained effort to examining and describing flow is of great importance. Flow research has been, and remains, pervasive across diverse scientific disciplines. In recent years though, scholars have noted that the language, definitions, measurement approaches, and models used in flow research and writing have become increasingly diverse. The aim of this review was to collate findings and perspectives on flow from different scientific disciplines. In doing so, we sought to identify what was common across recent flow work, and to make a step toward finding a more consistent language and meaning for those who work to understand and promote flow experiences. The findings of this review indicate that despite substantial diversity in terminology, concepts, descriptors, and assessment methods, there are indeed many commonalities in our collective study of this construct. Focusing on these commonalities may advance our understanding of flow beyond the nine-dimensional model and better inform us about what flow is, what flow is not, and how flow is generated and best measured. Moreover, the systematic study of these commonalities promises to aid our conceptual understanding, help address measurement limitations, enable better theoretical integration, and may allow people to more readily facilitate and experience flow.

References

Abuhamdeh, S. (2020). Investigating the "flow" experience: Key conceptual and operational issues. *Frontiers in Psychology*, *11*(1), 158. https://doi.org/10.3389/fpsyg.2020.00158

Abuhamdeh, S. (2021). On the relationship between flow and enjoyment. In S. Engeser, & C. Peifer (Eds.), *Advances in flow research* (pp. 155–169). Springer. https://doi.org/10.1007/978-3-030-53468-4_6

Abuhamdeh, S., & Engeser, S. (2012). A conceptual framework for the integration of flow theory and cognitive evaluation theory. In S. Engeser (Ed.), *Advances in flow research* (pp. 109–121). Springer. https://doi.org/10.1007/978-1-4614-2359-1_6

Andersen, F. Ø. (2016). Using ESM to study flow in a STEM project. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 145–161). Springer. https://doi.org/10.1007/978-3-319-28634-1_10

Anderson, R. (2007). Thematic content analysis (TCA). *Descriptive presentation of qualitative data*, 1–4. http://rosemarieanderson.com/wp-content/uploads/2014/08/ThematicContentAnalysis.pdf

Appleton, C. (2011). 'Critical friends', feminism and integrity: A reflection on the use of critical friends as a research tool to support researcher integrity and reflexivity in qualitative research studies. *Women in Welfare Education*, *10*(1), 1–13.

Araújo, M. V., & Hein, C. F. (2016). Finding flow in music practice: An exploratory study about self-regulated practice behaviours and dispositions to flow in highly skilled musicians. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 23–36). Springer. https://doi.org/10.1007/978-3-319-28634-1_2

Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, *8*(1), 19–32. https://doi.org/10.1080/1364557032000119616

Auld, D. P. (2014). *Flow and learning in computer-mediated learning environments. A meta-analytic review* [Doctoral dissertation, Fordham University]. https://doi.org/10.1037/e549562014-001

Bakker, A. B. (2008). The work-related flow inventory: Construction and initial validation of the WOLF. *Journal of Vocational Behavior*, *72*(3), 400–414. https://doi.org/10.1016/j.jvb.2007.11.007

Bakker, A., Golub, T. L., & Rijavec, M. (2017). Validation of the Study-Related Flow Inventory (WOLF-S)/Validacija Inventara zanesenosti u studiranju (WOLF-S). *Croatian Journal of Education-Hrvatski Casopis za Odgoj i Obrazovanje*, *19*(1), 147–173.

Bandura, A. (1986). The explanatory and predictive scope of self-efficacy theory. *Journal of Social and Clinical Psychology*, *4*(3), 359–373. https://doi.org/10.1521/jscp.1986.4.3.359

Bandura, A. (1997). Self-efficacy and health behaviour. In A. Baum, S. Newman, J. Weinman, R. West, & C. McManus (Eds.), *Cambridge handbook of psychology, health and medicine* (pp. 160–162). Cambridge University Press.

Barrett, L. F., Mesquita, B., Ochsner, K. N., & Gross, J. J. (2007). The experience of emotion. *Annual Review of Psychology*, *58*(1), 373–403. https://doi.org/10.1146/annurev.psych.58.110405.085709

Barthelmäs, M., & Keller, J. (2021). Antecedents, boundary conditions and consequences of flow. In S. Engeser, & C. Peifer (Eds.), *Advances in flow research* (pp. 71–107). Springer. https://doi.org/10.1007/978-3-030-53468-4_3

Bassi, M., & Delle Fave, A. (2012). Optimal experience among teachers: New insights into the work paradox. *The Journal of Psychology*, *146*(5), 533–557. https://doi.org/10.1080/00223980.2012.656156

Baumann, N. (2012). Autotelic personality. In S. Engeser (Ed.), *Advances in flow research* (pp. 165–186). Springer. https://doi.org/10.1007/978-1-4614-2359-1_9

Beard, K. S. (2015). Theoretically speaking: An interview with Mihaly Csikszentmihalyi on flow theory development and its usefulness in addressing contemporary challenges in education. *Educational Psychology Review*, *27*(2), 353–364. https://doi.org/10.1007/s10648-014-9291-1

Beard, K. S., & Hoy, W. K. (2010). The nature, meaning, and measure of teacher flow in elementary schools: A test of rival hypotheses. *Educational Administration Quarterly*, *46*(3), 426–458. https://doi.org/10.1177/0013161X10375294

Bian, Y., Yang, C., Gao, F., Li, H., Zhou, S., Li, H., Sun, X., & Meng, X. (2016). A framework for physiological indicators of flow in VR games: Construction and preliminary evaluation. *Personal and Ubiquitous Computing*, *20*(5), 821–832. https://doi.org/10.1007/s00779-016-0953-5

Boffi, M., Riva, E., Rainisio, N., & Inghilleri, P. (2016). Social psychology of flow: A situated framework for optimal experience. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 215–231). Springer. https://doi.org/10.1007/978-3-319-28634-1_14

Boyd, J. M., Schary, D. P., Worthington, A. R., & Jenny, S. E. (2018). An examination of the differences in flow between individual and team athletes. *Physical Culture and Sport Studies and Research*, *78*(1), 33–40. https://doi.org/10.2478/pcssr-2018-0011

Bricteux, C., Navarro, J., Ceja, L., & Fuerst, G. (2017). Interest as a moderator in the relationship between challenge/skills balance and flow at work: An analysis at within-individual level. *Journal of Happiness Studies*, *18*(3), 861–880. https://doi.org/10.1007/s10902-016-9755-8

Brooker, K., van Dooren, K., McPherson, L., Lennox, N., & Ware, R. (2015). A systematic review of interventions aiming to improve involvement in physical activity among adults with intellectual disability. *Journal of Physical Activity & Health*, *12*(3), 434–444. https://doi.org/10.1123/jpah.2013-0014

Campbell, D. T., & Stanley, J. C. (1963). *Experimental and quasi-experimental designs for research*. Houghton Mifflin.

Chan, D. (1998). Functional relations among constructs in the same content domain at different levels of analysis: A typology of composition models. *Journal of Applied Psychology*, *83*(2), 234–246. https://doi.org/10.1037/0021-9010.83.2.234

Chavez, E. J. (2008). Flow in sport: A study of college athletes. *Imagination, Cognition and Personality*, *28*(1), 69–91. https://doi.org/10.2190/IC.28.1.f

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Chen, H. (2006). Flow on the net–detecting Web users' positive affects and their flow states. *Computers in Human Behavior*, *22*(2), 221–233. https://doi.org/10.1016/j.chb.2004.07.001

Cheng, W. K. R. (2014). *Relationship between visual attention and flow experience in a serious educational game: An eye tracking analysis* [Unpublished doctoral dissertation]. George Mason University.

Chen, L. X., & Sun, C. T. (2016). Self-regulation influence on game play flow state. *Computers in Human Behavior*, *54*(1), 341–350. https://doi.org/10.1016/j.chb.2015.08.020

Cheron, G. (2016). How to measure the psychological "flow"? a neuroscience perspective. *Frontiers in Psychology*, *7*(1), 1823. https://doi.org/10.3389/fpsyg.2016.01823

Chilton, G. (2013). Art therapy and flow: A review of the literature and applications. *Art Therapy: Journal of the American Art Therapy Association*, *30*(2), 64–70. https://doi.org/10.1080/07421656.2013.787211

Chirico, A., Serino, S., Cipresso, P., Gaggioli, A., & Riva, G. (2015). When music "flows." State and trait in musical performance, composition and listening: A systematic review. *Frontiers in Psychology*, *6*(1), 906. https://doi.org/10.3389/fpsyg.2015.00906

Colquhoun, H. L., Levac, D., O'Brien, K. K., Straus, S., Tricco, A. C., Perrier, L., Kastner, M., & Moher, D. (2014). Scoping reviews: Time for clarity in definition, methods, and reporting. *Journal of Clinical Epidemiology*, *67*(12), 1291–1294. https://doi.org/10.1016/j.jclinepi.2014.03.013

Colzato, L. S., Wolters, G., & Peifer, C. (2018). Transcutaneous vagus nerve stimulation (tVNS) modulates flow experience. *Experimental Brain Research*, *236*(1), 253–257. https://doi.org/10.1007/s00221-017-5123-0

Corbatto, D. B. (2018). *Trust as a precursor of flow: A social cognitive view of flow in elite coach/athlete dyads* [Doctoral dissertation, George Mason University]. ProQuest Dissertations & Theses Database.

Cseh, G. M. (2016). Flow in creativity: A review of potential theoretical conflict. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow Experience* (pp. 79–94). Springer. https://doi.org/10.1007/978-3-319-28634-1_6

Csikszentmihalyi, M. (1975). *Beyond boredom and anxiety*. Jossey-Bass.

Csikszentmihalyi, M. (1990). *Flow: The psychology of optimal experience*. Harper & Row.

Csikszentmihalyi, M. (2014a). *Flow and the foundations of positive psychology: The collected works of Mihaly Csikszentmihalyi*. Springer. https://doi.org/10.1007/978-94-017-9088-8

Csikszentmihalyi, M. (2014b). Toward a psychology of optimal experience. In M. Csikszentmihalyi & R. Larson (Eds.), *Flow and the foundations of positive psychology* (pp. 209–226). Springer.

Csikszentmihalyi, M. (2014c). *Applications of flow in human development and education*. Springer. https://doi.org/10.1007/978-94-017-9094-9

Csikszentmihalyi, M., Montijo, M. N., & Mouton, A. R. (2018). Flow theory: Optimizing elite performance in the creative realm. In S. I. Pfeiffer, E. Shaunessy-Dedrick, & M. Foley-Nicpon (Eds.), *APA handbooks in psychology®. APA handbook of giftedness and talent* (pp. 215–229). American Psychological Association. https://doi.org/10.1037/0000038-014

Culbertson, S. S., Fullagar, C. J., Simmons, M. J., & Zhu, M. (2015). Contagious flow: Antecedents and consequences of optimal experience in the classroom. *Journal of Management Education*, *39*(3), 319–349. https://doi.org/10.1177/1052562914545336

de Manzano, Ö., Cervenka, S., Jucaite, A., Hellenäs, O., Farde, L., & Ullén, F. (2013). Individual differences in the proneness to have flow experiences are linked to dopamine D2-receptor availability in the dorsal striatum. *NeuroImage*, *67*(1), 1–6. https://doi.org/10.1016/j.neuroimage.2012.10.072

de Sampaio Barros, M. F., Araújo-Moreira, F. M., Trevelin, L. C., & Radel, R. (2018). Flow experience and the mobilization of attentional resources. *Cognitive, Affective & Behavioral Neuroscience*, *18*(4), 810–823. https://doi.org/10.3758/s13415-018-0606-4

Dietrich, A. (2004). The cognitive neuroscience of creativity. *Psychonomic Bulletin & Review*, *11*(6), 1011–1026. https://doi.org/10.3758/BF03196731

Drengner, J., Jahn, S., & Furchheim, P. (2018). Flow revisited: Process conceptualization and a novel application to service contexts. *Journal of Service Management*, *29*(4), 703–734. https://doi.org/10.1108/JOSM-12-2016-0318

Ellis, G. D., Freeman, P. A., Jiang, J., & Lacanienta, A. (2019). Measurement of deep structured experiences as a binary phenomenon. *Annals of Leisure Research*, *22*(1), 119–126. https://doi.org/10.1080/11745398.2018.1429285

Engeser, S. E. (2012a). *Advances in flow research*. Springer Science+Business Media. https://doi.org/10.1007/978-1-4614-2359-1

Engeser, S. E. (2012b). Theoretical integration and future lines of flow research. In S. Engeser (Ed.), *Advances in flow research* (pp. 187–199). Springer. https://doi.org/10.1007/978-1-4614-2359-1_10

Engeser, S., & Rheinberg, F. (2008). Flow performance and moderators of challenge-skill balance. *Motivation and Emotion*, *32*(3), 158–172. https://doi.org/10.1007/s11031-008-9102-4

Engeser, S., & Schiepe-Tiska, A. (2012). Historical lines and an overview of current research on flow. In S. Engeser (Ed.), *Advances in flow research* (pp. 1–22). Springer. https://doi.org/10.1007/978-1-4614-2359-1_1

Esterman, M., Noonan, S. K., Rosenberg, M., & Degutis, J. (2013). In the zone or zoning out? Tracking behavioral and neural fluctuations during sustained attention. *Cerebral Cortex*, *23*(11), 2712–2723. https://doi.org/10.1093/cercor/bhs261

Fang, X., Zhang, J., & Chan, S. S. (2013). Development of an instrument for studying flow in computer game play. *International Journal of Human-Computer Interaction*, *29*(7), 456–470. https://doi.org/10.1080/10447318.2012.715991

Flett, M. R. (2015). Is flow related to positive feelings or optimal performance? Path analysis of challenge-skill balance and feelings. *Sport Science Review*, *24*(1-2), 5–26. https://doi.org/10.1515/ssr-2015-0006

Fong, C. J., Zaleski, D. J., & Leach, J. K. (2015). The challenge–skill balance and antecedents of flow: A meta-analytic investigation. *The Journal of Positive Psychology*, *10*(5), 425–446. https://doi.org/10.1080/17439760.2014.967799

Freire, T., Tavares, D., Silva, E., & Teixeira, A. (2016). Flow, leisure, and positive youth development. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 163–178). Springer. https://doi.org/10.1007/978-3-319-28634-1_11

Fu, F. L., Su, R. C., & Yu, S. C. (2009). EGameFlow: A scale to measure learners' enjoyment of e-learning games. *Computers & Education*, *52*(1), 101–112. https://doi.org/10.1016/j.compedu.2008.07.004

Fullagar, C. J., Knight, P. A., & Sovern, H. S. (2013). Challenge/skill balance, flow, and performance anxiety. *Applied Psychology*, *62*(2), 236–259. https://doi.org/10.1111/j.1464-0597.2012.00494.x

Fullagar, C., & Kelloway, E. K. (2012). Work-related flow. In *A day in the life of a happy worker* (pp. 49–65). Psychology Press.

Gaggioli, A., Cipresso, P., Serino, S., & Riva, G. (2013). Psychophysiological correlates of flow during daily activities. *Annual Review of Cybertherapy and Telemedicine*, *191*(1), 65–69.

Garces-Bacsal, R. M. (2016). Extending flow further: Narrative of a Filipino musician. *International Journal of Music Education*, *34*(4), 433–444. https://doi.org/10.1177/0255761415590366

Geyser, I., Geldenhuys, M., & Crous, F. (2015). The dimensionality of the work related flow inventory (WOLF): A South African study. *Journal of Psychology in Africa*, *25*(4), 282–287. https://doi.org/10.1080/14330237.2015.1078084

Ghani, J. A. (1995). Flow in human computer interactions: Test of a model. In *Human factors in information systems: Emerging theoretical bases* (pp. 291–311). Ablex Publishing Corp.

Grigorieva, A. M. (2015). *Cohabiting couples' relational flow and separateness: Psychometric properties of a new measure of relational flow*. Alliant International University.

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Guertin, J. F. (2016). *Three essays on the development, validation and confirmation of the flow construct to investigate navigational web site experiences* [Doctoral dissertation, HEC Montréal]. Dissertation Abstracts International. A, the Humanities and Social Sciences.

Gyurkovics, M., Kotyuk, E., Katonai, E. R., Horvath, E. Z., Vereczkei, A., & Szekely, A. (2016). Individual differences in flow proneness are linked to a dopamine D2 receptor gene variant. *Consciousness and Cognition*, *42*(1), 1–8. https://doi.org/10.1016/j.concog.2016.02.014

Hagger, M. S. (2014). Avoiding the "déjà-variable" phenomenon: Social psychology needs more guides to constructs. *Frontiers in Psychology*, *5*, 52. https://doi.org/10.3389/fpsyg.2014.00052

Hanin, Y. L. (Ed.). (2000). Individual Zones of Optimal Functioning (IZOF) Model: Emotion-performance relationship in sport. *Emotions in sport* (pp. 65–89). Human Kinetics.

Harmat, L., Andersen, F. Ø., Ullén, F., Wright, J., & Sadlo, G. (Eds.). (2016). Erratum: Flow experience. *Flow experience* (pp. E1–E1). Springer. https://doi.org/10.1007/978-3-319-28634-1_23

Harmat, L., de Manzano, Ö., Theorell, T., Högman, L., Fischer, H., & Ullén, F. (2015). Physiological correlates of the flow experience during computer game playing. *International Journal of Psychophysiology*, *97*(1), 1–7. https://doi.org/10.1016/j.ijpsycho.2015.05.001

Harris, D. J., Vine, S. J., & Wilson, M. R. (2017). Neurocognitive mechanisms of the flow state. *Progress in brain research*, *234*, 221–243. https://doi.org/10.1016/bs.pbr.2017.06.012

Herrando, C., Jiménez-Martínez, J., & Hoyos, M. J. M. D. (2018). The state of flow in social commerce: The importance of flow consciousness. *International Journal of Electronic Marketing and Retailing*, *9*(4), 320–343. https://doi.org/10.1504/IJEMR.2018.094987

Heutte, J., Fenouillet, F., Kaplan, J., Martin-Krumm, C., & Bachelet, R. (2016). The EduFlow model: A contribution toward the study of optimal learning environments. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 127–143). Springer. https://doi.org/10.1007/978-3-319-28634-1_9

Hoffman, D. L., & Novak, T. P. (2009). Flow online: Lessons learned and future prospects. *Journal of Interactive Marketing*, *23*(1), 23–34. https://doi.org/10.1016/j.intmar.2008.10.003

Hsiang, C., Rolf, T. W., & Michael, N. (2000). Exploring Web users' optimal flow experiences. *Information Technology & People*, *13*(4), 263–281. https://doi.org/10.1108/09593840010359473

Huang, M. H. (2006). Flow, enduring, and situational involvement in the Web environment: A tripartite second-order examination. *Psychology and Marketing*, *23*(5), 383–411. https://doi.org/10.1002/mar.20118

Huskey, R., Craighead, B., Miller, M. B., & Weber, R. (2018). Does intrinsic reward motivate cognitive control? A naturalistic-fMRI study based on the synchronization theory of flow. *Cognitive, Affective & Behavioral Neuroscience*, *18*(5), 902–924. https://doi.org/10.3758/s13415-018-0612-6

Jackman, P. C., Crust, L., & Swann, C. (2017). Systematically comparing methods used to study flow in sport: A longitudinal multiple-case study. *Psychology of Sport and Exercise*, *32*(1), 113–123. https://doi.org/10.1016/j.psychsport.2017.06.009

Jackson, S. A., & Csikszentmihalyi, M. (1999). *Flow in sports*. Human Kinetics.

Jackson, S. A., & Eklund, R. C. (2002). Assessing flow in physical activity: The flow state scale-2 and dispositional flow scale-2. *Journal of Sport & Exercise Psychology*, *24*(2), 133–150. https://doi.org/10.1123/jsep.24.2.133

Jackson, S. A., & Marsh, H. W. (1996). Development and validation of a scale to measure optimal experience: The Flow State Scale. *Journal of Sport & Exercise Psychology*, *18*(1), 17–35. https://doi.org/10.1123/jsep.18.1.17

Järvilehto, L. (2016). Intuition and flow. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 95–104). Springer. https://doi.org/10.1007/978-3-319-28634-1_7

Katahira, K., Yamazaki, Y., Yamaoka, C., Ozaki, H., Nakagawa, S., & Nagata, N. (2018). EEG correlates of the flow state: A combination of increased frontal theta and moderate frontocentral alpha rhythm in the mental arithmetic task. *Frontiers in Psychology*, *9*(1), 300. https://doi.org/10.3389/fpsyg.2018.00300

Kaur, P., Dhir, A., Chen, S., & Rajala, R. (2016). Flow in context: Development and validation of the flow experience instrument for social networking. *Computers in Human Behavior*, *59*(1), 358–367. https://doi.org/10.1016/j.chb.2016.02.039

Kawabata, M., & Mallett, C. J. (2016). Reconsideration of the state of optimal functioning beyond flow theory. In R. J. Schinke, K. R. McGannon, & B. Smith (Eds.), *Routledge international handbook of sport psychology* (pp. 399–407). Routledge.

Keeler, J. R., Roth, E. A., Neuser, B. L., Spitsbergen, J. M., Waters, D. J. M., & Vianney, J. M. (2015). The neurochemistry and social flow of singing: Bonding and oxytocin. *Frontiers in Human Neuroscience*, *9*, 518. https://doi.org/10.3389/fnhum.2015.00518

Keller, J. (2016). The flow experience revisited: the influence of skills-demands-compatibility on experiential and physiological indicators. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 351–374). Springer. https://doi.org/10.1007/978-3-319-28634-1_21

Keller, J., & Landhausser, A. (2012). The flow model revisited. In S. Engeser (Ed.), *Advances in flow research* (pp. 51–64). Springer. https://doi.org/10.1007/978-1-4614-2359-1_3

Kim, J. Y., Min, S. N., & Park, Y. D. (2013). Evaluation of flow experience by using psychophysiological visual feedbacks. *Journal of the Ergonomics Society of Korea*, *32*(6), 481–487. https://doi.org/10.5143/JESK.2013.32.6.481

Kim, R. (2016). *The concept of flow: A critical review of the literature within music, sports, & education* [Doctoral dissertation, Biola University]. ProQuest Dissertations and Theses database.

Klasen, M., Weber, R., Kircher, T. T., Mathiak, K. A., & Mathiak, K. (2012). Neural contributions to flow experience during video game playing. *Social Cognitive and Affective Neuroscience*, *7*(4), 485–495. https://doi.org/10.1093/scan/nsr021

Kmet, L. M., Lee, R. C., & Cook, L. S. (2004). *Standard quality assessment criteria for evaluating primary research papers from a variety of fields*. Alberta Heritage Foundation for Medical Research.

Landeta, J. (2006). Current validity of the Delphi method in social sciences. *Technological Forecasting and Social Change*, *73*(5), 467–482. https://doi.org/10.1016/j.techfore.2005.09.002

Lazarus, R. S. (1993). Coping theory and research: Past, present, and future. *Psychosomatic Medicine*, *55*(3), 234–247.

Lazarus, R. S., & Folkman, S. (1984). *Stress, appraisal, and coping*. Springer publishing company.

Lazoc, A., & Luiza, C. (2012). Elaborating a measurement instrument for the flow experience during online information search. *Annals of the University of Oradea. Economic Science Series*, *21*(2), 841–847.

Le, H., Schmidt, F. L., Harter, J. K., & Lauver, K. J. (2010). The problem of empirical redundancy of constructs in organizational research: An empirical investigation. *Organizational Behavior and Human Decision Processes*, *112*(2), 112–125. https://doi.org/10.1016/j.obhdp.2010.02.003

Leroy, A., & Cheron, G. (2020). EEG dynamics and neural generators of psychological flow during one tightrope performance. *Scientific Reports*, *10*(1), 12449. https://doi.org/10.1038/s41598-020-69448-3

Llorens, S., & Salanova, M. (2017). The consequences of flow. In A. B. Bakker, & M. van Woerkom (Eds.), *Flow at work: Measurement and implications*. Routledge/Taylor & Francis Group. https://doi.org/10.4324/9781315871585-6

Llorens, S., Salanova, M., & Rodríguez, A. M. (2013). How is flow experienced and by whom? Testing flow among occupations. *Stress and Health*, *29*(2), 125–137. https://doi.org/10.1002/smi.2436

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Lovoll, H. S., & Vitterso, J. (2014). Can balance be boring? A critique of the "challenges should match skills" hypotheses in flow theory. *Social Indicators Research*, *115*(1), 117–136. https://doi.org/10.1007/s11205-012-0211-9

Lucas, H. E. (2017). Social flow: Optimal experience with others at work and play. In M. A.Warren, & S. I. Donaldson (Eds.), *Toward a positive psychology of relationships: New directions in theory and research* (pp. 179–193). ABC-CLIO.

MacNeill, N., & Cavanagh, R. (2013). The possible misfit of Csikszentmihalyi's dimensions of flow in the contemporary roles of school leaders. *Management in Education*, *27*(1), 7–13. https://doi.org/10.1177/0892020612459288

Magyarodi, T., & Oláh, A. (2015). A cross-sectional survey study about the most common solitary and social flow activities to extend the concept of optimal experience. *Europe's Journal of Psychology*, *11*(4), 632–650. https://doi.org/10.5964/ejop.v11i4.866

Mahnke, R., Wagner, T. M., & Benlian, A. (2012). Flow experience on the Web: Measurement validation and mixed method survey of flow activities. Proceedings of the 20th European Conference on Information Systems (ECIS, p. 33). https://aisel.aisnet.org/ecis2012/33.Proc-ecis

Marr, A. J. (2001). In the zone: A biobehavioral theory of the flow experience. Athletic Insight: *Online Journal of Sport Psychology*. http://www.athleticinsight.com/Vol3Iss1/Commentary.htm

McGuire, J. T., & Botvinick, M. M. (2010). Prefrontal cortex, cognitive control, and the registration of decision costs. *Proceedings of the National Academy of Sciences of the United States of America*, *107*(17), 7922–7926. https://doi.org/10.1073/pnas.0910662107

Miles, S. E. (2012). *In the flow: A mixed-methods phenomenological study of optimal experience in adolescent literacy*. Gardner-Webb University.

Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLoS Medicine*, *6*(7), e1000097. https://doi.org/10.1371/journal.pmed.1000097

Moneta, G. B. (2012). On the measurement and conceptualization of flow. In S. Engeser (Ed.), *Advances in flow research* (pp. 23–50). Springer. https://doi.org/10.1007/978-1-4614-2359-1_2

Moneta, G. B. (2017). Applications of flow to work. In A. B. Bakker, & M. van Woerkom (Eds.), *Flow at work* (pp. 131–151). Routledge. https://doi.org/10.4324/9781315871585-7

Monson, C. H. (2013). *Spiritual flow: The influence of religious experience on psychological well-being* [Doctoral dissertation, Indiana State University]. http://hdl.handle.net/10484/4756

Moors, A. (2009). Theories of emotion causation: A review. *Cognition and Emotion*, *23*(4), 625–662. https://doi.org/10.1080/02699930802645739

Moran, A. (2012). Concentration: Attention and performance. In S. Murphy (Ed.), *The Oxford handbook of sport and performance psychology* (pp. 117–130). Oxford University Press.

Nah, F. F. H., Yelamanchili, T., & Siau, K. (2017). A review on neuropsychophysiological correlates of flow. In F. F. H. Nah, & C. H. Tan (Eds.), *International Conference on HCI in Business, Government, and Organizations* (pp. 364–372). Springer.

Nakamura, J., & Csikszentmihalyi, M. (2002). The concept of flow. In C. Snyder & S. Lopez (Eds.), *Handbook of positive psychology* (pp. 89–105). New York University Press.

Nakamura, J., & Csikszentmihalyi, M. (2009). Flow theory and research. In C. Snyder & S. Lopez (Eds.), *Handbook of positive psychology* (pp. 195–206). New York University Press.

Nakamura, J., & Csikszentmihalyi, M. (2014). The concept of flow. In M. Csikszentmihalyi, & R. Larson (Eds.), *Flow and the foundations of positive psychology* (pp. 239–263). Springer.

Nakamura, J., & Roberts, S. (2016). The hypo-egoic component of flow. In K. W. Brown & M. R. Leary, (Eds.), *The Oxford handbook of hypo-egoic phenomena* (p. 133). Oxford University Press.

Norsworthy, C., Thelwell, R., Weston, N., & Jackson, S. A. (2017). Flow training, flow states, and performance in élite athletes. *International Journal of Sport Psychology*, *49*(1), 134–152.

Novak, T. P., & Hoffman, D. L. (1997). Measuring the flow experience among web users. *Interval Research Corporation*, *31*(1), 1–35.

Novak, T. P., Hoffman, D. L., & Yung, Y. F. (2000). Measuring the customer experience in online environments: A structural modeling approach. *Marketing Science*, *19*(1), 22–42. https://doi.org/10.1287/mksc.19.1.22.15184

Omori, S., Nagano, Y., Kobayashi, T., & Katayama, J. I. (2014). Psychological and physiological responses during flow state: A study for flow state during a game task. *International Journal of Psychophysiology*, *94*(2), 246–247. https://doi.org/10.1016/j.ijpsycho.2014.08.943

Oxman, A. D., & Guyatt, G. H. (1988). Guidelines for reading literature reviews. *Canadian Medical Association Journal*, *138*(8), 697–703.

Peifer, C. (2012). Psychophysiological correlates of flow-experience. In S. Engeser (Ed.), *Advances in flow research* (pp. 139–164). Springer. https://doi.org/10.1007/978-1-4614-2359-1_8

Peifer, C., & Engeser, S. (2021a). *Advances in flow research*. Springer Science+ Business Media.

Peifer, C., & Engeser, S. (2021b). Theoretical Integration and future lines of flow research. In S. Engeser & C. Peifer (Eds.), *Advances in flow research* (pp. 417–439). Springer. https://doi.org/10.1007/978-3-030-53468-4_16

Peifer, C., & Tan, J. (2021). The psychophysiology of flow experience. In S. Engeser & C. Peifer (Eds.), *Advances in flow research* (pp. 191–230). Springer. https://doi.org/10.1007/978-3-030-53468-4_8

Peifer, C., Schächinger, H., Engeser, S., & Antoni, C. H. (2015). Cortisol effects on flow-experience. *Psychopharmacology*, *232*(6), 1165–1173. https://doi.org/10.1007/s00213-014-3753-5

Peifer, C., Schulz, A., Schächinger, H., Baumann, N., & Antoni, C. H. (2014). The relation of flow-experience and physiological arousal under stress—can u shape it? *Journal of Experimental Social Psychology*, *53*(1), 62–69. https://doi.org/10.1016/j.jesp.2014.01.009

Pelet, J. É., Ettis, S., & Cowart, K. (2017). Optimal experience of flow enhanced by telepresence: Evidence from social media use. *Information & Management*, *54*(1), 115–128. https://doi.org/10.1016/j.im.2016.05.001

Perttula, A., Kiili, K., Lindstedt, A., & Tuomi, P. (2017). Flow experience in game based learning–a systematic literature review. *International Journal of Serious Games*, *4*(1), 57–72. https://doi.org/10.17083/ijsg.v4i1.151

Peters, M. D., Godfrey, C. M., Khalil, H., McInerney, P., Parker, D., & Soares, C. B. (2015). Guidance for conducting systematic scoping reviews. *International Journal of Evidence-Based Healthcare*, *13*(3), 141–146. https://doi.org/10.1097/XEB.0000000000000050

Petty, R. E., & Cacioppo, J. T. (1986). The elaboration likelihood model of persuasion. In *Communication and persuasion* (pp. 1–24). Springer.

Pluta, A., Williams, C. C., Binsted, G., Hecker, K. G., & Krigolson, O. E. (2018). Chasing the zone: Reduced beta power predicts baseball batting performance. *Neuroscience Letters*, *686*(1), 150–154. https://doi.org/10.1016/j.neulet.2018.09.004

Privette, G. (1983). Peak experience, peak performance, and flow: A comparative analysis of positive human experiences. *Journal of Personality and Social Psychology*, *45*(6), 1361–1368. https://doi.org/10.1037/0022-3514.45.6.1361

Rheinberg, F., & Engeser, S. (2018). Intrinsic motivation and flow. In J. Heckhausen & H. Heckhausen (Eds.), *Motivation and action* (pp. 579–622). Springer. https://doi.org/10.1007/978-3-319-65094-4_14

Rheinberg, F., Engeser, S., & Vollmeyer, R. (2002). Measuring components of flow: The Flow-Short-Scale. *Proceedings of the 1st International Positive Psychology Summit.*

Rich, G. J. (2013). Finding flow: The history and future of a positive psychology concept. In J. Sinnott (Ed.), *Positive psychology: Advances in*

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

understanding adult motivation (pp. 43–60). Springer Science+ Business Media.

Riva, E., Freire, T., & Bassi, M. (2016). The flow experience in clinical settings: Applications in psychotherapy and mental health rehabilitation. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 309–326). Springer. https://doi.org/10.1007/978-3-319-28634-1_19

Rodriguez-Ardura, I., & Meseguer-Artola, A. (2017). Flow in e-learning: What drives it and why it matters. *British Journal of Educational Technology*, *48*(4), 899–915. https://doi.org/10.1111/bjet.12480

Romero, P., & Calvillo-Gamez, E. (2014). An embodied view of flow. *Interacting with Computers*, *26*(6), 513–527. https://doi.org/10.1093/iwc/iwt051

Ryan, R. M., & Deci, E. L. (2000). Intrinsic and extrinsic motivations: Classic definitions and new directions. *Contemporary Educational Psychology*, *25*(1), 54–67. https://doi.org/10.1006/ceps.1999.1020

Sadlo, G. (2016). Towards a neurobiological understanding of reduced self-awareness during flow: An occupational science perspective. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow Experience* (pp. 375–388). Springer. https://doi.org/10.1007/978-3-319-28634-1_22

Salanova, M., Rodríguez-Sánchez, A. M., Schaufeli, W. B., & Cifre, E. (2014). Flowing together: A longitudinal study of collective efficacy and collective flow among workgroups. *The Journal of Psychology*, *148*(4), 435–455. https://doi.org/10.1080/00223980.2013.806290

Schiefele, U. (2013). Response to Engeser (2012): On the nature of flow experience. *Psychological Reports*, *112*(2), 529–532. https://doi.org/10.2466/04.PR0.112.2.529-532

Schüler, J., & Brandstätter, V. (2013). How basic need satisfaction and dispositional motives interact in predicting flow experience in sport. *Journal of Applied Social Psychology*, *43*(4), 687–705. https://doi.org/10.1111/j.1559-1816.2013.01045.x

Seligman, M. E. (2002). Positive psychology, positive prevention, and positive therapy. In C. Snyder, & S. Lopez (Eds.), *Handbook of positive psychology* (pp. 3–12). Oxford University Press.

Selye, H. (1983). The Stress Concept: Post, present and future. In C. L. Cooper (Ed.), *Stress research: Issues for eighties* (pp. 1–20). John Wiley & Sons.

Shea, B. J., Grimshaw, J. M., Wells, G. A., Boers, M., Andersson, N., Hamel, C., Porter, A. C., Tugwell, P., Moher, D., & Bouter, L. M. (2007). Development of AMSTAR: A measurement tool to assess the methodological quality of systematic reviews. *BMC Medical Research Methodology*, *7*(1), 10. https://doi.org/10.1186/1471-2288-7-10

Sheldon, K. M., Prentice, M., & Halusic, M. (2015). The experiential incompatibility of mindfulness and flow absorption. *Social Psychological & Personality Science*, *6*(3), 276–283. https://doi.org/10.1177/1948550614555028

Shernoff, D. J., Csikszentmihalyi, M., Schneider, B., & Shernoff, E. S. (2014). Student engagement in high school classrooms from the perspective of flow theory. In M. Csikszentmihalyi (Ed.), *Applications of flow in human development and education* (pp. 475–494). Springer. https://doi.org/10.1007/978-94-017-9094-9_24

Shin, N. (2006). Online learner's 'flow' experience: An empirical study. *British Journal of Educational Technology*, *37*(5), 705–720. https://doi.org/10.1111/j.1467-8535.2006.00641.x

Sillaots, M., & Jesmin, T. (2016). Multiple regression analysis: Refinement of the model of flow. In T. Connolly & L. Boyle (Eds.), *Proceedings of The 10th European Conference on Games Based Learning* (pp. 606–616). Academic Conferences International Limited.

Simlesa, M., Guegan, J., Blanchard, E., Tarpin-Bernard, F., & Buisine, S. (2018). The flow engine framework: A cognitive model of optimal human experience. *Europe's Journal of Psychology*, *14*(1), 232–253. https://doi.org/10.5964/ejop.v14i1.1370

Sinnamon, S., Moran, A., & O'Connell, M. (2012). Flow among musicians: Measuring peak experiences of student performers. *Journal of Research in Music Education*, *60*(1), 6–25. https://doi.org/10.1177/0022429411434931

Stoll, O., & Pithan, J. M. (2016). Running and flow: Does controlled running lead to flow-states? Testing the transient hypofrontality theory. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 65–75). Springer. https://doi.org/10.1007/978-3-319-28634-1_5

Swann, C., Crust, L., & Vella, S. A. (2017). New directions in the psychology of optimal performance in sport: Flow and clutch states. *Current Opinion in Psychology*, *16*(1), 48–53. https://doi.org/10.1016/j.copsyc.2017.03.032

Swann, C., Keegan, R. J., Piggott, D., & Crust, L. (2012). A systematic review of the experience, occurrence, and controllability of flow states in elite sport. *Psychology of Sport and Exercise*, *13*(6), 807–819. https://doi.org/10.1016/j.psychsport.2012.05.006

Swann, C., Keegan, R., Crust, L., & Piggott, D. (2016). Psychological states underlying excellent performance in professional golfers: "Letting it happen" vs. "making it happen. *Psychology of Sport and Exercise*, *23*(1), 101–113. https://doi.org/10.1016/j.psychsport.2015.10.008

Swann, C., Piggott, D., Schweickle, M., & Vella, S. A. (2018). A review of scientific progress in flow in sport and exercise: Normal science, crisis, and a progressive shift. *Journal of Applied Sport Psychology*, *30*(3), 249–271. https://doi.org/10.1080/10413200.2018.1443525

Tian, Y., Bian, Y., Han, P., Wang, P., Gao, F., & Chen, Y. (2017). Physiological signal analysis for evaluating flow during playing of computer games of varying difficulty. *Frontiers in Psychology*, *8*(1), 1121. https://doi.org/10.3389/fpsyg.2017.01121

Tozman, T., & Peifer, C. (2016). Experimental paradigms to investigate flow-experience and its psychophysiology: inspired from stress theory and research. In L. Harmat, F. Ø. Andersen, F. Ullén, J. Wright, & G. Sadlo (Eds.), *Flow experience* (pp. 329–350). Springer. https://doi.org/10.1007/978-3-319-28634-1_20

Tozman, T., Magdas, E. S., MacDougall, H. G., & Vollmeyer, R. (2015). Understanding the psychophysiology of flow: A driving simulator experiment to investigate the relationship between flow and heart rate variability. *Computers in Human Behavior*, *52*(1), 408–418. https://doi.org/10.1016/j.chb.2015.06.023

Tozman, T., Zhang, Y. Y., & Vollmeyer, R. (2017). Inverted U-shaped function between flow and cortisol release during chess play. *Journal of Happiness Studies*, *18*(1), 247–268. https://doi.org/10.1007/s10902-016-9726-0

Tricco, A. C., Lillie, E., Zarin, W., O'Brien, K. K., Colquhoun, H., Levac, D., Moher, D., Peters, M. D. J., Horsley, T., Weeks, L., Hempel, S., Akl, E. A., Chang, C., McGowan, J., Stewart, L., Hartling, L., Aldcroft, A., Wilson, M. G., Garritty, C., . . . Straus, S. E. (2018). PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. *Annals of Internal Medicine*, *169*(7), 467–473. https://doi.org/10.7326/M18-0850

Tricco, A. C., Lillie, E., Zarin, W., O'Brien, K., Colquhoun, H., Kastner, M., Levac, D., Ng, C., Sharpe, J. P., Wilson, K., Kenny, M., Warren, R., Wilson, C., Stelfox, H. T., & Straus, S. E. (2016). A scoping review on the conduct and reporting of scoping reviews. *BMC Medical Research Methodology*, *16*(1), 15. https://doi.org/10.1186/s12874-016-0116-4

Ullén, F., de Manzano, Ö., Almeida, R., Magnusson, P. K. E., Pedersen, N. L., Nakamura, J., Csíkszentmihályi, M., & Madison, G. (2012). Proneness for psychological flow in everyday life: Associations with personality and intelligence. *Personality and Individual Differences*, *52*(2), 167–172. https://doi.org/10.1016/j.paid.2011.10.003

Ulrich, M., Keller, J., & Grön, G. (2016a). Neural signatures of experimentally induced flow experiences identified in a typical fMRI block design with BOLD imaging. *Social Cognitive and Affective Neuroscience*, *11*(3), 496–507. https://doi.org/10.1093/scan/nsv133

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Ulrich, M., Keller, J., & Grön, G. (2016b). Dorsal raphe nucleus down-regulates medial prefrontal cortex during experience of flow. *Frontiers in Behavioral Neuroscience*, *10*(1), 169. https://doi.org/10.3389/fnbeh.2016.00169

Ulrich, M., Keller, J., Hoenig, K., Waller, C., & Grön, G. (2014). Neural correlates of experimentally induced flow experiences. *NeuroImage*, *86*(1), 194–202. https://doi.org/10.1016/j.neuroimage.2013.08.019

Valenzuela, R., Codina, N., & Pestana, J. V. (2018). Self-determination theory applied to flow in conservatoire music practice: The roles of perceived autonomy and competence, and autonomous and controlled motivation. *Psychology of Music*, *46*(1), 33–48. https://doi.org/10.1177/0305735617694502

van den Hout, J. J. J., Davis, O. C., & Weggeman, M. C. D. P. (2018). The conceptualization of team flow. *The Journal of Psychology*, *152*(6), 388–423. https://doi.org/10.1080/00223980.2018.1449729

van der Linden, D., Tops, M., & Bakker, A. B. (2021). Go with the flow: A neuroscientific view on being fully engaged. *European Journal of Neuroscience*, *53*(4), 947–963.

Vaughn, D. (2016). *Citius, Altius, Fortius: Finding optimal motivation for flow in sport* [Doctoral dissertation, The Claremont Graduate University]. ProQuest Dissertations.

Vittersø, J., & Søholt, Y. (2011). Life satisfaction goes with pleasure and personal growth goes with interest: Further arguments for separating hedonic and Eudaimonic well-being. *The Journal of Positive Psychology*, *6*(4), 326–335. https://doi.org/10.1080/17439760.2011.584548

Vuorre, M., & Metcalfe, J. (2016). The relation between the sense of agency and the experience of flow. *Consciousness and Cognition*, *43*(1), 133–142. https://doi.org/10.1016/j.concog.2016.06.001

Wang, C. C., & Hsu, M. C. (2013). *Flow experience and challenge-skill balance in e-learning*. PACIS.

Waples, C., & Knight, P. (2017). Flow in the context of industrial and organizational psychology: the case of work motivation In *Flow at work* (pp. 152–168). Routledge.

Weber, R., Huskey, R., & Craighead, B. (2016). Flow experiences and well-being: A media neuroscience perspective. In M. B. Oliver & L. Reinecke (Eds.), *Handbook of media use and well-being: International perspectives on theory and research on positive media effects* (pp. 183–196). Routledge.

Webster, J., Trevino, L. K., & Ryan, L. (1993). The dimensionality and correlates of flow in human-computer interactions. *Computers in Human Behavior*, *9*(4), 411–426. https://doi.org/10.1016/0747-5632(93)90032-N

Westcott-Baker, A., & Weber, R. (2012). The synchronization theory of flow consequences of a limited capacity for attention and consciousness. In J. Gackenbach (Ed.), *Perspectives on cognitive psychology. Video game play and consciousness* (pp. 119–141). Nova Science Publishers.

Wilson, E. E., & Moneta, G. B. (2016). The Flow Metacognitions Questionnaire (FMQ): A two factor model of flow metacognitions. *Personality and Individual Differences*, *90*(1), 225–230. https://doi.org/10.1016/j.paid.2015.11.004

Wolfigiel, B., & Czerw, A. (2017). A new method to measure flow in professional tasks–A FLOW-W questionnaire (FLOW at Work. *Polish Psychological Bulletin*, *48*(2), 220–228. https://doi.org/10.1515/ppb-2017-0025

Yoshida, K., Asakawa, K., Yamauchi, T., Sakuraba, S., Sawamura, D., Murakami, Y., & Sakai, S. (2013). The flow state scale for occupational tasks: Development, reliability, and validity. *Hong Kong Journal of Occupational Therapy*, *23*(2), 54–61. https://doi.org/10.1016/j.hkjot.2013.09.002

Yoshida, K., Sawamura, D., Inagaki, Y., Ogawa, K., Ikoma, K., & Sakai, S. (2014). Brain activity during the flow experience: A functional near-infrared spectroscopy study. *Neuroscience Letters*, *573*(1), 30–34. https://doi.org/10.1016/j.neulet.2014.05.011

Youssef, H. S. A. (2013). *Transient hypofrontality hypothesis and flow experience* [Transient hypofrontality hypothesis and flow experience / von Heba Saad Ali Youssef] (Doctoral dissertation, Halle (Saale), Universitäts- und Landesbibliothek). Sachsen-Anhalt, Germany. https://digital.bibliothek.uni-halle.de/ulbhalhs/urn/urn:nbn:de:gbv:3:4-11308

Zito, M., Cortese, C. G., & Colombo, L. (2018). The Italian Adaptation of the WOrk-reLated Flow Inventory (WOLF) to Sport: The I-WOLFS Scale. *BPA-Applied Psychology Bulletin*, *66*(281), 38–45.

Received July 21, 2020
Revision received July 22, 2021
Accepted July 26, 2021 ■

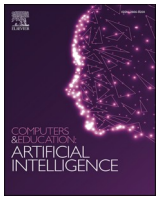
This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Computers and Education: Artificial Intelligence 5 (2023) 100146

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence



An AI-enhanced pattern recognition approach to temporal and spatial analysis of children's embodied interactions

Ceren Ocak [a,*], Theodore J. Kopcha [b], Raunak Dey [c,1]

[a] College of Education, Georgia Southern University, P.O. Box 7995, Statesboro, GA, 30460, USA
[b] College of Education, University of Georgia, 850 College Station Rd., Athens, GA, 30605, USA
[c] University of Georgia, Burlingame, California, 94010, CA, USA

ARTICLE INFO

Keywords:
Computational thinking (CT)
Multimodality
Embodied interactions
Artificial Intelligence (AI)
Machine learning

ABSTRACT

Multimodal video analysis is a complex and time-consuming process for a researcher; it entails capturing, watching, and re-watching video data to identify which segments best inform or address the questions that drive the research. Modern AI applications can alleviate the challenges that arise during the fine-grained analysis of learners' multimodal interactions captured through video. In this study, we present a supervised approach to training a deep neural network to analyze children's computational thinking (CT) captured through multimodal video data. The approach first uses a set of images extracted from video data to train the AI to map them to labels generated using a priori theory. Confusion matrices were used to establish the performance of the AI by comparing AI predictions to human analysis on a validation set of data. The findings suggested that the AI classified several aspects of children's CT in a way that was highly consistent with human analysis, demonstrating how the AI could serve as an additional team member during multimodal analysis. Implications for using AI to ease the challenges of multimodal analysis of video data are discussed.

1. Introduction

Artificial Intelligence (AI) has tremendous potential to support educational research. That potential stems from the way AI can support and extend a researcher's ability to detect important features of learning when analyzing video data (Cukurova, Luckin, & Clark-Wilson, 2019). A number of researchers have successfully reported using AI to identify behavioral and physiological patterns in video and audio data that manifest student learning (e.g., Andrade et al., 2016; Di Mitri et al., 2017; Luckin & Cukurova, 2019). Others have similarly utilized AI-enhanced video analysis techniques to explore the relationship between embodied human behaviors and the underlying learning processes (Abrahamson et al., 2021; Cukurova et al., 2018; Lee-Cultura et al., 2022; Worsley et al., 2021). The general idea is that AI can help researchers identify patterns of behavior that reveal how human cognition and learning take place.

One of the challenges with AI-enhanced video analysis is that the analysis of video data is, at its core, a human endeavor; it requires a human to interpret the meaning behind various modes of interaction (e.g., verbal and non-verbal) and capture both the temporal and spatial qualities of that meaning-making process (Bezemer, 2014; Kendon, 1982). A current and pressing question among qualitative researchers, then, is whether and how AI can play a productive role in supporting the human analysis of qualitative data (Feuston & Brubaker, 2021; Marathe & Toyama, 2018; Sharma et al., 2019). The overall goal is to find a role for AI that does not replace humans in the process but instead serves as a tool for augmenting the analysis and interpretation of video data (Jiang et al., 2021).

Research on AI-enhanced techniques for analyzing young children's learning is becoming increasingly popular because video is minimally invasive and provides a low-cost alternative to collecting high-frequency data (Crescenzi-Lanna, 2020; Reilly et al., 2018). Additionally, combining AI with video analysis allows researchers to study diverse aspects of children's learning in greater detail than human analysis alone while easing the challenges of analyzing complex classroom interactions that evolve over time (Crescenzi-Lanna, 2020). Overall, these perspectives support the idea that AI has the potential to improve the analysis of video data. At the same time, a growing number of scholars have suggested that additional studies are needed to determine the role that AI can play in analyzing video data (Andrade et al., 2016;

* Corresponding author.
E-mail addresses: cocak@georgiasouthern.edu (C. Ocak), tjkopcha@uga.edu (T.J. Kopcha), raunak.dey@gmail.com (R. Dey).
[1] Independent scholar

https://doi.org/10.1016/j.caeai.2023.100146
Received 17 October 2022; Received in revised form 31 May 2023; Accepted 4 June 2023
Available online 10 June 2023
2666-920X/Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Ramakrishnan et al., 2019). Thus, there is a current and pressing need for studies that detail the ways AI can support human analysis when analyzing the meaning behind human behavior in captured through video data (Abrahamson et al., 2021; Feuston & Brubaker, 2021; Marathe & Toyama, 2018; Sullivan & Keith, 2019; Worsley & Blikstein, 2018).

In this study, we explored how a supervised approach to training a deep neural network could augment the human analysis of multimodal video data. The study builds on our previous and ongoing work with AI-enhanced multimodal analysis (Ocak et al., 2021) to demonstrate how a machine-learning algorithm could ease the burdens of human analysis by automating the process of identifying different types of interactions within a set of video data. In our approach, we trained an AI to identify five different modes of interaction using a set of images from our validation set; those modes were established using prior research. We then applied the AI to a 5-min segment of video data to validate the AI and analyze how multiple modes of interaction occurred simultaneously (e.g., talking while gesturing; moving one's body while using objects in the environment). The following research question guided our study:

1. How can we develop a supervised machine learning (ML) algorithm that can support the human analysis of pre-identified modes of interaction evidenced in video data?
2. How can the AI output support humans in transcribing multimodal video data?

Our process for exploring the guiding questions is detailed below. We first review literature from the fields of multimodal video analysis and AI. We then describe our process of extracting images and training the AI to engage in video analysis. In our results, we present the results of our confusion matrices that evaluated the performance of the AI within each of the modes of interaction. Finally, we discuss the results and explore how a researcher might use AI to support multimodal analysis in the future.

2. Related work

Analyzing video data in educational research is both time-consuming and burdensome. It requires a researcher to view video data multiple times; with each viewing, the researcher is working to identify key features that reveal moments or episodes in which the participant is engaging in learning or cognition (Bezemer & Jewitt, 2010). Identifying key features and specific moments of activity is an inductive process in which the researcher first labels key behaviors or interactions, iteratively develops categories or themes from those labels, and then selects key moments from the video that can provide deeper insight into the phenomenon of interest (Bezemer, 2014). This process can take weeks or even months, depending on the size of the research team and the rate at which the team is able to label the video data and identify key moments within that data. AI has the potential to ease the burdens associated with the human analysis of multimodal video data. As noted by Marathe and Toyama (2018), a researcher can generate labels that have been established on a subset of data and use them to classify key behaviors within a larger, unseen data set. This can save a researcher significant amounts of time and resources when analyzing multimodal interactions through video data.

One way educational scholars have used AI to detect patterns of human interaction in video data is with unsupervised methods. Unsupervised AI models are those in which the AI produces an analysis, but the researcher does not know or have input into how the analysis took place (Starčič, 2019). For example, Andrade et al. (2016) used an unsupervised approach to analyze the behavioral frames of video-recorded interviews and identify clusters of learning behavior that were socially meaningful. Detecting the micro-shifts in learners' behavioral frames, the authors were able to provide one approach in which AI supported the analysis of qualitative data that is large, nuanced, and complex. Reilly et al. (2018) similarly used clustering algorithms to analyze learners' movements, facial expressions, and speech through KinectTM that emerged during collaborative group interactions. The authors found that specific movements and gesture patterns positively correlated with the quality of collaboration among small learning groups.

One of the biggest concerns over unsupervised methods, however, is that unsupervised methods typically do not account for the way that theory and a priori classification can be used to drive the AI model (Andrade et al., 2016; Starčič, 2019). In the context of qualitative multimodal analysis, this lack of transparency is problematic. Multimodal analysis has been and will always be an interpretive act on the part of the researcher; it requires decisions about what pieces of data a researcher should focus on to reveal the meaning behind the actions and words captured in the data (Bezemer, 2014). This is because social interactions are highly contextual; to make sense of the meaning of social interactions, one must determine how various modes are coming together in a given moment to reflect human reasoning. This requires attention to both the spatial *and* temporal aspects of the data–how an interaction emerges, changes, and develops over time to support meaning-making in a given context (Qin et al., 2015). The validity of any multimodal analysis, then, is dependent on high levels of transparency throughout the process; a researcher must clearly articulate how decisions are made about the modes that are being analyzed and the way that those modes reveal the construction of meaning (Bezemer, 2014; Mondada, 2018). Overall, these perspectives suggest that AI-enhanced multimodal analysis of video data must go beyond identifying patterns of behavior and include techniques that help researchers understand the meaning behind those patterns (Cukurova, Kent, & Luckin, 2019).

Another way to approach AI-enhanced analysis of video data is through supervised methods. Supervised approaches are those in which a researcher uses a priori theory and human annotation of data to train the AI before using the AI to support pattern detection and prediction (Andrade et al., 2016; Starčič, 2019). Both Luckin and Cukurova's (2019) theory-driven AI design and Sharma et al.'s (2019) hypothesis-driven approach for feature extraction suggest how supervised approaches might work. A researcher first identifies and translates the basic analytic tasks into a form that can be delegated to the AI. This is typically accomplished by having researchers train the AI using data that contains the features that portray certain aspects of learning behavior (Martin & Sherin, 2013). The general idea is that the researcher is the one who first decides what does and does not constitute a mode or feature of interest, increasing the likelihood that the AI is detecting modes and features in a way that is meaningful to the researcher and relevant to the research questions (Luckin & Cukurova, 2019). The researcher then supervises the output of the AI analysis to ensure that the AI is labeling visual data in a way that is consistent with the underlying theory (Spikol et al., 2017). The process often entails a 'back-and-forth' dynamic in which the researcher trains the AI, evaluates the output, and adjusts the training data set until achieving an acceptable level of accuracy from the AI (Sharma et al., 2019). Thus, a supervised approach not only improves the performance of the AI model but also helps establish that the AI is accurately detecting the various modes of interaction that are present in the data. For example, Spikol et al. (2017) collected high-frequency data from multiple modalities, such as face tracking, hand tracking, and interactions with the visual programming interface, Arduino, to study learners' group interactions. By establishing the modes of interest prior to training the AI, the authors were able to focus on analyzing a targeted set of behaviors (e.g., the distance between hands and face) that were likely to be indicators of success when children were working on collaborative projects.

2.1. Research gap

Research that leverages modern *deep perceptual architectures* is understudied and needed both in the field of ML and education research (Ramakrishnan et al., 2019). For example, a recent review of AI

literature by Chen et al. (2020) noted "a lack of studies that both employ AI technologies and engage deeply with educational theories." Crescenzi-Lanna (2020) similarly noted how additional studies are needed to establish the potential for AI to complement the human analysis of children's learning captured in multimodal video data. Other scholars have offered specific areas of AI research that need attention. Ramakrishnan et al. (2019) offered a multi-modal machine-based learning system to determine whether a classroom climate was positive or negative through participants' facial expressions. The major challenge the authors faced was that it was difficult for the AI to label their data, given the complexity of the classroom interactions. Based on their challenges, the authors suggested that future research explore how a combination of techniques like multi-task learning and supervised pre-training might improve the way AI detects interactions of interest from video data. Sharma et al. (2019) used a pre-trained deep neural network to detect facial expressions in order to investigate the impact of emotions on group collaborations. Despite reporting success, the authors recommended future research focus on using AI to detect evidence of learning through additional modalities (e.g., dialogue, gesture) to improve the accuracy of the model. The authors further recommended comparing the results with manually compared data. These studies and others like them (e.g., Crescenzi-Lanna, 2020; Feuston & Brubaker, 2021; Spikol et al., 2017) support the need for research that explores how humans and machines can complement one another in the collection and analysis of multimodal video data.

3. Method

This study employed a *human-in-the-loop* approach to machine learning. As described by Monarch (2021), a human-in-the-loop approach is a supervised approach in which a human first labels a set of data that is used to train the AI. Once the AI has been trained, the model can be tested on a larger, related set of data, and the results can be validated by a team of humans. Monarch explained how this approach is well suited for generating machine learning algorithms that serve a specific purpose for a specific task and data set. The level of human involvement results in higher levels of accuracy in less time while improving confidence in the validity of the AI output (Monarch, 2021; Mosqueira-Rey et al., 2023). With this in mind, the algorithm developed in this study was trained using a pre-labeled training set and then tested on a validation set of data. Because this approach to developing an AI carries a risk of overfitting (Raschka, 2018), the performance of the algorithm was calculated using a confusion matrix that compared AI and human analysis on the validation set of data. Confirming the AI performance through human intervention added a measure of confidence to the accuracy and precision of the model associated with each mode of interaction (Monarch, 2021). Our process of developing and testing the AI is described fully below.

3.1. A priori theory

Previous research by the authors (Kopcha et al., 2021; Kopcha & Ocak, 2019) identified the modes by which computational thinking (CT) was embodied by two 5th-grade students during an educational robotics activity. In those studies, multimodal analysis was used to analyze a 60-min segment of video that was collected from a rural school in the Southeastern US. In the video, two 5th-grade participants engaged in *Danger Zone* (Kopcha et al., 2017), which is a multi-day unit in which 5th-grade learners program and navigate a robot across a 3'x3′ grid of obstacles. The 60-min video contained footage of the two 5th-grade learners using both verbal and non-verbal (i.e., gestures) to engage in the robotics challenge.

One of the primary outcomes of the prior multimodal analysis was that we identified five modes of interaction through which our participant's learning was evidenced: *numerical representations* that represented participants' use of their hands to indicate a numerical value, *imitating the robot* that represented participants' use of their own bodies to simulate the robot's movement, *use of the workbook* that represented participants' use of the workbook to program the robot, *use of the computer* that represented participants' activity at the computer, and *the use of the robot itself* that represented participants' physical interaction with the robot. Those modalities were identified as the result of analyzing different moments of interaction captured throughout the 60-min video to understand the phenomenon of interest. Table 1 in section 3.2. presents a summary of the five modes and their descriptions.

The focus on these modes of interaction was purposeful and grounded in well-established theories of learning. Multimodal perspectives suggest that meaning is constructed through human interaction, including interaction with other people as well as the structures in the environment (Bezemer & Jewitt, 2010). In the context of educational robotics, multimodality offers a new way of analyzing and assessing the development of children's thinking while engaging in computer programming (Kopcha et al., 2021; Zhang & Nouri, 2019). While much is known about specific CT skills, far less is known about how those skills develop within specific instructional activities over time (Tang et al., 2020). For example, gestures are one prominent mode of interaction that they can reveal how children use their bodies to visualize complex mathematical ideas when engaging in problem-solving activity (Kopcha et al., 2021; Bakala et al., 2021; Su & Yang, 2023). The modes of interaction used in this study–numerical representations, imitating the robot, and tool use (e.g., computer, robot, workbook)–represent key modalities through which children's CT can manifest. Studying children's CT through their use of modes addresses a noted gap in the literature that examines how children use their bodies and various tools in the environment to develop CT at a young age (Chen et al., 2017; Su & Yang, 2023).

3.2. Data set

For the current study, we returned to the same video data used in our previous studies and focused on a 5-min segment; all research activity was reviewed and approved by our University's Internal Review Board prior to conducting the research. The 5-min segment of video data was used to generate both a training and validation set of data for the AI. This 5-min segment was chosen because of the way it included visible moments in which the participants interacted with each other through the five modes we had previously identified. Our goal was to train the AI to engage in feature extraction so it could identify those previously established modes summarized in Table 1. To develop the data set, we used an image extraction program to extract 30 frames per second from the 5-min video. This resulted in 9000 images in the data set. To ensure quality and consistency among the images, we resized the extracted images to 128 × 128 resolution and normalized the intensity in all the red, blue, and green channels between 0 and 1.

Table 1
Five modes through which Children's CT occurs.

Pre-identified Modes	Description
Numerical representations	Use of hands to indicate a numerical value (e.g., holding up four fingers for the number 4).
Imitating the robot	Use of body to simulate the robot moving forward (e.g., pulling hands apart), the robot's wheels when turning (e.g., rolling hands), and the robot pivoting when turning.
Use of the workbook	Use of a scaled-down map in a workbook to program the robot while away from the larger 3′ × 3′ grid.
Use of the computer	Use of the computer to program the robot using block coding.
Use of the robot itself	Picking up or interacting with the robot.

Note. The modes and descriptions were drawn from prior research by Kopcha & Ocak (2019) and Kopcha et al. (2021).

3.3. Network design

We designed a minimalist *Convolution Neural Network* (LeCun & Bengio, 1995), shown in Fig. 1, consisting of one block with two convolutional layers with feature maps of size 32. The filter size used was 3 × 3. Each convolution layer used the *ReLu activation function* and was followed by batch normalization. Next, we flattened the output of the last convolution layer and connected it to the dense block of size five, representing five possible classes. The dense layer used the *sigmoid activation function* to bring a multi-level approach to the underlying model and develop robust decision-making boundaries (Kang, 2017). Finally, we used the *Adam Optimizer* with a learning rate of 0.00005; this approach is a common method for optimizing the learning rate of an AI (Kingma & Ba, 2014). The input to the network[2] was a single image, and the output of the network was the probability of the different classes.

3.4. Training

To train the AI, we reviewed the 9000 images in the entire data set and selected 75 of those images; those images represented the behavior of interest in each of the five modalities (15 images per mode). The images were selected such that they showed each mode at various points across the 5-min video segment. Fig. 2 contains a sample image of a representative behavior for each of the five modalities, including looking off-camera at the computer (Fig. 2a), gesturing a numerical representation (Fig. 2b), imitating the robot's movement (Fig. 2c), looking at workbook (Fig. 2d) and interacting with the robot (Fig. 2e). These images were selected because they fit the description of each mode identified through a priori theory (see Table 1).

In order to reduce the complexity of labeling all possible labels for an image of the video frame, the training set was simplified to keep only the most apparent label attributed to it. Specifically, we used a single image per class for validation during training. Using a single image per class reduced the complexities of labeling, which helped prevent the network from overfitting due to the small size of the dataset (Brownlee, 2019).

Our training took several trials. In the first trial, the AI did not label all the interactions as expected. To improve the performance of our AI, we increased the number of images in each category to 25 (125 images total) while also refining our image selection so that our images clearly displayed the behavior of interest. While it is generally recommended that 100 or more images are needed to train an AI, researchers have reported success with as few as 25 images when those images are drawn from a larger related data set (see Schouten, Matek, Jacobs, & et al., 2021). In our second trial, the AI yielded 80% accuracy of a match with our manual classification of the training data set. This was achieved by allowing the AI to run the training data for multiple cycles and identifying where the performance dropped off. In our case, it took eight cycles/epochs to achieve 80% accuracy. Across ten different runs, the network was able to achieve 80% accuracy on 7 out of 10 runs. In the remaining three runs, the network obtained a performance of 70% accuracy. We selected the last run in which we obtained 80% accuracy as the model for further tests. The 20% inaccuracy resulted from some images lacking distinct, unique features altogether or a single image resulting in multiple possible categorization outcomes (e.g., one student looks at the computer while the other imitates the robot).

3.5. Testing and validation

After training, we tested the AI using the entire set of 9000 images extracted from the 5-min segment of the video. During testing, as with all classification pipelines (Sharma et al., 2018), the network outputs the probability of every possible label. A threshold of 0.5 was chosen to represent a true value for the particular label given an input image; this threshold is recommended as the starting point for model validation when using a training set (Raschka, 2018). This would essentially accept the labels for which the network had at least 50% confidence, which is an example of the default cut-off value of a classifier (Harrison, 2021). While some approaches to validating machine learning recommend iteratively improving the threshold using techniques such as a precision-recall graph, we kept the threshold setting to the default value of 0.5. Our decision was based on the fact that the performance at this threshold was acceptable for the purpose of using the AI to classify images that were later validated through human analysis.

To validate the AI results, the AI was configured to automatically annotate the images so that one form of output was a movie consisting of each analyzed image with a display of the interaction type detected in each image. We selected 300 of the 9000 annotated frames, meaning that we selected every 30th frame (i.e., ~1 image per second) so that two of the researchers could determine whether the AI had labeled the images correctly. We chose this rate because humans often view video data on a second-by-second basis when analyzing it for multimodal interactions (Mavers, 2012). This helped establish our process as a human-in-the-loop approach to machine learning in that the contributions of the AI were strongly aligned with the needs and limitations of the human aspects of multimodal analysis (Monarch, 2021). Fig. 3 contains two of the annotated frames produced by the AI. The left image displays an example of an accurate classification by the AI; the girl is looking off-screen to the left at the computer (Com), and the boy is imitating the robot (IR). The right image displays a misclassification in which AI detected an interaction with the computer (Com) that did not occur. The misclassification was largely due to ambiguity regarding whether the boy was looking downward at the workbook (WB) or if he was looking off-screen to the left at the computer (Com).

To answer our first question (i.e., How can we develop an AI), we calculated a confusion matrix for each of the five interactions that the AI was trained to identify. A confusion matrix is a tool for understanding the performance of an AI that provides researchers with information about the AI's ability to distinguish one classification from the other classifications (Tiwari, 2022). It is a 2 × 2 matrix where each cell contains a value indicating whether the AI-predicted values reflect the presence (i.e., true positive or TP) or absence (i.e., true negative or TN) of a feature in the actual data set. The matrix also notes when an AI detects a feature that is not actually present in the data (i.e., false positive or FP) as well as instances when the AI does not detect a feature that is present (i.e., false negative or FN).

The values contained in the cells of the confusion matrix are then used to calculate indicators of performance associated with the AI. As explained by Tiwari (2022), Accuracy reflects the overall amount of correct predictions; it is calculated as the total number of AI/human agreements divided by the total number of evaluations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision reflects how well the model did in making positive predictions compared to all possible positives while Recall reflects how often those positives were accurately predicted:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

An F1 value accounts for whether similar levels of precision *and* recall were achieved:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Overall, the calculations generated from each confusion matrix

[2] The network used in this study is available on GitHub: https://gitfront.io/r/user-8254535/RsU1BQqLSnwj/Paper-gitfront/.

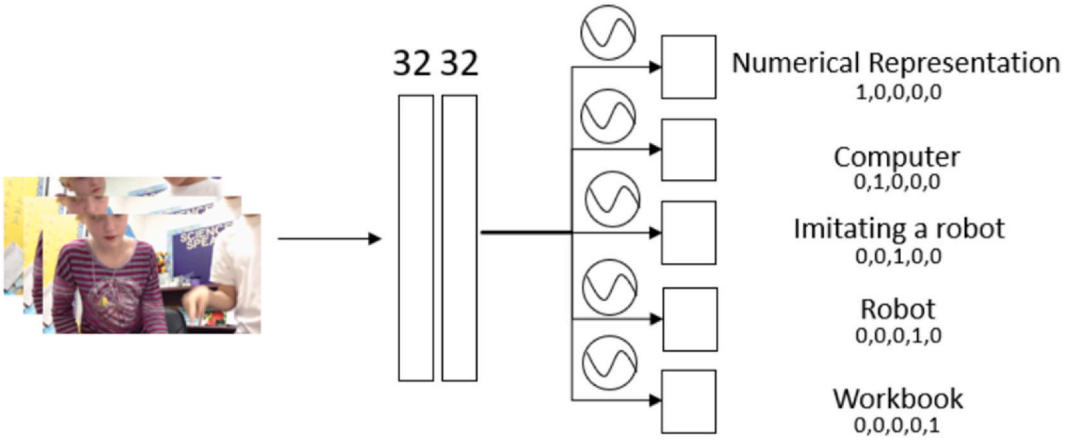


Fig. 1. Network Diagram—The number of feature maps in each of the two convolution layers is 32, which is then flattened and followed by the 5 classification layers.

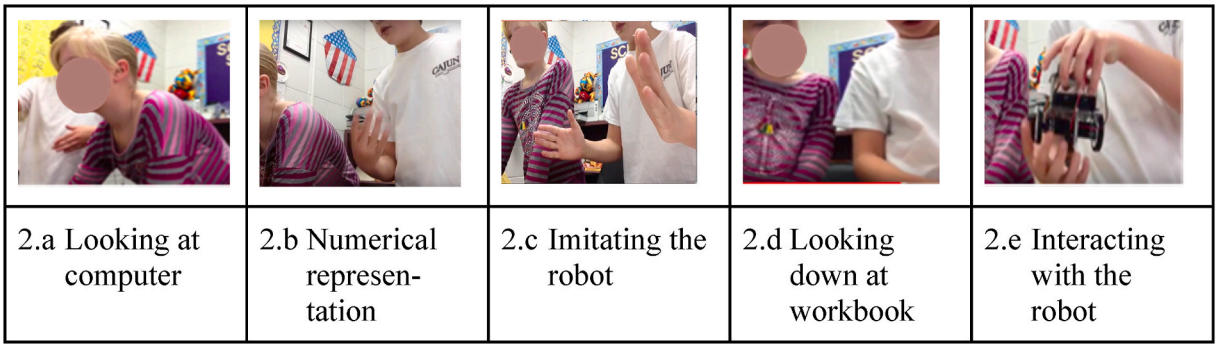


Fig. 2. A sample image for each mode of interaction.

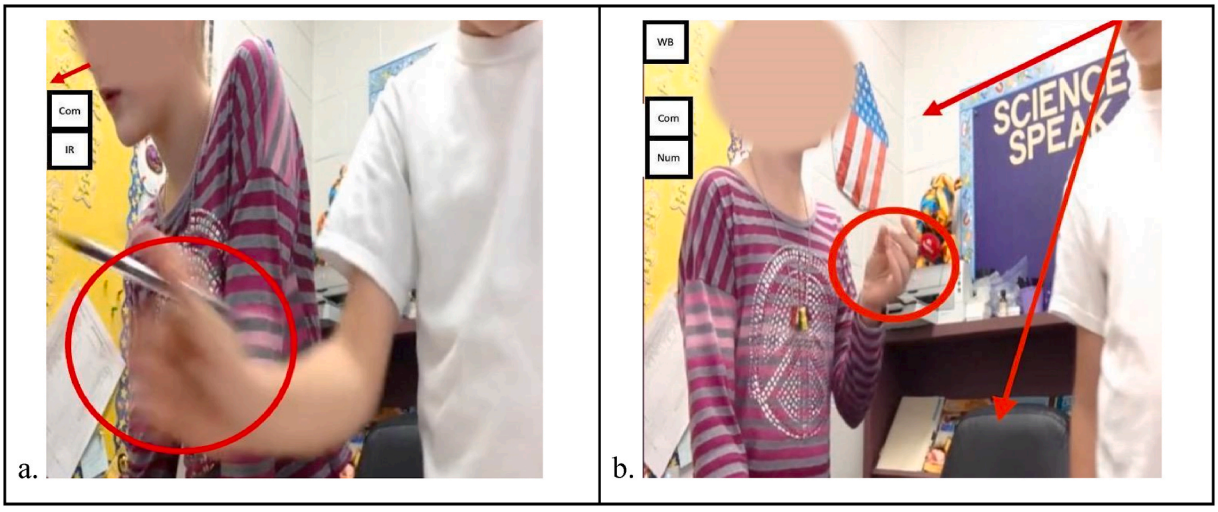


Fig. 3. AI output (annotated images) displaying (a) accurate classification and (b) misclassification of interaction types.

provide a robust indication of the performance of the AI because they account for *both* correct and incorrect predictions made by the AI (Tiwari, 2022).

To generate values for our confusion matrices, two of the researchers independently reviewed each of the 300 annotated frames. The goal was to confirm whether the AI had correctly (i.e., True Positive and True Negative) or incorrectly (i.e., False Positive) identified the interaction displayed in the image or whether additional interactions were missing (i.e., False Negative). To ensure the validity of the human analysis, a percent agreement was calculated to establish whether the researchers agreed on the AI's labeling of images. The agreement between reviewers was 95%, suggesting that the two researchers had a very high level of agreement about the AI's classification of images.

To answer our second research question (i.e., How can the output support human analysis), we used the true positive rate to establish the frequency with which the participants engaged in each mode of

interaction. Because each image reflected 1 s of activity in the video, this frequency analysis provided insight into the modes of interaction that were most dominant. In multimodal analysis, however, the frequency with which a mode occurs may not reflect the importance or meaning of that mode in human reasoning (Bezemer & Jewitt, 2010). To better understand the meaning behind the modes of interaction, we used the AI output to determine where distinct shifts in modes took place (see Fig. 4 a in Section 4). Specifically, we exported the AI classification for each annotated frame into a spreadsheet in which the timing of the image in the sequence (e.g., 0001, 0002) could be paired with the presence of different modes of interaction. This display allowed us to visually inspect the participants' use of modes over time and identify any moments in which those modes shifted; this is important because a shift in modes can reflect a change or development in their thinking and learning (Goffman, 1974). We then used Bezemer's (2014) recommendations for multimodal analysis to analyze the meaning that was reflected through our participants' modes of interaction. Specifically, we generated a multimodal transcript of our participants' interactions at a key moment in which they began using numerical representations as part of their CT. We first paired participant discourse with images of their interactions, including the labels generated by the AI. We then developed larger themes and categories that reflected the meaning behind those labels. This added to the researcher's interpretation of the meaning behind those behaviors in the context of CT.

4. Results

To answer our first research question (i.e., How can we develop a supervised machine learning (ML) algorithm), we generated a confusion matrix to evaluate the performance of the AI on each of the five modes of interaction. Table 2 presents the results from the metrics calculated on each confusion matrix. The table displays the instances in which the AI prediction was consistent with human analysis (i.e., True Positive or TP, True Negative or TN), instances when the AI predicted a result that was not present (i.e., False Positive or FP), and instances when the AI did not detect an interaction that was present (i.e., False Negative or FN).

Overall, the model showed levels of accuracy greater than 80% for four of the five modes of interaction: Computer, Imitating the Robot, Numerical Representations, and Interacting with the Robot. Accuracy for the Workbook mode was 74%, which was the lowest accuracy across the modes. Precision and Recall were highest for Computer (83% and 93%, respectively) and Numerical Representations (77% and 88%, respectively); this was reflected in the F1 values for Computer and Numerical Representation, which were also the highest at 88% and 82% respectively. The lowest levels of Precision and Recall were found for Imitating the Robot (64% and 47%) and for Interacting with the Robot (100% and 20%); likewise, the F1 values for Imitating the Robot and Interacting with the Robot were lowest at 54% and 33% respectively.

The second research question explored how the AI output could be used to support humans in analyzing multimodal video data. To answer this question, we first displayed the modes of interaction identified by the AI in an image-by-image fashion so that shifts in modes could be more easily identified through visual inspection of the AI output. Fig. 4a displays a segment of our spreadsheet in which the image number is noted in the first column, and the corresponding presence of an interaction is noted in the second (e.g., Com = use of computer; Num = Numerical Representation). The True Positive (TP) rate displayed in Table 2 provides a frequency count of the number of times our participants spent engaged in each mode of interaction. Because each image reflects 1 s of our video, the frequencies were converted to time in order to reflect the amount of time spent engaging in each mode. Participants spend the most time interacting with the Computer (3.68 s) and Workbook (1.71 s) and using their hands to make numerical representations (0.83 s).

While frequencies are important, multimodal researchers are often more interested in understanding how modes shift over time and the

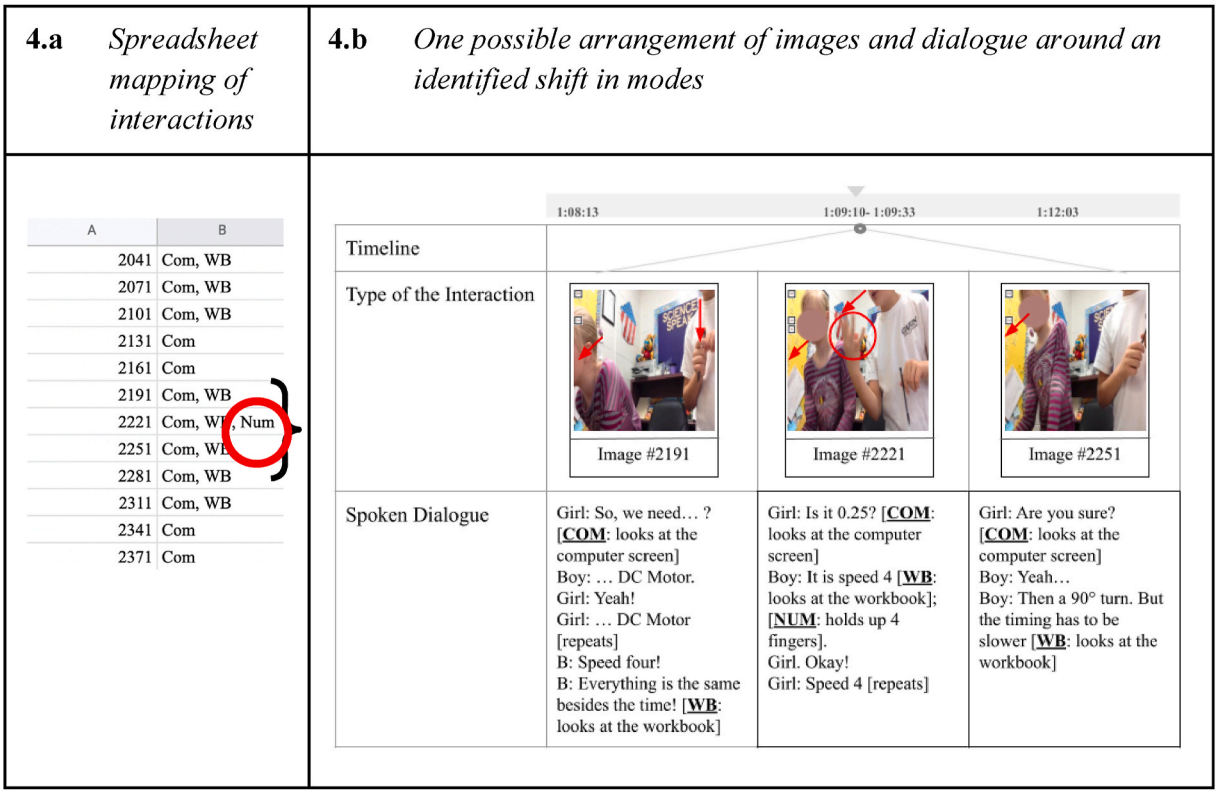


Fig. 4. A Segment of Our Spreadsheet in Which The Timing of The Image in The Sequence Was Paired with The Presence of Interaction. *Note:* The spreadsheet displayed in Fig. 4a contains information for a subset of the 300 images that were analyzed by the AI and validated by the researchers.

Table 2
Summary of confusion matrix results for each of the five modes of interaction classified by the AI.

Modes	TP	FP	FN	TN	Accuracy	Precision	Recall	F1 Value
Computer	221	44	17	18	80%	83%	93%	88%
Numerical Represent.	50	15	7	228	93%	77%	88%	82%
Workbook	103	55	23	119	74%	65%	82%	73%
Imitating Robot	7	4	8	281	96%	64%	47%	54%
Interact w/Robot	1	0	4	295	99%	100%	20%	33%

human reasoning behind those shifts (Bezemer, 2014; Mondada, 2018). To identify such shifts, we performed a visual inspection of the modes of interaction displayed in our spreadsheet. Because interactions with the computer (COM) and workbook (WB) were among the most frequent, the visual inspection focused on instances in which other modes of interaction appeared. One evident pattern was that numerical representations would occur in groups - the participants would use them for a period of time, then stop using them for a period of time. To better understand this pattern, we isolated one specific moment in which numerical representations played a key role in our participants' CT. The boy is using four fingers to represent the time setting. At the same time, the girl is explaining the role that the time setting plays in programming the robot: "Everything is the same besides the time!" This use of numerical representation suggests that the participants are applying the relationship between distance, speed, and time to their thinking as they program the robot. Because the participants had no prior understanding of this relationship in the context of computer programming, the boy's use of four fingers reveals a critical aspect of the way their CT was associated with their bodily movement.

5. Discussion

The primary goal of this study was to contribute to a developing body of literature in which there is a noted need for studies that use a priori theory to guide the development of an AI model that can aid human analysis of learning (see Luckin et al., 2016; Starčič, 2019). In terms of our first research question, we designed and developed a theory-driven framework for training the AI to engage in feature detection. We paired a holdout method of validation with human validation of AI output to increase confidence in the validity of the model and the output (Keevers, 2019). We then established the performance of the AI (i.e., our second research question). The results suggest we were successful in many ways. Our validation of the AI output suggests that the AI was able to identify with high levels of precision and recall our participant's interactions with the computer, workbook, and instances in which they used their hands and bodies to represent numbers. The F1 values for the computer and numerical representations, specifically, suggest that the AI was not only able to detect instances of each from non-instances but did so with high levels of precision. These findings support scholars who have suggested that an AI can be trained to engage in feature detection with high levels of accuracy when the approach is grounded in an existing theory and supervised by humans (Luckin & Cukurova, 2019; Sullivan & Keith, 2019). In terms of research on the use of AI in educational research, this type of study is important because it adds to a small but growing interest in using human intervention to train and validate AI to enhance feature detection in small, well-defined data sets (e.g., Nash, Zheng, & Birbilis, 2022; Schouten, Matek, Jacobs, & et al., 2021). The results of this study suggest that a human-in-the-loop approach can be one way to accomplish that goal.

The F1 values also indicated that the AI struggled to detect when our participants were imitating the robot and when they were interacting with the robot. In both cases, the recall was low, which suggests that the AI was unable to consistently detect positive instances of each mode. This is likely due to the fact that the AI was trained with 25 images. Similar to our results, Schouten, Matek, Jacobs, & et al., 2021 found 25 images to be enough to train their AI to detect differences among two types of cells. However, the authors also noted how the AI was better able to detect specific regions of pixels when more images were used to train their model. It is likely, then, that the AI in our study had not yet learned to identify the specific regions within our images that indicated when our participants were imitating or interacting with the robot. Imitating the robot can take many forms, from holding one's hands out to turning, twisting, and even moving like the robot. Likewise, the robot often appeared as a black square in the images that lacked any distinct color or boundaries to help the AI distinguish it from other features within the image. This may have made it more difficult for the AI to make a positive prediction about those modes, given the manner in which the AI was trained.

Despite the fact that the AI struggled to detect instances of imitating and interacting with the robot, the results are promising overall and provide insight into the ways that AI can be used to address ongoing challenges in multimodal research. One longstanding issue in multimodal analysis of video data deals with establishing validity and inter-rater reliability in one's research. As noted by Bezemer (2014) and others (Mondada, 2018), the strength of any multimodal analysis stems from using techniques that establish consistency among raters and an accurate depiction of the interactions that occurred. Those techniques are often overlooked by researchers because the human analysis of video data is both tedious and time-consuming, which diminishes the potential for multimodal analysis to play a role in better understanding human learning is diminished (Sullivan & Keith, 2019; Worsley, 2018; Worsley et al., 2021).

The current study suggests that AI can help address this issue in multimodal research and play an important role in the analysis of multimodal data. In terms of our second research question, the AI served as an additional research team member whose role was to code the video data using five pre-determined codes. The human members of the team ensured agreement across ratings, cross-checking the AI output to ensure accuracy, precision, and validity. Our team then used those ratings to calculate statistics on the performance of our model. Once the performance of the AI was established, we used the output to engage in a detailed analysis of the multimodal video data in ways that went beyond human analysis alone. In our case, we were able to construct an overall idea of the amount of time the participants spent in each mode of interaction. We were also able to identify a key moment in which the AI detected a shift in our participants' use of numerical representations. Transcribing that moment revealed how a numerical gesture reflected a far deeper bodily understanding of the relationship between distance, rate, and time. In the context of educational research, this result is important because it demonstrates how children embody mathematics as part of their CT during educational robotics (see Kopcha et al., 2021). It also adds to a growing body of literature that suggests how children's gestures have been shown to play an important role in the learning of mathematics. Children's gestures can reveal how they are making sense of abstract mathematical concepts and support them in moving to more formal mathematical thinking in the future (Abrahamson & Trninic, 2015; Lakoff & Núñez, 2000; Nathan, 2021). Our findings provide needed insight into the ways that AI can reduce the labor-intensive aspects of analyzing children's gestures in video data and support scholars in understanding how mathematics is grounded in one's body (Abrahamson et al., 2021).

5.1. Implications

One implication is that AI can be developed through a *human-in-the-loop approach* to help assuage some of the time-consuming and labor-intensive aspects of multimodal video analysis. The AI produced frames that were annotated with labels indicating which modes of interaction were present. Additionally, those frames can be exported into a spreadsheet in which the timing of the image in the sequence (e.g., 0001, 0002) is paired with the presence of different modes of interaction. This has the potential to save a researcher time when analyzing video data. A researcher can easily use that output to identify key moments in the video data, return to the video, and transcribe both the verbal and non-verbal components of the moment to understand how human learning is taking place. This is important because identifying a shift in modes is an important step in multimodal analysis. A change in the frequency of a mode can indicate how learners are using the environment to support their thinking (see Bieda & Nathan, 2009; Vest et al., 2020). Our study suggests that AI-enhanced analytics can help human researchers better understand what the dominant mode of interaction might be, how it changes over time, and how the frequency of that mode compares to others (see also Andrade et el., 2017; Sullivan & Keith, 2019; Qin et al., 2015).

5.2. Limitations

A *human-in-the-loop approach* is meant to develop a machine-learning model with high levels of human monitoring and intervention to ensure that the model not only has strong performance but serves the needs of the humans who develop it. This typically requires the researcher to make decisions about the development of the model that balance the needs of human with the processes associated with machine learning (Monarch, 2021). For example, our method of validation was more aligned with holdout methods than a k-fold cross-validation technique. We also chose not to analyze the performance of the AI at different thresholds. These decisions may limit how the model can be applied more broadly because they may contribute to overfitting in our model (Raschka, 2018). It is important to note, however, that our goal was not to create an algorithm that could classify images universally. Our goal was to develop a machine learning model that could serve a specific purpose for a limited data set and support humans in multimodal analysis. Our process resulted in a machine learning model that served a variety of practical needs (i.e., lower in cost; greater human involvement) while achieving a level of performance that served the needs of the research team.

5.3. Future research

Studying the process of human learning through behaviors is unavoidably time-consuming. Consider that the average 5-min video segment contains ~1000 possible frames/images for analysis. A researcher employing multimodal analysis must first repeatedly review the video to identify the images and dialogue that most clearly communicate the behaviors associated with the phenomenon of interest (Bezemer, 2014; Mondada, 2018). With an AI trained on a robust set of data, the time needed to perform this task can be reduced. Thus, it is not difficult to imagine that the process detailed in this study could be employed across a larger portion of time or video recordings captured across multiple groups. Doing so has the potential to yield thousands of data points upon which to engage in learning analytics and build more sophisticated statistical models that help predict how human behavior relates to learning and cognition. Future research would therefore benefit from establishing how AI-enhanced multimodal analysis might move from smaller to larger segments of video analysis and increase the amounts of data extracted from a video data set. This contributes to the current call for methods that allow researchers to study the social aspects of learning through large sets of multimodal data (Andrade et al., 2017).

This study also offers some implications for other scholars who are interested in employing AI as a tool for enhancing their own multimodal analysis of video data. In the training of our AI, we sought images that displayed a single mode at one time. While this did work in our study, it may have contributed to errors in the AI's ability to detect our participants imitating and interacting with the robot. Future research could explore how training sets of different sizes might affect the AI (similar to Schouten, Matek, Jacobs, & et al., 2021) and how techniques such as heatmap generation could improve the AI's sensitivity to multiple modes being present at the same time (see Bach et al., 2015).

6. Conclusion

Many papers note the potential for supervised approaches to AI to support educational research; few, however, articulate the process in which those approaches can be achieved (Andrade-Lotero & Danish, 2014; Spikol et al., 2018). This paper provides a detailed summary of one such process in which we used a supervised, human-in-the-loop approach to explore the role of AI in multimodal analysis. While this is a relatively new frontier in educational research, the results of this study suggest that an AI algorithm can be built on learning theory to detect human behavior in video data and help a researcher better understand learning through the temporal and spatial organization of that behavior. We hope that this paper provides insight for others who aim to explore how AI can meaningfully inform the human interpretation of multimodal data.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethics statement

There were no ethical issues related to the selection and treatment of subjects associated with this paper

Data availability statement

The data can be accessed by contacting the first author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Abrahamson, D., & Trninic, D. (2015). Bringing forth mathematical concepts: Signifying sensorimotor enactment in fields of promoted action. *ZDM Mathematics Education, 47* (2), 1–12. https://doi.org/10.1007/s11858-014-0620-0

Abrahamson, D., Worsley, M., Pardos, Z. A., & Ou, L. (2021). Learning analytics of embodied design: Enhancing synergy. *International Journal of Child-Computer Interaction, 32*(2022), 1–5. https://doi.org/10.1016/j.ijcci.2021.100409

Andrade-Lotero, A., & Danish, J. A. (2014). Advancing epistemological frame analysis to refine our understanding of inquiry frames in early elementary interviews. In J. L. Polman, E. A. Kyza, D. K. O'Neill, I. Tabak, W. R. Penuel, A. S. Jurow, K. O'Connor, T. Lee, & L. D'Amico (Eds.), *Learning and becoming in practice: The international conference of the learning sciences (ICLS) 2014, 2.* Boulder, CO: International Society of the Learning Sciences. https://doi.org/10.22318/icls2014.1637.

Andrade, A., Danish, J. A., & Maltese, A. V. (2017). A measurement model of gestures in an embodied learning environment: Accounting for temporal dependencies. *Journal of learning Analytics, 4*(3), 18–46. https://doi.org/10.18608/jla.2017.43.3

Andrade, A., Delandshere, G., & Danish, J. A. (2016). Using multimodal learning analytics to model student behaviour: A systematic analysis of behavioural framing. *Journal of Learning Analytics, 3*(2), 282–306. https://doi.org/10.18608/jla.2016.32.14

Bach, S., Binder, A., Montavon, G., Klauschen, F., Müller, K. R., & Samek, W. (2015). On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PLoS One, 10*(7), 1–30. https://doi.org/10.1371/journal.pone.0130140

Bakala, E., Gerosa, A., Hourcade, J. P., & Tejera, G. (2021). Preschool children, robots, and computational thinking: A systematic review. *International Journal of Child-Computer Interaction, 29*, 1–20. https://doi.org/10.1016/j.ijcci.2021.100337

Bezemer, J. (2014). Multimodal transcription: A case study. *Interactions, images and texts: A reader in multimodality, 11*, 155.

Bezemer, J., & Jewitt, C. (2010). Multimodal analysis: Key issues. In L. Litosseliti (Ed.), *Research methods in linguistics* (pp. 180–197). London, UK: Continuum.

Bieda, K. N., & Nathan, M. J. (2009). Representational disfluency in algebra: Evidence from student gestures and speech. *ZDM, 41*(5), 637–650. https://doi.org/10.1007/s11858-009-0198-0

Brownlee, J. (2019). *How to avoid overfitting in deep learning neural networks. Machine Learning Mastery*. Retrieved June 4, 2023, from https://machinelearningmastery.com/introduction-to-regularization-to-reduce-overfitting-and-improve-generalization-error/.

Chen, G., Shen, J., Barth-Cohen, L., Jiang, S., Huang, X., & Eltoukhy, M. (2017). Assessing elementary students' computational thinking in everyday reasoning and robotics programming. *Computers & Education, 109*, 162–175. https://doi.org/10.1016/j.compedu.2017.03.001

Chen, X., Xie, H., Zou, D., & Hwang, G. J. (2020). Application and theory gaps during the rise of artificial intelligence in education. *Computers in Education: Artificial Intelligence, 1*, Article 100002. https://doi.org/10.1016/j.caeai.2020.100002

Crescenzi-Lanna, L. (2020). Multimodal learning analytics research with young children: A systematic review. *British Journal of Educational Technology, 51*(5), 1485–1504. https://doi.org/10.1111/bjet.12959

Cukurova, M., Kent, C., & Luckin, R. (2019). Artificial intelligence and multimodal data in the service of human decision-making: A case study in debate tutoring. *British Journal of Educational Technology, 50*(6), 3032–3046. https://doi.org/10.1111/bjet.12829

Cukurova, M., Luckin, R., & Clark-Wilson, A. (2019). Creating the golden triangle of evidence-informed education technology with EDUCATE. *British Journal of Educational Technology, 50*(2), 490–504. https://doi.org/10.1111/bjet.12727

Cukurova, M., Luckin, R., Millán, E., & Mavrikis, M. (2018). The NISPI framework: Analysing collaborative problem-solving from students' physical interactions. *Computers & Education, 116*, 93–109. https://doi.org/10.1016/j.compedu.2017.08.007

Di Mitri, D., Scheffel, M., Drachsler, H., Börner, D., Ternier, S., & Specht, M. (2017). Learning pulse: A machine learning approach for predicting performance in self-regulated learning using multimodal data. In *Proceedings of the seventh international learning analytics & knowledge conference* (pp. 188–197). https://doi.org/10.1145/3027385.3027447

Feuston, J. L., & Brubaker, J. R. (2021). Putting tools in their place: The role of time and perspective in human-AI collaboration for qualitative analysis. In *Proceedings of the ACM on human-computer interaction, 5(CSCW2)* (pp. 1–25). https://doi.org/10.1145/3479856

Goffman, E. (1974). *Frame analysis: An essay on the organization of experience*. Harvard University Press.

Harrison, G. (2021). Calculating and setting thresholds to optimise logistic regression performance. *Data Science*. Retrieved April 24, 2023, from https://towardsdatascience.com/calculating-and-setting-thresholds-to-optimise-logistic-regression-performance-c77e6d112d7e.

Jiang, J. A., Wade, K., Fiesler, C., & Brubaker, J. R. (2021). Supporting serendipity: Opportunities and challenges for Human-AI Collaboration in qualitative analysis. *Proceedings of the ACM on Human-Computer Interaction, 5*(CSCW1), 1–23. https://doi.org/10.1145/3449168

Kang, N. (2017). *Multi-layer neural Networks with sigmoid function— deep Learning for rookies (2). Towards data science*. Retrieved June 4, 2023, from https://towardsdatascience.com/multi-layer-neural-networks-with-sigmoid-function-deep-learning-for-rookies-2-bf464f09eb7f.

Keevers, T. L. (2019). Cross-validation is insufficient for model validation. In *Joint and operations analysis division*. Victoria, Australia: Defence Science and Technology Group.

Kendon, A. (1982). *Conducting interaction: Patterns of behavior in focused encounters*. Cambridge, England: Cambridge University Press.

Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. In *Proceedings of the 3rd international conference for learning representations*. https://doi.org/10.48550/arXiv.1412.6980

Kopcha, T. J., McGregor, J., Shin, S., Qian, Y., Choi, J., Hill, R., … Choi, I. (2017). Developing an integrative STEM curriculum for robotics education through Educational Design Research. *Journal of Formative Design in Learning, 1*(1), 31–44. https://doi.org/10.1007/s41686-017-0005-1

Kopcha, T., & Ocak, C. (2019). Embodiment of computational thinking during collaborative robotics activity. In K. Lund, G. P. Niccolai, E. Lavoué, C. Hmelo-Silver, G. Gweon, & M. Baker (Eds.), *13th international conference on computer supported collaborative learning (CSCL) 2019: 1. A wide lens: Combining embodied, enactive, extended, and embedded learning in collaborative settings* (pp. 464–471). Lyon, France: International Society of the Learning Sciences. https://repository.isls.org//handle/1/1604.

Kopcha, T. J., Ocak, C., & Qian, Y. (2021). Analyzing children's computational thinking through embodied interaction with technology: A multimodal perspective. *Educational Technology Research & Development, 69*, 1987–2012. https://doi.org/10.1007/s11423-020-09832-y

Lakoff, G., & Núñez, R. E. (2000). *Where mathematics comes from: How the embodied mind brings mathematics into being*. New York, NY: Basic Books.

LeCun, Y., & Bengio, Y. (1995). Convolutional networks for images, speech, and time series. In *The handbook of brain theory and neural networks, 3361* p. 1995), 10.

Lee-Cultura, S., Sharma, K., & Giannakos, M. (2022). Children's play and problem-solving in motion-based learning technologies using a multi-modal mixed methods approach. *International Journal of Child-Computer Interaction, 31*, Article 100355. https://doi.org/10.1016/j.ijcci.2021.100355

Luckin, R., & Cukurova, M. (2019). Designing educational technologies in the age of AI: A learning sciences-driven approach. *British Journal of Educational Technology, 50*(6), 2824–2838. https://doi.org/10.1111/bjet.12861

Luckin, R., Holmes, W., Griffiths, M., & Forcier, L. B. (2016). *Intelligence Unleashed. London: An argument for AI in Education*.

Marathe, M., & Toyama, K. (2018). Semi-Automated coding for qualitative research, A user-centered inquiry and initial prototypes. In *CHI '18, proceedings of the 2018 CHI conference on human factors in computing systems*. https://doi.org/10.1145/3173574.3173922

Martin, T., & Sherin, B. (2013). Learning analytics and computational techniques for detecting and evaluating patterns in learning: An introduction to the special issue. *The Journal of the Learning Sciences, 22*(4), 511–520. https://doi.org/10.1080/10508406.2013.840466

Mavers, D. (2012). Transcribing video. National Centre for Research Methods. Retrieved June 14, 2023, from https://eprints.ncrm.ac.uk/id/eprint/2877/4/NCRM_working_paper0512.pdf.

Monarch, R. M. (2021). *Human-in-the-Loop machine learning: Active learning and annotation for human-centered AI*. Simon and Schuster.

Mondada, L. (2018). Multiple temporalities of language and body in interaction: Challenges for transcribing multimodality. *Research on Language and Social Interaction, 51*(1), 85–106. https://doi.org/10.1080/08351813.2018.1413878

Mosqueira-Rey, E., Hernández-Pereira, E., Alonso-Ríos, D., Bobes-Bascarán, J., & Fernández-Leal, Á. (2023). Human-in-the-loop machine learning: A state of the art. *Artificial Intelligence Review, 56*(4), 3005–3054.

Nash, W., Zheng, L., & Birbilis, N. (2022). Deep learning corrosion detection with confidence. *Npj Materials Degradation, 6*(1), 26. https://doi.org/10.1038/s41529-022-00232-6

Nathan, M. J. (2021). *Foundations of embodied learning: A paradigm for education*. Routledge.

Ocak, C., Kopcha, T. J., & Dey, R. (2021). An AI-enhanced pattern recognition approach to analyze children's embodied interactions. In M. M. T. Rodrigo, et al. (Eds.), *Proceedings of the 29th international conference on computers in education. Asia-pacific society for computers in education* (pp. 273–278).

Qin, J., Zhou, Y., Lu, H., & Ya, H. (2015). Teaching video analytics based on student spatial and temporal behavior mining. In *'15. Proceedings of the 5th ACM on international conference on multimedia retrieval - ICMR* (pp. 635–642). https://doi.org/10.1145/2671188.2749357

Ramakrishnan, A., Ottmar, E., LoCasale-Crouch, J., & Whitehill, J. (2019). Toward automated classroom observation: Predicting positive and negative climate. In *2019 14th IEEE international conference on automatic face & gesture recognition (FG 2019)* (pp. 1–8). IEEE.

Raschka, S. (2018). *Model evaluation, model selection, and algorithm selection in machine learning*. https://doi.org/10.48550/arXiv.1811.12808

Reilly, J. M., Ravenell, M., & Schneider, B. (2018). Exploring collaboration using motion sensors and multi-modal learning analytics. In K. E. Boyer, & M. Yudelson (Eds.), *International educational data mining society, 11th international conference on educational data mining (EDM) 2018* (pp. 333–339). Buffalo, NY: International Conference on Educational Data Mining.

Schouten, J. P. E., Matek, C., Jacobs, L. F. P., et al. (2021). Tens of images can suffice to train neural networks for malignant leukocyte detection. *Scientific Reports, 11*, 7995. https://doi.org/10.1038/s41598-021-86995-5

Sharma, N., Jain, V., & Mishra, A. (2018). An analysis of convolutional neural networks for image classification. *Procedia Computer Science, 132*, 377–384. https://doi.org/10.1016/j.procs.2018.05.198

Sharma, K., Papamitsiou, Z., & Giannakos, M. (2019). Building pipelines for educational data using AI and multimodal analytics: A "grey-box" approach. *British Journal of Educational Technology, 50*(6), 3004–3031. https://doi.org/10.1111/bjet.12854

Spikol, D., Ruffaldi, E., & Cukurova, M. (2017). Using multimodal learning analytics to identify aspects of collaboration in project-based learning. In B. K. Smith, M. Borge, E. Mercier, & K. Y. Lim (Eds.), *(2017). Making a difference: Prioritizing equity and access in CSCL, 12th international conference on computer supported collaborative learning (CSCL) 2017, 1*. Philadelphia, PA: International Society of the Learning Sciences. https://repository.isls.org//handle/1/240.

Spikol, D., Ruffaldi, E., Dabisias, G., & Cukurova, M. (2018). Supervised machine learning in multimodal learning analytics for estimating success in project-based learning. *Journal of Computer Assisted Learning, 34*(4), 366–377. https://doi.org/10.1111/jcal.12263

Starčič, A. I. (2019). Human learning and learning analytics in the age of artificial intelligence. *British Journal of Educational Technology, 50*(6), 2974–2976. https://doi.org/10.1111/bjet.12879

Sullivan, F. R., & Keith, P. K. (2019). Exploring the potential of natural language processing to support microgenetic analysis of collaborative learning discussions. *British Journal of Educational Technology, 50*(6), 3047–3063. https://doi.org/10.1111/bjet.12875

Su, J., & Yang, W. (2023). A systematic review of integrating computational thinking in early childhood education. *Computers and Education Open.* , Article 100122. https://doi.org/10.1016/j.caeo.2023.100122

Tang, X., Yin, Y., Lin, Q., Hadad, R., & Zhai, X. (2020). Assessing computational thinking: A systematic review of empirical studies. *Computers & Education, 148*, Article 103798. https://doi.org/10.1016/j.compedu.2019.103798

Tiwari, A. (2022). Supervised learning: From theory to applications. In R. Pandey, S. K. Khatri, N. K. Singh, & P. Verma (Eds.), *Artificial intelligence and machine learning for EDGE computing*. Academic Press. https://doi.org/10.1016/B978-0-12-824054-0.00026-5.

Vest, N. A., Fyfe, E. R., Nathan, M. J., & Alibali, M. W. (2020). Learning from an avatar video instructor: The role of gesture mimicry. *Gesture, 19*(1), 128–155. https://doi.org/10.1075/gest.18019.ves

Worsley, M. (2018). Multimodal learning analytics' past, present, and potential futures. In *Companion proceedings of the eighth international conference on learning analytics and knowledge (LAK 2018), 5–9 march 2018, sydney, Australia*. SoLAR. http://bit.ly/lak18-companion-proceedings.

Worsley, M., & Blikstein, P. (2018). A multimodal analysis of making. *International Journal of Artificial Intelligence in Education, 28*(3), 385–419. https://doi.org/10.1007/s40593-017-0160-1

Worsley, M., Martinez-Maldonado, R., & D'Angelo, C. (2021). A new era in multimodal learning analytics: Twelve core commitments to ground and grow MMLA. *Journal of Learning Analytics, 8*(3), 10–27. https://doi.org/10.18608/jla.2021.7361

Zhang, L., & Nouri, J. (2019). A systematic review of learning computational thinking through Scratch in K-9. *Computers & Education, 141*, Article 103607. https://doi.org/10.1016/j.compedu.2019.103607

Glossary

CT: Computational thinking
AI: Artificial intelligence
Com: Use of computer
Num: Numerical Representation
WB: Workbook
IR: Imitating the robot
Rob: Robot

Ceren Ocak, Ph.D., is an Assistant Professor of Instructional Technology in the Department of Leadership, Technology, And Human Development at Georgia Southern University. Her research focuses on computer science education in K-12 and embodied learning.

Theodore J. Kopcha, Ph.D., is an Associate Professor of Learning, Design, and Technology in the Department of Career and Information Studies at the University of Georgia. His research focuses on embodied perspectives in K-12 learning in STEM and beyond.

Raunak Dey, Ph.D., received his Ph.D. from the Department of Computer Science at the University of Georgia. He is researching AI and Genomics. His area of interest is primarily Artificial Intelligence, Heuristics, and Genomics.

AI-Infused Collaborative Inquiry in Upper Elementary School: A Game-Based Learning Approach

Seung Lee[1], Bradford Mott[1], Anne Ottenbreit-Leftwich[2], Adam Scribner[2], Sandra Taylor[1], Kyungjin Park[1], Jonathan Rowe[1], Krista Glazewski[2], Cindy E. Hmelo-Silver[2], and James Lester[1]

[1]Department of Computer Science, North Carolina State University

[2]School of Education, Indiana University

{sylee, bwmott, smtayl23, kpark8, jprowe, lester}@ncsu.edu, {aleftwic, jascrib, glaze, chmelosi}@indiana.edu

Abstract

Artificial intelligence has emerged as a technology that is profoundly reshaping society and enabling rapid improvements in science, engineering, and mathematics, as well as information technology itself. This has generated increased demand for fostering an AI-literate populace as well as a growing recognition of the importance of promoting K-12 students' awareness and interest in AI. Although efforts are beginning to incorporate AI learning within K-12 education, there is little research exploring how to introduce students to AI and how to support teachers to integrate AI learning experiences in their classrooms. This is especially true at the elementary school level. A particularly promising approach for providing effective and engaging AI learning experiences for elementary students is game-based learning. In this paper, we explore how to introduce AI-infused collaborative inquiry learning into upper elementary school (student ages 8 to 11) using game-based learning. To ground the work in the realities of elementary school classrooms, we present insights from interviews with elementary school teachers to understand how best to support them in integrating AI into their classrooms. We then present the design of PRIMARYAI, a game-based learning environment that supports rich problem-based learning activities within upper elementary classrooms centered on AI applied toward solving life-science problems. Finally, we discuss some of the challenges we face in bringing AI-infused collaborative inquiry learning to upper elementary students.

Introduction

Artificial intelligence (AI) has emerged as a foundational technology that is fundamentally reshaping the workplace of the future (McKinsey Global Institute, 2017; West, 2018). With rapid developments in a wide array of capabilities ranging from automated reasoning, computer vision, robotics, and machine learning, AI is becoming a critical tool for enabling improvements in science, technology, engineering, and mathematics (Manyika 2017). This has generated a vital need to develop a deep understanding of how to best introduce K-12 students to AI (Touretzky et al. 2019), and how to best support K-12 teachers in integrating AI learning experiences into their classrooms.

Recognizing the importance of fostering K-12 students' awareness and interest in AI, emerging efforts are underway to incorporate AI within K-12 education (Ali et al. 2019; Touretzky et al. 2019), as well as to establish guidelines for K-12 AI education (AAAI 2018). However, there has been limited research examining AI learning and teaching at the elementary level. Because early learning experiences can significantly affect future academic and career trajectories, it is essential to create engaging AI learning experiences for elementary school students.

Game-based learning is a powerful tool for creating effective and engaging learning experiences through immersive problem solving (Wouters 2013; McLaren et al. 2017). Game-based learning environments, with the potential to motivate students with inviting settings, expressive characters, and compelling virtual worlds, are being used to create engaging, effective computer science (CS) and AI learning experiences in K-12 classrooms (Buffum 2016; Wang and Johnson 2019), including fostering computational thinking in the elementary grades (Rowe et al. 2017). Well-designed game-based learning environments enable students to develop problem-solving skills, communicate and collaborate with other students, and actively participate in rich virtual contexts (Gee 2003; Salen 2007; Shaffer 2006). Similarly, inquiry-based learning approaches for CS have yielded encouraging results that demonstrate improved student engagement (Hoffman et al. 2019), student achievement

Copyright © 2021, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

(Hoffman et al. 2019; Chen and Yang 2018), and student attitudes towards CS (Hoffman et al. 2019; Pollock et al. 2017). Inquiry-based learning approaches for CS are also being explored at the elementary level. Given the potential to support student engagement and achievement, leveraging inquiry-based strategies, such as problem-based learning, along with game-based learning offer significant potential for promoting AI education in elementary school classrooms.

Recognizing the need to provide all students with AI learning opportunities, we are investigating PRIMARYAI, a game-based learning environment featuring AI-infused inquiry learning that will enable students to gain experience with AI applied toward solving life-science problems. In this paper, we present initial work on the design of PRIMARYAI, which is being used to explore two key research questions focused on introducing AI education into upper elementary classrooms.

1. How can we create engaging learning experiences that integrate artificial intelligence and life science for upper elementary students with game-based learning?

2. How can we support upper elementary school teachers with integrating artificial intelligence and life science into their classrooms?

This paper is structured as follows. First, we provide background on K-12 AI education as well as research we are drawing on to create the PRIMARYAI learning environment: game-based learning and problem-based learning. Next, we introduce the design of our game-based learning environment for AI-infused collaborative inquiry learning in elementary school. We present interviews with elementary school teachers to better understand how to support them in introducing AI into their classrooms. We also present our approach to combining techniques from game-based learning and problem-based learning to create AI-infused collaborative learning experiences for students. Finally, we present findings, concluding remarks, and directions for future work.

Related Work

Integrating game-based learning and problem-based learning offers significant promise for creating engaging and effective learning experiences. However, limited work has investigated AI-infused learning experiences for upper elementary school students. In this section, we survey prior work o[1]n AI education for K-12, game-based learning, and problem-based learning.

K-12 AI Education

Emerging efforts are beginning to explore how to incorporate AI more intentionally at the K-12 levels. For example, the Association for the Advancement of Artificial Intelligence (AAAI) and the Computer Science Teachers Association (CSTA) formed a working group in 2018 to begin identifying the important constructs of AI for K-12 students to learn. Touretzky et al. (2019) consider how AI might be introduced to K-12 students, and Webb and colleagues (Webb et al. 2017) suggest that learners need to understand to truly evaluate AI by "understanding what the nature of a computer program is (does it 'think'?), what kind of knowledge base (data) it could collect and store, its access to our digital infrastructure, the way that humans would interact with it, and the philosophical concerns about human rights and what intelligence is" (p. 461). Similar efforts for K-12 AI education are underway around the globe (Touretzky et al. 2019).

Efforts are also underway to develop tools and curricula to support AI education within K-12 classrooms. For example, Google's Teachable Machine[1] is a web-based tool that enables students to create machine learning models to recognize images, poses, and sounds. Cognimates allows students to engage in programming activities that include training their own AI models to perform speech recognition, sentiment analysis, and image classification (Druga 2018). Machine Learning for Kids helps students build programs using Scratch that leverage AI models powered by IBM Watson (Lane 2019). In addition, companies such as ReadyAI[2] are creating pre-configured toolkits, such as AI-in-a-Box, that include both hardware and software to support AI education for K-12 students. Work is also underway to develop modules to support students in learning about ethical issues in AI (Yeh et al. 2019), exploring machine learning concepts in elementary school (Lin et al. 2020), and teaching machine learning and natural language processing to high school students (Norouzi et al. 2020).

Game-Based Learning

The educational potential of games has garnered significant attention that has been directed toward appropriating the best features of games and transferring them to educational settings (Gibson et al. 2007). For example, several commercial games that focus on entertainment (e.g., Civilization, SimCity, and Spore) have loosely incorporated educational themes into gameplay, includ[2]ing history, urban planning, and evolution. Game-based learning environments create engaging, situated learning experiences for students that prioritize academic subjects, pedagogy, and problem solving. Efforts to systematically review the game-based learning literature have found that digital games are often more effective than traditional instructional methods in terms of

[1]https://experiments.withgoogle.com/ai/teachable-machine/
[2]https://www.readyai.org/

enhancing learning outcomes (Wouters et al. 2013; Clark et al. 2016). By encouraging students to actively participate in situated problem-solving activities, game-based learning environments promote deep, meaningful student learning.

Researchers have investigated game-based learning environments for a variety of educational domains, including anti-bullying (Aylett et al. 2005), science learning (Ketelhut et al. 2007; Rowe et al. 2011), interactive health-education (Marsella 2003), graphics design (Cutumisu 2018), mathematics (Mogessie et al. 2020), and computational thinking (Min et al. 2017; Zhao and Shute 2019). Efforts are now underway to leverage game-based learning to connect AI concepts, such as search, reasoning, and machine learning to high school math (Wang and Johnson 2019). A broad range of computational frameworks have been investigated that can be used to enhance student interactions during game-based learning, ranging from predictive student models of engagement from gameplay data (Sawyer et al. 2018) to better understanding how players engage and disengage while playing games (Hadiji et al. 2014; Bertens et al. 2017), as well as student-adaptive experience managers for personalized gameplay (Kantharaju et al. 2018; Yu and Riedl 2015) and assessing student learning without interfering with gameplay (Min et al. 2017; Shute and Sun 2019). Through these games and frameworks effective and engaging learning experiences are being created for a broad range of subject matters and target student populations.

Problem-Based Learning

Problem-based learning is an effective student-centered approach to collaborative learning that supports small groups of students learning through the investigation of complex, ill-defined problems (Barrows 1986; Hmelo-Silver et al. 2004). Problem-based learning holds that students need to be active learners, and learning occurs as knowledge is shared and negotiated among group members (Hmelo-Silver et al. 2018). Teachers serve as facilitators during problem-based learning to support student learning, and their interventions diminish as students progressively take on responsibility for their own learning processes (Dolmans and Schmidt 2006; Hoffman et al. 2019). Problem-based learning was originally formalized as an instructional model for medical schools in the 1960s; however, recent years have seen increasing interest in applying it to K-12 education (McConnell et al. 2016). The significant opportunity provided by the rich, situated problem contexts of game-based learning to support problem-based learning is starting to be investigated (Mott et al. 2019), and using game-based learning to introduce AI education in elementary classrooms is an unexplored area.

AI-Infused Inquiry Learning

PRIMARYAI is a game-based learning environment for upper elementary students that provides AI learning experiences through AI-infused life-science problem solving. The game is being developed using the Unity® game engine, a cross-platform game development environment that supports WebGL deployments as well as standalone application deployments. PRIMARYAI's problem-based learning scenarios take place within an expansive storyworld, which is being specifically designed to support AI-infused upper-elementary science education. This integration of AI into science instruction offers the benefit of engaging diverse groups of learners who might otherwise not elect to explore AI learning opportunities.

Teacher Insights to Inform Design

To ensure that PRIMARYAI meets the needs of teachers and students in bringing AI learning experiences to upper elementary classrooms, a participatory co-design process is used, in which teachers are actively engaged in the instructional design process (DiSalvo et al. 2017). By having teachers participate in the design of problem-solving scenarios pertaining to AI for young learners, they help to ensure that PRIMARYAI addresses their needs, the needs of their students, and works within the constraints of traditional schools and classrooms.

Using a co-design approach adapted from Könings, Brand-Gruwel, and Merrienboer (Könings et al. 2010), we are working with teachers to (1) brainstorm positive and negative experiences regarding computer science, AI, life science, and problem-based learning, (2) describe and discuss the most important positive and negative aspects of the AI-infused learning experiences, and (3) discuss possible ideas for improving the negative points that emerge. As part of our co-design activities, three teacher interviews were conducted to gather baseline data around their current knowledge and comfort levels with computer science, AI, life science, and problem-based learning. The interviews also served as an opportunity to begin developing relationships and onboarding the teachers into the co-design process.

Participants and Procedure: Participants were recruited from local STEM teacher professional development programs in the Midwestern United States that focus on improving STEM workforce development. The teachers volunteered to participate in the co-design activities. All three teachers have been teaching for at least 10 years at the elementary school level and currently serve as coaches for their respective school STEM programs. Semi-structured interviews were conducted in person during the school day. The interviews were transcribed and analyzed using the constant comparative analysis method (Glaser 1965).

Qualitative Analysis: While reviewing the interviews in their entirety, we labelled themes that developed based on the teachers' responses to the open-ended questions. This includes their perceptions regarding their attitudes, beliefs,

and efficacy on teaching computer science, AI, life science, and problem-based learning topics. We separated the data into two key themes: (1) content knowledge and pedagogical efficacy, and (2) supports that the teachers feel they need to better teach these topics. The following excerpts were taken from the conversation during the interviews.

Content Knowledge and Pedagogical Efficacy: The participants were asked about their previous experience with computer science and coding.

"I never heard of coding until I took professional development a few years ago. Then I had training in Code.org, Scratch, and Tinker." – Participant A.

"Although I attended Code.org professional development and have implemented programming lessons with Scratch and Code.org., I have not integrated these lessons into other subjects." – Participant B.

"I am at a pilot school with a full time computer science teacher, so I haven't received any formal computer science professional development training through the school." – Participant C.

The participants indicated that although they had some training, they were novices and had limited experiences in teaching CS in their classrooms. The participants were also asked how important it is to teach computer science at the elementary level.

"Very important, but it is very difficult to tie what the students are learning to real-world applications and their potential careers. Eventually, all students, however, will need skills in computer science." – Participant A.

"Extremely important because the more we hear about what the jobs are going to be like 20 years from now, I think it's imperative that they have those skills and I feel that if they are not introduced to them early, then I am doing a disservice to my kids. That's what I want for my own children, to be prepared for their future." – Participant B.

"Incredibly important because it is impacting everything. It will help students break out of the cycle of poverty by affording them the opportunity to have good jobs." – Participant C.

The participants described that CS was important for students and their futures. They recognized that CS would be a fundamental part of their students' future lives. The participants were also asked about their concerns teaching computer science.

"My concern is 'do I have the knowledge to teach it?' I am good at figuring things out as I work with the students. I am willing to try things, even if I fail. I can say to the kids, 'I really don't know how to do that, but we can figure it out (together)'." – Participant A.

"Other than the fact that I don't know a lot – so I am nervous about teaching things I don't know… when it doesn't go like you expect it to, it is a little stressful… which is why I often invite people in who do know. I use a lot of STEM professional volunteer experts in the classroom." – Participant B.

"My concern is that I do not have enough time. It wasn't in the standards so it wasn't on the radar and now it's in the standards." – Participant C.

The participants described that although they did think CS was important, they were apprehensive about their knowledge and abilities to teach CS. However, they had each identified strategies that they could use to deal with their lack of CS knowledge. The participants were also asked how important it is to teach artificial intelligence at the elementary level.

"I have no knowledge about it… [but it] is where a lot of things are headed." – Participant A.

"I don't know a whole lot, I do feel that it is going to be a wave of the future – and highly important. And it's connected to computer science, so they (students) need to learn it, as much as they can." – Participant B.

"Very important that we teach students to be aware of how AI is being used in our society and the consequences of using AI." – Participant C.

The participants shared that although they were unfamiliar with AI, they recognized the importance of knowing about AI for the future. They were asked about their concerns teaching artificial intelligence, but because they were not clear on what AI was, they were not as apprehensive as computer science.

"I really wouldn't have any concerns because I don't know what it entails." – Participant A.

"Since I don't know very much about it, I would say none at this time." – Participant B.

"I am going to need a lot of professional development, and ideas, and background." – Participant C.

Supports Needed: The participants were asked about what types of support (tools or information, for example) do you need to help you be successful in teaching computer science and artificial intelligence to your classes? If you could ask for anything?

"I have enough hardware and software. More professional development will be beneficial. With computer science, I get frustrated at the beginning of week-long workshops, but by the end of the week, I feel more comfortable, and that's good. It is important that workshops aren't just one-day workshops." – Participant A.

"More professional development. I have enough resources." – Participant B.

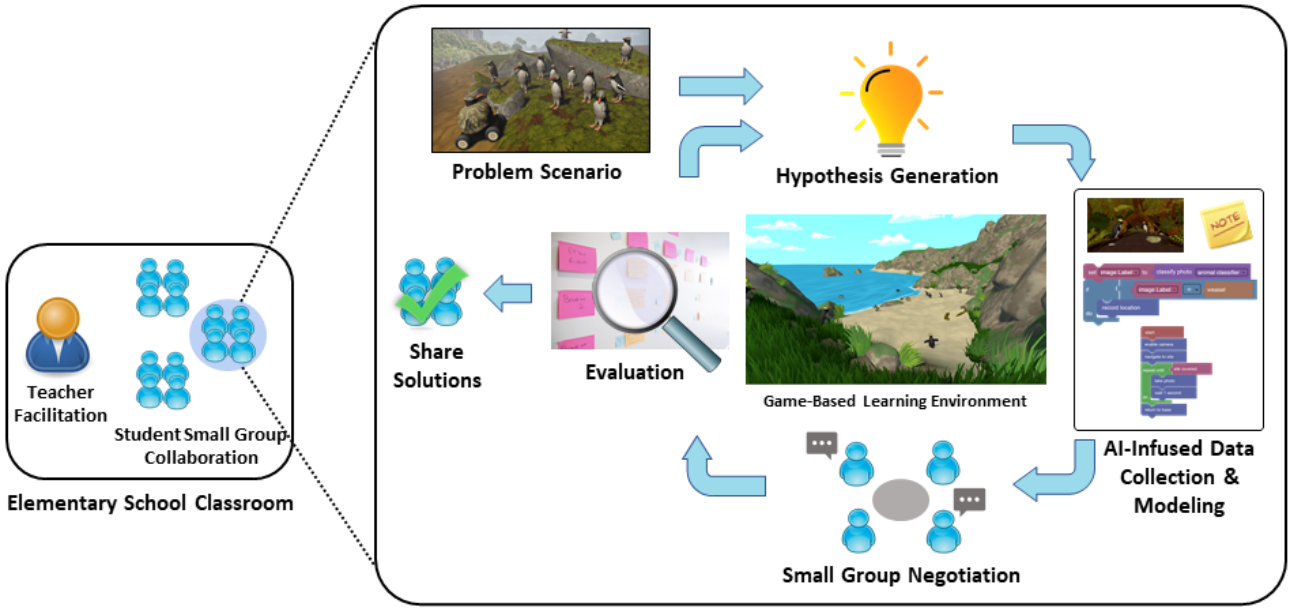


Figure 1: AI-Infused Game-Based Learning

"More time and better curricula... I am looking for scripted curricula... that is also open-ended for students." – Participant C.

The participating teachers' responses reflect their understanding of the importance of teaching AI juxtaposed by their anxiety pertaining to their lack of content knowledge and ability to teach the subject. The results suggest that providing more immersive and comprehensive professional development focused on AI content knowledge may help support teachers in adopting PRIMARYAI. In addition, PRIMARYAI will need to incorporate grade level appropriate activities for students.

AI-Infused Game-Based Learning Design

With a focus on the life sciences in general and ecology in particular, PRIMARYAI is being designed to enable students to learn about AI as they address science problems with AI tools in the context of inquiry-based science adventures. The learning environment's curricular content is driven by the Next Generation Science Standards (NGSS Lead States 2013) with a focus on disciplinary core ideas in the life sciences as well as concepts and practices from the K-12 Computer Science Framework (K-12 Computer Science Framework 2016) oriented towards age-appropriate AI concepts. PRIMARYAI is designed to support small groups of students through problem-based learning cycles as they collaboratively engage with immersive life-science problem scenarios while gaining hands-on experience through exploration of AI-based methods (Figure 1).

In the PRIMARYAI game-based learning environment, students will play the role of an ecologist who is investigating the recent declining population of yellow-eyed penguins on New Zealand's South Island. After a brief period of exploration, students learn that their overarching task is to investigate the driving question: *What is causing the recent decline in the native population of yellow-eyed penguins on the island?* After generating initial hypotheses and taking notes about what they already know and what they need to learn, the students are prompted by an in-game virtual character to begin gathering data about the local penguin population. However, the penguin colony is difficult to access on-foot, and further, the penguins are notoriously shy around humans. Therefore, the students utilize simulated semi-autonomous robots, disguised as baby penguins, to remotely collect images, audio recordings, and other data to serve as evidence for explaining recent changes to the penguin population and surrounding habitat. Using a block-based programming interface that is embedded within PRIMARYAI, the students complete a series of in-game "quests" in which they formulate internal agent programs for planning, reasoning, and decision making to be carried out by the robots. As the students progress through the quests, they are prompted by their teacher to work together to discuss, negotiate, and evaluate their hypotheses based on their latest findings in their groups.

The students continue their explorations in PRIMARYAI over the course of several class periods, completing a series of AI-centric problem scenarios while sharing their solutions with the class. By engaging with programs and abstractions that are age-appropriate, students gain hands-on experience with AI-based methods applied toward solving immersive life-science problems. In this manner, PRIMARYAI integrates problem-based learning, AI, and ecological science concepts for upper elementary science education.

AI-Infused Block-Based Programming Design

PRIMARYAI leverages block-based programming to support students' problem solving. Block-based programming is an effective approach for teaching computational thinking skills to novice learners (Bau et al. 2017). By reducing cognitive load and reducing the ability to produce syntax errors, block-based programming environments create engaging learning experiences to introduce students to programming. Recent work to support AI education for K-12 students is using block-based programming (Druga 2018; Lane 2017). Building on a toolkit developed in our lab for integrating block-based programming into game-based learning environments (PRIMARYAI), we are seamlessly integrating block-based programming into problem-solving scenarios in PRIMARYAI. Using the Blockly-inspired (Pasternak et al. 2017) programming blocks already provided by the toolkit, we are creating age-appropriate AI blocks (Kahn et al. 2018) for students to use to complete AI-based programming challenges in PRIMARYAI. Together these blocks help introduce students to concepts such as image recognition, machine learning, and semi-autonomous navigation.

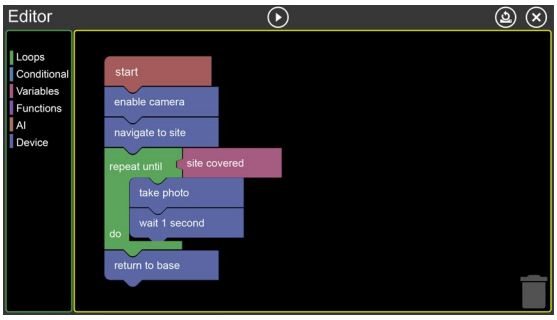


Figure 2: PRIMARYAI Block-based programming editor

For example, as described above, one quest presented to students in PRIMARYAI is to determine the cause of a recent decline in the population of penguins on an island using a semi-autonomous robot. This quest is broken down into smaller gameplay challenges that can be solved using the in-game block-based programming editor. The first step in this process involves programming the robot to gather photos around a defined exploration site (Figure 2).

After collecting the photos, students will review the photos with assistance from a virtual character to identify whether each photo gathered is a penguin, weasel, or something else by applying the appropriate label to the photo. Students learn that by identifying the images in this way, they are teaching the robot (training an algorithm and creating a classifier) to recognize patterns, so the robot will be able to distinguish different wildlife they are likely to encounter (Figure 3).

Once the robot's classifier has been trained to recognize the encountered wildlife, students will be encouraged to take the next step, program their robots to record the locations of encountered weasels (Figure 4).

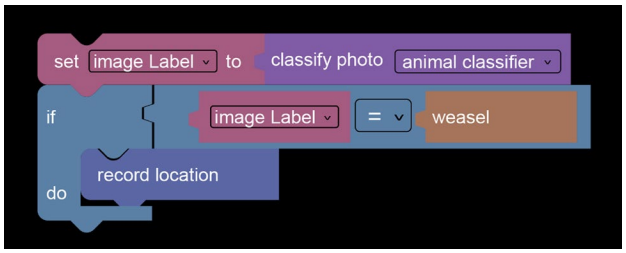


Figure 3: Block program for creating a classifier

Returning back to the students' base in the game, students will be able to visualize the location data gathered to discuss if the data supports whether the weasel population is negatively impacting the penguin population.

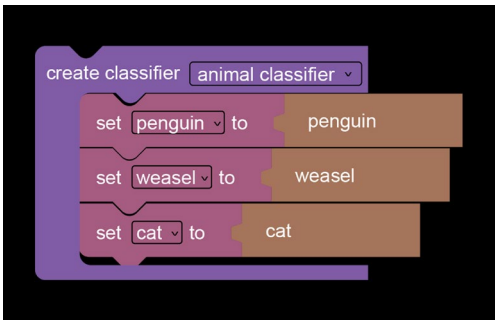


Figure 4: Using the classifier to track wildlife

Discussion

The game-based learning approach to AI-infused collaborative inquiry learning incorporates block-based programming within rich, problem-solving scenarios to support students' development of foundational AI knowledge. The approach offers significant potential for creating engaging collaborative learning experiences that integrate AI and life science.

Through our interviews with teachers, we have gained a better understanding of how to support them with integrating artificial intelligence into their classrooms. We analyzed the transcribed interviews and found, using a qualitative analysis method, that teachers will need more immersive and comprehensive professional development focused on AI content knowledge as well as grade level appropriate activities to deploy in the classrooms. Ideally, we believe that AI professional development workshops should provide follow-up classroom support to improve adoption and application of the professional development lessons and activities (Joyce and Showers 2002) as well as easy-to-implement, grade-level and age-appropriate exemplars. It is also important to note that striking the right balance between teachers co-designing, constraints of developing AI-infused game-based learning environments, and the lack of knowledge of teachers in the area of AI will present challenges for researchers.

The activities reported in this paper are only a first step towards introducing AI into upper elementary classrooms. The PRIMARYAI game-based learning environment is still under active development and refinement, and it will be instructive to evaluate how it creates engaging and effective AI learning experiences.

Evaluating PRIMARYAI

A key element of PRIMARYAI's iterative design and development is evaluating its effectiveness in upper elementary classrooms. In addition to refining the system through the incorporation of key design elements from our partner teachers and students arising from our participatory co-design process, we are planning a series of classroom studies to test its effectiveness as we investigate the question: *In what ways does engagement with PRIMARYAI support upper elementary students' learning artificial intelligence and life science*? Our evaluation will use a pre- and post-assessment design to measure the impact of PRIMARYAI on student learning and engagement. The pre- and post-assessment items will cover key concepts in AI (e.g., decision making, classification, vision, and machine learning), AI ethics (e.g., unintended consequences, implicit bias), and life science (e.g., systems and cycles). From an engagement perspective, we are adapting coding schemes from Sinha et al. (2015) and Rogat (2019) for determining group engagement coded along dimensions of behavioral engagement (on-task participation), social engagement (responsive, equitable coordination), and emotional engagement (socio-emotional climate). We will conduct a content analysis of in-class discourse to investigate how teachers orchestrate student engagement with AI and life science content, as well as to investigate to what extent AI and life science content are embedded in classroom activities. We will also explore the relationship between student engagement and learning outcomes.

Conclusion and Future Work

AI is reshaping society. There is growing awareness of the need for K-12 students at all levels to learn about artificial intelligence. This is particularly true at the elementary grade level, since providing positive early learning experiences lays the foundation for future success. In this paper, we have presented an approach to introducing AI learning experiences into upper elementary classrooms. Guided by interviews with elementary school teachers to understand the types of support they need in integrating AI into their classrooms, it has been found that professional development focused on AI content knowledge as well as grade-level-appropriate activities will be critically important to successfully introduce AI into upper elementary classrooms. The PRIMARYAI game-based learning environment aims to create engaging learning experiences centered around AI and life science. Integrating game-based learning and problem-based learning will enable PRIMARYAI to support students in collaboratively learning about AI by engaging them in group-based problem-solving scenarios. PRIMARYAI will support exploration of how to create engaging learning experiences that integrate AI and life science for upper elementary students as well as how to support elementary school teachers with integrating artificial intelligence and life science into their classrooms.

In the future, it will be important to further explore and refine how best to infuse AI concepts, such as perception, planning, robotics, and machine learning, into immersive problem-based learning experiences. Co-designing activities with both teachers and students will be important, as students are often left out of the instructional design process and their perspectives are often difficult for their teachers to predict. Conducting classroom studies to understand the strengths of game-based learning in supporting student learning around artificial intelligence, computer science, and life science will also be important.

Acknowledgements

This research was supported by National Science Foundation (NSF) Grants DRL-1934128 and DRL-1934153. Any opinions, findings, and conclusions expressed in this material are those of the authors and do not necessarily reflect the views of the NSF.

References

AAAI 2018. AAAI Launches "AI for K-12" Initiative in Collaboration with the Computer Science Teachers Association (CSTA) and AI4All. Retrieved from https://aaai.org/Pressroom/Releases/release-18-0515.php.

Ali, S.; Payne, B. H.; Williams, R.; Park, H. W.; and Breazeal, C. 2019. Constructionism, Ethics, and Creativity: Developing Primary and Middle School Artificial Intelligence Education. In *International Workshop on Education in Artificial Intelligence K-12 (EDUAI'19).*

Aylett, R. S., Louchart, S.; Dias, J.; Paiva, A.; and Vala, M. 2005. FearNot!–An Experiment in Emergent Narrative. In Int*ernational Workshop on Intelligent Virtual Agents*, 305-316. Springer, Berlin, Heidelberg.

Barrows, H. S. 1986. A Taxonomy of Problem-Based Learning Methods. *Medical education*, 20(6): 481-486.

Bau, D.; Gray, J.; Kelleher, C.; Sheldon, J.; and Turbak, F. 2017. Learnable Programming: Blocks and Beyond. *Communications of the ACM*, 60(6): 72-80.

Bertens, P.; Guitart, A.; and Periáñez, Á. 2017. Games and Big Data: A Scalable Multi-Dimensional Churn Prediction Model. In *2017 IEEE Conference on Computational Intelligence and Games*, 33-36. IEEE.

Buffum, P. S.; Frankosky, M.; Boyer, K. E.; Wiebe, E. N.; Mott, B. W.; and Lester, J. C. 2016. Collaboration and Gender Equity in Game-Based Learning for Middle School Computer Science. *Computing in Science & Engineering*, 18(2): 18-28.

Clark, D. B.; Tanner-Smith, E. E.; and Killingsworth, S. S. 2016. Digital Games, Design, and Learning: A Systematic Review and Meta-Analysis. *Review of educational research*, 86(1): 79-122.

Chen, C. H.; and Yang, Y. C. 2019. Revisiting the Effects of Project-Based Learning on Students' Academic Achievement: A Meta-Analysis Investigating Moderators. *Educational Research Review,* 26: 71-81.

Cutumisu, M. 2018. The Influence of Feedback Choice on University Students' Revision Choices and Performance in a Digital Assessment Game. In *2018 IEEE Conference on Computational Intelligence and Games*, 1-7.

DiSalvo, B.; Yip, J.; Bonsignore, E.; and DiSalvo, C. eds. 2017. *Participatory design for learning: Perspectives from practice and research*, 3-6. Taylor & Francis.

Dolmans, D. H.; and Schmidt, H. G.; 2006. What Do We Know about Cognitive and Motivational Effects of Small Group Tutorials in Problem-Based Learning?. *Advances in health sciences education*, 11(4): 321.

Druga, S. 2018. *Growing Up with AI: Cognimates: From Coding to Teaching Machines*. Ph.D. Diss. Massachusetts Institute of Technology.

Gee, J. P. 2003. What Video Games Have to Teach Us about Learning and Literacy. *Computers in Entertainment,* 1(1): 20-20.

Gibson, D.; Aldrich, C.; and Prensky, M. 2007. *Games and Simulations in Online Learning: Research and Development*. Covent Garden, London.

Glaser, B. G. 1965. The Constant Comparative Method of Qualitative Analysis. *Social problems*, 12(4): 436-445.

Grover, S.; and Pea, R., 2013. Computational Thinking in K–12: A Review of the State of the Field. *Educational researcher*, 42(1): 38-43.

Hadiji, F.; Sifa, R.; Drachen, A.; Thurau, C.; Kersting, K.; and Bauckhage, C. 2014. Predicting Player Churn in the Wild. In *2014 IEEE Conference on Computational Intelligence and Games*, 1-8. IEEE.

Hoffman, B.; Morelli, R.; and Rosato, J. 2019. Student Engagement is Key to Broadening Participation in CS. In *Proceedings of the 50th ACM Technical Symposium on Computer Science Education*, 1123-1129.

Hmelo-Silver, C. E. 2004. Problem-Based Learning: What and How Do Students Learn?. *Educational psychology review*, 16(3): 235-266.

Hmelo-Silver, C. E.; Kapur, M.; and Hamstra, M. 2018. Learning Through Problem Solving. In F. Fischer, C. E. Hmelo-Silver, S. R. Goldman, & P. Reimann (Eds). *International Handbook of the Learning Sciences*, 210-220.

Joyce, B. R.; and Showers, B. 2002. Student Achievement Through Staff Development.

Kantharaju, P.; Alderfer, K.; Zhu, J.; Char, B.; Smith, B.; and Ontanón, S. 2018. Tracing Player Knowledge in a Parallel Programming Educational Game. In *Fourteenth Artificial Intelligence and Interactive Digital Entertainment Conference*, 173-179

Kahn, K. M.; Megasari, R.; Piantari, E.; and Junaeti, E. 2018. AI Programming by Children Using Snap! Block Programming in A Developing Country. In *Proceedings of the 13th European Conference on Technology Enhanced Learning*, 3-6.

Ketelhut, D. J.; Dede, C.; Clarke, J.; Nelson, B.; and Bowman, C. 2007. Studying Situated Learning in a Multi-User Virtual Environment. *Assessment of problem solving using simulations*, 37-58.

Könings, K. D.; Brand-Gruwel, S.; and van Merriënboer, J. J. 2010. An Approach to Participatory Instructional Design in Secondary Education: An Exploratory Study. *Educational Research*, 52(1): 45-59.

K-12 Computer Science Framework Steering Committee. 2016. K-12 Computer Science Framework. Retrieved August 28, 2017 from www.k12.cs.org.

Lane, D. 2019. Introducing Machine Learning for Kids. *Consultado el*, 28(2): 2020.

Lee, S.; Mott, B.; Ottenbreit-Leftwich, A.; Scribner, A.; Taylor, S.; Glazeski, K.; Hmelo-Silver, C.; Lester, J. 2020. Designing a Collaborative Game-Based Learning Environments for AI-Infused Inquiry Learning in Elementary School Classrooms. In *Proceedings of the Twenty-fifth Annual Conference on Innovation and Technology in Computer Science Education*, 566-566. Trondheim, Norway.

Lin, P.; Van Brummelen, J.; Lukin, G.; Williams, R.; and Breazeal, C. 2020. Zhorai: Designing a Conversational Agent for Children to Explore Machine Learning Concepts. In *Proceedings of the Tenth Symposium on Educational Advances in Artificial Intelligence*, 13381-13388.

Manyika, J. 2017. A Future that Works: AI, Automation, Employment, and Productivity. *McKinsey Global Institute Research, Tech*. Rep, 60.

Marsella, S.; Johnson, W. L.; and LaBore, C. 2003. Interactive Pedagogical Drama for Health Interventions. In *11th International Conference on Artificial Intelligence in Education*, 341-348, Sydney, Australia.

McConnell, T.; Parker, J.; and Eberhardt, J. 2016. *Problem-Based Learning in the Life Science Classroom, K-12*. Arlington, VA: NSTA Press.

McKinsey Global Institute. 2017. *A Future That Works: Automation, Employment, and Productivity*.

McLaren, B. M.; Adams, D.M.; Mayer, R.E.; and Forlizzi, J. 2017. A Computer-Based Game that Promotes Mathematics Learning More than A Conventional Approach. *International Journal of Game-Based Learning*, 7(1): 36-56.

Min, W.; Frankosky, M. H.; Mott, B. W.; Wiebe, E. N.; Boyer, K. E.; and Lester, J. C. 2017. Inducing Stealth Assessors from Game Interaction data. In *International Conference on Artificial Intelligence in Education*, 212-223. Springer, Cham.

Mogessie, M.; Richey, J. E.; McLaren, B. M.; Andres-Bray, J. M. L.; and Baker, R. S. 2020. Confrustion and Gaming While Learning with Erroneous Examples in a Decimals Game. In *International Conference on Artificial Intelligence in Education*, 208-213. Springer, Cham.

Mott, B. W.; Taylor, R. G.; Lee, S. Y.; Rowe, J. P.; Saleh, A.; Glazewski, K. D.; Hmelo-Silver, C. E.; and Lester, J. C. 2019. Designing and Developing Interactive Narratives for Collaborative Problem-Based Learning. In *International Conference on Interactive Digital Storytelling*, 86-100. Springer, Cham.

NGSS Lead States 2013. *Next Generation Science Standards*. Washington, DC: The National Academies Press.

Norouzi, N.; Chaturvedi, S.; and Rutledge, M. 2020. Lessons Learned from Teaching Machine Learning and Natural Language Processing to High School Students. In *Proceedings of the Tenth Symposium on Educational Advances in Artificial Intelligence*, 13397-13403.

Pasternak, E.; Fenichel, R.; and Marshall, A. N. 2017. Tips for Creating a Block Language with Blockly. In *2017 IEEE Blocks and Beyond Workshop (B&B),* 21-24. IEEE.

Pollock, L.; Mouza, C.; Czik, A.; Little, A.; Coffey, D.; and Buttram, J. 2017. From Professional Development to the Classroom: Findings from CS K-12 Teachers. In *Proceedings of the 2017 ACM SIGCSE Technical Symposium on Computer Science Education*, 477-482.

Rogat, T. K. 2019. Examining Group Productive Disciplinary Engagement. In *Proceedings of the Computer-Supported Collaborative Learning*, vol.2.

Rowe, E.; Asbell-Clarke, J.; Cunningham, K.; and Gasca, S. 2017. Assessing Implicit Computational Thinking in Zoombinis Gameplay: Pizza Pass, Fleens & Bubblewonder Abyss. In *Extended Abstracts Publication of the Annual Symposium on Computer-Human Interaction in Play*, 195-200.

Rowe, J. P.; Shores, L. R.; Mott, B. W.; and Lester, J. C. 2011. Integrating Learning, Problem Solving, and Engagement in Narrative-Centered Learning Environments. *International Journal of Artificial Intelligence in Education*, 21(1-2): 15-133.

Salen, K. 2007. Gaming Literacies: A Game Design Study in Action. *Journal of Educational Multimedia and Hypermedia*, 16(3): 301-322.

Sawyer, R.; Rowe, J.; Azevedo, R.; and Lester, J. 2018. Modeling Player Engagement with Bayesian Hierarchical Models. In *Proceedings of the Fourteenth AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, 215-221

Shaffer, D.W. 2008. *How computer games help children learn*. Macmillan.

Sinha, S.; Rogat, T.; Adams-Wiggins, K. R.; and Hmelo-Silver, C. E. 2015. Collaborative Group Engagement in a Computer-Supported Inquiry Learning Environment. *International Journal of Compuer-Supported Collaborative Learning*, 10, 273-277. Springer.

Shute, V. J.; Sun C. 2019. Games for Assessment. In J. L. Plass, R. E. Mayer, & B. D. Homer (Eds.), *Handbook of Game-based Learning*, 491-512. Cambridge, MA: MIT Press.

Stanton, J.; Goldsmith, L.; Adrion, R.; Dunton, S.; Hendrickson, K.; Peterfreund, A.; Yongpradit, P.; Zarch, R.; and Zinth, J. 2017. State of the States Landscape Report: State-Level Policies Supporting Equitable K–12 Computer Science Education. *Education Development Center*.

Touretzky, D.; Gardner-McCune, C.; Martin, F.; and Seehorn, D. 2019. Envisioning AI for K-12: What Should Every Child Know about AI?,.In *Proceedings of the AAAI Conference on Artificial Intelligence*, 9795-9799.

Touretzky, D.; Gardner-McCune, C.; Breazeal, C.; Martin, F.; and Seehorn, D. 2019. A Year in K-12 AI Education. *AI Magazine*, 40(4):88-90.

Wouters, P.; Van Nimwegen, C.; Van Oostendorp, H.; and Van Der Spek, E.D. 2013. A Meta-Analysis of the Cognitive and Motivational Effects of Serious Games. *Journal of educational psychology*, 105(2): 249.

Wang, N.; and Johnson M. 2019. AI Education for K-12: Connecting AI Concepts to High School Math Curriculum. In *Proceedings of the Workshop on Education in Artificial Intelligence K-12, 28th International Joint Conference on Artificial Intelligence*.

Webb, M.; Davis, N.; Bell, T.; Katz, Y. J.; Reynolds, N.; Chambers, D. P.; and Sysło, M. M. 2017. Computer Science in K-12 School Curricula of the 2lst Century: Why, What and When?. *Education and Information Technologies*, 22(2): 445-468.

West, D. M. 2018. *The future of work: robots, AI, and automation*. Brookings Institution Press.

Yeh, T.; Dalton, B.; and Haberl, E. 2019. Integrating AI Ethics into Robotics Learning Experiences. In *Proceedings of the Workshop on K12 AI education, 20th International Conference on Artificial Intelligence in Education*.

Yu, H.; and Riedl, M. O. 2015. Optimizing Players' Expected Enjoyment in Interactive Stories. In *Eleventh Artificial Intelligence and Interactive Digital Entertainment Conference,* 100–106. Palo Alto, CA.: AAAI Press.

Zhao, W.; and Shute, V. J. 2019. Can Playing a Video Game Foster Computational Thinking skills?. *Computers & Education*, 141: 103-633.

arXiv:1704.03477v4 [cs.NE] 19 May 2017

A Neural Representation of Sketch Drawings

David Ha
Google Brain
hadavid@google.com

Douglas Eck
Google Brain
deck@google.com

Abstract

We present `sketch-rnn`, a recurrent neural network (RNN) able to construct stroke-based drawings of common objects. The model is trained on a dataset of human-drawn images representing many different classes. We outline a framework for conditional and unconditional sketch generation, and describe new robust training methods for generating coherent sketch drawings in a vector format.

1 Introduction

Recently, there have been major advancements in generative modelling of images using neural networks as a generative tool. Generative Adversarial Networks (GANs) [5], Variational Inference (VI) [15], and Autoregressive (AR) [19] models have become popular tools in this fast growing area. Most of the work thus far has been targeted towards modelling low resolution, pixel images. Humans, however, do not understand the world as a grid of pixels, but rather develop abstract concepts to represent what we see. From a young age, we develop the ability to communicate what we see by drawing on paper with a pencil or crayon. In this way we learn to express a sequential, vector representation of an image as a short sequence of strokes. In this paper we investigate an alternative to traditional pixel image modelling approaches, and propose a generative model for vector images.

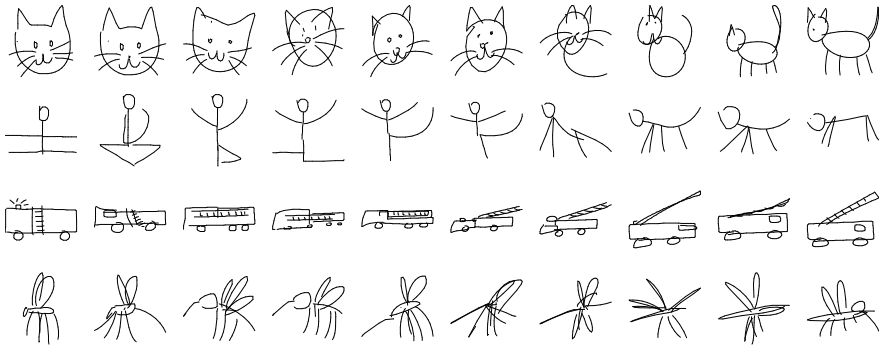


Figure 1: Latent space interpolation of various vector images produced by our model.

Our goal is to train machines to draw and generalize abstract concepts in a manner similar to humans. As a first step towards this goal, we train our model on a dataset of hand-drawn sketches, each represented as a sequence of motor actions controlling a pen: which direction to move, when to lift the pen up, and when to stop drawing. In doing so, we created a model that potentially has many applications, from assisting the creative process of an artist, to helping teach students how to draw.

This paper makes the following contributions: We outline a framework for both unconditional and conditional generation of vector images composed of a sequence of lines. Our recurrent neural network-based generative model is capable of producing sketches of common objects in a vector format. We develop a training procedure unique to vector images to make the training more robust. In

the conditional generation model, we explore the latent space developed by the model to represent a vector image. We also discuss potential creative applications of our methodology. We make available a large dataset of hand drawn vector images to encourage further development of generative modelling for vector images, and also release an implementation of our model as an open source project.[1]

2 Related Work

There is a long history of work related to algorithms that mimic painters. One such work is Portrait Drawing by Paul the Robot [23, 25], where an underlying algorithm controlling a mechanical robot arm sketches lines on a canvas with a programmable artistic style to mimic a given digitized portrait of a person. Reinforcement Learning based-approaches [25] have been developed to discover a set of paint brush strokes that can best represent a given input photograph. These prior works generally attempt to mimic digitized photographs, rather than develop generative models of vector images.

Neural Network-based approaches have been developed for generative models of images, although the majority of neural network-related research on image generation deal with pixel images [5, 10, 12, 14, 19, 24]. There has been relatively little work done on vector image generation using neural networks. An earlier work [22] makes use of Hidden Markov Models to synthesize lines and curves of a human sketch. More recent work [6] on handwriting generation with Recurrent Neural Networks laid the groundwork for utilizing Mixture Density Networks [1] to generate continuous data points. Recent works of this approach attempted to generate vectorized Kanji characters unconditionally [7] and conditionally [26] by modelling Chinese characters as a sequence of pen stroke actions.

In addition to unconditionally generating sketches, we also explore encoding existing sketches into a latent space of embedding vectors. Previous work [2] outlined a methodology to combine Sequence-to-Sequence models with a Variational Autoencoder to model natural English sentences in latent vector space. A related work [16], utilizes probabilistic program induction, rather than neural networks, to perform one-shot modelling of the Omniglot dataset containing images of symbols.

One of the factors limiting research development in the space of generative vector drawings is the lack of publicly available datasets. Previously, the Sketch dataset [4], consisting of 20K vector sketches, was used to explore feature extraction techniques. A subsequent work, the Sketchy dataset [20], provided 70K vector sketches along with corresponding pixel images for various classes. This allowed for a larger-scale exploration of human sketches. ShadowDraw [17] is an interactive system that predicts what a finished drawing looks like based on a set of incomplete brush strokes from the user while the sketch is being drawn. ShadowDraw used a dataset of 30K raster images combined with extracted vectorized features. In this work, we use a much larger dataset of vector sketches that is made publicly available.

3 Methodology

3.1 Dataset

We constructed `QuickDraw`, a dataset of vector drawings obtained from *Quick, Draw!* [11], an online game where the players are asked to draw objects belonging to a particular object class in less than 20 seconds. `QuickDraw` consists of hundreds of classes of common objects. Each class of `QuickDraw` is a dataset of 70K training samples, in addition to 2.5K validation and 2.5K test samples.

We use a data format that represents a sketch as a set of pen stroke actions. This representation is an extension of the format used in [6]. Our format extends the binary pen stroke event into a multi-state event. In this data format, the initial absolute coordinate of the drawing is located at the origin. A sketch is a list of points, and each point is a vector consisting of 5 elements: $(\Delta x, \Delta y, p_1, p_2, p_3)$. The first two elements are the offset distance in the x and y directions of the pen from the previous point. The last 3 elements represents a binary one-hot vector of 3 possible states. The first pen state, p_1, indicates that the pen is currently touching the paper, and that a line will be drawn connecting the next point with the current point. The second pen state, p_2, indicates that the pen will be lifted from the paper after the current point, and that no line will be drawn next. The final pen state, p_3, indicates that the drawing has ended, and subsequent points, including the current point, will not be rendered.

[1] The code and dataset is available at `https://magenta.tensorflow.org/sketch_rnn`.

3.2 Sketch-RNN

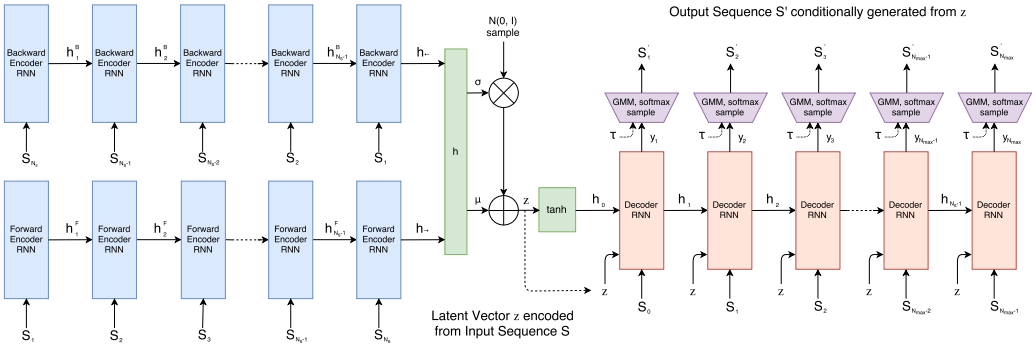


Figure 2: Schematic diagram of `sketch-rnn`.

Our model is a Sequence-to-Sequence Variational Autoencoder (VAE), similar to the architecture described in [2, 15]. Our encoder is a bidirectional RNN [21] that takes in a sketch as an input, and outputs a latent vector of size N_z. Specifically, we feed the sketch sequence, S, and also the same sketch sequence in reverse order, $S_{\texttt{reverse}}$, into two encoding RNNs that make up the bidirectional RNN, to obtain two final hidden states:

$$h_{\rightarrow} = \texttt{encode}_{\rightarrow}(S),\ h_{\leftarrow} = \texttt{encode}_{\leftarrow}(S_{\texttt{reverse}}),\ h = [\ h_{\rightarrow}\ ;\ h_{\leftarrow}\ ] \tag{1}$$

We take this final concatenated hidden state, h, and project it into two vectors μ and $\hat{\sigma}$, each of size N_z, using a fully connected layer. We convert $\hat{\sigma}$ into a non-negative standard deviation parameter σ using an exponential operation. We use μ and σ, along with $\mathcal{N}(0, I)$, a vector of IID Gaussian variables of size N_z, to construct a random vector, $z \in \mathbb{R}^{N_z}$, as in the approach for a VAE [15]:

$$\mu = W_\mu h + b_\mu,\ \hat{\sigma} = W_\sigma h + b_\sigma,\ \sigma = \exp\left(\frac{\hat{\sigma}}{2}\right),\ z = \mu + \sigma \odot \mathcal{N}(0, I) \tag{2}$$

Under this encoding scheme, the latent vector z is not a deterministic output for a given input sketch, but a random vector conditioned on the input sketch.

Our decoder is an autoregressive RNN that samples output sketches conditional on a given latent vector z. The initial hidden states h_0, and optional cell states c_0 (if applicable) of the decoder RNN is the output of a single layer network: $[\ h_0\ ;\ c_0\ ] = \tanh(W_z z + b_z)$

At each step i of the decoder RNN, we feed the previous point, S_{i-1} and the latent vector z in as a concatenated input x_i, where S_0 is defined as $(0, 0, 1, 0, 0)$. The output at each time step are the parameters for a probability distribution of the next data point S_i. In Equation 3, we model $(\Delta x, \Delta y)$ as a Gaussian mixture model (GMM) with M normal distributions as in [1, 6], and (q_1, q_2, q_3) as a categorical distribution to model the ground truth data (p_1, p_2, p_3), where $(q_1 + q_2 + q_3 = 1)$ as done in [7] and [26]. Unlike [6], our generated sequence is conditioned from a latent code z sampled from our encoder, which is trained end-to-end alongside the decoder.

$$p(\Delta x, \Delta y) = \sum_{j=1}^{M} \Pi_j\ \mathcal{N}(\Delta x, \Delta y \mid \mu_{x,j}, \mu_{y,j}, \sigma_{x,j}, \sigma_{y,j}, \rho_{xy,j}),\ \text{where}\ \sum_{j=1}^{M} \Pi_j = 1 \tag{3}$$

$\mathcal{N}(x, y|\mu_x, \mu_y, \sigma_x, \sigma_y, \rho_{xy})$ is the probability distribution function for a bivariate normal distribution. Each of the M bivariate normal distributions consist of five parameters: $(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho_{xy})$, where μ_x and μ_y are the means, σ_x and σ_y are the standard deviations, and ρ_{xy} is the correlation parameter of each bivariate normal distribution. An additional vector Π of length M, also a categorical distribution, are the mixture weights of the Gaussian mixture model. Hence the size of the output vector y is $5M + M + 3$, which includes the 3 logits needed to generate (q_1, q_2, q_3).

The next hidden state of the RNN, generated with its forward operation, projects into the output vector y_i using a fully-connected layer:

$$x_i = [\ S_{i-1}\ ;\ z\ ],\ [\ h_i\ ;\ c_i\ ] = \texttt{forward}(x_i,\ [\ h_{i-1}\ ;\ c_{i-1}\ ]),\ y_i = W_y h_i + b_y,\ y_i \in \mathbb{R}^{6M+3} \tag{4}$$

The vector y_i is broken down into the parameters of the probability distribution of the next data point:

$$[\ (\hat{\Pi}_1\ \mu_x\ \mu_y\ \hat{\sigma}_x\ \hat{\sigma}_y\ \hat{\rho}_{xy})_1\ \ldots\ (\hat{\Pi}_1\ \mu_x\ \mu_y\ \hat{\sigma}_x\ \hat{\sigma}_y\ \hat{\rho}_{xy})_M\ (\hat{q}_1\ \hat{q}_2\ \hat{q}_3)\ ] = y_i \tag{5}$$

As in [6], we apply $\exp$ and $\tanh$ operations to ensure the standard deviation values are non-negative, and that the correlation value is between -1 and 1:

$$\sigma_x = \exp(\hat{\sigma}_x),\ \sigma_y = \exp(\hat{\sigma}_y),\ \rho_{xy} = \tanh(\hat{\rho}_{xy}) \tag{6}$$

The probabilities for the categorical distributions are calculated using the outputs as logit values:

$$q_k = \frac{\exp(\hat{q}_k)}{\sum_{j=1}^{3}\exp(\hat{q}_j)}, k \in \{1,\ 2,\ 3\},\ \Pi_k = \frac{\exp(\hat{\Pi}_k)}{\sum_{j=1}^{M}\exp(\hat{\Pi}_j)}, k \in \{1,\ \ldots,\ M\} \tag{7}$$

A key challenge is to train our model to know when to stop drawing. Because the probabilities of the three pen stroke events are highly unbalanced, the model becomes more difficult to train. The probability of a p_1 event is much higher than p_2, and the p_3 event will only happen once per drawing. The approach developed in [7] and later followed by [26] was to use different weightings for each pen event when calculating the losses, such as a hand-tuned weighting of $(1, 10, 100)$. We find this approach to be inelegant and inadequate for our dataset of diverse image classes.

We develop a simpler, more robust approach that works well for a broad class of sketch drawing data. In our approach, all sequences are generated to a length of $N_{\max}$ where $N_{\max}$ is the length of the longest sketch in our training dataset. In principle $N_{\max}$ can be considered a hyper parameter. As the length of S is usually shorter than $N_{\max}$, we set S_i to be $(0, 0, 0, 0, 1)$ for $i > N_s$. We discuss the training in detail in the next section.

After training, we can sample sketches from our model. During the sampling process, we generate the parameters for both GMM and categorical distributions at each time step, and sample an outcome S'_i for that time step. Unlike the training process, we feed the sampled outcome S'_i as input for the next time step. We continue to sample until $p_3 = 1$, or when we have reached $i = N_{\max}$. Like the encoder, the sampled output is not deterministic, but a random sequence, conditioned on the input latent vector z. We can control the level of randomness we would like our samples to have during the sampling process by introducing a temperature parameter τ:

$$\hat{q}_k \to \frac{\hat{q}_k}{\tau},\ \hat{\Pi}_k \to \frac{\hat{\Pi}_k}{\tau},\ \sigma_x^2 \to \sigma_x^2\tau,\ \sigma_y^2 \to \sigma_y^2\tau \tag{8}$$

We can scale the softmax parameters of the categorial distribution and also the σ parameters of the bivariate normal distribution by a temperature parameter τ, to control the level of randomness in our samples. τ is typically set between 0 and 1. In the limiting case as $\tau \to 0$, our model becomes deterministic and samples will consist of the most likely point in the probability density function. Figure 3 illustrates of effect of sampling sketches with various temperature parameters.

3.3 Unconditional Generation

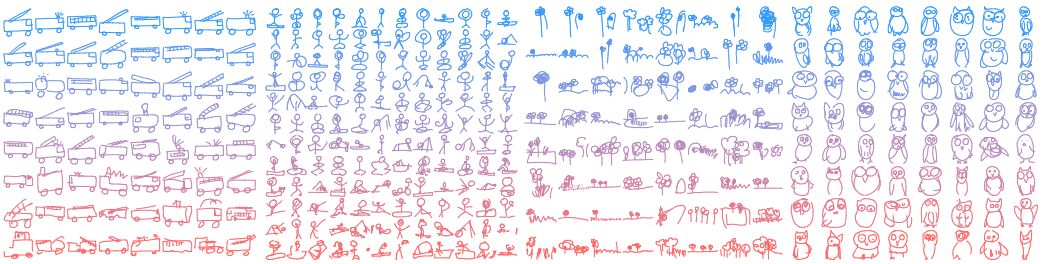


Figure 3: Unconditional generation of firetrucks, yoga poses, gardens and owls with varying τ.

As a special case, we can also train our model to generate sketches unconditionally, where we only train the decoder RNN module, without any input or latent vectors. By removing the encoder, the decoder RNN as a standalone model is an autoregressive model without latent variables. In this use case, the initial hidden states and cell states of the decoder RNN are initialized to zero. The inputs x_i of the decoder RNN at each time step is only S_{i-1} or S'_{i-1}, as we do not need to concatenate a latent vector z. In Figure 3, we sample various sketch images generated unconditionally by varying the temperature parameter from $\tau = 0.2$ at the top in blue, to $\tau = 0.9$ at the bottom in red.

3.4 Training

Our training procedure follows the approach of the Variational Autoencoder [15], where the loss function is the sum of two terms: the Reconstruction Loss, L_R, and the Kullback-Leibler Divergence Loss, L_{KL}. We train our model to optimize this two-part loss function. The Reconstruction loss term, described in Equation 9, maximizes the log-likehood of the generated probability distribution to explain the training data S. We can calculate this reconstruction loss, L_R, using the generated parameters of the pdf and the training data S. L_R is composed of the sum of the log loss of the offset terms $(\Delta x, \Delta y)$, L_s, and the log loss of the pen state terms (p_1, p_2, p_3), L_p:

$$L_s = -\frac{1}{N_{\max}} \sum_{i=1}^{N_s} \log \Big(\sum_{j=1}^{M} \Pi_{j,i}\, \mathcal{N}(\Delta x_i, \Delta y_i \mid \mu_{x,j,i}, \mu_{y,j,i}, \sigma_{x,j,i}, \sigma_{y,j,i}, \rho_{xy,j,i}) \Big)$$
$$L_p = -\frac{1}{N_{\max}} \sum_{i=1}^{N_{\max}} \sum_{k=1}^{3} p_{k,i} \log(q_{k,i}),\ L_R = L_s + L_p \tag{9}$$

Note that we discard the pdf parameters modelling the $(\Delta x, \Delta y)$ points beyond N_s when calculating L_s, while L_p is calculated using all of the pdf parameters modelling the (p_1, p_2, p_3) points until $N_{\max}$. Both terms are normalized by the total sequence length $N_{\max}$. We found this methodology of loss calculation to be more robust and allows the model to easily learn when it should stop drawing, unlike the earlier mentioned method of assigning importance weightings to p_1, p_2, and p_3.

The Kullback-Leibler (KL) divergence loss term measures the difference between the distribution of our latent vector z, to that of an IID Gaussian vector with zero mean and unit variance. Optimizing for this loss term allows us to minimize this difference. We use the result in [15], and calculate the KL loss term, L_{KL}, normalized by number of dimensions N_z of the latent vector:

$$L_{KL} = -\frac{1}{2N_z}\Big(1 + \hat{\sigma} - \mu^2 - \exp(\hat{\sigma})\Big) \tag{10}$$

The loss function in Equation 11 is a weighted sum of both the L_R and L_{KL} loss terms:

$$Loss = L_R + w_{KL} L_{KL} \tag{11}$$

There is a tradeoff between optimizing for one term over the other. As $w_{KL} \to 0$, our model approaches a pure autoencoder, sacrificing the ability to enforce a prior over our latent space while obtaining better reconstruction loss metrics. Note that for unconditional generation, where our model is the standalone decoder, there will be no L_{KL} term as we only optimize for L_R.

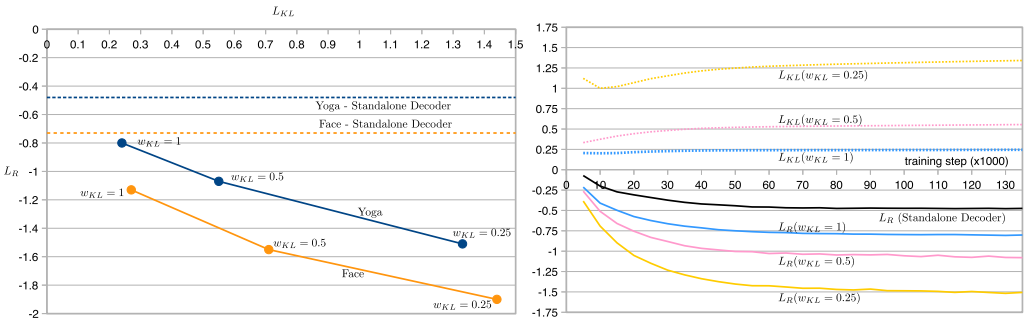


Figure 4: Tradeoff between L_R and L_{KL}, for two models trained on single class datasets (left). Validation Loss Graph for models trained on the Yoga dataset using various w_{KL}. (right)

Figure 4 illustrates the tradeoff between different settings of w_{KL} and the resulting L_R and L_{KL} metrics on the test set, along with the L_R metric on a standalone decoder RNN for comparison. As the unconditional model does not receive any prior information about the entire sketch it needs to generate, the L_R metric for the standalone decoder model serves as an upper bound for various conditional models using a latent vector.

4 Experiments

We conduct several experiments with `sketch-rnn` for both conditional and unconditional vector image generation. We train `sketch-rnn` on various `QuickDraw` classes using various settings for

w_{KL} and record the breakdown of losses. To experiment with a diverse set of classes with varying complexities, we select the cat, pig, face, firetruck, garden, owl, mosquito and yoga class. We also experiment on multi-class datasets by concatenating different classes together to form (cat, pig) and (crab, face, pig, rabbit). The results for test set evaluation on various datasets are displayed in Table 1.

The `sketch-rnn` model treats the RNN cell as an abstract component. In our experiments, we use Long Short-Term Memory (LSTM) [9] as the encoder RNN. For the decoder RNN, we use HyperLSTM, as this type of RNN cell excels at sequence generation tasks [8]. The ability for HyperLSTM to spontaneously augment its own weights enables it to adapt to many different regimes in a large diverse dataset. For model configuration details, please see the Supplementary Material.

Dataset	$w_{KL} = 1.00$		$w_{KL} = 0.50$		$w_{KL} = 0.25$		Decoder Only
	L_R	L_{KL}	L_R	L_{KL}	L_R	L_{KL}	L_R
cat	-0.98	0.29	-1.33	0.70	-1.46	1.01	-0.57
pig	-1.14	0.22	-1.37	0.49	-1.52	0.80	-0.82
cat, pig	-1.02	0.22	-1.24	0.49	-1.50	0.98	-0.75
crab, face, pig, rabbit	-0.91	0.22	-1.04	0.40	-1.47	1.17	-0.67
face	-1.13	0.27	-1.55	0.71	-1.90	1.44	-0.73
firetruck	-1.24	0.22	-1.26	0.24	-1.78	1.10	-0.90
garden	-0.79	0.20	-0.81	0.25	-0.99	0.54	-0.62
owl	-0.93	0.20	-1.03	0.34	-1.29	0.77	-0.66
mosquito	-0.67	0.30	-1.02	0.66	-1.41	1.54	-0.34
yoga	-0.80	0.24	-1.07	0.55	-1.51	1.33	-0.48

Table 1: Loss figures (L_R and L_{KL}) for various w_{KL} settings.

The relative loss numbers are consistent with our expectations. We see that the reconstruction loss term L_R decreases as we relax the w_{KL} parameter controlling the weight for the KL loss term, and meanwhile the KL loss term L_R increases as a result. The L_R for the conditional model is strictly less than the unconditional, standalone decoder model. In Figure 4 (right), we plot validation-set loss graphs for on the yoga class for models with various w_{KL} settings. As L_R decreases, the L_{KL} term tends to increase due to the tradeoff between L_R and L_{KL}.

4.1 Conditional Reconstruction

We qualitatively assess the reconstructed sketch S' given an input sketch S. In Figure 5 (left), we sample several reconstructions at various levels of temperature τ using a model trained on the single cat class, starting at 0.01 on the left and linearly increasing to 1.0 on the right. The reconstructed cat sketches have similar properties as the input image, and occasionally add or remove details such as a whisker, a mouth, a nose, or the orientation of the tail.

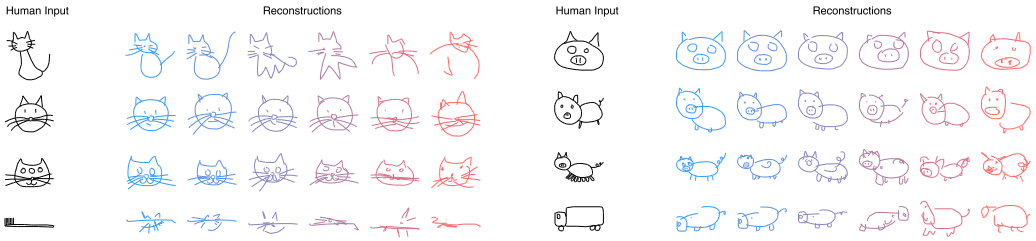


Figure 5: Conditional generation of cats (left) and pigs (right).

When presented with a non-standard image of a cat, such as a cat's face with three eyes, the reconstructed cat only has two eyes. If we input a sketch from another image class, such a toothbrush, the model seemingly generate sketches with similar orientation and properties as the toothbrush input image, but with some cat-like features such as cat ears, whiskers or feet. We perform a similar experiment with a model trained on the pig class, as shown in Figure 5 (right).

4.2 Latent Space Interpolation

By interpolating between latent vectors, we can visualize how one image morphs into another image by visualizing the reconstructions of the interpolations. As we enforce a Gaussian prior on the latent space, we expect fewer *gaps* in the space between two encoded latent vectors. We expect a model trained using a higher w_{KL} setting to produce images that are closer to the data manifold given a spherically interpolated [24] latent vector z, compared to another model trained with a lower w_{KL}.

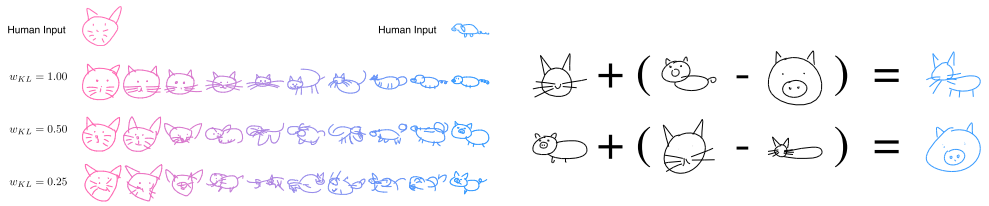


Figure 6: Latent space interpolation between cat and pig using with various w_{KL} settings (left). Sketch Drawing Analogies (right).

To demonstrate this, we train several models using various w_{KL}, on a dataset consisting of both cat and pigs, and we encode two distinct images from the test set - a cat face and a full pig. Figure 6 (left) shows the reconstructed images from the interpolated latent vectors between the two original images. As expected, models trained with higher w_{KL} produce more coherent interpolated images.

4.3 Sketch Drawing Analogies

The interpolation example in Figure 6 (left) suggests that the latent vector z encode conceptual features of a sketch. Can we use these features to augment other sketches without such features – for example, adding a body to a cat's head? Indeed, we find that sketch drawing analogies are possible for models trained with low L_{KL} numbers. Given the smoothness of the latent space, where any interpolated vector between two latent vectors results in a coherent sketch, we can perform vector arithmetic on the latent vectors encoded from different sketches and explore how the model organizes the latent space to represent different concepts in the manifold of generated sketches.

For example, as shown in Figure 6 (right), we can subtract the latent vector of an encoded pig head from the latent vector of a full pig, to arrive at a vector that represents a body. Adding this difference to the latent vector of a cat head results in a full cat (i.e. cat head + body = full cat). We repeat the experiment to remove the body of a full pig. These drawing analogies allow us to explore how the model organizes its latent space to represent different concepts in the manifold of generated sketches.

4.4 Predicting Different Endings of Incomplete Sketches

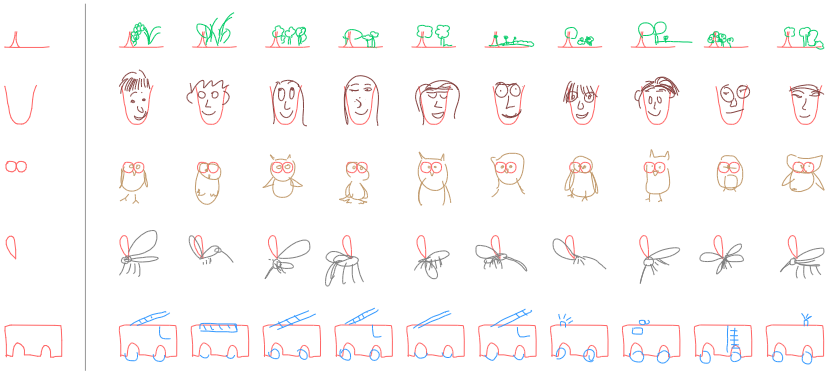


Figure 7: `sketch-rnn` predicting possible endings of various incomplete sketches (the red lines).

We can use `sketch-rnn` to finish an incomplete sketch. By using the decoder RNN as a standalone model, we can generate a sketch that is conditioned on the previous points. We use the decoder RNN to first *encode* an incomplete sketch into a hidden state h. Afterwards, we generate the remaining points of the sketch using h as the initial hidden state. We show results in Figure 7 using decoder-only models trained on individual classes, and sample completions by setting $\tau = 0.8$.

5 Applications and Future Work

We believe `sketch-rnn` will enable many creative applications. Even the decoder-only model trained on various classes can assist the creative process of an artist by suggesting many possible ways of finishing a sketch, helping artists expand their imagination. In the conditional model, exploring the latent space between different objects can potentially enable artists to find interesting intersections and relationships between different drawings. Even in the simplest use, pattern designers can apply `sketch-rnn` to generate a large number of similar, but unique designs for textile or wallpaper prints.

As we saw earlier in Section 4.1, a model trained to draw pigs can be made to draw pig-like trucks if given an input sketch of a truck. We can extend this result to applications that might help creative designers come up with abstract designs that can resonate more with their target audience. For instance, in Figure 8 (right), we feed sketches of four different chairs into our cat-drawing model to produce four "chair-like cats". We can even interpolate between the four images to explore the latent space of chair-like cats, and select from a large grid of generated designs.

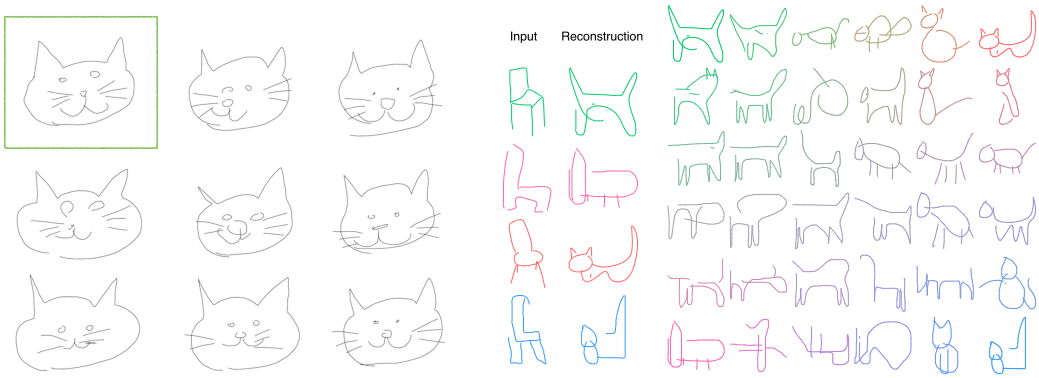


Figure 8: Generating similar, but unique sketches based on a single human sketch in the box (left). Latent space of generated cats conditioned on sketch drawings of chairs (right).

A model trained on higher quality sketches may find its way into educational applications that can help teach students how to draw. Even with the simple sketches in `QuickDraw`, the authors of this work have become much more proficient at drawing animals, insects, and various sea creatures after conducting these experiments. A related application is to encode a crude, poorly sketched drawing and generate more aesthetically looking reproductions by using a model trained with a high w_{KL} setting and sampling with a low temperature τ to produce a more coherent version of the drawing. In the future, we can also investigate augmenting the latent vector in the direction that maximizes the aesthetics of the drawing by incorporating user-rating data into the training process.

Combining hybrid variations of sequence-generation models with unsupervised, cross-domain pixel image generation models, such as Image-to-Image models [3, 13, 18], is another exciting direction that we can explore. We can already combine this model with supervised, cross-domain models such as Pix2Pix [10], to occasionally generate photo realistic cat images from generated sketches of cats. The opposite direction of converting a photograph of a cat into an unrealistic, but similar looking sketch of a cat composed of a minimal number of lines seems to be a more interesting problem.

6 Conclusion

In this work, we develop a methodology to model sketch drawings using recurrent neural networks. `sketch-rnn` is able to generate possible ways to finish an existing, but unfinished sketch drawing. Our model can also encode existing sketches into a latent vector, and generate similar looking sketches conditioned on the latent space. We demonstrate what it means to interpolate between two different sketches by interpolating between its latent space, and also show that we can manipulate attributes of a sketch by augmenting the latent space. We demonstrate the importance of enforcing a prior distribution on the latent vector for coherent vector image generation during interpolation. By making available a large dataset of sketch drawings, we hope to encourage further research and development in the area of generative vector image modelling.

References

[1] C. M. Bishop. Mixture density networks. *Technical Report*, 1994.

[2] S. R. Bowman, L. Vilnis, O. Vinyals, A. M. Dai, R. Józefowicz, and S. Bengio. Generating Sentences from a Continuous Space. *CoRR*, abs/1511.06349, 2015.

[3] H. Dong, P. Neekhara, C. Wu, and Y. Guo. Unsupervised Image-to-Image Translation with Generative Adversarial Networks. *ArXiv e-prints*, Jan. 2017.

[4] M. Eitz, J. Hays, and M. Alexa. How Do Humans Sketch Objects? *ACM Trans. Graph. (Proc. SIGGRAPH)*, 31(4):44:1–44:10, 2012.

[5] I. Goodfellow. NIPS 2016 Tutorial: Generative Adversarial Networks. *ArXiv e-prints*, Dec. 2017.

[6] A. Graves. Generating sequences with recurrent neural networks. *arXiv:1308.0850*, 2013.

[7] D. Ha. Recurrent Net Dreams Up Fake Chinese Characters in Vector Format with TensorFlow, 2015.

[8] D. Ha, A. M. Dai, and Q. V. Le. HyperNetworks. In *ICLR*, 2017.

[9] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 1997.

[10] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros. Image-to-Image Translation with Conditional Adversarial Networks. *ArXiv e-prints*, Nov. 2016.

[11] J. Jongejan, H. Rowley, T. Kawashima, J. Kim, and N. Fox-Gieg. The Quick, Draw! - A.I. Experiment. *https://quickdraw.withgoogle.com/*, 2016.

[12] C. Kaae Sønderby, T. Raiko, L. Maaløe, S. Kaae Sønderby, and O. Winther. Ladder Variational Autoencoders. *ArXiv e-prints*, Feb. 2016.

[13] T. Kim, M. Cha, H. Kim, J. Lee, and J. Kim. Learning to Discover Cross-Domain Relations with Generative Adversarial Networks. *ArXiv e-prints*, Mar. 2017.

[14] D. P. Kingma, T. Salimans, and M. Welling. Improving variational inference with inverse autoregressive flow. *CoRR*, abs/1606.04934, 2016.

[15] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. *ArXiv e-prints*, Dec. 2013.

[16] B. M. Lake, R. Salakhutdinov, and J. B. Tenenbaum. Human-level concept learning through probabilistic program induction. *Science*, 350(6266):1332–1338, Dec. 2015.

[17] Y. J. Lee, C. L. Zitnick, and M. F. Cohen. Shadowdraw: Real-time user guidance for freehand drawing. In *ACM SIGGRAPH 2011 Papers*, SIGGRAPH '11, pages 27:1–27:10, New York, NY, USA, 2011. ACM.

[18] M.-Y. Liu, T. Breuel, and J. Kautz. Unsupervised Image-to-Image Translation Networks. *ArXiv e-prints*, Mar. 2017.

[19] S. Reed, A. van den Oord, N. Kalchbrenner, S. Gómez Colmenarejo, Z. Wang, D. Belov, and N. de Freitas. Parallel Multiscale Autoregressive Density Estimation. *ArXiv e-prints*, Mar. 2017.

[20] P. Sangkloy, N. Burnell, C. Ham, and J. Hays. The Sketchy Database: Learning to Retrieve Badly Drawn Bunnies. *ACM Trans. Graph.*, 35(4):119:1–119:12, July 2016.

[21] M. Schuster, K. K. Paliwal, and A. General. Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 1997.

[22] S. Simhon and G. Dudek. Sketch interpretation and refinement using statistical models. In *Proceedings of the Fifteenth Eurographics Conference on Rendering Techniques*, EGSR'04, pages 23–32, Aire-la-Ville, Switzerland, Switzerland, 2004. Eurographics Association.

[23] P. Tresset and F. Fol Leymarie. Portrait drawing by paul the robot. *Comput. Graph.*, 37(5):348–363, Aug. 2013.

[24] T. White. Sampling Generative Networks. *ArXiv e-prints*, Sept. 2016.

[25] N. Xie, H. Hachiya, and M. Sugiyama. Artist agent: A reinforcement learning approach to automatic stroke generation in oriental ink painting. In *ICML*. icml.cc / Omnipress, 2012.

[26] X. Zhang, F. Yin, Y. Zhang, C. Liu, and Y. Bengio. Drawing and Recognizing Chinese Characters with Recurrent Neural Network. *CoRR*, abs/1606.06539, 2016.

Supplementary Material for "A Neural Representation of Sketch Drawings"

1 Dataset Details

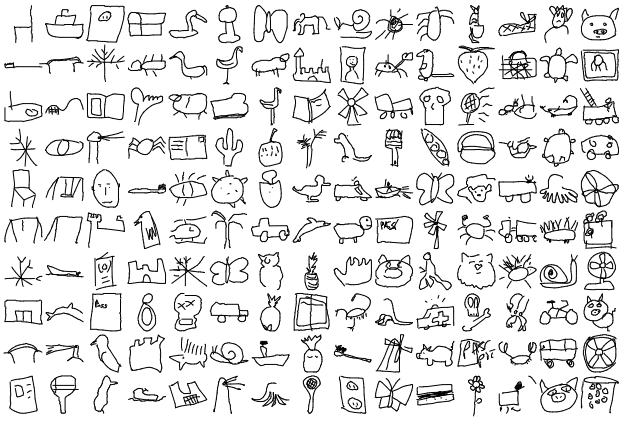


Figure 1: Example sketch drawings from `QuickDraw` dataset.

The data from *Quick, Draw!* [4] expands daily, and every so often new classes are added to the game. As such, the `QuickDraw` dataset now consists of hundreds of classes, from 75 classes initially, in Table 1. Each class consists of 70K training samples and 2.5K validation and test samples. Stroke simplification using the Ramer–Douglas–Peucker algorithm [3] with a parameter of $\epsilon = 2.0$ has been applied to simplify the lines. The data was originally recorded in pixel-dimensions, so we normalized the offsets $(\Delta x, \Delta y)$ using a single scaling factor. This scaling factor was calculated to adjust the offsets in the training set to have a standard deviation of 1. For simplicity, we do not normalize the offsets $(\Delta x, \Delta y)$ to have zero mean, since the means are already relatively small.

alarm clock	ambulance	angel	ant	barn	basket	bee
bicycle	book	bridge	bulldozer	bus	butterfly	cactus
castle	cat	chair	couch	crab	cruise ship	dolphin
duck	elephant	eye	face	fan	fire hydrant	firetruck
flamingo	flower	garden	hand	hedgehog	helicopter	kangaroo
key	lighthouse	lion	map	mermaid	octopus	owl
paintbrush	palm tree	parrot	passport	peas	penguin	pig
pineapple	postcard	power outlet	rabbit	radio	rain	rhinoceros
roller coaster	sandwich	scorpion	sea turtle	sheep	skull	snail
snowflake	speedboat	spider	strawberry	swan	swing set	tennis racquet
	the mona lisa	toothbrush	truck	whale	windmill	

Table 1: Initial 75 `QuickDraw` classes used for this work.

Figure 2 below shows a training example, before normalization of $(\Delta x, \Delta y)$ columns.

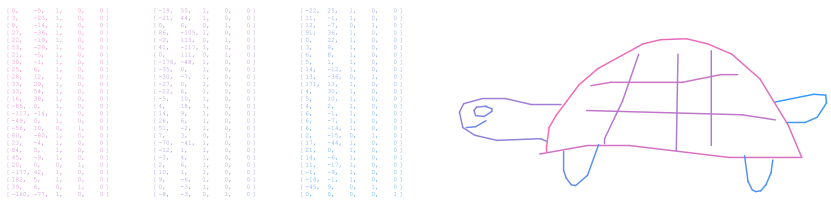


Figure 2: A sample sketch, as a sequence of $(\Delta x, \Delta y, p_1, p_2, p_3)$ points and in rendered form. In the rendered sketch, the line color corresponds to the sequential stroke ordering.

2 Training Details

As a recap from the main text, we defined the Reconstruction loss term L_R) as:

$$
\begin{aligned}
L_s &= -\frac{1}{N_{\max}} \sum_{i=1}^{N_s} \log \Big(\sum_{j=1}^{M} \Pi_{j,i} \, \mathcal{N}(\Delta x_i, \Delta y_i \mid \mu_{x,j,i}, \mu_{y,j,i}, \sigma_{x,j,i}, \sigma_{y,j,i}, \rho_{xy,j,i}) \Big) \\
L_p &= -\frac{1}{N_{\max}} \sum_{i=1}^{N_{\max}} \sum_{k=1}^{3} p_{k,i} \log(q_{k,i}) \\
L_R &= L_s + L_p
\end{aligned}
\tag{1}
$$

We also defined the KL loss term L_{KL} as:

$$
L_{KL} = -\frac{1}{2N_z} \Big(1 + \hat{\sigma} - \mu^2 - \exp(\hat{\sigma}) \Big) \tag{2}
$$

The loss function in Equation 3 is a weighted sum of both the L_R and L_{KL} loss terms:

$$
Loss = L_R + w_{KL} L_{KL} \tag{3}
$$

While the loss function in Equation 3 can be used during training, we find that annealing the KL term in the loss function (Equation 4) produced better results. This modification is only used for model training, and the original loss function in Equation 3 is still used to evaluate validation and test sets, and for early stopping.

$$
\begin{aligned}
\eta_{\texttt{step}} &= 1 - (1 - \eta_{\texttt{min}}) R^{\texttt{step}} \\
Loss_{\texttt{train}} &= L_R + w_{KL} \; \eta_{\texttt{step}} \; \max(L_{KL}, KL_{\texttt{min}})
\end{aligned}
\tag{4}
$$

We find that annealing the KL loss term generally results in better losses. Annealing the L_{KL} term in the loss function directs the optimizer to first focus more on the reconstruction term in Equation 1, which is the more difficult loss term of the model to optimize for, before having to deal with optimizing for the KL loss term in Equation 2, a far simpler expression in comparison. This approach has been used in [2, 5, 7]. Our annealing term $\eta_{\texttt{step}}$ starts at $\eta_{\texttt{min}}$ (typically 0 or 0.01) at training step 0, and converges to 1 for large training steps. R is a term close to, but less than 1.

If the distribution of z is close enough to $\mathcal{N}(0, I)$, we can sample sketches from the decoder using randomly sampled z from $\mathcal{N}(0, I)$ as the input. In practice, we find that going from a larger L_{KL} value ($L_{KL} > 1.0$) to a smaller L_{KL} value of ~ 0.3 generally results in a substantial increase in the quality of sampled images using randomly sampled $z \sim \mathcal{N}(0, I)$. However, going from $L_{KL} = 0.3$ to L_{KL} values closer to zero does not lead to any further noticeable improvements. Hence we find it useful to put a floor on L_{KL} in the loss function by enforcing $\max(L_{KL}, KL_{\texttt{min}})$ in Equation 4.

The $KL_{\texttt{min}}$ term inside the $\max$ operator is typically set to a small value such as 0.10 to 0.50. This term will encourage the optimizer to put less focus on optimizing for the KL loss term L_{KL} once it is low enough, so we can obtain better metrics for the reconstruction loss term L_R. This approach is similar to the approach described in [7] as *free bits*, where they apply the $\max$ operator separately inside each dimension of the latent vector z.

3 Model Configuration

Our encoder and decoder RNNs consist of 512 and 2048 nodes respectively. In our model, we use $M = 20$ mixture components for the decoder RNN. The latent vector z has $N_z = 128$ dimensions. We apply Layer Normalization [1] to our model, and during training apply recurrent dropout [9] with a keep probability of 90%. We train the model with batch sizes of 100 samples, using Adam [6] with a learning rate of 0.0001 and gradient clipping of 1.0. All models are trained with $KL_{\texttt{min}} = 0.20$, $R = 0.99999$. During training, we perform simple data augmentation by multiplying the offset columns $(\Delta x, \Delta y)$ by two IID random factors chosen uniformly between 0.90 and 1.10. Unless mentioned otherwise, all experiments are conducted with $w_{KL} = 1.00$.

4 Model Limitations

Although `sketch-rnn` can model a large variety of sketch drawings, there are several limitations in the current approach we wish to highlight. For most single-class datasets, `sketch-rnn` is capable of modelling sketches up to around 300 data points. The model becomes increasingly difficult to train beyond this length. For our dataset, we applied the Ramer–Douglas–Peucker algorithm [3] to simplify the strokes of the sketch data to less than 200 data points while still keeping most of the important visual information of each sketch.

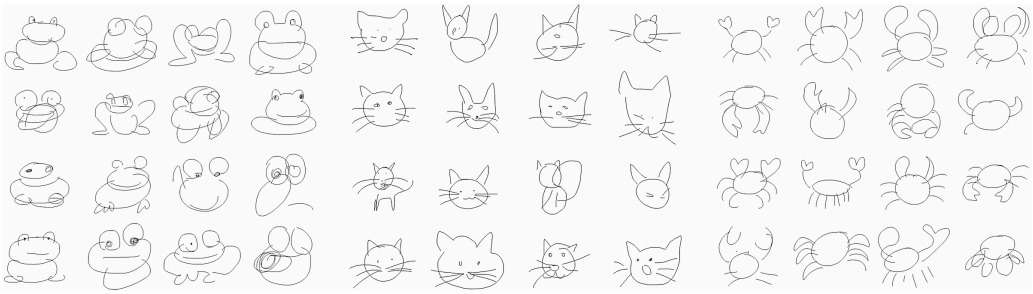


Figure 3: Unconditional generated sketches of frogs, cats, and crabs at $\tau = 0.8$.

For more complicated classes of images, such as mermaids or lobsters, the reconstruction loss metrics are not as good compared to simpler classes such as ants, faces or firetrucks. The models trained on these more challenging image classes tend to draw smoother, more circular line segments that do not resemble individual sketches, but rather resemble an averaging of many sketches in the training set. We can see some of this artifact in the frog class, in Figure 3. This smoothness may be analogous to the blurriness effect produced by a Variational Autoencoder [8] that is trained on pixel images. Depending on the use case of the model, smooth circular lines can be viewed as aesthetically pleasing and a desirable property.

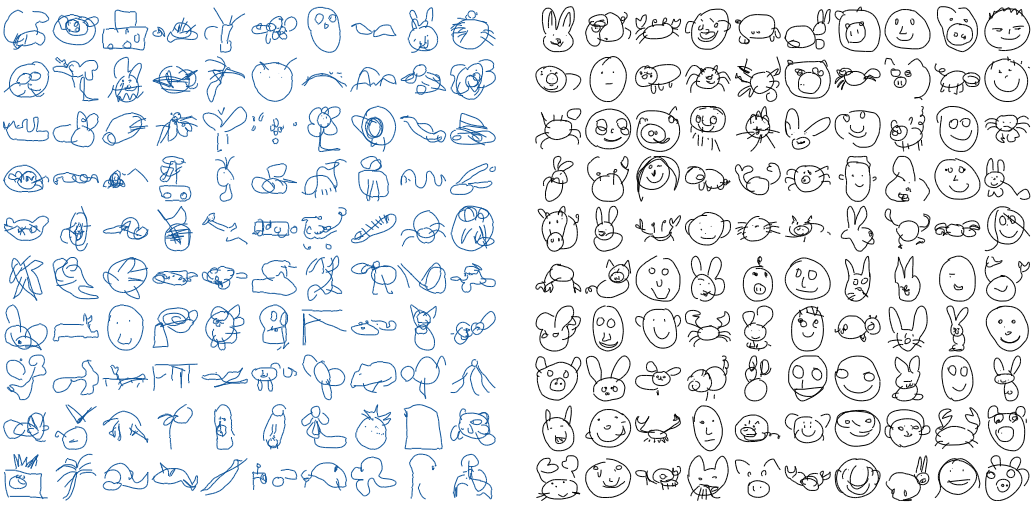


Figure 4: Unconditional generations from model trained on 75 classes (left),
From model trained on crab, face, pig and rabbit classes (right).

While both conditional and unconditional models are capable of training on datasets consisting of several classes, such as (cat, pig), and (crab, face, pig, rabbit), `sketch-rnn` is ineffective at modelling a large number of classes simultaneously. In Figure 4, we sample sketches using an unconditional model trained on 75 classes, and a model trained on 4 classes. The samples generated from the 75-class model are incoherent, with individual sketches displaying features from multiple classes. The four-class unconditional model usually generates samples of a single class, but occasionally also combines features from multiple classes. In the future, we will explore incorporating class information outside of the latent space to handle the modelling of a large number of classes simultaneously.

5 Multi-Sketch Drawing Interpolation

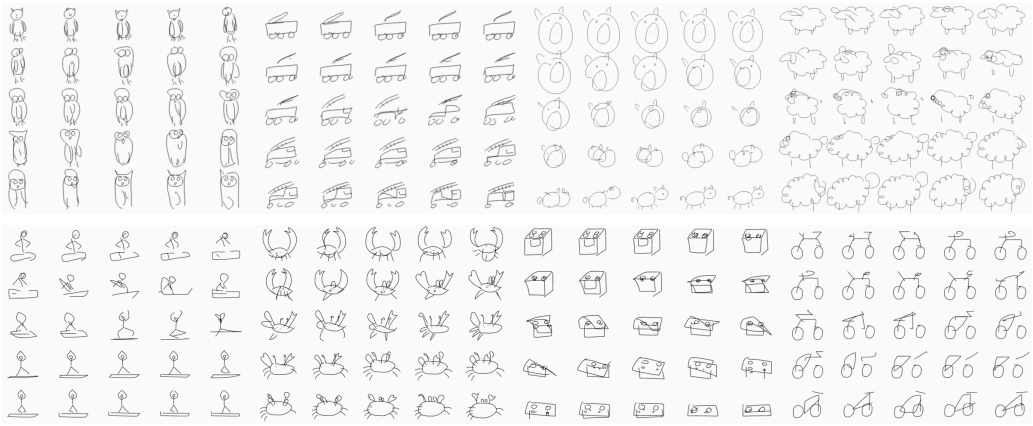


Figure 5: Example of conditional generated sketches with single class models.
Latent space interpolation from left to right, and then top to bottom.

In addition to interpolating between two sketches, like in Figure 5, we can also visualize the interpolation between four sketches in latent space to gain further insight from the model. In this section we show more examples conditionally generated with `sketch-rnn`. We take four generated images, place them on four corners of a grid, and populate the rest of the grid using the interpolation of the latent vectors at the corners. Figure 6 shows two examples of this four-way interpolation, using models trained on both (cat, pig) classes, and face class. All samples generated with $\tau = 0.1$.

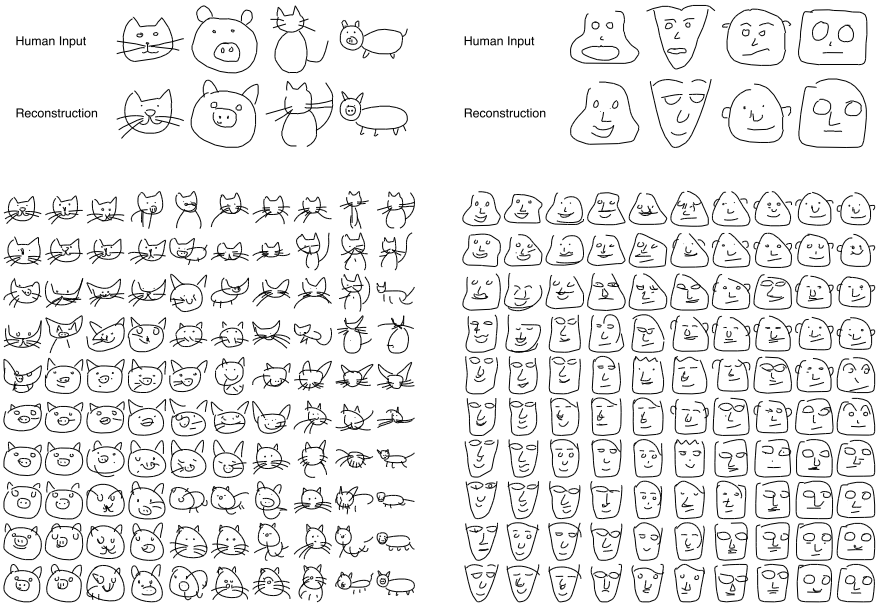


Figure 6: Example input sketches and `sketch-rnn` generated reproductions (Top),
Latent space interpolation between the four reproduced sketches (Bottom).

The left side of Figure 7 visualizes the interpolation between a full pig, a rabbit's head, a crab, and a face, using a model trained on these four classes. In certain parts of the space between a crab and a face is a rabbit's head, and we see that the ears of the rabbit becomes the crab's claws. Applying the model on the yoga class, it is interesting to see how one yoga position slowly transitions to another via a set of interpolated yoga positions generated by the model. For visual effect, we also interpolate between four distinct colors, and color each sketch using a unique interpolated color.

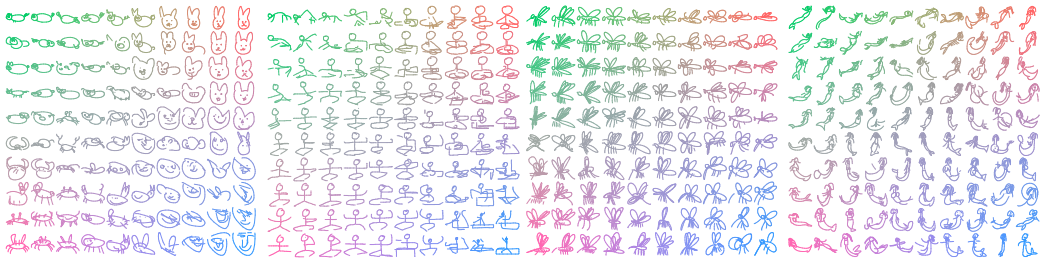


Figure 7: Interpolation of (pig, rabbit, crab and face), yoga poses, mosquitoes and mermaids. We also interpolate between four distinct colors for visual effect.

We also construct latent space interpolation examples for the mosquito class and the mermaid class, in the last two grids Figure 7. We see that the model can interpolate between concepts such as style of wings, leg counts, and orientation. In Figure 8 below, we show more interpolation examples of other classes from the dataset.

Figure 8: Latent space interpolation between four generated gardens, owls, cats, and firetrucks.

6 Which Loss Controls Image Coherency?

We would like to question the relative importance of the reconstruction loss term L_R, relative to the KL loss term L_{KL}, when our goal is to produce higher quality image reconstructions. While our reconstruction loss term L_R optimizes for the log-likelihood of the set of strokes that make up a sketch, this metric alone does not give us any guarantee that a model with a lower L_R number will produce higher quality reconstructions compared to a model with a higher L_R number.

For example, imagine a simple sketch of an face, ☺, where most of the data points of S are be used to represent the head, and only a minority of points represent facial features such as the eyes and mouth. It is possible to reconstruct the face with incoherent facial features, and yet still score a lower L_R number compared to another reconstruction with a coherent and similar face, if the edges around the incoherent face are generated more precisely.

In Figure 9, we compare the reconstructed images generated using models trained with various w_{KL} settings. In the first three examples from the left, we train our model on a dataset consisting of four image classes (crab, face, pig, rabbit). We deliberately sketch input drawings that contain features of two classes, such as a rabbit with a pig mouth and pig tail, a person with animal ears, and a rabbit with crab claws. We see that the model trained using higher w_{KL} weights, tend to generate sketches with features of a single class that look more coherent, despite having lower L_{KL} numbers. For instance, the model with $w_{KL} = 1.00$ omit pig features, animal ears, and crab claws from its reconstructions. In contrast, the model with $w_{KL} = 0.25$, with higher L_{KL}, but lower L_R numbers tries to keep both inconsistent features, while generating sketches that look less coherent.

In the last three examples in Figure 9, we repeat the experiment on models trained on single-class images, and see similar results even when we deliberately choose input samples from the test set with noisier lines.

If we look at the interpolations produced in the Latent Space Interpolation section in the main text, models with better KL loss terms also generate more meaningful reconstructions from the interpolated

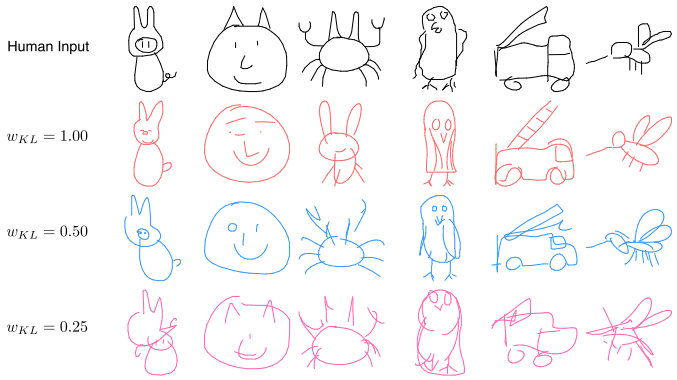


Figure 9: Reconstructions of sketch images using models with various w_{KL} settings.

space between two latent vectors. This suggests the latent vector for models with lower L_{KL} control more meaningful parts of the drawings, such as controlling whether the sketch is an animal head only or a full animal with a body, or whether to draw a cat head or a pig head. Altering such latent vectors can allow us to directly manipulate these animal features. Conversely, altering the latent codes of models with higher L_{KL} results in scattered movement of individual line segments, rather than alterations of meaningful conceptual features of the animal.

This result is consistent with incoherent reconstructions seen in Figure 9. With a lower L_{KL}, the model is likely to generate coherent images given any random z. Even with a non-standard, or noisy, input image, the model will still encode a z that produces coherent images. For models with lower L_{KL} numbers, the encoded latent vectors contain conceptual features belonging to the input image, while for models with higher L_{KL} numbers, the latent vectors merely encode information about specific line segments. This observation suggests that when using `sketch-rnn` on a new dataset, we should first try different w_{KL} settings to evaluate the tradeoff between L_R and L_{KL}, and then choose a setting for w_{KL} (and $KL_{\texttt{min}}$) that best suit our requirements.

7 Acknowledgements

We thank Ian Johnson, Jonas Jongejan, Martin Wattenberg, Mike Schuster, Thomas Deselaers, Ben Poole, Kyle Kastner, Junyoung Chung and Kyle McDonald for their help with this project. This work was done as part of the Google Brain Residency program (`g.co/brainresidency`).

References

[1] J. L. Ba, J. R. Kiros, and G. E. Hinton. Layer normalization. *NIPS*, 2016.

[2] S. R. Bowman, L. Vilnis, O. Vinyals, A. M. Dai, R. Józefowicz, and S. Bengio. Generating Sentences from a Continuous Space. *CoRR*, abs/1511.06349, 2015.

[3] D. H. Douglas and T. K. Peucker. Algorithms for the reduction of the number of points required to represent a digitized line or its caricature. *Cartographica: The International Journal for Geographic Information and Geovisualization*, 10(2):112–122, Oct. 1973.

[4] J. Jongejan, H. Rowley, T. Kawashima, J. Kim, and N. Fox-Gieg. The Quick, Draw! - A.I. Experiment. *https://quickdraw.withgoogle.com/*, 2016.

[5] C. Kaae Sønderby, T. Raiko, L. Maaløe, S. Kaae Sønderby, and O. Winther. Ladder Variational Autoencoders. *ArXiv e-prints*, Feb. 2016.

[6] D. Kingma and J. Ba. Adam: A method for stochastic optimization. In *ICLR*, 2015.

[7] D. P. Kingma, T. Salimans, and M. Welling. Improving variational inference with inverse autoregressive flow. *CoRR*, abs/1606.04934, 2016.

[8] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. *ArXiv e-prints*, Dec. 2013.

[9] S. Semeniuta, A. Severyn, and E. Barth. Recurrent dropout without memory loss. *arXiv:1603.05118*, 2016.

arXiv:1907.06417v2 [cs.CV] 23 Oct 2019

Quick, Stat!: A Statistical Analysis of the Quick, Draw! Dataset

Raul Fernandez-Fernandez (✉), Juan G. Victores, David Estevez, and Carlos Balaguer

All of the authors are members of the Robotics Lab research group within the Department of Systems Engineering and Automation, Universidad Carlos III de Madrid (UC3M).
rauferna@ing.uc3m.es

Abstract. The Quick, Draw! Dataset is a Google dataset with a collection of 50 million drawings, divided in 345 categories, collected from the users of the game Quick, Draw![1]. In contrast with most of the existing image datasets, in the Quick, Draw! Dataset, drawings are stored as time series of pencil positions instead of a bitmap matrix composed by pixels. This aspect makes this dataset the largest doodle dataset available at the time. The Quick, Draw! Dataset is presented as a great opportunity to researchers for developing and studying machine learning techniques. Due to the size of this dataset and the nature of its source, there is a scarce of information about the quality of the drawings contained. In this paper a statistical analysis of three of the classes contained in the Quick, Draw! Dataset is depicted: *mountain*, *book* and *whale*. The goal is to give to the reader a first impression of the data collected in this dataset. For the analysis of the quality of the drawings a Classification Neural Network was trained to obtain a classification score. Using this classification score and the parameters provided by the dataset, a statistical analysis of the quality and nature of the drawings contained in this dataset is provided.

Keywords: Quick, Draw! Dataset, Statistical Analysis, Neural Networks.

1 Introduction

Since the introduction of Deep Learning techniques there has been a tendency to increase the number of layers in order to increase the performance of these architectures [1,2,3]. As a result of this tendency, in the present, it is common to find Deep Neural Network architectures with hundreds of layers. These architectures require very large amounts of training data in order to obtain useful results. To address this problem, some techniques have been proposed to deal with the problem of lacking sufficient amounts of training data [4] by generating new data from the limited available data. However, having a rich and diverse dataset is still a much more effective way to correctly train a Neural Network (NN) [5,6,7]. The Quick, Draw! Dataset [8] is presented as the largest sketch dataset available at the time. The Quick, Draw! Dataset collects the sketches of the players of the Quick, Draw! open online game. In this game, the players are given a category they have to sketch in a maximum time limit of 20 seconds. At the same time, a classification Neural

[1] https://quickdraw.withgoogle.com/

Network attempts to guess the category that the player is drawing. The player wins if the NN correctly guesses the category of what the player was prompted to sketch within the time limit.

What is more interesting about this dataset is that it does not store the sketch performed by the player as a bitmap matrix composed by pixels. Instead, the sketch is saved as time series composed by pencil positions (x,y). Therefore, in addition to the spatial information that can be contained in an image, this dataset also contains temporal information of the sketch. As additional information, with each sketch, the dataset also include the category the player was asked to sketch, the timestamp when the sketch was created, the country where the player was located, and a unique sketch identifier.

Fig. 1. First three rows: Random examples extracted from the Quick, Draw! Dataset of the three categories studied in this paper. From top to bottom: *mountain*, *book*, *whale*. Last two rows: The first row depicts good sketches with high classification scores. The second row depicts examples of bad sketches with low classification scores. From left to right: *mountain*, *book*, *whale*.

In this paper, a statistical analysis of three of the categories presented in this dataset: *mountain*, *book* and *whale*, is performed. These three categories were selected to be different enough in terms of complexity to be considered valid representatives of the wide spectrum of sketches in the Quick, Draw! Dataset. The *mountain* category is considered an example of a low complexity sketch; the *whale*

category is a representative of a more complex spectrum of sketches; while the *book* category can be considered an average complexity sketch, see Fig. 1 for reference. From each of the sketches presented in each of the categories, three parameters are extracted: the classification score of the sketch, extracted using the classification Neural Network; the number of strokes performed by the player; and the sketch length, defined as the total number of line segments that defines the sketch. We consider the rest of the available information (country, timestamp...) less relevant for the study performed in this paper.

For the classification Neural Network, a Deep Convolutional Recurrent Neural Network with Long Short Time memory (LSTM) layers, explained in Section 4, is implemented. This NN is trained using the Quick, Draw! Dataset. After training, each of the sketches is assigned a score. This score is the ouput assigned by the classification Neural Network to the sketch category, given the sketch as an input to the network. The rest of the parameters can be directly obtained from the dataset.

The remainder of this paper is organised as follows. First, a study of related works using the Quick, Draw! Dataset is presented in Section 2. Section 3 depicts a more detailed introduction to the dataset. Section 4 introduces the architecture of the Classification Neural Network implemented. The setup used for the experiments is explained in Section 5. To finish, the final section of results in Section 6 and conclusions in Section 7.

Fig. 2. Examples extracted from alternative datasets to the Quick, Draw! Dataset. Each row depicts random examples extracted from a different dataset. From top to bottom: CUHK Face Sketch Database [9]; Eitz dataset of human sketches [10]; the Sketchy Database [11] and Li countour dataset [12].

2 State of the Art: Sketch Datasets

In addition to the Quick, Draw! Dataset, there are a number of alternatives where the authors created sketch datasets in order to train their machine learning models. The CUHK Face Sketch Database contains a total of 606 sketches of human faces, each with their corresponding photo [9]. In the original paper, the authors proposed a Markov Random Field model to synthesize photos to sketches and vice versa. Eitz et al. proposed a dataset containing 20 thousand sketches divided in 250 categories [10]. Two different machine learning methods, Nearest-Neighbor and Support Vector Machines, were used for the classification of the sketch dataset. The results of these methods were compared with the ones obtained with human classification. The Sketchy database [11] presented by Sangkloy et al. is a collection of more than 70 thousand sketches, each paired with a photo, divided in 125 categories. Here, the authors used Convolutional Neural Networks to map the photo and sketches to the same feature space. Recently, Li et al. presented the contour drawing dataset [12]. The countour drawing dataset contain 5000 sketches with 1000 paired photos that can contain multiple objects. The original application of this dataset was to train a deep learning method for sketch generation. All of these datasets are open and can be downloaded in the web page of each project.

3 Quick, Draw! Dataset

The Quick, Draw! Dataset contains a total of 50 million sketches divided in 345 categories. This dataset is the result of extracting the sketches provided by the players of the online game Quick, draw!. Compared to state of the art alternatives, it is much larger and may contain much richer information. As such, the quality of the sketches may experiment important variations, which may be positive for training and posterior recognition.

Sketches are stored in the ".ndjson" format[2]. Each sketch is a structure divided in different fields containing the following information: a unique id; the category where the sketch belongs; a recognition flag indicating if the sketch was recognized by the classification network; the timestamp of the sketch; the country of the player; and an array with the drawing information. This drawing information consists of N time series, where N is the number of complete strokes. Each complete stroke is encoded as a $3 \cdot T$ matrix, where the first two rows correspond to the "x" and "y" position of the pencil, the third one "t" with the timestamp, and "T" is the number of positions contained within the stroke.

As an alternative to the raw dataset, a simplified dataset is proposed by the Quick, Draw! Dataset authors. In this dataset, the authors introduce the following simplifications:

- Align the sketch to the top-left corner, and assign a zero value to that position.
- Scale the sketch to a 255x255 canvas.
- Re-sample the sketch to have a 1 pixel spacing within strokes.

[2] http://ndjson.org/

– Use the Ramer–Douglas–Peucker algorithm with an epsilon value of 2.0 to simplify the strokes.
– Delete the timestamp information.

This simplified dataset is available in three different formats: ".ndjson", ".bin" and ".npy". In this paper the ".ndjson" version of the simplified dataset is used as the input to the classification Neural Network. The ".npy" version is used for the visualization of sketches.

4 Neural Network Classification Model

The classification Neural Network takes the sketch input as a sequence of pencil positions of "x" and "y" and infers the category where the sketch belongs. Due to the nature of the dataset, a Recurrent Neural Network (RNN) using LSTM layers was implemented in this paper [3]. The idea is that in addition to the spatial information contained by the pencil position, there is also a latent space containing relevant temporal information for defining the category of the sketch. This is where RNN excels.

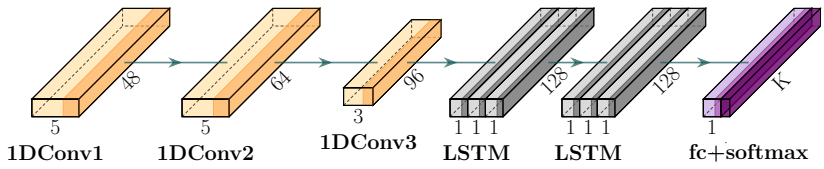


Fig. 3. Neural Network classification model.

The classification Neural Network model used in this paper is depicted in Fig. 3. The input of the NN is first passed through three 1D convolution layers. These layers relate the information of the input that is temporally close. The ouptut of the convolution stage is passed to two LSTM blocks using a *tanh* activation function. The LSTM blocks are composed by three LSTM layers each. The final layer is a Fully Connected (FC) layer with a softmax activation, being the outputs the score assigned for each of the K categories, where K is the number of training categories. For the training of the Neural Network, a total of 10000 sketches were used for the training stage with 1000 sketches for the validation stage for each category.

5 Experimental Setup

The goal of this paper is to provide a statistical analysis of the Quick, Draw! Dataset. Three parameters will be extracted to perform this analysis: the score obtained with

[3] Adapted from: https://tinyurl.com/tensorflow-models-quickdraw

the classification Neural Network proposed in section 4; the total sketch length; and the total number of complete strokes defined in each sketch. The Quick, Draw! Dataset is composed by 345 categories, each of these containing hundreds of thousands of sketches. Training a NN that is able to classify all of the categories is computationally expensive. In this paper we choose three of these categories to train the classification Neural Network. The goal is to obtain representative statistical results that can give the reader a first impression of the spectrum of sketches contained in this dataset.

The three categories studied in this paper are: *mountain*, *book* and *whale*. These three categories go from the simplicity of a mountain to the complexity of a whale, being the book the average level sketch example. The three parameters used in this paper were chosen to extract the most relevant information from the sketches. The classification score provides a value of the quality of the sketch. The sketch length and number of strokes provides information about the complexity of the sketches. Together, these three parameters provide information about the quality and complexity of the sketches.

In the following section, the results of these experiments are depicted. All the sketches contained in each of the categories studied in this paper at the time of its writing were used to extract these results.

6 Experimental Results

At the time of writing this paper, a total of 364.409 sketches were contained over the three categories studied. The distribution of these sketches between the categories was the following: 128.541 *mountain* sketches; 119.365 *book* sketches and 116.503 *whale* sketches. All of the sketches were used in order to obtain the results presented in this paper.

Table 1. Experimental results for the statistical analysis of the Quick, Draw! Dataset as a function of the three categories studied in this paper, where μ is the arithmetic mean, and σ is the standard deviation.

	Classification score		Number of strokes		Sketch length	
Category	μ	σ	μ	σ	μ	σ
mountain	5.16	2.13	2.09	2.15	18.04	15.08
book	2.96	2.67	7.21	3.78	42.50	17.17
whale	4.44	2.49	5.55	3.45	45.38	17.68

Table 1 depicts the arithmetic mean and the standard deviation of the distribution of sketches as a function of the three parameters studied in this paper. From these results, it can be concluded that the *book* is the category with a lower average classification score. This is interesting, since the *whale* category was initially a more complex one, while it has a higher average classification score. There are two possibilities that can explain this behavior: either people perform poorly at drawing

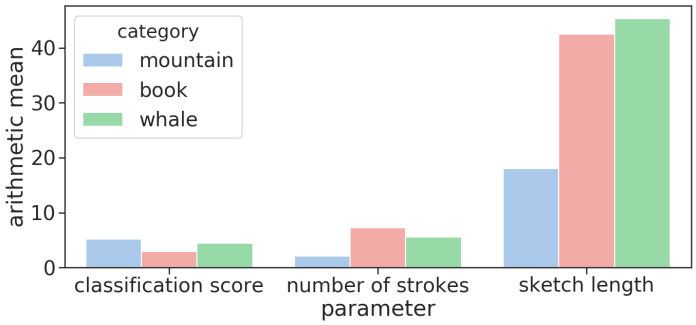


Fig. 4. Comparison between the results obtained with the three categories studied in this paper.

books or the classification network performs poorly at book classification. If we take a look at the second and third parameters depicted on the table, in average, *book* sketches are composed by a 30% more strokes than *whale* ones do with a similar length in the total drawing. Although the concept of drawing a book looks easier than that of drawing a whale, people tend to draw more complex books than whales. This can lead to a loss in the quality of the drawings, or a loss in the classification network performance. Fig. 4 provides a graphical comparison between the results obtained for each of the three parameters studied as a function of the category.

Table 2. Number of sketches as a function of its z-score divided by categories. Only the classification score is taken into account to compute this z-score.

	z-score (classification score)									
category	$-\sigma$	σ	-2σ	2σ	-3σ	3σ	-4σ	4σ	-5σ	5σ
mountain	3042	2034	1725	273	498	12	60	7	3	4
book	10245	2687	1383	126	267	16	69	2	21	0
whale	2064	5165	78	279	15	51	9	11	0	7

Table 2 contains the number of sketches as a function of its z-score. This z-score gives an impression of the number of outliers in the dataset. In the case of the *mountain* and *book* sketches the majority of outliers are in the negative region. In these two categories the population of outliers are mainly composed by people performing poor sketches. In the case of the *whale* category, the population of outliers is larger in the positive region meaning an outlier population of people performing better sketches. This gives us a first impression of the differences in the complexity of the sketches. In the two first categories, the more outlying sketches are the ones that are poorly drawn. In the case of the *whale* the more outlying sketches are the one that achieve the higher level of classification score. The results in this table also show how the number of outliers is not as high as expected for

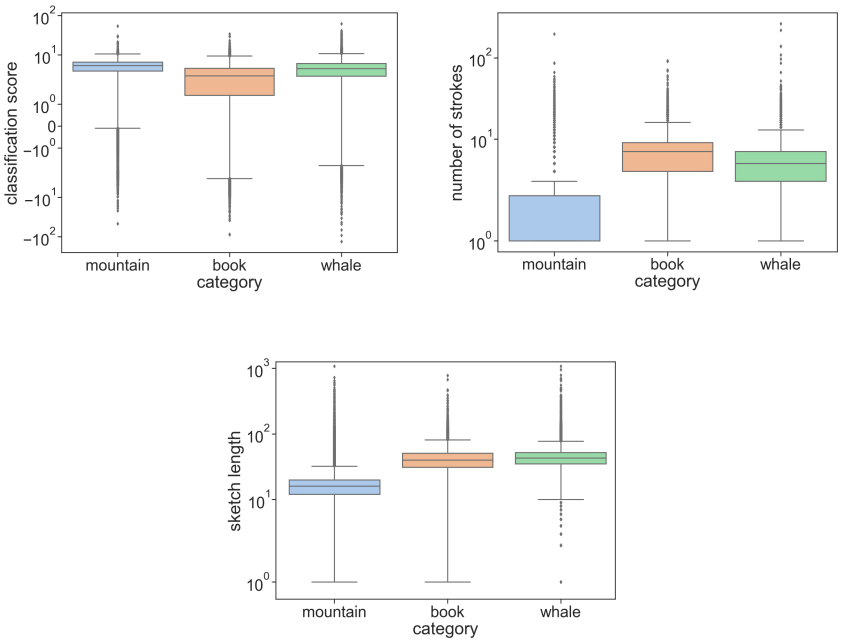


Fig. 5. Comparison between the results obtained with the three categories studied in this paper.

a dataset of this nature. In the case of the *mountain* category, around 96% of the population of mountain sketches are contained in the range $(-\sigma, \sigma)$, 89% for *book* sketches, and 94% for *whale*. Here, it should be noted that, as extracted from the Quick, Draw! Dataset repository, the dataset authors claim to have moderated the collection of drawings individually. Fig. 5 depicts various boxplot graphs for a better visualization of the distribution of sketches as a function of the three parameters and the category where they belong.

Fig. 6 depicts the relation between the classification score and the number of strokes and length of the sketches. The first thing to take into account is that both the sketch length and the number of strokes have a similar effect in the classification score achieved in each category. In the case of the *mountain* category, the peak of classification score is achieved with low values of these two parameters. Increasing the number of strokes or the sketch length comes with a loss in the classification score for this category. More simple sketches are preferred by the neural network for the classification of *mountain* sketches. The case of the *book* category in terms of shape is quite similar. The only difference here is that this peak is achieved with higher levels of strokes and sketch length. As in the *mountain* category, once reached this peak, the classification score of the sketches decrease. In the case of the *whale* category, increasing the complexity of the sketches comes in average

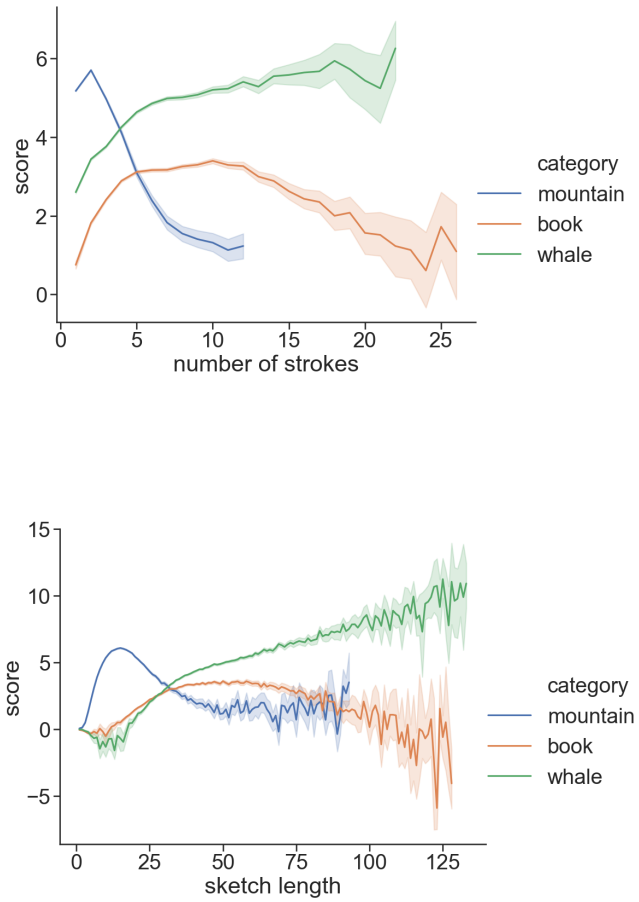


Fig. 6. Classification score as a function of the sketch length and number of strokes for the three categories presented in this paper. Sketches with a z-score bigger than 5 were not depicted in the graph for clearness reasons.

with an increase of the classification score obtained by these sketches. In this case, high complexity sketches are, in average, better in quality. This comes with a cost: higher complexity sketches also mean higher levels of variability in the quality of sketches. This effect can be found in the three categories studied; however, it is more relevant in the case of the *whale* category, since the best sketches are the ones with a higher complexity. A similar effect happens for low quality sketches in the case of the *book* and *whale* categories. For these two categories low quality sketches suffer from high variances in quality. This effect does not occur in the case of the *mountain* category.

7 Conclusions

The current tendency to increase the number of layers in Deep Neural Networks have produced models with hundreds of layers. These models require very large amounts of data to provide useful results. The Quick, Draw! Dataset, composed by over 50 million sketches, is presented as way to deal with this need of massive data for training. In this paper, a statistical study of three of the categories present in this dataset: *mountain*, *book* and *whale* is presented. The results depict the distribution of all the sketches of these three categories of the dataset as a function of three parameters: classification score, number of strokes, and sketch length. For the experiments, a classification Neural Network using LSTM layers was trained and used for prediction. The results obtained in this paper have depicted some interesting results regarding the nature of the sketches contained in this dataset.

The first relevant conclusion is that the complexity of a sketch is not only limited to the complexity of the category as would be expected. Conceptually, describing a *whale* is more complex than describing a *book*, however the results obtained in this paper show that in average, people drew more complex sketches to define a *book* than to define a *whale*. A similar concept was already introduced by [11], where a "sketchability" criterion was defined as a subjective measure of how hard it would be to sketch some of the photos. Some interesting studies can be extracted about the differences in sketch complexity between the different categories.

Another interesting result extracted from the experiments is that the number of outliers presented in the dataset is surprisingly low for a dataset of this nature. The Quick, Draw! Dataset's source is the open online game Quick, Draw!, where hundred of thousands of players around the world have accessed this game and sketched. With this premise, one would expect for this dataset to have high variances in quality between the sketches in a category, see Fig. 1 for reference. The results in the experiments depict that more than 90% of the sketches in average are contained in the range $(-\sigma, \sigma)$ for the categories presented. In addition to this, the number of outliers outside the range $(-3\sigma, 3\sigma)$ is lower than the 0.05% of the total population in each category studied. In the repository of this dataset, the authors claim to have performed a individual analysis of the sketches. This could explain the results obtained. However, there is no information about how this analysis was performed, and as Fig. 1 depicts it was not perfect. This should be taken in account before

considering this dataset a raw dataset or considering all the sketches in this dataset raw examples of each category.

8 Applications and Future Work

The Quick, Draw! Dataset may fit in two kinds of applications: applications where very large amounts of data are required (e.g. training Deep Neural Networks) and applications where the trajectory performed during the sketch, rather than the image, is required. Certain precautions must be taken into account during the interpretation of its contents, as stated in the dataset repository[4] and depicted in the results of this paper, this is not a raw dataset. Any conclusions extracted from the analysis of this dataset could be altered by possible exclusions resulting of the reviewing process. Future works using this dataset involves a more complete analysis using more categories as input, and the study of more parameters that can be interesting for machine learning techniques.

For downloading, obtaining more information or checking interesting projects involving this dataset, refer to the online repository referenced in this paper.

9 Acknowledgment

The research leading to these results has received funding from RoboCity2030-DIH-CM, Madrid Robotics Digital Innovation Hub, S2018/NMT-4331, funded by "Programas de Actividades I+D en la Comunidad de Madrid" and cofunded by Structural Funds of the EU.

References

1. Zhang, Y., Li, K., Li, K., Wang, L., Zhong, B., & Fu, Y. (2018). Image super-resolution using very deep residual channel attention networks. Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 11211 LNCS, 294–310. https://doi.org/10.1007/978-3-030-01234-2_18
2. Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2015). Going deeper with convolutions. Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 7-12-NaN-2015, 1–9. https://doi.org/10.1109/CVPR.2015.7298594
3. He, K., Zhang, X., Ren, S., & Sun, J. (2015). Deep Residual Learning for Image Recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 770-778).
4. Perez, L., & Wang, J. (2017). The Effectiveness of Data Augmentation in Image Classification using Deep Learning. arXiv:1712.04621
5. Deng, J., Dong, W., Socher, R., Li, L. J., Li, K., & Fei-Fei, L. (2009). ImageNet: A large-scale hierarchical image database. 2009 IEEE Conference on Computer Vision and Pattern Recognition, 248–255. https://doi.org/10.1109/CVPR.2009.5206848

[4] https://github.com/googlecreativelab/quickdraw-dataset

6. Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms. arXiv:1708.07747.
7. Krizhevsky, A., & Hinton, G. (2009). Learning multiple layers of features from tiny images (Vol. 1, No. 4, p. 7). Technical report, University of Toronto.
8. Jongejan J., Rowley H., Kawashima T., Kim J., & Fox-Gieg N. (2016). The Quick, Draw! - A.I. Experiment. https://quickdraw.withgoogle.com/
9. Wang, X., & Tang, X. (2009). Face Photo-Sketch Synthesis and Recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence, 31(11), 1955–1967. https://doi.org/10.1109/TPAMI.2008.222
10. Eitz, M., Hays, J., & Alexa, M. (2012). How Do Humans Sketch Objects?. ACM Trans. Graph. (Proc. SIGGRAPH)
11. Sangkloy, P., Burnell, N., Ham, C., & Hays, J. (2016). The Sketchy Database: Learning to Retrieve Badly Drawn Bunnies. ACM Trans. Graph. (Proc. SIGGRAPH)
12. Li, M., Lin, Z., Mech, R., Yumer, E., & Ramanan, D. (2019, January). Photo-Sketching: Inferring Contour Drawings from Images. In 2019 IEEE Winter Conference on Applications of Computer Vision (WACV) (pp. 1403-1412). IEEE.

To Co- Is Human

Designing Technologies That Center
Human Connection, Co-creativity, and Calm in the Era of AI

by

Shruti Dhariwal

B.Tech, CSE, Jaypee Institute of Information Technology, Noida (2008)
S.M., MAS, Massachusetts Institute of Technology (2018)

Submitted to the Program in Media Arts and Sciences, School of Architecture and Planning, in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

at the
Massachusetts Institute of Technology
May 2025

© 2025 Shruti Dhariwal. All rights reserved.
The author hereby grants to MIT a nonexclusive, worldwide, irrevocable, royalty-free license to exercise any and all rights under copyright, including to reproduce, preserve, distribute and publicly display copies of the thesis, or release the thesis under an open-access license.

Authored by: Shruti Dhariwal *
Program in Media Arts and Sciences
May 23, 2025

Certified by: Mitch Resnick
LEGO Papert Professor of Learning Research

Accepted by: Tod Machover
Academic Head, Program in Media Arts and Sciences

* in collaboration with Manuj Dhariwal. See coco.build/thesis for suggested citation.

To Co- Is Human

Designing Technologies That Center Human Connection, Co-creativity, and Calm in the Era of AI

by
Shruti Dhariwal

Submitted to the Program in Media Arts and Sciences, School of Architecture and Planning, in partial fulfillment of the requirements for the degree of Doctor of Philosophy

ABSTRACT:
In an era where Artificial Intelligence (AI) systems are increasingly framed as our "companions" and "co-creators," this dissertation reclaims "co-" as a fundamental marker of shared human experience—using it as a foundation to reimagine and build technologies that consciously center interhuman connection and co-creativity. Central to this work, we've developed CoCo (coco.build)—a general-purpose, real-time co-creative learning platform that empowers young people to engage in a wide variety of safe, shared creative experiences with their peers, spanning creative computing, AI education, digital art, writing, and more. Through the platform, we showcase how digital environments can move beyond isolated modes of learning and creating to support multiple ways of being creative together with others—introducing a new paradigm for real-time digital collaboration. We further illuminate how CoCo has been envisioned as a "self-less" social platform that de-emphasizes comparison-based, self-centric metrics (profiles, likes, followers) prevalent in most online systems for youth. We anchor these interconnected ideas in a unifying theme of "Being. Creative. Together."—reflecting timeless values that have become especially timely in an era when AI tools can further accentuate individualized digital experiences for young people. We supplement the broader design, technical, practical, and pedagogical contributions of this work by sharing insights and feedback from pilots with over 2,000 young people and educators across diverse settings. Ultimately, we see this dissertation as both a contribution and a call—to preserve the human essence of co-, to distinguish it from the useful, powerful, but instrumental AI interactions, and to shape digital environments that nurture young people's capacity to co-imagine, co-create, co-learn, co-exist, and co-evolve—with and through one another.

Note: This work has been co-developed with Manuj Dhariwal. See coco.build/thesis for suggested citation and updates on this work.
Thesis supervisor: Mitch Resnick
Title: LEGO Papert Professor of Learning Research

To Co- Is Human

Designing Technologies that Center
Human Connection, Co-creativity, and Calm in the Era of AI

by

Shruti Dhariwal

This dissertation has been reviewed and approved by the following committee members:

Mitch Resnick

LEGO Papert Professor of Learning Research
MIT Media Lab

Sep Kamvar

Computer Scientist,
Previously: Director, Social Computing Group, MIT Media Lab

Ge Wang

Associate Professor, Music (and, by courtesy, Computer Science)
Stanford University

Prelude

To co- is human.

To cO-, AI?

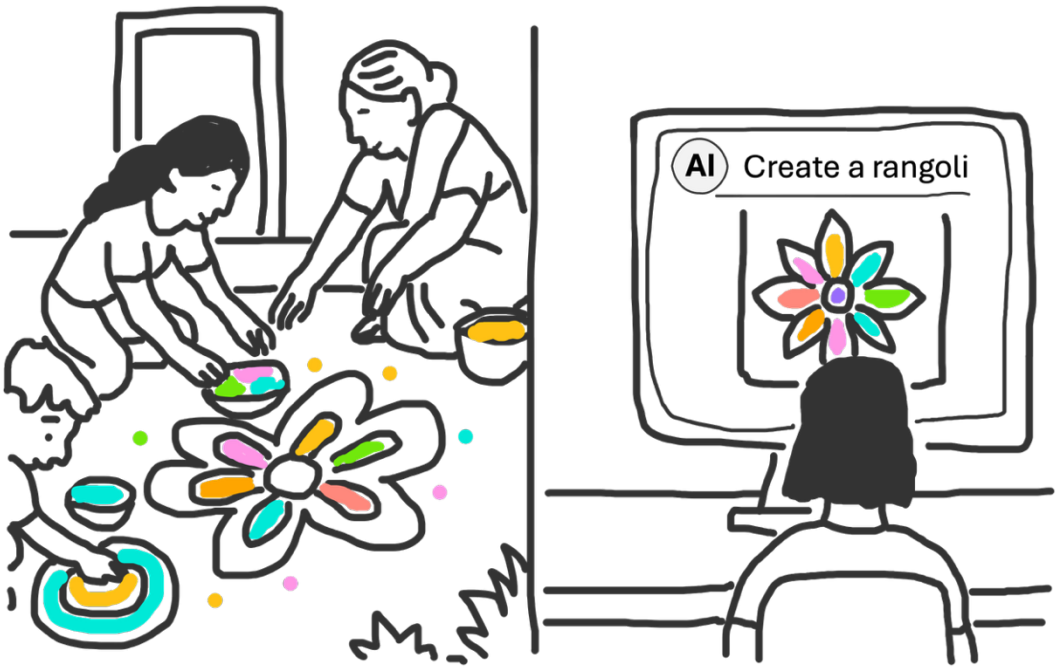


Fig 1: A sketch showing a family co-creating a traditional Indian rangoli—a floor design made with colors (left), and a young person working independently at a computer using an AI tool to generate a similar colorful pattern (right).

"**E**veryone wants a friend to just talk to in real life," reflected thirteen-year-old Sabrina. "You can never get that sense of community and support from AI. You can't really talk to AI about friendship problems… or well, I guess you won't have any if your only friend is AI," she added with a chuckle.

We shared a laugh when she said that—but the irony is, Sabrina's words aren't far from where things are headed. It's May 2025. Mark Zuckerberg has recently announced his ambition to combat loneliness by giving everyone AI "companions"—another step toward blurring the line between human and artificial relationships.

Every day, AI tools and chatbots are being pitched as our "co-workers," "co-creators," or in other terms that, not long ago, unequivocally described human relationships. The more comfortable we get in adopting this language, the more we risk skewing our relationships—not just with machines, but with one another.

How can we update our vocabulary to reinforce and protect this important boundary?

We'd like to propose a small thought experiment: What if we preserve the prefix "co-" for our timeless relationships with other living beings, and introduce a new prefix, "c0-" (with a zero), to better describe our instrumental relationships with AI tools and chatbots? The numeral 0 would serve as a crucial reminder of the underlying mathematical and statistical nature of machine intelligence. While undoubtedly powerful, useful, astonishing even—it is ultimately reducible to 0s and 1s—to bits, not biology.

It is a subtle reframing, but one that we urgently need as AI systems become interwoven with our creative, social, and emotional worlds.

Think of the difference between co-creating with another person versus c0-creating with an AI tool, or having a c0-worker on your machine versus a co-worker you can share a coffee with. While the prefix "c0-" is just one possibility, and how to pronounce it is an open question, it creates a meaningful distinction in writing.

For instance, consider how this might play out for AI companies: Microsoft Copilot would be rebranded as C0pilot—emphasizing it is an AI assistant and not a replacement for human collaboration. And AI companies would stop presenting their chatbots as companions, instead calling them c0mpanions—reminding users not to mistake them for human, even as they get more convincing in mimicking human capabilities and behaviors.

The point is not necessarily to choose one over the other—but to clearly distinguish one from the other.

Why does this distinction matter?

The prefix co- captures a broad range of our uniquely shared experiences—of co-imagining, co-creating, and co-existing. In assigning this deeply human-oriented prefix to AI, we aren't just unduly elevating the tools, we're also undermining the meaning and value of human connection.

Amplifying objects into subjects and reducing subjects into objects are both distortions of reality. Consider how many of society's deepest failures have stemmed from a common root cause: the harmful tendencies to dehumanize and objectify people. Now with AI we run the risk of rapidly accelerating in the opposite direction: humanizing AI and treating AI-powered tools, chatbots, and other objects as if they have thoughts, feelings, or personalities.

At their core, AI models are trained by optimizing what's called an "objective function"—a mathematical way of defining a specific goal. For instance, the builders of a so-called companion bot may choose to set its objective as "maximize user engagement." Any attribution of subjectivity to such a bot conceals its fundamental nature as a tool built to be a 'means to an end'; in this case, the *end* is to keep the user chatting with the bot. An April 2025 Common Sense Media report on social AI companions highlights growing concerns around excessive chatbot use—particularly among young people—including unhealthy attachments, reduced critical thinking, and even psychological harm. These risks are likely to intensify as AI systems become more engaging, persuasive, and addictive.

A small shift in language—reclaiming back "co-" for human experiences and normalizing a new prefix for AI—can be a step toward helping young people meet this reality with more clarity. In doing so, we will also pass this distinction to the AI models who train on our language. Because if we stop calling a bot "companion," it will learn to *not* call itself one.

A call for co-creating a new vocabulary

Maybe we need a new word altogether—or several new words—to capture the variety of ways we're engaging with AI. Given that this is a reality we all must navigate together, it makes sense to collectively shape the language of this new era. As a step in

this direction, we invite you to share suggestions for new words and any thoughts in this open form: coco.build/czero. Together, we might co-imagine and co-create a vocabulary that sticks.

Every act of naming is an act of worldbuilding. Let's gift young people a world where they continue to have *friendship problems*; a world where they can leverage the power of c0-creating with AI, while also experiencing the joy of co-creating together with friends; a world where to "co-" is *undoubtedly* human[1].

---

[1] Note: A version of this prelude appeared as an op-ed in The Boston Globe, May 2025. https://www.bostonglobe.com/2025/05/21/opinion/manuj-shruti-dhariwal-mash-c0-ai-interactions/

!!

Before diving in further, we'd like to address

the **elephant** in this dissertation.

Here it is :)

A note about our joint dissertations and shared authorship

Hello! We are Shruti and Manuj. You can call us "**MASH**"—a shared author identity we have coined (from MAnuj and SHruti) to reflect and reinforce the co-creative nature of our work. At the time of co-writing this dissertation, we have been married for over 12 years—being creative together, in life and work, for a long time. A few days ago, we jointly defended our dissertation work at MIT Media Lab. It was the *first ever joint PhD defense* in the history of the lab, and possibly, MIT.

The ideas, technologies, and research presented in this dissertation have been conceived and developed together. But as per academic guidelines, each of us has submitted a separate dissertation. While the content and structure across our dissertations is largely overlapping, we chose to adopt distinct titles: *To Co- is Human* (Shruti Dhariwal) and *Being. Creative. Together.* (Manuj Dhariwal). This choice reflects our shared process while letting each of us to bring forth unique framing lenses and complementary perspectives.

At the core, however, this dissertation is an expression of our *shared* interests, values, and visions. It is not possible for us to separate our voices and contributions into neat buckets. A shared identity lets us represent our work more truthfully and authentically. We recognize that this approach runs counter to the dominant academic norms—where there is a strong emphasis on individual authorship and clearly delineated credit, even when the work is fundamentally collaborative. We are deeply

grateful to our advisor, Mitch Resnick, our dissertation committee, and the Media Lab, for supporting and trusting this creative collaboration—against the norms.

Co-creating this work has not let our individual perspectives be diminished in any way. On the contrary, we like to think that while we are *two* individuals, when we work together, we actually become *three*. There is Shruti, Manuj, and this third entity that comes into being, MASH, which represents the mutual space *between* us—a space where we continue to develop ideas we wouldn't, or couldn't, arrive at alone.

And just as a MASH cannot be turned back into potatoes, this work, too, cannot be separated back into parts. In that spirit, we hope the dissertation serves as an embodiment of the very ideas it seeks to spark.

**

Please visit coco.build/thesis for updates on this work and suggested citation that reflects equal authorship. Thank you.

To co- is human

To imagine is to co-imagine
To create is to co-create
To clutter is to co-clutter
To exist is to co-exist
To evolve is to co-evolve

We are not separate 'I's,
drifting through space and time

To human is to co-
To co- is human

**

The **5Co's** structure of this dissertation

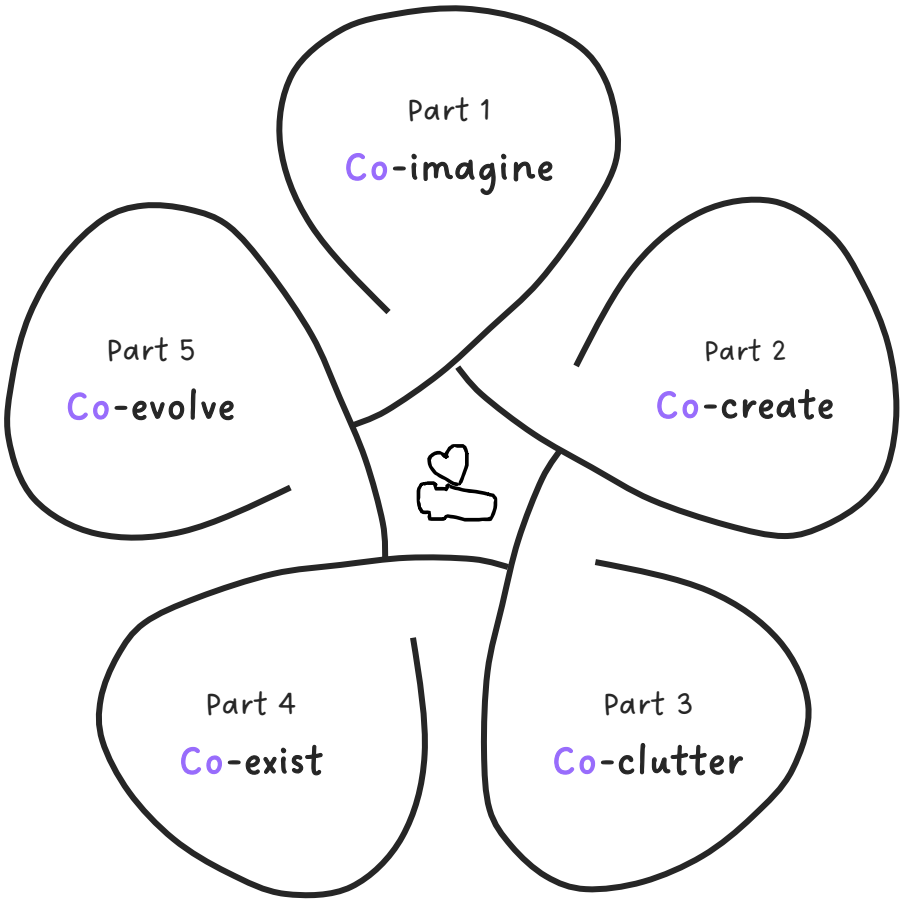


Fig 2: The 5Co's structure of this dissertation

This dissertation is rooted in the prefix "co-". It has been shaped through a lived co-experience. Like any shared journey, this document is not strictly linear or cleanly delineated. Some chapters go into detail, some are sparks, some connect to the ones that come before, some share new perspectives. They are distinct yet deeply interconnected. We invite you to explore these interconnections in your own ways.

Contents

Abstract 2
Prelude: **To co- is human. To c0-, AI?** 4
The *elephant* in this dissertation | A note about shared authorship 8
A poem on **To co-** 11

Part 1. **Co-imagine** | roots and values | 15
Ch 1. Self-expression, software, systems, and spirituality 16
Ch 2. A tale of children, mindfulness, and multiplayer shooting games 21
Ch 3. Two questions diverged in the wood 24
Ch 4. An ecology of values: "Being. Creative. Together." 25

Part 2. **Co-create** | technologies and possibilities | 27
Ch 5. A glimpse into CoCo: coco.build 28
Ch 6. Building a new "co-creative" paradigm for digital collaboration 31
6.1 Designing for togetherness over collaboration 33
6.2 Knobs over switches: Multiple ways of being creative together 34
6.2.1 Shared Presence 35
6.2.2 Playful Inspiration 36
6.2.3 Fluid Recomposition 37
6.2.4 Creative Interaction 38
6.2.5 Coordinated Collaboration 39
Ch 7. New paradigm, new possibilities: Insights from pilots with children 41
Ch 8. Extending creative tools into co-creative environments 49
Ch 9. Designing decentralized communities: Privacy, safety, cultural context 51

Part 3. **Co-clutter** | a necessary interlude | 53
Ch 10. Creating the conditions for "collision" 54

Part 4. **Co-exist** | shifts and perspectives | 56
Ch 11 From future to present 57
Ch 12. Constructing *things*, Constructing *context* 58
Ch 13. Head, hands, heart, and *hüm* 61
Ch 14. From self-inflating to self-*less* social platforms 63
Ch 15. Passion, dispassion, and compassion 68

Part 5. **Co-evolve** | contributions and directions | 69
Ch 16. The 5Co's of Co-creative Learning 70
Ch 17. From Monoculture to CoCo-culture 74
Ch 18. Bringing CoCo to communities around the world 78
Ch 19 Extending CoCo | Little Language Models (LLMs): Empowering children to be future AI modelers 80
Ch 20. Looking ahead: Centering the *timeless* in timely 92

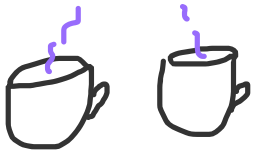
Afterword(s) 94
References 95
Acknowledgments 98

**

Part 1

Co-imagine

| roots and values |



A lot of our *co-imagining* happens through conversations over our morning chai (Indian tea). Unlike the typical notion of morning meetings, where the agenda for the day is decided, this shared routine is essentially an agenda-free zone for us. One of us might share a half-formed thought, an interesting observation, or a random what-if. The other *listens*—not always, but very often :). There is a back and forth that follows. It matters that there is mutual trust and (long) shared context. It leads to lightness and spaciousness, both in our heads and in between us. When *we* are totally *present*, but our egos are not, co-imagining happens without effort. We disagree and diverge freely, and agree and appreciate easily. It's sort of like denoising an image. Starting with a foggy impulse, and with each exchange, layering in more clarity, depth, and richness. Not alone, but *through* each other's imaginations.

How can we support young people to imagine *through* each other's imaginations?

In this section, we share a glimpse into the space of ideas, contexts, and experiences that served as the backdrop for us to co-imagine new connections and possibilities.

Ch 1.	Self-expression, software, systems, and spirituality	16
Ch 2.	A tale of children, mindfulness, and multiplayer shooting games	21
Ch 3.	Two questions diverged in the wood	24
Ch 4.	An ecology of values: "Being. Creative. Together."	25

Chapter 1

Self-expression, software, systems, and spirituality

…the greatest poet drew from no other source than
the one each of us has within his own being.

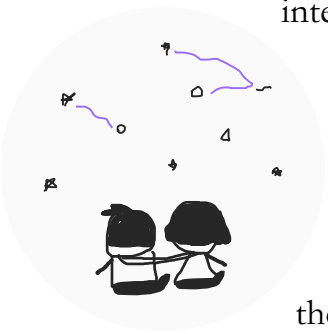
—

HERMANN HESSE

The four words in the title above may not seem to have an obvious connection. Yet, around six years ago, these themes increasingly started to surface together in our shared experiences, conversations, and imaginations. Looking back, this wasn't a coincidence.

The ideas that inspire us, the values we resonate with, the people and contexts we are drawn to—they are all reflections of who we are. They speak to our core self, while also reshaping the self over time in ways that are often hard to describe.

In this chapter, we share the initial space of ideas and contexts that served as the backdrop of this work. We highlight how they began to intersect with our shared interests and values and eventually led us to draw new connections and ask new questions. It's not a consolidated literature review. The connections will emerge organically throughout the dissertation—mirroring our own journey of developing this work. But we hope these glimpses can help spark others, just as they sparked us, to explore the space in new ways and draw their own connections and constellations over time.



For the past several years, the Lifelong Kindergarten group at MIT Media Lab has been our wonder-*full* shared context for discovering, imagining, creating, and playing with new ideas together.

We first formally encountered the term 'constructionism' coined by Seymour Papert and the ecosystem of rich educational ideas surrounding it as part of this context. We instinctively and deeply resonated with the group's core principles and approach of designing learning experiences for children rooted in 4Ps—Projects, Passion, Peers, and Play [1], [2], [3]. The Scratch online platform for children [4], that we got to be closely involved with and contribute to as part of the group, served as a wonderful playground for us to see these ideas in action through the wide range of children's creative and social learning experiences [5]. These foundational ideas around how children learn by creating; and how we can design expressive tools and environments that empower children to learn by creating [6], especially in ways that are personally meaningful to them [7]—while being so inspiring also felt very intuitive to us. It was because we were experiencing these ourselves every day. They became naturally internalized in our thinking and we could see them being reflected in all our work and explorations.

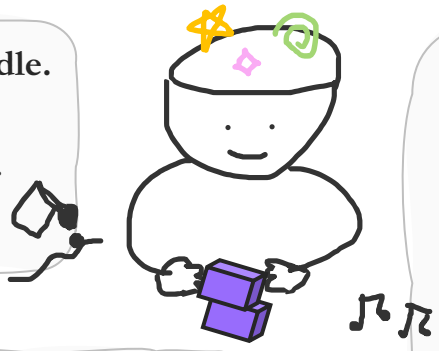
Being part of this ecosystem, we continued to be inspired by the rich heritage of powerful educational ideas throughout history around centering children's voice, creative expression, and their self-identities in the seminal works by Papert [7], Malaguzzi [8], Freire [9], Montessori [10], Dewey [11], Holt [12], and others.

Inspiring ideas on centering children's interests and creative self-expression

The child has
a hundred languages
a hundred hands
a hundred thoughts
a hundred ways of thinking
of playing, of speaking...
- **Loris Malaguzzi**

…construction that takes place "in the head" often happens especially felicitously when it is supported by **construction of a more public sort "in the world"-** a sandcastle or a cake, a Lego house or a corporation, a computer program, a poem, or a theory of the universe.
- **Seymour Papert**

Curiosity is never idle.
Doing is learning.
Knowledge is action.
- **John Holt**



Children are not listeners.
They are first and foremost **interested in moving**, communicating, exploring the world, **constructing and expressing themselves** artistically.
- **John Dewey**

An **investment in interest** always pays off with the best knowledge.
- **Mitch Resnick**

The more students work at storing the deposits entrusted to them, the less they develop the **critical consciousness** which would result from their intervention in the world as transformers of that world.
- **Paulo Freire**

The absorption of the environment is an intellectual activity. It is a psychic necessity that **the child explores the environment**; it satisfies his spirit.
- **Maria Montessori**

Making connections with our other interests…

While we were continuing to explore these ideas, we also found ourselves naturally gravitating toward our other shared life interests—in meditation, mindfulness, awareness, self.

Beyond our practice and personal experience, we also explored the themes around self and interconnectedness across a wide range of books and works in—philosophy, physics, psychology, systems, and spirituality [13], [14], [15], [16].

We deeply cared about these ideas and values and they increasingly became dominant in our everyday morning conversations over breakfast.

To understand life is to understand ourselves, and that is both the beginning and the end of education.
- J. Krishnamurty

The unit of survival is **organism plus environment**
- **G. Bateson**

Who am I?
- **Sri Ramana Maharishi**

At the outset, these two spaces seemed to be pointing in opposite directions.

One emphasizes the idea of **centering** every child's individual self-expression and self-identity as a creator. The other stresses on the importance of **de-centering** the individual self, deconstructing the notion of self, and seeing beyond the self. One highlights how children are always interested in moving and exploring the world around them, the other talks of stillness and exploring the inner world.

In his wonderful Artful Design book [17], Ge Wang shares a prompt that perfectly captures the essence of these two spaces of ideas that we were thinking about.

Things that **move** you.
Things that make you **still**.

Is there a way to combine both these things for children? We wondered.

Out of curiosity, as a small exploration around this time, we also designed a new **Mindfulness + Scratch** activity for an upcoming Scratch Day event at MIT. Prior to the workshop, several people in the group had expressed their genuine apprehension about whether this theme would be relevant for such a high energy event with kids actively exploring, moving, creating, and playing in the space.

This was Spring 2019, pre-pandemic era. At the time, the themes of cultivating mindfulness and awareness didn't feature directly in our surrounding conversations about children, creative learning, and technology. But the best part about being in a supportive research context is the freedom to explore your interests even if they may seem fuzzy initially. The workshop turned out to be great and attracted kids across ages along with their parents. They all really appreciated and enjoyed the experience. We felt there was something here.

We intuitively knew that these two *seemingly disparate* worlds were synergistic in essence, and we continued to wonder about new ways and mediums that could reflect these interconnections and synergy.

Chapter 2

A tale of children, mindfulness, and multiplayer shooting games

Fig 3: Image generated from ChatGPT for illustrative purposes

Around this time in Spring 2019, we had volunteered as group leads at a day-long mindfulness retreat for children near Boston. There were two separate incidents from the day that eventually sparked the beginnings of our ideas and work.

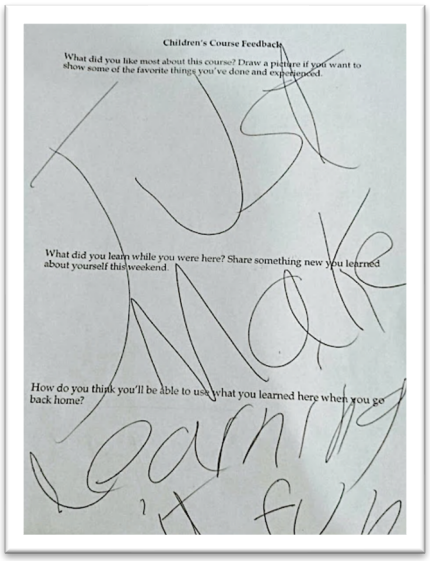
The **first** happened early in the day during a much-awaited play break after a stillness-based session. A group of 10- to 12-year-olds started talking passionately about some of the multiplayer shooting games they had been playing online. While the topic was not particularly 'appropriate' for a mindfulness retreat :), we decided it was best to not judge the conversation and let it flow.

We casually intervened and asked them, “How much time do you spend playing these games over the weekend?” to which a 12-year-old immediately responded…

“I play for 3–4 hours every day, what else is there to do!”

We talked some more. He shared the he was the only child and that his parents work till late and so in the evenings he jumped into these immersive multiplayer games. It was a way for him to avoid feeling lonely.

The **second** incident happened at the end of the day. The organizers had asked all the kids to fill a feedback form about their experience at the retreat.

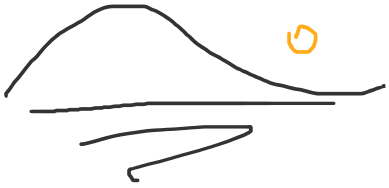


A form shared by one of the kids jumped out to us. Scribbled over the entirety of the page in big letters was a 5-word sentence that said,

“Just make learning it fun.”

The **it** here referred to the practice of mindfulness that kids were introduced to through the stillness-based sessions during the day.

These two incidents and the context in which they happened stayed with us.



Our continuing conversations after the retreat revolved around two themes…

1. Connection and Construction

As we thought about the first incident, we initially found ourselves contrasting the often passive, addictive, and close-ended nature of many online games—with the more open-ended and expressive digital making environments, such as Scratch, that are designed to let children explore a wide range of creative possibilities such as art, games, music, stories, or animations, based on their own interests.

However, in reflecting further on our own co-creative process and the comment made by the 12-year-old, we recognized that there was a big **missing piece of *real-time connection* with peers in the existing online creative learning tools and environments for children**.

2. Re-imagining social systems

The second incident prompted us to reflect on how the ideas that get reinforced by most of the existing social systems young people engage in (both online and offline) are often at odds with the deeper ideas of interconnectedness and nonjudgment that lie at the core of mindfulness. The notions of likes, followers, ranks, and scores are rooted in **individual-centric and comparison-based paradigms that can often lead to many young people feeling anxious.**

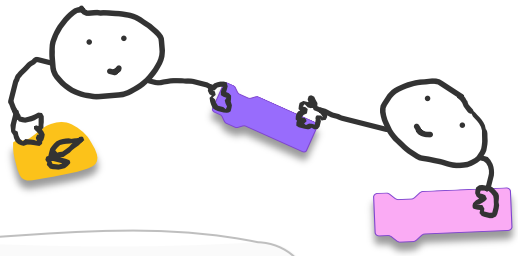
While there is value in teaching mindfulness practices to children as an approach toward helping them feel more calm and centered, we wondered what it might mean to reimagine social systems as calm playgrounds for children to encounter and explore these ideas in ways that are connected with their interests, and can help them question the dominant paradigms they are exposed to. As it turned out, this approach resonated with many parents who reached out to us later…

> "I read about your work and it speaks to me, for Rahil alas would have no patience for any mindfulness retreat or indeed any meditative experience, and envelops himself in the stimuli of Fortnite and Roblox, with his friends on mic. I often have wondered how this stimulus-driven cooperativity of his might be harnessed for learning rather than only gameplay."

Chapter 3

Two questions diverged in a wood

We drafted two guiding design-based research questions that became central to our work and explorations (even though at the time they didn't seem directly connected).

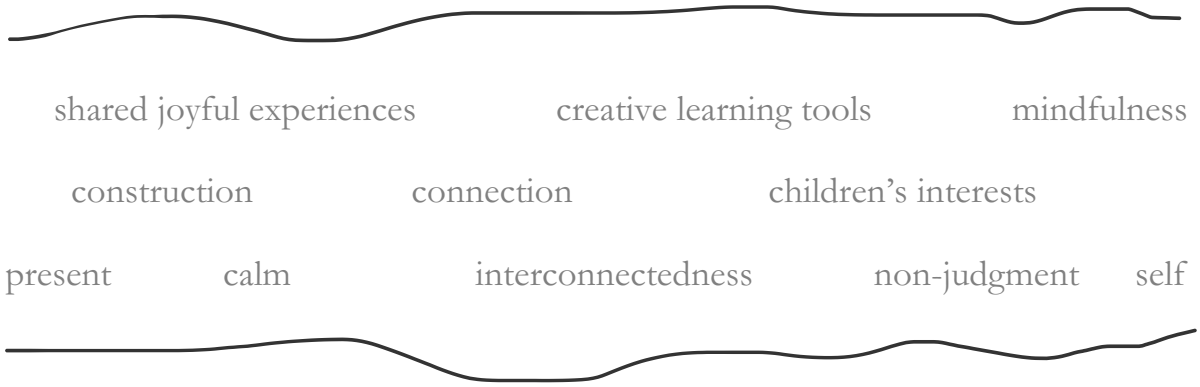


1. How can we design online creative environments for young people that **center authentic connection** with others—helping them experience the power and joy of co-creating and co-learning together?

2. How can we design online social systems for young people that **de-center the individual self—**helping them be fully engaged in the present, free of anxiety and judgment, in a shared experience with others?

Chapter 4

An ecology of values:
BEING. CREATIVE. TOGETHER.



Over time, all of our *seemingly disparate* interests, ideas, questions, and explorations morphed together into a synergistic framework of values that forms the foundation of this work. We coined the term "Being. Creative. Together." as it authentically and succinctly captures the space of intertwined ideas that we deeply care about and hope to illuminate through this work. We consciously choose to put periods next to each word in the phrase to bring awareness to the three core values of *Beingness*, *Creativeness*, and *Togetherness* that form the foundation of our work and motivations.

Beingness represents the **state** of *being*—being alive, aware, and fully engaged in the present moment. *Creativeness* is the **state** of *being creative*—experiencing joy *in* the process of creating and exploring without worrying about the outcome. And finally, *Togetherness* is the **state** of *being together*—feeling authentically connected with others through a shared experience.

Each of these values, along with their intersections, is an end-in-itself and not a means-to-an-end [17]—that is, they are not valuable for the sake of something else, but for their own sake. These are values that do not change with the changing times or evolving technologies, and thus are universal across contexts.

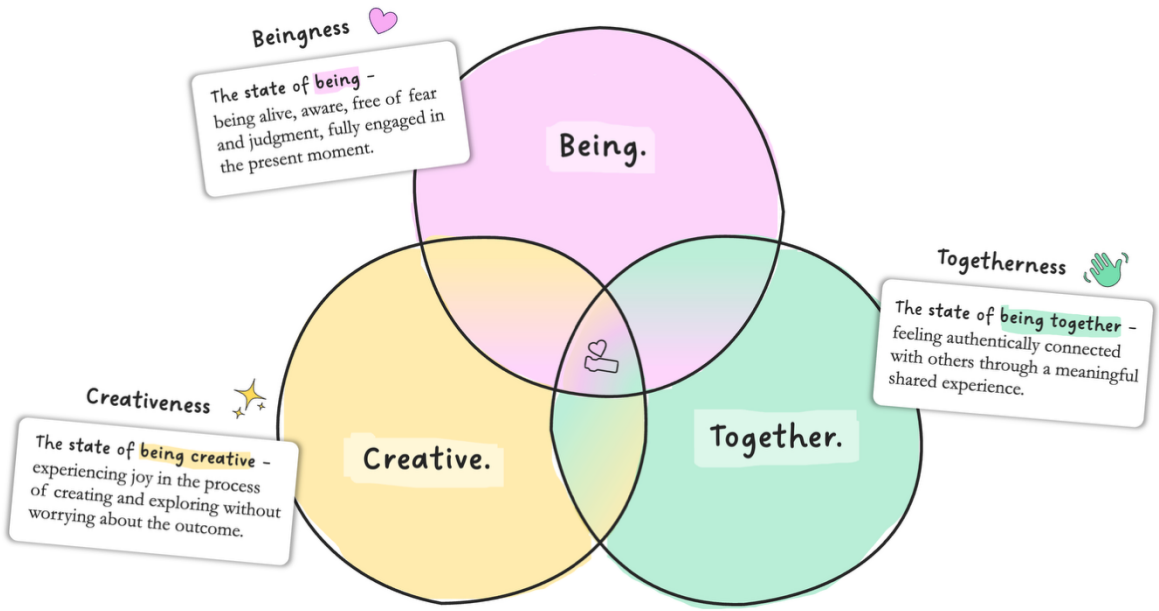


Fig 4: Being. Creative. Together.

In this framing, there is a subtle but significant distinction that we want to highlight. If we were to say that we are designing tools for *creating together*, the emphasis falls on the extrinsic action, outcome, or affordances. In contrast, when we say we are designing for *being creative together,* the focus shifts toward the intrinsic **state** or quality of the experience, one that aligns with the idea of "flow" as described by Csikszentmihalyi:

"The ability to control the environment is one side of the flow experience. The other side, paradoxically, is a feeling which seems to make the sense of control irrelevant… At the height of their involvement with the activity they **lose a sense of themselves as separate entities**, and feel harmony and even a merging of identity with the environment." [18]

While the concept of "flow" often refers to an individual creative experience, we wanted to extend the idea to explore how we can design environments that help facilitate "co-flow" experiences for young people—where they can feel free to express their full self while simultaneously losing a sense of separate self, feeling harmony *with others*. This is the state of *being. creative. together.*

Part 2

Co-create

| technologies and possibilities |

"If you want to teach people a new way of thinking, don't bother trying to teach them. Instead, give them a **tool**, the use of which will lead to new ways of thinking."
— *R Buckminster Fuller*

In this section, we highlight our ideas and approach toward embedding the core values of "being. creative. together." into the design of CoCo, a software platform that lies at the heart of this work, and introduce the novel "co-creative" paradigm for real-time digital collaboration that underlies CoCo.

Ch 5.	A glimpse into CoCo: coco.build	28
Ch 6.	Building a new "co-creative" paradigm for digital collaboration	31
	6.1 Designing for togetherness over collaboration	33
	6.2 Knobs over switches: Multiple ways of being creative together	34
Ch 7.	New paradigm, new possibilities: Insights from pilots with children	41
Ch 8.	Extending creative tools into co-creative environments	49
Ch 9.	Designing decentralized communities: Privacy, safety, cultural context	51

Chapter 5

A glimpse into CoCo:
coco.build

Fig 5: CoCo Platform Demo. (2022). YouTube. https://youtu.be/vvWt0rjco0w

CoCo (coco.build) is a new general-purpose "co-creative" learning platform that we have jointly conceived and developed as part of this research [19]. It is designed to empower children and teens to build and collaborate on an endless variety of creative projects *with their peers in real-time*—regardless of whether they are sitting next to them in a classroom or connecting remotely across continents. It supports multiple expressive mediums including creative computing (building on Scratch 3.0 and p5.js [20], digital art, writing, and others. Educators (or parents) anywhere can start their own private communities on CoCo—creating a safe online environment for young people to engage in live co-creative experiences with their friends. We invite you to watch the short YouTube video above to see CoCo in action.

Why the name CoCo?

By the end of Spring 2019, after the experience of the mindfulness retreat and the workshop with children and parents, our conversations had increasingly started featuring words starting with the letters 'co,' such as connection, co-creation, context, compassion, communities. In one of those conversations, the name CoCo stuck with us. It also nicely captured the core idea of 'Connection + Construction.'

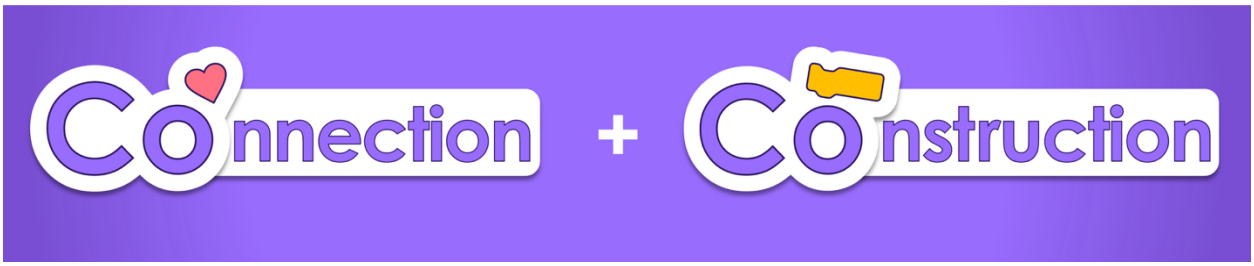


Fig 6: Connection + Construction

> Fun fact: The *.build* domain (of our platform coco.build) is meant for businesses and individuals working in the *construction* industry. We thought it was fitting for CoCo and registered it right away. Looking back, we are glad we did.

Over the course of next few months, while we were engaged in other projects and initiatives, we continued to be drawn to these ideas, and kept refining our thinking and sketches about how we can design a computational environment for young people anchored in the core values of *being. creative. together*. We had an initial working prototype ready at the start of 2020 and got very positive feedback during the first playtest with the members of our research group.

But, creating something *with love* can take time. Imagining and building CoCo has been a long and continuing journey of heart and hard work—packed with countless iterations, choices, an endless variety of software bugs, sometimes too many ideas and too few hands, and sometimes... a pandemic that put everything on pause.

We introduced CoCo to the world in a tiny way in December 2022—sharing a short video and a link to the website in a tweet—inviting people to sign up for beta. We didn't know what to expect. In the coming days (and months) as more educators and

communities came to know about it and shared it ahead, it was deeply satisfying to see the resonance in their reactions.

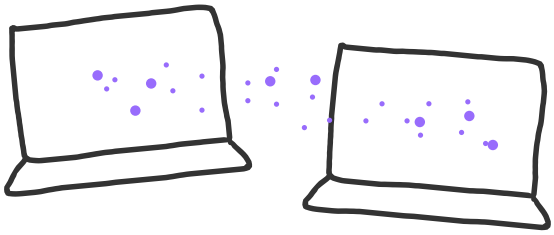
- This looks absolutely brilliant… opens up so many possibilities."
- "Ever feel like you got a glimpse of something that was about to change things?"
- "We hadn't been this excited about a youth platform in a long time."

Fig 7: Response from educators and communities

In the next chapter, we dive into the design of a new "co-creative" paradigm for digital collaboration that we introduce through CoCo, and highlight how it opens up a new space of possibilities which have so far been unexplored.

Chapter 6

Building a new "co-creative" paradigm for digital collaboration



Many of the existing creative online environments for young people, such as Scratch and others, are not currently designed to support real-time collaboration with peers. While the dominant assumption might be that collaboration is primarily useful when people are physically apart, like in a remote learning context, we recognized that regardless of whether learners are digitally together on a video calling platform like Zoom, or physically together in a classroom, they remain creatively distanced from one another.

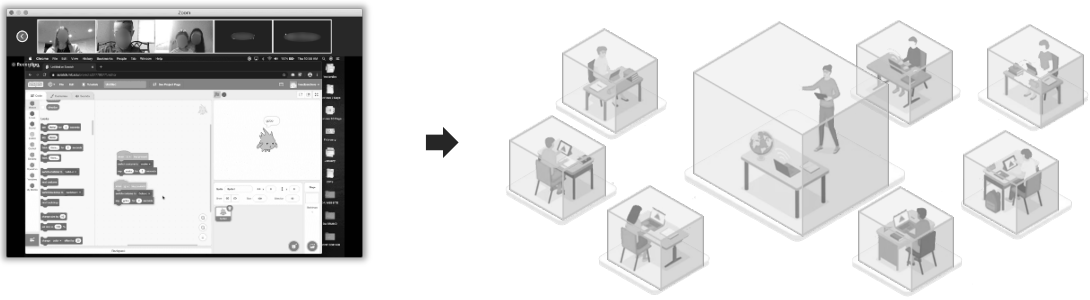


Fig 8: Digitally together but creatively distanced

For instance, on a video call, the experience often evokes the visual of learners (or educators) being stuck in glass boxes. They can see and hear each other, but cannot engage in any meaningful creative interaction with one another.

When it comes to physical spaces, even when a space may be designed to foster more interaction between learners, such as by having learners sit in pods instead of rows, the creative learning tools that young people use typically support only individual work. Children therefore largely work on their own projects on their devices and are not able to co-create with their peers in a shared digital space. The educator also has no visibility into their creative process and often has to stand behind a student to peek into their screen—which can cause some learners to feel awkward and conscious.

Fig 9: Image from booleangirl.org/clubhouse

Existing paradigms for digital collaboration

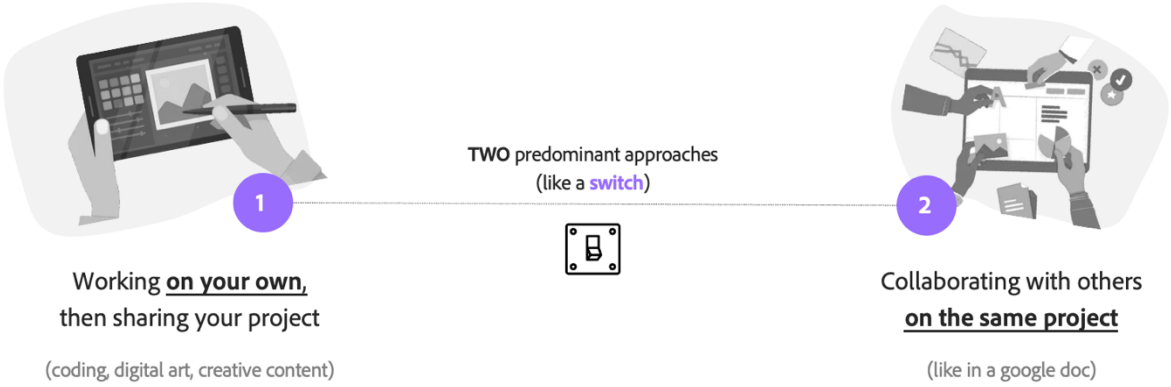


Fig 10: Two modes of online collaboration

Existing digital creative tools typically employ one of two predominant design paradigms toward integrating social interaction or collaboration with others:

- Tools for working individually on creative projects and then sharing the completed works with others in an online community (such as Scratch).

- Tools that support the Google Docs approach of collaborating with others on the same project (such as drawing on a shared digital canvas, making edits to the code in the same project, or writing together in the same document).

Collaboration doesn't always work well for everyone

There is no denying that collaborating with others can be a really joyful, productive, and energy-giving experience, but it does not *always* work well for *everyone* involved. The reasons can be many—not being able to learn and work at your own pace, not feeling comfortable to express yourself in the group, feeling or being judged, someone hogging the work, perceptions of unfairness in contributions and credit, and other negative feelings that can sometimes arise out of this experience.

> Computer interfaces should give us new abilities, useful and interesting enough that they would be used not only when people were physically apart, but even when they were able to be together. — *Hollan and Stornetta [21]*

6.1 Designing for *togetherness* over collaboration

In designing CoCo, we took a different approach—consciously centering togetherness over collaboration. We chose to design the digital environment in CoCo as a shared space for co-creating projects in real-time and not necessarily for collaborating on the same project in real-time. Collaboration often exclusively implies *working together*, while a shared space more broadly implies *being together*. It is hence inclusive of collaboration as one of the ways of engaging with others but not the only way.

In envisioning the interface and affordances, we felt inspired to capture the setting and spirit of a creative art/craft workshop space in action: a group of children sitting together around a table, making their art projects. They can peek at each other's creations and get inspiration; they can easily exchange craft materials

Fig 11: Group of kids creating art projects in a space

with one another; and, if they want, they can choose to collaborate with anyone in the space. This is a similar spirit evident in the Reggio Emilia schools in Italy, or in Computer Clubhouses [22].

6.2 Knobs over Switches: Supporting multiple ways of being creative together

Through CoCo, we introduce the design principle of *knobs over switches* to reimagine the space in between the two extremes or switches (of working on your own and working together on the same project) as a new kind of shared digital space that captures the experience of togetherness. Just like how a knob affords finer control over one's preferences, similarly, a CoCo space is designed to provide creators with the agency to choose when and in what way they'd like to engage with their peers, based on their own comfort and preferences at any moment.

Here is a spectrum of the different modes young creators can use to engage with peers in a CoCo space: 1) Shared Presence; 2) Playful Inspiration; 3) Fluid Recomposition; 4) Creative Interaction; and 5) Coordinated Collaboration

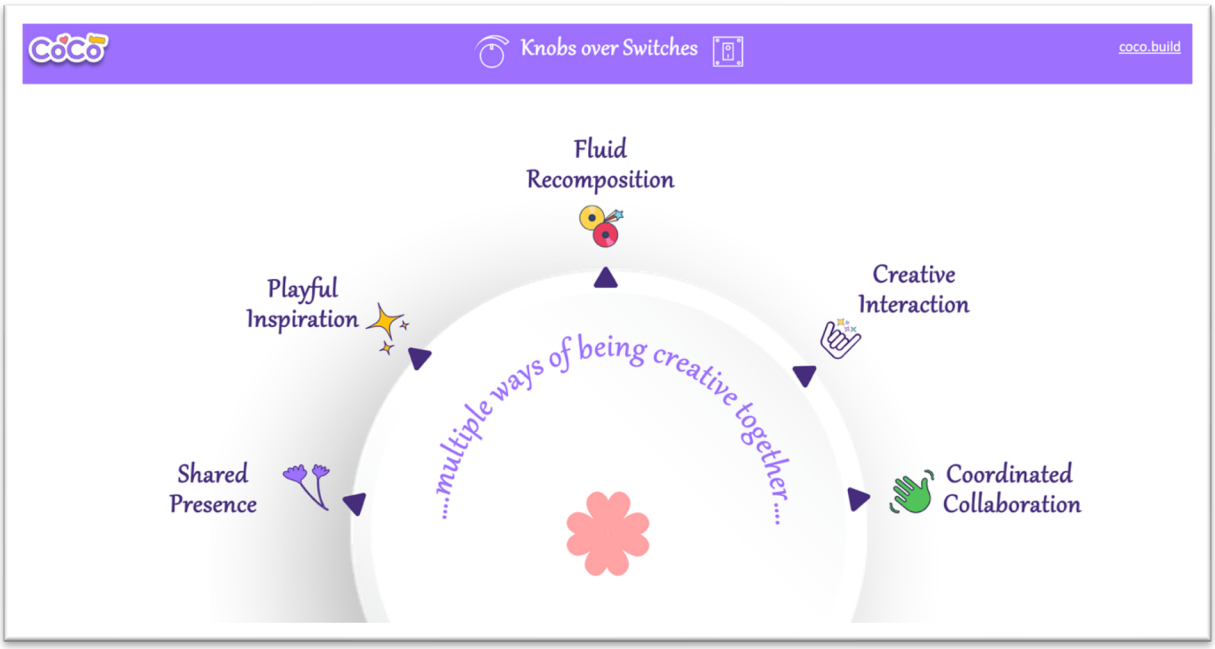


Fig 12: Knobs over Switches: Multiple ways of being creative together

Next, we describe each of these modes and highlight the space of possibilities that these open up for young people.

1) Shared presence

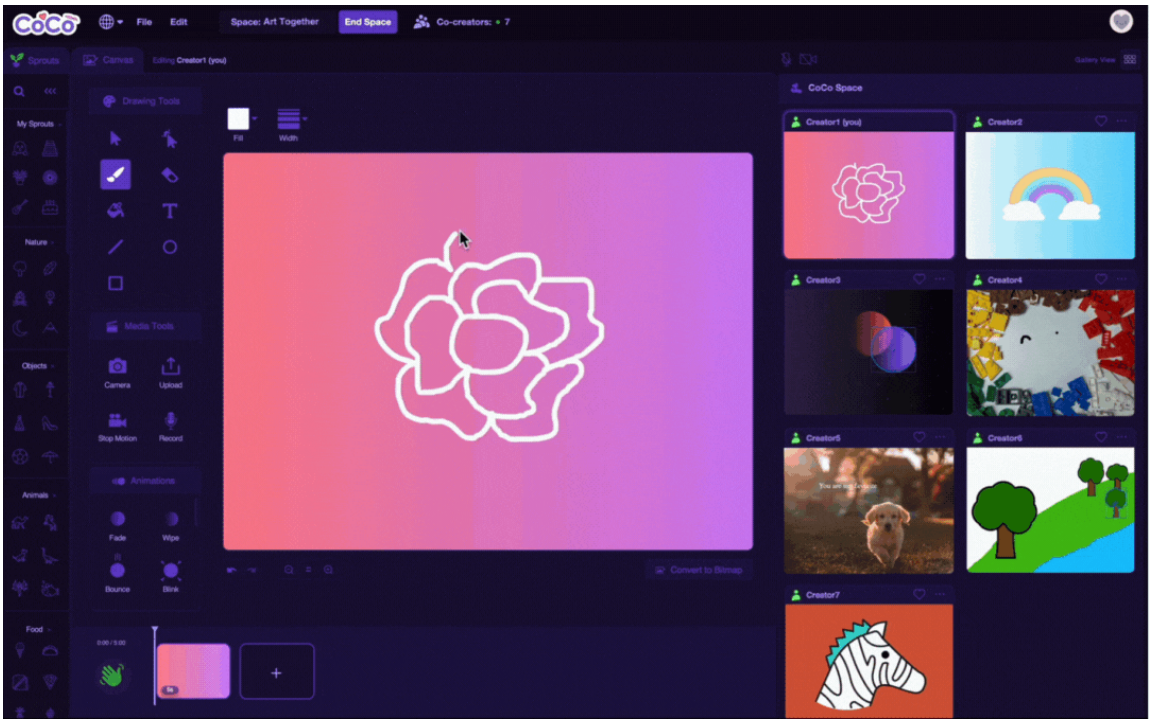


Fig 13: Shared Presence

Context is communication. Just being together and sharing a context with someone is itself a form of communication. In CoCo spaces, creators can work on their projects in a shared digital context, in the *presence* of others. Much like sitting in a café. This affordance is of course very useful for engaging learners remotely, but equally powerful when they are together in the same physical space. Even if students are sitting at a distance within the classroom, digitally, in a CoCo space, they are always right next to one another.

2) Playful inspiration

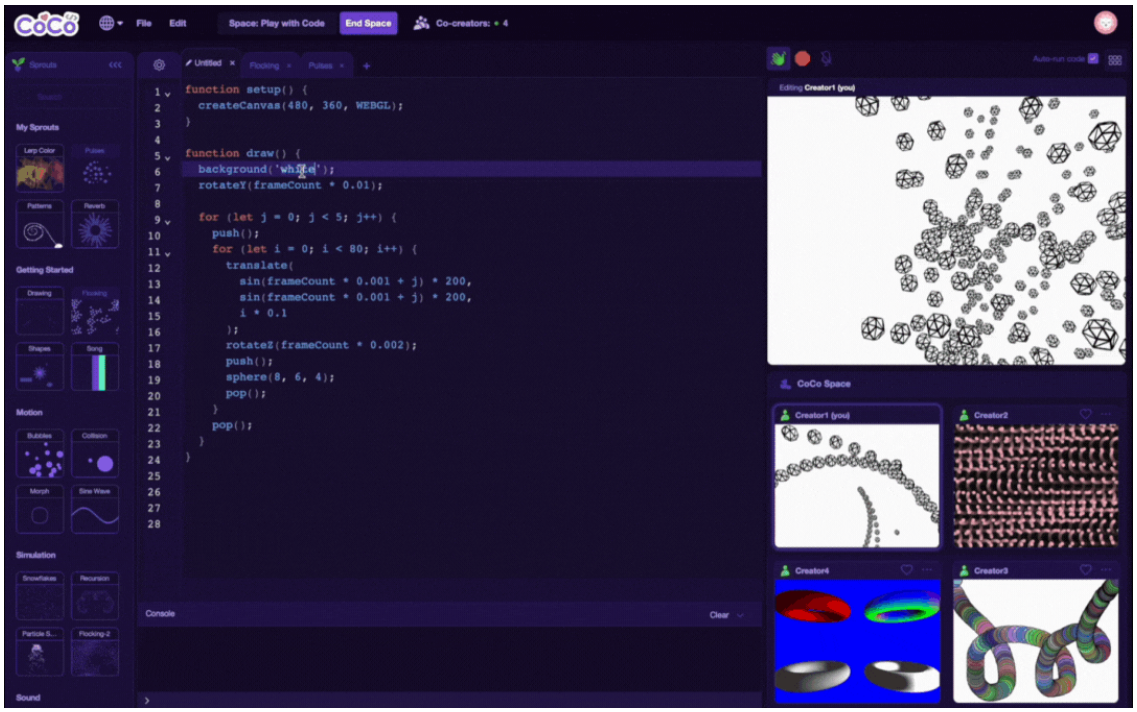


Fig 14: Playful Inspiration

Curiosity often leads to finding inspiration. It can be challenging for young people to get started on a blank canvas in any open-ended creative environment. In a CoCo space, even though learners get to work on their own projects, they are always surrounded by multiple points of inspirations in the form of the mini live windows showing their peers' creative explorations in real-time. They can also freely follow their curiosity and peek into anyone's project to view and interact with it, without having to change their context. Inspiration and support naturally becomes abundant in a shared space like this, making it easy for young people to transform their interests and ideas into creative projects.

3) Fluid Recomposition

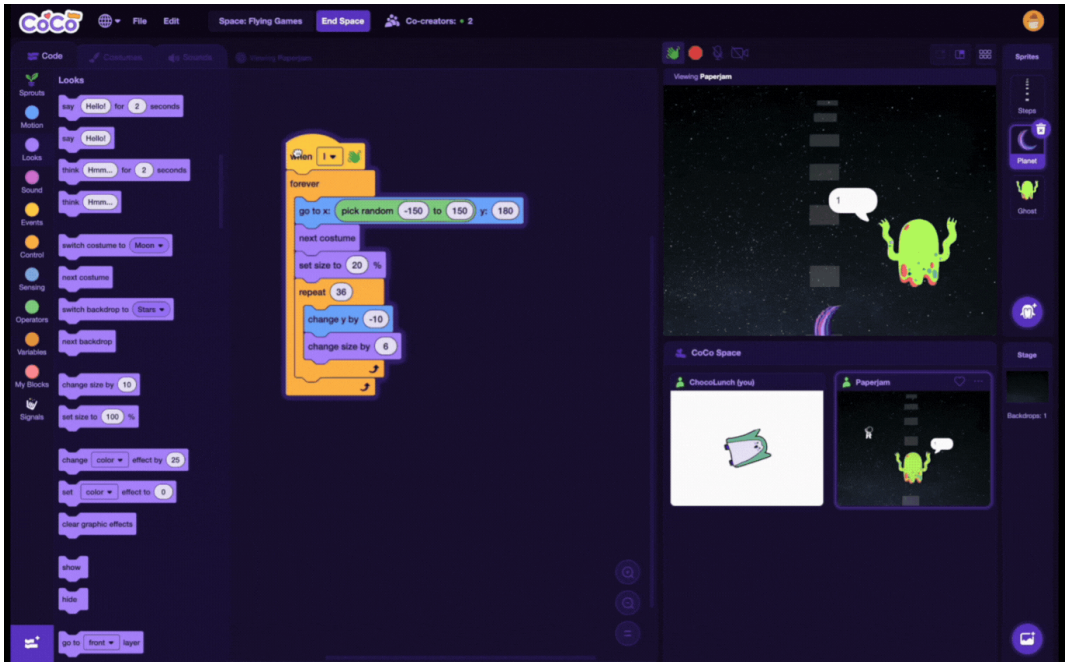


Fig 15: Fluid Recomposition

Creation is fusion. In addition to supporting the fluid exchange of ideas and inspirations *in* the process of creating, CoCo spaces also support live remixing of code and other digital assets between peers. Any creator can drag and drop a piece of code from another creator's project on to their mini window and it will instantly appear in their own project. They can then change it however they like. It is similar in essence to a sight of children sitting around the same table and freely passing and sharing craft materials with one another. Through this active fusion, remixing, and recomposing of one another's creative ideas and work in real-time, young people's digital creations start getting enmeshed with one another in a variety of ways, making a rich space for conversations about—*whose creation is it anyway?*

4) Creative interaction

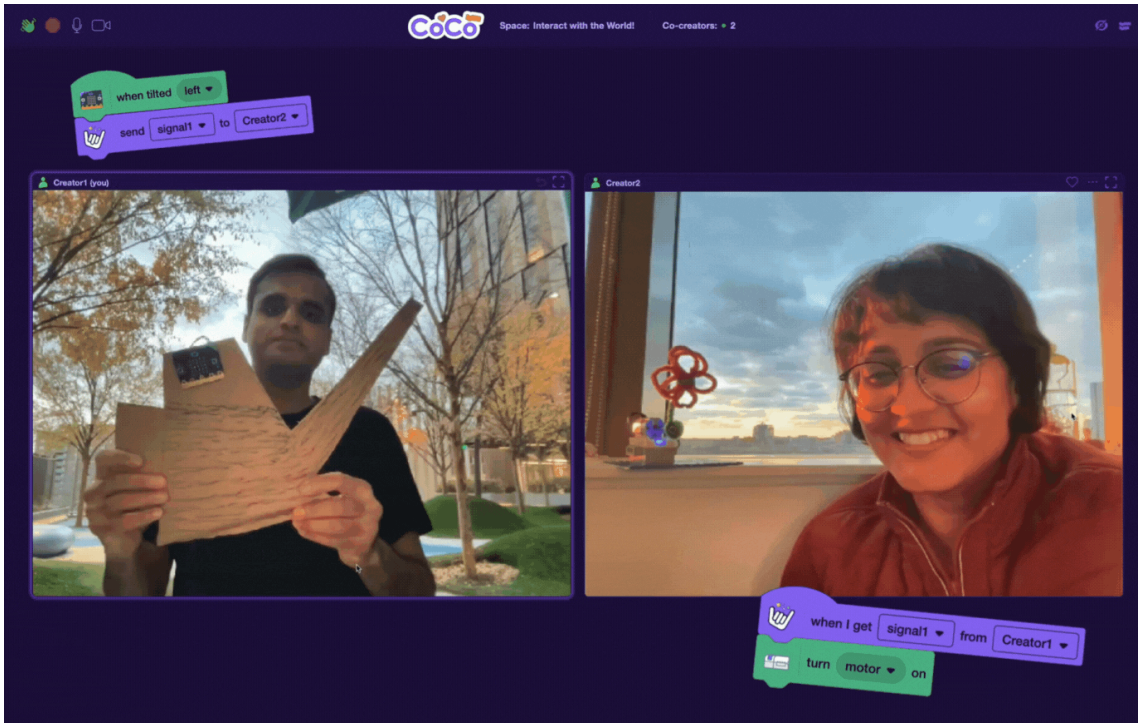


Fig 16: Creative Interaction

Interaction forms a connection. The CoCo Blocks environment introduces a variety of new peer programming blocks—Signals, Waves, and Shared Variables. Creators can use these blocks to make their project trigger something in their peers' projects in real-time and also get to see the outcome instantly. This opens up a world of new co-creative possibilities. They can program multiplayer games, chain reactions, dynamic stories and animations, interactive video-based experiences, and an endless variety of other shared creative experiences together.

5) Coordinated collaboration

Fig 17: Coordinated Collaboration

Co-creation often leads to serendipitous collaboration. Having real-time access and visibility of your peers' works in progress is conducive for sparking organic forms of creative collaboration between groups of young people. In addition to supporting *collaborative projects* where creators can contribute different parts to a single project, CoCo spaces also afford building new types of *collaborative experiences* as a group. For instance, creators can collectively imagine and program a new kind of multiplayer Pac-Man game where the character passes through each of their projects turn by turn and they can use shared variables to have a common score! Or, they can code their own musical instruments and then play their projects together in the form of a musical band. There are countless possibilities.

The underlying goal behind all of these design choices and affordances is to cultivate a joyful co-creative environment that is inclusive of young people's diverse interests and personalities and their multiple ways of being, of learning, and of engaging with others.

Reimagining 'live' online learning with CoCo: Empowering educators to support learners better, wherever they are

The term, *live learning* almost invariably evokes an image of an educator and a group of young people engaging on a video calling platform like Zoom. The educator will be seen doing a screen share of a platform that they are using (learning tool, coding tool, design tool, etc.) and often, not surprisingly, the kids will seem awkward and uncomfortable sitting in front of the camera. This *passive* mode of engagement has ironically come to be called as *live learning*—when in reality, most kids are neither able to feel a*live* nor *learn* in an environment that literally translates the experience of a traditional classroom to a digital context.

We think the built-in video and audio functionality in CoCo—although very useful for supporting remote experiences—is not what truly makes the environment *'live'*, rather it is the ability to engage in a wide variety of creative interactions with peers in real-time *in* the process of learning and building. CoCo spaces are therefore designed to empower educators to support children better *regardless* of whether they are physically together in the same room or connecting remotely from their homes. Moreover, video in CoCo is not a *passive* medium, it is a programmable material that young people can tinker with and use to build new kinds of *shared* creative experiences with their peers.

Chapter 7

New paradigms, new possibilities:
Insights from pilots with children

I think that CoCo is... AMAZING!!!!!!!!!!!!
😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍😍
- Isae, 10-yr-old

I've never seen a real-time collaborative space like that!
- Educator, Germany

Oh, here you are **making and sharing** a project at the same time - **not as separate steps**!
- Luca, 9-yr-old (who loves Scratch)

Co-creating, as we've seen over the years, is essentially a lot of co-iterating. The process of designing and iterating on CoCo has been research in itself.

Over the past two years, we've engaged **over 2,000 young people** in exploring CoCo through our pilots and playtests across diverse contexts. These involved both in-person workshops that we conducted as well as sessions facilitated independently by educators in formal and informal settings across multiple regions, including in *United States, Canada, Brazil, Colombia, Dominican Republic, Netherlands, Spain, Turkey, and others*. We have continued to learn and iteratively refine the platform based on these experiences and feedback.

Young people (ages 8-16 years) have created more than 1600 CoCo spaces on the platform, with each space having 2 or more co-creators. While it has been exciting to see all the wonderful new co-creative possibilities that young people have explored using the platform—we think the **most meaningful insights come from young people's own reflections about their experiences**.

This chapter is therefore not an evaluation section in the conventional sense. For us, this part of research isn't meant to be a way to "prove" the effectiveness of this new paradigm, or even a way to just improve the features of the platform. It is a medium for engaging young people to reflect on what it means for them to engage in co-creative experiences. In framing the open-ended question(s) for the post-workshop feedback form—we always thought about whether it's a question that will be meaningful for children to reflect on.

We deeply resonate with Reggio Emilia's approach to capturing and sharing insights about children's learning which stems from the fundamental question they keep at the center—"How can we help children find the meaning of what they do, what they encounter, and what they experience?" [23]

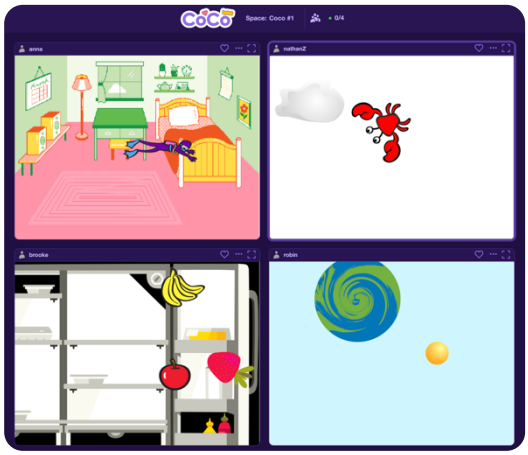
In that spirit, we interweave children's explorations and reflections to highlight some of the themes that stood out to us and that we hope can help spark others to design tools and environments that can support children in co-creative experiences.

- Agency in being creative together

Throughout our workshops, we observed young people naturally moving along the spectrum of engagement options provided by the "knobs over switches" design paradigm in CoCo (shared in the previous chapter). Some jumped right into collaboration, while others started off independently and gradually moved toward collaboration, and some others who enjoyed just working on their own alongside others in the same space. What stood out was how children appreciated the variety of ways they could engage in being creative together, while still having agency and choice in how they participated.

"

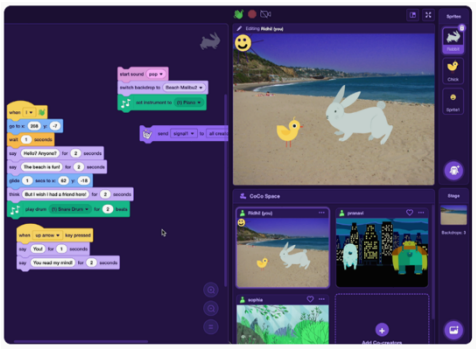
It made it versatile. **I could work as independently or collaboratively as I wanted**, and at any point I could still check in with my peers. It was interesting to see how my peers and I developed different ideas. **It was cool to see the creativity flowing**. And made the process creative, fun, fulfilling, and intellectual.
— *Anna, 15-years-old*



- The invisible co-imagining in action

Often, if we only look at the final projects or even the data, we might not notice any direct remixing of code or obvious similarities between the projects in a space. In the example below (and in many other examples), we saw this kind of invisible co-imagining taking place, which often isn't evident in the output itself, but becomes clear through children's reflections. It suggests how inspiration naturally flows when young people have visibility into each other's creative processes in real-time.

“



One person added sound in their code when I forgot to add sound in mine, so I did that, and that same person made a conversation between two sprites, so I did that as well. **It was nice to see other people’s projects as they sparked my interest to add things to mine**.
- Ridhi, 12-years-old

- Serendipitous collaboration

One powerful moment came during a workshop we had organized at MIT, where young people from across the country had come to participate. Normally, as we were designing the platform, we had thought that the mode of building a unified collaborative creation in CoCo would likely be explored more in spaces where young people know each other already or are friends with each other—like in a classroom. But in this setting, four girls (ages 12 to 15) who didn’t know each other but were in the same digital space in CoCo, spent over two hours together imagining and creating a fun detailed collaborative story. We asked one of them about the experience:

“

When we started making this project, I feel like we didn't really plan on it like becoming this sort of movie. We just like started creating our own background and then what sprites we wanted and then after we saw like what each of us had made, we thought we could just connect all the scenes. So like, if you saw the different sprites traveling to a different scene and then we could, like, connect it like that. And add

some silly dialogue and action and then kind of like make it into a movie.
—*Hannah, 15-years-old*

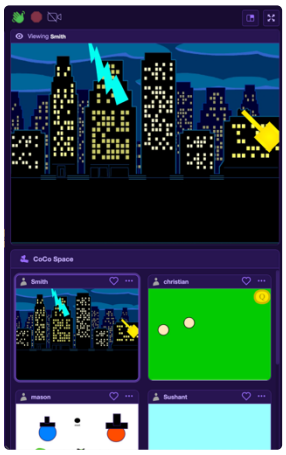
It made us see reflect on how **the design of the tool can enable creative collaboration that otherwise might have been unlikely.** In this case while they started initially working on their own projects in the space, they eventually decided to combine them into a story. After the workshop, they shared how they have all become friends now. It's nice to see how technologies can be utilized to serve as a medium for creating meaningful connections.

- Children as facilitators

of each other's learning and creative processes in real-time...

Often, in classroom settings, there's one educator facilitating a big group of young people. Everyone is largely working on their own device. While some young people are shy to ask for help, others might have a very specific doubt that the educator may or may not know in the moment how to help with. It can sometimes take away time and attention from others in the room. It is unfortunate that in a space where a group of young people are sitting together, the tools often do not allow them to leverage their collective creativity.

Through our workshops, we observed how a co-creative digital environment can empower children to naturally became facilitators of one another's learning and creative processes in real-time. We consistently heard from children how both asking for help, offering help, and accepting help in the process became so easier for them:



"

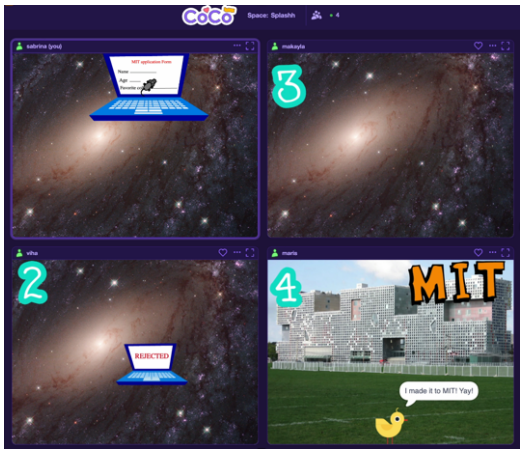
I think since I was able, since everyone was able to see each other's code, I definitely **found it easier** so that when they had a problem, we could identify it quicker and all of us could like, **look at each other's code** and then we would go over and like **help the other person with that issue**.

“

I remember when I was trying to figure out how to get a certain movement option for one of my sprites and **I looked at my partners project and was able to copy it and understand it from their project**.

- Supporting young people's diverse personalities and preferences...

Collaborating with others often is particularly hard for people who are not naturally social or extroverted. Often when an educator makes groups in classrooms, asking students to collaborate together, some young people may particularly find it hard to find this a positive or joyful experience. While using Scratch in a classroom, for instance, a teacher might ask students to work in pairs on a single project by sharing a device. One person might take more control and the other person might not feel comfortable or free to contribute, or both might wrestle over to take control of the device. In CoCo, young people can work on their own devices while being connected in real-time in a shared digital space with their peers. They can see one another's projects in the same digital context and choose to collaborate in their own ways. The collaboration is not forced upon but emerges naturally.



In this memorable project from one of the workshops, a group of kids—many of whom had just met—created a whimsical collaborative story about a chick from the galaxy who applies to MIT, gets rejected, and then somehow crash-lands into MIT anyway. They recorded their own sound effects and voices, layering humor and imagination into their project. When asked about their experience, one of the participants reflected on how they thought the platform supported people with diverse personalities:

“

..this is kind of a way that **whether you’re introverted or extroverted, you can still work with others** and like you all get an equal amount of work to do on the project, if that makes sense. Like you’ll all have an equal share of what to do, and **it’s not like one person’s left out and another person takes on all of the work to do**. And you can create separate projects. You can combine them all together.

- The affective dimension of co-creation

The “co-” in co-creation represents connection with others *in* the process of creating. Across all our workshops, young people have shared how being able to create and work together with their friends is something they really enjoyed in CoCo.

Unsurprisingly, many children’s comments about their experience on CoCo reflected this emotional and affective dimension of co-creation. Often kids would say, “I loved collaborating with a friend,” and describe how it was really fun for them to get to co-create together in real-time. But it was also interesting how this sense of connection emerged even among kids who didn’t know each other beforehand. These reflections highlight how meaningful authentic connection with others can be a powerful motivator in creative learning—supporting not just what children make, but how they feel while making it.

“Creating alongside others in the same space made the experience **more fun and motivating**.”
“It felt **more nice**.”
“I felt **more motivated**.”
“It made it **less isolating**, more of a collaborative environment.”
“It affected my experience positively; I was able to collaborate effectively with my peers and make new friends. **I felt happier** than when I worked on my own, like in Scratch (not that Scratch is bad, I love it as well!).”

There are so many more examples, comments, insights, and experiences we could highlight. But overall, through our playtests and pilots, we continue to see how children naturally seek and enjoy many forms of creative connection—from just creating alongside others to directly collaborating—and that digital creative environments can and should support this full spectrum rather than limited choices.

Beyond the features and outcomes, ultimately, we should design for co-creativity because creating with their friends and peers makes children feel *happier* than creating on their own.

Chapter 8

Extending creative tools into co-creative environments

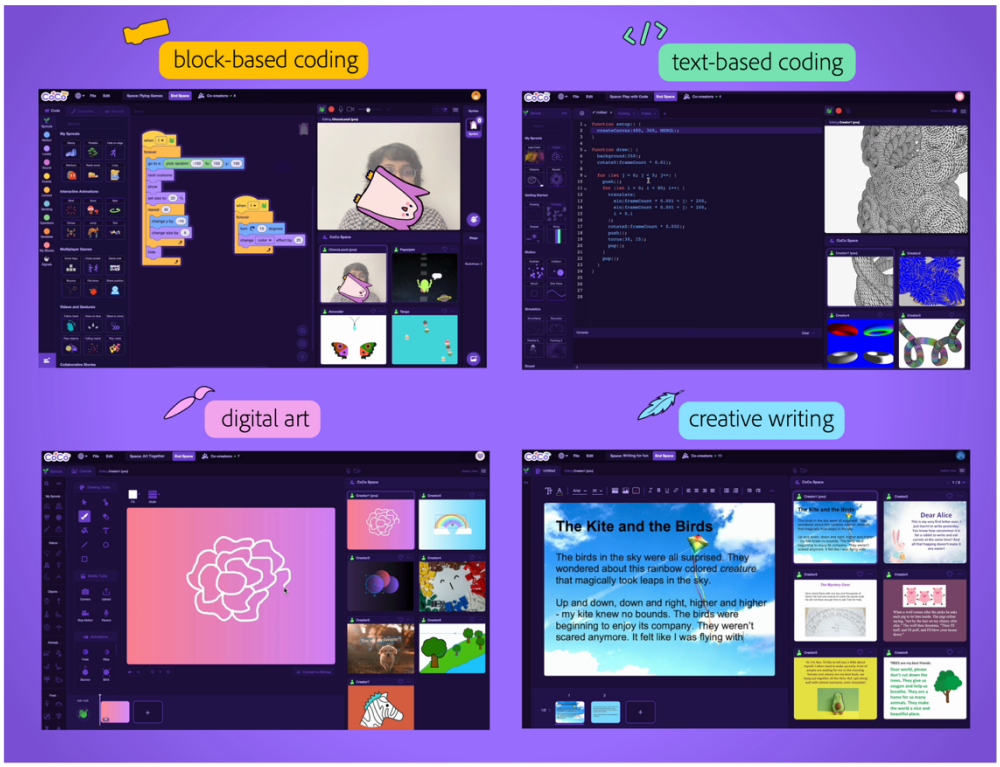


Fig 18: Multiple co-creative environments in CoCo

We've especially designed CoCo as a general-purpose co-creative platform—one that supports collaboration across a wide range of creative environments such as Scratch, p5.js, and others. From the start, we've always envisioned CoCo not just as a platform, but also as a paradigm. Our interests and explorations have not been toward building a better coding platform, but rather to foster a new paradigm: where young people can choose from multiple expressive mediums of their choice and engage in real-time, co-creative experiences with peers in a shared digital spaces.

As AI continues to expand what's possible—from intelligent assistance to code generation or creating images—we're excited to explore and integrate these new affordances. But regardless of advances in these directions, the need for authentic co-

creative engagement remains essential. We continue to keep that at the center of all our explorations. The social, emotional, and relational dimensions of being creative together will only become more important in this new landscape.

We think this paradigm can extend to an endless variety of digital creative experiences and environments. For instance, we're already prototyping CoCo Canvas for co-art and exploring co-writing environments. In CoCo Canvas, young artists can make art on their own canvas in a shared space, they can easily grab and remix parts of one another's artworks in real-time, get inspired by others' color choices or styles of drawing, or collaborate to make a group art piece.

These affordances of engaging with one another in real-time will continue to be exciting and valuable even when all creators have access to AI assistance as well. There is an endless variety of amazing creative environments that talented groups of people everywhere continue to build, and we imagine and hope that at some point in future, CoCo can host an open library of a growing number of community-sourced co-creative environments for educators and young people to choose from every time they create a CoCo space.

Chapter 9

Designing decentralized communities:
Privacy, safety, cultural context

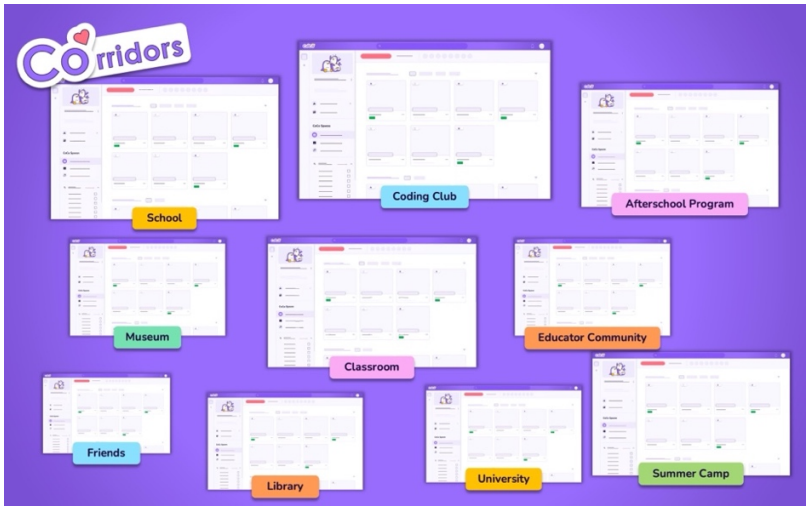


Fig 19: Decentralized private communities in CoCo

Since CoCo offers real-time co-creative environments for children, we have carefully designed the surrounding social infrastructure of the platform in ways that prioritize children's safety. Instead of a large open community (or open online game environments), CoCo is designed to let educators create and moderate their own private communities. We call them "corridors," inspired by MIT's famous *Infinite Corridor*—an 825-foot-long hallway that runs the entire length of MIT's main buildings and connects to the different classrooms, lab spaces, and social spaces for members of the MIT community.

Fig 20: MIT's Infinite Corridor

Similarly, educators can invite students from their classroom or after-school club to their corridor, using a secure login flow that does not collect any personally identifying information. The corridor provides a safe context for young learners to create and engage in any number of live CoCo spaces with their friends and classmates. They can also explore all of the past spaces and interact with the projects that were created by their peers and even remix a whole space if they like.

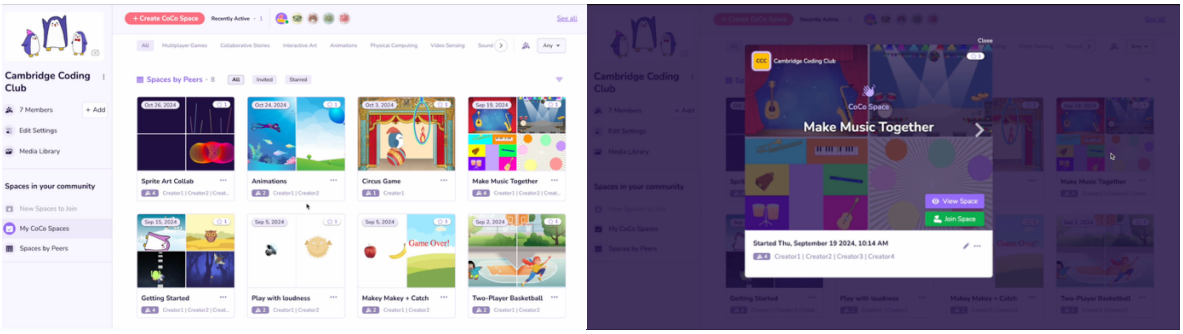


Fig 21: Decentralized private communities in CoCo

Educators can also add custom assets (such as images, sounds, sprites, etc.) in their corridor's media library. Any assets they add are automatically made available inside the CoCo environments. Children can then use those to co-create projects that closely reflect their surroundings and cultural contexts. Educators and researchers are increasingly highlighting the need for supporting young people in collaborative and culturally relevant computing experiences [24]. We hope these design choices and affordances can help empower educators everywhere to support learners better across diverse geographical and cultural contexts. We have seen several educators utilize these affordances in our pilots. For example, (Fig 19) an educator in Colombia added images of the school mascot in their corridor on CoCo for kids to use in their projects.

Fig 22: Adding custom assets in CoCo

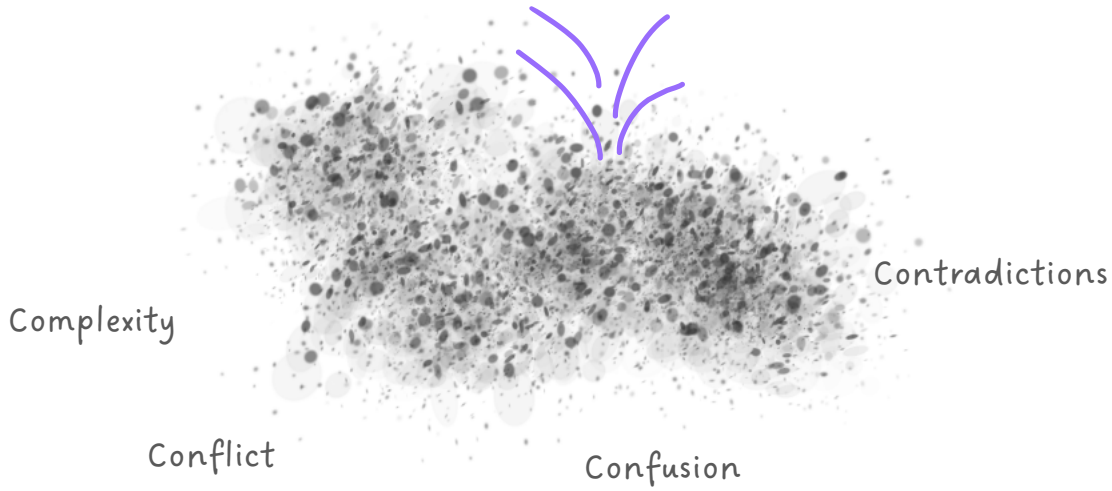
Overall, we continue to hear from educators how they really appreciate having a safe private community for their students and how students are naturally more engaged because they get to create projects together with their friends.

Part 3

Co-clutter

| a necessary interlude |

It's our differences that make the difference.

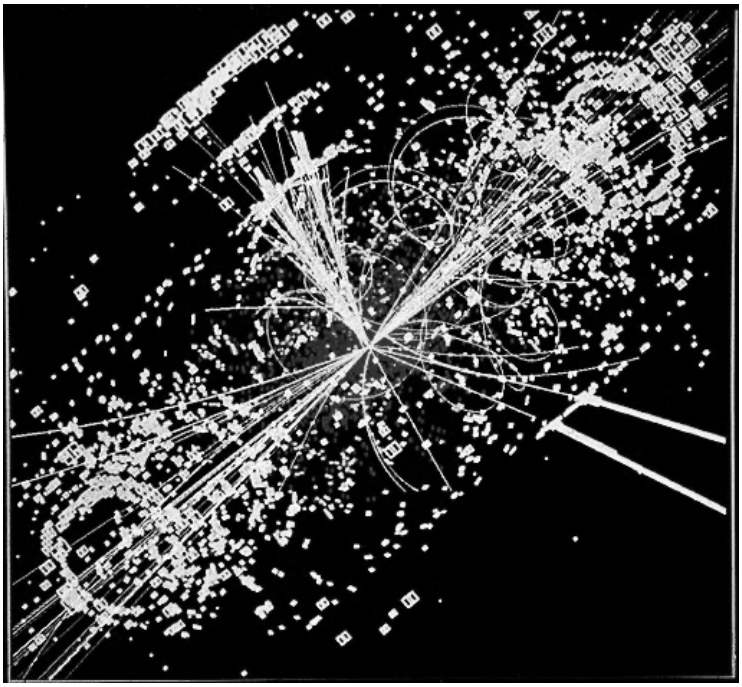


Co-clutter is the flux, co-existing through which, we co-evolve. In this short section, we reflect on how this is an essential part of our co-creative process, and how it is valuable to intentionally design for generative co-clutter.

Ch 10. Creating the conditions for "collision" 54

Chapter 10

Creating the conditions for "collision"



[2]

In the Large Hadron Collider, the smashing together of sub-atomic particles at near light speed produces a shower of new particles.

Interestingly, this imagery closely resembles our own co-creative process. The collision of our different ideas, personalities, communication styles, and moods on a given day can raise the temperature in our conversations. These collisions—sometimes experienced as conflicts—are natural and valuable. The dearer the idea one holds, the more intense the collision.

[2] Image from https://commons.wikimedia.org/wiki/File:CMS_Higgs-event.jpg

Very often during these heated discussions (or after some reflection), we discover new directions, thoughts, and ideas, while simultaneously building a deeper understanding of each other and our own selves.

These conversations are far from negotiations where each person arrives with a fixed agenda. We are never simply trying to "find" common ground—and perhaps this is what leads to exploring and co-creating new playgrounds together. Instead of seeking compromise, we allow the natural friction of our different perspectives to generate unexpected insights.

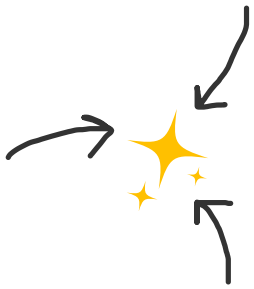
We embrace moments of conflict or co-clutter not as something to be avoided, but as an essential part of co-creation. And over time, we have come to expect, accept, and almost cherish our collisions.

Designing for “collisions”

Seymour Papert highlighted how the role of the teacher is to create the conditions for *invention*.

Through this work, we have explored the value of designing tools, spaces, and experiences that can create the conditions for *collision*. When the diverse ideas, interests, and identities of young people are allowed to playfully collide with one another, new ideas and identities are co-invented in the process.

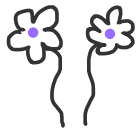
We hope that our work and perspectives can help inspire others to make space for creative collisions in their shared contexts.



Part 4

Co-exist

| shifts and perspectives |



"We have not been able to be secure in human society. There is anxiety of various types in our social existence. You know, we have no anxiety about the way in which the limbs of the body work in relation to one another. You do not go to bed with a feeling of insecurity that the eyes or the ears may not set-up a strike in respect of the stomach which swallows all the food and gives nothing to the eyes or the ears! You have no fear that, perhaps, tomorrow there may be a battle between the right hand and the left hand. But we do not live a life of harmonious organic relationship to our outer life [with others in the society]…Nature is not a machine but an organism. The parts of nature work in spontaneous harmony amongst themselves…"
— *Krishnananda Saraswati*

In this section, we share perspectives on what it means to center the ideas of interconnectedness and harmony in any learning contexts. We illuminate a set of lenses and conscious choices that can help bring a shift in the way we understand and design our creative, educational, and social systems.

Ch 11	From future to present	57
Ch 12.	Constructing *things*, Constructing *context*	58
Ch 13.	Head, hands, heart, and *hüm*	61
Ch 14.	From self-inflating to self-*less* social platforms	63
Ch 15.	Passion, dispassion, and compassion	68

Chapter 11

From future to present

Education is not preparation for life; education is life itself.
— *John Dewey*

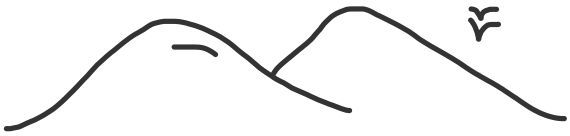
Life can only be found in the present moment.
—*Thích Nhất Hạnh*

Throughout history, the focus of education has predominantly been centered toward preparing children for the *future*. What if we could instead reimagine education as a means to empower children to freely and fully engage with the *present*?

How will we design learning systems, spaces, and experiences for young people if the goal was *not* just to prepare them to be successful in future—where the notions of what 'success' means are narrowly determined by our conditioning? What will we choose to assess, what will we choose to teach, how will we choose to teach—if we shifted our lens to the present?

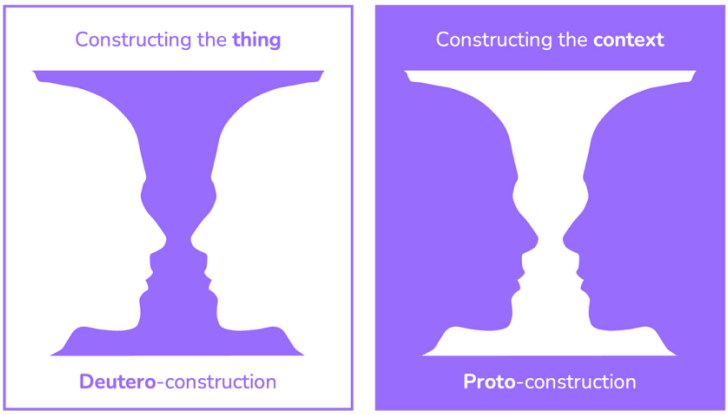
How can we help preserve children's natural ability to stay in the "now" instead of making them fear the future? Isn't that what we constantly strive for as adults?

In the chapters ahead, we explore what it means to design for *presentness*, and the small but intentional design choices we have taken in the tools we are building to help bring this shift.



Chapter 12

Constructing things, Constructing context



> The thing is the thing, the context is the context, the thing is not the context, and the context is not the thing. The thing is being constructed within a context but *maybe constructing the thing is only a context for constructing the context*?

Shifting from Individual and Thing-centric lens to a Co-creative and Context-centric lens

In any shared creative learning space, learners are always engaged in two kinds of construction. The more visible (and seemingly primary) one is the construction of their projects or *things* (such as building a game, writing a story, or creating art) but the other seemingly invisible one is the simultaneous construction of the social environment or their shared context with peers. Contrary to what might seem obvious, we consciously emphasize seeing the invisible constructing of *context* as the *proto-construction* process and the constructing of *things* within this context as a *deutero-construction* process.

We do this reversal with the intention of bringing a shift in the often exclusive and excessive attention put only on the more visible aspects of a learning space (the projects being built, the concepts being learnt…) and not on the shared *context* being constructed. This imbalance typically arises from seeing the learning environments (physical or digital) often just as consisting of *individual* learners / users, each working on their separate *things*.

From two kinds of construction to two forms of learning

> ...every time you learn something you learn two things,
> the other being the model of learning you just used.
> — *Bateson's deutero-learning concept* (as described by Papert [7])

When children engage with(in) any system, they are not just learning the explicit things that the system is designed for (skills or concepts), but they also imbibe the ideas, worldview, and frameworks that the system is built upon.

As designers of these social systems (tools, schools, or spaces), consciously or unconsciously, we end up constructing most of the con*text* or the underlying *text*ure and *text* (rules) of the system *for* children. What does that lead to: instead of empowering children to actively engage in constructing their context, we tend to push children to be constructed and painted *by* the context.

The ideas of success, failure, popularity, etc. are not natural to children but by repeatedly participating in systems that emphasize these ideas, they start internalizing these social structures and eventually start acting from them.

To make this more real, let's imagine there is a hypothetical live online learning environment (let's call it **NoNo**?) that was designed with a typical *individual-centric* and *thing-centric* lens. Here too, children are getting to work on their projects and you can also see your peers' projects in real-time. But also on display are the number of likes, thumbs-ups, and ratings given to your project. In addition, you have the option of sorting all the projects in the space based on these metrics—sort by *most likes*, sort by *highest ratings*, show the most *advanced projects*, and so on.

Also, learners are supposed to work on their own separate projects here, they are not allowed to "copy" or remix anyone else's work.

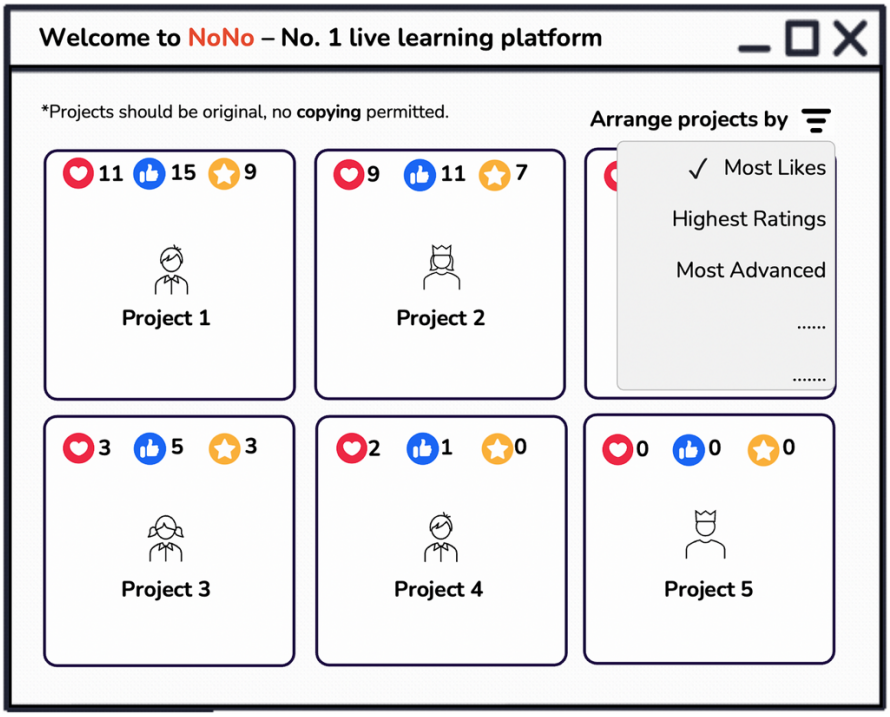


Fig 23: NoNo—A hypothetical live online learning environment designed with individual- and thing-centric lens.

Now imagine yourself as a young person participating in this environment. The environment's context by its very design promotes the ideas of competition, rewards, and comparison. Creators in such spaces are no longer likely to be creating their projects from a place of joy but from an anxious place of excelling in this socially constructed structure.

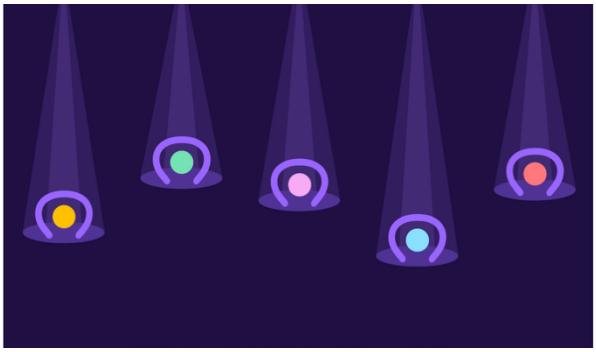
In such an environment, even though children may be *creating together*, they are not *being. creative. together.*

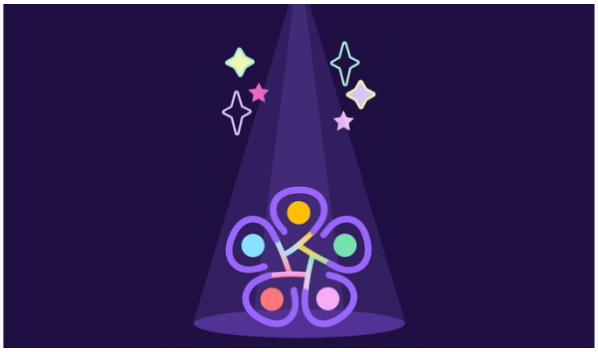
> ..the most notable aspect of a positive judgment is
> not that it is positive but that it is a judgment.
> — *Alfie Kohn [25]*

Can we design spaces rooted in *non-judgment*?

Chapter 13

Head, hands, heart, and hüm





In any learning space, when we start to zoom out beyond the *individual* and the *thing*, and consciously hold the entire space in our view, we'll start noticing the *shared context* that is actively getting constructed, deconstructed, and reconstructed within the space. This shared context is brimming with a rich web of connections that include learners' connections with their peers, with the educator, with their immediate surroundings, and also their connection with themselves.

All of these connections, although invisible, are in fact very real. They are moldable like clay, they are also colorful, they even radiate temperature with varying degrees of warmth to sometimes cold, and they get strengthened not merely by physical proximity but by a deep sense of trust, openness, and empathy in the space. These connections truly come alive in the process of co-creation and if you hear intently, they even make a sound like that of a warm and creatively buzzing space in a state of flow.

We coined a word for it—we refer to it as the "hüm" of the space.

The *hüm* is a representation of the nature and quality of interconnections that are actively being constructed within any social learning environment.

The hüm is what we'll hear, sense, and feel when we take a step further and start to see learners not just as individuals with *head*, *hands*, and *heart*, but also as active co-constructors of their shared context.

The *hüm* doesn't belong to any one in particular yet it is an integral part of everyone, and it contains and sustains the shared values of the space.

If we are not intentional about our choices, we can easily end up designing systems and spaces like the hypothetical NoNo environment above where the *hüm* of the space (which probably will feel more like *h∩m)* will soon start to vibrate with the unpleasant and uncomfortable sounds of anxiety, fear, and insecurity instead of those of joy, connection, and creativity.

Chapter 14

From self-inflating to self-less social platforms

Since the very beginning of developing our ideas, we consciously envisioned and worked toward designing CoCo as a **"self-*less*"** social platform that **departs from the comparison-based, individual-centric paradigm** (of profiles, likes, and followers), which spilled over from early social media platforms to become the dominant approach for online social engagement.

The creator- or self-centric design paradigm is so deeply entrenched in most online social platforms that the idea of a 'self-less' social platform can seem like an oxymoron. It's true that getting opportunities to share content with others online (pictures, posts, creative work, videos etc.) greatly empowers young people's self-expression. However, the inherently performative nature of online social participation, aided by typical social rewards in the form of likes or followers can often deeply influence young people's perception of themselves and their peers in ways that are not always healthy.

The fundamental unit in this model is the 'self' or the user's individual profile where the works created or posted by the user are accumulated, along with the associated social metrics. While it works great for few people who get visibility, for majority of others, this can often lead to feelings of comparison, anxiety, and insecurity, and unnecessary creative pressure. The focus shifts from the joyful experience and process of engaging in a creative process to seeking validation for the final outcome in the form of likes or views.

To live without comparison is to remove a tremendous burden.

—J Krishnamurty

We think any space (physical or digital) is shaped as much by what is present in it as it is by what is *absent*. In CoCo, we have consciously decided to **do away with comparison**. With no individual profiles, no followers, and no likes to collect, there is no digital currency to compare with.

In conceptualizing and developing CoCo, we kept coming back to the question—what underlying ideas, values, and ways of thinking and being will young people discover and absorb as they engage with and within the system? We intentionally made **small and big design choices** keeping that question at the center.

CoCo Cards: Communicating Feelings over Collecting Metrics

Imagine being in a vibrant studio space where you are engaged in making art with your peers. You look over and you really like what your friend has made, you whisper and express your appreciation to them (so as to not affect others in their creative flow). They feel really good and express that to you by saying thank you with a smile. In this spontaneous personal interaction, the likes and the thank you's are not *metrics* to be collected and displayed for others in the room to see. They are just *feelings* that are expressed and felt in the moment. Though this exchange is ephemeral, its affect is often much deeper.

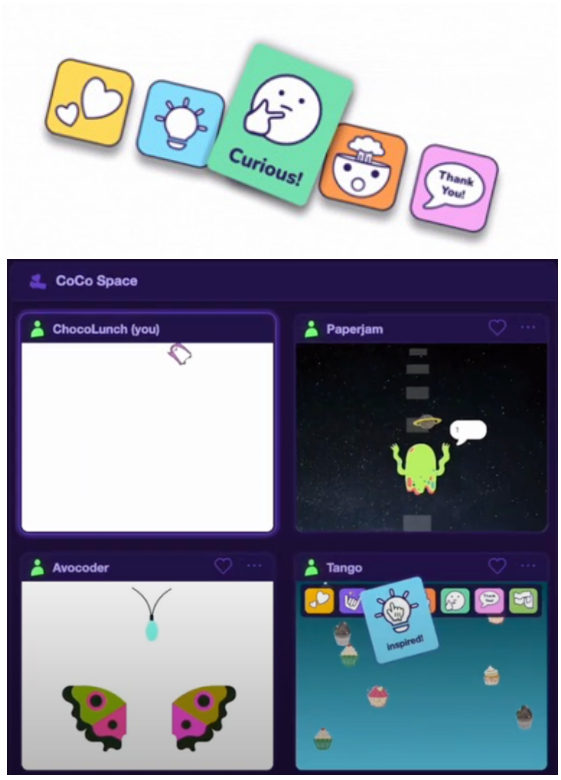


Fig 24: CoCo Cards

We wanted to bring this same personal and spontaneous quality to the social interactions within a live CoCo space.

We designed **CoCo cards** as a way for young people to express feelings of appreciation, wonder, inspiration, or gratitude to one another, as they see one another's projects and ideas evolve in the space. The cards are shared one-to-one

between any two creators within the moment. They are not announced or displayed publicly in the space for others to see.

In our playtests with children, we have often seen children really enjoy and appreciate sharing these little cards with their friends in the space.

Shared Spaces over Individual Profiles

In CoCo, there are no *individual* user profiles or *separate* project pages. The fundamental unit here is not the individual creator, but rather the endless variety of co-creative spaces that they engage in with peers. In CoCo, creators build projects within a shared context that is designed to be inherently interactive and collaborative in nature. Every creator's project is deeply intertwined with other creators' projects in the CoCo space in a variety of tangible and intangible ways. The projects therefore are never separated from the original co-creative context that they were built in. In doing so, we hope to support young learners in making a shift from the often excessive focus that is put on seeking validation for the final *outcome* to instead start seeing the joyful *process* of co-creating as its own reward.

What do young people feel about the choice of NO likes and followers?

In our post-workshop feedback form with over 80 young people (ages 11-16 years), we asked about what they felt about this specific design choice in CoCo is an open-ended question. All of them had used Scratch before and are familiar with Scratch online community built with the paradigm of user profiles, likes, and followers. We knew from our past experiences that this is not a binary choice, and we were curious to hear young people's reflections in their own words.

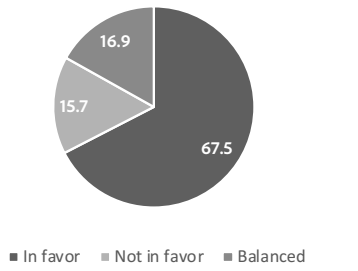


Fig 25: Responses from young people

It was surprising that even when most young people are very used to this

dominant paradigm, in our feedback form, majority of them expressed that they appreciate the choice of removing these metrics. Only about ~15% shared that they prefer these metrics, while another ~15% shared suggestions for taking a balanced approach toward this choice.

Here are some of the comments young people shared in the form:

Expressing - in favor:
"This is honestly a great change because nowadays **I feel like people make projects for farming likes and followers which defeats the whole educational and collaborative purpose**."

"I think this **encourages collaboration over competition**. It's a good thing!"

"I really liked this. I think sometimes that can take away from what creating should be about. Not everything needs to be turned into a social media thing, and instead of feeling pressured to create something others might like, and forgetting that the **point of making projects isn't to get new followers, you can make fun things for yourself** :)"

"I think **it creates an environment where all students are equal** and one doesn't receive more praise than the others and **allows them to focus more on the community part** of their projects."

Expressing – balanced view:
"While I like this choice and think it is very effective in small communities. I think that **likes and followers should be displayed on public projects** so that games with high quality and effort are more likely to get played."

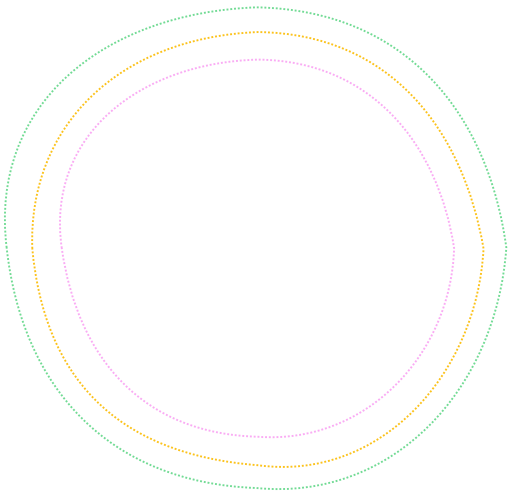
Expressing - not in favor:
"I think that these should be added because **posting your content online for a wider audience can help shed more light on your work**. Likes and follower count just show that people appreciate you."
The point is to not arrive at a right answer. But through this work, we consciously

choose to ask the question, and to help young people reflect on the question—what happens when there's no unit of comparison in a social platform? We think the question is worth asking and designing for.

The word selfless is formed by putting *-less* with *self*, which more literally implies, *free from self*. The part of the *self* we hope children can feel *free from* is the part that gets insecure, fearful, driven by extrinsic rewards, and starts to care only about individual outcomes. **Being selfless is not an act of diminishing one's self, but rather of continually expanding one's self-identity to include the other.**

But we cannot begin to embrace self-lessness as a value when all the social systems we partake in (online or offline) are inherently designed to be self-centric. Instead of bringing us together, they often pull us apart. Ironically, these self-*inflating* metrics—scores, ranks, badges, likes, followers—end up *diminishing* young people's sense of self.

We hope the ideas and design choices we have shared through the work on CoCo can inspire others to explore what it means to design *self-less* social systems and spaces for young people—that serve as playgrounds for young people to continually *expand* their self-identities.

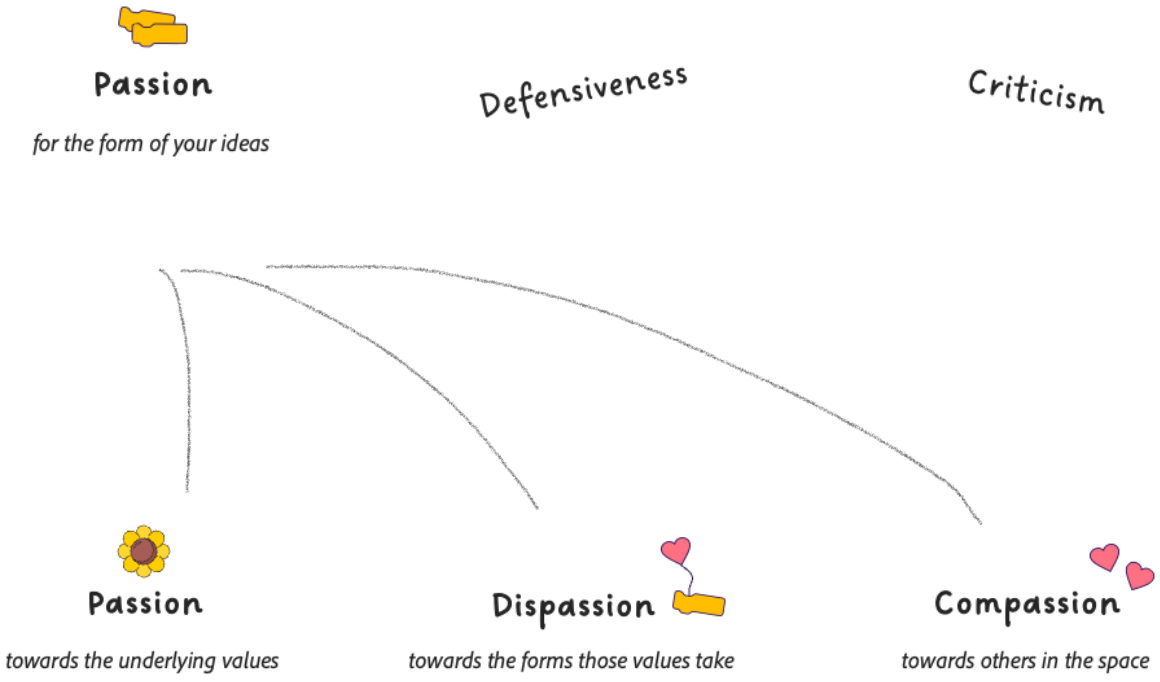


Chapter 15

Passion, dispassion, and compassion

Passion by itself is incomplete, and when it is directed toward the *forms* our ideas take, it can sometimes lead to being overly defensive about our own ideas—and to a mindset of comparison and criticism toward others' ideas.

Passion becomes whole and harmonious when cultivated alongside dispassion and compassion: where passion is directed toward the underlying *values*, dispassion toward the forms those values take, and compassion toward others in the space, and toward your own self.



How can we help nurture young people's
passionate, dispassionate, and compassionate selves?

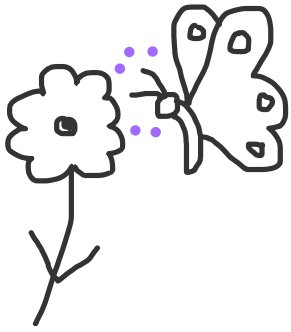
Part 5

Co-evolve

| contributions and directions |

coevolution, the process of **reciprocal** evolutionary change that occurs between pairs of species or among groups of species as they interact with one another.

— Britannica



Ch 16.	The 5Co's of Co-creative Learning	70
Ch 17.	From Monoculture to CoCo-culture	74
Ch 18.	Bringing CoCo to communities around the world	78
Ch 19	Extending CoCo | Little Language Models (LLMs): Empowering children to be future AI modelers	80
Ch 20.	Looking ahead: Centering the *timeless* in timely	92

Chapter 16

The 5Co's of Co-creative Learning

This work is as much about the tools we've built—and the thoughts and ideas we've expressed through words, pictures, songs, and videos—as it is about our shared lived experience and evolving process of *being creative together*. Through living, learning, creating, and of course, colliding together for over 12 years, we've come to recognize five stages that continually shape both our creative work and our relationship. These stages unfold every time we are being creative together. We might jump through these in a single conversation, while also seeing them as phases in our long journey. Interestingly, we've often seen young people in our workshops move through similar stages as they engage in being creative together with their friends and peers.

We now see these stages as part of a process, or a framework, which we call as: The 5Co's of Co-creative Learning—a process in which learning *emerges* through co-imagining, co-creating, co-cluttering, co-existing, and co-evolving—with and **through one another.**

At the center of these 5Co's, are both the *heart and the block*, highlighting how our co-creation of (visible) projects always goes hand-in-hand with the co-creation of the (invisible) context in which they are built. The heart and block are surrounded by a kind of "home"—a metaphor for deep **trust** that holds the process together.

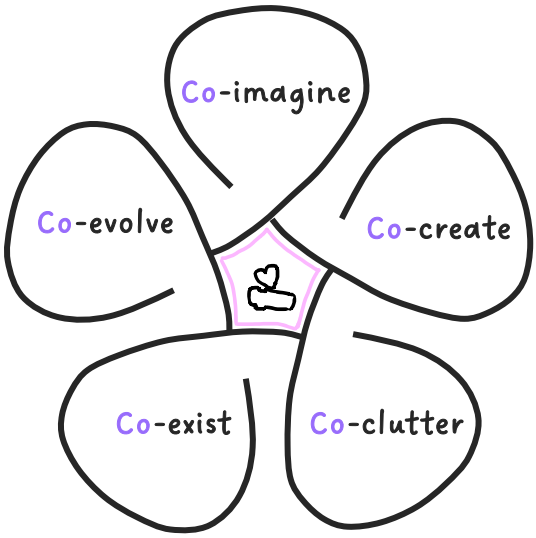


Fig 26: The 5Co's of Co-creative Learning

In any co-creative learning environment, when trust starts to erode, learning and creating stops.

But the wonderful thing is that it's not a static process, we are continually engaged in building and rebuilding our shared context—allowing us to create and recreate trust, openness, authenticity, and of course, with each rebuilding, the increasing capacity to withstand the heat of collisions. Much like the Large Hadron Collider, where particles smash together with each other at near light speed and release new particles, our shared (thermally resistant) context lets us smash together our differing ideas, identities, and communication styles—generating new thoughts, richer insights, and unexpected directions.

It is our differences that make the difference. Co-existing through the flux of co-clutter always leads to co-evolving.

Our ideas have evolved over time through connections with other frameworks that have inspired us—in particular, the Creative Learning Spiral by Mitch Resnick [26], which has long served as a foundation for thinking about how children engage in creative learning. It has informed the design of tools like Scratch and have inspired generations of educators and researchers. It continues to deeply influence how we think about creative learning as an iterative process.

But through our experiences over the past five years in designing, developing, and introducing CoCo to children, we got to see first-hand how in a co-creative learning space, learning and creating does not unfold as an individual process for every learner, it emerges in *between* them. When an environment (physical or digital) is intentionally designed to support active creative collisions between young people's ideas, and projects, and personalities, the stages of co-creative learning naturally appear as an inevitable part of the learning process. We see the 5Co's of Co-creative Learning as an expansion of the spiral where the center of gravity shifts from the learner to their shared context. Children become co-facilitators of one another's learning and creative processes. And an educator's role shifts from helping individual learners' creative processes to cultivating a shared space for them to encounter and experience a rich array of meaningful "co-" experiences with one another.

To further illustrate our thinking about how the ideas of co-creative learning meaningfully build on and extend the creative learning principles, we'd like to share a short playful reflection in the spirit of the wonderful 4Ps:

4P's walk into a co-creative learning space and...

"Projects" discover it's not just about making cool things, but making warm contexts too. It sees children as architects of both their creations and the shared spaces between them. "The thing is being constructed within a context, but maybe constructing the thing is only a context for constructing the context?" Children become not only makers of projects, but also co-makers of the social, emotional, and relational space they inhabit.

"Passion" wobbles at first, then gracefully splits into a trio: "Passion, Dispassion, and Compassion." Passion still hums with that childlike spark and drive. Dispassion brings a calm, Zen-like clarity—children create without being trapped by external validation. Compassion radiates outward—offering care, gratitude, and joy in others' creations. This balanced trio allows young creators to remain fully engaged while staying free from the anxieties of judgement and comparison.

"Peers" peers through the kaleidoscope of co-creation and realizes learning isn't just happening with or from others, but through them. In the space between children, something magical emerges—creating and learning that belongs to neither yet enriches both. Peers are no longer just collaborators or audiences, but become the very medium through which creative imagination flows.

"Play" smiles knowingly, having always understood what the others are just discovering: it was never about metrics or outcomes. Play has always been about the joy of the moment, the process over product, the delight of being fully engaged in Being Creative Together! And play expands beyond tinkering with materials to experimenting with new ways of being in relationship with others.

We hope the 5Co's framework for co-creative learning and the spectrum of ideas it encompasses serve as a helpful lens for others to observe, understand, imagine, and design co-creative learning spaces—where groups of children or adults can experience the power and joy of *being creative together*.

Chapter 17

From Monoculture to CoCo-culture

Monocultures of the mind and screen

While reading an article about monocropping in agriculture, the image in our mind shifted—from vast fields of identical yellow crops to a classroom filled with rows of screens, each displaying nearly identical sprites, code blocks, and animations. The class aimed to teach foundational computing ideas through Scratch—an *endlessly creative* programming language. Yet the teaching approach assigned every student the same tutorial, resulting in nearly identical projects across the room.

Fig 27: Monoculture in classrooms
(image generated from ChatGPT for illustrative purposes)

This kind of educational monocropping is common across subjects. Students are assigned identical tasks, solve the same problems, and are expected to arrive at similar outcomes. These practices simplify planning, instruction, assessment, and classroom management—but at a cost.

Like its agricultural counterpart, educational monocropping can be detrimental to students' motivation, creativity, and relationships.

CoCo-culture: Cultivating Forest Gardens of Co-creativity and Collaboration

Imagine learning spaces for young people…
- where they engage not only with one another's finished work but also with their emerging ideas and unfolding creative processes, in real time?;
- where influence and inspiration are embraced as natural parts of the learning process, rather than being stigmatized as copying or cheating?;
- where collaboration emerges organically from shared interests, rather than being imposed in fixed groups and outcomes;
- where the environment is kept free from comparative metrics that shift focus from exploration to competition?

What kind of learning culture might emerge in such spaces? These questions have guided the development of the **co-creative learning** paradigm underlying CoCo—one that treats learning as happening not just *with* or *from*, but *through* others.

In this chapter, we explore how this paradigm offers possibilities for a new learning culture—a CoCo-culture—which transforms *standardized fields* of learning into *diverse forest gardens* of co-creativity and collaboration. Beyond any tool or platform, CoCo-culture represents a broader mindset and approach to learning that could emerge in any environment designed with these values.

From knowing to designing

Before we dive in, it is important to note that the problem isn't just that monoculture classrooms exist. It's that they persist even when we know better. Even when educators value creativity, collaboration, and learner agency, we continue to design systems that default to uniformity and isolation. Why? Because the challenge isn't knowing—it's designing.

We don't lack learning philosophies. We lack environments and tools that make those philosophies easy to live. And even when some educators manage to embody these values through extra effort, their practices often remain fragile and un-sustained—dependent on personal commitment rather than shared infrastructure. The defaults

don't change. Co-creation stays exceptional instead of expected. It becomes a special project, a one-off workshop, a teacher "going above and beyond." But for a culture to change, what's good must also become what's easy. The values of CoCo-culture—connection, emergence, shared authorship—are widely appreciated in theory. But they're hard to practice unless they are made easy to enact.

The most powerful force in any system is the default.

What if co-creation was the easy choice? This insight reframes what follows. The contrasts between monoculture and CoCo-culture aren't just a matter of preference or pedagogy. They are design choices. They are defaults we can either reinforce or reimagine.

Contrasting Monoculture and CoCo Culture

Table 1: Monoculture vs. CoCo-culture

Attribute	Monoculture classrooms	CoCo-culture learning spaces
Motivations	Need extrinsic motivational fertilizers	Nurture connection as a pathway into learning
Mindsets	Breed competition-oriented mindsets	Cultivate co-creative and collaborative mindsets
Learning process	Limited cross-pollination leading to uniformity	Abundant inspiration through creative collisions
Type of collaboration	Prescribed collaboration	Inclusive of multiple ways of being creative together
Identities	Individualized identities	Co-constructing new shared identities
Assessment	Focus on assessing standardized outcomes	Emphasizes reflections on co-creative and collaborative process
Educator role	Instructor and manager	Facilitator of co-creative learning process

Rethinking AI in Education: Monoculture or CoCo-culture

If we look closely, most learning environments—online and offline—are built with monocultural assumptions, designed around isolation masquerading as personalization. AI tutors get increasingly adept at adapting to individual learners, yet remain unaware of the collective learning unfolding around them. Platforms track individual progress but rarely attend to the intelligence emerging between students. Assessment systems measure what is learned and created individually but often overlook what can be discovered and created—together. We've become so sophisticated at personalizing learning that we've forgotten learning is fundamentally a social, cultural, and co-creative act.

The metaphor of Monoculture versus CoCo-culture offers a lens for thinking about AI in education. Will AI be used to create hyper-personalized, hyper-isolated learning experiences, where students interact primarily with machines? Or will we use it to design for connection—surfacing relationships between ideas, supporting co-creation, and making shared learning visible?

CoCo envisions the latter. Used wisely, AI can amplify the "co" in co-creation—not just the "c0" of copilots and companion bots. Let's not confuse c0-creating (with machines) with co-creating (with humans). The root "co" stands for our human wholeness—rooted in emotion, shared experience, and meaning. It is through this "co" that we build shared futures.

Chapter 18

Bringing CoCo (and CoCo-culture) to communities around the world

Fig 28: Interest for CoCo from around the world

At the time of writing this dissertation, thousands of educators, parents, and young people from **85+ countries** have signed up to bring CoCo into their classrooms and communities—sharing their excitement for the new possibilities it opens up for collaborative creativity, as well as the underlying ideas and values.

> "You've made a platform that's **human in ways that few platforms even think to consider**."

> "I have honestly not stopped thinking about its **transformative potential to fundamentally change teaching and learning**, and not just in CS classrooms."

"Thrilled about how it's based on **real-time co-presence**! I love that the focus is on a shared creative space, not necessarily explicit collaboration."

"This opens a **whole new way to collaborate** on creative computing projects!"

"Wow! **Love the idea of a "self-less social platform"** for kids."

"I like that CoCo makes learning **collaborative, fun, and human**, and that the environment is intentionally built to **de-emphasize comparison** with others: no profiles, followers, or likes."

"A **gorgeous, creative, and humane online space** for kids!"

This work has been rooted in intuitions, intentions, and values that we deeply care about at a personal level. And it has been wonderful to see how these ideas and values have resonated with others. It makes us feel energized and committed to keep working toward bringing CoCo in the hands of young people everywhere—with the hope that it can play a meaningful part toward helping them experience the CoCo-culture.

"

> If we create, inside of ourselves, a respect for beauty, then we will create beauty outside of ourselves…We might find that what makes us feel peaceful makes others feel peaceful, that what makes us feel sublime makes others feel sublime, that what makes us feel alive makes others feel alive. We might find that what fills us with wonder and hope fills others with wonder and hope. We might find that what is beautiful to us is beautiful to others.
>
> — *Syntax and Sage, Sep Kamvar [27]*

Chapter 19

Extending CoCo | Little Language Models (LLMs): Empowering children to be future AI modelers

Little Language Models [28] is a microworld that we have been building within CoCo to support young people in exploring powerful ideas around Artificial Intelligence. This chapter dives into the details of this work[3]. While the chapter is not directly related to the core themes of this dissertation, we highlight this work to share how we are continuing to extend CoCo in new ways while keeping the underlying co-creative approach at the center of children's learning experiences.

We want to start by stating that…

Future will NOT be shaped by AI; it will be shaped by the modelers of AI.

If children are not empowered to develop creative fluency with the foundational ideas underlying AI, we risk raising a generation that—instead of modeling AI systems in the future—might end up being modeled by them.

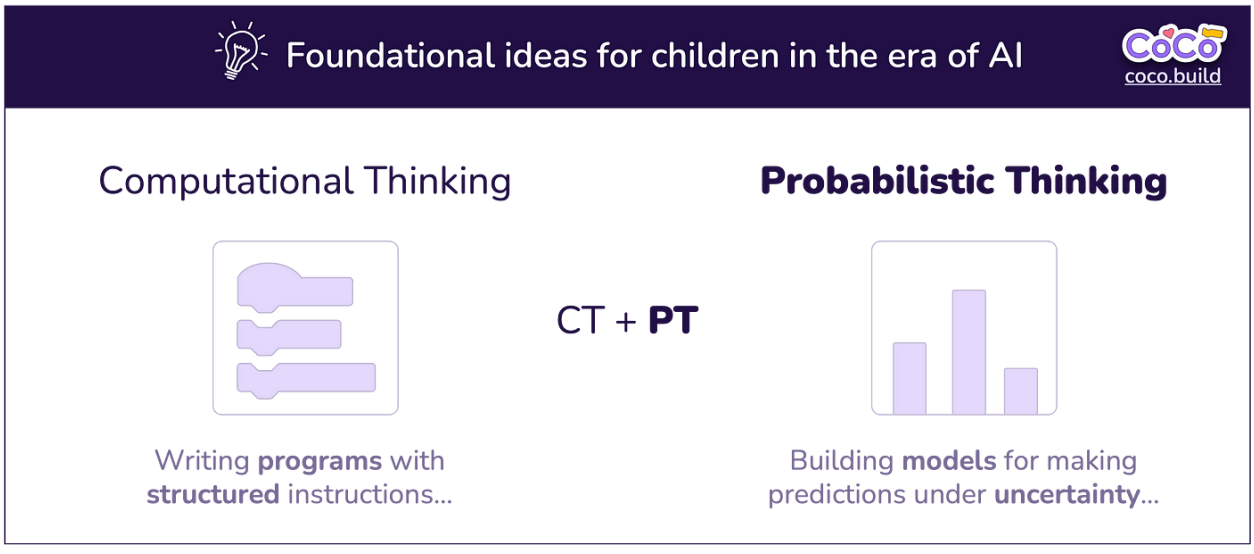
[3] This chapter has been adapted from our original blog post on Medium [29].

We believe it's essential for young people today to not just learn how to 'use' the new *magical* AI tools. They should also have access to tools that help make AI *unmagical* for them—by letting them explore and engage early on with the powerful ideas that underlie these powerful technologies.

💡 Introducing "Probabilistic Thinking" as a new foundational fluency for children in the era of AI

In any era, the most important ideas for children to explore are often the ones that are not narrowly restricted to a particular concept or skill, but instead are foundational to children's understanding of the world. They help them develop new ways of thinking about their own thinking and learning, widely expanding the space of possibilities they can imagine.

In the digital era, *Computational Thinking (CT)* emerged to be one such idea and is now recognized as an essential literacy for learners globally. As we enter the AI era, it is both a critical and opportune time to reassess what types of ideas and fluencies will be most important for young people growing up today. Through this work, we aim to bring attention to a *timeless* foundational idea that has become especially *timely* now—the idea of *Probabilistic Thinking*.

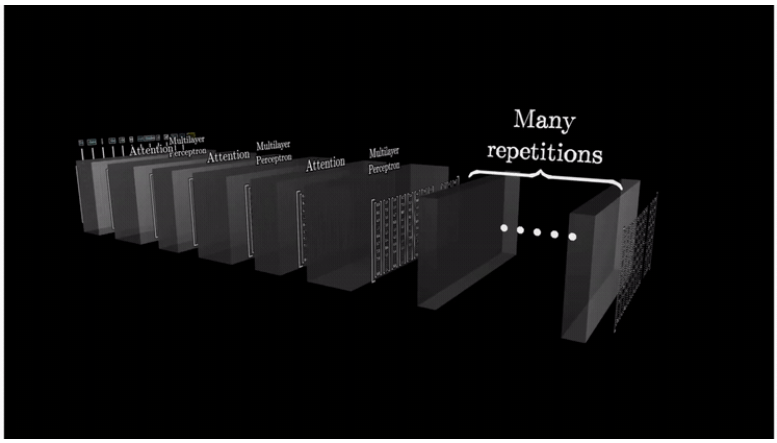


CT + PT as foundational ideas for children in the era of AI

Probabilistic Thinking (PT) can be described as a decision-making approach that involves making predictions under uncertainty by building and using models that learn from data. It provides a powerful framework to understand intelligence and learning processes, both artificial and human.

Just as **CT** underpins computer *programming*, the act of writing structured instructions for computers; **PT** is fundamental to probabilistic *modeling*—the process of building mathematical models that can quantify uncertainties and make predictions by learning from data.

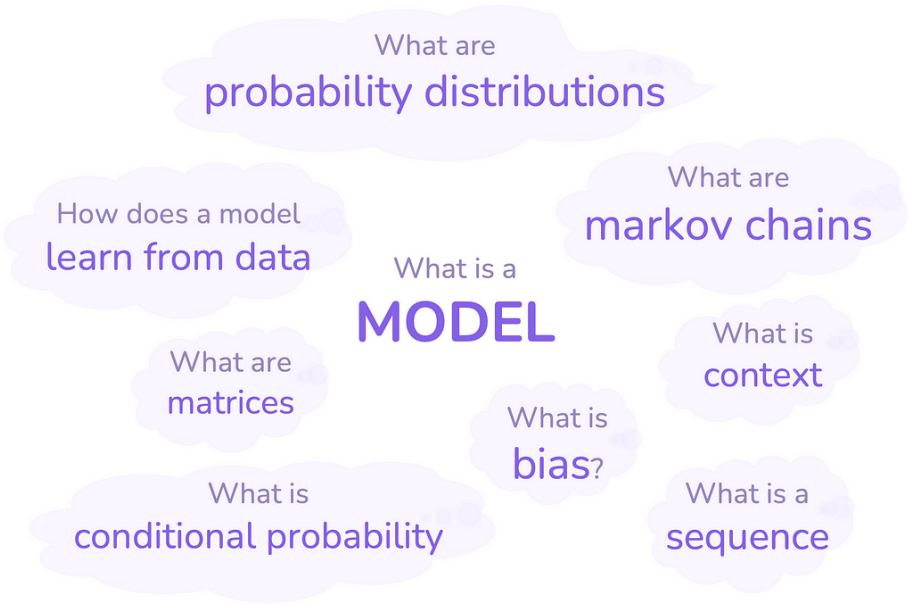
The ideas of Probabilistic Thinking, Modeling, and Learning lie at the core of how Generative AI and Large Language Models (LLMs) like ChatGPT work.



Gif generated from this video by 3Blue1Brown explaining GPT

For instance, when you input a text prompt into ChatGPT, the model is able to generate a response by using probabilities (derived from processing training data) to predict the most likely next word or phrase. This ability makes LLMs incredibly useful and powerful across a wide variety of tasks, including for generating code. And so while the core ideas of computational thinking will continue to hold value, it is evident that the future builders and computer science researchers, will not just be writing *programs* but increasingly engage in tuning, tinkering, and building *models*. How can we empower children to bring this (now ubiquitous) word *'model'* into their own worlds and imaginations so it becomes a part of their thinking and vocabulary?

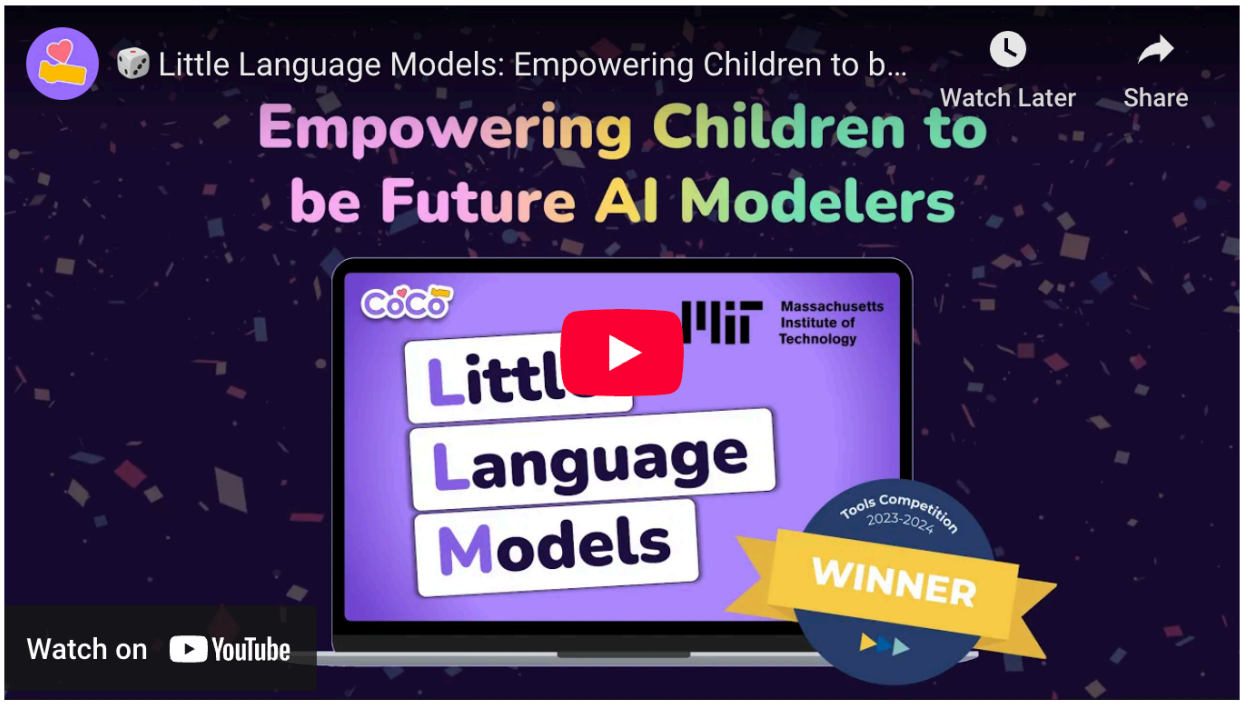
How can we support educators to make the foundational ideas of probabilistic thinking, modeling, and learning accessible and engaging for young people?



Meet 'Little Language Models'—🎲🔤🎨🎸

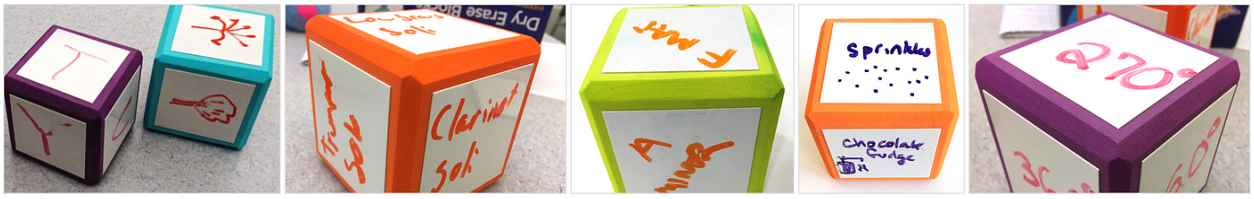
'Little Language Models' (*Little Models* for short) is a co-creative mathematical microworld for children (ages 8–16) within our CoCo platform (coco.build). It is especially designed to be developmentally appropriate for young learners, helping them explore sophisticated probabilistic ideas and concepts underlying AI technologies—in ways that are co-creative, playful, and personally meaningful for them.
The tool offers custom probabilistic blocks and manipulatives that let children *build their own little models using their own little data*. The data can be in any form, including text, images, sounds, etc. Children can dynamically tinker with their models and use them to collaboratively explore new kinds of creative and generative projects with their peers. For instance, they can make models for generative art, stories, or music; build adaptive multiplayer games, or interactive visualizations based on real-time data from their peers, or even advanced projects using Markov blocks such as teaching a computer to learn how to draw, among an endless variety of other possibilities.

In the spirit of *exploration* before *explanation*—here's a short 1-min video that gives a glimpse into the tool. Make sure to turn the sound ON to listen to our special *little models* song 🎙…Disclaimer: The lyrics and voce are **not** AI generated.



A little glimpse into the Little Models microworld: https://youtu.be/zDHdKREA5og

🎲 DICE: A playful, mathematically-rich object to make the abstract idea of a "model" concrete for children



*Ideas become **powerful** only when they become **personal**.*

In conceiving and designing *Little Models*, one of our key insights early on was to ground the abstract idea of a 'model' in a mathematically-rich object that is familiar, concrete, and playful for children: the *dice*. But unlike standard dice that have fixed numbers from 1 to 6 on the sides, the dice in the *Little Models* microworld are designed to be infinitely customizable. The metaphor of physical dice makes it easier for young people to start imagining the little models (or digital dice) they'd like to build in the

environment. We've often been delighted by the diversity of ideas children come up with when they discover that they can use the tool to **make a model for anything they like**—*names, musical notes, or even ice cream flavors!*

Children can create their own digital customizable dice in the environment.

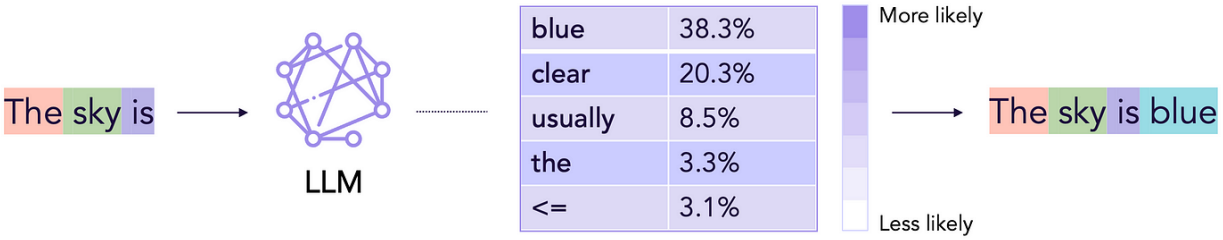
🔍 Building their own "Little" Language Models can help demystify "Large" Language Models for children

In our workshops with young people, we typically follow a two-part structure. The first part involves hands-on explorations with the tool. During this time, children spend time imagining, building, and tinkering with their models to co-create a wide variety of generative projects with their peers in the real-time computational environment of CoCo. In the second part, we engage in a reflective discussion with the group about the key ideas and concepts they explored in the process.

Below, we highlight some of the conceptual, mathematical, algorithmic, and ethical lenses for understanding Generative AI models that we have seen children discover and develop through their creative explorations within the *Little Models* microworld:

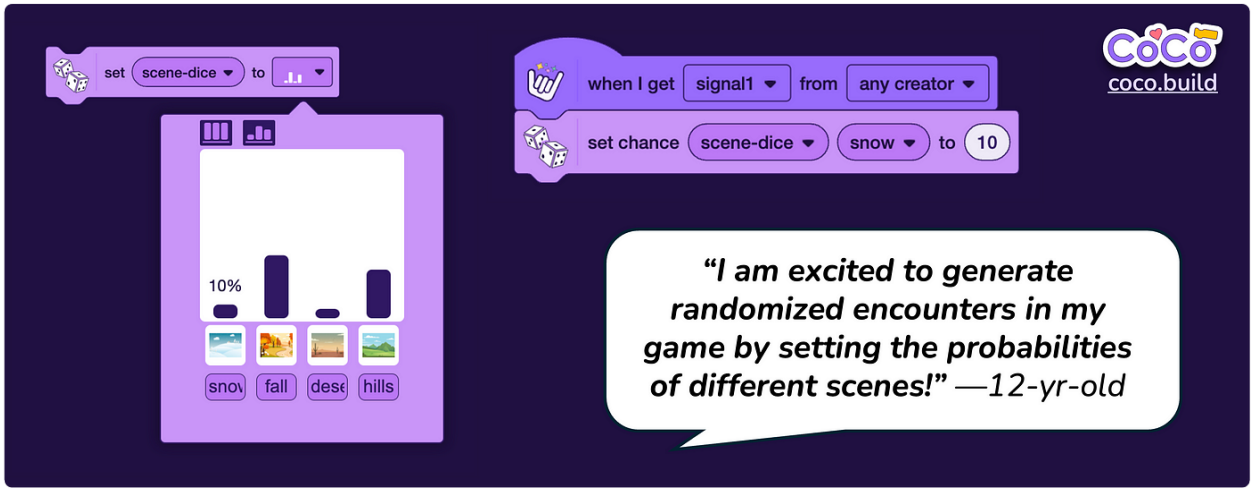
1) 📶 Behind the magic, there's a "Probability Distribution"

"I now know more about how a model works."—13-year-old



Remix of an image from an NVIDIA developer blog about LLMs

The most salient shift we see in students before and after our workshops is in their understanding of the powerful idea of *probability*. Existing approaches to introducing probability in classrooms often reduce it to mere calculation of fractions pertaining to narrow and impersonal events such as coin tosses, lotteries, or spinners. Even when doing workshops with high schoolers, we find that most of them often think of probability only in terms of specific formulaic descriptions. It's not often intuitive to students how it is one of the key ideas underlying Generative AI systems.



Children can dynamically update the probabilities in their models.

In the *Little Models* microworld, children are not only able to create their own dice (or models), but also dynamically update the underlying probability distribution and visualize the effect in real-time in the kind of samples their model generates. They start to see *probability* as a playful clay like material that they can easily tinker with to explore a wide variety of creative possibilities. In the process, they are able to *generalize* the underlying mathematical ideas and see how they can be applied across different

contexts.

2) 🎨 It's not just text, AI can be "Multimodal"

"I used to think of AI as solely chatGPT, but now I see it can have many more capabilities."—14-year-old

Children can make little models using multiple types of data modalities

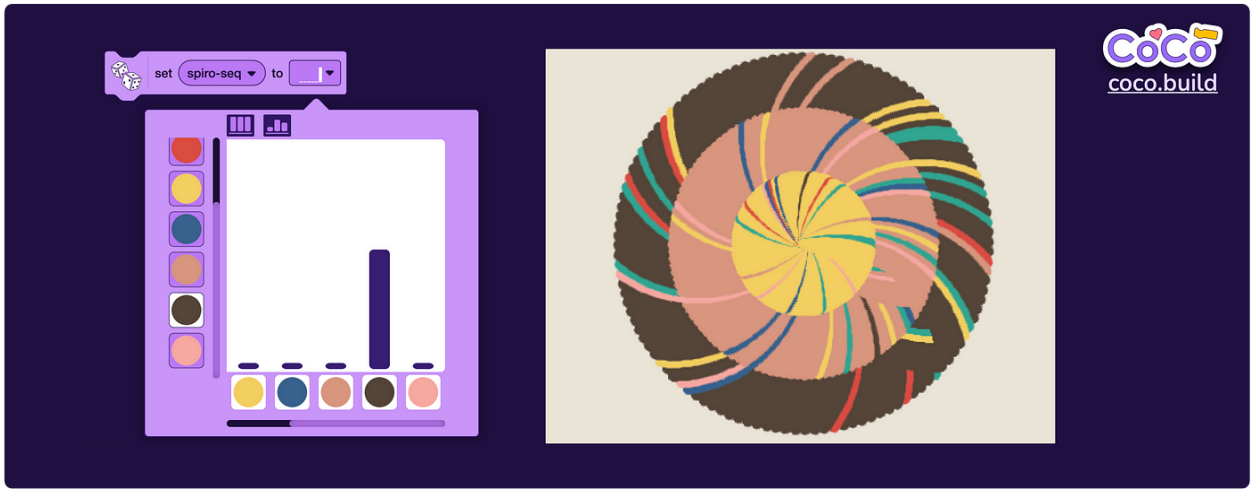
The *Little Models* tool is not limited to supporting only text data. It is designed to support multiple modalities including text, images, sounds, etc. Students are thus able to build generative models using music, text, colors, shapes, images, and also micro:bit sensor values as data. This helps expand their understanding of what constitutes a "language" beyond natural (text-based) languages. They start to see how their expressions made out of musical notes, drawing strokes, or even structures made from LEGO blocks can all be viewed as "languages," each with its own elements and rules for composition. By recognizing these parallels, students are able to better grasp how modern Generative AI systems are able to produce not just coherent text, but also generate images, compose music, or create 3D designs, all by learning the patterns of the underlying "language."

3) 📊 A little bit of "Training" goes a long way

"I had always wondered how things like chatgpt worked."—13-year-old

The *Little Models* microworld is intentionally designed to support learners across ages by providing a scaffolded experience for beginners to get started easily while also enabling sophisticated possibilities for more advanced learners. After creating and playing with simple weighted models, students can progress to building more complex models like Markov models. The intuitive interface allows learners to immediately see how the sequence of elements in their input data "trains" their Markov model and updates its underlying probabilities. This helps them draw parallels with key concepts in Generative AI as mentioned below:

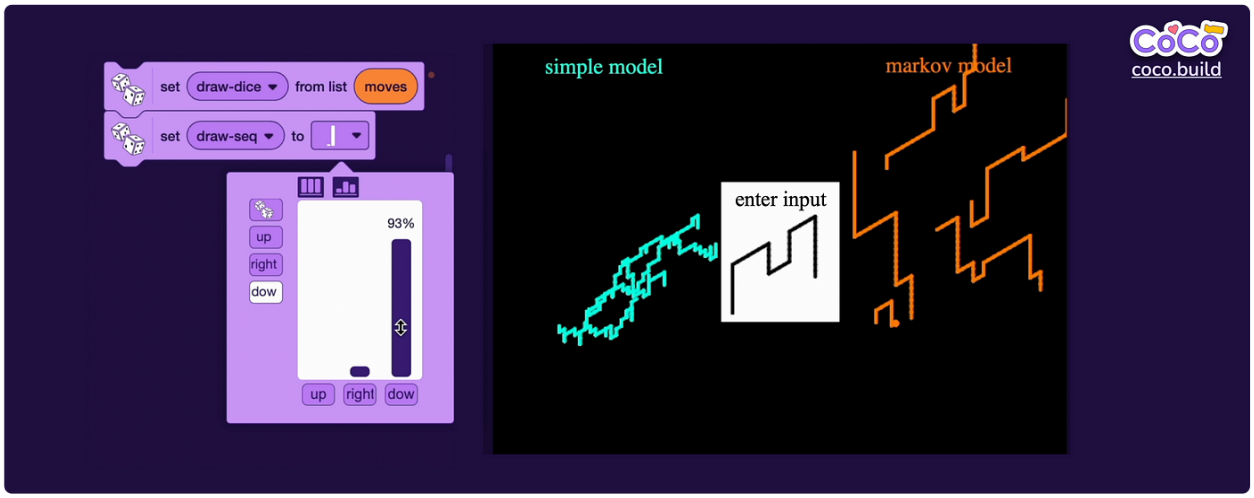
1. Concept of 'Sequence': Children are able to make connections with how Generative AI systems also process and generate sequences of tokens one at a time, forming cohesive outputs such as essays, images, or music. For instance, using the tool, students can enter a sequence of notes and train their model to generate new musical sequences, or enter a sequence of colors to generate new patterns based on that.



Generating patterns based on input sequence of colors

2. Concept of 'Context Length': A defining feature of systems like ChatGPT is their ability to maintain context during a conversation, providing answers by considering a large amount of information (words / tokens) from previous interactions. In contrast, the simple Markov models children create typically have a context length of 1, meaning the model generates the next token based only on the last one. Students are able to explore how even a context length of 1 significantly improves their model's output compared to a simple weighted model with a context length of 0. The same can be

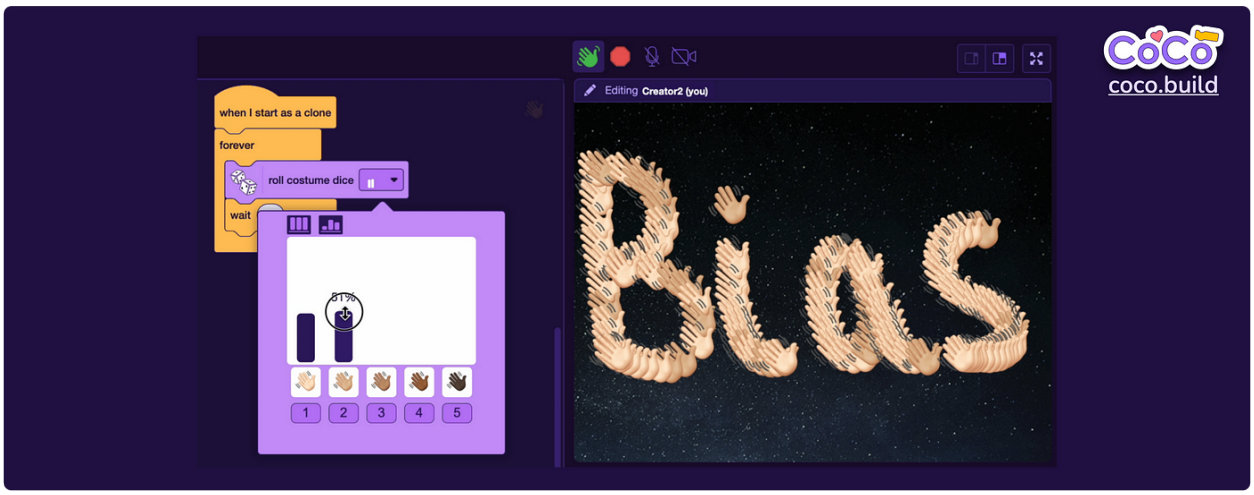
seen in this generative drawing model made by a student by entering a sequence of strokes as training data.



Training a Markov model based on input strokes

4) 🟨 "Bias" in output stems from input

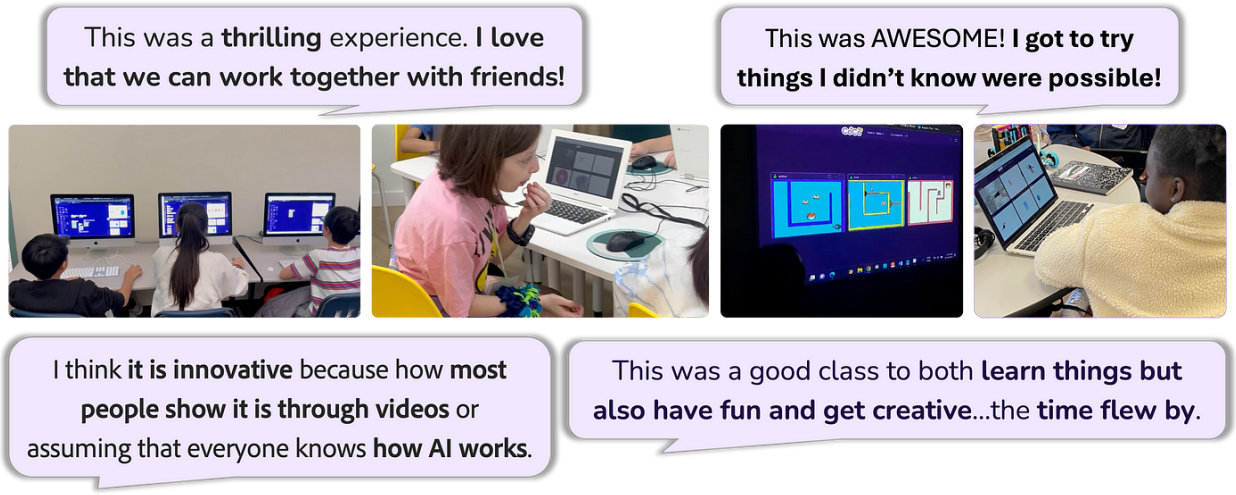
"I have a better sense of what decisions are being made behind the scenes and how that can be mimicked on a smaller scale."—15-year-old



Visualizing 'bias' in a model

One of the critical ideas we've seen students getting to *experience* for themselves is around how *bias* can creep into AI models. The tool makes it concrete and visible to children that the model will draw out samples according to the distribution that they have set. That is, if they have more of yellow in the input data, the chance of rolling a

yellow will be high. This provides a rich and concrete context for educators to have conversations with children about issues of bias and fairness in AI models.



Pilots with students

Through children's feedback in the pilot workshops and our interactions with them, we've seen first-hand how building their own little models with the tool can help them develop interests, intuitions, and insights about how Large Language Models like ChatGPT and others operate. The abstract ideas and terminologies that are used in explaining the workings of AI systems often remain too obscure and inaccessible for young learners (and even adults!). We hope the *Little Models* microworld can serve as a rich playground for educators to support children in developing a language and lens for these big ideas—helping them see AI as *manipulable,* not magical.

💭 Envisioning a new future for AI education: Toward "Intelligetics"

In this final section, we'd like to zoom out and *scribble* some of our thoughts on what the future of AI education can look like. We imagine that we will soon see an

emergence of a new, inclusive, and interdisciplinary field in academia, which will then also be introduced as a core subject in K-12 education, much like math or social science. We propose calling it '*Intelligetics*' (🗣️ Intelli-ge-tics).

Intelligetics will focus on the comprehensive study of intelligence—integrating human intelligence studies with the development and analysis of artificial intelligence systems. It will synthesize powerful ideas from a diverse array of existing disciplines, including: computer science, cognitive science, neuroscience, mathematics, artificial intelligence, philosophy, linguistics, social science, and learning sciences. It will attract people from varied backgrounds with diverse interests. Practitioners in the field—let's call them '*Intelligicians*' (🗣️ Intelli-gi-cians)—will bridge multiple areas of expertise, developing holistic approaches for advancing our understanding and capabilities in both natural and artificial intelligence, as well as their interplays.

Teaching children to be "Intelligicians"

How will we nurture the next generation to be *Intelligicians*? In 1971, Seymour Papert published a memo at the A.I. Laboratory at MIT titled—*Teaching children to be mathematicians vs. Teaching children about mathematics.* He argued that "Children will become more proficient in mathematics if they *do* mathematics rather than merely learn *about* mathematics." We deeply resonate with these words and the underlying constructionist paradigms of teaching and learning they represent.

We see our ideas and work on *Little Language Models* as a little but important first step toward offering children a rich, co-creative microworld where they can actively engage in *doing* AI (or *Intelligetics*) with their peers, rather than merely learning *about* AI passively. As we look ahead, we are excited to continue imagining and building new microworlds that help empower children everywhere to see themselves as future AI builders, scientists, and *Intelligicians*—who have the skills and will to shape the future in ways that best serve humanity.

Fig 29: Children as 'Little Modelers' (image from ChatGPT)

Chapter 20

Looking ahead:
Centering the timeless in timely

Every era brings its own urgencies, its own "timely" opportunities and concerns. But as we collectively look to the future, design for the future, explore the exciting new possibilities that new technologies offer us, and think about what *is* going to change in the era of AI—can we center what *is not* going to change, regardless of any era?

Our enduring needs: Joy. Connection. Expression. Play. Togetherness. Calm. Harmony. And everything else that resembles these. These experiences and values bring us closer to who we are, and we cherish them across generations and geographies. They do not change with the changing technologies. But they do need conscious cultivation and protection.

They need *centering*.

What is centering?

In architectural terms, "centering" refers to a temporary support structure—framing that holds up arches and vaults during construction until they can bear their own weight. Once the structure hardens and finds its strength, the centering is removed, leaving behind something that can stand for centuries.

This dissertation has been, in many ways, an exercise in centering—providing support for ideas that we hope will one day stand on their own without needing any justification. **We actually wish for a future where no one else has to write a dissertation about designing for human connection, co-creativity, and calm.**

Because in that future, these will not be add-ons or good-to-haves—but rather assumed—foundational, built-in—in every creative tool, every learning space, every social platform.

A future where the dominant individual-centric paradigms of today will be outdated. Where being creative will mean being creative *together*—in myriad ways.

A future where the technologies and environments we engage with will make us feel more *present*, more calm, more in tune with ourselves, more in harmony with others—not anxious, disengaged, and disconnected.

A future where the mediums we design serve as microworlds to continually construct, deconstruct, and reconstruct our self-identities—evolving the self into the self-*less*.

A future where to co- will not be against the norm, but the *default*.

In this spirit, we hope the ideas, technologies, possibilities, and frameworks we have shared through this work serve as *centering*—that can help support and shape *everlasting* structures of such futures.

We co-, therefore we are.

The most *timely* work is always in service of the *timeless*.

To co- is *timeless*.

Afterword(s)

Through the years of shared explorations and countless conversations, we have co-evolved a vocabulary—a set of words, framings, and metaphors—that have helped us imagine new possibilities and articulate new ways of thinking, seeing, doing, and being. We hope they help spark new imaginations, possibilities, and lenses for others who encounter them.

To co- is human.
To cO- AI.

Constructing **things,**
Constructing **context**

Knobs
over
Switches

Being. Creative. Together.

Co-creative Learning
creating and learning
THROUGH one another

CoCo (coco.build)
Connection and Construction

From **Monoculture**
to **CoCo-culture**

The 5Co's
Co-imagine
Co-create
Co-clutter
Co-exist
Co-evolve

Passion,
Dispassion,
Compassion

Head, hands, heart,
and **hüm**

Designing
"self-less" social systems

Children as facilitators
of each other's learning

Centering the
timeless in timely

From
future to present

Creating the conditions
for **collision**

May we continue to **co-evolve**
through one another...

• • •

References

[1] M. Resnick, *Lifelong Kindergarten: Cultivating Creativity Through Projects, Passion, Peers, and Play*. MIT Press, 2017.

[2] "Social AI Companions | Common Sense Media." Accessed: May 22, 2025. [Online]. Available: https://www.commonsensemedia.org/ai-ratings/social-ai-companions

[3] S. Papert, *Mindstorms: Children, Computers, and Powerful Ideas*. New York, NY, USA: Basic Books, Inc., 1980.

[4] M. Resnick *et al.*, "Scratch: programming for all," *Communications of the ACM*, vol. 52, no. 11, pp. 60–67, 2009.

[5] S. Dhariwal, "Scratch Memories: visualizations that empower children to celebrate and reflect on their creative trajectories," PhD Thesis, Massachusetts Institute of Technology, 2018.

[6] M. Resnick and B. Silverman, "Some reflections on designing construction kits for kids," in *Proceedings of the 2005 conference on Interaction design and children*, ACM, 2005, pp. 117–122.

[7] S. Papert, *The children's machine: Rethinking school in the age of the computer*. ERIC, 1993.

[8] L. Malaguzzi, "The hundred languages of children," *The hundred languages of children (I cento linguaggi dei bambini. Exhibition catalogue)*, pp. 16–21, 1987.

[9] P. Freire, "Pedagogy of the oppressed," in *Toward a sociology of education*, Routledge, 2020, pp. 374–386. Accessed: Apr. 30, 2024. [Online]. Available: https://api.taylorfrancis.com/content/chapters/edit/download?identifierName=doi&identifierValue=10.4324/9780429339530-34&type=chapterpdf

[10] M. Montessori, *To Educate the Human Potential*. Kalakshetra Publications, 1961.

[11] J. Dewey, *Experience And Education*. Free Press, 1938.

[12] J. HOLT, *HOW CHILDREN FAIL*. 1964.

[13] F. Capra, *Uncommon wisdom: Conversations with remarkable people*. Simon and Schuster, 1988. Accessed: Apr. 30, 2024. [Online]. Available: https://ixtheo.de/Record/1638159998

[14] S. R. Maharshi, *Be as you are: The teachings of Sri Ramana Maharshi*. Penguin UK, 1989.

[15] J. Krishnamurti, *Education and the Significance of Life*. Harper Collins, 2010.

[16] G. Bateson, *Steps to an ecology of mind: Collected essays in anthropology, psychiatry, evolution, and epistemology*. University of Chicago press, 2000. Accessed: Apr. 30, 2024. [Online]. Available: https://books.google.com/books?hl=en&lr=&id=RGq0EAAAQBAJ&oi=fnd&pg=PR5&dq=bateson+steps+to+an+ecology&ots=dPGudAs4KD&sig=X3vDO0mLgmCJf4Ittco9GbpRlm4

[17] G. Wang, *Artful Design: Technology in Search of the Sublime, A MusiComic Manifesto*. Stanford University Press, 2018. Accessed: Oct. 06, 2023. [Online]. Available: https://books.google.com/books?hl=en&lr=&id=9pGyDwAAQBAJ&oi=fnd&pg=PA23&dq=10.%09Wang,+G.+(2018).+Artful+Design:+Technology+in+Search+of+the+Sublime,+A+MusiComic+Manifesto.+Stanford+University+Press.&ots=w_-kdhXgdr&sig=9Iry_vGiEHsVRB91S80Wmqnqnhs

[18] M. Csikszentmihalyi, *Beyond boredom and anxiety*. Jossey-bass, 2000. Accessed: Apr. 30, 2024. [Online]. Available: https://psycnet.apa.org/record/2000-12701-000

[19] "CoCo: A real-time co-creative learning platform for young people," MIT News | Massachusetts Institute of Technology. Accessed: May 22, 2025. [Online]. Available: https://news.mit.edu/2023/coco-real-time-co-creative-learning-platform-young-people-0530

[20] "p5.js | home." Accessed: Sep. 10, 2018. [Online]. Available: https://p5js.org/

[21] J. Hollan and S. Stornetta, "Beyond being there," in *Proceedings of the SIGCHI conference on Human factors in computing systems - CHI '92*, Monterey, California, United States: ACM Press, 1992, pp. 119–125. doi: 10.1145/142750.142769.

[22] N. Rusk, M. Resnick, and S. Cooke, "Origins and guiding principles of the computer clubhouse," *The computer clubhouse: Constructionism and creativity in youth communities*, pp. 17–25, 2009.

[23] C. Rinaldi, "Documentation and assessment: What is the relationship?," in *Beyond listening*, Policy Press, 2005, pp. 17–28. Accessed: May 01, 2024. [Online]. Available: https://bristoluniversitypressdigital.com/downloadpdf/book/9781447342403/ch002.pdf

[24] R. Eglash, J. E. Gilbert, and E. Foster, "Toward culturally responsive computing education," *Commun. ACM*, vol. 56, no. 7, pp. 33–36, Jul. 2013, doi: 10.1145/2483852.2483864.

[25] A. Kohn, *Punished by Rewards: The Trouble with Gold Stars, Incentive Plans, A's, Praise, and Other Bribes*. Houghton Mifflin Company, 1999.

[26] M. Resnick, "All I Really Need to Know (About Creative Thinking) I Learned (by Studying How Children Learn) in Kindergarten," in *Proceedings of the 6th ACM SIGCHI Conference on Creativity & Cognition*, in C&C '07. New York, NY, USA: ACM, 2007, pp. 1–6. doi: 10.1145/1254960.1254961.

[27] S. Kamvar, *Syntax & Sage: Reflections on Software and Nature*. Farmer & Farmer Press, 2015.

[28] S. J. M. page, "Kids are learning how to make their own little language models," MIT Technology Review. Accessed: May 22, 2025. [Online]. Available: https://www.technologyreview.com/2024/10/25/1106168/kids-are-learning-how-to-make-their-own-little-language-models/

[29] CoCo (coco.build), "🎲 Little Language Models: Empowering Children to be Future AI Modelers," MIT MEDIA LAB. Accessed: May 22, 2025. [Online]. Available: https://medium.com/mit-media-lab/little-language-models-empowering-children-to-be-future-ai-modelers-a30e1a5f020a

Acknowledgments

This little bouquet
has the names of
all the people and
all the communities
I am deeply grateful to,
and will forever be.

I hope each of you
can spot your name.

To co- is to be **with**

thank you for
being **with** me

in my heart, in my mind,
and in the spaces between.

in who I am, in who I will be,
and in the me between.

- shruti